Tracts in Mathematics 26

Vincent Guedj
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Degenerate Complex Monge–Ampère Equations



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25 Yves Cornulier and Pierre de la Harpe, *Metric Geometry of Locally Compact Groups*
27 Nicolas Raymond, *Bound States of the Magnetic Schrödinger Operator*

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Introduction

The study of complex Monge–Ampère equations on compact Kähler manifolds is an important part of the interface between complex analysis and differential geometry. Prescribing the Ricci curvature, constructing Kähler–Einstein metrics, or finding metrics of constant scalar curvature are problems that boil down to (or use in an essential way) solving a complex Monge–Ampère equation. Consider for example

$$(\omega + dd^c\varphi)^n = e^{F(\varphi)}\mu,$$

where ω is a Kähler form, $n = \dim_{\mathbb{C}} X$, $\mu = e^h\omega^n$ is a smooth volume form, and F a smooth function: solving the Calabi conjecture boils down to solving this equation when $F \equiv 0$, while constructing Kähler–Einstein metrics corresponds to $F(\varphi) = -\lambda\varphi$, where λ is a real number whose sign is that of the first Chern class of the underlying manifold.

In local coordinates, this equation for an unknown function φ can be written

$$\det\left(\omega_{\alpha\beta} + \frac{\partial^2\varphi}{\partial z_\alpha \partial \overline{z}_\beta}\right) = e^{F(\varphi)+h}\det\left(\omega_{\alpha\beta}\right),$$

where the Kähler form ω is locally written as

$$\omega = \sum_{\alpha,\beta=1}^{n} \omega_{\alpha\beta}\, i dz_\alpha \wedge d\overline{z}_\beta.$$

It has been realized a few decades ago that it is necessary to deal with mildly singular complex algebraic varieties in order to study the birational classification of smooth complex algebraic manifolds in dimension ≥ 3. It is thus desirable to construct canonical metrics on varieties with mild singularities.

If the singularities are sufficiently mild, one can still make sense of the key objects of study (canonical bundle, Ricci curvature, etc). The search for canonical metrics leads one to study degenerate complex Monge–Ampère equations where ω is merely semi-positive and $\mu = fdV$ is absolutely continuous with respect to Lebesgue measure, with density $f \in L^p$, $p > 1$, that might vanish or be unbounded.

The attempt to solve these degenerate complex Monge–Ampère equations runs into severe analytic difficulties, as the solutions are no longer smooth. One needs to introduce new tools, study weak solutions, and establish partial regularity of the latter. This is the main purpose of this book.

What you will find in this book

The book is divided in four parts and sixteen chapters. The *first part* deals with pluripotential theory in domains of $\mathbb{C}^n$, giving a self-contained presentation of Bedford–Taylor Theory initiated in [BT76, BT82]. This theory allows one to define generalized solutions of complex Monge–Ampère equations and has many applications in complex analysis and dynamics. We haven't tried to address all of these; rather, we merely present in detail those results that we use in the sequel.

In the *second part* we transfer and adapt this theory to the context of compact Kähler manifolds. Since there are no plurisubharmonic functions, quasi-plurisubharmonic functions play the leading role. We also introduce and study finite-energy classes, which play a crucial role in the sequel, following [GZ07] (with complements from [BBEGZ11]). Similar notions were developed by Cegrell in the local setting [Ceg98].

The *third part* is devoted to solving degenerate complex Monge–Ampère equations in various ways. We develop a variational technique in finite-energy classes following [BBGZ13], we explore a viscosity approach [EGZ11], we present a detailed proof of several deep a priori L^∞ estimates that extend Yau's and Kolodziej's cornerstone results [Yau78, Kol98], and we eventually establish the smoothness of solutions to some complex Monge–Ampère equations in Zariski open sets.

We give many applications of these results in the *fourth part*, obtaining Yau's solution to the Calabi conjecture [Yau78] as well as its singular extensions, and showing how to construct (singular) Kähler–Einstein metrics on varieties with mild singularities [EGZ09].

There has been an enormous amount of work on pluripotential theory since Bedford and Taylor laid down the foundations fourty years ago. There have also been many important works on Kähler geometry since Yau's resolution of the Calabi conjecture at about the same time. These fields are vast and developing very rapidly, so it should be made clear that this book is not an attempt to survey these developments. There are good surveys on these, including [Bed93, Kis00, Kol05, PS10, PSS12, G12, Dem13, BEG13].

Our selection of material reflects our own taste and limitations. The heart of the book is Part 3, where we solve various degenerate complex Monge–Ampère equations by using pluripotential tools. We have tried to organize the first two parts in such a way as to quickly reach the point where we can efficiently solve these equations.

There are many interesting subjects that we haven't had the time to cover, notably the recent resolution by Chen, Donaldson, and Sun of the Yau–Tian–Donaldson conjecture in the Kähler–Einstein setting [CDS1, CDS2, CDS3]; it would require at least another book to cover this single result!

Prerequisites

This book is an extended version of lecture notes for a graduate course given by the authors at Université Paul Sabatier (Toulouse, France) in 2011–12. There were sixteen four-hour lectures roughly corresponding to the contents of the chapters.

Of course we could not cover all the material presented in this text. On the average, two thirds of each chapter were covered.

There are plenty of *Exercises* at the end of each chapter. But for a few exceptions, their purpose is either to complete the proof of a result which is only sketched in the text, or to add extra information to the material presented in the chapter. These exercises are not easy and are mostly a pretext to encourage the reader to think further and have complementary readings.

We have succeeded in presenting the major part of this material in a year-long graduate course which corresponds to the fifth year of University studies in France (M2R), but the book is so organized that it should be equally convenient for one (or more) course(s) for PhD students having some interest in one of the following topics

- Complex analysis/geometry.
- Non-linear PDEs of geometric origin.
- Geometric analysis and/or differential geometry.

In the book we briefly recall some fundamental facts from complex analysis and Kähler geometry, but it would certainly be preferable (although not absolutely necessary) for the reader either to have some basic knowledge in, or complement with lectures on

- Complex algebraic geometry [GH, Laz, V].
- Minimal Model Program [Deb01, Kollar].
- Riemannian and complex differential geometry [GHL, T].
- Several complex variables [Dem, Horm90].

We recall several notions from complex algebraic and Kähler manifolds in Chapter 7, together with the various definitions of curvatures and canonical metrics in Kähler geometry. We are aware that this chapter is rather heavy for those who have no background in these fields. Apart from a few striking facts like the dd^c-lemma, they are not used intensively in the sequel, which focuses instead mainly on the study of (degenerate) complex Monge–Ampère equations.

Apart from these exceptions, the first three parts of the book are rather self-contained. The only other tools that we take for granted are Hörmander's solution of $\overline{\partial}$ with L^2-estimates (which we use only on a few occasions) and Schauder's theory for linear elliptic equations of second order (as can be found in any standard PDE textbook, e.g., [GT83]). The fourth part is of a more expository nature.

How are complex Monge–Ampère equations solved?

To solve degenerate complex Monge–Ampère equations of the type, say $(\omega+dd^c\varphi)^n = \mu$, one might first try to treat the case when μ is a linear combination of Dirac masses, and then proceed by approximation.

While this approach works quite well for the *real* Monge–Ampère equation [Gut01, Theorem 1.6.2], it fails miserably in the complex case. In general, it's not even clear how to treat the case of a single Dirac mass [CG09]! The difficulty lies in the lack of regularity of quasi-plurisubharmonic functions: while the real Monge–Ampère equation deals with convex functions which are locally Lipschitz, the complex Monge–Ampère equation concerns (quasi-)plurisubharmonic functions, and these are not necessarily continuous (or even bounded). In particular, the complex Monge–Ampère operator is not well defined for all quasi-plurisubharmonic functions, and it is not continuous for the weak topology.

We list below different methods that we develop in the book, starting with the continuity method. It was advocated by Calabi [Cal57] as an approach to solve the Calabi conjecture, and was successfully implemented by Yau [Yau78].

Continuity method The continuity method is a classical tool to try and solve non-linear PDEs. It consists in deforming the PDE of interest into a simpler one for which one already knows the existence of a solution. For the Calabi conjecture, one can use the following path,

$$(\mathrm{CY})_t \qquad (\omega + dd^c\varphi_t)^n = \left[te^h + (1-t)\right]\omega^n,$$

where $0 \leq t \leq 1$ and φ_t is a Kähler potential (i.e., $\omega + dd^c\varphi_t$ is a Kähler form) normalized so that $\int_X \varphi_t\, \omega^n = 0$ (to guarantee uniqueness). The equation of interest corresponds to $t = 1$, while $(\mathrm{CY})_0$ admits the obvious (and unique) solution $\varphi_0 \equiv 0$.

The goal is then to show that the set $S \subset [0,1]$ of parameters for which there is a (smooth) solution is both open and closed in $[0,1]$: since $[0,1]$ is connected and $0 \in S$, it will then follow that $S = [0,1]$, hence $1 \in S$.

The openness follows by linearizing the equation (this involves the Laplace operator associated to $\omega_t = \omega + dd^c\varphi_t$) and using the inverse function theorem. One then needs to establish various a priori estimates to show that S is closed. This step constitutes the core of the proof. We establish these estimates in Chapters 12 and 14 and use them to construct Kähler–Einstein metrics in Chapter 15.

The situation is a bit more delicate when ω is merely semi-positive and big (i.e., with positive volume $\int_X \omega^n > 0$). The Laplace operator Δ_ω is for instance no longer invertible. One can approximate ω by $\omega + \varepsilon\omega_X$, where ω_X is Kähler and $\varepsilon \searrow 0$. When the left-hand side $\mu = f dV$ is moreover degenerate, one can regularize it, use Yau's solution to the Calabi conjecture, and try and pass to the limit. This is the approach used in [EGZ09] to solve the singular Calabi conjecture on mildly singular varieties, as we explain in Chapter 16.

Variational approach For φ a Kähler potential we set

$$\mathcal{F}_\lambda(\varphi) := E(\varphi) + \frac{1}{\lambda}\log\left[\int_X e^{-\lambda\varphi - h}\frac{\omega^n}{V}\right],$$

where $V = \int_X \omega^n = \mathrm{Vol}_\omega(X)$ and

$$E(\varphi) := \frac{1}{(n+1)V}\sum_{j=0}^{n}\int_X \varphi(\omega + dd^c\varphi)^j \wedge \omega^{n-j}$$

is a primitive of the complex Monge–Ampère operator. A function φ is a critical point for $\mathcal{F}_\lambda$ if and only if it satisfies the Euler–Lagrange equation

$$\frac{1}{V}(\omega + dd^c\varphi)^n = \frac{e^{-\lambda\varphi - h}\omega^n}{\int_X e^{-\lambda\varphi - h}\omega^n}.$$

Observe that $\mathcal{F}_\lambda(\varphi + C) = \mathcal{F}_\lambda(\varphi)$, for all $C \in \mathbb{R}$, thus $\mathcal{F}_\lambda$ can be thought of as a functional acting on the metrics $\omega_\varphi := \omega + dd^c\varphi$.

The variational approach consists in trying to extremize the functional $\mathcal{F}_\lambda$ in order to solve the above complex Monge–Ampère equation. This usually requires $\mathcal{F}_\lambda$ to be proper, i.e., $\mathcal{F}(\varphi_j) \to -\infty$ whenever φ_j is a sequence of normalized Kähler potentials with diverging energies $E(\varphi_j) \to -\infty$.

It turns out that $\mathcal{F}_\lambda$ is always proper when $\lambda \leq 0$, but might not be proper when $\lambda > 0$. This difference partially explains why it is more difficult to construct Kähler–Einstein metrics of positive curvature.

When X is a Fano manifold with no holomorphic vector fields and $\omega \in c_1(X)$, a theorem of Tian [Tian97] (with complements by Phong, Song, Sturm, and Weinkove [PSSW08]) states that there exists a Kähler–Einstein metric if and only if the functional $\mathcal{F}_1$ is proper.

We will explain a partial generalization of Tian's theorem in Chapter 11 following [BBGZ13]. This generalization has interesting applications in the study of the long-term behavior of the Kähler–Ricci flow.

Viscosity techniques A standard PDE approach to second-order degenerate elliptic equations is the method of viscosity solutions (see [CIL92] for a survey). This method is local in nature, and solves existence and uniqueness problems for weak solutions very efficiently.

Whereas the viscosity theory for real Monge–Ampère equations has been developed by P.L. Lions and others (see e.g., [IL90]), the complex case was not studied until very recently (see [HL09] for a viscosity approach to the Dirichlet problem for the complex Monge–Ampère equation on hyperconvex domains).

A viscosity approach for complex Monge–Ampère equations on compact complex manifolds has been developed in [EGZ11]. Combining pluripotential and vis-

cosity techniques, this approach yields the existence and uniqueness of continuous solutions to complex Monge–Ampère equations of the type

$$(\omega + dd^c\varphi)^n = v,$$

where X is a compact complex manifold in the Fujiki class, v is a semi-positive probability measure with L^p-density, $p > 1$, and $\omega \geq 0$ is a smooth closed semipositive $(1,1)$-form such that $\int_X \omega^n = 1$.

This method gives an alternative proof of Kolodziej's C^0-theorem which does not depend on [Yau78]. It also easily produces the unique negatively curved singular Kähler–Einstein metric in the canonical class of a projective manifold of general type, a result obtained first in [EGZ09].

We explain this viscosity approach first in a local setting in Chapter 6, and then on compact Kähler manifolds in Chapter 13.

Precise Contents

We now describe more precisely the contents of each chapter.

Chapter 1 We start by reviewing the basic properties of harmonic and subharmonic functions in the complex plane $\mathbb{C}$ (mean value, maximum principle, Dirichlet problem in the unit disc, logarithmic potentials).

We then introduce and study the basic properties of plurisubharmonic functions and give many examples and constructions (infinite series, gluing techniques).

We establish important (Montel type) compacity properties. They will play a key role in the sequel, showing that properly normalized families of (quasi-)plurisubharmonic functions are relatively compact for any of the equivalent L^p-topologies, $p \geq 1$.

Chapter 2 We develop in this chapter the basic theory of positive closed currents, as introduced by Lelong [Lel57, Lel67, Lel69].

We first recall a few facts of de Rham theory of currents. The latter can be seen as differential forms with coefficient distributions.

We define positive forms and their duals, positive currents. Fundamental examples of closed positive currents are currents of integration along (the regular part of) analytic subsets, as well as closed and positive differential forms (smooth currents).

We give several equivalent definitions of Lelong numbers of plurisubharmonic functions and present Siu's fundamental analyticity result. We also explain an important minimum principle due to Kiselman, and a (uniform) integrability result of Skoda. We use these results on several occasions in the book.

Chapter 3 In this chapter we develop the first steps of the theory of weak complex Monge–Ampère operators due to Bedford and Taylor [BT76, BT82].

We establish Chern–Levine–Nirenberg type inequalities that allow to define various currents of Monge–Ampère type. These inequalities eventually lead to the definition of the complex Monge–Ampère operator for bounded plurisubharmonic functions.

We show that the complex Monge–Ampère operator is continuous along monotonic sequences, but it is discontinuous for the L^1-topology.

We establish various maximum (comparison and domination) principles. The basic reference here is [BT82], but we have freely used subsequent simplifications by Demailly [Dem91], Cegrell [Ceg88, Ceg04], Blocki [Blo02] and Kolodziej [Kol05]. We propose a simplified approach to the continuity along increasing sequences.

Chapter 4 The main focus in this chapter is the Monge–Ampère capacity, an important concept again due to Bedford and Taylor.

We review the Choquet theory of generalized capacities, which are set functions generalizing measures, in that they do not satisfy the additivity property.

We study relative extremal functions, i.e., Perron type envelopes, which facilitates the (rarely explicit) computation of the Monge–Ampère capacity.

The Monge–Ampère capacity characterizes pluripolar sets: a set has zero outer Monge–Ampère capacity if and only if it is pluripolar (i.e., contained in the $-\infty$ locus of a plurisubharmonic function).

We then show that plurisubharmonic functions are quasi-continuous with respect to the Monge–Ampère capacity (they are continuous outside a set of arbitrarily small capacity).

The standard reference here is [BT82], but we have also benefited from various lecture notes (e.g. [Kol05]).

Chapter 5 We study in Chapter 5 various Dirichlet problems. A landmark reference here is [BT76].

We start by solving the Dirichlet problem for the Laplace equation and use it to obtain a useful characterization of subharmonic functions (the inspiration here comes from viscosity theory, but the latter is only developed in Chapter 6).

We then study the Perron–Bremermann envelope, showing that it is continuous up to the boundary (with a precise control on the modulus of continuity). This envelope is the maximal subsolution to the Dirichlet problem. To show that it is actually a solution, we first treat the case of the unit ball, and then use a classical balayage technique, following [BT76].

We study several additional Dirichlet problems, allowing the right-hand side to depend on the unknown function and treating more general densities. This study depends heavily on stability estimates (works of Cegrell and Kolodziej).

Chapter 6 A standard PDE approach to second-order degenerate elliptic equations is the method of viscosity solutions introduced in [Lio83] (see [CIL92] for a survey).

This method is local in nature and efficiently solves existence and uniqueness problems for weak solutions.

We develop the viscosity approach for complex Monge–Ampère equations on domains of $\mathbb{C}^n$. We first explain the general strategy and then develop the appropriate notions in the complex case. An important difference with the real case is a lack of symmetry between subsolutions and supersolutions.

The key step is to establish the (local) comparison principle for complex Monge–Ampère equations.

Chapter 7 This chapter reviews, without proof, some of the most important concepts and results from complex algebraic and Kähler geometry. We give various definitions and examples (of complex manifolds, tangent bundles, holomorphic vector bundles, sections and metrics, blow-ups, positive currents, Kähler forms, ample line bundle) and say just a few words about Hodge theory.

Among all these important notions, the *canonical bundle* and the $\partial\overline{\partial}$*-lemma* will be used over and over in the sequel. We only say a few words about Hermitian geometry, normal coordinates, canonical connections and curvatures; the reader may wish to complement this chapter with other readings.

The material surveyed here is classical. More information can be found in [GH, Wells, Dem, T, V].

Chapter 8 We establish in Chapter 8 the first basic properties of quasi-plurisubharmonic functions (uniform integrability, compactness, approximation, regularization, extension).

The material presented here uses the local properties of plurisubharmonic functions and Skoda's integrability theorem, which have been studied in the first part, as well as the Ohsawa–Takegoshi L^2-extension theorem, which is used to show that one can weakly approximate a positive (singular) metric of an ample line bundle by (logarithm of modulus of) holomorphic sections of powers of the latter.

This approximation result is not used in the rest of the book, but the Ohsawa–Takegoshi theorem is one of the most important tools in complex analytic geometry. We highly recommend [Bern10, Dem13] for a good survey and [GZhou15] for the most recent developments.

Chapter 9 We introduce in this chapter two families of global intrinsic capacities: the Monge–Ampère and the Alexander–Taylor capacities. As in the local setting, they both characterize pluripolar sets and can be understood through the use of *extremal functions*, envelopes of special subfamilies of quasi-plurisubharmonic functions.

The only prerequisite here is the local theory of Bedford and Taylor. More precisely one needs the following facts studied in the first part of the book:

- continuity, along monotone sequences, of complex Monge–Ampère operators acting on bounded plurisubharmonic functions,
- solution of the homogeneous Dirichlet problem in a ball,

- the maximum principle $\mathbf{1}_{\{v<u\}}MA(\max(u,v)) = \mathbf{1}_{\{v<u\}}MA(u)$,
- pluripolar sets are the zero sets of the Monge–Ampère capacity.

The material presented here essentially comes from [GZ05].

Chapter 10 We extend the definition of the complex Monge–Ampère operator to a class of unbounded quasi-plurisubharmonic functions, the class $\mathcal{E}(X,\omega)$ of ω-psh functions with finite energy.

We define and study various intermediate classes of finite weighted energy, which interpolate between the class of bounded ω-psh functions and those of finite (unweighted) energy.

We show that the class $\mathcal{E}(X,\omega)$ is the largest class of ω-psh functions for which the complex Monge–Ampère operator is well-defined and the maximum principle holds.

The material here is taken from [GZ07], [CGZ08] and [BBGZ13]. An important source of inspiration was the work of Cegrell [Ceg98, Ceg04], who has developed similar objects in domains of $\mathbb{C}^n$.

Chapter 11 Here we start to solve degenerate complex Monge–Ampère equations. We use a variational method, i.e., we try to extremize various functionals whose Euler–Lagrange equations are the complex Monge–Ampère equations we want to solve.

This requires the functionals to have semi-continuity properties (a delicate issue in some cases) and to be *proper*, a property that is possibly lacking on Fano manifolds and explains the non-existence of Kähler–Einstein metrics, as was proved by Tian in [Tian97].

As we build here weak solutions, another delicate issue is to verify that extremizers are critical points. This is done by using an important property of upper envelopes (the projection theorem) whose idea goes back to the fundamental work of Alexandrov [Ale38].

The material of this chapter comes from [BBGZ13], except for the proof of the projection theorem, which is one of the main results of [BeBo10] by Berman and Boucksom. The uniqueness of solutions in its most general form is due to Dinew [Din09].

Chapter 12 We study here when finite energy solutions of complex Monge–Ampère equations are actually locally bounded (or even continuous) off a divisor. We do this by establishing a uniform a priori estimate due to Kolodziej, which generalizes the celebrated a priori estimate of Yau [Yau78].

Refinements of the arguments (which rely on the maximum principle and the comparison of previously introduced capacities) allow one to show that the solutions are continuous and *stable*: if the Monge–Ampère measures converge one to another (in a reasonably strong sense), so do the (normalized) solutions.

The material here is essentially due to Kolodziej [Kol98], with complements from [EGZ09, GZ12] and [DiNL14].

The arguments to show (Hölder) continuity require the reference cohomology class to be Kähler. The generalized capacities introduced by Di Nezza and Lu provide a useful generalization of Kolodziej's technique. It allows us to handle situations where the potentials are not globally bounded and yields a simple proof of the domination principle.

Chapter 13 In this chapter we develop the viscosity approach to the equation

$$(\mathrm{DMA}_v^\varepsilon) \qquad (\omega + dd^c\varphi)^n = e^{\varepsilon\varphi} v,$$

where ω is a closed smooth real $(1,1)$-form on a n-dimensional connected compact complex manifold X, v is a volume form with non-negative continuous density, and $\varepsilon \in \mathbb{R}_+$.

The global comparison principle lies at the heart of the viscosity approach. Once it is established, Perron's method can be applied to produce viscosity solutions. We establish the global comparison principle for $(\mathrm{DMA}_v^\varepsilon)$ and use it to solve the above degenerate complex Monge–Ampère equation.

We eventually show that many solutions of degenerate complex Monge–Ampère equations obtained in previous chapters are continuous on the ample locus of the reference cohomology class. The material here is taken from [EGZ11].

Chapter 14 Here we establish higher-order a priori estimates for the solutions of complex Monge–Ampère equations. We complete here the proof of the existence of smooth solutions to non-degenerate complex Monge–Ampère equations, a celebrated result due to Yau [Yau78].

We also explain a generalization of Yau's result, due to Szekelyhidi and Tosatti [SzTo11] and Di Nezza and Lu [DiNL14], using an elliptic method which requires a refined $\mathcal{C}^2$-estimate due to Păun [Pau08] (the approach of [SzTo11] uses the Kähler–Ricci flow).

The presentation of the material follows closely the ones in [Blo13, BBEGZ11, DiNL14].

Chapter 15 The purpose of this chapter is to introduce and study the canonical metrics of Kähler geometry: extremal, constant scalar curvature, and Kähler–Einstein metrics.

The problem of constructing the latter boils down to solving certain complex Monge–Ampère equations that were studied in the previous chapters. We thus explain how to solve the Calabi Conjecture (following Yau [Yau78]), and how to construct Kähler–Einstein metrics.

We discuss a Riemannian structure on the infinite-dimensional space of Kähler metrics, as introduced by Mabuchi [Mab87] and revisited by Semmes and Donaldson [Sem92, Don99]. We mention without proof the main results of Chen [Che00, CC02] on weak geodesics and discuss how weak geodesics were used by Berndtsson and Berman [Bern15, BerBer14] to show the uniqueness of constant scalar curvature

Kähler metrics, generalizing works of several authors, notably Bando and Mabuchi [BM87] and Donaldson [Don01].

We state the Yau–Tian–Donaldson conjecture and (very) briefly discuss its resolution by Chen, Donaldson, and Sun [CDS1, CDS2, CDS3] in the Kähler–Einstein Fano case.

Chapter 16 The classification of higher-dimensional complex algebraic manifolds (the so called Mori program or MMP=Minimal Model Program) requires one to work on varieties that are "mildly singular". Fortunately, in the presence of such mild singularities, one can still make sense of the canonical bundle, the Ricci curvature, and the Kähler–Einstein equation.

The latter again boils down to solving a complex Monge–Ampère equation which is degenerate, due to the presence of singularities. One can use a log-resolution of singularities, and so work on a smooth manifold, but a price is paid because the reference cohomology class is then merely semi-positive.

In this chapter we present the various notions of positivity for cohomology classes, and many notions of singularities (notably canonical and Kawamata log terminal singularities) that we are going to consider. We then apply the techniques developed in previous chapters to show the existence of singular Kähler–Einstein metrics.

Basic references for this section are [Deb01, Dem13, KM, Laz] and [EGZ09] for the applications to singular Kähler–Einstein metrics.

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Bonne lecture!

Contents

I The local theory . . . 1

1 Plurisubharmonic functions . . . 3
1.1 Harmonic functions . . . 3
1.1.1 Definitions and basic properties . . . 3
1.1.2 Poisson formula and Harnack's inequalities . . . 5
1.1.3 The maximum principle . . . 6
1.1.4 The Dirichlet problem in the disc . . . 7
1.2 Subharmonic functions . . . 8
1.2.1 Definitions and basic properties . . . 8
1.2.2 The maximum principle . . . 11
1.2.3 The Riesz representation formula . . . 13
1.2.4 Poisson–Jensen formula . . . 17
1.3 Plurisubharmonic functions . . . 18
1.3.1 Basic properties . . . 19
1.3.2 Submean value inequalities . . . 20
1.3.3 Differential characterization . . . 24
1.4 Hartogs lemma and the Montel property . . . 27
1.4.1 Hartogs lemma . . . 27
1.4.2 Comparing topologies . . . 30
1.5 Exercises . . . 33

2 Positive currents . . . 39
2.1 Currents in the sense of de Rham . . . 39
2.1.1 Forms with distributions coefficients . . . 39
2.1.2 Closed currents . . . 40
2.1.3 Bidegree . . . 41
2.2 Positive currents . . . 43
2.2.1 Positive forms . . . 43
2.2.2 Positive currents . . . 46
2.2.3 Examples . . . 49
2.2.4 The trace measure of a positive current . . . 50
2.3 Lelong numbers . . . 52
2.3.1 Lelong numbers of a plurisubharmonic function . . . 52
2.3.2 Invariance properties . . . 55
2.3.3 Siu's Theorem . . . 60
2.4 Skoda's integrability theorem . . . 64
2.4.1 Projective mass . . . 64
2.4.2 The Jensen–Lelong formula . . . 66
2.4.3 A uniform version of Skoda's integrability theorem . . . 69
2.5 Exercises . . . 72

3 The complex Monge–Ampère operator . . . 75
3.1 Currents of Monge–Ampère type . . . 75
3.1.1 Definitions . . . 75
3.1.2 Localization principle . . . 78
3.2 The complex Monge–Ampère measure . . . 80
3.2.1 Chern–Levine–Nirenberg inequalities . . . 80
3.2.2 Symmetry of Monge–Ampère operators . . . 82
3.2.3 Integrability with respect to Monge–Ampère measures . . . 84
3.2.4 Compact singularities . . . 85
3.3 Continuity of the complex Monge–Ampère operator . . . 86
3.3.1 Continuity along decreasing sequences . . . 87
3.3.2 Continuity along increasing sequences . . . 89
3.3.3 Discontinuity of the Monge–Ampère operator . . . 92
3.4 Maximum principles . . . 92
3.4.1 The comparison principle . . . 92
3.4.2 The Lelong class . . . 95
3.5 Exercises . . . 96

4 The Monge–Ampère capacity . . . 101
4.1 Choquet capacity theory . . . 101
4.1.1 Choquet capacities . . . 101
4.1.2 Choquet's capacitability theorem . . . 104
4.1.3 The Monge–Ampère capacity . . . 108
4.2 Continuity of the complex Monge–Ampère operator . . . 110
4.2.1 Quasi-continuity of plurisubharmonic functions . . . 110
4.2.2 Convergence in capacity . . . 112
4.2.3 Continuity of the Monge–Ampère operator . . . 114
4.3 The relative extremal function . . . 117
4.3.1 Definition and Choquet's lemma . . . 117
4.3.2 Capacity and relative extremal function . . . 119
4.4 Small sets . . . 123
4.4.1 Pluripolar sets and extremal functions . . . 123
4.4.2 Negligible sets . . . 125
4.5 Exercises . . . 127

5 The Dirichlet problem . . . 131
5.1 The Dirichlet problem for the Laplace operator . . . 131
5.1.1 The maximum principle . . . 132
5.1.2 Green functions . . . 133
5.1.3 A characterization of subharmonic functions . . . 134
5.2 The Perron–Bremermann envelope . . . 135
5.2.1 A characterization of plurisubharmonicity . . . 135
5.2.2 Perron envelopes . . . 138
5.2.3 Continuity of the envelope . . . 141
5.3 The case of the unit ball . . . 143

5.3.1 $\mathcal{C}^{1,1}$-regularity . . . 143
5.3.2 Solution to the Dirichlet problem . . . 147
5.4 Strictly pseudo-convex domains . . . 149
5.4.1 Continuous densities . . . 149
5.4.2 More general densities . . . 150
5.4.3 Stability estimates . . . 153
5.4.4 More general right-hand side . . . 154
5.4.5 Further results . . . 155
5.5 Exercises . . . 156

6 Viscosity solutions . . . 161
6.1 The viscosity approach to degenerate non-linear PDEs . . . 161
6.1.1 Classical solutions . . . 161
6.1.2 Examples . . . 162
6.1.3 Viscosity concepts . . . 164
6.1.4 Jet differentials . . . 167
6.1.5 Jensen's maximum principle . . . 169
6.2 Degenerate complex Monge–Ampère equations . . . 170
6.2.1 Viscosity subsolutions of $(dd^c\varphi)^n = \mu$ . . . 170
6.2.2 Viscosity subsolutions of $(dd^c\varphi)^n = e^{\varepsilon\varphi}\mu$ . . . 173
6.2.3 Supersolutions of the equation $(dd^c\varphi)^n = e^{\varepsilon\varphi}\mu$ . . . 177
6.3 The viscosity comparison principle . . . 179
6.4 Perron's method . . . 182
6.5 Exercises . . . 184

II Pluripotential theory on compact manifolds . . . 189

7 Compact Kähler manifolds . . . 191
7.1 Examples of compact complex manifolds . . . 191
7.1.1 Tangent spaces . . . 191
7.1.2 Quotients . . . 192
7.1.3 Blow-ups . . . 193
7.1.4 Meromorphic maps . . . 194
7.2 Holomorphic vector bundles . . . 196
7.2.1 Holomorphic line bundles and divisors . . . 196
7.2.2 Holomorphic sections . . . 199
7.2.3 Kodaira embedding theorem . . . 200
7.2.4 Canonical bundle and Kodaira dimension . . . 201
7.3 Hodge theory . . . 203
7.3.1 Positive forms and currents . . . 203
7.3.2 Hodge decomposition theorem . . . 206
7.3.3 The $\partial\overline{\partial}$-lemma . . . 207
7.4 Curvatures of metrics . . . 207
7.4.1 Hermitian and Kähler metrics . . . 207

7.4.2 The Chern–Levi-Civita connection . . . 208
7.4.3 Curvatures . . . 209
7.5 Exercises . . . 211

8 Quasi-plurisubharmonic functions . . . 215
8.1 Topology . . . 215
8.1.1 Definition . . . 215
8.1.2 Compactness properties . . . 217
8.2 Examples . . . 218
8.2.1 Positive metrics of holomorphic lines bundles . . . 218
8.2.2 Extension property . . . 220
8.2.3 Approximation by holomorphic sections . . . 221
8.3 Integrability properties . . . 223
8.3.1 Skoda's integrability theorem . . . 223
8.3.2 Gradient of a qpsh function . . . 225
8.4 Regularization . . . 227
8.4.1 Elementary smoothing . . . 228
8.4.2 Demailly estimates . . . 228
8.4.3 Regularity of upper envelopes . . . 230
8.5 Exercises . . . 231

9 Envelopes and capacities . . . 235
9.1 Complex Monge–Ampère operators . . . 235
9.1.1 Dirichlet problem . . . 235
9.1.2 Maximum principles . . . 236
9.1.3 Choquet's lemma . . . 237
9.2 The Monge–Ampère capacity . . . 237
9.2.1 Chern–Levine–Nirenberg inequalities . . . 237
9.2.2 Quasicontinuity . . . 240
9.3 Outer regularity . . . 242
9.3.1 The relative extremal function . . . 242
9.3.2 A twisted formula . . . 243
9.4 Alexander–Taylor capacity . . . 245
9.4.1 Global extremal functions . . . 245
9.4.2 Alexander–Taylor capacity . . . 248
9.4.3 Chebyshev constants . . . 250
9.4.4 The weighted theory . . . 251
9.5 Exercises . . . 253

10 Finite energy classes . . . 257
10.1 The class $\mathcal{E}(X,\omega)$ . . . 257
10.1.1 Definition . . . 257
10.1.2 Comparison principles . . . 259
10.2 Weighted energies . . . 263
10.2.1 The classes $\mathcal{E}_\chi(X,\omega)$ . . . 263

10.2.2 Continuity of twisted Monge–Ampère operators 266
10.2.3 Mixed Monge–Ampère measures 267
10.2.4 Compactness properties 268
10.2.5 Integrability properties 269
10.3 The class $\mathcal{E}^1(X,\omega)$ 270
10.3.1 Primitive of the Monge–Ampère operator 271
10.3.2 Capacity of sublevel sets 272
10.3.3 Convergence in energy 273
10.4 Defining the Monge–Ampère operator 278
10.4.1 The local domain of definition 279
10.4.2 The global domain of definition 280
10.4.3 A characterization of $\mathcal{E}(X,\omega)$ 281
10.5 Exercises 282

III Solving complex Monge–Ampère equations 287

11 The variational approach 289
11.1 The $\mathcal{F}_\lambda$-functionals 289
11.1.1 Euler–Lagrange equations 289
11.1.2 Upper semi-continuity 290
11.2 Solutions in $\mathcal{E}^1(X,\omega)$ 293
11.2.1 Maximizers are critical points 293
11.2.2 Properness of the functionals 295
11.2.3 Solutions in $\mathcal{E}^1(X,\omega)$ 296
11.2.4 The projection theorem 298
11.3 More solutions 299
11.3.1 A properness criterion when $\lambda > 0$ 299
11.3.2 Non-pluripolar measures 302
11.3.3 The classes $\mathcal{E}^p(X,\omega)$ 304
11.4 Uniqueness 305
11.4.1 Calabi's trick 305
11.4.2 Uniqueness in $\mathcal{E}^1(X,\omega)$ 306
11.4.3 Uniqueness in $\mathcal{E}(X,\omega)$ 307
11.5 Exercises 311

12 Uniform a priori estimates 315
12.1 Bounded solutions 315
12.1.1 Comparison of capacities 315
12.1.2 Josefson's theorem 317
12.1.3 Domination by capacity 318
12.1.4 Proof of Theorem 12.1 319
12.2 Continuity of the solutions 321
12.2.1 Refinements of previous results 321
12.2.2 Continuity 323

12.2.3 Relative bounds . . . 324
12.3 Stability of the solutions . . . 327
12.3.1 Qualitative stability . . . 327
12.3.2 Strong stability . . . 330
12.4 Further results . . . 332
12.4.1 Hölder continuous solutions . . . 332
12.4.2 More capacities . . . 335
12.5 Exercises . . . 338

13 The viscosity approach . . . 343
13.1 Definitions . . . 343
13.1.1 Subsolutions . . . 344
13.1.2 (Super)solutions . . . 345
13.2 The global viscosity comparison principle . . . 345
13.2.1 A warm up . . . 345
13.2.2 The global maximum principle . . . 346
13.2.3 Perron's method . . . 351
13.3 Regularity of potentials of singular K-E metrics . . . 352
13.3.1 The case $\lambda = -\varepsilon < 0$ . . . 352
13.3.2 The case $\lambda = \varepsilon = 0$ . . . 355
13.3.3 The case $\lambda > 0$ . . . 357
13.4 Concluding remarks . . . 357
13.4.1 More continuous solutions . . . 357
13.4.2 More comparison principles . . . 358
13.5 Exercises . . . 360

14 Smooth solutions . . . 363
14.1 The continuity method . . . 363
14.1.1 Openness . . . 364
14.1.2 The strategy . . . 365
14.2 A priori $\mathcal{C}^2$ estimates . . . 366
14.2.1 Preliminaries . . . 366
14.2.2 A warm up . . . 369
14.2.3 The refined estimate . . . 370
14.3 Higher-order estimates . . . 371
14.3.1 Complex Evans–Krylov method . . . 371
14.3.2 Schauder estimates . . . 376
14.3.3 Proof of Theorem 14.1 . . . 378
14.3.4 More general equations . . . 379
14.4 The case of semi-positive forms . . . 379
14.4.1 Uniform estimate . . . 380
14.4.2 Laplacian estimate . . . 381
14.5 Exercises . . . 383

IV Singular Kähler–Einstein metrics . . . 387

15 Canonical metrics . . . 389
15.1 Canonical metrics . . . 389
15.1.1 Extremal metrics . . . 389
15.1.2 The Calabi–Yau theorem . . . 392
15.1.3 Kähler–Einstein metrics . . . 393
15.2 The Riemannian space of Kähler metrics . . . 395
15.2.1 The Riemannian structure . . . 396
15.2.2 Dirichlet problem for the Monge–Ampère equation . . . 398
15.2.3 The Mabuchi distance . . . 401
15.2.4 Convexity of functionals along geodesics . . . 403
15.3 Fano manifolds . . . 406
15.3.1 Classification . . . 406
15.3.2 Bochner formula . . . 406
15.3.3 Holomorphic vector fields . . . 408
15.4 The Yau–Tian–Donaldson conjecture . . . 411
15.4.1 Del Pezzo surfaces . . . 411
15.4.2 K-stability . . . 411
15.4.3 The Kähler–Einstein case . . . 413
15.5 Exercises . . . 414

16 Singularities and the Minimal Model Program . . . 419
16.1 Positivity properties of cohomology classes . . . 419
16.1.1 Algebraic notions . . . 419
16.1.2 Metric notions . . . 421
16.1.3 Positivity of the canonical bundle . . . 422
16.1.4 Examples . . . 423
16.2 Mildly singular varieties . . . 426
16.2.1 The Minimal Model Program . . . 426
16.2.2 $\mathbb{Q}$-factorial varieties . . . 427
16.2.3 Canonical and log terminal singularities . . . 430
16.2.4 Pairs . . . 431
16.2.5 Surfaces . . . 432
16.3 Kähler geometry on normal spaces . . . 433
16.3.1 Kähler metrics on normal spaces . . . 433
16.3.2 Adapted volume forms . . . 435
16.3.3 Estimating Lelong numbers in log-resolutions . . . 439
16.4 Singular Kähler–Einstein metrics . . . 442
16.4.1 Singular Ricci curvature . . . 442
16.4.2 Varieties of general type . . . 443
16.4.3 Calabi–Yau varieties and pairs . . . 444
16.4.4 Fano varieties . . . 447
16.5 Exercises . . . 449

Bibliography 451

Index 469

Part I

The local theory

Plurisubharmonic functions have been introduced independently in 1942 by Pierre Lelong in France and Kiyoshi Oka in Japan.

Oka used them to define pseudoconvex sets and solved the Levi problem in dimension two. Lelong established their first properties and asked influential questions, some of which remained open for decades. These problems have been eventually solved by Bedford and Taylor in two landmark papers [BT76, BT82] which laid down the foundations of the now called *pluripotential theory*.

Our purpose in this first part is to develop the first steps of Bedford–Taylor Theory. We haven't tried to make an exhaustive presentation, we merely present those results that we use in the sequel of the book, when adapting this theory to the setting of compact Kähler manifolds.

The readers already familiar with pluripotential theory in domains of $\mathbb{C}^n$ can skip this first part. We encourage those who wish to learn more about it to consult the excellent surveys that are available, notably [Sad81, Bed93, Ceg88, Dem91, Kis00, Klim, Blo02, Kol05].

1 Plurisubharmonic functions

Plurisubharmonic functions are in many ways analogous to convex functions. They relate to subharmonic functions of one complex variable like convex functions of several variables do to convex functions of one real variable. On the other hand, plurisubharmonic functions can have singularities (they are not necessarily continuous, nor even locally bounded). This makes questions of local regularity much trickier than for convex functions.

One needs infinitely many (submean value) inequalities to define plurisubharmonic functions; this is best expressed in terms of currents (differential forms with distribution coefficients): a function φ is plurisubharmonic if and only if the current $dd^c\varphi$ is positive.

In this chapter we establish basic properties of plurisubharmonic functions. We first recall that harmonic functions are characterized by mean value equalities, briefly review the definition and properties of subharmonic functions in the plane, and then move on to study plurisubharmonic functions (defined as upper semi-continuous functions whose restriction to any complex line is subharmonic).

We give several examples and establish important compactness and integrability properties of families of plurisubharmonic functions.

1.1 Harmonic functions

1.1.1 Definitions and basic properties Let $\Omega \subset \mathbb{R}^2 \simeq \mathbb{C}$ be a domain.Recall that a function $h : \Omega \to \mathbb{R}$ is *harmonic* if h is C^2-smooth and satisfies the Laplace equation

$$\Delta h = 0$$

in Ω, where

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4\frac{\partial^2}{\partial z \partial \bar{z}},$$

is the Laplace operator in $\mathbb{C} = \mathbb{R}^2$.

It follows from the Cauchy–Riemann equations that if $f : \Omega \to \mathbb{C}$ is a holomorphic function, then its real part $h = \Re f$ is harmonic. The Cauchy formula shows that f has the mean value property: for any closed disc $\overline{\mathbb{D}}(a, r) \subset \Omega$,

$$f(a) = \int_0^{2\pi} f(a + re^{i\theta})\frac{d\theta}{2\pi}. \tag{1.1.1}$$

Conversely, any harmonic function is locally the real part of a holomorphic function, hence harmonic functions satisfy the mean value property. The latter actually characterizes harmonic functions:

Proposition 1.1. *Let $h : \Omega \to \mathbb{R}$ be a continuous function in Ω. The following properties are equivalent:*

(i) *h is harmonic in Ω;*

(ii) *for any $a \in \Omega$ and any disc $\mathbb{D}(a,r) \subset \Omega$ there is a holomorphic function f in the disc $\mathbb{D}(a,r)$ such that $h \equiv \Re f$ in $\mathbb{D}(a,r)$;*

(iii) *h satisfies the mean value property (1.1.1) at each point $a \in \Omega$ and for any $r > 0$ such that $\overline{\mathbb{D}}(a,r) \subset \Omega$;*

(iv) *h satisfies the mean value property (1.1.1) at each point $a \in \Omega$, for $r > 0$ small enough.*

In particular, harmonic functions are real analytic hence C^∞-smooth.

Proof. We first show the implication (i) $\Longrightarrow$ (ii). We need to prove that for a fixed disc $D = \mathbb{D}(a,r) \Subset \Omega$ there exists a smooth function g in D such that $h + ig$ is holomorphic in D. This boils down to solving the equation

$$dg = -\frac{\partial h}{\partial y}dx + \frac{\partial h}{\partial x}dy =: \alpha$$

in D. The 1-form α is closed in Ω since h is harmonic,

$$d\alpha = \Delta h\, dx \wedge dy \equiv 0.$$

The existence of g therefore follows from Poincaré's lemma.

The implication (ii) $\Longrightarrow$ (iii) follows from the Cauchy formula as we have already observed, while the implication (iii) $\Longrightarrow$ (iv) is obvious.

It remains to show (iv) $\Longrightarrow$ (i). We first prove that h is actually smooth in Ω. Let $\rho : \mathbb{C} \to \mathbb{R}^+$ be a radial function with compact support in the unit disc $\mathbb{D}$ such that $\int_{\mathbb{C}} \rho(z)\, d\lambda(z) = 2\pi \int_0^1 \rho(r) r dr = 1$. We consider, for $\varepsilon > 0$,

$$\rho_\varepsilon(z) := \varepsilon^{-2}\rho(z/\varepsilon), \quad \text{so that} \quad \int_{\mathbb{C}} \rho_\varepsilon(z) d\lambda(z) = 1.$$

Set $h_\varepsilon := h \star \rho_\varepsilon$ for $\varepsilon > 0$ small enough. These functions are smooth and we claim that $h_\varepsilon = h$ in $\Omega_\varepsilon = \{z \in \Omega \,|\, \operatorname{dist}(z, \partial\Omega) > \varepsilon\}$ for $\varepsilon > 0$ small enough. Indeed, integrating in polar coordinates and using the mean value property for h we get

$$h_\varepsilon(a) = \int_0^1 r\rho(r)dr \int_0^{2\pi} h(a + \varepsilon r e^{i\theta})d\theta = 2\pi h(a) \int_0^1 r\rho(r)dr = h(a).$$

Therefore, $h = h_\varepsilon$ is smooth in Ω_ε.

Fix now $a \in \Omega$ and use the Taylor expansion of h in a neighborhood of a: for $|z - a| = r \ll 1$:

$$h(z) = h(a) + \Re P(z-a) + \frac{r^2}{2}\Delta h(a) + o(r^2),$$

where P is a quadratic polynomial in z such that $P(0) = 0$. Thus,

$$\frac{1}{2\pi}\int_0^{2\pi} h(a + re^{i\theta})d\theta = h(a) + \frac{r^2}{2}\Delta h(a) + o(r^2),$$

hence

$$\Delta h(a) = \lim_{r\to 0^+} \frac{2}{r^2}\left(\int_0^{2\pi} h(a + re^{i\theta})\frac{d\theta}{2\pi} - h(a)\right) = 0,$$

by the mean value property. Thus h is harmonic in Ω. □

Let $\mathcal{D}(\Omega)$ denote the space of smooth functions with compact support in Ω, and $\mathcal{D}'(\Omega)$ the space of distributions (continuous linear forms on $\mathcal{D}(\Omega)$). Recall that a function $f \in L^1_{\text{loc}}(\Omega)$ defines a distribution $T_f \in \mathcal{D}'(\Omega)$ by

$$T_f : \chi \in \mathcal{D}(\Omega) \longmapsto \int_\Omega \chi f d\lambda \in \mathbb{R},$$

where $d\lambda$ denotes the Lebesgue measure in Ω.

Weyl's lemma shows that harmonic distributions are induced by harmonic functions:

Lemma 1.2. *Let $T \in \mathcal{D}'(\Omega)$ be a harmonic distribution on Ω. Then there is a unique harmonic function h in Ω such that $T = T_h$.*

Proof. Consider radial mollifiers $(\rho_\varepsilon)_{\varepsilon>0}$ as above and set $T_\varepsilon := T \star \rho_\varepsilon$. Then T_ε is a smooth function in Ω_ε which satisfies $\Delta T_\varepsilon = (\Delta T) \star \rho_\varepsilon = 0$ in Ω_ε, hence it is a harmonic function in Ω_ε.

The proof of the previous proposition shows that for $\varepsilon, \eta > 0$,

$$T_\varepsilon = T_\varepsilon \star \rho_\eta = T_\eta \star \rho_\varepsilon = T_\eta$$

weakly in $\Omega_{\varepsilon+\eta}$. Letting $\varepsilon \to 0$ we obtain $T = T_\eta$ in the weak sense of distributions in Ω_η. Therefore, as $\eta \to 0^+$ the functions T_η glue into a unique harmonic function h in Ω such that $T = T_h$ in Ω. □

1.1.2 Poisson formula and Harnack's inequalities

The *Poisson formula* is a reproducing formula for harmonic functions:

Proposition 1.3. (*Poisson formula*). *Let $h : \overline{\mathbb{D}} \to \mathbb{R}$ be a continuous function which is harmonic in $\mathbb{D}$. Then for all $z \in \mathbb{D}$*

$$h(z) = \frac{1}{2\pi}\int_0^{2\pi} h(e^{i\theta})\frac{1 - |z|^2}{|e^{i\theta} - z|^2}d\theta.$$

Proof. We reduce to the case when h is harmonic in a neighborhood of $\overline{\mathbb{D}}$ by considering $z \mapsto h(rz)$ for $0 < r < 1$ and letting r increase to 1 in the end.

For $z = 0$ the formula above is the mean value property in the unit disc. Fix $a \in \mathbb{D}$ and let f_a be the automorphism of $\mathbb{D}$ sending 0 to a,

$$f_a(z) := \frac{z + a}{1 + \bar{a}z}.$$

The function $h \circ f_a$ is harmonic in a neighborhood of $\overline{\mathbb{D}}$, hence

$$h(a) = h \circ f_a(0) = \int_{\partial\mathbb{D}} h \circ f_a(z) d\sigma$$

The change of variables $\zeta = f_a(z)$ yields $z = f_{-a}(\zeta)$ and

$$h(a) = \frac{1}{2\pi} \int_0^{2\pi} h(e^{i\theta}) \frac{1 - |a|^2}{|1 - \bar{a}e^{i\theta}|^2} d\theta,$$

as desired. □

The following are called Harnack's inequalities:

Corollary 1.4. *Let $h : \overline{\mathbb{D}} \to \mathbb{R}^+$ be a non-negative continuous function which is harmonic in $\mathbb{D}$. For all $0 < \rho < 1$ and $z \in \mathbb{D}$ with $|z| = \rho$, we have*

$$\frac{1 - \rho}{1 + \rho} h(0) \leq h(z) \leq \frac{1 + \rho}{1 - \rho} h(0). \qquad \square$$

Proof. Fix $z \in \mathbb{D}$ such that $|z| = \rho$ and observe that

$$\frac{1 - \rho}{1 + \rho} \leq \frac{1 - |z|^2}{|e^{i\theta} - z|^2} \leq \frac{1 + \rho}{1 - \rho}.$$

Since $h \geq 0$ in $\partial\mathbb{D}$, we can multiply these inequalities by $h(e^{i\theta})$ and integrate over the unit circle. Poisson formula and the mean value property for h thus yield

$$\frac{1 - \rho}{1 + \rho} h(0) \leq h(z) \leq \frac{1 + \rho}{1 - \rho} h(0). \qquad \square$$

1.1.3 The maximum principle

Harmonic functions satisfy the following fundamental maximum principle:

Theorem 1.5. *Let $h : \Omega \to \mathbb{R}$ be a harmonic function.*

1. *If h admits a local maximum at some point $a \in \Omega$, then h is constant in a neighborhood of a.*
2. *For any bounded subdomain $D \Subset \Omega$ we have*

$$\max_{\bar{D}} h = \max_{\partial D} h.$$

Moreover, $h(z) < \max_{\partial D} h$ for all $z \in D$ unless h is constant.

Proof. 1. Assume there is a disc $\mathbb{D}(a,r) \subset \Omega$ s.t. $h(z) \leq h(a)$ for all $z \in \mathbb{D}(a,r)$. Fix $0 < s < r$ and note that $h(a) - h(a + se^{i\theta}) \geq 0$ for all $\theta \in [0, 2\pi]$. The mean value property yields

$$\int_0^{2\pi} \Big(h(a) - h(a + se^{i\theta}) \Big) \, d\theta = 0.$$

Since h is continuous we infer $h(a) - h(a + se^{i\theta}) = 0$ for all $\theta \in [0, 2\pi]$, hence h is constant, as claimed.

2. By compactness, there exists $a \in \bar{D}$ such that $h(a) = \max_{\bar{D}} h$. If $a \in D$, the previous case shows that h is constant in a neighborhood of a. Therefore the set $A := \{z \in D; h(z) = h(a) = \max_{\bar{D}} h\}$ is open, non-empty and closed (by continuity). We infer $A = D$, hence h is constant in D. □

1.1.4 The Dirichlet problem in the disc Let $\Omega \Subset \mathbb{C}$ be a bounded domain and $\phi : \partial\Omega \to \mathbb{R}$ a continuous function (the boundary data). The Dirichlet problem for the homogeneous Laplace equation consists in finding a harmonic function $h : \Omega \to \mathbb{R}$ that is a solution of the following linear PDE with prescribed boundary values:

$$\text{DirMA}(\Omega, \phi) \quad \begin{cases} \Delta h = 0 \text{ in } \Omega, \\ h|_{\partial\Omega} = \phi. \end{cases}$$

By the maximum principle, if a solution exists, it is unique. We only treat here the case when Ω is the unit disc

$$\mathbb{D} := \{\zeta \in \mathbb{C} | \, |\zeta| < 1\}.$$

The solution can be expressed by using the Poisson formula:

Proposition 1.6. *Assume $\phi \in \mathcal{C}^0(\partial\mathbb{D})$. The function*

$$z \longmapsto h_\phi(z) := \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |z|^2}{|z - e^{i\theta}|^2} \phi(e^{i\theta}) d\theta$$

is harmonic in $\mathbb{D}$ and continuous up to the boundary, where it coincides with ϕ. Thus h_ϕ is the unique solution to $\text{DirMA}(\Delta, \phi)$.

Proof. Observe that for fixed $\zeta = e^{i\theta} \in \partial\mathbb{D}$, the Poisson kernel is the real part of a holomorphic function in $\mathbb{D}$,

$$P(\zeta, z) := \frac{1 - |z|^2}{|z - \zeta|^2} = \Re \left(\frac{\zeta + z}{z - \zeta} \right).$$

Thus h_ϕ is harmonic in $\mathbb{D}$ as an average of harmonic functions.

We now establish the continuity property. Fix $\zeta_0 = e^{i\theta_0} \in \partial\mathbb{D}$ and $\varepsilon > 0$. Since ϕ is continuous at ζ_0, we can find $\delta > 0$ such that $|\phi(\zeta) - \phi(\zeta_0)| < \varepsilon/2$ whenever $\zeta \in \partial\mathbb{D}$ and $|\zeta - \zeta_0| < \delta$. Observing that the Poisson formula for $h \equiv 1$ implies

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{1-|z|^2}{|z-e^{i\theta}|^2} d\theta \equiv 1,$$

we infer

$$|h_\phi(z) - \phi(\zeta_0)| \leq \varepsilon/2 + M \int_{|e^{i\theta}-\zeta_0|\geq\delta} \frac{1-|z|^2}{|z-e^{i\theta}|^2} d\theta,$$

where $2\pi M = \sup_{\partial\mathbb{D}} |\phi|$. Note that $|z - e^{i\theta}| \geq \delta/2$ if z is close enough to ζ_0 and $|e^{i\theta} - \zeta_0| \geq \delta$. The latter integral is therefore bounded from above by $4(1-|z|^2)/\delta^2$, hence converges to zero as z approaches the unit circle. □

1.2 Subharmonic functions

We now recall some basic facts concerning subharmonic functions in $\mathbb{R}^2 \simeq \mathbb{C}$. These are characterized by submean value inequalities.

1.2.1 Definitions and basic properties

Let $\Omega \subset \mathbb{C}$ be a domain.

Definition 1.7. *A function $u : \Omega \to [-\infty, +\infty[$ is subharmonic if it is upper semi-continuous in Ω and for all $a \in \Omega$ there exists $0 < \rho(a) < \operatorname{dist}(a, \partial\Omega)$ such that, for all $0 < r < \rho(a)$,*

$$u(a) \leq \frac{1}{2\pi}\int_0^{2\pi} u(a + re^{i\theta}) d\theta. \tag{1.2.1}$$

Recall that a function u is upper semi-continuous (u.s.c. for short) in Ω if and only if for all $c \in \mathbb{R}$ the sublevel sets $\{u < c\}$ are open subsets of Ω. Note that harmonic functions are subharmonic; the class of subharmonic functions is however much larger.

Subharmonicity is a local concept. By semi-continuity, a subharmonic function is bounded from above on any compact subset $K \subset \Omega$ and attains its maximum on K. It can however take the value $-\infty$ at some points. With our definition the function which is identically $-\infty$ is subharmonic in Ω.

We will soon show that if u is subharmonic in a domain Ω and $u \not\equiv -\infty$, then $u \in L^1_{\mathrm{loc}}(\Omega)$, hence the set $\{u = -\infty\}$ has zero Lebesgue measure in $\mathbb{C}$. It is called the *polar set* of u.

Observe that the maximum of two subharmonic functions is subharmonic; so is a convex combination of subharmonic functions. Here are some further recipes to construct subharmonic functions:

Proposition 1.8. *Let $\Omega \subset \mathbb{C}$ be a domain in $\mathbb{C}$.*

(1) *If $u : \Omega \to [-\infty, +\infty[$ is subharmonic in Ω and $\chi : I \to \mathbb{R}$ is a convex increasing function on an interval I containing $u(\Omega)$, then $\chi \circ u$ is subharmonic in Ω.*

(2) *Let $(u_j)_{j\in\mathbb{N}}$ be a decreasing sequence of subharmonic functions in Ω. Then $u := \lim u_j$ is subharmonic in Ω.*

(3) *Let $(u_j)_{j\in\mathbb{N}}$ be a sequence of subharmonic functions in Ω, which is locally bounded from above in Ω, and let $(\varepsilon_j) \in \mathbb{R}_+^{\mathbb{N}}$ be such that $\sum_{j\in\mathbb{N}} \varepsilon_j < +\infty$. Then $u := \sum_{j\in\mathbb{N}} \varepsilon_j u_j$ is subharmonic in Ω.*

(4) *Let $(X, \mathcal{T})$ be a measurable space, μ a positive measure on $(X, \mathcal{T})$, and $E(z, x) : \Omega \times X \to \mathbb{R} \cup \{-\infty\}$ a function s.t.*

(i) *for μ-a.e. $x \in X$, $z \mapsto E(z, x)$ is subharmonic in Ω,*

(ii) *for all $z_0 \in \Omega$, there exists D a neighborhood of z_0 in Ω and $g \in L^1(\mu)$ s.t. $E(z, x) \le g(x)$ for all $z \in D$ and μ-a.e. $x \in X$.*

Then $z \mapsto U(z) := \int_X E(z, x)d\mu(x)$ is subharmonic in Ω.

Proof. (1) is an immediate consequence of Jensen's convexity inequality.

(2) It is clear that $u = \inf\{u_j ; j \in \mathbb{N}\}$ is usc in Ω. The submean value inequality is a conseqence of the monotone convergence theorem.

(3) The statement is local, hence it is enough to prove that u is subharmonic in any subdomain $D \Subset \Omega$. By assumption, there exists $C > 0$ such that $\sup_D u_j \le C$ for all $j \in \mathbb{N}$. Write

$$u = \sum_{j\in\mathbb{N}} \varepsilon_j (u_j - C) + C \sum_{j\in\mathbb{N}} \varepsilon_j .$$

The first sum is the limit of a decreasing sequence of subharmonic functions, hence it is subharmonic, and so is u.

(4) The upper semi-continuity of U is a consequence of Fatou's lemma. The submean value property is a consequence of the Tonelli–Fubini theorem. □

We now give some examples of subharmonic functions.

Examples 1.9.

1. *Fix $a \in \mathbb{C}$ and $c > 0$. The function $z \mapsto c \log|z - a|$ is subharmonic in $\mathbb{C}$ and harmonic in $\mathbb{C} \setminus \{a\}$.*

2. *Let $(a_j) \in \mathbb{C}^{\mathbb{N}}$ be a bounded sequence and let $\varepsilon_j > 0$ be positive reals such that $\sum_j \varepsilon_j < +\infty$. The function*

$$z \longmapsto u(z) := \sum_j \varepsilon_j \log|z - a_j|$$

is a locally integrable subharmonic function in $\mathbb{C}$. If the sequence (a_j) is dense in a domain Ω, it follows by a Baire category argument that the polar set $(u = -\infty)$

is uncountable, but has zero Lebesgue measure. Any point $a \in \Omega \setminus \{u = -\infty\}$ is a point of discontinuity of u: the function u is finite at a, but not locally bounded near a.

We generalize the first example above:

Proposition 1.10. *Let $f : \Omega \to \mathbb{C}$ be a holomorphic function with $f \not\equiv 0$ in Ω. Then $\log|f|$ is a subharmonic function in Ω which is harmonic in the domain $\Omega \setminus f^{-1}(0)$. In particular, for any $\alpha > 0$, $|f|^\alpha$ is a subharmonic function in Ω.*

Proof. Observe that $\{u = -\infty\} = \{f = 0\}$. It is clear that u is u.s.c. in Ω, since for every $c \in \mathbb{R}$, $\{u < c\} = \{|f| < e^c\}$ is open.

If $a \in \Omega$ and $u(a) = -\infty$, the submean value inequality (1.2.1) is trivially satisfied. If $a \in \Omega$ and $u(a) > -\infty$, then $f(a) \neq 0$. By continuity, $f(z) \neq 0$ for $|z - a| < r$, where $r > 0$ is small enough. It follows that $\log f$ has a continuous branch that is holomorphic in the disc $\mathbb{D}(a, r)$. Therefore, $u = \Re(\log f)$ is harmonic in $\mathbb{D}(a, r)$, hence it satisfies the mean value equality.

The last statement follows from the fact that $|f|^\alpha = \chi(\log|f|)$, where $\chi(t) := \exp(\alpha t)$ is a convex increasing function in $\mathbb{R} \cup \{-\infty\}$. □

Proposition 1.11. *Let $u : \Omega \to \mathbb{R}$ be a convex function. Then u is a continuous subharmonic function in Ω.*

Proof. When u is smooth in Ω we see that Δu is the trace of the real hessian of u, which is a semi-positive quadratic form by convexity. Hence $\Delta u \geq 0$ pointwise in Ω, which proves that u is subharmonic in Ω. For the general case we use regularization by convolution with radial mollifiers to conclude. □

Remark 1.12. *Observe that convex functions are continuous but this is not the case for subharmonic functions, as Examples 1.9 show. This is an important source of difficulty when studying fine properties of subharmonic functions.*

The converse of the above proposition is thus false: there are subharmonic functions that are not convex. On the other hand, if $u : D \times I \subset \mathbb{R}^2 \to \mathbb{R}$ is a function which depends only on the real part of z, i.e., $u(z) = v(x)$ for any $z = x + iy \in D \times I$, then u is subharmonic in $D \times I$ iff v is convex in D, as the reader will check.

The mean value of a subharmonic function has an important monotonicity property:

Proposition 1.13. *Let u be a subharmonic function, $a \in \Omega$ and set $\delta(a) := \operatorname{dist}(a, \partial\Omega)$. The mean value*

$$r \mapsto M(a, r) := \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}) d\theta,$$

is increasing and continuous in $[0, \delta(a)[$; it converges to $u(a)$ as $r \to 0$.

Proof. Fix $0 < r < \delta(a)$ and let h be a continuous function on the unit circle $\partial\mathbb{D}$ such that $u(a + re^{i\theta}) \leq h(e^{i\theta})$ for all $e^{i\theta} \in \partial\mathbb{D}$. Let H be the unique harmonic function in $\mathbb{D}$ such that $H = h$ on $\partial\mathbb{D}$. The classical maximum principle ensures that $u(a + r\zeta) \leq H(\zeta)$ for $\zeta \in \mathbb{D}$.

If $0 < s < r$, it follows from the mean value property for harmonic functions that

$$\int_0^{2\pi} u(a + se^{i\theta})d\theta \leq \int_0^{2\pi} H(se^{i\theta})d\theta = \int_0^{2\pi} H(re^{i\theta})d\theta.$$

Therefore, $\int_0^{2\pi} u(a + se^{i\theta})d\theta \leq \int_0^{2\pi} h(re^{i\theta})d\theta$ for any continuous function h such that $u(a + r\zeta) \leq h(\zeta)$ on $\partial\mathbb{D}$.

Since u is upper semi-continuous, there exists a decreasing sequence h_j of continuous functions on the circle $\partial\mathbb{D}$ that converges to the function $\zeta \mapsto u(a + r\zeta)$ on $\partial\mathbb{D}$ (see Exercise 1.1). The monotone convergence theorem yields

$$\int_0^{2\pi} u(a + se^{i\theta})d\theta \leq \int_0^{2\pi} u(a + re^{i\theta})d\theta. \qquad \square$$

Corollary 1.14. *If u is subharmonic in Ω, $a \in \Omega$, and $0 < r < \delta(a)$, then*

$$u(a) \leq \frac{1}{\pi r^2} \int_{\mathbb{D}(a,r)} u(z)dV(z),$$

where dV is the Lebesgue measure on $\mathbb{R}^2$. For any $a \in \Omega$,

$$u(a) = \lim_{r \to 0^+} \frac{1}{\pi r^2} \int_{\mathbb{D}(a,r)} u(z)dV(z).$$

In particular, if u and v are subharmonic functions in Ω such that $u \leq v$ almost everywhere in Ω, then $u \leq v$ everywhere in Ω.

1.2.2 The maximum principle The maximum principle is one of the most powerful tools in Potential Theory:

Theorem 1.15. *Assume u is subharmonic in Ω.*

1. *If u admits a local maximum at some point $a \in \Omega$, then u is constant in a neighborhood of a,*
2. *For any bounded subdomain $D \Subset \Omega$ we have*

$$\max_{\bar{D}} u = \max_{\partial D} u.$$

Moreover, $u(z) < \max_{\partial D} u$ for all $z \in D$ unless u is constant on D.

Proof. The proof follows the same lines as in the case of harmonic functions, with some modifications due to the fact that u is not necessarily continuous.

1. By hypothesis, there is a disc $\mathbb{D}(a,r) \Subset \Omega$ such that $u(z) \leq u(a)$ for any $z \in \mathbb{D}(a,r)$. Fix $0 < s \leq r$ and observe that

$$u(a) - u(a + se^{i\theta}) \geq 0 \text{ for all } \theta \in [0, 2\pi].$$

Integrating in polar coordinates gives

$$\int_{\mathbb{D}(a,r)} (u(a) - u(z)) dV(z) \geq 0,$$

while the submean value property shows that the above integral is negative. Therefore,

$$\int_{\mathbb{D}(a,r)} (u(z) - u(a)) dV(z) = 0.$$

We infer that $u(z) - u(a) = 0$ almost everywhere in $\mathbb{D}(a,r)$, hence everywhere in $\mathbb{D}(a,r)$, since u is subharmonic. This proves that u is constant in a neighborhood of a.

2. By compactness and upper semi-continuity, we can find $a \in \bar{D}$ such that $u(a) = \max_{\bar{D}} u$. If $a \in D$, then by the previous case u is constant in a neighborhood of a, therefore the set

$$A := \{z \in D \,|\, u(z) = u(a) = \max_{\bar{D}} u\} = \{z \in D \,|\, u(z) \geq \max_{\bar{D}} u\}$$

is open, non-empty and closed by upper semi-continuity, hence u is constant in D. □

Corollary 1.16. *Let $\Omega \Subset \mathbb{C}$ a bounded domain and u a subharmonic function in Ω. Assume that $\limsup_{z \to \zeta} u(z) \leq 0$ for all $\zeta \in \partial\Omega$. Then $u \leq 0$ in Ω.*

Proof. Fix $\varepsilon > 0$. By the compactness and upper semi-continuity of u, there exists a compact subset $K \subset \Omega$ such that $u \leq \varepsilon$ in $\Omega \setminus K$. Take a subdomain $D \Subset \Omega$ such that $K \subset D$ and apply Theorem 1.15 to conclude that $u \leq \varepsilon$ in D. Therefore, $u \leq \varepsilon$ in Ω and the conclusion follows since $\varepsilon > 0$ is arbitrary. □

The following consequence is known as the *comparison principle*:

Corollary 1.17. *Let $\Omega \Subset \mathbb{C}$ be a bounded domain and u, v subharmonic functions in $L^1_{\text{loc}}(\Omega)$ such that the following holds:*
(i) *For all $\zeta \in \partial\Omega$, $\liminf_{z \to \zeta}(u(z) - v(z)) \geq 0$.*
(ii) *$\Delta u \leq \Delta v$ in the weak sense of Radon measures in Ω.*
Then $u \geq v$ in Ω.

Proof. Set $w := v - u$ and observe that w is well defined at all points in Ω where the two functions do not take the value $-\infty$ at the same time, hence almost everywhere in Ω, and $w \in L^1_{\text{loc}}(\Omega)$ (see Proposition 1.34). From the condition (ii) it follows that $\Delta w \geq 0$ in the sense of distributions in Ω.

We infer that w is equal almost everywhere to a subharmonic function W in Ω (see Proposition 1.19). Therefore $v = u + W$ almost everywhere in Ω, hence everywhere in Ω.

We claim that $\limsup_{z\to\zeta} W(z) \leq 0$. Indeed let $(z_j) \in \Omega^{\mathbb{N}}$ be a sequence converging to $\zeta \in \partial\Omega$. Since $\limsup_{z_j\to\zeta}(v(z_j) - u(z_j)) \leq 0$, it follows that for $j > 1$ large enough $v(z_j) - u(z_j) < +\infty$, hence $W(z_j) = v(z_j) - u(z_j)$ and $\limsup_{z_j\to\zeta} W(z_j) \leq 0$. This proves our claim. It follows from Corollary 1.16 that $W \leq 0$ in Ω as desired. □

1.2.3 The Riesz representation formula In this section we lay down the foundations of *Logarithmic potential theory*. We associate a canonical (Riesz) measure to any subharmonic function and show how to reconstruct the function from its boundary values and its Riesz measure.

Definition 1.18. *We let $SH(\Omega)$ denote the set of all subharmonic functions in the domain Ω which are not identically $-\infty$.*

The set $SH(\Omega)$ is a convex positive cone contained in $L^1_{\mathrm{loc}}(\Omega)$. The proof of this fact will be given in Proposition 1.34, in the more general context of plurisubharmonic functions.

Proposition 1.19. *If $u \in SH(\Omega)$, then the distribution $\Delta u \geq 0$ is a non-negative distribution: for any positive test function $\varphi \in \mathcal{D}^+(\Omega)$,*

$$\langle \Delta u, \varphi\rangle = \int_\Omega u\Delta\varphi \, dV \geq 0.$$

Conversely, if $T \in \mathcal{D}'(\Omega)$ is a distribution such that $\Delta T \geq 0$, then there is a unique function $u \in SH(\Omega)$ such that $T_u = T$.

Proof. Fix $u \in SH(\Omega)$. Assume first that u is smooth in Ω and fix $a \in \Omega$. It follows from Taylor's formula that

$$\Delta u(a) = \lim_{r\to 0^+} \frac{2}{r^2}\left(\frac{1}{2\pi}\int_0^{2\pi} u(a + re^{i\theta})d\theta - u(a)\right).$$

Since u is subharmonic, the right-hand side is non-negative, hence $\Delta u(a) \geq 0$ pointwise in Ω.

We now get rid of the regularity assumption. It follows from Proposition 1.34 that $u \in L^1_{\mathrm{loc}}(\Omega)$, hence we can regularize u by convolution setting $u_\varepsilon = u \star \rho_\varepsilon$ for $\varepsilon > 0$, using radial mollifiers.

The functions u_ε are subharmonic as convex combinations of subharmonic functions. Since u_ε is moreover smooth, we infer $\Delta u_\varepsilon \geq 0$, hence $\Delta u \geq 0$, since $u_\varepsilon \to u$ in L^1_{loc}.

We note for later use that $\varepsilon \mapsto u_\varepsilon$ is non-decreasing: this follows from the mean value inequalities and the fact that we use radial and non-negative mollifiers. In particular, u_ε decreases to u as ε decreases to zero (cf Proposition 1.13 and Corollary 1.14).

Let now T be a distribution in Ω and $(\rho_\varepsilon)_{\varepsilon>0}$ be mollifiers as above. Then $v_\varepsilon := T \star \rho_\varepsilon$ is a smooth function such that $\Delta v_\varepsilon = (\Delta T) \star \rho_\varepsilon \geq 0$ in Ω_ε, thus v_ε is subharmonic in Ω_ε.

We claim that $\varepsilon \mapsto v_\varepsilon$ is non-decreasing. Indeed, for $\varepsilon > 0$ small enough, the map $\eta \mapsto (v_\varepsilon \star \rho_\eta)$ is non-decreasing since v_ε is subharmonic in Ω_ε. By the definition of convolution, we have $v_\varepsilon \star \rho_\eta = v_\eta \star \rho_\varepsilon$ in $\Omega_{\varepsilon+\eta}$ for $\varepsilon, \eta > 0$ small enough. Therefore, for any fixed $\eta > 0$ small enough, the map $\varepsilon \mapsto v_\varepsilon \star \rho_\eta$ is also non-decreasing for small ε. Since $v_\varepsilon \star \rho_\eta \to v_\varepsilon$ as $\eta \to 0^+$, the claim follows.

Now since v_ε is non-decreasing in $\varepsilon > 0$ it converges to a subharmonic function u as ε decreases to zero. The function u cannot be identically $-\infty$ since $T_u = T$ as distributions (this follows from the monotone convergence theorem). The uniqueness follows again from Corollary 1.14: two subharmonic functions which coincide almost everywhere are actually equal. □

Recall that a positive distribution always extends to a positive Borel measure (see Exercise 1.11). Therefore, if $u \in SH(\Omega)$, then the positive distribution $(1/2\pi)\Delta u$ can be extended as a positive Borel measure μ_u on Ω which we call the Riesz measure of u.

Definition 1.20. *The Riesz measure of $u \in SH(\Omega)$ is*

$$\mu_u = \frac{1}{2\pi}\Delta u.$$

Using the complex coordinate $z = x + iy$, the real differential operator acting on smooth functions $f : \Omega \to \mathbb{C}$ by

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

splits as $d = \partial + \overline{\partial}$, where the complex differential operators ∂ and $\overline{\partial}$ are defined by

$$\partial f = \frac{\partial f}{\partial z}dz \quad \text{and} \quad \overline{\partial} f = \frac{\partial f}{\partial \overline{z}}d\overline{z}.$$

We extend these differential operators to distributions: if f is a distribution, then df is defined as above, but it has to be understood in the sense of currents of degree 1 on Ω: it is a differential form of degree 1 with distribution coefficients.

Observe that the volume form in $\mathbb{C}$ can be writen as

$$dx \wedge dy = \frac{i}{2}dz \wedge d\overline{z}.$$

We define the real operator d^{c} by

$$d^{\mathrm{c}} := \frac{1}{2i\pi}(\partial - \overline{\partial}),$$

so that for $u \in SH(\Omega)$ we obtain

$$dd^{\mathrm{c}}u = \frac{1}{2\pi}\Delta u\, dx \wedge dy = \mu_u\, dx \wedge dy,$$

where μ_u is the Riesz measure of u and the notation $\mu_u\, dx \wedge dy$ is understood in the sense of currents of degree 2: it is a differential form of degree 2 with distribution coefficients.

Example 1.21. *Fix $a \in \Omega$. The function $z \mapsto \ell_a(z) := \log|z-a|$ is subharmonic and satisfies*

$$dd^{\mathrm{c}}\ell_a = \frac{1}{2\pi}\Delta\ell_a = \delta_a, \tag{1.2.2}$$

in the sense of distributions, where δ_a denotes the Dirac mass at the point a. In particular, ℓ_0 is a fundamental solution of the linear differential operator $dd^{\mathrm{c}} = (2\pi)^{-1}\Delta$ in $\mathbb{C}$.

The following result connects logarithmic potential theory to the theory of holomorphic functions in one complex variable:

Proposition 1.22. *Let $f : \Omega \to \mathbb{C}$ be a holomorphic function such that $f \not\equiv 0$. Then $\log|f| \in SH(\Omega)$. Moreover,*

$$dd^{\mathrm{c}}\log|f| = \sum_{a\in Z_f} m_f(a)\delta_a,$$

where $Z_f := f^{-1}(0)$ is the zero set of f in Ω and $m_f(a)$ is the order of vanishing of f at the point a.

Observe that since $f \not\equiv 0$, the zero set Z_f is discrete in Ω, hence the sum is locally finite.

Proof. On any subdomain $D \Subset \Omega$ the set $A := Z_f \cap D$ is finite and there exists a non-vanishing holomorphic function g such that

$$f(z) = \prod_{a\in A}(z-a)^{m_f(a)}g(z)$$

for $z \in D$. Since g is zero free, $\log|g|$ is harmonic hence

$$dd^{\mathrm{c}}\log|f| = \sum_{a\in A} m_f(a)dd^{\mathrm{c}}\log|z-a| = \sum_{a\in A} m_f(a)\delta_a$$

in the sense of distributions. □

Let μ be a Borel measure with compact support on $\mathbb{C}$, then

$$z \longmapsto U_\mu(z) := \int_{\mathbb{C}} \log|z-\zeta| d\mu(\zeta) = \mu \star \ell_0(z)$$

is subharmonic in $\mathbb{C}$. Moreover, if $z \in \mathbb{C} \setminus \mathrm{Supp}(\mu)$, then

$$U_\mu(z) \geq \log \mathrm{dist}(z, \mathrm{Supp}(\mu)) > -\infty,$$

and

$$U_\mu(z) \leq \mu(\mathbb{C}) \log^+ |z| + C(\mu),\ z \in \mathbb{C}.$$

It follows that $U_\mu \in SH(\mathbb{C})$.

Definition 1.23. *The function $U_\mu : z \mapsto \int_{\mathbb{C}} \log|z-\zeta| d\mu(\zeta)$ is called the logarithmic potential of the measure μ.*

Observe that

$$\frac{1}{2\pi} \Delta U_\mu = \left(\frac{1}{2\pi} \Delta \ell_0 \right) \star \mu = \mu,$$

in the sense of distributions on $\mathbb{C}$. This implies that U_μ is subharmonic in $\mathbb{C}$ and harmonic (hence real analytic) in $\mathbb{C} \setminus \mathrm{Supp}(\mu)$.

We can now derive the *Riesz representation formula*:

Proposition 1.24. *Fix $u \in SH(\Omega)$ and $D \Subset \Omega$ a subdomain. Then*

$$u(z) = \int_D \log|z-\zeta| d\mu_u(\zeta) + h_D(z),\ \ z \in D,$$

where $\mu_u := \frac{1}{2\pi} \Delta u$ and h_D is a harmonic function in D.

Proof. Apply the last construction to the measure $\mu_D := \mathbf{1}_D \cdot \mu_u$, which is a Borel measure with compact support on $\mathbb{C}$: the function

$$v(z) := \int_D \log|z-\zeta| d\mu_u(\zeta) = \mu_D \star \ell_0(z)$$

is subharmonic in $\mathbb{C}$ and satisfies

$$\Delta v = \mu_D \star \Delta(\ell_0) = \mathbf{1}_D(\Delta u)$$

in the sense of Borel measures in $\mathbb{C}$. Therefore, $h := u - v$ is a locally integrable function which satisfies $\Delta h = 0$ in the weak sense of distributions in D. It follows from Weyl's lemma that h coincides almost everywhere in D with a harmonic function, denoted by h_D. This implies that $u = v + h_D$ almost everywhere in D, hence everywhere in D. □

This result shows that a subharmonic function u coincides locally (up to a harmonic function which is smooth) with the logarithmic potential of its *Riesz measure* μ_u. In particular, the information on the singularities of u (discontinuities, polar points, etc) is contained within its potential U_μ.

The study of fine properties of subharmonic functions is therefore reduced to that of logarithmic potentials of compactly supported Borel measures on $\mathbb{C}$, hence the name *Logarithmic Potential Theory*.

1.2.4 Poisson–Jensen formula The *Poisson–Jensen formula* is a generalization of the Poisson formula for harmonic functions in the unit disc. It is a precise version of the Riesz representation formula that takes into account the boundary values of the function:

Proposition 1.25. *Assume $u \in SH(\Omega)$, where Ω is a domain containing the closed unit disc $\overline{\mathbb{D}}$. Then for all $z \in \mathbb{D}$,*

$$u(z) = \int_0^{2\pi} u(e^{i\theta}) \frac{1-|z|^2}{|z-e^{i\theta}|^2} \frac{d\theta}{2\pi} + \int_{|\zeta|<1} \log \frac{|z-\zeta|}{|1-z\bar{\zeta}|} d\mu(\zeta),$$

where $\mu = \frac{1}{2\pi}\Delta u$ is the Riesz measure of u.

When u is harmonic in $\mathbb{D}$ we recover the Poisson formula (Theorem 1.6). The first term is called the Poisson transform of u in $\mathbb{D}$. This is the least harmonic majorant of u in $\mathbb{D}$. The second term is a non-positive subharmonic function encoding the singularities of u; it is called the *Green potential* of the measure μ.

Proof. Fix $w \in \mathbb{D}$ and set for $z \in \mathbb{D}$,

$$G_{\mathbb{D}}(z,w) = G_w(z) := \log \frac{|z-w|}{|1-z\bar{w}|}.$$

Observe that G_w is subharmonic in $\mathbb{D}$, $dd^c G_w = \delta_w$ in the weak sense of distributions in $\mathbb{D}$, and $G_w \le 0$ in $\mathbb{D}$ with G_w identically 0 on $\partial\mathbb{D}$.

It follows from the comparison principle (Corollary 1.17) that G_w is the unique function having these properties. It is called the Green function of the unit disc with logarithmic singularity at the point w.

For fixed $w \in \mathbb{C}$, we set

$$H_w(z) := \int_{\partial\mathbb{D}} \log|\xi - w| P_{\mathbb{D}}(z,\xi) d\sigma(\xi),$$

where $d\sigma$ is the normalized Lebesgue measure on $\partial\mathbb{D}$ and

$$P_{\mathbb{D}}(z,\xi) := \frac{1-|z|^2}{|\xi - z|^2},$$

is the Poisson kernel for the unit disc. We claim that

$$H_w(z) = \log|z-w| - G_w(z), \quad \text{if } w \in \mathbb{D}, \tag{1.2.3}$$
$$H_w(z) = \log|z-w|, \quad \text{if } w \in \mathbb{C}\setminus\mathbb{D}. \tag{1.2.4}$$

Indeed, if $w \in \mathbb{D}$, then by Proposition 1.6, the function H_w is harmonic in $\mathbb{D}$ and continuous up to the boundary, where it coincides with the function $z \mapsto \log|z-w|$. Therefore, the function $g(z) := \log|z-w| - H_w(z)$ is harmonic in $\mathbb{D}\setminus\{w\}$, subharmonic in $\mathbb{D}$ with a logarithmic singularity at w, and is 0 on $\partial\mathbb{D}$. The maximum principle thus yields

$$G_w(z) = \log|z-w| - H_w(z), \tag{1.2.5}$$

for any $z \in \mathbb{D}$, which proves (1.2.3).

If $w \in \mathbb{C}\setminus\overline{\mathbb{D}}$, the function $z \mapsto \log|z-w|$ is harmonic in $\mathbb{D}$ and continuous in $\overline{\mathbb{D}}$. Therefore, (1.2.4) follows from the Poisson formula.

By the Riesz representation formula, if $\mathbb{D}'$ is a disc containing $\overline{\mathbb{D}}$ so that u is subharmonic on $\overline{\mathbb{D}'}$, we have $u = U_\mu + h$ in $\mathbb{D}'$, where $\mu := \mu_{\mathbb{D}'}$ and h is a harmonic function in $\mathbb{D}'$. Fubini's theorem and (1.2.3), (1.2.4) yield

$$\begin{aligned}
\int_{\partial\mathbb{D}} u(\xi)P_{\mathbb{D}}(z,\xi)d\sigma(\xi) &= \int_{\partial\mathbb{D}}\left(\int_{\mathbb{D}'}\log|\xi-\zeta|d\mu(\zeta)+h(\xi)\right)P_{\mathbb{D}}(z,\xi)d\sigma(\xi)\\
&= \int_{\mathbb{D}'}\left(\int_{\partial\mathbb{D}}\log|\xi-\zeta|P_{\mathbb{D}}(z,\xi)d\sigma(\xi)\right)d\mu(\zeta)+h(z)\\
&= \int_{\mathbb{D}'}\log|z-\zeta|d\mu(\zeta) - \int_{\mathbb{D}}G_\zeta(z))d\mu(\zeta)+h(z)\\
&= u(z) - \int_{\mathbb{D}}G_\zeta(z))d\mu(\zeta),
\end{aligned}$$

which is the required formula. □

Remark 1.26. *The Poisson–Jensen formula suggests to consider the following general Dirichlet problem: given a finite Borel measure on the disc $\mathbb{D}$ and a continuous function ϕ in $\partial\mathbb{D}$, find $u \in SH(\mathbb{D})$ which extends to the boundary, such that*

$$\mathrm{DirMA}(\mathbb{D},\phi,\mu)\ \begin{cases}\Delta u = \mu \text{ in } \mathbb{D},\\ u|_{\partial\mathbb{D}} = \phi.\end{cases}$$

We shall come back to this problem in Chapter 5.

1.3 Plurisubharmonic functions

We now introduce the fundamental objects that we are going to study in the sequel. The notion of plurisubharmonic function is the pluricomplex counterpart of the notion of subharmonic function.

1.3.1 Basic properties

We fix Ω a domain of $\mathbb{C}^n$.

Definition 1.27. *A function $u : \Omega \to [-\infty + \infty[$ is plurisubharmonic if it is upper semi-continuous and for all complex lines $\Lambda \subset \mathbb{C}^n$, the restriction $u|\Omega \cap \Lambda$ is subharmonic in $\Omega \cap \Lambda$.*

The latter property can be reformulated as follows: for all $a \in \Omega$, $\xi \in \mathbb{C}^n$ with $|\xi| = 1$, and $r > 0$ such that $\bar{B}(a, r) \subset \Omega$,

$$u(a) \leq \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}\xi) d\theta. \tag{1.3.1}$$

All basic results that we have established for subharmonic functions are also valid for plurisubharmonic functions. We state them and leave the proofs to the reader:

Proposition 1.28.

(1) *If $u : \Omega \to [-\infty, +\infty[$ is plurisubharmonic in Ω and χ is a real convex increasing function on an interval containing the image $u(\Omega)$ of u, then $\chi \circ u$ is plurisubharmonic in Ω.*
(2) *Let $(u_j)_{j\in\mathbb{N}}$ be a decreasing sequence of plurisubharmonic functions in Ω. Then $u := \lim_{j\to+\infty} u_j$ is plurisubharmonic in Ω.*
(3) *Let $(X, \mathcal{T})$ be a measurable space, μ a positive measure on $(X, \mathcal{T})$, and $E(z, x) : \Omega \times X \to \mathbb{R} \cup \{-\infty\}$ be such that*
   (i) *for μ-a.e. $x \in X$, $z \mapsto E(z, x)$ is plurisubharmonic,*
   (ii) *$\forall z_0 \in \Omega$ there exist $r > 0$ and $g \in L^1(\mu)$ such that for all $z \in B(z_0, r)$ and μ-a.e. $x \in X$, $E(z, x) \leq g(x)$.*

   Then $z \mapsto V(z) := \int_X E(z, t) d\mu(x)$ is plurisubharmonic.

Recall that a function $f : z = (z_1, \ldots, z_n) \in \Omega \mapsto f(z) \in \mathbb{C}$ is holomorphic if it satisfies the Cauchy–Riemann equations $\partial f/\partial \overline{z}_j = 0$ for all $1 \leq j \leq n$.

Proposition 1.29. *Let $\Omega \subset \mathbb{C}^n$ be a domain in $\mathbb{C}^n$ and f a holomorphic function such that $f \not\equiv 0$ in Ω. Then $\log|f|$ is plurisubharmonic in Ω and pluriharmonic in $\Omega \setminus \{f = 0\}$. Moreover, for any $\alpha > 0$, $|f|^\alpha$ is plurisubharmonic in Ω.*

A function is pluriharmonic if it satifies the linear equations

$$\frac{\partial^2 f}{\partial z_j \partial \overline{z}_k} = 0$$

for all $1 \leq j, k \leq n$. One shows (like in complex dimension 1) that a function is pluriharmonic if and only if it is locally the real part of a holomorphic function.

We add one more recipe known as the *gluing construction*:

Proposition 1.30. *Let u be a plurisubharmonic function in a domain Ω. Let v be a plurisubharmonic function in a relatively compact subdomain $\Omega' \subset \Omega$. If $u \geq v$ on $\partial\Omega'$, then the function*

$$z \longmapsto w(z) = \begin{cases} \max[u(z), v(z)], & \text{if } z \in \Omega', \\ u(z), & \text{if } z \in \Omega \setminus \Omega', \end{cases}$$

is plurisubharmonic in Ω.

Proof. The upper semi-continuity property is clear. Replacing v by $v-\varepsilon$, one gets that u strictly dominates $v-\varepsilon$ in a neighborhood of $\partial\Omega'$ and the corresponding function w_ε is then clearly plurisubharmonic. Now w is the increasing limit of w_ε as ε decreases to zero, so it satisfies the appropriate submean value inequalities. □

1.3.2 Submean value inequalities

The following result follows from its analogue in one complex variable.

Proposition 1.31. *Let $u : \Omega \to [-\infty + \infty[$ be a plurisubharmonic function. Fix $a \in \Omega$ and set $\delta(a) := dist(a, \partial\Omega)$. Then*

(i) *the spherical submean value inequality holds: for $0 < r < \delta(a)$,*

$$u(a) \leq \int_{|\xi|=1} u(a + r\xi)\, d\sigma(\xi), \tag{1.3.2}$$

where $d\sigma$ is the normalized area measure on the unit sphere $S^{2n-1} \subset \mathbb{C}^n$;

(ii) *the spatial submean value inequality holds: for $0 < r < \delta(a)$ and any increasing right continuous function γ on $[0, r]$ with $\gamma(0) = 0$,*

$$u(a) \leq \frac{1}{\gamma(r)} \int_0^r d\gamma(s) \int_{|\xi|=1} u(a + s\xi)\, d\sigma(\xi); \tag{1.3.3}$$

(iii) *the toric submean value inequality holds: for $0 < r < \delta(a)/\sqrt{n}$,*

$$u(a) \leq \int_{\mathbb{T}^n} u(a + r\zeta)\, d\tau_n(\zeta), \tag{1.3.4}$$

where $d\tau_n$ is the normalized Lebesgue measure on the torus $\mathbb{T}^n$.

All these integrals make sense in $[-\infty, +\infty[$. We will soon see that they are usually finite (cf Proposition 1.34).

Proof. The first inequality follows from (1.3.1) by integration over the unit sphere in $\mathbb{C}^n$ and the second inequality follows from the first one by integration over $[0, r]$ against the measure $d\gamma$. The third inequality follows from (1.3.1) by integration on the torus. □

Remark 1.32. *Let u be a plurisubharmonic in Ω. Using polar coordinates we can write*

$$\int_{|\zeta|<1} u(a+r\zeta)d\lambda(\zeta) = \int_0^r t^{2n-1}dt \int_{|\xi|=1} u(a+t\xi)d\sigma(\xi).$$

It follows from (1.3.3) that

$$u(a) \leq \frac{1}{\kappa_{2n}} \int_{|\zeta|<1} u(a+r\zeta)\, dV(\zeta), \tag{1.3.5}$$

where κ_{2n} denotes the volume of the unit ball in $\mathbb{C}^n$.

The function u, considered as a function on $2n$ real variables, is thus subharmonic in Ω considered as a domain in $\mathbb{R}^{2n}$.

Definition 1.33. *$PSH(\Omega)$ denotes the convex cone of plurisubharmonic functions u in Ω such that $u|\Omega \not\equiv -\infty$.*

The submean value inequalities imply the following important integrability result:

Proposition 1.34.

$$PSH(\Omega) \subset L^1_{\mathrm{loc}}(\Omega).$$

Moreover, the restriction of $u \in PSH(\Omega)$ to any Euclidean sphere (resp. any torus $\mathbb{T}^n$) contained in Ω is integrable with respect to the area measure of the sphere (resp. the torus).

In particular, the polar set $P(u) := \{u = -\infty\}$ has volume zero in Ω and its intersection with any Euclidean sphere (resp. any torus $\mathbb{T}^n$) has measure zero with respect to the corresponding area measure.

Definition 1.35. *A set is called (locally) pluripolar if it is (locally) included in the polar set $\{u = -\infty\}$ of a function $u \in PSH(\Omega)$.*

It follows from previous proposition that pluripolar sets are somehow small. We will provide more precise information on their size in the next chapters.

Proof. Fix $u \in PSH(\Omega)$ et let G denote the set of points $a \in \Omega$ such that u is integrable in a neighborhood of a. We are going to show that G is a non-empty open and closed subset of Ω. It will follow that $G = \Omega$ (by connectedness) and $u \in L^1_{\mathrm{loc}}(\Omega)$.

Note that G is open by definition. If $a \in \Omega$ and $u(a) > -\infty$, the volume submean value inequalities yield, for all $0 < r < \mathrm{dist}(a, \partial\Omega)$,

$$-\infty < \kappa_{2n} r^{2n} u(a) \leq \int_{\mathbb{B}(a,r)} u(z)\, dV(z).$$

Since u is bounded from above on $\mathbb{B}(a,r) \Subset \Omega$, it follows that u is integrable on $\mathbb{B}(a,r)$. In particular, if $u(a) > -\infty$, then $a \in G$, hence $G \neq \emptyset$, since $u \not\equiv -\infty$.

We finally prove that G is closed. Let $b \in \Omega$ be a point in the closure of G and $r > 0$ so that $\mathbb{B}(b,r) \Subset \Omega$. By definition, there exists $a \in G \cap \mathbb{B}(b,r)$. Since u is locally integrable in a neighborhood of a, there exists a point a' close to a in $\mathbb{B}(b,r)$ such that $u(a') > -\infty$. Since $b \in \mathbb{B}(a',r) \Subset \Omega$ and u is integrable on $\mathbb{B}(a',r)$, it follows that $b \in G$.

The other properties are proved similarly, replacing volume submean inequalities by spherical (resp. toric) ones (Proposition 1.31). □

Proposition 1.36. *Fix $u \in PSH(\Omega)$, $a \in \Omega$ and set $\delta_\Omega(a) := \operatorname{dist}(a, \partial\Omega)$. Fix γ a non-decreasing right continuous function such that $\gamma(0) = 0$. Then*

$$r \longmapsto M_\gamma(a,r) := \frac{1}{\gamma(r)} \int_0^r d\gamma(s) \int_{|\xi|=1} u(a+s\xi)\, d\sigma(\xi),$$

is a non-decreasing continuous function in $[0, \delta_\Omega(a)[$ which converges to $u(a)$ as $r \to 0$.

Proof. This property has already been established when $n = 1$. Fix $0 < r < \delta_\Omega(a)$ and observe that for all $e^{i\theta} \in \mathbb{T}$,

$$\int_{|\xi|=1} u(a+s\xi) d\sigma(\xi) = \int_{|\xi|=1} u(a+se^{i\theta}\cdot\xi) d\sigma(\xi),$$

since the area measure on the sphere is invariant under the action of the circle $\mathbb{T}$. Integrating on the circle and using the one-dimensional case yields the required property. □

Corollary 1.37. *For $u \in PSH(\Omega)$, $a \in \Omega$, and $0 < r < \delta_\Omega(a)$,*

$$u(a) \le \frac{1}{\kappa_{2n}} \int_{|\xi|\le 1} u(a+r\zeta) d\lambda(\zeta) \le \int_{|\xi|=1} u(a+r\xi)\, d\sigma(\xi).$$

Corollary 1.38. *If two plurisubharmonic functions coincide almost everywhere, then they are equal.*

Proof. Assume $u, v \in PSH(\Omega)$ are equal a.e. Then for all $a \in \Omega$,

$$\begin{aligned} u(a) &= \lim_{r\to 0} \frac{1}{\kappa_{2n}} \int_{|\xi|\le 1} u(a+r\zeta) d\lambda(\zeta) \\ &= \lim_{r\to 0} \frac{1}{\kappa_{2n}} \int_{|\xi|\le 1} v(a+r\zeta) d\lambda(\zeta) = v(a). \end{aligned}$$

□

We endow the space $PSH(\Omega)$ with the L^1_{loc}-topology. The following property will be used on several occasions in the sequel:

Proposition 1.39. *The evaluation functional*

$$(u, z) \in PSH(\Omega) \times \Omega \longmapsto u(z) \in \mathbb{R} \cup \{-\infty\}$$

is upper semi-continuous.

In particular, if $\mathcal{U} \subset PSH(\Omega)$ is a compact family of plurisubharmonic functions, its upper envelope

$$U := \sup\{u \,|\, u \in \mathcal{U}\}$$

is upper semi-continuous, hence plurisubharmonic in Ω.

Proof. Fix $(u, z_0) \in PSH(\Omega) \times \Omega$. Let (u_j) be a sequence in $PSH(\Omega)$ converging to u and let $r > 0$ and $\delta > 0$ be small enough.

We observe first that (u_j) is locally uniformly bounded from above. Indeed, if $B(a, 2r) \subset \Omega$, the submean value inequalities yield

$$u_j(z) \leq \frac{1}{\kappa_2 r^{2n}} \int_{B(z,r)} u_j(w) dV(w) \leq \frac{1}{\kappa_2 r^{2n}} \int_{B(a,2r)} |u_j(w)| dV(w)$$

for $|z - a| < r$. Thus

$$\sup_{B(z_0,r)} u_j \leq \frac{1}{\kappa_2 r^{2n}} \int_{B(a,2r)} |u_j(w)| dV(w) \leq C,$$

since $\mathcal{U}$ is compact hence bounded.

We can thus assume without loss of generality that $u_j \leq 0$. The submean value inequalities again yield, for $|z - z_0| < \delta$,

$$u_j(z) \leq \frac{1}{\kappa_{2n}(r + \delta)^{2n}} \int_{B(z,r+\delta)} u_j d\lambda \leq \frac{1}{\kappa_{2n}(r + \delta)^{2n}} \int_{B(z_0,r)} u_j d\lambda.$$

Taking the limit in both j and z, we obtain

$$\limsup_{(j,z)\to(+\infty,z_0)} u_j(z) \leq \frac{1}{\kappa_{2n}(r + \delta)^{2n}} \int_{B(z_0,r)} u d\lambda.$$

Now let $\delta \to 0^+$ and then $r \to 0^+$ to obtain

$$\limsup_{(j,z)\to(+\infty,z_0)} u_j(z) \leq u(z_0),$$

which proves the desired semi-continuity at point (u, z_0).

It follows that the envelope U is upper semi-continuous. Since it clearly satisfies the mean value inequalities on each complex line, we infer that U is plurisubharmonic. □

The upper envelope of a family of plurisubharmonic functions which is merely relatively compact is not necessarily upper semi-continuous. It turns out that its upper semi-continuous regularization is plurisubharmonic:

Proposition 1.40. *Let $(u_i)_{i\in I}$ be a family of plurisubharmonic functions in a domain Ω, which is locally uniformly bounded from above in Ω, and let $u := \sup_{i\in I} u_i$ be its upper envelope. The upper semi-continuous regularization*

$$z \longmapsto u^*(z) := \limsup_{\Omega\ni z'\to z} u(z') \in \mathbb{R}\cup\{-\infty\}$$

is plurisubharmonic in Ω and $\{u < u^*\}$ has Lebesgue measure zero.

Proof. It follows from Choquet's lemma (see Chapter 4) that there exists an increasing sequence (v_j) of plurisubharmonic functions such that

$$u^* = \left(\lim v_j\right)^*.$$

Set $v = \lim \nearrow v_j$. This function satisfies various mean value inequalities, but it is not necessarily upper semi-continuous. Let χ_ε be standard radial mollifiers. Observe that, for $\varepsilon > 0$ fixed, $v_j \star \chi_\varepsilon$ is an increasing sequence of plurisubharmonic functions, thus its continuous limit $v \star \chi_\varepsilon$ is plurisubharmonic and $\varepsilon \mapsto v \star \chi_\varepsilon$ is increasing.

We let w denote the limit of $v \star \chi_\varepsilon$ as ε decreases to zero. The function w is plurisubharmonic as a decreasing limit of plurisubharmonic functions. It satisfies, for all $\varepsilon > 0$, $w \le u \star \chi_\varepsilon$ since $v_j \star \chi_\varepsilon \le u \star \chi_\varepsilon$.

On the other hand for all $\varepsilon > 0$, $u \le v^* \le v \star \chi_\varepsilon$ hence

$$u \le u^* = v^* \le w \le u \star \chi_\varepsilon.$$

Since $u \star \chi_\varepsilon$ converges to u in L^1_{loc}, we conclude that $u^* = w$ is plurisubharmonic. Note that the set $\{u < u^*\}$ has Lebesgue measure zero. □

Example 1.41. *If $f : \Omega \to \mathbb{C}$ is a holomorphic function such that $f \not\equiv 0$ and $c > 0$, then $c \log|f| \in PSH(\Omega)$. Conversely, one can show, when Ω is pseudoconvex, that the cone $PSH(\Omega)$ is the closure (in $L^1_{\mathrm{loc}}(\Omega)$) of the set of functions*

$$\{c \log|f|;\, f \in \mathcal{O}(\Omega),\, f \not\equiv 0, c > 0\}.$$

This can be shown by using Hörmander's L^2-estimates [Horm90].

1.3.3 Differential characterization

We show in this section that plurisubharmonicity can be characterized by differerential inequalities.

1.3.3.1 Plurisubharmonic smoothing

We first explain how any $u \in PSH(\Omega)$ can be approximated by a decreasing family of smooth plurisubharmonic functions (on any subdomain $D \Subset \Omega$).

Let $\rho(z) \ge 0$ be a smooth radial function on $\mathbb{C}^n$ with compact support in the unit ball $\mathbb{B} \subset \mathbb{C}^n$ such that $\int_{\mathbb{C}^n} \rho(z) d\lambda(z) = 1$. We set

$$\rho_\varepsilon(\zeta) := \varepsilon^{-2n}\rho(\zeta/\varepsilon),$$

for $\varepsilon > 0$. The functions ρ_ε are smooth with compact support in $\mathbb{B}(0,\varepsilon)$ and $\int_{\mathbb{C}^n} \rho_\varepsilon d\lambda = 1$; they approximate the Dirac mass at the origin.

Let $u : \Omega \to \mathbb{R} \cup \{-\infty\}$ be a L^1_{loc}-function. We set

$$\Omega_\varepsilon := \{z \in \Omega | \operatorname{dist}(z, \partial\Omega) > \varepsilon\}$$

and consider, for $z \in \Omega_\varepsilon$,

$$u_\varepsilon(z) := \int_\Omega u(\zeta)\rho_\varepsilon(z-\zeta)d\lambda(\zeta).$$

These functions are smooth and converge to u in L^1_{loc}.

Proposition 1.42. *If $u \in PSH(\Omega)$, then the smooth functions u_ε are plurisubharmonic and decrease to u as ε decreases to 0^+.*

Proof. The functions u_ε are plurisubharmonic as (convex) average of plurisubharmonic functions. By definition, if $z \in \Omega_\varepsilon$ we have

$$u_\varepsilon(z) = \int_{|\zeta|<1} u(z+\varepsilon\zeta)\rho(\zeta)d\lambda(\zeta),\ z \in \Omega_\varepsilon.$$

Integrating in polar coordinates, we get

$$u_\varepsilon(z) = \int_0^1 r^{2n-1}\rho(r)dr \int_{|\xi|=1} u(z+\varepsilon r\xi)d\sigma(\xi).$$

The monotonicity property now follows from Proposition 1.36. □

We let the reader check that a function u of class C^2 is plurisubharmonic in Ω iff for all $a \in \Omega$ and $\xi \in \mathbb{C}^n$,

$$\mathcal{L}_u(a;\xi) := \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 u}{\partial z_i \partial \overline{z}_j}(a)\xi_i\overline{\xi}_j \geq 0. \tag{1.3.6}$$

In other words, the Hermitian form $\mathcal{L}_u(a,.)$ (the *Levi form* of u at the point a) should be semi-positive in $\mathbb{C}^n$.

For non-smooth plurisubharmonic functions this positivity condition has to be understood in the sense of distributions:

Proposition 1.43. *If $u \in PSH(\Omega)$, then for any $\xi \in \mathbb{C}^n$,*

$$\sum_{1\leq j,k\leq n} \xi_j\bar{\xi}_k \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \geq 0$$

is a positive distribution in Ω.

Conversely, if $U \in \mathcal{D}'(\Omega)$ is a distribution such that for all $\xi \in \mathbb{C}^n$, the distribution $\sum \xi_j\bar{\xi}_k \dfrac{\partial^2 U}{\partial z_j \partial \bar{z}_k}$ is positive, then there exists a unique $u \in PSH(\Omega)$ such that $U \equiv T_u$.

Each distribution $\dfrac{\partial^2 u}{\partial z_i \partial \bar{z}_j}$ extends to a complex Borel measure $\mu_{j,\bar{k}}$ on Ω so that the matrix $(\mu_{j,\bar{k}})$ is Hermitian semi-positive.

Proof. The proof follows that of Proposition 1.19. Fix $\xi \in \mathbb{C}^n$ and consider the linear operator with constant coefficients

$$\Delta_\xi := \sum_{1 \le j,k \le n} \xi_j \bar{\xi}_k \frac{\partial^2}{\partial z_j \partial \bar{z}_k}.$$

Assume first that u is smooth in Ω and fix $a \in \Omega$. The one-variable function $u_\xi : \zeta \mapsto u(a + \zeta \cdot \xi)$ is defined on a small disc around 0. For $\zeta \in \mathbb{C}$ small enough observe that

$$\frac{\partial^2 u_\xi}{\partial \zeta \partial \bar{\zeta}}(\zeta) = \Delta_\xi u(a + \zeta \cdot \xi).$$

This yields the first claim of the proposition in this smooth setting.

We proceed by regularization to treat the general case. Since Δ_ξ is linear with constant coefficients, it commutes with convolution,

$$\Delta_\xi u_\varepsilon = (\Delta_\xi u)_\varepsilon,$$

and we can pass to the limit to conclude.

For the converse we proceed as in the proof Proposition 1.19 to show that the distributions $\dfrac{\partial^2 u}{\partial z_j \partial \bar{z}_j}$ are non-negative in Ω. Thus they extend to non-negative Borel measures in Ω. The mixed complex derivatives are controlled by using polarization identities for Hermitian forms. □

1.3.3.2 Invariance properties Plurisubharmonicity is invariant under holomorphic changes of coordinates, hence it makes sense on complex manifolds. More generally we have the following:

Proposition 1.44. *Fix $\Omega \subset \mathbb{C}^n$ and $\Omega' \subset \mathbb{C}^m$. If $u \in PSH(\Omega)$ and $f : \Omega' \to \Omega$ is a holomorphic map, then $u \circ f \in PSH(\Omega')$.*

Proof. Using convolutions we reduce to the case when u is smooth. Fix $z_0 \in \Omega'$ and set $w_0 = f(z_0)$. By the chain rule,

$$\frac{\partial^2 (u \circ f)}{\partial z_i \partial \bar{z}_j}(z_0) = \sum_{1 \le k,l \le n} \frac{\partial f_k}{\partial z_i} \overline{\frac{\partial f_l}{\partial z_j}}(z_0) \frac{\partial^2 u}{\partial w_k \partial \bar{w}_l}(w_0),$$

for $i, j = 1, \ldots, m$. Thus for all $\xi \in \mathbb{C}^n$,

$$\sum_{i,j} \xi_j \bar{\xi}_j \frac{\partial^2 (u \circ f)}{\partial z_i \partial \bar{z}_j}(z_0) = \sum_{k,l} \eta_k \bar{\eta}_l \frac{\partial^2 u}{\partial w_k \partial \bar{w}_l}(w_0),$$

where $\eta_k := \sum_{i=1}^m \xi_i \dfrac{\partial f_k}{\partial z_i}(z_0)$ for $k = 1, \ldots, n$.

This means in terms of the Levi forms that $\mathcal{L}_{u\circ f}(z_0;\xi) = \mathcal{L}_u(w_0;\eta)$, where $\eta := \partial f(z_0)\cdot\xi$ and $w_0 := f(z_0)$. The result follows. □

We have observed that plurisubharmonic functions are subharmonic when considered as functions of $2n$ real variables, identifying $\mathbb{C}^n \simeq \mathbb{R}^{2n}$. Conversely, one can characterize plurisubharmonic functions as those subharmonic functions in $\mathbb{R}^{2n} \simeq \mathbb{C}^n$ which are invariant under complex linear transformations of $\mathbb{C}^n$:

Proposition 1.45. *Let $\Omega \subset \mathbb{C}^n \simeq \mathbb{R}^{2n}$ be a domain and $u : \Omega \to [-\infty, +\infty[$ an upper semi-continuous function in Ω. Then u is plurisubharmonic in Ω iff for any complex affine transformation $S : \mathbb{C}^n \to \mathbb{C}^n$, $u \circ S$ is subharmonic in the domain $S^{-1}(\Omega) \subset \mathbb{R}^{2n}$.*

Proof. One implication follows from Proposition 1.44: if u is plurisubharmonic, then $u \circ S$ is subharmonic for any complex affine transformation $S : \mathbb{C}^n \to \mathbb{C}^n$.

We now prove the converse. Let $r > 0$ be small enough and $z \in \Omega_r$. By assumption, for all $0 < \varepsilon < 1$, the function $\xi \mapsto u(z_1 + r\xi_1, z' + r\varepsilon\xi')$ is subharmonic in ξ in a neighborhood of the unit ball $\{|\xi| \le 1\}$ in $\mathbb{R}^{2n}$. The spherical submean value inequality yields

$$u(z) \le \int_{|\xi|=1} u(z_1 + r\xi_1, z' + r\varepsilon\xi')d\sigma(\xi),$$

where $d\sigma$ denotes the normalized Lebesgue measure on the unit sphere. Since u is upper semi-continuous and locally bounded from above, by Fatou's lemma, we obtain as $\varepsilon \to 0$

$$u(z) \le \int_{|\xi|=1} u(z_1 + r\xi_1, z')d\sigma(\xi).$$

This means that the function of one complex variable $\zeta \to u(\zeta, z')$ is subharmonic in its domain. The subharmonicity on other lines follows from the invariance under complex transformations. □

1.4 Hartogs lemma and the Montel property

1.4.1 Hartogs lemma

Theorem 1.46. *Let (u_j) be a sequence of functions in $PSH(\Omega)$ which is locally uniformly bounded from above in Ω.*

1. *If (u_j) does not converge to $-\infty$ locally uniformly on Ω, then it admits a subsequence which converges to some $u \in PSH(\Omega)$ in $L^1_{\mathrm{loc}}(\Omega)$.*

2. *If $u_j \to U$ in $\mathcal{D}'(\Omega)$, then the distribution U is defined by a unique function $u \in PSH(\Omega)$. Moreover*

- $u_j \to u$ *in* $L^1_{\mathrm{loc}}(\Omega)$*;*
- $\limsup u_j(z) \le u(z)$ *for all* $z \in \Omega$*, with equality a.e. in* Ω*;*
- *for any compact set* K *and any continuous function* h *on* K*,*

$$\limsup \max_K (u_j - h) \le \max_K (u - h).$$

The last item is usually called *Hartogs lemma*. When the compact set K is "regular", we actually have an equality:

$$\lim \max_K (u_j - h) = \max_K (u - h).$$

We are not going to study this notion any further here. The reader will check in Exercise 1.20 that if a compact set is the closure of an open set with smooth boundary, then it is regular.

Proof. The statement is local, so we can assume that $\Omega \Subset \mathbb{C}^n$ and $u_j \le 0$ in Ω for all $j \in \mathbb{N}$ (substracting a constant if necessary).

Since (u_j) does not converge uniformly towards $-\infty$, we can find a compact set E and $C > 0$ such that

$$\limsup_{j \to +\infty} \max_E u_j \ge -C > -\infty.$$

Thus there exist an increasing sequence (j_k) of integers and a sequence of points (x_k) in E such that the sequence $u_{j_k}(x_k)$ is bounded from below by $-2C$. Extracting again we can assume that $x_k \to a \in E$. Set for simplicity $v_k := u_{j_k}$ for $k \in \mathbb{N}$.

We know that $v_k \in L^1_{\mathrm{loc}}(\Omega)$ and we claim that if $B \Subset \Omega$ is a ball around a, the sequence $\int_B v_k d\lambda$ is bounded. Indeed, for k large enough there is a ball B_k centered at x_k such that $B \subset B_k \Subset \Omega$, hence

$$\int_B v_k d\lambda \ge \int_{B_k} v_k d\lambda \ge \lambda(B_k) v_k(x_k) \ge -2C\lambda(\Omega).$$

By the same reasoning as in the proof of Proposition 1.34, we deduce from this that the set X of points $x \in \Omega$ which have a neighborhood $W \subset \Omega$ such that the sequence $\int_W v_k d\lambda$ is bounded below is closed. Since it is open (by definition) and not empty (by assumption), we infer from connectedness that $X = \Omega$.

The sequence (v_k) is therefore bounded in $L^1_{\mathrm{loc}}(\Omega)$, i.e., the sequence of nonnegative measures $\mu_k := (-v_k)\lambda$ is bounded in the weak topology of Radon measures on Ω. Thus it admits a subsequence which converges weakly (in the sense of Radon measures), hence the first assertion is now a consequence of the second one.

We now prove the second statement. Assume that $u_j \rightharpoonup U$ in the weak sense of distributions on Ω. It follows from Proposition 1.43 that $U = T_u$ is defined by a plurisubharmonic function u.

We want to show that $u_j \to u$ in $L^1_{\mathrm{loc}}(\Omega)$. Fix (ρ_ε) mollifiers as earlier. We observe that the sequence $(u_j \star \rho_\varepsilon)_{j\in\mathbb{N}}$ is equicontinuous in Ω_ε, since (u_j) is bounded in $L^1_{\mathrm{loc}}(\Omega)$. Indeed, fix $a \in \Omega_\varepsilon$, $0 < \eta < \varepsilon/2$. Then for $x, y \in B(a,\eta)$,

$$|u_j \star \rho_\varepsilon(x) - u_j \star \rho_\varepsilon(y)| \leq \sup_{|t|\leq\varepsilon} |\rho(x-t) - \rho(y-t)| \cdot \|u_j\|_{L^1(B(a,\varepsilon))}.$$

Fix $K \subset \Omega$ a compact set and χ a continuous test function in Ω such that $\chi \equiv 1$ on K and $0 \leq \chi \leq 1$ on Ω. Then

$$\begin{aligned}\int_K |u_j - u| d\lambda &\leq \int (u_j \star \rho_\varepsilon - u_j)\chi d\lambda + \int \chi |u_j \star \rho_\varepsilon - u \star \rho_\varepsilon| d\lambda \\ &+ \int (u \star \rho_\varepsilon - u)\chi d\lambda.\end{aligned}$$

We use here the key fact that $u_j \star \rho_\varepsilon - u_j \geq 0$.

By weak convergence, the fist term converges to $\int (u\star\rho_\varepsilon - u)\chi d\lambda$ and by equicontinuity, $u_j \star \rho_\varepsilon \to u \star \rho_\varepsilon$ uniformly on K as $j \to +\infty$. Hence

$$\limsup_{j\to+\infty} \int_K |u_j - u| d\lambda \leq 2 \int (u \star \rho_\varepsilon - u)\chi d\lambda.$$

The monotone convergence theorem ensures that the right-hand side converges to 0 as $\varepsilon \searrow 0$.

Since $u_j \star \rho_\varepsilon \to u \star \rho_\varepsilon$ locally uniformly in Ω_ε and $u_j \leq u_j \star \rho_\varepsilon$, it follows that $\limsup u_j \leq u \star \rho_\varepsilon$ in Ω, hence $\limsup u_j \leq u$ in Ω.

Fatou's lemma ensures that for any fixed compact set $K \subset \Omega$,

$$\int_K u d\lambda = \lim_j \int_K u_j d\lambda \leq \int_K (\limsup_j u_j) d\lambda \leq \int_K u d\lambda$$

As $u - \limsup u_j \geq 0$ in Ω and $\int_K (u - \limsup u_j) d\lambda = 0$, we infer that $u - \limsup_j u_j = 0$ almost everywhere in K.

To prove the last property, we observe that

$$\max_K (u_j - h) \leq \max_K (u_j \star \rho_\varepsilon - h) \longrightarrow \max_K (u \star \rho_\varepsilon - h),$$

where the last convergence follows from the equicontinuity of the family $(u_j \star \rho_\varepsilon - h)$ for fixed $\varepsilon > 0$. □

The following consequence is a kind of "Montel property" of the convex set $PSH(\Omega)$:

Corollary 1.47. *The space $PSH(\Omega)$ is a closed subset of $L^1_{\mathrm{loc}}(\Omega)$ for the L^1_{loc}-topology which has the Montel property: every bounded subset in $PSH(\Omega)$ is relatively compact.*

1.4.2 Comparing topologies Plurisubharmonic functions have rather good integrability properties: they belong to the spaces L^p_{loc} for all $1 \le p < +\infty$ and their gradients are in L^q_{loc} for all $1 \le q < 2$:

Theorem 1.48. *Let (u_j) be a sequence of functions in $PSH(\Omega)$ converging in L^1_{loc} to $u \in PSH(\Omega)$. Then*

1. *the sequence is locally uniformly bounded from above;*
2. *$u_j \to u$ in L^p_{loc} for all $p \ge 1$;*
3. *the gradients Du_j converge in L^q_{loc} to Du for all $q < 2$.*

Together with Theorem 1.46, this result shows that

$$PSH(\Omega) \subset L^p_{\mathrm{loc}}(\Omega) \subset \mathcal{D}'(\Omega)$$

and the weak topology of distributions on Ω and the L^p_{loc}-topology coincide on the space $PSH(\Omega)$ for all $p \ge 1$.

Proof. Step 1. We first show that (u_j) is locally uniformly bounded in L^p_{loc}, for all $p \ge 1$. Assume first that $n = 1$, $\bar{\mathbb{D}} \subset \Omega$, and $u(0) > -\infty$. We can assume without loss of generality that $u \le 0$. The Poisson–Jensen formula yields

$$u(z) = \int_{\partial\mathbb{D}} u(\zeta)\frac{1-|z|^2}{|z-\zeta|^2}d\sigma(\zeta) + \int_{|\zeta|<1} \log\frac{|z-\zeta|}{|1-z\bar{\zeta}|}d\mu(\zeta), \tag{1.4.1}$$

where $d\sigma(\zeta) := \dfrac{|d\zeta|}{2\pi}$ is the normalized length measure on $\partial\mathbb{D}$ and $\mu := \dfrac{\Delta u}{2\pi}$ is the Riesz measure of u in $\mathbb{D}$. In particular,

$$u(0) = \int_0^{2\pi} u(e^{i\theta})\frac{d\theta}{2\pi} + \int_{|\zeta|<1} \log|\zeta| d\mu(\zeta).$$

Set

$$h(z) = \int_{\partial\mathbb{D}} u(\zeta)\frac{1-|z|^2}{|z-\zeta|^2}d\sigma(\zeta).$$

This is a negative harmonic function in the unit disc. It follows from Harnack inequalities that $3h(0) \le h(z) \le 3^{-1}h(0)$ for $|z| < 1/2$. Thus, for $p \ge 1$,

$$\left(\int_{|z|<1/2} |h(z)|^p d\lambda(z)\right)^{1/p} \le 3(\pi/4)^{1/p}|h(0)|.$$

We claim that there is $C_p > 0$ such that, for all $a \in \mathbb{D}$,

$$\left(\int_{|z|<1/2} \left(-\log\frac{|z-a|}{|1-z\bar{a}|}\right)^p d\lambda(z)\right)^{1/p} \le -C_p \log|a|. \tag{1.4.2}$$

Indeed, set $h_a(z) := -\log \dfrac{|z-a|}{|1-z\bar{a}|}$. This is a positive harmonic function in $\mathbb{D} \setminus \{a\}$ with a logarithmic singularity at a. Moreover,

$$0 \le h_a(z) \le \log \frac{2}{|z-a|},$$

for $|z| < 1$, hence for $|a| < 1$,

$$\int_{|z|<1/2} |h_a(z)|^p \, d\lambda(z) \le \Gamma(p+1).$$

The inequality (1.4.2) is thus valid when $|a| \le 3/4$ with C_p such that

$$C_p \ge \Gamma(p+1)^{1/p}/(\log(4/3))^p.$$

Assume now $|a| > 3/4$. Then h_a is a positive harmonic function near $\overline{D}(3/4)$. It follows again from Harnack inequalities that

$$0 \le h_a(z) \le 5h_a(0) = 5\log(1/|a|),$$

for $|z| \le 1/2$. Therefore

$$\int_{|z|<1/2} |h_a(z)|^p \, d\lambda(z) \le 5^p \, (\log(1/|a|))^p \, .$$

This proves our claim with $C_p := \max\{\Gamma(p+1)^{1/p}/(\log(4/3)), 5\}$. It follows now from Minkowski's inequality that

$$\left(\int_{|z|<1/2} |u|^p \, d\lambda(z) \right)^{1/p} \le C_p(|h(0)| + \int_{|\zeta|<1} \log(1/|\zeta|) d\mu(\zeta) = C_p|u(0)|.$$

In higher dimension we use this inequality n-times: if u is plurisubharmonic, $u \le 0$ near $a + \overline{\mathbb{D}}_R^n \subset \Omega$, and $u(a) > -\infty$, then

$$\int_{\mathbb{D}_{1/2}^n} |u(a+Rz)|^p \, d\lambda(z) \le C_p^{np} |u(a)|^p. \tag{1.4.3}$$

Using the same reasoning as in the proof of Proposition 1.34 we deduce from (1.4.3) that the set of points where $|u|^p$ is locally integrable is a non-empty, open, and closed in Ω, hence $u \in L^p_{\mathrm{loc}}(\Omega)$.

Recall now that $u_j \to u$ in L^1_{loc}, thus (u_j) does not converge uniformly to $-\infty$ on any compact set $K \Subset \Omega$ and it is locally bounded from above in Ω. Arguing as in the proof of Proposition 1.34, we infer from (1.4.3) that the set of points where the sequence $|u_j|^p$ is locally uniformly integrable is a non-empty open and closed set in Ω, hence (u_j) is locally bounded in $L^p_{\mathrm{loc}}(\Omega)$.

Step 2. We now show that $u_j \to u$ in $L^p_{\mathrm{loc}}(\Omega)$. Fix a compact set $K \Subset \Omega$ and assume that $u_j \le 0$ in K for all $j \in \mathbb{N}$.

Assume first that the sequence is locally uniformly bounded. There exists $M > 0$ such that for any $j \in \mathbb{N}$, $-M \le u_j \le 0$ on K. We fix a subsequence (v_j) of (u_j) such that $v_j \to u$ almost everywhere. The Lebesgue convergence theorem ensures that $v_j \to u$ in $L^p(K)$. This implies that u is the unique limit point of the sequence (u_j) in $L^p_{\mathrm{loc}}(\Omega)$.

To treat the general case, we set for $m \ge 1$ and $j \in \mathbb{N}$

$$u_j^m := \sup\{u_j, -m\}, \quad u^m := \sup\{u, -m\}.$$

Minkowski's inequality yields

$$\|u_j - u\|_{L^p(K)} \le \|u_j - u_j^m\|_{L^p(K)} + \|u_j^m - u^m\|_{L^p(K)} + \|u^m - u\|_{L^p(K)}.$$

By the monotone convergence theorem, the last term converges to 0 as $m \to +\infty$. By the previous case, for a fixed m the second term converges to 0 as $j \to +\infty$. To conclude it is thus enough to show that the first term converges to 0 uniformly in j as $m \to +\infty$. Markov's inequality yields, for $m \ge 1$ and $j \in \mathbb{N}$,

$$\begin{aligned} \int_K |u_j - u_j^m|^p d\lambda &= 2\int_{K\cap\{u_j \le -m\}} |u_j|^p d\lambda \\ &\le \frac{2}{m}\int_K |u_j|^{p+1} d\lambda, \end{aligned}$$

which allows us to conclude, since (u_j) is bounded in $L^{p+1}(K)$.

Step 3. We now establish local uniform bounds on the gradient of u in $L^p(\Omega)$ for $1 \le p < 2$. Assume first that $n = 1$. It suffices to consider the case when $2\mathbb{D} \Subset \Omega$ and $u(0) > -\infty$ and get a uniform estimate on $\mathbb{D}_{1/2}$.

The Poisson–Jensen formula (1.4.1) shows that for $z \in \mathbb{D}$,

$$2\partial_z u(z) = 2\partial_z h(z) + \int_{\mathbb{D}} \frac{1-|\zeta|^2}{(z-\zeta)(1-z\bar\zeta)} d\mu(\zeta).$$

Since h is harmonic, the representation formula yields

$$\partial_z h(z) = \int_{\partial\mathbb{D}} u(\zeta)\partial_z P(z,\zeta) d\sigma(\zeta)$$

when $z \in \mathbb{D}$. Further, since

$$\partial_z P(z,\zeta) = \frac{-\bar z}{|z-\zeta|^2} - \frac{(|1-|z|^2)(\bar\zeta - \bar z)}{|z-\zeta|^4},$$

it follows that for $|z| \le 1/2$,

$$|\partial_z h(z)| \le 6\int_{\partial\mathbb{D}} |u(\zeta)| d\sigma(\zeta) \le 6\|u\|_{L^1(\mathbb{D})}.$$

We have used here (1.37) and the fact that $u \le 0$.

We need a uniform estimate for the second term, which we denote by $g(z)$. From the expression of g we get

$$|g(z)| \leq 2 \int_{\mathbb{D}} \frac{d\mu(\zeta)}{|z-\zeta|}.$$

Using Minkowski's inequality we deduce that

$$\begin{aligned} \|g\|_{L^p(\{|z|<1/2\})} &\leq 2 \int_{\mathbb{D}} \left(\int_{\{|z|<1/2\}} \frac{d\lambda(z)}{|z-\zeta|^p} \right)^{1/p} d\mu(\zeta) \\ &\leq 2\pi \frac{2^{2-p}}{2-p} \int_{\mathbb{D}} d\mu. \end{aligned}$$

Since $2\mathbb{D} \Subset \Omega$ we apply Stokes' formula to get

$$\int_{\mathbb{D}} d\mu = \int_{\mathbb{D}} dd^c u \leq \frac{1}{3} \int_{\mathbb{D}} (4-|z|^2) dd^c u \leq \frac{1}{3} \int_{2\mathbb{D}} (-u) d\lambda(z).$$

Adding all these inequalities, we get a uniform bound on the gradient of u in the disc $|z| \leq 1/2$,

$$\|\partial_z u\|_{L^p(\frac{1}{2}\mathbb{D})} \leq c_p \|u\|_{L^1(2\mathbb{D})},$$

where c_p is a uniform constant depending only on p.

Using these inequalities n times we get a local uniform bound for the gradient of any function $u \in PSH(\Omega)$. In particular, for any $p < 2$, we have $PSH(\Omega) \subset W^{1,p}_{\mathrm{loc}}(\Omega)$ and the inclusion operator takes bounded sets into bounded sets.

Step 4. We finally prove that this inclusion is continuous. Let $(u_j) \in PSH(\Omega)^{\mathbb{N}}$ be a sequence converging to u in L^1_{loc}. Since plurisubharmonic functions are subharmonic in $\mathbb{R}^{2n}$, it follows from Exercise 1.21 that $Du_j \to Du$ in L^1_{loc}, hence almost everywhere (up to extracting and relabelling). The local uniform bounds for $||Du_j||_{L^q}$, $q < 2$, allow to conclude as above that $Du_j \to Du$ in L^q_{loc} for all $q < 2$. □

1.5 Exercises

Exercise 1.1.

1) *Show that a (real valued) function u (on a metric space) is upper semi-continuous iff* $\limsup_{z\to a} u(z) = u(a)$ *at all points a.*

2) *Let $u : X \to \mathbb{R} \cup \{-\infty\}$ be an upper semi-continuous function on a metric space (X, d) which is bounded. Show that*

$$x \longmapsto u_k(x) := \sup\{u(y) - k d(x, y) \mid y \in X\}$$

are Lipschitz functions which decrease to u as k increases to $+\infty$,

3) *Same question if u is merely bounded from above (replace u by $(\sup\{u, -j\})_{j\in\mathbb{N}}$ and use previous question).*

Exercise 1.2. *Let $u_1, \dots, u_s$ be subharmonic functions in a domain $\Omega \subset \mathbb{C}$. Show that*

$$v := \log\left[e^{u_1} + \cdots + e^{u_s}\right]$$

defines a subharmonic function in Ω.

Deduce from this, by a rescaling argument, that $\max(u_1, \dots, u_s)$ *is subharmonic as well.*

Exercise 1.3. *Let I be an open subset of $\mathbb{R}$. Show that $f : I \to \mathbb{R}$ is convex if and only if*

$$\limsup_{h \to 0} \left[\frac{f(x+h) + f(x-h) - 2f(x)}{h^2} \right] \geq 0.$$

Exercise 1.4. *Let I be an open subset of $\mathbb{R}$ and $f : I \to \mathbb{R}_*^+$ a positive function. Show that* $\log f : I \to \mathbb{R}$ *is convex if and only if for all $c \in \mathbb{R}$, $t \in I \mapsto e^{ct} f(t) \in \mathbb{R}$ is convex.*

Exercise 1.5. *Let $f_j : \mathbb{R} \to \mathbb{R}$ be a sequence of convex functions which converge pointwise towards a function $f : \mathbb{R} \to \mathbb{R}$. Show that f is convex and that (f_j) converges to f uniformly on each compact subset of $\mathbb{R}$.*

Exercise 1.6.

1. *Compute the Laplacian in polar coordinates in $\mathbb{C}$.*
2. *Let $u(z) = \chi(|z|)$, χ a smooth function in $[0, R[$. Show that*

$$\Delta u(z) = \chi''(r) + \frac{1}{r}\chi'(r).$$

3. *Describe all harmonic radial functions in $\mathbb{C}$.*
4. *Show that u is subharmonic in a disc $\mathbb{D}(0, R)$ iff χ is a convex increasing function of $t = \log r$ in the interval $]-\infty, \log R[$.*

Exercise 1.7. *Let $h : \mathbb{R}^2 \to \mathbb{R}$ be a harmonic function. Assume there exists $C, d > 0$ such that*

$$|h(x)| \leq C\left[1 + ||x||\right]^d, \quad \text{for all } x \in \mathbb{R}^2.$$

Show that h is a polynomial of degree at most d.

Exercise 1.8. *Let $\phi : \partial\Delta \to \mathbb{R}$ be a continuous function and let u_ϕ denote the Poisson envelope of ϕ in the unit disc $\Delta \subset \mathbb{C}$.*

i) *Show that u_ϕ is Hölder continuous on $\overline{\Delta}$ if and only if ϕ is Hölder continuous. Is the Hölder exponent preserved?*

ii) *By considering $\phi(e^{i\theta}) = |\sin\theta|$, show that ϕ being Lipschitz on $\partial\Delta$ does not necessarily imply that u_ϕ is Lipschitz on $\overline{\Delta}$.*

Exercise 1.9. *Let μ be a probability measure in $\mathbb{C}$.*

1) *Show that*

$$\varphi_\mu(z) := \int_{w\in\mathbb{C}} \log|z-w| d\mu(w)$$

defines a subharmonic function with logarithmic growth in $\mathbb{C}$.

2) *Show that if φ is a subharmonic function with logarithmic growth in $\mathbb{C}$, then there exists $c \in \mathbb{R}$ such that $\varphi = \varphi_\mu + c$, where $\mu = \Delta\varphi/2\pi$.*

3) *Approximating μ by Dirac masses, show that every subharmonic function with logarithmic growth in $\mathbb{C}$ can be approximated in L^1 by functions of the type $j^{-1}\log|P_j|$, where P_j is a polynomial of degree j.*

Exercise 1.10. *Let $(a_j) \in \mathbb{C}^{\mathbb{N}}$ be a bounded sequence which is dense in the unit disc and let $\varepsilon_j > 0$ be positive reals such that $\sum_j \varepsilon_j < +\infty$. Show that the function*

$$z \longmapsto u(z) := \sum_j \varepsilon_j \log|z-a_j|$$

belongs to $SH(\mathbb{C})$ and has an uncoutable polar set ($u = -\infty$).

Check that u is discontinuous almost everywhere in the unit disc.

Exercise 1.11. *Let $T \in \mathcal{D}'(\Omega)$ be a non-negative distribution, i.e.,*

$$\langle T, \chi\rangle \geq 0$$

for all non-negative test functions $0 \leq \chi \in \mathcal{D}(\Omega)$. Show that T is of order zero, i.e., it can be extended as a continuous (non-negative) linear form on the space of continuous functions with compact support in Ω (in other words T extends as a Radon measure).

Exercise 1.12. *Let φ be a subharmonic function in $\mathbb{R}^{2n} \simeq \mathbb{C}^n$, i.e., an upper semi-continuous function which is locally integrable and satisfies*

$$\Delta\varphi := \frac{1}{4}\sum_{i=1}^{n} \frac{\partial^2\varphi}{\partial z_i \partial\overline{z}_i} \geq 0$$

in the sense of distributions. Show that φ is pluri-subharmonic if and only if for all $A \in \mathrm{GL}(n,\mathbb{C})$, the function

$$\varphi_A : z \in \mathbb{C}^n \longmapsto \varphi(A\cdot z) \in \mathbb{R}$$

is subharmonic.

Exercise 1.13. *Show that any convex function $f : \mathbb{R} \to \mathbb{R}$ that is bounded from above is constant. Use this to prove that any plurisubharmonic function $\varphi : \mathbb{C}^n \to \mathbb{R}$ that is bounded from above is constant.*

Exercise 1.14. *Let $\Omega \subset \mathbb{C}^n$ be a domain. Show that $\varphi : \Omega \to \mathbb{R}$ is pluriharmonic if and only if it is locally the real part of a holomorphic function.*

Exercise 1.15.

1) *Let $\varphi : \mathbb{C}^n \to \mathbb{R}$ be a plurisubharmonic function. Show that*

$$\varphi|_{\mathbb{R}^n} \in L^1_{\mathrm{loc}}(\mathbb{R}^n).$$

2) *Let φ_j be a sequence of plurisubharmonic functions in $\mathbb{C}^n$ such that $\varphi_j \to \varphi$ in $L^1_{\mathrm{loc}}(\mathbb{C}^n)$. Show that*

$$\varphi_{j\,|\mathbb{R}^n} \longrightarrow \varphi|_{\mathbb{R}^n} \text{ in } L^1_{\mathrm{loc}}(\mathbb{R}^n).$$

Exercise 1.16. *What is the limit of $\varphi_j(z) = j \log ||z||$ in $\mathbb{C}^n$? Is it in contradiction with the compactness criteria we have established?*

Exercise 1.17. *Let $\Omega \subset \mathbb{C}^n$ be a domain and $F = \{u = -\infty\}$ a closed complete pluripolar set (the $-\infty$ locus of a plurisubharmonic function u). Let φ be a plurisubharmonic function in $\Omega \setminus F$ which is locally bounded near F. Show that φ uniquely extends through F as a plurisubharmonic function.*

Exercise 1.18. *Let $\Omega \subset \mathbb{C}^n$ be a domain and $A \subset \mathbb{C}^n$ an analytic subset of complex codimension ≥ 2. Let φ be a plurisubharmonic function in $\Omega \setminus A$. Show that φ uniquely extends through A as a plurisubharmonic function (see* [Cirka] *for some help).*

Exercise 1.19. *Let $f : \Omega \to \Omega'$ be a proper surjective holomorphic map between two domains $\Omega \subset \mathbb{C}^n$, $\Omega' \subset \mathbb{C}^k$. Let u be a plurisubharmonic function in Ω and set, for $z' \in \Omega'$,*

$$v(z') := \max\{u(z)\ ,\ f(z) = z'\}.$$

Show that v is plurisubharmonic in Ω'.

Exercise 1.20. *Let u be a plurisubharmonic function in $\mathbb{C}^n$.*

1) *Show that for any ball B, $\sup_B u = \sup_{\overline{B}} u$.*

2) *Generalize* 1) *to bounded open sets Ω with smooth boundary.*

3) *Deduce that $K = \overline{\Omega}$ is a regular set: if u_j is a sequence of plurisubharmonic functions which converge to u in L^1_{loc}, then*

$$\sup_K u_j \longrightarrow \sup_K u.$$

4) *Using the Riemann mapping theorem, show that a connected compact set $K \subset \mathbb{C}$ is regular.*

Exercise 1.21. *Let $(u_j) \in SH(\mathbb{R}^k)^{\mathbb{N}}$ be a sequence of subharmonic functions converging to u in L^1_{loc}. Using the linearity of the Laplace operator, show that*

$$Du_j \longrightarrow Du \ \text{ in } \ L^q_{\mathrm{loc}} \quad \text{for all} \quad q < k/(k-1).$$

Exercise 1.22. *Let $\Omega \subset \mathbb{C}^n$ be a domain. For $K \subset \Omega$ denote by*

$$\widehat{K} := \{z \in \Omega \mid u(z) \leq \sup_K u, \ \forall u \in PSH(\Omega)\}$$

the plurisubharmonic hull of K. Say that Ω is pseudoconvex if $\widehat{K}$ is relatively compact in Ω whenever K is.

1) *Describe $\widehat{K}$ when $n = 1$ and show that any Ω is pseudoconvex.*

2) *Show that $\Omega := \{z \in \mathbb{C}^n \mid 1 < ||z|| < 2\}$ is not pseudoconvex.*

3) *Show that Ω is pseudoconvex iff*

$$z \in \Omega \longmapsto -\log \operatorname{dist}(z, \partial\Omega) \in \mathbb{R}$$

is plurisubharmonic.

2 Positive currents

2.1 Currents in the sense of de Rham

In this section Ω will denote an open subset of $\mathbb{R}^N$.

2.1.1 Forms with distributions coefficients A differential form α of degree p on Ω with locally integrable coefficients,

$$\alpha = \sum_{|I|=p} \alpha_I \, dx_I,$$

acts as a linear form on the space of continuous test forms of complementary degree $q = N - p$: if $\psi = \chi dx_K$ is a continuous test form of degree q with compact support then

$$\langle \alpha, \psi \rangle = \sum_{|I|=p} \varepsilon_{I,K} \int_\Omega \chi \alpha_I \, dV,$$

where $\varepsilon_{I,K}$ is such that $dx_I \wedge dx_K = \varepsilon_{I,K} \, dV$, with $\varepsilon_{I,K} = 0$ unless $K = I^{\mathrm{c}}$ complements I in $[1, N]$, in which case $\varepsilon_{I,K} = \pm 1$.

Definition 2.1. *A current S of degree p is a continuous linear form on the space $\mathcal{D}_{N-p}(\Omega)$ of test forms (i.e., smooth differential forms with compact support) of degree $N - p$ on Ω.*

We let $\mathcal{D}'_{N-p}(\Omega)$ denote the space of currents of degree p. The action of S on a test form $\Psi \in \mathcal{D}_{N-p}(\Omega)$ is denoted by $\langle S, \Psi \rangle$.

If α is a smooth form of degree q, the wedge product of α and S is defined as follows:

Definition 2.2. *For a test form Ψ of degree $N - p - q$, we set*

$$\langle S \wedge \alpha, \Psi \rangle := \langle S, \alpha \wedge \Psi \rangle.$$

We also define $\alpha \wedge S := (-1)^{pq} S \wedge \alpha$.

Observe that the current $S \wedge \alpha \wedge \Psi$ is a current of maximal degree with compact support, and it can be identified with a distribution with compact support. One can similarly interpret a current of degree p as a differential form of degree p with distribution coefficients.

2.1.2 Closed currents If S is a smooth form of degree p in Ω and Ψ is a test form of degree $N-p-1$, then

$$d(S\wedge\Psi)=dS\wedge\Psi+(-1)^pS\wedge d\Psi.$$

Since $S\wedge\Psi$ is a differential form with compact support, it follows from Stokes formula that $\int_\Omega d(S\wedge\Psi)=0$, hence

$$\int_\Omega dS\wedge\Psi=(-1)^{p+1}\int_\Omega S\wedge d\Psi.$$

This suggests the following definition:

Definition 2.3. *If S is a current of degree p, then dS is the current of degree $(p+1)$ defined by*

$$\langle dS,\psi\rangle=(-1)^{p+1}\langle S,d\psi\rangle,$$

for any test form ψ of degree $N-p-1$.

This definition allows one to extend differential calculus on forms to currents. It follows from the definition that a current of degree p is a differential p-form $T=\sum_{|I|=p}T_I\,dx_I$ with distribution coefficients. We set

$$dT=\sum_{|I|=p}\sum_{1\le j\le N}\frac{\partial T_I}{\partial x_j}dx_j\wedge dx_I,$$

where $\frac{\partial T_I}{\partial x_j}$ is the partial derivative of the distribution T_I, acting on test functions according to Stokes formula by

$$\left\langle\frac{\partial T_I}{\partial x_j},\psi\right\rangle=-\left\langle T_I,\frac{\partial\psi}{\partial x_j}\right\rangle.$$

The reader can check that this is consistent with the above definition of dT.

Most properties valid in the differential calculus on forms extend to currents. In particular, if T is a current of degree p and α is a differential form of degree m, then

$$d(T\wedge\alpha)=dT\wedge\alpha+(-1)^pT\wedge d\alpha,$$

as the reader will check in Exercise 2.1.

The following version of Stokes' formula for currents will be used on several occasions:

Lemma 2.4. *Let S be a current of degree $N-1$ with compact support in Ω. Then*

$$\int_\Omega dS=0.$$

Proof. Let χ be a smooth cut-off function in Ω such that $\chi \equiv 1$ in a neighborhood of K, a compact subset of Ω containing the support of S. Then

$$\int_\Omega dS = \int_\Omega \chi dS = \langle S, d\chi \rangle = 0,$$

since $d\chi = 0$ in a neighborhood of the support of S. □

2.1.3 Bidegree Assume now $\Omega \subset \mathbb{C}^n$ is a domain in the complex Hermitian space $\mathbb{C}^n$. The complex structure induces a splitting of differential forms into types. The space of test forms of bidegree (q,q) will be denoted by $\mathcal{D}_{q,q}(\Omega)$, where $0 \leq q \leq n$.

Definition 2.5. *A current T of bidegree (p, p) is a differential form of bidegree (p, p) with distribution coefficients, i.e.,*

$$T = i^{p^2} \sum_{|I|=|J|=p} T_{I,J} dz_I \wedge d\bar{z}_J,$$

where $T_{I,J} \in \mathcal{D}'(\Omega)$.

A current of bidegree (p, p) acts on the space $\mathcal{D}_{q,q}(\Omega)$, $q := n - p$, of test forms of bidegree (q,q) as follows: if

$$\Psi = i^{q^2} \sum_{|K|=|L|=q} \psi_{K,L} dz_K \wedge d\bar{z}_L,$$

where $\psi_{K,L} \in \mathcal{D}(\Omega)$, then

$$\langle T, \Psi \rangle = \sum_{|I|=p,|J|=q} \langle T_{I,I'}, \psi_{J,J'} \rangle,$$

since $i^{p^2} dz_I \wedge d\bar{z}_J \wedge i^{q^2} dz_K \wedge d\bar{z}_L = \varepsilon_{I,J}\varepsilon_{K,L} i dz_1 \wedge d\overline{z_1} \wedge \cdots \wedge i dz_n \wedge d\overline{z_n}$.

One defines similarly currents of bidegree (p,q).

Remark 2.6. *We shall also say that a current of bidegree (p, p) is a current of bidimension $(n - p, n - p)$, since it acts on forms of bidegree $(n - p, n - p)$.*

Recall the decomposition $d = \partial + \overline{\partial}$. We have defined the differential dS of a current; we can similarly define the derivatives ∂S and $\overline{\partial} S$ as follows. If S is a smooth differential form of bidegree (p, p) and $\Psi \in \mathcal{D}_{(n-p-1,n-p)}(\Omega)$, observe that

$$\partial S \wedge \Psi = dS \wedge \Psi = d(S \wedge \Psi) - S \wedge d\Psi = d(S \wedge \Psi) - S \wedge \partial\Psi,$$

hence $\int_\Omega \partial S \wedge \Psi = -\int_\Omega S \wedge \partial\Psi$. This suggests the following:

Definition 2.7. *Let S be a current of bidegree (p, p). The current ∂S is the current of bidegree $(p+1, p)$ defined by*

$$\langle \partial S, \Psi \rangle = -\langle S, \partial \Psi \rangle$$

for all $\Psi \in \mathcal{D}_{(n-p-1,n-p)}(\Omega)$. We define $\bar{\partial} S$ similarly.

We set $d^c := (i/2\pi)(\overline{\partial} - \partial)$. Observe that d and d^c are real differential operators of order one and

$$d d^c = \frac{i}{\pi} \partial \overline{\partial}$$

is a real differential operator of order 2. These operators act naturally on differential forms, their actions are extended to currents by duality.

Lemma 2.8. *Let S be a current of bidegree (p, p), $0 \leq p \leq n-1$, and Ψ a smooth form of bidegree (q, q), $0 \leq q \leq n-p-1$. If α is a smooth form of bidegree $(n-p-q-1, n-p-q-1)$, then*

$$dS \wedge d^c \Psi \wedge \alpha = d\Psi \wedge d^c S \wedge \alpha$$

and

$$S \wedge d d^c \Psi \wedge \alpha - d d^c S \wedge \Psi \wedge \alpha = d(S \wedge d^c \Psi - \Psi \wedge d^c S) \wedge \alpha.$$

If Ψ is a smooth test form of bidegree $(n-p-1, n-p-1)$ then

$$\langle d d^c S, \Psi \rangle = \langle S \wedge d d^c \Psi, 1 \rangle.$$

Proof. Observe that

$$\begin{aligned} d\Psi \wedge d^c S &= \left(\partial \Psi + \overline{\partial} \Psi\right) \wedge \frac{i}{2\pi} \left(\overline{\partial} S - \partial S\right) \\ &= \frac{i}{2\pi} \left(\partial \Psi \wedge \overline{\partial} S + \partial S \wedge \overline{\partial} \Psi + \overline{\partial} \Psi \wedge \overline{\partial} S - \partial \Psi \wedge \partial S\right). \end{aligned}$$

Similarly

$$dS \wedge d^c \Psi = \frac{i}{2\pi} \left(\partial S \wedge \overline{\partial} \Psi + \partial \Psi \wedge \overline{\partial} S + \overline{\partial} S \wedge \overline{\partial} \Psi - \partial S \wedge \partial \Psi\right),$$

in the weak sense of currents on Ω.

The currents $d\Psi \wedge d^c S$ and $dS \wedge d^c \Psi = -d^c \Psi \wedge dS$ have the same $(p+q+1, p+q+1)$-part, hence

$$d\Psi \wedge d^c S \wedge \alpha = dS \wedge d^c \Psi \wedge \alpha,$$

since $\partial S \wedge \partial \Psi \wedge \alpha = 0 = \overline{\partial} \Psi \wedge \overline{\partial} S \wedge \alpha$. On the other hand,

$$d(\Psi \wedge d^c S - S \wedge d^c \psi) = d\Psi \wedge d^c S + \Psi \wedge d d^c S - dS \wedge d^c \Psi - S \wedge d d^c \Psi.$$

The last formula is obtained taking $\alpha = 1$ and applying Lemma 2.4. □

2.2 Positive currents

2.2.1 Positive forms Let V be a complex vector space of complex dimension $n \geq 1$. Consider a basis $(e_j)_{1\leq j\leq n}$ of V and denote by $(e_j^*)_{1\leq j\leq n}$ the dual basis of V^*.Then any $v \in V$ can be written as

$$v = \sum_{1\leq j\leq n} e_j^*(v)e_j.$$

Since we are mainly interested in the case where $V = T_x X$ is the complex tangent space to a complex manifold X of dimension n, we use complex differential notations. A vector $v \in V$ acts as a derivation on germs of smooth functions in a neighborhood of the origin in V by

$$v \cdot f(0) := D_v f(0).$$

If $z = (z_1, \ldots, z_n)$ are complex coordinates identifying V with $\mathbb{C}^n$, then $e_j = \frac{\partial}{\partial z_j}$ is the partial derivative with respect to z_j and $e_j^* = dz_j$.

The exterior algebra of V is

$$\Lambda V_{\mathbb{C}}^* := \oplus \Lambda^{p,q} V^*, \;\; \Lambda^{p,q} V^* := \Lambda^p V^* \otimes \Lambda^q \overline{V^*},$$

where $\Lambda^p V^*$ is the complex vector space of alternating $\mathbb{C}$-linear p-forms. A complex basis of the space $\Lambda^p V^*$ is given by the $dz_{k_1} \wedge \ldots \wedge dz_{k_p}$ where $K = (k_1, \ldots, k_p)$ varies in the set of ordered multi-indices of length $|K| = p$. Thus $\dim_{\mathbb{C}} \Lambda^p V^* = \binom{n}{p}$.

The complex vector space V has a canonical orientation given by the (n,n)-form

$$\beta_n(z) := \frac{i}{2} dz_1 \wedge d\overline{z}_1 \wedge \ldots \wedge \frac{i}{2} dz_n \wedge d\overline{z}_n = dx_1 \wedge dy_1 \ldots \wedge dx_n \wedge dy_n,$$

where $z_j = x_j + iy_j$, $j = 1, \ldots, n$. If $(w_1, \ldots, w_n)$ is another (complex) coordinate system on V, then $dw_1 \wedge \ldots \wedge dw_n = \det(\frac{\partial w_j}{\partial z_k}) dz_1 \wedge \ldots \wedge dz_n$ so that

$$\beta_n(w) = |\det(\partial w_j / \partial z_k)|^2 \beta_n(z).$$

In particular, any complex manifold inherits a canonical orientation induced by its complex structure.

Definition 2.9.

1) *A (n,n)-form $v \in \Lambda^{n,n} V^*$ is positive if in some local coordinate system $(z_1, \ldots, z_n)$ it can be written as $v = \lambda(z)\beta_n(z)$, with $\lambda(z) \geq 0$.*

2) *A (p,p)-form $f \in \Lambda^{p,p} V^*$ $(0 \leq p \leq n)$ is strongly positive if it is a linear combination with positive coefficients of a finite number of decomposable (p,p)-forms, i.e., forms of the type*

$$i\alpha_1 \wedge \overline{\alpha}_1 \wedge \ldots i\alpha_p \wedge \overline{\alpha}_p,$$

where $\alpha_1, \ldots, \alpha_p$ are $(1,0)$-forms on V.

3) *A (p,p)-form $u \in \Lambda^{p,p}V^*(1 \le p \le n-1)$ is (weakly) positive if for all $(1,0)$-forms $\alpha_j \in \Lambda^{1,0}V^*$, $(1 \le j \le q := n-p)$, the (n,n)-form $u \wedge i\alpha_1 \wedge \overline{\alpha}_1 \wedge \ldots i\alpha_q \wedge \overline{\alpha}_q$ is positive.*

Examples 2.10.

1) *For all $(1,0)$-forms $\gamma_1, \ldots, \gamma_p \in \Lambda^{1,0}V^*$, the (p,p)-form*

$$i\gamma_1 \wedge \overline{\gamma}_1 \wedge \ldots i\gamma_p \wedge \overline{\gamma}_p = i^{p^2}\gamma \wedge \overline{\gamma},$$

is positive, where $\gamma := \gamma_1 \wedge \ldots \wedge \gamma_p$.

2) *For all $\alpha \in \Lambda^{p,0}V^*$, the (p,p)-form $i^{p^2}\alpha \wedge \overline{\alpha}$ is positive, hence any strongly positive (p,p)-form is (weakly) positive. If $\alpha \in \Lambda^{p,0}V^*$ and $\beta \in \Lambda^{q,0}V^*$, then*

$$i^{p^2}\alpha \wedge \overline{\alpha} \wedge i^{q^2}\beta \wedge \overline{\beta} = i^{(p+q)^2}\alpha \wedge \beta \wedge \overline{\alpha} \wedge \overline{\beta}.$$

In particular, if $p+q=n$, then $\alpha \wedge \beta = \lambda dz_1 \wedge \ldots \wedge dz_n$, $\lambda \in \mathbb{C}$, hence

$$i^{n^2}\alpha \wedge \beta \wedge \overline{\alpha} \wedge \overline{\beta} = |\lambda|^2 \beta_n(z) \ge 0.$$

The following lemma will be useful in the sequel.

Lemma 2.11. *Let $(z_1, \ldots, z_n)$ be a coordinate system in V. The complex vector space $\Lambda^{p,p}V^*$ is generated by the strongly positive forms*

$$\gamma := i\gamma_1 \wedge \overline{\gamma}_1 \wedge \ldots i\gamma_p \wedge \overline{\gamma}_p, \tag{2.2.1}$$

where the $(1,0)$-forms γ_l are of the type $dz_j \pm dz_k$ or $dz_j \pm i dz_k$.

Proof. The proof relies on the following polarization identities for the forms $dz_j \wedge d\bar{z}_k$:

$$\begin{aligned} 4dz_j \wedge d\overline{z}_k = {} & (dz_j + dz_k) \wedge \overline{(dz_j + dz_k)} - (dz_j - dz_k) \wedge \overline{(dz_j - dz_k)} \\ & + i(dz_j + idz_k) \wedge \overline{(dz_j + idz_k)} - i(dz_j - idz_k) \wedge \overline{(dz_j - idz_k)}. \end{aligned}$$

Since

$$dz_J \wedge d\overline{z}_K = dz_{j_1} \wedge \ldots \wedge dz_{j_p} \wedge d\overline{z}_{k_1} \wedge \ldots \wedge d\overline{z}_{k_p} = \pm \bigwedge_{1 \le s \le p} dz_{j_s} \wedge d\overline{z}_{k_s},$$

it follows that the (p,p)-forms γ_s of type (2.2.1) generate the space $\Lambda^{p,p}V^*$ over $\mathbb{C}$. □

Corollary 2.12.

1. *All positive forms $u \in \Lambda^{p,p}V^*$ are real, i.e., $\bar{u} = u$, and if $u = i^{p^2}\sum_{|I|=|J|=p} u_{I,J}\, dz_I \wedge d\bar{z}_J$, then the coefficients satisfy the Hermitian symmetry relation $\bar{u}_{I,J} = u_{J,I}$ for all I, J.*

2. *A form $u \in \Lambda^{p,p}V^*$ is positive if and only if its restriction to any complex subspace $W \subset V$ of dimension p is a positive form of top degree on W.*

Proof. Let (θ_s) be the basis of $\Lambda^{p,p}V^*$ dual to the basis of strongly positive forms (γ_s) of $\Lambda^{n-p,n-p}V^*$ given by Lemma 2.11. Observe that strongly positive forms are real. If $\alpha \in \Lambda^{p,p}V^*$ is positive, we decompose it as $\alpha = \sum_s c_s\theta_s$ with $c_s = \alpha\wedge\gamma_s \geq 0$ for any s. Thus $\overline{\alpha} = \alpha$.

Suppose that $\alpha = i^{p^2}\sum_{|I|=p,|J|=p}\alpha_{I,J}dz_I \wedge d\overline{z}_J$; then

$$\overline{\alpha} = (-1)^{p^2}i^{p^2}\sum_{|I|=p,|J|=p}\overline{\alpha_{I,J}}d\overline{z}_I \wedge dz_J.$$

Since $d\overline{z}_I \wedge dz_J = (-1)^{p^2}dz_J \wedge d\overline{z}_I$, it follows that $\overline{\alpha}_{I,J} = \alpha_{J,I}$, $\forall I, J$.

If $W \subset V$ is a complex subspace of dimension p, there exists a system of complex coordinates $(z_1, \ldots, z_n)$ such that

$$W = \{z_{p+1} = \ldots = z_n = 0\}.$$

Thus $u|_W = c_W\frac{i}{2}dz_1 \wedge d\overline{z}_1 \wedge \ldots \frac{i}{2}dz_p \wedge d\overline{z}_p$, where c_W is given by

$$\alpha \wedge \frac{i}{2}dz_{p+1} \wedge d\overline{z}_{p+1} \wedge \ldots \frac{i}{2}dz_n \wedge d\overline{z}_n = c_W\beta_n(z).$$

Therefore, if α is positive, then $\alpha|_W \geq 0$ for any complex subspace $W \subset V$ of dimension p. The converse is true since the $(n-p, n-p)$-forms $\wedge_{j>p}idz_j \wedge d\overline{z}_j$ generate the cone of strongly positive (p, p)-forms when W varies among all complex subspaces W of dimension p. □

Corollary 2.13. *A $(1,1)$-form $\omega = i\sum_{j,k}\omega_{jk}dz_j \wedge d\bar{z}_k$ is positive if and only if the matrix (ω_{jk}) is a Hermitian semi-positive matrix, i.e.,*

$$\sum_{j,k}\omega_{jk}\xi_j\bar{\xi}_k \geq 0 \quad \textit{for all} \quad \xi \in \mathbb{C}^n.$$

Proof. Indeed, if $W = \mathbb{C}\cdot\xi$ is a complex line generated by the vector $\xi \neq 0$, then $\omega|_W = (\sum_{j,k}\omega_{j,k}\xi_j\overline{\xi_k})idt \wedge d\overline{t}$. □

Remark 2.14. *There is a canonical correspondence between Hermitian forms and real $(1,1)$-forms on V. Indeed, in a system of complex coordinates $(z_1, \ldots, z_n)$, a Hermitian form can be written as*

$$h = \sum_{1\leq j,k\leq n}h_{j,k}dz_j \otimes d\overline{z}_k.$$

The associated $(1,1)$-form

$$\omega_h := \frac{i}{2}\sum_{1\leq j,k\leq n}h_{j,k}dz_j \wedge d\overline{z}_k$$

is a real $(1,1)$*-form on* V*. This correspondence does not depend on the system of complex coordinates, since for all* $\xi, \eta \in V$*,*

$$\omega_h(\xi,\eta) = \frac{i}{2} \sum_{1\le j,k\le n} h_{j,k}(\xi_j \overline{\eta_k} - \eta_j \overline{\xi_k}) = -\Im h(\xi,\eta),$$

and $h_{k,j} = \overline{h_{j,k}}$*. Moreover*

$$\omega_h(i\xi,\eta) = -\Im h(i\xi,\eta) = \Re h(\xi,\eta),$$

for all $(\xi,\eta) \in V$*, which proves that* h *is completely determined by* ω_h*.*

Observe finally that h *is a positive Hermitian form on* V *if and only if the* $(1,1)$*-form* ω_h *is positive.*

The notions of positivity (strong and weak) usually differ, but they do coincide in bidegree $(1,1)$:

Proposition 2.15. *A* $(1,1)$*-form is strongly positive if and only if it is weakly positive. In particular, if* $\alpha \in \Lambda^{p,p}V^*$ *is a weakly positive form on* V*, then for all positive* $(1,1)$*-forms* $\omega_1, \dots \omega_q$ *with* $p+q \le n$*, the* $(p+q, p+q)$*-form* $\alpha \wedge \omega_1 \wedge \ldots \wedge \omega_q$ *is weakly positive.*

Proof. Let $\omega \in \Lambda^{1,1}V^*$ be a positive $(1,1)$-form on V. Diagonalizing the Hermitian form h associated to ω, we see that

$$\omega = \sum_{1\le j\le r} i\,\gamma_j \wedge \overline{\gamma_j},$$

where $\gamma_j \in V^*$ for $1 \le j \le r$. Thus ω is strongly positive. □

We finally define the positivity of differential forms as follows:

Definition 2.16. *A smooth differential* (q,q)*-form* $\phi \in \mathcal{D}_{q,q}(\Omega)$ *in an open set* $\Omega \subset \mathbb{C}^n$ *is positive* (*resp. strongly positive*) *if for all* $x \in \Omega$*, the* (q,q)*-form* $\phi(x) \in \Lambda^{q,q}\mathbb{C}^n$ *is positive* (*resp. strongly positive*)*.*

2.2.2 Positive currents The duality between positive and strongly positive forms enables us to define the corresponding positivity notions for currents:

Definition 2.17. *A current* T *of bidimension* (q,q) *is* (*weakly*) *positive if* $\langle T, \phi\rangle \ge 0$ *for all strongly positive differential test forms* ϕ *of bidegree* (q,q)*.*

It follows from the definitions that T is positive iff for all $\alpha_1, \dots, \alpha_q \in \mathcal{D}_{1,0}(\Omega)$, $T \wedge i\alpha_1 \wedge \bar{\alpha_1} \wedge \cdots \wedge i\alpha_q \wedge \bar{\alpha_q} \ge 0$ as a distribution on Ω.

Here is an important consequence of this definition.

Proposition 2.18. *Let $T \in \mathcal{D}'_{p,p}(X)$ be a positive current and set $q := n - p$. Then T can be extended as a real current of order 0, i.e.,*

$$T = i^{p^2} \sum_{|I|=|J|=q} T_{I,J}\, dz_I \wedge d\overline{z}_J,$$

where the coefficients $T_{I,J}$ are complex measures in Ω satisfying the Hermitian symmetry $\overline{T_{I,J}} = T_{J,I}$ for any multi-indices $|I| = |J| = q$.

Moreover for any I, $T_{I,I} \geq 0$ is a positive Borel measure in Ω and the (local) total variation measure

$$\|T\| := \sum_{|I|=|J|=q} |T_{I,J}|$$

of the current T is bounded from above by the trace measure:

$$\|T\| \leq c_{p,n} \sum_{|K|=q} T_{K,K}, \tag{2.2.2}$$

where $c_{p,n} > 0$ are universal constants.

Proof. Since positive forms are real, it follows by duality that every positive current is real, as the current $\overline{T}$ is defined by the formula $\overline{T}(\phi) := \overline{T(\overline{\phi})}$ for $\phi \in \mathcal{D}_{n-p,n-p}(X)$.

It follows from Lemma 2.11 that any form $\varphi \in \mathcal{D}_{n-p,n-p}(U)$ can be written as $\phi = \sum_s c_s \gamma_s$, where (γ_s) is a basis of strongly positive $(n-p, n-p)$-forms

If ϕ is real, the functions c_s are real C^∞-smooth with compact support in Ω. Writing c_s as a difference of non-negative C^∞-smooth functions with compact support, the real form ϕ can be written as a difference of strongly positive forms, hence $T(\phi)$ is a difference of two positive reals. We infer $\overline{T_{I,J}} = T_{J,I}$ for all multi- indices $|I| = |J| = q$.

Observe now that

$$T_{I,I}\beta_n = T \wedge \frac{i^{q^2}}{2^q} dz_{I'} \wedge d\overline{z}_{I'} \geq 0,$$

while the proof of Lemma 2.11 yields

$$T_{I,J} 2^q \beta_n = T \wedge i^{q^2} dz_{I'} \wedge d\overline{z}_{J'} = \sum_{\nu \in \{0,1,2,3\}} \varepsilon_\nu T \wedge \gamma_\nu,$$

where $\varepsilon_\nu = \pm 1, \pm i$,

$$\gamma_\nu = \bigwedge_{1 \leq s \leq q} i\,\ell_{\nu,s} \wedge \overline{\ell_{\nu,s}},$$

and $\ell_{\nu,s}$ are $\mathbb{C}$-linear forms on $\mathbb{C}^n$. Since $T \wedge \gamma_a$ is a positive measure on Ω, the distributions $T_{I,J}$ are complex measures in Ω such that

$$2^q |T_{I,J}| \beta_n \leq \sum_\nu T \wedge \gamma_\nu.$$

The only terms that matter here are those for which

$$\gamma_\nu = \bigwedge_{1\le s\le q} i\,\ell_{\nu,s}\wedge\overline{\ell_{\nu,s}} \neq 0.$$

We can thus assume that the $\mathbb{C}$-linear forms $\ell_{\nu,1},\dots,\ell_{\nu,s}$ are linearly independent. Fix such ν and set $\ell_s = \ell_{\nu,s}$. There exists a unitary transformation $A : \mathbb{C}^n_z \to \mathbb{C}^n_w$ such that the direct image $A_\star$ sends the subspace of $(\mathbb{C}^n)^*$ generated by the $\mathbb{C}$-linear 1-forms $(\ell_s)_{1\le s\le q}$ onto the subspace generated by the 1-forms $(dw_s)_{1\le s\le q}$. Therefore,

$$A_\star(\bigwedge_{1\le s\le q}\ell_s\wedge\overline{\ell}_s) = \bigwedge_{1\le s\le q}A_\star(\ell_s) = |\det(\partial\tilde{\ell}_s/\partial w_k)|^2\bigwedge_{1\le k\le q}dw_k\wedge d\bar{w}_k.$$

Hence

$$A_\star(T\wedge\gamma_\nu) = a_\nu A_\star(T)\wedge\bigwedge_{1\le k\le q}\frac{i}{2}dw_k\wedge d\bar{w}_k = a_\nu A_\star(T)\frac{i^q}{2^q}dw_K\wedge d\overline{w}_K,$$

where $K := (1,2,\dots,q)$ and for all $\nu, a_\nu \ge 0$ is a constant which does not depend on T. Thus

$$A_\star(T\wedge\gamma_\nu)\le a_\nu A_\star(T)\wedge\beta_n(w)$$

and

$$T\wedge\gamma_\nu \le a_\nu T\wedge(A^{-1})_\star(\beta_n) = a_\nu T\wedge\beta_q,$$

since β is invariant under unitary transformations on $\mathbb{C}^n$. Observe that

$$\beta_q = \sum_{|K|=q}\left(\frac{i}{2}\right)^q\bigwedge_{1\le s\le q}dz_{k_s}\wedge d\bar{z}_{k_s} = \sum_{|K|=q}\frac{i^{q^2}}{2^q}dz_K\wedge d\bar{z}_K,$$

hence $T\wedge\beta_q = \sum_{|L|}T_{L,L}$ and $T\wedge\gamma_\nu \le a_\nu\sum_{|L|=q}T_{L,L}$ for all ν. It follows that there exists a uniform constant $c_{p,q} > 0$ such that

$$|T_{I,J}| \le c_{p,q}\sum_{|L|=q}T_{L,L}. \qquad \square$$

Corollary 2.19. *Let T be a positive current of bidegree (p,p) and v a continuous strongly positive (m,m)-form, $p+m\le n$. Then the current $T\wedge v$ is positive.*

In particular for all continuous positive $(1,1)$-forms $\alpha_1,\dots,\alpha_m$,

$$T\wedge\alpha_1\wedge\cdots\wedge\overline{\alpha_m}\ge 0$$

is a positive current.

2.2.3 Examples

2.2.3.1 Currents of bidegree (1, 1) Let X be a (connected) complex manifold of dimension n. We let $PSH(X)$ denote the convex cone of plurisubharmonic functions in X which are not identically $-\infty$.

Proposition 2.20. *If $u \in PSH(X)$, then the current $T_u := dd^c u$ is a closed positive current of bidegree $(1,1)$ on X.*

Proof. The property is local, so we can assume $X = \Omega$ is an open subset of $\mathbb{C}^n$. Let first $u \in PSH(\Omega) \cap C^2(\Omega)$. Then

$$dd^c u = \frac{i}{\pi} \sum_{1 \le j,k \le n} \frac{\partial^2 u}{\partial z_j \partial \overline{z_k}} dz_j \wedge \partial \overline{z_k}$$

is a strongly positive $(1,1)$-form, since for all $z \in \Omega$ and all $\xi \in \mathbb{C}^n$,

$$\sum_{1 \le j,k \le n} \frac{\partial^2 u(z)}{\partial z_j \partial \overline{z_k}} \xi_j \overline{\xi_k} \ge 0.$$

To treat the general case we regularize u by using radial mollifiers and set $u_\varepsilon := u \star \rho_\varepsilon$. This is a smooth plurisubharmonic function in the open set $\Omega_\varepsilon := \{z \in \Omega | \operatorname{dist}(z; \partial\Omega) > \varepsilon\}$. Since $u_\varepsilon \to u$ in $L^1_{\mathrm{loc}}(\Omega)$, it follows that $dd^c u_\varepsilon$ converges to $dd^c u$ in the weak sense of currents, hence $dd^c u$ is a positive current in Ω. □

Conversely, one can show that a closed positive current T of bidegree $(1,1)$ can be locally written $T = dd^c u$, where u is a (local) plurisubharmonic function (see Exercise 2.4). Such a function is called a local potential of T. Observe that two local potentials differ by a pluriharmonic function h, i.e., a smooth function such that $dd^c h = 0$ in Ω.

2.2.3.2 Current of integration over a complex analytic set

Complex submanifolds Let $Z \subset X$ be a complex submanifold of X of dimension $m \ge 1$. Its complex structure induces a natural orientation on Z, we can integrate a smooth test form ψ of top degree $2m$ on Z. Thus using a partition of unity we may assume that the support of Ψ lies in a coordinate chart (D, z). Thus $\psi(z) = f(z)\beta_m(z)$ in D, where f is a test function in D and by definition

$$\int_D \psi = \int_D f(z_1, \ldots, z_m) \frac{i}{2} dz_1 \wedge d\overline{z}_1 \wedge \ldots \wedge \frac{i}{2} dz_m \wedge d\overline{z}_m.$$

This formula does not depend on the local coordinates, as follows from the change of variables formula.

Definition 2.21. *The current of integration over Z, denoted by $[Z]$, is the positive current of bidimension (m,m) defined by*

$$\langle [Z], \varphi \rangle := \int_Z j^*(\varphi)$$

for $\varphi \in \mathcal{D}_{m,m}(X)$, where $j : Z \to X$ denotes the embedding of Z in X.

If Φ is a strongly positive test form of bidegree (m,m), then $j^*(\Phi)$ is a positive volume form on Z, hence $\int_Z j^*(\Phi) \geq 0$, which shows that $[Z]$ is a positive current of bidimension (m,m) on X.

If Z if a *closed* complex submanifold of X (no boundary), Stokes' formula shows that for all $\Psi \in \mathcal{D}_{2m-1}(X)$,

$$\langle d[Z], \Psi \rangle = -\langle [Z], d\Psi \rangle = -\int_Z j^*(d\Psi) = -\int_Z dj^*(\Psi) = 0,$$

thus $[Z]$ is a closed positive current on X.

Analytic subsets Let Z be a (closed) complex analytic subset of X of pure dimension m, $1 \leq m \leq n$. We refer the reader to [Cirka] for basics on analytic sets. One can consider the positive current $[Z]$ of integration over the complex manifold Z_{reg} of regular points of Z, i.e., for any test form Φ of bidegree $(n-m, n-m)$ we set

$$\langle [Z], \Phi \rangle := \int_{Z_{\mathrm{reg}}} j^*\Phi,$$

where $j : Z \to X$ is the canonical embedding.

It is not obvious that this integral converges, since $j^*\Phi$ is not compactly supported in Z_{reg}. The result was proved by Lelong, who showed the following:

Theorem 2.22. *The current $[Z]$ is a closed positive current of bidegree (m,m) on X.*

We refer the reader to [Cirka, Theorem 14.1] for a proof. In particular, if f is a holomorphic function in X which is not identically 0, then its zero locus $Z(f)$ defines a positive closed current on X which satifies the Poincaré–Lelong equation

$$dd^{\mathrm{c}} \log |f| = [Z(f)]$$

in the sense of currents.

2.2.4 The trace measure of a positive current

Let X be a Hermitian manifold; for each $x \in X$ the complex tangent space $T_x X$ is endowed with a positive definite Hermitian scalar product $h(x)$ which depends smoothly on x. We let ω denote its fundamental $(1,1)$-form.

Definition 2.23. *Let $T \in \mathcal{D}'_{p,p}$ be a positive current of bidegree (p, p), $1 \leq p \leq n$. The trace measure of T is*

$$\sigma_T := \frac{1}{(n-p)!} T \wedge \omega^{n-p}.$$

In a local coordinates $(U, z_1, \ldots, z_n)$ we can write

$$h := \sum_{j,k} h_{j,k} dz_j \otimes dz_k,$$

where $(h_{j,k})$ is a positive definite Hermitian matrix with smooth entries in U. For $x \in U$, the complex cotangent space $T^*_x X$ can also be endowed with a natural Hermitian scalar product: applying the Hilbert–Schmidt orthonormalization process to the the basis $(dz_1(x), \ldots, dz_n(x))$, we construct an orthonormal basis $(\zeta_1(x), \ldots, \zeta_n(x))$ of $T^*_x X$.

Thus $(\zeta_1, \ldots, \zeta_n)$ is a system of smooth differential $(1,0)$-forms in U such that $(\zeta_1(x), \ldots, \zeta_n(x))$ is an orthonormal basis of $T^*_x X$. We say that $(\zeta_1, \ldots, \zeta_n)$ is a local orthonormal frame (with respect to the Hermitian product h) of the cotangent bundle $T^*(X)$ over U. Writing

$$h = \sum_{1 \leq j \leq n} \zeta_j \otimes \overline{\zeta}_j,$$

we get

$$\omega = \frac{i}{2} \sum_{1 \leq j \leq n} \zeta_j \wedge \overline{\zeta_j}$$

and

$$\frac{\omega^q}{q!} = \frac{i^{q^2}}{2^q} \sum_{|K|=q} \zeta_K \wedge \overline{\zeta}_K.$$

Fix $T \in \mathcal{D}'^{+}_{p,p}$ and set $q := n - p$. We can decompose T as

$$T = \sum_{|I|=|J|=q} i^{q^2} T_{I,J} \zeta_I \wedge \overline{\zeta}_J,$$

where $T_{I,J} \in \mathcal{D}(U)$ and $\zeta_I := \zeta_{i_1} \wedge \ldots \wedge \zeta_{i_q}$. A simple computation yields

$$\sigma_T = \left(\sum_{|I|=q} T_{I,I} \right) \bigwedge_{1 \leq j \leq n} \frac{i}{2} \zeta_j \wedge \overline{\zeta_j},$$

i.e., $\sigma_T = \sum_{|I|=q} T_{I,I}$, identifying currents of top degree and distributions. We let the reader check that if X is a domain in $\mathbb{C}^n$ equipped with the standard Euclidean

metric $\sum_{1\le j\le n} dz_j \otimes \overline{dz_j}$ and if $T \in \mathcal{D}'_{p,p}(X)$, then

$$\sigma_T = \sum_{|I|=q} T_{I,I}.$$

Proposition 2.18 can now be reformulated as follows:

Corollary 2.24. *Let $T \in \mathcal{D}'_{p,p}(X)$ be a positive current. If we decompose T locally, $T = \sum_{|I|=|J|=p} i^{p^2} T_{I,J} dz_I \wedge d\overline{z_J}$, the total variation $\|T\| = \sum_{|I|=|J|=p} |T_{I,J}|$ of T is dominated by the trace measure σ_T:*

$$\sigma_T \le \|T\| \le c_{n,p}\sigma_T,$$

where $c_{n,p}$ is an absolute constant independent of T.

In particular, the topology of weak convergence in the sense of distributions coincides, for positive currents, with the weak convergence in the sense of Radon measures.

Example 2.25. *Let $u \in PSH(\Omega)$, where $\Omega \subset \mathbb{C}^n$ is a domain. The trace measure of the closed positive current $T := dd^c u$ coincides with the Riesz measure of u, i.e.,*

$$\sigma_T := dd^c u \wedge \beta_{n-1} = \frac{1}{2\pi}\Delta u.$$

This formula can be generalized to any plurisubharmonic function on a complex Hermitian manifold (X, ω), replacing β by ω and Δ by Δ_ω, the Laplace operator associated to the Hermitian metric ω.

2.3 Lelong numbers

2.3.1 Lelong numbers of a plurisubharmonic function We consider several natural quantities measuring the local size of a plurisubharmonic function u.

The *spherical mean values* of u are defined by

$$S_u(z,r) := \frac{1}{\sigma_{2n-1}} \int_{|\xi|=1} u(z + r\xi) d\sigma(\xi),$$

where $d\sigma$ is the area measure on the unit sphere in $\mathbb{C}^n$, σ_{2n-1} is its area.

The *mean value* of u on a ball $\bar{B}(z,r) \subset \Omega$ is defined by

$$V_u(z,r) := \frac{1}{\kappa_{2n} r^{2n}} \int_{\bar{B}(z,r)} u d\lambda = \frac{1}{\kappa_{2n}} \int_{|w|\le 1} u(z + rw) d\lambda(w),$$

where $d\lambda$ is the lebesgue measure on $\mathbb{C}^n$ and $\kappa_{2n} = 2n\sigma_{2n-1}$ is the volume of the unit ball in $\mathbb{C}^n$.

The *maximum value* of u on a ball $\bar{B}(z,r) \subset \Omega$ is

$$M_u(z,r) := \max_{|w|=1} u(z+rw).$$

Observe that

$$V_u(z,r) = \frac{1}{\kappa_{2n} r^{2n}} \int_0^r t^{2n-1} S_u(z,t)dt. \tag{2.3.1}$$

Lemma 2.26. *Consider* $\tilde{\Omega} := \{(z,\tau) \in \Omega \times \mathbb{C} |\ |\tau| < \delta_\Omega(z)\}$. *The function*

$$(z,\tau) \longmapsto \tilde{S}_u(z,\tau) = \frac{1}{\sigma_{2n-1}} \int_{|\xi|=1} u(z+\tau\xi)d\sigma(\xi)$$

is plurisubharmonic in $\tilde{\Omega}$. *It depends only on* $|\tau|$ *and satisfies*

$$\tilde{S}_u(z,\tau) = S_u(z,|\tau|) \quad \textit{for all} \quad (z,\tau) \in \tilde{\Omega}.$$

In particular, for all $z \in \Omega$, $r \mapsto S_u(z,r)$ *is a convex non-decreasing function of* $t = \log r$ *in the interval* $]-\infty, \log \delta_\Omega(z)[$, *hence the following limit exists in* $[0,+\infty[$:

$$\nu(u,z) := \lim_{r\to 0} \frac{S_u(z,r)}{\log r} = \lim_{r\to 0^+} r\partial_r^- S_u(z,r).$$

Proof. For all $\xi \in \mathbb{C}^n$ with $|\xi| = 1$ the function

$$(z,\tau) \longmapsto u(z+\tau\xi)$$

is plurisubharmonic in $\tilde{\Omega}$, hence so is the function $\tilde{S}_u$ as an average of plurisubharmonic functions.

Since the area measure on the sphere is invariant under the action of the circle $\mathbb{T}$, it follows that for $(z,\tau) \in \tilde{\Omega}$,

$$\tilde{S}_u(z,\tau) = S_u(z,|\tau|).$$

Therefore $\tau \mapsto \tilde{S}_u(z,\tau)$ is subharmonic in the disc $\{|\tau| < \delta_\Omega(z)\}$ and depends only on $|\tau|$. We infer that for a fixed $z \in \Omega$, the function $r \mapsto S_u(z,r)$ is convex and non-decreasing in $\log r$ by Proposition 1.13. Moreover, by invariance we also have

$$S_u(z,r) = \frac{1}{\sigma_{2n-1}} \int_{|\xi|=1} \left(\int_0^{2\pi} u(z+re^{i\theta}\xi)d\theta/2\pi \right) d\sigma(\xi).$$

The slopes of the increasing convex function $\log r \mapsto S_u(z,r)$ are therefore increasing and positive. It follows that for any fixed $0 < r_0 < \delta_\Omega(z)$, the following limit exists in $[0,+\infty[$:

$$\lim_{r\to 0} \frac{S_u(z,r) - S_u(z,r_0)}{\log r - \log r_0} = \lim_{r\to 0} \frac{S_u(z,r)}{\log r} = \lim_{r\to 0^+} r\partial_r^- S_u(z,r). \qquad \square$$

Definition 2.27. *The number $\nu(u,z)$ is called the Lelong number of the plurisubharmonic function u at the point z.*

We will also use the notation $\nu_u(z)$.

Lemma 2.28. *Fix $z \in \Omega$. The functions $r \mapsto S_u(z,r)$, $r \mapsto V_u(z,r)$, and $r \mapsto M_u(z,r)$ are convex increasing in the variable $t = \log r$.*

Moreover, if $0 < r < R < \delta_\Omega(z)$ and $u \le 0$ in $B(z,R)$, then

$$u(z) \le V_u(z,r) \le S_u(z,r) \le M_u(z,r) \le (1 - r/R)^{2n} V_u(z, R-r). \qquad (2.3.2)$$

In particular, if a plurisubharmonic function is bounded from above in $\mathbb{C}^n$, then it is constant.

The last property is known as Liouville's Theorem for plurisubharmonic functions.

Proof. Let $f(z,r)$ denote any of the above functions and observe that $(z,\tau) \mapsto f(z,|\tau|)$ is plurisubharmonic in the variables (z,τ) in $\tilde{\Omega}$ and only depends on $|\tau|$. Thus $r \mapsto f(z,r)$ is a convex increasing function in the variable $t = \log r$ for any fixed $z \in \Omega$.

The first inequality in (2.3.2) follows from submean value property, the second one is a consequence of (2.3.1), and the third one is obvious.

It thus remains to prove the last inequality in (2.3.2). We can assume that $z = 0$ and $R > 0$ is such that $\bar{B}(0,R) \subset \Omega$ and $u \le 0$ in $\bar{B}(0,R)$. Fix $z \in \mathbb{C}^n$ with $|z| \le r$ and observe that $\mathbb{B}(0, R-r) \subset \mathbb{B}(z,R)$. The submean value inequality yields

$$u(z) \le \frac{1}{\kappa_{2n} R^{2n}} \int_{\mathbb{B}(z,R)} u(\zeta) d\lambda(\zeta) \le \frac{1}{\kappa_{2n} R^{2n}} \int_{\mathbb{B}(0,R-r)} u(\zeta) d\lambda(\zeta),$$

since $u \le 0$ in $B(z,R)$. Consequently,

$$M_u(0,r) \le \frac{(R-r)^{2n}}{R^{2n}} V_u(0, R-r).$$

Assume that u is plurisubharmonic and bounded from above in $\mathbb{C}^n$. We can assume without loss of generality that $u(0) > -\infty$. Then the function $\log r \mapsto M_u(0,r)$ is convex, non-decreasing, and bounded from above in $\mathbb{R}$, hence it is constant and equal to its limit when $r \to 0^+$, which is equal to $u(0)$. This implies that u attains its local maximum at the origin on any ball $B(0,r)$. The maximum principle now shows that u is constant. □

Corollary 2.29. *For all $z \in \Omega$,*

$$\nu(u,z) = \lim_{r \to 0} \frac{V_u(z,r)}{\log r} = \lim_{r \to 0} \frac{M_u(z,r)}{\log r}. \qquad (2.3.3)$$

Proof. By (2.3.2), we have

$$\nu_u(z) = \lim_{r\to 0} \frac{S_u(z,r)}{\log r} \le \lim_{r\to 0} \frac{V_u(z,r)}{\log r}.$$

Fix $0 < s < 1$ and observe that if $r = sR$, then (2.3.2) yields

$$M_u(z,r) \le (1-s)^{2n} V_u((1/s-1)r).$$

Since $\log((1/s-1)r)/\log r \to 1$ as $r \to 0^+$ and $S_u \le M_u$, it follows that

$$(1-s)^{2n} \lim_{r\to 0} \frac{V_u(z,r)}{\log r} \le \lim_{r\to 0} \frac{M_u(z,r)}{\log r} \le \nu_u(z),$$

which proves our statement by letting $s \to 0^+$. □

The previous result shows that if $u \in PSH(\Omega)$ and $a \in \Omega$, then

$$u(z) \le \max_{|z-a|=r_0} u(z) + \nu(u,a)(\log|z-a| - \log r_0)$$

for all $r < r_0 < \delta_\Omega(a)$ with $|z-a| = r$ This shows that u has at worst logarithmic singularities.

Examples 2.30.

1. *Let $u \in PSH(\Omega)$ and $z \in \Omega$. If $u(z) > -\infty$, then*

$$-\infty < u(z) \le S_u(z,r)$$

for all $r < \delta_\Omega(z)$, hence $\nu(u,z) = 0$.

2. *Assume $u(z) = \alpha \log|z| + O(1)$ near the origin in $\mathbb{C}^n$ with $\alpha > 0$. Then $\nu(u,0) = \alpha$.*

Remark 2.31. *The reader will check in Exercise 2.7 that the functional $u \in PSH(\Omega) \mapsto \nu(u,z) \in \mathbb{R}^+$ is additive and positively homogeneous.*

2.3.2 Invariance properties

2.3.2.1 Lelong formula We now show that the Lelong number of a plurisubharmonic function u only depends on the current $dd^c u$. Set

$$\beta := \frac{i}{2} \sum_{1\le j\le n} dz_j \wedge d\bar z_j = \frac{i}{2}\partial\bar\partial|z|^2,$$

where $|z|^2 := \sum_{j=1}^n |z_j|^2$. For $1 \le p \le n$ we set

$$\beta_p := \frac{\beta^p}{p!}$$

and for $z \in \Omega$ and $0 < r < \delta_\Omega(z)$,

$$\nu_u(z,r) := \frac{1}{\kappa_{2n-2} r^{2n-2}} \int_{B(z,r)} dd^c u \wedge \beta_{n-1} = \frac{\mu_u(B(z,r))}{\kappa_{2n-2} r^{2n-2}},$$

where $\mu_u = dd^c u \wedge \beta_{n-1} = \frac{1}{2\pi}\Delta u$ is the Riesz measure of u and $B(z,r)$ is the Euclidean ball of center z and radius r.

Theorem 2.32. *For all $z \in \Omega$ and $0 < r < \delta_\Omega(z)$,*

$$\nu_u(z,r) = r\partial_r^- S_u(z,r) = \partial^-_{\log r} S_u(z,r), \tag{2.3.4}$$

where $\partial^-_{\log r} = r\partial_r^-$ is the left derivative operator with respect to $\log r$.

In particular, the function $r \mapsto \nu_u(z,r)$ is non-decreasing and

$$\nu_u(z) = \lim_{r \to 0^+} \nu_u(z,r).$$

This formula shows that the Lelong number depends on the positive current $dd^c u$ rather than on the function u itself.

Proof. We can assume without loss of generality that $z = 0$ and u is plurisubharmonic in a neighborhood of the Euclidean ball $\bar{B}(0,R)$ for some $R > 0$.

Let χ be a smooth radial function with compact support in $B(0,R)$. Integrating by parts and passing to spherical coordinates we get

$$\begin{aligned}
\int_{B(0,R)} \chi(\zeta) d\mu_u(\zeta) &= \frac{1}{2\pi} \int_{B(0,R)} \Delta\chi(\zeta) u(\zeta) d\lambda(\zeta) \\
&= \kappa_{2n-2} \int_0^R S_u(z,r) r^{2n-1} (\chi"(r) + (2n-1)\chi'(r)/r) dr \\
&= \kappa_{2n-2} \int_0^R S_u(z,r) d(r^{2n-1}\chi'(r)) \\
&= -\kappa_{2n-2} \int_0^R \chi'(r) r^{2n-1} \partial_r^- S_u(z,r) dr.
\end{aligned}$$

We let $\chi(r)$ tend to the characteristic function of the interval $[0, R[$. The function $\chi'(r)$ thus converges weakly to the negative of the measure of integration on the positively oriented boundary of $[0, R]$, denoted by $[0] - [R]$, which is the difference of the two Dirac masses at the end points of the interval. Therefore,

$$\mu_u(B(0,R)) = \kappa_{2n-2} R^{2n-1} \partial_r^- S_u(z,R) = \kappa_{2n-2} R^{2n-2} \partial^-_{\log r} S_u(z,R).$$

Therefore $\nu_u(z) = \lim_{r \to 0^+} \nu_u(z,r)$. □

Example 2.33. *In dimension $n = 1$ the Lelong number $\nu_u(z)$ is the point mass of the Riesz measure at z, i.e., $\nu_u(z) = \mu_u(\{z\})$.*

We deduce the so called Poisson–Jensen formula:

Corollary 2.34. *Fix $z \in \Omega$. For all $0 < r_1 \leq r_2 < \delta_\Omega(z)$,*

$$S_u(z, r_2) - S_u(z, r_1) = \int_{r_1}^{r_2} \nu_u(z, r) \frac{dr}{r} = \int_{r_1}^{r_2} \nu_u(z, r) d \log r.$$

In particular, for all $0 < R < \delta_\Omega(z)$,

$$u(z) = \frac{1}{\sigma_{2n-1}} \int_{|\xi|=1} u(z + R\xi) d\sigma(\xi) - \int_0^R \nu_u(z, r) d \log r.$$

Proof. The first formula follows from (2.3.4). To obtain the second formula we can fix $r_2 = R$ and let $r_1 \to 0$ in the first one. □

Example 2.35. *If $u = \log |f|$, f a non-zero holomorphic function, then $dd^c \log |f|$ is the current of integration over the analytic set $Z_f = (f = 0)$ and μ_u is the area measure on the analytic set Z_f.*

In this case the number $\nu_{\log|f|}(z, r)$ is the quotient of the area of the set $B(z, r) \cap Z_f$ in the analytic set Z_f by the volume of the ball of the same radius r and the same dimension.

Thus $\nu(u, z)$ can be thought of as a $(2n-2)$-dimensional density of the measure μ_u at the point z. It is the vanishing order of f at z.

2.3.2.2 Non-integrability and multiplicity

Corollary 2.36. *Fix $u \in PSH(\Omega)$ and $z_0 \in \Omega$. If $\nu_u(z_0) \geq 2n$, then e^{-u} is not locally integrable in a neighborhood of z_0.*

Proof. We may assume that $z_0 = 0$. It follows from (2.3.3) that there exist $r_0 > 0$ small enough and $C > 0$ such that

$$S_u(0, r) \leq \nu_u(0) \log r + C,$$

for $0 < r \leqslant r_0$. Using Jensen's convexity inequality we infer

$$\begin{aligned} \int_{|\xi|=1} e^{-u(r\xi)} d\sigma(\xi)/\sigma_{2n-1} &\geq e^{(-S_u(0,r))} \\ &\geq e^{-C} r^{-\nu_u(0)}. \end{aligned}$$

Therefore,

$$\int_{|\zeta|<\rho} e^{-u(\zeta)} d\lambda(\zeta) \geq \sigma_{2n-1} e^{-C} \int_0^\rho r^{2n-1-\nu_u(0)} dr,$$

hence e^{-u} is not integrable near 0 if $\nu_u(0) \geq 2n$. □

Lelong numbers are invariant under local changes of coordinates:

Theorem 2.37. *Fix $u \in PSH(\Omega)$, $z \in \Omega$, and $\zeta \in \mathbb{C}^n \setminus \{0\}$. Then*

$$\nu_u(z) \leq \lim_{r\to 0^+} \frac{1}{\log r} \int_0^{2\pi} u(z + e^{i\theta}\zeta) d\theta/2\pi, \tag{2.3.5}$$

with equality for almost every $\zeta \in \mathbb{C}^n$.

If $f : \Omega' \to \Omega$ is a holomorphic map, then for $z \in \Omega'$,

$$\nu_u(f(z)) \leq \nu_{u\circ f}(z),$$

with equality if f is a local biholomorphism at z.

Proof. We may assume that $z = 0$. Fix $\delta > 0$ small enough and assume that $u(\zeta) < 0$ when $|\zeta| \leq \delta$. Consider, for $r < \delta$ and $|\zeta| < \delta/r$,

$$\phi(\zeta, r) := \int_0^{2\pi} u(re^{i\theta}\zeta) \frac{d\theta}{2\pi}.$$

For each $r > 0$, $\phi(\cdot, r)$ is a continuous negative plurisubharmonic function in the ball $|\zeta| < \delta/r$. For fixed ζ, $\phi(\zeta, \cdot)$ is a convex non-decreasing function in $\log r$. This shows the existence of

$$\psi(\zeta) := \lim_{r\to 0^+} \frac{\phi(\zeta, r) - \phi(\zeta, \delta)}{\log \delta - \log r} = \lim_{r\to 0^+} \frac{\phi(\zeta, r)}{-\log r},$$

Note that $\psi(\zeta)$ is equal to the mass at the origin of the Laplacian of the one variable function $u_\zeta : \tau \mapsto u(\tau\zeta)$, unless this function is identically $-\infty$ in the intersection of the line $L_\zeta := \{\tau\zeta; \tau \in \mathbb{C}\}$ with the ball $|\zeta| < \delta/r$.

Since $u \not\equiv -\infty$ in a neighborhood of the origin, we infer that ψ^* is negative plurisubharmonic in $\mathbb{C}^n$, hence it is constant by Lemma 2.28. We conclude that $\psi(\zeta) = \psi^*(\zeta) = \psi^*(0)$ for almost all $\zeta \in \mathbb{C}^n$ with $|\zeta| = 1$, since ψ is constant on each complex line through the origin.

Observe that

$$\frac{S_u(0, r)}{\log r} = -\frac{1}{\sigma_{2n-1}} \int_{|\zeta|=1} \frac{\phi(\zeta, r)}{\log r} d\sigma(\zeta)$$

and apply the Lebesgue convergence theorem to deduce (2.3.5):

$$\nu_u(0) = -\psi^*(0)/\sigma_{2n-1} = \int_{|\zeta|=1} -\psi(\zeta) d\sigma(\zeta)/\sigma_{2n-1} \leq -\psi(\zeta).$$

We infer that $\nu_{u\circ f} = \nu_u \circ f$ for any complex affine bijection f. To prove the same result for an arbitrary holomorphic bijection, it is enough to assume that $f(0) = 0$ and $f'(0) = \mathrm{Id}$ is the identity in $\mathbb{C}^n$. It follows from formula (2.3.3) that

$$\nu_u(0) = \lim_{r\to 0} \frac{1}{\kappa_{2n} r^{2n} \log r} \int_{|\zeta|<r} u(\zeta) d\lambda(\zeta).$$

Since $u < 0$, we obtain by the change of variables $\zeta \mapsto f(\zeta)$

$$\begin{aligned} \nu_u(0) &= \lim_{r\to 0^+} \frac{1}{\kappa_{2n} r^{2n} \log r} \int_{|f(\zeta)|<r} u \circ f(\zeta) |J_f(\zeta)| d\lambda(\zeta) \\ &\le \sup_{|f(\zeta)|<r} |J_f(\zeta)| \lim_{r\to 0^+} \frac{1}{\kappa_{2n} r^{2n} \log r} \int_{|f(\zeta)|<r} u \circ f(\zeta) d\lambda(\zeta) \\ &= \nu_{u\circ f}(0). \end{aligned}$$

Thus $\nu_u(0) \le \nu_{u\circ f}(0)$. If f is a biholomorphism in a neighborhood of 0, then $\nu_u(0) = \nu_{u\circ f\circ f^{-1}}(0) \ge \nu_{u\circ f}(0)$, whence the equailty. □

Remark 2.38. *In the proof above the function ψ is constant on any complex line through the origin in $\mathbb{C}^n$. Thus (2.3.5) is an equality for almost every ζ in the unit sphere $|\zeta| = 1$. This shows that the Lelong number of u is invariant under restriction to almost all complex directions in $\mathbb{C}^n$. In particular,*

$$\nu_u(0) = \min\left\{\nu_{u_\zeta}(0) |\ \zeta \in \mathbb{C}^n \setminus \{0\}\right\}, \tag{2.3.6}$$

where $u_\zeta(\tau) := u(\tau\zeta)$ is the restriction of u to the complex line $\mathbb{C}\zeta$ ($\nu_{u_\zeta}(0) = +\infty$ if $u_\zeta \equiv -\infty$).

Example 2.39. *Let $f = (f_1, \dots, f_N) : \Omega \to \mathbb{C}^N$ be a holomorphic mapping such that $f \not\equiv 0$. Set $Z := \{z \in \Omega |\ f(z) = 0\}$. Then*

$$\varphi := \frac{1}{2} \log(|f_1|^2 + \cdots + |f_N|^2)$$

is plurisubharmonic in Ω and $\nu_\varphi(z) = 0$ if $a \notin Z$. If $a \in Z$, then

$$\nu_\varphi(a) = \mathrm{mult}(f, a)$$

is the vanishing multiplicity of f at a, i.e., the largest integer m such that all the f_j's vanish at order m at the point a. We may indeed assume by translation, that $a = 0$. For each j ($1 \le j \le N$), we have $f_j(z) = P_j(z) + O(|z|^{m_j+1})$, where P_j is a non-zero homogeneous polynomial of degree $m_j \ge m$. By definition, $m = \min m_j$, without loss of generality $m = m_1$. Thus for $\zeta \in \mathbb{C}^n$,

$$\begin{aligned} \varphi(\tau\zeta) &= m \log|\tau| + \frac{1}{2} \log\left(|P_1(\zeta)|^2 + \sum_{j=2}^N |\tau^{m_j - m} P_j(z)|^2 \right) \\ &= m \log|\tau| + O(1), \end{aligned}$$

when $P_1(\zeta) \not\equiv 0$. Since $P_1 \not\equiv 0$, if follows that the set $\{P_1 = 0\}$ is of Lebesgue measure 0, hence $\nu_\varphi(0) = m$ by (2.3.6).

2.3.3 Siu's Theorem Let u be a plurisubharmonic function in a domain Ω. We let the reader check in Exercise 2.7 that the function $z \mapsto \nu(u, z)$ is upper semi-continuous in Ω. We prove in this section, following [Siu74, Kis78], that $z \mapsto \nu(u, z)$ is also upper semi-continuous with respect to the (analytic) Zariski topology, i.e., the level sets

$$E_c(\varphi) = \{x \in \Omega |\ \nu_u(x) \geq c\}$$

are analytic subsets whenever $c > 0$.

2.3.3.1 Hörmander's L^2-estimates Recall that a domain $\Omega \subset \mathbb{C}^n$ is *pseudoconvex* if the function $-\log \operatorname{dist}(\cdot, \partial\Omega)$ is plurisubharmonic in Ω [Horm90]. It follows that

$$\rho(z) := |z|^2 - \log \operatorname{dist}(\cdot, \partial\Omega)$$

is a continuous plurisubharmonic function which is an exhaustion for the domain Ω, i.e., for all $c \in \mathbb{R}$,

$$\{z \in \Omega |\ \rho(z) < c\} \Subset \Omega.$$

The converse also holds: if a domain admits a plurisubharmonic exhaustion function, then it is pseudoconvex. One can moreover show that the exhaustion function can be chosen smooth and strictly plurisubharmonic in Ω.

Theorem 2.40. *Let $\Omega \subset \mathbb{C}^n$ be a pseudoconvex domain and $\varphi \in PSH(\Omega)$. Let $\eta = \sum_{j=1}^n \eta_j\, d\bar{z}_j$ be a $(0,1)$-form on Ω with coefficients in $L^2_{\mathrm{loc}}(\Omega)$ such that $\bar{\partial}\eta = 0$ in the sense of currents on Ω. Then for any $\varepsilon > 0$ the equation $\bar{\partial} h = \eta$ has a solution h in $L^2_{\mathrm{loc}}(\Omega)$ such that*

$$\varepsilon \int_\Omega |h|^2 e^{-\varphi(z)} (1 + |z|^2)^{-\varepsilon} d\lambda(z) \leq \int_\Omega |\eta(z)|^2 e^{-\varphi(z)} (1 + |z|^2)^{2-\varepsilon} d\lambda(z).$$

We refer the reader to [Dem, Horm90] for a proof of this fundamental result. We need the following consequence of Theorem 2.40, known as the Bombieri–Skoda–Hörmander theorem.

Theorem 2.41. *Let $\Omega \subset \mathbb{C}^n$ be a pseudoconvex domain, $z_0 \in \Omega$, and $u \in PSH(\Omega)$ such that e^{-u} is locally integrable in a neighborhood of z_0. For each $\varepsilon > 0$, there exists a holomorphic function f in Ω such that $f(z_0) = 1$ and*

$$\int_\Omega |f(z)|^2 (1 + |z|^2)^{-n-\varepsilon} e^{-u(z)} d\lambda(z) < +\infty. \tag{2.3.7}$$

Proof. We may assume that $z_0 = 0$ and choose a ball $B(0, r) \Subset \Omega$ such that $e^{-u} \in L^1(B(0, r))$. Let χ be smooth function with compact support in $B(0, r)$ and such that $\chi \equiv 1$ in a small neighborhood of 0 holds. We want to correct χ by adding a smooth function h in Ω such that $f := \chi + h$ is holomorphic in Ω, $f(z_0) = 1$, and the estimate (2.3.7).

We solve the Cauchy–Riemann equation $\bar\partial h = -\bar\partial\chi$ in Ω with the plurisubharmonic weight function $\varphi(z) := u(z) + 2n\log|z|$. By Theorem 2.40, for any $\varepsilon > 0$ there exists a solution to the equation $\bar\partial h = -\bar\partial\chi$ in Ω and

$$\varepsilon\int_\Omega \frac{|h|^2 e^{-u(z)}}{|z|^{2n}(1+|z|^2)^{-\varepsilon}}d\lambda(z) \le \int_\Omega \frac{|\bar\partial\chi(z)|^2 e^{-u(z)}}{|z|^{2n}(1+|z|^2)^{\varepsilon-2}}d\lambda(z). \tag{2.3.8}$$

The right-hand side of (2.3.8) is finite, since $\bar\partial\chi$ vanishes in a neighborhood of 0 and has compact support in $B(0,r)$, where the function e^{-u} is integrable.

By construction, $\bar\partial f = \bar\partial(h+\chi) = 0$ in Ω, hence f is holomorphic and h is smooth in Ω. The fact that the left-hand side of (2.3.8) is finite and u is bounded from above in $B(0,r)$ implies that $h(0) = 0$, since h is continuous. Therefore $f(0) = 1$ and the proof is complete. □

The following consequence will be useful in the sequel.

Corollary 2.42. *The set*

$$NI(u) := \{z \in \Omega |\ e^{-u} \notin L^1_{\mathrm{loc}}(z)\},$$

is an analytic subset of Ω.

Here $L^1_{\mathrm{loc}}(z)$ means locally integrable in a neighborhood of z.

Proof. Fix $\varepsilon > 0$ and let Z be the zero locus of the family of all holomorphic function f in Ω for which the estimate (2.3.7) holds. By Theorem 2.41, $Z \subset NI(u)$. The converse is clear. □

Remark 2.43. *A closed analytic subset in a pseudoconvex domain can be globally defined as the zero locus of a finite number of holomorphic functions. This is again a consequence of the L^2-estimates of Hörmander (see* [Horm90]*).*

2.3.3.2 Kiselman's attenuation principle We now explain a useful method for attenuating the singularities of plurisubharmonic functions, known as *Kiselman's minimum principle*.

Let Ω be a pseudoconvex domain in $\mathbb{C}^n$. We set

$$\widetilde\Omega := \{(z,w) \in \Omega\times\mathbb{C} |\ |w| < \delta_\Omega(z)\}.$$

The reader can check that $\widetilde\Omega$ is a pseudoconvex domain in $\mathbb{C}^{n+1}$. Since $\widetilde\Omega$ is S^1-invariant (rotations in the w-variable), we can choose an S^1-invariant exhaustion plurisubharmonic function $\rho(z,w)$.

Theorem 2.44. *Fix $u \in PSH(\Omega)$ and $\alpha > 0$. The function*

$$z \longmapsto u_\alpha(z) := \inf_{0<r<\delta_\Omega(z)} \{S_u(z,r) + \rho(z,r) - \alpha\log r\}$$

is plurisubharmonic in Ω. *It satisfies* $u_\alpha(z) > -\infty$ *if* $\nu_u(z) < \alpha$ *and*

$$\nu(u_\alpha, z) = \max\{\nu(u, z) - \alpha, 0\} \quad \textit{for all} \quad z \in \Omega.$$

Note that an infimum of plurisubharmonic functions is usually not plurisubharmonic: the S^1-invariance is here crucial.

Proof. The function

$$(z, \tau) \longmapsto U(z, \tau) := S_u(z, e^\tau) + \rho(z, e^\tau) - \alpha \Re\tau.$$

is continuous and plurisubharmonic in the domain $\widetilde{\Omega}$.

For $z \in \Omega$ fixed, it only depends on $t := \Re\tau$, hence it is a convex function in t on $]-\infty, \log\delta_\Omega(z)[$. It follows that u_α is upper semi-continuous in Ω. Fix $z \in \Omega$ such that $u(z) > -\infty$ and observe that

$$U(z, t) \geq u(z) + \rho(z, e^t) - \alpha t$$

tends to $+\infty$ when t tends to either $-\infty$ or $\log\delta_\Omega(z)$. Thus, the infimum is attained on a compact interval $[t_0(z), t_1(z)] \subset]-\infty, \log\delta_\Omega(z)[$, hence $u_\alpha(z) > -\infty$.

Recall that $\partial_t^- S_u(z, r) = r\partial_r^- S_u(z, r)$ converges to $\nu_u(z)$ as $t = \log r \to -\infty$. Hence if $\nu_u(z) < \alpha$, the function $t \mapsto S_u(z, e^t) - \alpha t$ is decreasing near $-\infty$. Thus there is $t_0 < \log\delta_\Omega(z)$ such that $S_u(z, e^t) - \alpha t \geq S_u(z, e^{t_0}) - \alpha t_0$ for $t \leq t_0$. It follows that $U_\alpha(z, \tau)$ is bounded from below in $t = \Re\tau$ on the interval $]-\infty, \log\delta_\Omega(z)[$, hence $u_\alpha(z) > -\infty$. Therefore, $\nu_{u_\alpha}(z) = 0$ if $\nu_u(z) < \alpha$.

To prove that u_α is plurisubharmonic, we consider, for $\varepsilon > 0$,

$$U^\varepsilon(z, \tau) := U(z, \tau) + \varepsilon e^{\Re\tau}.$$

Observe that U^ε is continuous and plurisubharmonic in the domain $\widetilde{\Omega}$ and only depends on $(z, \Re\tau)$, hence it is a convex function in $t = \Re\tau$ on the interval $]-\infty, \log\delta_\Omega(z)[$. It is even strictly convex in t thanks to the extra term εe^t. We set

$$u^\varepsilon(z) := \inf\{U^\varepsilon(z, t) |\, t < \log\delta_\Omega(z)\}$$

and observe that u^ε decreases to u_α in Ω as ε decreases to 0.

It is thus enough to show that u^ε is plurisubharmonic in Ω. We first assume that u is smooth. In this case the function U^ε is smooth in $\widetilde{\Omega}$. Fix $z \in \Omega$ and $\xi \in \mathbb{C}^n \setminus \{0\}$ and consider the one-variable function

$$\phi(\sigma) := u^\varepsilon(z + \sigma\xi) = \inf\{\Phi(\sigma, t) |\, t < \log\delta(z + \sigma\xi)\},$$

defined in a disc $\mathbb{D}(0, \delta)$ for $\delta > 0$ small enough, where

$$\Phi(\sigma, \tau) := U^\varepsilon(z + \sigma\xi, \tau),$$

is a smooth plurisubharmonic function in the domain

$$\{(\sigma, \tau) \in \mathbb{D}(0, \delta) \times \mathbb{C} |\, \Re\tau < \log\delta(z + \sigma\xi).$$

For $\sigma \in \mathbb{D}(0,\delta)$ fixed, it is strictly convex in $t = \Re\tau$ and tends to $+\infty$ when $t \to -\infty$ or $\log \delta(z+\sigma\xi)$. Therefore the infimum is achieved at a unique point

$$t(\sigma) \in]-\infty, \log \delta(z+\sigma\xi)[,$$

characterized by the condition

$$v = \frac{\partial \Phi}{\partial t}(\sigma, t(\sigma)) = 0.$$

Since $\partial_t^2 \Phi(\sigma, t(\sigma)) > 0$, we can apply the implicit function theorem to conclude that the function $\sigma \mapsto t(\sigma)$ is smooth in $\mathbb{D}(0,\delta)$, hence $\phi(\sigma) = \Phi(\sigma, t(\sigma))$ is a smooth function in $\mathbb{D}(0,\delta)$. Therefore we can compute its Laplacian with respect to the variable σ as follows

$$\begin{aligned}
\frac{\partial^2 \phi}{\partial\sigma\partial\bar{\sigma}}(\sigma) &= \frac{\partial}{\partial\sigma}\left(\frac{\partial\Phi}{\partial\bar{\sigma}}(\sigma,t(\sigma)) + \partial_t\Phi(\sigma,t(\sigma))\frac{\partial t}{\partial\bar{\sigma}}(\sigma)\right) \\
&= \frac{\partial^2\Phi}{\partial\sigma\partial\bar{\sigma}}(\sigma,t(\sigma)) + \frac{\partial^2\Phi}{\partial t\partial\sigma}(\sigma,t(\sigma))\frac{\partial t}{\partial\bar{\sigma}}(\sigma) \\
&= \frac{\partial^2\Phi}{\partial\sigma\partial\bar{\sigma}}(\sigma,t(\sigma)) \geq 0.
\end{aligned}$$

We use here twice the fact that $\sigma \mapsto \Phi(\sigma,\tau)$ is subharmonic for τ fixed and $\partial_t\Phi(\sigma, t(\sigma)) = 0$. The conclusion follows when u is smooth.

In the general case we choose smooth plurisubharmonic functions u^ε decreasing to u, where u^ε is smooth in $\Omega_{a(\varepsilon)}$ with $a(\varepsilon)$ increasing to $+\infty$ and $\Omega_{a(\varepsilon)}$ increases to Ω as ε decreases to 0.

Fix a domain $\Omega' \Subset \Omega$. When $z \in \Omega'$, the infimum in the definition of $(u^\varepsilon)_\alpha(z)$ is attained in $\Omega_{a(\varepsilon)}$ when ε is small enough, so $(u^\varepsilon)_\alpha(z)$ is plurisubharmonic in Ω'. Since $(u^\varepsilon)_\alpha$ decreases to u_α, it follows that u_α is plurisubharmonic in Ω', hence in Ω. □

2.3.3.3 Analyticity of superlevel sets of Lelong numbers Let u be a plurisubharmonic function in a pseudoconvex domain $\Omega \subset \mathbb{C}^n$ and consider, for $c > 0$, the superlevel set of Lelong numbers

$$\Lambda(u,c) := \{z \in \Omega |\ \nu_u(u,z) \geq c\}.$$

We also consider the non-integrability set of u

$$NI(u) := \{z \in \Omega |\ e^{-u} \notin L^1_{\mathrm{loc}}(\{z\})\},$$

and the polar set of u

$$P(u) := \{z \in \Omega |\ u(z) = -\infty\}.$$

It follows from Theorem 2.41 and Corollary 2.36 that

$$\Lambda(u,2n) \subset NI(u) \subset P(u).$$

The following result has been established by Siu in [Siu74]:

Theorem 2.45. *For all $c > 0$ the sets $\Lambda(u, c)$ are closed analytic subsets of Ω.*

Proof. Using the functions u_α constructed in Theorem 2.44 we obtain

$$\Lambda(u, c + 2n) = \Lambda(u_c, 2n) \subset NI(u_c) \subset P(u_c) \subset \Lambda(u, c).$$

If $\alpha > 2n/c$, we infer

$$\Lambda(u, c) = \Lambda(\alpha u, \alpha c - 2n + 2n) \subset NI((\alpha u)_{\alpha c - 2n}) \subset \Lambda(u, c - 2n/\alpha).$$

Therefore

$$\Lambda(u, c) = \bigcap_{\alpha > 2n/c} NI((\alpha u)_{\alpha c - 2n}),$$

which is an analytic set by Corollary 2.36. □

2.4 Skoda's integrability theorem

2.4.1 Projective mass We establish here a formula due to Lelong which expresses the mass of the Riesz measure associated to a plurisubharmonic function u in terms of the projective mass of the current $dd^c u$. Recall that

$$\nu_u(z, r) := \frac{\mu_u(B(z, r))}{\kappa_{2p} r^{2p}},$$

where $\mu_u = dd^c u \wedge \beta_{n-1}$ is the trace measure of the current $dd^c u$ and $B(z, r)$ is the Euclidean open ball of center z and radius r.

Lemma 2.46. *If $0 < r < R < \operatorname{dist}(a, \partial\Omega)$, then*

$$\nu_u(a, R) = \nu_u(a, r) + \int_{r \le |z-a| < R} dd^c u \wedge (dd^c \log |z - a|)^{n-1}.$$

In particular,

$$\nu_u(a, R) = \nu_u(a) + \int_{0 < |z-a| < R} dd^c u \wedge (dd^c \log |z - a|)^{n-1}.$$

The Lelong number $\nu_u(a)$ coincides with the projective point mass of the current $dd^c u$ at a, i.e.,

$$\nu_u(a) = \int_{\{a\}} dd^c u \wedge (dd^c \log |z - a|)^{n-1}, \tag{2.4.1}$$

hence

$$\nu_u(a, R) = \int_{|z-a| < R} dd^c u \wedge (dd^c \log |z - a|)^{n-1}. \tag{2.4.2}$$

Proof. For $a \in \mathbb{C}^n$ we consider

$$\alpha_a(z) := dd^c \log|z-a| = \frac{i}{2\pi}\partial\bar\partial \log|z-a|^2.$$

This is a smooth differential $(1,1)$-form in $\mathbb{C}^n \setminus \{a\}$ with a logarithmic singularity at the point a. An easy computation shows that in $\mathbb{C}^n \setminus \{a\}$

$$\alpha_a(z) = |z-a|^{-2}(1/\pi)\beta - |z-a|^{-4}\frac{i}{2\pi}\partial|z-a|^2 \wedge \bar\partial|z-a|^2,$$

thus for $1 \le p \le n$ and $z \in \mathbb{C}^n \setminus \{a\}$ we obtain

$$\alpha_a(z)^p = |z-a|^{-2p}\pi^{-p}\beta^p - p|z-a|^{-2p-2}\frac{i}{2\pi}\partial|z-a|^2 \wedge \bar\partial|z-a|^2 \wedge \beta^{p-1}.$$

Observe that $\alpha_a(z)^p = |z-a|^{-2p}\pi^{-p}\beta^p$ when $|z-a| = r$. Thus, for $0 < r < R < \delta_\Omega(a)$,

$$\nu_u(a,R) - \nu_u(a,r) = \int_{r\le|z-a|<R} dd^c u \wedge \alpha_a^{n-1}.$$

This follows from the Stokes formula when u is smooth and by approximation in the general case. Letting $r \to 0^+$, we obtain

$$\nu_u(a,R) = \nu_u(a) + \int_{0<|z-a|<R} dd^c u \wedge (dd^c \log|z-a|)^{n-1}. \tag{2.4.3}$$

We consider for $s > 0$,

$$\ell_{a,s}(z) := \sup\{\log|z-a|, \log s\}.$$

Observe that if $0 < s < R$, then $\ell_{a,s}(z) = \log|z-a|$ in a neighborhood of the sphere $\{|z-a| = R\}$. Hence, by the Stokes theorem,

$$\int_{|z-a|<R} dd^c u \wedge (dd^c \ell_{a,s})^{n-1},$$

is independent of s: this is the projective mass of the current $dd^c u$ in the ball $B(a,R)$, i.e.,

$$\int_{|z-a|<R} dd^c u \wedge \alpha_a^{n-1} = \int_{|z-a|<R} dd^c u \wedge (dd^c \ell_{a,s})^{n-1}.$$

This definition will be justified in the next chapter when studing the Monge–Ampère operator. The formulas (2.4.1) and (2.4.2) are then consequences of the formula (2.4.3). □

2.4.2 The Jensen–Lelong formula We establish in this section the *Jensen–Lelong formula*, a representation formula for plurisubharmonic functions in terms of their boundary values and associated projective current.

Lemma 2.47. *The Poincaré form*

$$\eta(z) := d^{\mathrm{c}} \log|z| \wedge (d\,d^{\mathrm{c}} \log|z|)^{n-1}$$

is a smooth differential form of degree $2n-1$ *in* $\mathbb{C}^n \setminus \{0\}$. *Its coefficients are locally integrable in* $\mathbb{C}^n$ *and* η *satisfies*

$$d\eta = \delta_0$$

in the sense of currents on $\mathbb{C}^n$.

Moreover, on any sphere $\mathbb{S}(0,R) := \{\xi \in \mathbb{C}^n \mid |\xi| = R\}$ *the restriction* $\eta|_{\mathbb{S}(0,R)}$ *coincides with the normalized* $(2n-1)$*-volume form on* $\mathbb{S}(0,R)$.

Here δ_0 denotes the Dirac mass at the origin.

Proof. It is clear that η is smooth in $\mathbb{C}^n \setminus \{0\}$. A straightfoward computation yields

$$\eta(z) = 2^{-n}|z|^{-2n} d^{\mathrm{c}}|z|^2 \wedge (d\,d^{\mathrm{c}}|z|^2)^{n-1},$$

which is smooth off the origin and locally integrable in a neighborhood of the origin. It follows from the Stokes theorem that

$$\int_{|z|=R} \eta = 2^{-n} R^{-2n} \int_{|z|<R} (d\,d^{\mathrm{c}}|z|^2)^n = 1,$$

for we have normalized d^{c} so that $d\,d^{\mathrm{c}}|z|^2 = \frac{2}{\pi}\beta$, hence

$$(d\,d^{\mathrm{c}}|z|^2)^n = n!\frac{2^n}{\pi^n}\beta_n,$$

where β_n is the Euclidean volume form.

Observe that a defining function on the Euclidean sphere $\mathbb{S}(0,R)$ of radius R is $\rho_R(z) := |z|^2 - R^2$ and the orientation induced by the embedding of $\mathbb{S}(0,R)$ into $\mathbb{C}^n$ is defined by $d\rho_R = d|z|^2$. Thus a $(2n-1)$-form η on the sphere $\mathbb{S}(0,R)$ is positive if and only if $d\rho_R \wedge \eta(z)$ is a positive form on $\mathbb{C}^n$ for each $z \in \mathbb{S}(0,R)$.

Since $d\rho_R \wedge \eta = (d\,d^{\mathrm{c}}|z|^2)^n$ is indeed a positive form on $\mathbb{C}^n$, it follows that the restriction of η to the sphere $\mathbb{S}(0,R)$ is a positive form on the sphere. Observe that η is invariant under unitary transformation of $\mathbb{C}^n$ and its total mass on any sphere is 1. Therefore, the restriction of η to any sphere $\mathbb{S}(0,R)$ coincides with the normalized area form on $\mathbb{S}(0,R)$. On the other hand, for all $z \in \mathbb{C}^n \setminus \{0\}$,

$$d\eta(z) = 2^{-n}\left(|z|^{-2n}(d\,d^{\mathrm{c}}|z|^2)^n - |z|^{-2n-2} d|z|^2 \wedge d^{\mathrm{c}}|z|^2 \wedge (d\,d^{\mathrm{c}}|z|^2)^{n-1}\right) = 0.$$

The current $d\eta$ is thus supported at the origin. Let χ be a smooth test function in a neighborhood of 0 with support in some ball $|z| < R$. Since η is locally integrable, Stokes theorem yields

$$\begin{aligned}\langle d\eta, \chi\rangle &= -\int_{|z|<R} d\chi \wedge \eta = -\lim_{\varepsilon\to 0^+}\int_{\varepsilon<|z|<R} d\chi\wedge\eta\\ &= \lim_{\varepsilon\to 0^+}\int_{|z|=\varepsilon}\chi\eta\\ &= \chi(0) + \lim_{\varepsilon\to 0^+}\int_{|z|=\varepsilon}(\chi(z)-\chi(0))\eta\\ &= \chi(0).\end{aligned}$$

Therefore $d\eta = \delta_0$ in the sense of currents in $\mathbb{C}^n$. □

We can now prove the Jensen–Lelong formula.

Theorem 2.48. *Fix $u \in PSH(\Omega)$, $a \in \Omega$, $0 < R < \delta_\Omega(a)$. Then*

$$\begin{aligned}u(a) = &\int_{|z-a|=R} u(z)\, d^c\log|z-a| \wedge (dd^c\log|z-a|)^{n-1}\\ &+\int_{|z-a|<R}\log(|z-a|/R)\, dd^c u\wedge(dd^c\log|z-a|)^{n-1}\end{aligned}\tag{2.4.4}$$

Proof. The previous lemma yields

$$\int_{|\xi|=1} u(a+R\xi)d\sigma(\xi)/\sigma_{2n-1} = \int_{|\zeta|=R} u(a+\zeta)\eta(\zeta).$$

It follows from the Poisson–Jensen formula that

$$\begin{aligned}\frac{1}{\sigma_{2n-1}}\int_{|\xi|=1} u(a+R\xi)d\sigma(\xi) - \frac{1}{\sigma_{2n-1}}\int_{|\xi|=1} u(a+r\xi)d\sigma(\xi)&\\ = \int_r^R \nu_u(a,t)d\log t.&\end{aligned}$$

Observe that the restriction of $d^c\log|z-a|\wedge(dd^c\log|z-a|)^{n-1}$ to the sphere $\partial\mathbb{B}(a,R)$ is the normalized Lebesgue measure. Letting $r\to 0^+$, we thus obtain

$$u(a) = \int_{|z-a|=R} u(z)d^c\log|z-a|\wedge(dd^c\log|z-a|)^{n-1} - \int_0^R \nu_u(a,t)d\log t.$$

Thus the integral $\int_r^R \nu_u(a,t)d\log t$ is finite if and only if $u(a) > -\infty$. The Lelong formula yields

$$\int_0^R \nu_u(a,t)d\log t = \int_0^R dt\int_{|z-a|<t} dd^c u\wedge(dd^c\log|z-a|)^{n-1}.$$

Using the Fubini–Tonelli theorem, we obtain

$$\int_0^R \nu_u(a,t)d\log t = \int_{|z-a|<R} \log(|z-a|/R)dd^c u \wedge (dd^c \log|z-a|)^{n-1}.$$

These identities imply the Jensen–Lelong formula. □

We now derive a representation formula for plurisubharmonic function in the unit ball $\mathbb{B}$ of $\mathbb{C}^n$, similar to the Poisson–Jensen formula for the unit disc.

For $z \in \mathbb{B}$, we denote by Φ_z the automorphism of the unit ball $\mathbb{B}$ which sends $z \in \mathbb{B}$ to the origin, and consider

$$G_z(\zeta) := \log|\Phi_z(\zeta)|\,,\ (z,\zeta) \in \mathbb{B}\times\mathbb{B}.$$

This is a plurisubharmonic function in $\mathbb{B}$ which is smooth (except at the point z) up to the boundary, where it vanishes identically and has a logarithmic pole at the point z. It is called the pluricomplex Green function of the ball $\mathbb{B}$ with a logarithmic pole at the point z.

Observe that $G_z = \Phi_z^*(\ell_0)$, where $\ell_0(\zeta) = \log|\zeta|$ satisfies the equation $(dd^c\ell_0)^n = \delta_0$ in the sense of currents in $\mathbb{C}^n$. Therefore, G_z satisfies

$$(dd^c G_z)^n = \delta_z$$

in the sense of currents on $\mathbb{B}$.

From the Jensen–Lelong formula proved above we can now deduce the following representation formula:

Corollary 2.49 (Poisson–Szegö formula). *Let u be plurisubharmonic in a neighborhood of the closed unit ball $\overline{\mathbb{B}}$. Then for all $z \in \mathbb{B}$,*

$$u(z) = \int_{\partial\mathbb{B}} u(\zeta)d^c G_z(\zeta) \wedge (dd^c G_z(\zeta))^{n-1} + \int_{\mathbb{B}} G_z dd^c u \wedge (dd^c G_z)^{n-1}.$$

Proof. We proceed as in the one-dimensional case. Fix a point $z \in \mathbb{B}$ and apply the formula (2.4.4) to the plurisubharmonic function $v := u \circ \Phi_z^{-1}$ (with $a = 0$ and $R = 1$) to get

$$\begin{aligned} u(z) = v(0) &= \int_{|\xi|=1} v(\xi)\, d^c \log|\xi| \wedge (dd^c \log|\xi|)^{n-1} \\ &+ \int_{|\xi|<1} \log|\xi|\, dd^c v \wedge (dd^c \log|\xi|)^{n-1}. \end{aligned}$$

Make the change of variables $\zeta := \Phi_z(\xi)$ to conclude. □

2.4.3 A uniform version of Skoda's integrability theorem We have observed that if a plurisubharmonic function u has a large Lelong number $\nu(u,a) > 2n$ at some point a, then e^{-u} is not locally integrable near a. Skoda established in [Sko72] a partial converse: if $\nu(u,a) < 2$, then e^{-u} is locally integrable in a neighborhood of a.

We prove here a uniform version of Skoda's theorem. If $K \subset \Omega$ is a compact set and $\mathcal{U} \subset PSH(\Omega)$ is a compact class of plurisubharmonic functions, we set

$$\nu(K,\mathcal{U}) := \sup\{\nu(u,a) \,|\, a \in K, u \in \mathcal{U}\}.$$

It follows from the upper semi-continuity property of $(u,a) \mapsto \nu(u,a)$ (see Exercise 2.7) that the number $\nu(K,\mathcal{U})$ is finite.

Theorem 2.50. *Fix $0 < \alpha < 2/\nu(K,\mathcal{U})$. There exist an open neighborhood E of K and a uniform constant $C = C(K,\mathcal{U},\alpha) > 0$ such that, for all $u \in \mathcal{U}$,*

$$\int_E e^{-\alpha u} d\lambda_{2n} \leq C.$$

Proof. It follows from the submean value inequality that the class $\mathcal{U}$ is locally uniformly bounded from above in Ω. Substracting a uniform constant, we can thus assume that for a fixed neighborhood $D \Subset \Omega$ of K we have $\sup_D u \leq 0$ for all $u \in \mathcal{U}$.

By homogeneity we can assume that $\nu = \nu(K,\mathcal{U}) < 2$ and we have then to establish the uniform integrability of e^{-u}. By compactness of K, we can assume (after translating and rescaling) that D is the closed unit ball $\overline{\mathbb{B}} \subset \Omega$.

The Poisson–Szegö formula applies to all $u \in \mathcal{U}$: for $z \in \mathbb{B}$,

$$u(z) = \int_{\partial\mathbb{B}} u(\zeta) d^c G_z(\zeta) \wedge (dd^c G_z(\zeta))^{n-1} + \int_{\mathbb{B}} G_z dd^c u \wedge (dd^c G_z)^{n-1}.$$

We thus set $u = u_1 + \tilde{u}_1$, with

$$u_1(z) := \int_{\partial\mathbb{B}} u(\zeta) d^c G_z(\zeta) \wedge (dd^c G_z(\zeta))^{n-1}$$

and

$$\tilde{u}_1(z) := \int_{\mathbb{B}} G_z dd^c u \wedge (dd^c G_z)^{n-1}.$$

We are going to estimate each term separately.

The first term is easier to handle. Indeed, the Poisson–Szegö kernel $d^c G_z(\zeta) \wedge (dd^c G_z(\zeta))^{n-1}$ is smooth and positive on $\partial\mathbb{B} \times \mathbb{B}$, hence it is uniformly bounded from below by a positive constant $C_1 > 0$ on $\partial\mathbb{B} \times \{|z| \leq 1/2\}$. Since $u \leq 0$ on $\overline{\mathbb{B}}$, we infer that

$$u_1(z) \geq C_1 \int_{\partial\mathbb{B}} u(\zeta) d\sigma(\xi), \text{ for } |z| \leq 1/2.$$

We now estimate the second term $\tilde{u}_1$. Consider the pseudo-ball of center $z \in \mathbb{B}$ and radius $r \in]0,1[$ defined by

$$D(z,r) := \{\zeta \in \mathbb{B} \mid |\Phi_z(\zeta)| < r\},$$

and write for $z \in \mathbb{B}$,

$$\begin{aligned} \tilde{u}_1(z) &= \int_{D(z,r)} G_z dd^c u \wedge (dd^c G_z)^{n-1} + \int_{\mathbb{B}\setminus D(z,r)} G_z dd^c u \wedge (dd^c G_z)^{n-1} \\ &= u_2(z) + u_3(z), \end{aligned}$$

where

$$u_2(z) := \int_{D(z,r)} G_z dd^c u \wedge (dd^c G_z)^{n-1}$$

and

$$u_3(z) := \int_{\mathbb{B}\setminus D(z,r)} G_z dd^c u \wedge (dd^c G_z)^{n-1}.$$

We first estimate u_3 fom below. Since $G_z(\zeta) = (1/2)\log|\Phi_z(\zeta)|^2$, an easy computation shows that

$$\begin{aligned} dd^c u \wedge (dd^c G_z)^{n-1} &\leq \frac{dd^c u \wedge (dd^c|\Phi_z|^2)^{n-1}}{2^{n-1}|\Phi_z|^{2n-2}} \\ &\leq c_3 \frac{dd^c u \wedge (dd^c|\zeta|^2)^{n-1}}{|\Phi_z|^{2n-2}}. \end{aligned} \tag{2.4.5}$$

Thus for $|z| \leq 1/2$ and $0 < r \leq 1/2$ we have

$$u_3(z) \geq -2^{2n-2}\log 2 \int_{\mathbb{B}} \Delta u. \tag{2.4.6}$$

We now estimate $\exp(-u_2(z))$ by applying Jensen's inequality. We need to compute the total mass of the measure $\sigma_z := dd^c u \wedge (dd^c G_z)^{n-1}$, on the domain $D(z,r)$ for $|z| << 1$. This is, for $z \in \mathbb{B}$ and $0 < r < 1$,

$$\vartheta_u(z,r) := \int_{D(z,r)} dd^c u \wedge (dd^c G_z)^{n-1}.$$

Observe that

$$\vartheta_u(z,r) = \int_{\mathbb{B}_r} dd^c u \circ \Phi_z \wedge (dd^c \log|\zeta|)^{n-1} = \nu_{u\circ\Phi_z}(0,r),$$

hence for all $z \in \mathbb{B}$

$$\vartheta_u(z) := \lim_{r\to 0} \vartheta_u(z,r) = \nu_{u\circ\Phi_z}(0),$$

since Φ_z is an automorphism sending the origin to z.

Since $\sup_{\mathcal{U}} \vartheta_u(0) < \nu < 2$, there exists $\delta > 0$ such that for all $u \in \mathcal{U}$

$$\vartheta_u(z,\delta) < \nu < 2, \quad \text{for } |z| \le \delta.$$

Set $\rho(z) := |z|^2 - 1$, on $\mathbb{B}$. The function ρ is negative, smooth, and strictly plurisubharmonic in the ball $\mathbb{B}$ with

$$\vartheta_\rho(z,\delta) = \int_{\mathbb{B}_r} dd^c|\Phi_z(\zeta)|^2 \wedge (dd^c \log|\zeta|)^{n-1}.$$

Since Φ_z depends smoothly on z, the right-hand side of the previous equation is a continuous non-vanishing function on $\mathbb{B}$, hence there exists $\eta_1, \eta_2 > 0$ such that

$$\eta_1 \le \vartheta_\rho(z,\delta) \le \eta_2, \quad \text{for } |z| \le \delta.$$

Replacing u by $\tilde{u} := u + \varepsilon\rho$, on $\mathbb{B}$ with $\varepsilon > 0$ small enough, we can assume that, for some $\eta > 0$,

$$\eta \le \vartheta_u(z,\delta) < \nu < 2, \quad \text{for } |z| \le \delta.$$

We now apply Jensen's inequality and use the last inequality to obtain, for $|z| < \delta$,

$$\exp(-u_2(z)) \le \frac{1}{\eta} \int_{D(z,\delta)} \exp(-\nu G_z) dd^c u \wedge (dd^c G_z)^{n-1}.$$

Using (2.4.5) it follows that, for $|z| < \delta$,

$$\exp(-u_2(z)) \le \frac{1}{\eta\, 2^{n-1}} \int_{D(z,\delta)} \frac{dd^c u \wedge (dd^c|\Phi_z|^2)^{n-1}}{|\Phi_z|^{2n-2+\nu}}.$$

Since Φ_z depends smoothly on z and $\Phi_z(z) = 0$, there exists a uniform constant $c_4 > 0$ such that for $|z| \le 1/2$,

$$c_4^{-1}|z-\zeta| \le |\Phi_z(\zeta)| \le c_4|z-\zeta|.$$

Taking $\delta > 0$ small enough, there exists uniform constants $c_5 > 0$ and $c_6 > 0$ such that for $|z| \le \delta$,

$$\exp(-u_2(z)) \le c_5 \int_{\mathbb{B}(z,c_6\delta)} \frac{dd^c u \wedge (dd^c|\zeta|^2)^{n-1}}{|z-\zeta|^{2n-2+\nu}}.$$

If λ is a finite Borel measure, then by the Fubini–Tonelli theorem

$$\begin{aligned}\int_{\{|z|<\delta\}} \exp(-u_2(z)) d\lambda(z) &\le c_5 \int_{\{|\zeta|<c_7\delta\}} d\mu(\zeta) \int_{\{|z|<\delta\}} \frac{d\lambda(z)}{|z-\zeta|^{2n-2+\nu}} \\ &\le c_5 \int_{\{|\zeta|<c_7\delta\}} d\mu(\zeta) \int_{\{|\zeta-z|<c_8\delta\}} \frac{d\lambda(z)}{|z-\zeta|^{2n-2+\nu}},\end{aligned}$$

where $\mu := dd^c u \wedge (dd^c|\zeta|^2)^{n-1}$ is the Riesz measure associated to μ.

The second integral in the right-hand side is uniformly bounded as long as $\nu < 2$, it thus remains to bound integrals of the type

$$I(r) := \int_{\{|\zeta|<r\}} d\mu(\zeta).$$

Let χ be a non-negative test function with compact support in $\{|\zeta| < 2r\}$ such that $\chi \equiv 1$ in $\{|\zeta| < r\}$. Then $dd^c\chi \geq -Add^c|\zeta|^2$ for some positive constant depending only on r. Since $u \leq 0$ we get

$$\begin{aligned} I(r) &\leq \int_{\{|\zeta|<2r\}} \chi dd^c u \wedge (dd^c|\zeta|^2)^{n-1} \\ &= \int_{\{|\zeta|<2r\}} u dd^c\chi \wedge (dd^c|\zeta|^2)^{n-1} \\ &\leq A\int_{\{|\zeta|<2r\}} (-u)(dd^c|\zeta|^2)^n. \end{aligned}$$

By the compactness of $\mathcal{U}$, the last integral is uniformly bounded. □

2.5 Exercises

Exercise 2.1. *Let T be a current of degree p and α a differential form of degree m. Show that*

$$d(T \wedge \alpha) = dT \wedge \alpha + (-1)^p T \wedge d\alpha.$$

Exercise 2.2. *Let $T : \mathcal{D}(\Omega) \to \mathbb{C}$ be a positive $\mathbb{C}$-linear functional on $\mathcal{D}(\Omega)$. Show that there exists a positive Borel measure μ_T on Ω such that for any smooth test function $\chi \in \mathcal{D}(\Omega)$,*

$$\langle T, \chi\rangle = \int_\Omega \chi(x) d\mu_T(x).$$

Exercise 2.3. *Give an example of a (weakly) positive form of bidegree $(2,2)$ in a domain of $\mathbb{C}^n$ that is not strongly positive. How large should n be?*

Exercise 2.4. *Let T be a closed positive current of bidegree $(1,1)$ in the unit ball of $\mathbb{C}^n$. Show that there exists a plurisubharmonic function u in $\mathbb{B}$ such that $T = dd^c u$.*

Exercise 2.5. *Let T be a positive current in $\mathbb{C}^n$ with compact support. Show that $T \equiv 0$ when $n \geq 2$.*

Exercise 2.6. *Let f be a non-zero holomorphic function in a domain $\Omega \subset \mathbb{C}^n$. Show that*

$$dd^c \log|f| = [Z_f]$$

is the current of integration along the zero set $Z_f = \{f = 0\}$.

Exercise 2.7. *Show that the Lelong number functional*

$$\nu : PSH(\Omega) \times \Omega \longrightarrow \mathbb{R}^+$$

has the following properties:

(i) *it is additive and positively homogenous, i.e., for all* $u, v \in PSH(\Omega)$ *and* $z \in \Omega$, $\alpha > 0$,

$$\nu(\alpha u + v, z) = \alpha \nu(u, z) + \nu(v, z);$$

(ii) *if* $u, v \in PSH(\Omega)$, *then* $\nu_{\max(u,v)} = \inf\{\nu_u, \nu_v\}$;

(iii) $(u, z) \mapsto \nu(u, z)$ *is upper semi-continuous for the product topology* (L^1_{loc}-*topology on* $PSH(\Omega)$ *and Euclidean topology on* Ω).

Exercise 2.8. *Fix* $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{R}^n_+$, *positive real numbers, and consider*

$$z \in \mathbb{C}^n \longmapsto u_\alpha(z) := \max_{1 \le j \le n} \{\alpha_j \log |z_j|\}.$$

Show that $\nu_{\varphi_\alpha}(0) = \min \alpha_j$.

Exercise 2.9. *Let* $f = (f_1, \ldots, f_k) : \Omega \to \mathbb{C}^k$ *be a holomorphic map from a domain* $\Omega \subset \mathbb{C}^n$ *into* $\mathbb{C}^k$ *such that* $f \not\equiv 0$ *in* Ω. *Fix* $\alpha := (\alpha_1, \ldots, \alpha_n) \in \mathbb{R}^n_+$ *positive reals. Show that the function*

$$\varphi := \frac{1}{2} \log(|f_1|^{2\alpha_1} + \cdots + |f_1|^{2\alpha_N})$$

is plurisubharmonic and compute its Lelong numbers.

Exercise 2.10. *Let* $u \in PSH(\Omega)$ *and* χ *be a convex increasing function in some interval containing* $u(\Omega)$. *Compute the Lelong numbers of the plurisubharmonic function* $\chi \circ u$. *Show that* $\nu_{\chi \circ u} \equiv 0$ *if* $\chi'(-\infty) = 0$.

Exercise 2.11. *Let* u *be plurisubharmonic function in the unit ball of* $\mathbb{C}^n$ *such that* $u \ge 1$.

1) *Show that* u^2 *is plurisubharmonic and prove that the current* $d\,d^c u^2$ *dominates* $d\,d^c u$.

2) *Prove that* $\nabla u \in L^2_{\text{loc}}$.

3) *Show that* $u = \log |z_1|$ *is an unbounded plurisubharmonic function whose gradient does not belong to* L^2_{loc}.

Exercise 2.12. *Consider*

$$u(z, w) = \log |z^2 - w^3|$$

1) *Compute the Lelong numbers of* u *at all points of* $\mathbb{C}^2$.

2) *Compute the critical integrability index of* u *at the origin, i.e.,*

$$c(u) := \sup\{c > 0\,;\, e^{-2cu} \in L^1_{\text{loc}}((0, 0))\}$$

3) *Is* $e^{-2c(u)u}$ *locally integrable at the origin?*

Exercise 2.13. *Let Ω be a pseudoconvex domain in $\mathbb{C}^n$. Show that*

$$\tilde{\Omega} := \{(z, w) \in \Omega \times \mathbb{C} \,|\, |w| < \delta_\Omega(z)\}.$$

is a pseudoconvex domain in $\mathbb{C}^{n+1}$ by exhibiting an S^1-invariant (rotations in the w-variable) exhaustion plurisubharmonic function $\rho(z, w)$.

Exercise 2.14. *Let μ be a Borel measure in some domain Ω. Assume there exists $\alpha \in]0, 2]$ and $C > 0$ such that for any $a \in \Omega$ and $r > 0$,*

$$(\star) \qquad \mu(B(a, r)) \leq C r^{2n-2+\alpha}.$$

1) *Prove that if $u \in PSH(\Omega)$ and $\nu(u, a) < \alpha$ at some point $a \in \Omega$, then e^{-u} is locally μ-integrable in a neighborhood of a.*

2) *Give a uniform version of this result.*

3) *Let $u \in PSH(\Omega)$ be Hölder continuous of order $\alpha > 0$. Prove that the Riesz measure Δu satisfies condition $(\star)$.*

3 The complex Monge–Ampère operator

The complex Monge–Ampère operator is the operator

$$\varphi \longmapsto (dd^c\varphi)^n$$

which associates to each given plurisubharmonic function φ a non-negative Radon measure. When φ is $\mathcal{C}^2$-smooth, this Monge–Ampère measure is

$$(dd^c\varphi)^n = c_n \det\left(\frac{\partial^2\varphi}{\partial z_i \partial \overline{z}_j}\right) dV,$$

where dV denotes the Lebesgue measure and $c_n > 0$ is a normalizing constant chosen so that the Monge–Ampère measure of the plurisubharmonic function $\varphi(z) = \frac{1}{2}\log[1 + \|z\|^2]$ has total mass 1 in $\mathbb{C}^n$.

The above formula still makes sense almost everywhere for plurisubharmonic functions φ which belong to the Sobolev space $W^{2,n}_{\rm loc}$. It is however crucial for applications to consider plurisubharmonic functions which are far less regular.

In this chapter (and the rest of Part 1) we explain how to define and study the complex Monge–Ampère operator acting on plurisubharmonic functions which are locally bounded, following the pioneering work of Bedford and Taylor [BT82]. We show that it is continuous along monotone sequences, but that it is not continuous for the $L^1_{\rm loc}$-convergence.

In the whole chapter Ω denotes a smoothly bounded strictly pseudoconvex subset of $\mathbb{C}^n$, usually equipped with a smooth strictly plurisubharmonic defining function ρ, i.e., $\Omega = \{z \in \mathbb{C}^n\,;\, \rho(z) < 0\}$.

The whole theory could be developed in the slightly more general context of hyperconvex domains but we shall not need this in the book. The reader may even like to think that Ω is the unit ball (and $\rho(z) = \|z\|^2 - 1$), which is sufficient for most applications we have in mind.

3.1 Currents of Monge–Ampère type

3.1.1 Definitions Let T be a closed positive (p, p)- current in a domain $\Omega \subset \mathbb{C}^n$, $0 \le p \le n-1$. It can be decomposed as

$$T = i^{p^2} \sum_{|I|=p,|J|=p} T_{I,J}\, dz_I \wedge d\overline{z}_J,$$

where the coefficients $T_{I,J}$ are complex Borel measures.

A locally bounded Borel function u is locally integrable with respect to the coefficients of T, hence uT is a well-defined current of order 0, setting

$$\langle uT, \Psi\rangle = \langle T, u\Psi\rangle,$$

Ψ a continuous form of bidegree $(n-p, n-p)$ with compact support.

If u is a smooth function the current $dd^c u \wedge T$ is a $(p+1, p+1)$-current in Ω defined by

$$\langle dd^c u \wedge T, \Psi\rangle := \langle T, dd^c u \wedge \Psi\rangle$$

for $\Psi \in \mathcal{D}_{q,q}(\Omega)$, $q = n-p-1$. Observe that $\Theta = d^c u \wedge \Psi - u d^c \Psi$ is a smooth form with compact support such that $dd^c u \wedge \Psi - u dd^c \Psi = d\Theta$. Since $dT = 0$, it follows that

$$\begin{aligned}\langle T, dd^c u \wedge \Psi\rangle &= \langle T, u dd^c \Psi\rangle + \langle T, d\Theta\rangle \\ &= \langle T, u dd^c \Psi\rangle = \langle dd^c(uT), \Psi\rangle.\end{aligned}$$

This motivates the following:

Definition 3.1. *Let T be a closed positive current of bidegree (p,p) in a domain $\Omega \subset \mathbb{C}^n$ and $u \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$. The current $dd^c u \wedge T$ is the $(p+1, p+1)$-current defined by*

$$dd^c u \wedge T := dd^c(uT),$$

i.e.,

$$\langle dd^c u \wedge T, \Psi\rangle = \langle uT, dd^c \Psi\rangle,$$

for all test forms Ψ of bidegree (q,q), $q := n-p-1$.

We also need to consider currents of the type $du \wedge d^c u \wedge T$, where $u \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$. Recall that du is locally in $L^p_{\mathrm{loc}}(\Omega)$ (with respect to the Lebesgue measure) for any $p < 2$. When u is locally bounded du actually belongs to $L^2_{\mathrm{loc}}(\Omega)$, as we now explain.

Since u is (locally) bounded from below, we can add a large constant and assume that $u \geq 0$. It follows that u^2 is plurisubharmonic, hence $dd^c u^2 \wedge T$ is a well-defined closed positive current if u is smooth, with

$$dd^c u^2 = 2u dd^c u + 2du \wedge d^c u.$$

This motivates the following:

Definition 3.2. *Let $u \in PSH(\Omega) \cap L^\infty(\Omega)$. We set*

$$du \wedge d^c u \wedge T := \frac{1}{2} dd^c (u-m)^2 \wedge T - (u-m) dd^c u \wedge T,$$

where m is a lower bound for u in Ω.

This is a well-defined closed current in Ω which does not depend on m (note that $du = d(u-m)$). These currents (a priori of order 2) are of order zero, since they are positive:

Proposition 3.3. *Let (u_j) be locally bounded plurisubharmonic functions in Ω which decrease to $u \in PSH(\Omega) \cap L^\infty_{\rm loc}$. Then*

$$dd^c u_j \wedge T \longrightarrow dd^c u \wedge T, \quad du_j \wedge d^c u_j \wedge T \longrightarrow du \wedge d^c u \wedge T,$$

in the weak sense of currents on Ω.

In particular $dd^c u \wedge T$ and $du \wedge d^c u \wedge T$ are closed positive currents, hence their coefficients extend as complex Borel measures on Ω.

Proof. By definition, $dd^c u \wedge T = dd^c(uT)$ is a closed current. By assumption, the sequence (u_j) converges in $L^1_{\rm loc}(\Omega, \sigma_T)$, hence $u_j T \to uT$ in the weak sense of currents in Ω. The operator dd^c is continuous for the weak convergence of currents, hence $dd^c u_j \wedge T \to dd^c u \wedge T$ in the sense of currents in Ω.

The second statement follows from the first one and the fact that $u_j dd^c u_j \wedge T$ converges to $u dd^c u \wedge T$. This latter convergence relies on several technical results that we establish below: it follows from the Chern–Levine–Nirenberg inequalities (Theorem 3.9) that the currents $u_j dd^c u_j \wedge T$ have uniformly bounded masses. We can thus consider a cluster point σ. The reader will check in Exercise 3.1 that

$$\sigma \leq u dd^c u \wedge T,$$

since (u_j) is decreasing and $dd^c u_j \wedge T \to dd^c u \wedge T$. To prove the reverse inequality we can use the localization principle (Proposition 3.6) to ensure that u_j and u coincide near the boundary of Ω, and then use Proposition 3.7 to show that the currents σ and $u dd^c u \wedge T$ have the same total mass.

It remains to prove the positivity. Since this is a local property, we can reduce to the case where u is plurisubharmonic in a neighborhood of $\overline{\Omega}$ and $0 \leq u \leq M$ on Ω. We can then approximate u by a decreasing family of smooth plurisubharmonic functions, $u_\varepsilon := u \star \rho_\varepsilon$, using standard mollifiers.

We know that $dd^c u_\varepsilon \wedge T \to dd^c u \wedge T$ and $du_\varepsilon \wedge d^c u_\varepsilon \wedge T \to du \wedge d^c u \wedge T$ in the sense of currents in Ω. Since $dd^c u_\varepsilon$ is a positive form of bidegree $(1,1)$, we infer that $dd^c u_\varepsilon \wedge T$ and $du_\varepsilon \wedge d^c u_\varepsilon \wedge T$ are positive currents in Ω, hence so are their limits. □

Iterating this process we define by induction the intersection of currents of the above type.

Definition 3.4. *If $u_1, \ldots, u_k \in PSH(X) \cap L^\infty_{\rm loc}(X)$, and T is a closed positive current of bidimension (m,m), we define the current $dd^c u_1 \wedge \ldots \wedge dd^c u_k \wedge T$ by*

$$dd^c u_1 \wedge \ldots \wedge dd^c u_k \wedge T := dd^c \left(u_1 dd^c u_2 \wedge \ldots \wedge dd^c u_k \wedge T \right).$$

We similarly define $du_1 \wedge d^c u_1 \wedge \ldots \wedge du_k \wedge d^c u_k \wedge T$.

It turns out that these definitions are symmetric in the u_j's, as we shall soon show.

In particular, if $V \in PSH(\Omega)$ and $u_1, \ldots, u_k \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$, the current $dd^c u_1 \wedge \ldots \wedge dd^c u_k \wedge dd^c V$ is a well-defined closed positive current on Ω.

Definition 3.5. *The complex Monge–Ampère measure of a locally bounded plurisubharmonic function is*

$$(dd^c u)^n := dd^c u \wedge \ldots \wedge dd^c u.$$

It remains to make sure that this is a *good* definition.

3.1.2 Localization principle A technical point that we are going to use on several occasions is that we can arbitrarily modify a bounded plurisubharmonic function near the boundary of a pseudoconvex domain without changing it on a given compact subset.

Fix $\Omega = \{\rho < 0\} \Subset \mathbb{C}^n$, a bounded strictly pseudoconvex domain, with ρ smooth and strictly plurisubharmonic in a neighborhood of $\overline{\Omega}$.

Proposition 3.6. *Fix a compact set $K \subset \Omega$ and $M > 0$. There exist $C > 0$, depending only on K and Ω, a compact subset $E \subset \Omega$ such that $K \subset E^\circ$ and for any $u \in PSH(\Omega) \cap L^\infty(\Omega)$ with $u < 0$ in Ω, there exist $A > 0$ and a bounded plurisubharmonic function $\tilde{u}$ in a neighborhood of $\bar{\Omega}$ such that*

(i) $\tilde{u} = u$ *in a neighborhood of* K,
(ii) $\tilde{u} = A\rho$ *in* $\Omega \setminus E$, *with* $A \leq C \, \|u\|_{L^\infty(\Omega)}$,
(iii) $u \leq \tilde{u} \leq A\rho$ *on* Ω.

In particular, $\|\tilde{u}\|_{L^\infty(D)} \leq C \, \|u\|_{L^\infty(\Omega)}$.

Proof. Consider, for $c > 0$, $D_c = \{z \in \Omega | \, \rho(z) < -c\}$, and choose $a > 0$ such that $K \subset D_a$. Set $M := \|u\|_{L^\infty(\Omega)}$ and $A := M/a$ so that

$$u \geq A\rho \text{ on } \partial D_a.$$

Pick $b > 0$ so small so that $a < b$ and $A\rho \geq u$ on ∂D_b. The gluing lemma for plurisubharmonic functions shows that the function

$$\tilde{u}(z) = \begin{cases} u(z), & \text{for} \quad z \in D_a, \\ \max\{u(z), A\rho(z)\}, & \text{for} \quad z \in D_b \setminus D_a, \\ A\rho(z), & \text{for} \quad z \in \Omega \setminus D_b, \end{cases}$$

is plurisubharmonic in Ω and satisfies all our requirements with $E := \overline{D_b}$ and $C := \max\{\max_{\bar{\Omega}} |\rho|/a, 1\}$. □

We now establish an integration by parts formula due to U. Cegrell [Ceg04]:

Proposition 3.7. *Let T be a closed positive current of bidimension $(1,1)$ in Ω. Let $u, v \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$ be such that $u, v \le 0$, $\lim_{z\to\partial\Omega} u(z) = 0$ and $\int_\Omega dd^c v \wedge T < +\infty$. Then*

$$\int_\Omega v dd^c u \wedge T \le \int_\Omega u dd^c v \wedge T,$$

where the inequality holds in $[-\infty, 0[$. If $\lim_{z\to\partial\Omega} v(z) = 0$, then

$$\int_\Omega v dd^c u \wedge T = \int_\Omega u dd^c v \wedge T, \tag{3.1.1}$$

provided that $\int_\Omega dd^c u \wedge T < +\infty$.

Proof. For $\varepsilon > 0$ set $u_\varepsilon := \sup\{u, -\varepsilon\}$ and observe that u_ε is plurisubharmonic in Ω and increases to 0 as ε decreases to 0. The monotone convergence theorem yields

$$\int_\Omega u dd^c v \wedge T = \lim_{\varepsilon\to 0} \int_\Omega (u - u_\varepsilon) dd^c v \wedge T.$$

Set $\Omega_\varepsilon := \{u < -\varepsilon\}$. Then $K := \overline{\Omega}_\varepsilon \subset \Omega$ is a compact subset such that $u_\varepsilon = u$ on $\Omega \setminus K$. Let $D_1 \Subset \Omega$ be a domain close to Ω such that $K \subset D_1$. Let $(\rho_\eta)_{\eta>0}$ be standard mollifiers. For $\eta > 0$ small enough, the smooth function $(u - u_\varepsilon) \star \rho_\eta$ has compact support contained in the η-neighborhood $K_\eta \subset \Omega$ of K and converges to $u - u_\varepsilon$ on Ω as η decreases to 0. It follows from Lebesgue's convergence theorem that

$$\int_{D_1} (u - u_\varepsilon) dd^c v \wedge T = \lim_{\eta\to 0} \int_{D_1} (u - u_\varepsilon) \star \rho_\eta dd^c v \wedge T.$$

Since $(u - u_\varepsilon) \star \rho_\eta$ is smooth and has a compact support in D_1, we infer

$$\int_{D_1} (u-u_\varepsilon)\star\rho_\eta dd^c v \wedge T = \int_{D_1} v dd^c((u-u_\varepsilon)\star\rho_\eta) \wedge T \ge \int_{D_1} v dd^c(u\star\rho_\eta) \wedge T.$$

Assume first that v is continuous in a neighborhood of $\bar{D}_1$ and observe that we can choose D_1 so that the positive Borel measure $(-v)dd^c u \wedge T$ has no mass on ∂D_1. Thus

$$(-v)dd^c(u \star \rho_\eta) \wedge T \longrightarrow (-v)dd^c u \wedge T$$

weakly in the sense of Radon measures in Ω. Therefore

$$\lim_{\eta\to 0} \int_{D_1} v\, dd^c(u \star \rho_\eta) \wedge T = \int_{D_1} v\, dd^c u \wedge T.$$

It follows that

$$\int_{D_1} u\, dd^c v \wedge T \ge \int_{D_1} v\, dd^c u \wedge T - \varepsilon \int_{D_1} dd^c v \wedge T. \tag{3.1.2}$$

If v is not lower semi-continuous we take a decreasing sequence (v_j) of negative continuous plurisubharmonic functions in Ω which converges to v in a neighborhood of $\overline{D}_1$ and apply the last inequality to each function v_j. Choose a domain D_2 such that $D_2 \Subset D_1 \Subset \Omega$. By the upper semi-continuity on the compact set $L := \overline{D_2} \subset D_1$, it follows from (3.1.2) that

$$\begin{aligned}\int_L u\,dd^c v \wedge T &\geq \limsup_j \int_L u\,dd^c v_j \wedge T \\ &\geq \lim_j \int_{D_1} v_j\,dd^c u \wedge T - \varepsilon \liminf_j \int_{D_1} dd^c v_j \wedge T \\ &\geq \int_{D_1} v\,dd^c u \wedge T - \varepsilon \int_{\bar{D}_1} dd^c v \wedge T.\end{aligned}$$

Letting D_1 and D_2 increase to Ω and taking the limit as $\varepsilon \to 0$, we obtain the required inequality provided that $\int_\Omega dd^c v \wedge T < +\infty$. □

Simple examples show that the total mass of the current $dd^c u \wedge T$ in Ω need not be finite:

Example 3.8. *Fix* $0 < \alpha < 1$ *and consider*

$$z \in \mathbb{D} \longmapsto u(z) := -(1 - |z|^2)^\alpha \in \mathbb{R}$$

This is a smooth and bounded subharmonic function in the unit disc $\mathbb{D}$*, which extends as a Hölder continuous function up to the boundary. The reader can check* (*Exercise* 3.3) *that the total mass of* Δu *is infinite,*

$$\int_{\mathbb{D}} dd^c u = +\infty.$$

3.2 The complex Monge–Ampère measure

3.2.1 Chern–Levine–Nirenberg inequalities The following inequalities are due to Chern, Levine, and Nirenberg [CLN69]. They are the first step towards defining the complex Monge–Ampère operator on bounded plurisubharmonic functions:

Theorem 3.9. *Let* T *be a closed positive current of bidimension* (k,k) *in* Ω *and* $u_1, \ldots, u_k \in PSH(\Omega) \cap L^\infty_{\rm loc}(\Omega)$*. Then for all open subsets* $\Omega_1 \Subset \Omega_2 \Subset \Omega$*, there exists a constant* $C = C_{\Omega_1,\Omega_2} > 0$ *such that for any compact subsets* $K \subset \Omega_1$

$$\int_{\Omega_1} dd^c u_1 \wedge \ldots \wedge dd^c u_k \wedge T \leq C \|u_1\|_E \cdots \|u_k\|_E \|T\|_E, \tag{3.2.1}$$

and

$$\int_{\Omega_1} du_1 \wedge d^c u_1 \wedge dd^c u_2 \wedge \ldots \wedge dd^c u_k \wedge T \leq C \|u_1\|_E^2 \|u_2\|_E \cdots \|u_k\|_E \|T\|_E, \tag{3.2.2}$$

where $E := (\Omega_2 \setminus \Omega_1) \cap \mathrm{Supp}(T)$.

Here (and in the sequel) we let $\|u\|_E$ denote the L^∞-norm of the function u on the Borel set E.

Proof. It is enough to prove inequality (3.2.1) for $k = 1$ and then use induction on k. Set $u = u_1$. We can always assume that $u \leq 0$ on Ω_2, since the function $v := u - \sup_{\Omega_2} u$ is negative on Ω_2 and satisfies $dd^c v = dd^c u$ and $\|v\|_{\Omega_2 \setminus \Omega_1} \leq 2\|u\|_{\Omega_2 \setminus \Omega_1}$.

Let $\chi \in \mathcal{D}(\Omega_2)$ be a non-negative test function with $\chi = 1$ in Ω_1. Then

$$\int_{\Omega_1} T \wedge dd^c u \leq \int_{\Omega_2} \chi T \wedge dd^c u.$$

Since $dd^c \chi = 0$ in Ω_1, we obtain

$$\int_{\Omega_2} \chi dd^c u \wedge T = \int_{\Omega_2} u dd^c \chi \wedge T = \int_{\Omega_2 \setminus \Omega_1} u dd^c \chi \wedge T.$$

Fixing $A > 0$ such that $dd^c \chi \leq A\beta$ on Ω, we infer that

$$\int_{\Omega_2} \chi dd^c u \wedge T \leq A\|u\|_{\Omega_2 \setminus \Omega_1} \int_{\Omega_2 \setminus \Omega_1} T \wedge \beta.$$

We now prove the second inequality (3.2.2). We can assume without loss of generality that $u := u_1 \geq 0$ in Ω_2. Then u^2 is plurisubharmonic and

$$dd^c u^2 = 2u dd^c u + 2du \wedge d^c u,$$

in the sense of currents in Ω_2. Thus (3.2.2) follows from (3.2.1) applied with u_1 replaced bu u_1^2,

$$\int_{\Omega_1} du_1 \wedge d^c u_1 \wedge dd^c u_2 \wedge \ldots \wedge dd^c u_k \wedge T \leq C \|u_1\|_E^2 \|u_2\|_E^2 \cdots \|u_k\|_E \|T\|_E. \quad \square$$

Corollary 3.10. *For all subdomains* $\Omega_1 \Subset \Omega_2 \Subset \Omega$, *there exists a constant* $C = C_{\Omega_1,\Omega_2} > 0$ *such that if* $V \in PSH(\Omega)$ *and* $u_1, \ldots, u_k \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$, *then*

$$\int_{\Omega_1} dd^c u_1 \wedge \ldots \wedge dd^c u_k \wedge dd^c V \wedge \beta_q \leq C \|u_1\|_E \cdots \|u_k\|_E \|V\|_{L^1(E)}, \tag{3.2.3}$$

where $E := \Omega_2 \setminus \Omega_1$ *and* $q = n - (k+1)$.

Proof. It suffices to find an upper bound for the mass of the current $T = dd^c V \wedge \beta_{n-1}$ on Ω_1 and apply previous inequalities.

We use the same notations as in the previous proof and assume first that $V < 0$ in Ω_2. Then

$$\int_{\Omega_1} dd^c V \wedge \beta_{n-1} \leq \int_{\Omega_2} \chi dd^c V \wedge \beta_{n-1} = \int_{\Omega_2} V dd^c \chi \wedge \beta_{n-1}.$$

Since $dd^c \chi \wedge \beta_{n-1} = \Delta \chi \beta_n$, it follows that

$$\int_K dd^c V \wedge \beta_{n-1} \leq \|\Delta \chi\|_{\Omega_2 \setminus \Omega_1} \int_{\Omega_2 \setminus \Omega_1} |V| \beta_n,$$

which proves the required inequality.

We now treat the general case. The submean value inequality ensures that we can find an open set Ω' such that $\Omega_1 \Subset \Omega' \Subset \Omega_2$ and a constant $C > 0$ such that $\max_{\bar{\Omega}'} V \leq C \int_{\Omega_2 \setminus \Omega_1} V_+ \beta_n$. We can now apply the last inequality to $V - \sup_{\Omega'} V$ in Ω' and obtain the required estimate. □

3.2.2 Symmetry of Monge–Ampère operators

Proposition 3.11. *Let T be a closed positive current on Ω of bidimension (m, m). Let (u_j) be bounded plurisubharmonic functions in Ω which decrease to $u \in PSH(\Omega) \cap L^\infty_{\text{loc}}$. Then for any continuous plurisubharmonic function h on Ω,*

$$h dd^c u_j \wedge T \longrightarrow h dd^c u \wedge T$$

and

$$dd^c h \wedge dd^c u_j \wedge T \longrightarrow dd^c h \wedge dd^c u \wedge T$$

in the weak sense of Radon measures on Ω.

Proof. We already know that $dd^c u_j \wedge T \to dd^c u \wedge T$ in the weak sense of currents, by Proposition 3.3.

The Chern–Levine–Nirenberg inequalities ensure that the currents $dd^c u_j \wedge T$ have locally uniformly bounded masses in Ω, hence the weak convergence holds in the sense of Radon measures on Ω. Thus

$$h dd^c u_j \wedge T \longrightarrow h dd^c u \wedge T$$

in the weak sense of Radon measures.

Since the operator dd^c is continuous for the weak convergence of currents, it follows that $dd^c h \wedge dd^c u_j \wedge T \to dd^c h \wedge dd^c u \wedge T$ in the sense of currents in Ω.

Now these are positive currents (h is plurisubharmonic), hence the convergence actually holds in the sense of Radon measures. □

This shows that these operators are symmetric:

Corollary 3.12. *Let T (resp. S) be a positive closed current of bidimension $(2,2)$ (resp. (k,k)) and u, v be locally bounded plurisubharmonic functions in a domain $\Omega \subset \mathbb{C}^n$. Then*

$$dd^c u \wedge dd^c v \wedge T = dd^c v \wedge dd^c u \wedge T.$$

More generally, the Monge–Ampère type operator

$$(u_1, \ldots, u_k) \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega) \longmapsto dd^c u_1 \wedge \ldots \wedge dd^c u_k \wedge S$$

is symmetric.

Proof. The formula is clear when both functions are smooth. Assume now that u is smooth and take a sequence of smooth plurisubharmonic functions v_j which locally decrease to u. Then

$$dd^c u \wedge dd^c v_j \wedge T = dd^c v_j \wedge dd^c u \wedge T,$$

hence the previous convergence result yields the required identity.

We now treat the general case. Since the property is local, we can use the localization principle and assume that u, v are negative in a ball $\mathbb{B} \Subset \Omega$ and equal to 0 on $\partial\mathbb{B}$. Let ρ be a strictly plurisubharmonic defining function of $\mathbb{B}$. It follows from Proposition 3.7 that

$$\int_{\mathbb{B}} \rho dd^c u \wedge dd^c v \wedge T = \int_{\mathbb{B}} u dd^c \rho \wedge dd^c v \wedge T.$$

As ρ is smooth, the first part of the proof yields $dd^c \rho \wedge dd^c v \wedge T = dd^c v \wedge dd^c \rho \wedge T$ in the sense of currents. Thus

$$\int_{\mathbb{B}} \rho dd^c u \wedge dd^c v \wedge T = \int_{\mathbb{B}} u dd^c v \wedge dd^c \rho \wedge T.$$

Using again the formula (3.1.1) and the fact that $dd^c \rho \wedge dd^c u \wedge T = dd^c u \wedge dd^c \rho \wedge T$, we get

$$\begin{aligned} \int_{\mathbb{B}} \rho dd^c u \wedge dd^c v \wedge T &= \int_{\mathbb{B}} v dd^c u \wedge dd^c \rho \wedge T \\ &= \int_{\mathbb{B}} v dd^c \rho \wedge dd^c u \wedge T \\ &= \int_{\mathbb{B}} \rho dd^c v \wedge dd^c u \wedge T. \end{aligned}$$

Observe finally that any smooth test function χ with compact support in $\mathbb{B}$ can be written as the difference of two defining functions for $\mathbb{B}$. Indeed $A dd^c \rho > -dd^c \chi$ for $A > 1$ large enough, thus $\chi = \rho_1 - \rho_2$, where $\rho_1 := \chi + A\rho$ and $\rho_2 := A\rho$. □

Here is a first example of an explicit non-smooth complex Monge–Ampère measure:

Example 3.13. *Fix $r > 0$. The function*

$$u_r(z) := \log^+(|z|/r) = \max\{\log|z|, \log r\} - \log r$$

is a continuous plurisubharmonic function in $\mathbb{C}^n$ which is smooth in $\mathbb{C}^n \setminus \{|z| = r\}$, where it satisfies $(dd^c u)^n = 0$.

The Borel measure $(dd^c u_r)^n$ is invariant under unitary transformations and has total mass 1 in $\mathbb{C}^n$ (see Proposition 3.34). It coincides with the normalized Lebesgue measure σ_r on the sphere $|z| = r$.

3.2.3 Integrability with respect to Monge–Ampère measures

Theorem 3.14. *Fix $V \in PSH(\Omega)$ and $u_1, \ldots, u_n \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$. For any subdomain $D \Subset \Omega$ and any compact subset $K \subset D$, there exists a constant $C = C(K, D) > 0$ such that*

$$\int_K |V| dd^c u_1 \wedge \ldots \wedge dd^c u_n \leq C \|u_1\|_D \ldots \|u_n\|_D \|V\|_{L^1(D)}.$$

In particular, the Monge–Ampère measure $dd^c u_1 \wedge \ldots \wedge dd^c u_n$ does not charge the polar set $P = \{V = -\infty\}$.

Proof. Since K is compact, we can cover it by finitely many small balls. Using the localization principle we can thus assume that there are Euclidean balls such that $K \Subset \mathbb{B}_1 \Subset \mathbb{B} \Subset \Omega$ and that u_k coincides in a neighborhood of $\mathbb{B} \setminus \mathbb{B}_1$ with $A_k \rho$. Here ρ denotes a defining function for the ball $\mathbb{B}$ and $A_k \leq C\|u_k\|_{L^\infty(\mathbb{B})}$ with a uniform constant C.

We first assume that $V < 0$ on $\mathbb{B}$ and set $V_j := \sup\{V, j\rho\}$, for $j \in \mathbb{N}$. The V_j's are bounded plurisubharmonic functions on $\mathbb{B}$ with boundary values 0 which converge to V. It follows from Proposition 3.7 that

$$\begin{aligned} \int_K (-V_j) \bigwedge_{1\leq k\leq n} dd^c u_k &\leq \int_{\mathbb{B}} (-V_j) \bigwedge_{1\leq k\leq n} dd^c u_k \\ &= \int_{\mathbb{B}} (-u_1) dd^c V_j \wedge dd^c u_2 \ldots \wedge dd^c u_n. \end{aligned}$$

The Chern–Levine–Nirenberg inequalities (3.2.1) yield

$$\int_K (-u_1) dd^c V_j \wedge dd^c u_2 \ldots \wedge dd^c u_n \leq C_n \prod_{1\leq k\leq n} \|u_k\|_{\mathbb{B}} \int_{\mathbb{B}} |V_j| d\lambda,$$

where $C_n > 0$ is a uniform constant.

On the other hand, formula (3.1.1) yields, setting $A = A_1 \cdots A_n$,

$$\begin{aligned}\int_{\mathbb{B}\setminus\mathbb{B}_1}(-u_1)dd^cV_j \wedge dd^cu_2 \ldots \wedge dd^cu_n &= A\int_{\mathbb{B}\setminus\mathbb{B}_1}(-\rho)dd^cV_j \wedge \beta_{n-1}\\ &= A\int_{\mathbb{B}}(-V_j)dd^c\rho \wedge \beta_{n-1}\\ &\leq A\int_{\mathbb{B}}(-V_j)\beta_n.\end{aligned}$$

Altogether this yields

$$\int_K(-V_j)dd^cu_1 \wedge \ldots \wedge dd^cu_n \leq (C+A)\int_{\mathbb{B}}(-V_j)\beta_n.$$

Since (V_j) decreases to $V \in L^1(\mathbb{B})$, the monotone convergence theorem implies

$$\int_K(-V)dd^cu_1 \wedge \ldots \wedge dd^cu_n \leq (C+A)\int_{\mathbb{B}}(-V)\beta_n < +\infty.$$

We can finally replace V by $V' := V - \sup_{\mathbb{B}} V$, note that $dd^cV = dd^cV'$, and use the submean value inequality to obtain

$$\|V'\|_{L^1(\mathbb{B})} \leq C_0\|V\|_{L^1(\mathbb{B}_2)},$$

where $\mathbb{B} \Subset \mathbb{B}_2 \Subset \Omega$. This proves the required estimate. □

3.2.4 Compact singularities The estimates (3.2.1) and (3.2.2) only require a control on the functions near the boundary of Ω. We can thus improve the last estimate as follows:

Proposition 3.15. *Let T be a closed positive current of bidimension (p,p) and $\varphi \in PSH(\Omega)$. Assume there exist a compact set E and a strictly pseudoconvex domain D such that $E \subset D \subset \Omega$, and a constant $M > 0$ such that $-M \leq \varphi \leq 0$ on $\Omega \setminus E$. Then there exists a constant $C > 0$ which does not depend on φ and M, such that*

$$\int_D |\varphi|T \wedge \beta_p \leq CM\int_{D\setminus E} T \wedge \beta_p.$$

In particular

$$dd^c\varphi \wedge T := dd^c(\varphi T),$$

is a well-defined closed positive current in Ω.

Proof. Let ρ be a defining function for D which is stricly plurisubharmonic in a neighborhood of $\bar{D}$. We can assume that $\beta \leq dd^c\rho$ on D. Set $\varphi_j := \max\{\varphi, -j\}$

and observe that φ_j is plurisubharmonic, bounded on D, and $\varphi_j = \varphi$ on $D \setminus E$ for $j > M$. It follows from Proposition 3.7 that

$$\begin{aligned}\int_D (-\varphi_j) T \wedge \beta_p &\leq \int_D (-\varphi_j) dd^c\rho \wedge T \wedge \beta_{p-1} \\ &= \int_D (-\rho) dd^c\varphi_j \wedge T \wedge \beta_{p-1} \\ &\leq \sup_D(-\rho) \int_D dd^c\varphi_j \wedge T \wedge \beta_{p-1}.\end{aligned}$$

Since $\varphi_j = \varphi$ on $\Omega \setminus D$ for $j \geq M$, the Chern–Levine–Nirenberg inequalities show that the last integral is bounded from above by $C_1 \cdot M$. Letting $j \to \infty$ yields the required estimate. □

Corollary 3.16. *The Monge–Ampère measure*

$$dd^c\varphi_1 \wedge \ldots \wedge dd^c\varphi_n$$

is well-defined for all plurisubharmonic functions which are locally bounded near the boundary of Ω.

Such plurisubharmonic functions are called plurisubharmonic functions with *compact singularities*.

Example 3.17. *The function*

$$\ell(z) := \log|z|, \quad z \in \mathbb{C}^n$$

is plurisubharmonic in $\mathbb{C}^n$ *and has an isolated singularity at the origin. Its Monge–Ampère measure* $(dd^c\ell)^n$ *is therefore well-defined. The reader will check in Exercise 3.15 that*

$$(dd^c\ell)^n = \delta_0,$$

is the Dirac mass at the origin, hence ℓ *is a fundamental solution of the Monge–Ampère operator. This latter notion is however of limited interest, as this operator is non-linear when* $n \geq 2$ *and there are very many fundamental solutions!*

3.3 Continuity of the complex Monge–Ampère operator

When the (complex) dimension is $n = 1$ thecomplex Monge–Ampère operator is nothing but the Laplace operator dd^c. It is then a linear operator which is well-defined on all subharmonic functions and is continuous for the weak (L^1_{loc}) convergence.

The situation is much more delicate in dimension $n \geq 2$. The complex Monge–Ampère operator is then non-linear and cannot be defined for all plurisubharmonic functions. It is moreover discontinous for the the L^1_{loc}-convergence. We will nevertheless show that it is continuous for the monotone convergence.

3.3.1 Continuity along decreasing sequences We first show that the complex Monge–Ampère operators is continuous along decreasing sequences of plurisubharmonic functions.

Theorem 3.18. *Let T be a closed positive current of bidegree (p,p) and let $(u_0^j)_{j\in\mathbb{N}},\dots,(u_q^j)_{j\in\mathbb{N}}$ be decreasing sequences of plurisubharmonic functions which converge to $u_0,\dots,u_q \in PSH(\Omega)\cap L^\infty_{\rm loc}(\Omega)$, $p+q\le n$. Then*

$$u_0^j\, dd^c u_1^j \wedge \ldots \wedge dd^c u_q^j \wedge T \longrightarrow u_0\, dd^c u_1 \wedge \ldots \wedge dd^c u_q \wedge T$$

in the weak sense of currents.

Proof. The proof proceeds by induction on q. For $q=0$ the theorem is a consequence of the monotone convergence theorem. Fix $1\le q\le n-p$ and assume that the theorem is true for $q-1$, so that

$$S^j := \bigwedge_{1\le k\le q} dd^c u_k^j \wedge T \longrightarrow S := \bigwedge_{1\le k\le q} dd^c u_k \wedge T.$$

It follows from the Chern–Levine–Nirenberg inequalities (3.2.1) that the sequence $(u_0^j S^j)$ is relatively compact for the weak topology of currents. Up to extracting and relabelling, it suffices to show that if the sequence $(u_0^j S^j)$ converges weakly to a current Θ, then $\Theta = u_0 S$.

By upper semi-continuity we already know that for all elementary positive $(n-p-q,n-p-q)$-forms Γ, $\Theta\wedge\Gamma \le u_0 S\wedge\Gamma$, hence $u_0S-\Theta$ is a positive current on Ω. It thus remains to prove that

$$\int_\Omega u_0 S\wedge\beta^{n-p-q} \le \int_\Omega \Theta\wedge\beta^{n-p-q}.$$

The problem being local, it is enough to prove that the total mass of the positive current $u_0S-\Theta$ on each ball $\mathbb{B}=\mathbb{B}(a,R)\Subset\Omega$ is zero. By the localization principle, we can assume that the functions u_0^j coincide with the function $\rho(z)=A(|z-a|^2-R^2)$ in a neighborhood of $\partial\mathbb{B}$, where $A>1$ is a large constant, and that $-1\le u_0^j<0$ in an open neighborhood $\Omega''\Subset\Omega$. Integrating by parts (using (3.1.1)), we infer

$$\begin{aligned}\int_{\mathbb{B}} u_0 \bigwedge_{1\le i\le q} dd^c u_i\wedge T\wedge\beta^{n-p-q} &\le \int_{\mathbb{B}} u_0^j \bigwedge_{1\le i\le q} dd^c u_i\wedge T\wedge\beta^{n-p-q}\\ &= \int_{\mathbb{B}} u_1\wedge dd^c u_0^j \bigwedge_{2\le i\le q} dd^c u_i\wedge T\wedge\beta^{n-p-q}\\ &\le \int_{\mathbb{B}} u_0^j \bigwedge_{1\le i\le q} dd^c u_i^j\wedge T\wedge\beta^{n-p-q}.\end{aligned}$$

We use here the symmetry of the wedge products (see Corollary 3.12).

Since the positive measures $(-u_0^j)\bigwedge_{1\le i\le q} dd^c u_i^j \wedge T\wedge\beta^{n-p-q}$ converge weakly to $-\Theta\wedge\beta^{n-p-q}$, it follows from the lower semi-continuity that

$$\liminf_{j\to+\infty}\int_{\mathbb{B}}(-u_0^j)\bigwedge_{1\le i\le q} dd^c u_i^j\wedge T\wedge\beta^{n-p-q}\ge\int_{\mathbb{B}}-\Theta\wedge\beta^{n-p-q},$$

which proves the theorem. □

The reader can use decreasing sequences of smooth approximants to compute the following complex Monge–Ampère measure:

Example 3.19. *Consider*

$$\psi : z\in\mathbb{C}^n\longmapsto\max\{\log^+|z_j|\ |1\le j\le n\}\in\mathbb{R}.$$

The function ψ is Lipschitz continuous and plurisubharmonic in $\mathbb{C}^n$. Its Monge–Ampère measure $(dd^c\psi)^n$ coincides with the normalized Lebesgue measure τ_n on the torus

$$\mathbb{T}^n=\{z\in\mathbb{C}^n|\ |z_1|=\cdots=|z_n|=1\}.$$

(*See Exercise* 3.9).

Corollary 3.20. *If u, v are plurisubharmonic and locally bounded, then*

$$(dd^c[u+v])^n=\sum_{j=0}^{n}\binom{n}{j}(dd^c u)^j\wedge(dd^c v)^{n-j}.$$

Proof. The formula is clear if u, v are smooth. The general case follows by approximation by smooth decreasing sequences. □

The following estimates are due to Blocki [Blo93]:

Proposition 3.21. *Fix $u, v, w\in PSH^-(\Omega)\cap L^\infty(\Omega)$ such that $\lim_{z\to\partial\Omega}(w(z)-v(z))=0$. Then*

$$\int_\Omega(w-v)_+^{n+1}(dd^c u)^n\le(n+1)!M^{n+1}\int_\Omega(w-v)_+\,(dd^c v)^n,$$

where $M=\sup_\Omega u-\inf_\Omega u$ and $(w-v)_+:=\sup\{w-v,0\}$.

Proof. We can assume without loss of generality that $\sup_\Omega u=0$. Observe that for $\varepsilon>0$ the function $w_\varepsilon:=\sup\{v,w-\varepsilon\}$ is a bounded plurisubharmonic function on Ω such that $w_\varepsilon\nearrow\sup\{v,w\}$ as $\varepsilon\searrow 0$ and $w_\varepsilon=v$ near $\partial\Omega$. By the monotone convergence theorem we may thus assume that $w=v$ near $\partial\Omega$.

Set $h:=(w-v)_+$ on Ω and fix a compact set $K\subset\Omega$ such that $h=0$ in $\Omega\setminus K$. Consider smooth approximants $h_\varepsilon:=h\star\rho_\varepsilon$ of h. These functions are smooth in a

neighborhood Ω'' of K with compact support in the ε-neighborhood K_ε of K. By definition, $dd^c u \wedge T := dd^c(uT)$, hence

$$\int_\Omega h_\varepsilon^p dd^c u \wedge T = \int_{\Omega''} u dd^c h_\varepsilon^p \wedge T.$$

On the other hand, if $p \geq 1$,

$$dd^c h_\varepsilon^p = p h_\varepsilon^{p-1} dd^c h_\varepsilon + p(p-1) h_\varepsilon^{p-2} dh_\varepsilon \wedge d^c h_\varepsilon \geq p h_\varepsilon^{p-1} dd^c h_\varepsilon,$$

thus $u\, dd^c h_\varepsilon^p \leq p u h_\varepsilon^{p-1} dd^c h_\varepsilon$ and

$$\begin{aligned} \int_{\Omega''} h_\varepsilon^p dd^c u \wedge T &\leq \int_{\Omega''} p u h_\varepsilon^{p-1} dd^c h_\varepsilon \wedge T \\ &\leq pM \int_{\Omega''} h_\varepsilon^{p-1} (-dd^c h_\varepsilon) \wedge T \\ &\leq pM \int_{\Omega''} h_\varepsilon^{p-1} dd^c v_\varepsilon \wedge T. \end{aligned}$$

The last inequality follows from the observation that

$$h = \max\{w - v, 0\} = \max\{w, v\} - v,$$

hence $-dd^c h_\varepsilon \leq dd^c v_\varepsilon$.

We can use this argument $n+1$ times and obtain

$$\int_{\Omega''} h_\varepsilon^{n+1} (dd^c u)^n \leq (n+1)! M^{n+1} \int_{\Omega''} h_\varepsilon (dd^c v_\varepsilon)^n.$$

Since $h = \sup\{w, v\} - v$, $h_\varepsilon = (\sup\{w, v\})_\varepsilon - v_\varepsilon$ is a difference of decreasing sequences of bounded plurisubharmonic functions, we can use the continuity of the Monge–Ampère operator along decreasing sequences and apply Lebesgue's convergence theorem to obtain

$$\int_{\Omega''} h^{n+1} (dd^c u)^n \leq (n+1)! M^{n+1} \int_{\Omega''} h (dd^c v)^n,$$

as claimed. □

3.3.2 Continuity along increasing sequences In this section we show that complex Monge–Ampère operators are continuous along increasing sequences.

To this end we need the following technical result which relies on the quasicontinuity of plurisubharmonic functions:

Lemma 3.22. *Let $\mathcal{P}$ be a family of plurisubharmonic functions which are locally uniformly bounded. Let $\mathcal{T}$ denote the set of currents of the form $T := \bigwedge_{1 \leq i \leq p} dd^c u_i$, where $u_1, \ldots, u_n \in \mathcal{P}$.*

If $(T_j)_{j \in \mathbb{N}}$ is a sequence of currents in $\mathcal{T}$ converging weakly to a current $T \in \mathcal{T}$, then for all locally bounded plurisubharmonic functions f, the currents $f T_j$ weakly converge to $f T$.

The proof of the quasi-continuity will be given in the next chapter, so we postpone the proof of this Lemma as well.

Theorem 3.23. *Let $(u_0^j), \ldots, (u_q^j)$ be sequences of locally bounded plurisubharmonic functions which increase almost everywhere towards $u_0, \ldots, u_q \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$. Then*

$$u_0^j dd^c u_1^j \wedge \ldots \wedge dd^c u_q^j \longrightarrow u_0 dd^c u_1 \wedge \ldots \wedge dd^c u_q$$

in the weak sense of currents.

Proof. We proceed by induction on q. The case $q = 0$ follows from the monotone convergence theorem.

Suppose that the theorem is true for $q-1$. By the continuity of dd^c, we infer that the currents $S_j := dd^c u_1^j \wedge \ldots \wedge dd^c u_q^j$ converge to the current $S := dd^c u_1 \wedge \ldots \wedge dd^c u_q$.

The Chern–Levine–Nirenberg inequalities ensure that the currents $(u_0^j S_j)$ form a relatively compact sequence. We need to show it has a unique limit point. Extracting and relabelling, we need to show that if $u_0^j S_j \to \Theta$ weakly, then $\Theta = u_0 S$ on Ω.

The problem is local, so it suffices to prove the convergence in a ball $\mathbb{B} = \mathbb{B}(a; r) \Subset \Omega$. We can modify the functions in a neighborhood of $\partial\mathbb{B}$ so that they all coincide with $\rho(z) = A(|z-a|^2 - R^2)$ near $\partial\mathbb{B}$, where $A > 1$ is a uniform constant.

Note that $u_0^j S_j \leq u_0 S_j$ since (u_0^j) is increasing. It follows therefore from the upper semi-continuity that $\Theta \leq u_0 S$. We now show that $\Theta = u_0 S$ by proving that

$$\int_{\overline{\mathbb{B}}} \Theta \wedge \beta_{n-q} \geq \int_{\overline{\mathbb{B}}} u_0 S \wedge \beta_{n-q}.$$

Indeed, for $j \geq k \geq 0$, the integration by parts formula yields

$$\int_{\mathbb{B}} u_0^j S_j \wedge \beta_{n-q} \geq \int_{\mathbb{B}} u_0^k \bigwedge_{1 \leq i \leq q} dd^c u_i^j \wedge \beta_{n-q}.$$

The induction hypothesis and Lemma 3.22 yield

$$\begin{aligned} \liminf_j \int_{\mathbb{B}} u_0^j S_j \wedge \beta_{n-q} &\geq \int_{\mathbb{B}} u_0^k \bigwedge_{1 \leq i \leq q} dd^c u_i \wedge \beta_{n-q} \\ &= \int_{\mathbb{B}} u_1 dd^c u_0^k \bigwedge_{2 \leq i \leq q} dd^c u_i \wedge \beta_{n-q}. \end{aligned}$$

Applying Lemma 3.22 and Stokes' theorem again, we get

$$\begin{aligned}\lim_k \int_{\mathbb{B}} u_1 dd^c u_0^k \bigwedge_{2\le i\le q} dd^c u_i \wedge \beta_{n-q} &= \int_{\mathbb{B}} u_1 dd^c u_0 \bigwedge_{2\le i\le q} dd^c u_i \wedge \beta_{n-q} \\ &= \int_{\mathbb{B}} u_0 \bigwedge_{1\le i\le q} dd^c u_i \wedge \beta_{n-q},\end{aligned}$$

hence

$$\liminf_{j\to+\infty} \int_{\mathbb{B}} u_0^j S_j \wedge \beta_{n-q} \ge \int_{\mathbb{B}} u_0 \bigwedge_{1\le i\le q} dd^c u_i \wedge \beta_{n-q},$$

and the proof is complete. □

The following corollary will allow us to show in the next chapter that "negligible sets are pluripolar", answering a celebrated question of P. Lelong.

Corollary 3.24. *Let (V_j) be plurisubharmonic functions increasing almost everywhere to $V \in PSH(\Omega)$. Then the exceptional set*

$$N := \{\sup_j V_j < V\}$$

has measure 0 with respect to all Monge–Ampère measures of the type $dd^c u_1 \wedge \ldots dd^c u_n$, where $u_1, \ldots, u_n \in PSH(\Omega) \cap L^\infty(\Omega)$.

We have already seen (see Theorem 1.46) that

$$\sup_j V_j(x) = \limsup V_j(x) = V(x)$$

for almost every x with respect to Lebesgue measure. This corollary gives much more precise information.

Proof. Set $V_s := \sup\{V, s\}$ and $V_{j,s} = \sup\{V_j, s\}$, and observe that

$$N \subset \bigcup_{s\in\mathbb{Q}} N_s, \text{ where } N_s := \{\sup_j V_{j,s} < V_s\}.$$

By the subadditivity property of Borel measures, we can therefore assume that all the functions (V_j) are locally bounded.

Set $W := \sup_j V_j$. Then W is a locally bounded Borel function such that $W \le V$, with equality almost everywhere. Again by subadditivity, it is enough to prove that

$$\int_{K\cap N} dd^c u_1 \wedge \ldots dd^c u_n = 0,$$

where $K \subset \Omega$ is a compact subset. Pick a non-negative cutoff function χ such that $\chi \equiv 1$ near K. The previous convergence theorem yields

$$\begin{aligned}\int_\Omega \chi W dd^c u_1 \wedge \ldots dd^c u_n &= \lim_j \int_\Omega \chi V_j dd^c u_1 \wedge \ldots dd^c u_n \\ &= \int_\Omega \chi V dd^c u_1 \wedge \ldots dd^c u_n,\end{aligned}$$

which proves the required result since $W \leq V$. □

3.3.3 Discontinuity of the Monge–Ampère operator We have proved that the complex Monge–Ampère operator is well-defined on the set $PSH(\Omega) \cap L^\infty_{\rm loc}$ and is continuous under monotone convergence. This operator is however not continuous under the weaker $L^1_{\rm loc}$ convergence, as was first emphasized by Cegrell [Ceg84]. Here is a simple example:

Example 3.25. *The functions*

$$u_j(z_1, z_2) := \frac{1}{2j} \log\left[|z_1^j + z_2^j|^2 + 1\right]$$

are smooth and plurisubharmonic in $\mathbb{C}^2$. *They form a locally bounded sequence which converges in* $L^1_{\rm loc}(\mathbb{C}^2)$ *to*

$$u(z_1, z_2) = \log \max[1, |z_1|, |z_2|].$$

Observe that $(dd^c u_j)^2 = 0$, *while* $(dd^c u)^2$ *is the Lebesgue measure on the real torus* $\{|z_1| = |z_2| = 1\}$.

We leave the details as an Exercise 3.10. Another example is given in Exercise 3.7. This discontinuity is actually rather common as was observed by Lelong [Lel83], who showed the following:

Proposition 3.26. *Every locally bounded plurisubharmonic function can be approximated in* $L^1_{\rm loc}$ *by locally bounded plurisubharmonic functions with vanishing Monge–Ampère measures.*

Such plurisubharmonic functions are natural generalizations of harmonic functions, they are called *maximal plurisubharmonic* functions.

3.4 Maximum principles

The comparison principle is one the most effective tools in pluripotential theory. It is a non-linear version of the classical maximum principle.

3.4.1 The comparison principle We establish in this section several types of maximum principles, starting with the following *local maximum principle*:

Theorem 3.27. *Set* $T = dd^c w_1 \wedge \ldots \wedge dd^c w_{n-p}$, *where* $0 \leq p \leq n-1$ *and* $w_i \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$. *Then for all* $u, v \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$,

$$\mathbf{1}_{\{u>v\}}(dd^c \max\{u,v\})^p \wedge T = \mathbf{1}_{\{u>v\}}(dd^c u)^p \wedge T,$$

in the sense of Borel measures on Ω.

Proof. Set $D := \{u > v\}$. If u is continuous, then D is an open subset of Ω and $\max\{u,v\} = u$ in D, hence

$$(dd^c \max\{u,v\})^p \wedge T = (dd^c u)^p \wedge T,$$

so our claim is easy in this case.

We now treat the general case. Let (u_j) be a sequence of continuous plurisubharmonic functions decreasing to u. Since the problem is local, we can assume that Ω is a ball and all functions are bounded and plurisubharmonic on a fixed neighborhood of $\overline{\Omega}$. We know that

$$\mathbf{1}_{\{u_j>v\}}(dd^c \max\{u_j,v\})^p \wedge T = \mathbf{1}_{\{u_j>v\}}(dd^c u_j)^p \wedge T.$$

Set $f_j := (u_j - v)^+$ and $f = (u-v)^+$. The previous identity yields

$$f_j(dd^c \max\{u_j,v\})^p \wedge T = f_j(dd^c u_j)^p \wedge T,$$

in the weak sense of measures on Ω.

Observe that $f_j = \max\{u_j, v\} - v$, $f = \max\{u,v\} - v$, and the sequence $(\max\{u_j, v\})$ decreases to $\max\{u,v\}$. It follows from Theorem 3.18 that

$$f(dd^c \max\{u,v\})^p \wedge T = f(dd^c u)^p \wedge T$$

in the sense of Borel measures on Ω.

Fix $\varepsilon > 0$. Since $1/(f+\varepsilon)$ is a bounded Borel function, we get

$$\frac{f}{f+\varepsilon}(dd^c \max\{u,v\})^p \wedge T = \frac{f}{f+\varepsilon}(dd^c u)^p \wedge T.$$

Now let $\varepsilon \searrow 0$ and observe that $f/(f+\varepsilon) \nearrow \mathbf{1}_{\{u>v\}}$ to conclude. □

Corollary 3.28. *With the same hypotheses as in the theorem,*

$$(dd^c \max\{u,v\})^p \wedge T \geq \mathbf{1}_{\{u\geq v\}}(dd^c u)^p \wedge + \mathbf{1}_{\{u<v\}}(dd^c v)^p \wedge T$$

in the sense of Borel measures on Ω.

The following is often called the comparison principle:

Theorem 3.29. *Assume* $u, v \in PSH(\Omega) \cap L^\infty(\Omega)$ *are such that* $\liminf_{z\to\partial\Omega}(u(z) - v(z)) \geq 0$. *Then*

$$\int_{\{u<v\}} (dd^c v)^n \leq \int_{\{u<v\}} (dd^c u)^n.$$

Proof. By assumption, we can find a compact subset $K \subset \Omega$ and an arbitrarily small $\varepsilon > 0$ such that $\sup\{u, v-\varepsilon\} = u$ in $\Omega \setminus K$. Fix a domain Ω' such that $K \subset \Omega' \Subset \Omega$. Then

$$\int_{\Omega'} (dd^c \sup\{u, v-\varepsilon\})^n = \int_{\Omega'} (dd^c u)^n. \tag{3.4.1}$$

Indeed, set $w := \sup\{u, v-\varepsilon\}$ and observe that $(dd^c w)^n - (dd^c u)^n = dd^c S$, where $S := w(dd^c w)^{n-1} - u(dd^c u)^{n-1}$ is a current of bidimension $(1,1)$. Since $w = u$ in $\Omega \setminus K$, we get $S = 0$ there, i.e., the support of the current S is contained in K. Pick a smooth test function χ on Ω' such that $\chi \equiv 1$ in a neighborhood of K. Then

$$\int_{\Omega'} dd^c S = \int_{\Omega'} \chi dd^c S = \int_{\Omega'} S \wedge dd^c \chi = 0,$$

since $dd^c \chi = 0$ on the support of S. This proves (3.4.1).

We now apply Theorem 3.27 and (3.4.1) in Ω' to get

$$\begin{aligned}
\int_{\{u<v-\varepsilon\}} (dd^c v)^n &= \int_{\{u<v-\varepsilon\}} (dd^c \sup\{u, v-\varepsilon\})^n \\
&= \int_{\Omega'} (dd^c \sup\{u, v-\varepsilon\})^n - \int_{\{u\geq v-\varepsilon\}} dd^c \sup\{u, v-\varepsilon\})^n \\
&\leq \int_{\Omega'} (dd^c u)^n - \int_{\{u>v-\varepsilon\}} dd^c \sup\{u, v-\varepsilon\})^n \\
&= \int_{\Omega'} (dd^c u)^n - \int_{\{u>v-\varepsilon\}} (dd^c u)^n \\
&= \int_{\{u\leq v-\varepsilon\}} (dd^c u)^n \leq \int_{\{u<v\}} (dd^c u)^n.
\end{aligned}$$

The conclusion follows by letting $\varepsilon \searrow 0$. □

We now deduce the following *global maximum principle*:

Corollary 3.30. *Assume $u, v \in PSH(\Omega) \cap L^\infty(\Omega)$ are such that $\liminf_{z\to\partial\Omega}(u(z) - v(z)) \geq 0$. If $(dd^c u)^n \leq (dd^c v)^n$, then $v \leq u$ in Ω.*

Proof. For $\varepsilon > 0$ we set $v_\varepsilon := v + \varepsilon\rho$, where $\rho(z) := |z|^2 - R^2$ is choosen so that $\rho < 0$ on Ω. Then $\{u < v_\varepsilon\} \subset \{u < v\} \Subset \Omega$. The comparison principle yields

$$\int_{\{u<v_\varepsilon\}} (dd^c v_\varepsilon)^n \leq \int_{\{u<v_\varepsilon\}} (dd^c u)^n.$$

It follows from Corollary 3.20 that

$$(dd^c v_\varepsilon)^n \geq (dd^c v)^n + \varepsilon^n (dd^c \rho)^n \geq (dd^c u)^n + \varepsilon^n (dd^c \rho)^n,$$

hence $\int_{\{u<v_\varepsilon\}} (dd^c \rho)^n = 0$. We infer that the set $\{u < v_\varepsilon\}$ has Lebesgue measure 0. Since $\{u < v\} = \bigcup_{j\geq 1} \{u < v_{1/j}\}$, it follows that the set $\{u < v\}$ has Lebesgue measure 0 as well, hence $v \leq u$ on Ω by the sub-mean value inequalities. □

We now prove the domination principle.

Corollary 3.31. *Fix $u, v \in PSH(\Omega) \cap L^\infty(\Omega)$ such that $v \le u$ on $\partial\Omega$. Assume that $v \le u$ a.e. in Ω with respect to the measure $(dd^c u)^n$. Then $v \le u$ in Ω.*

Proof. For $\varepsilon > 0$ we set $v_\varepsilon := v + \varepsilon\rho$, where $\rho(z) := |z|^2 - R^2$ is choosen so that $\rho < 0$ on Ω. Then $\{u < v_\varepsilon\} \subset \{u < v\} \Subset \Omega$. The comparison principle yields

$$\int_{\{u<v_\varepsilon\}} (dd^c v_\varepsilon)^n \le \int_{\{u<v_\varepsilon\}} (dd^c u)^n \le \int_{\{u<v\}} (dd^c u)^n = 0.$$

Since $(dd^c v_\varepsilon)^n \ge \varepsilon^n (dd^c \rho)^n$, it follows that the set $\{u < v_\varepsilon\}$ has volume zero in Ω hence it is empty by the sub-mean value inequality. Therefore $\{u < v\}$ is empty, i.e., $v \le u$ in Ω. □

3.4.2 The Lelong class

We defined above the complex Monge–Ampère measure of plurisubharmonic functions that are bounded or with compact singularities.

One cannot expect to define the Monge–Ampère measure of *any* unbounded plurisubharmonic function, as the following example due to Kiselman [Kis83] shows:

Example 3.32. *Set*

$$\varphi(z) := (-\log|z_1|)^{1/n} \left(|z_2|^2 + \cdots + |z_n|^2 - 1\right).$$

We let the reader check in Exercise 3.8 that φ is a smooth plurisubharmonic function in $\mathbb{B}^n \setminus \{z_1 = 0\}$ such that

$$(dd^c\varphi)^n = c_n \frac{1 - \frac{1}{n} - \sum_{\ell=2}^n |z_\ell|^2}{|z_1|^2 |\log|z_1||} dV_{\mathrm{Leb}}$$

and that this measure has infinite mass in $\mathbb{B}^n \setminus \{z_1 = 0\}$.

We now introduce an important class of plurisubharmonic functions in $\mathbb{C}^n$ for which such a phenomenon cannot occur.

Definition 3.33. *The Lelong class $\mathcal{L}(\mathbb{C}^n)$ is the class of plurisubharmonic functions u in $\mathbb{C}^n$ with logarithmic growth, i.e., for which there exists $C_u \in \mathbb{R}$ such that for all $z \in \mathbb{C}^n$,*

$$u(z) \le \log^+|z| + C_u.$$

The reader will check in Exercise 3.13 that a non-constant plurisubharmonic function in $\mathbb{C}^n$ has at least logarithmic growth. This class of functions will play an important role in this book, as it leads to the model class of quasi-plurisubharmonic functions on the complex projective space $\mathbb{CP}^n$ (see Chapter 8).

We also consider

$$\mathcal{L}^+(\mathbb{C}^n) := \{u \in \mathcal{L}(\mathbb{C}^n) \mid \exists C'_u \text{ s.t. } -C'_u + \log^+|z| \le u(z),\ \forall z \in \mathbb{C}^n\}.$$

We let the reader check in Exercise 3.14 that locally bounded functions from the Lelong class have finite total Monge–Ampère mass:

Proposition 3.34. *If u belongs to $\mathcal{L}(\mathbb{C}^n) \cap L^\infty_{\rm loc}(\mathbb{C}^n)$, then*

$$\int_{\mathbb{C}^n} (dd^c u)^n \leq 1.$$

Moreover, if u belongs to $\mathcal{L}^+(\mathbb{C}^n)$, then

$$\int_{\mathbb{C}^n} (dd^c u)^n = 1.$$

The proof of these facts relies on the following result of independent interest:

Lemma 3.35. *Let u, v be locally bounded plurisubharmonic functions in $\mathbb{C}^n$ such that $u(z) \to +\infty$ as $z \to \infty$. Assume that $v(z) \leq u(z) + o(u(z))$ as $z \to +\infty$. Then*

$$\int_{\mathbb{C}^n} (dd^c v)^n \leq \int_{\mathbb{C}^n} (dd^c u)^n.$$

Proof. Fix $\varepsilon > 0$. Our assumption ensures that for $R > 1$ large enough, $v(z) \leq (1+\varepsilon)u(z)$ if $|z| \geq R$. The comparison principle yields

$$\begin{aligned} \int_{B_R \cap \{(1+\varepsilon)u < v\}} (dd^c v)^n &\leq (1+\varepsilon)^n \int_{B_R \cap \{(1+\varepsilon)u < v\}} (dd^c u)^n \\ &\leq (1+\varepsilon)^n \int_{\mathbb{C}^n} (dd^c u)^n. \end{aligned}$$

Letting $R \to +\infty$ and $\varepsilon \to 0$, we obtain the required inequality. □

For classes of plurisubharmonic functions with prescribed Monge–Ampère mass, one might hope that it is easier to define the complex Monge–Ampère measure. This is however a delicate problem. The local domain of definition of the complex Monge–Ampère operator has been characterized by Blocki and Cegrell in [Ceg04, Blo04, Blo06]. We shall say a few more words about this problem in Chapter 10.

3.5 Exercises

Exercise 3.1. *Let $(f_j)_{j\in\mathbb{N}}$ be a decreasing sequence of upper semi-continuous functions in a domain Ω converging to f. Let $(\mu_j)_{j\in\mathbb{N}}$ be a sequence of positive Borel measures in Ω which converges weakly to a Borel measure μ. Show that any limit point ν of the sequence of measures $\nu_j := f_j \cdot \mu_j$ satisfies the inequality $\nu \leq f \cdot \mu$ in the weak sense of Radon measures on Ω, i.e.,*

$$\limsup_j f_j \mu_j \leq f\mu.$$

Exercise 3.2. *Let $(\mu_j)_{j\in\mathbb{N}}$ be a sequence of positive Borel measures on Ω which converges weakly to a positive Borel measure μ on Ω. Show that for any compact set $K \subset \Omega$ and any open subset $D \subset \Omega$,*

$$\limsup_j \mu_j(K) \le \mu(K) \quad \textit{and} \quad \liminf_j \mu_j(D) \ge \mu(D).$$

Exercise 3.3. *Fix $0 < \alpha < 1$ and consider*

$$\mathbb{D} \ni z \longmapsto u(z) := -(1-|z|^2)^\alpha \in \mathbb{R}.$$

Show that u is a smooth subharmonic function in the unit disc $\mathbb{D}$, which is Hölder continuous up to the boundary. Check that

$$\frac{\partial^2 u}{\partial z \partial \bar z}(z) = \alpha(1-|z|^2)^{\alpha-1} + \alpha(1-\alpha)|z|^2(1-|z|^2)^{\alpha-2},$$

and conclude that $\int_{\mathbb{D}} dd^c u = +\infty$. Is this in contradiction with Chern–Levine–Nirenberg inequalities?

Exercise 3.4. *Let u be a locally bounded plurisubharmonic function in a domain Ω and $\chi : I \to \mathbb{R}$ a smooth convex non-decreasing function with $u(\Omega) \subset I$.*

1) *Check that $\varphi := \chi \circ u$ is plurisubharmonic in Ω, with*

$$dd^c\varphi = \chi'(u)\, dd^c u + \chi''(u)\, du \wedge d^c u,$$

and

$$(dd^c\varphi)^n = \chi'(u)^n\, (dd^c u)^n + \chi''(u)\cdot \chi'(u)^{n-1}\, du \wedge d^c u \wedge (dd^c u)^{n-1}.$$

2) *Fix $0 < \alpha < 1$ and consider*

$$\mathbb{B} \ni z \longmapsto u(z) := -(1-|z|^2)^\alpha \in \mathbb{R}.$$

Prove that u is is plurisubharmonic in the unit ball $\mathbb{B}$, Hölder continuous up to the boundary and satisfies $\int_{\mathbb{B}} (dd^c u)^n = +\infty$.

Exercise 3.5. *Let Ω, Ω' be domains in $\mathbb{C}^n$, $F : \Omega' \to \Omega$ a holomorphic map and $u \in PSH(\Omega) \cap L^\infty(\Omega)$.*

1) *If $u \in C^2(\Omega)$, show that*

$$(dd^c u \circ F)^n(\zeta) = |J_F(\zeta)|^2 (dd^c u)^n(F(\zeta))$$

as differential forms in Ω', where J_F denotes the Jacobian of F.

2) *Check that if F is proper, then*

$$F_*(dd^c u \circ F)^n = (dd^c u)^n$$

in the sense of currents.

3) *Deduce that if F is a biholomorphism, then*

$$(F^{-1})_*(dd^c u)^n = (dd^c u \circ F)^n.$$

Exercise 3.6. *Consider, for* $1 \le j \le n$.

$$\mathbb{C}^n \ni z \longmapsto u_j(z) := (\operatorname{Im} z_j)^+ \in \mathbb{R}.$$

1) *Check that these are continuous plurisubharmonic functions s.t.*

$$dd^c u_1 \wedge \ldots \wedge dd^c u_n = (4\pi)^{-n} \iota_* \lambda_n,$$

in the sense of Borel measures on $\mathbb{C}^n$, *where* λ_n *is the Lebesgue measure on* $\mathbb{R}^n$ *and* $\iota : \mathbb{R}^n \to \mathbb{C}^n$ *is the embedding induced by* $\mathbb{R} \to \mathbb{C}$.

2) *Using Theorem* 3.14 *deduce that the restrictions of plurisubharmonic functions to* $\mathbb{R}^n$ *are locally integrable with respect to the* n*-dimensional Lebesgue measure* λ_n *on* $\Omega \cap \mathbb{R}^n$.

3) *Consider similarly* $v : \mathbb{C}^n \ni z \mapsto \sum_{j=1}^n (\operatorname{Im} z_j)^+ \in \mathbb{R}^n$. *Show that* v *is a Lipschitz continuous plurisubharmonic function in* $\mathbb{C}^n$ *such that*

$$(dd^c v)^n = (n!/2\pi)^n \iota_* \lambda_n.$$

Exercise 3.7. *For* $n \ge 2$ *we set*

$$u_j(z) = \log(|z_1 \cdots z_n|^2 + 1/j) \text{ and } v_j(z) = \sum_{\ell=1}^n \log(|z_\ell|^2 + 1/j).$$

Show that these sequences of bounded plurisubharmonic functions both decrease to $\varphi(z) = 2\log|z_1 \cdots z_n|$ *and that* $(dd^c u_j)^n = 0$, *while* $(dd^c v_j)^n$ *converge to a positive multiple of the Dirac mass at the origin.*

Exercise 3.8. *Set*

$$\varphi(z) := (-\log|z_1|)^{1/n} \left(|z_2|^2 + \cdots + |z_n|^2 - 1\right).$$

1) *Prove that* φ *is a smooth plurisubharmonic function in* $\mathbb{B}^n \setminus \{z_1 = 0\}$ *such that*

$$(dd^c \varphi)^n = c_n \frac{1 - \frac{1}{n} - \sum_{\ell=2}^n |z_\ell|^2}{|z_1|^2 |\log|z_1||} dV_{\text{Leb}}$$

and that this measure has infinite mass in $\mathbb{B}^n \setminus \{z_1 = 0\}$.

2) *Observe that* $\varphi(z) = u \circ L(z)$, *where*

$$L : (\mathbb{C}^*)^n \ni z \longmapsto (\log|z_1|, \ldots, \log|z_n|) \in \mathbb{R}^n$$

and u *is an appropriate convex function. Express* $(dd^c \varphi)^n$ *in terms of the real Monge–Ampère measure of* u *and give an alternative proof of the fact that* $(dd^c \varphi)^n$ *has infinite mass near* $\{z_1 = 0\}$.

Exercise 3.9. *Fix* $r := (r_1, r_2, \cdots, r_n) \in]0, +\infty[^n$ *and consider*

$$\psi_r : \mathbb{C}^n \ni z \longmapsto \max\{\log^+(|z_j|/r_j)|\ 1 \le j \le n\} \in \mathbb{R}.$$

1) *Prove that* ψ_r *is a Lipschitz continuous plurisubharmonic function and* $(dd^c\psi_r)^n$ *is supported on the torus*

$$\mathbb{T}^n(r) := \{z \in \mathbb{C}^n \,|\, |z_1| = r_1, \ldots, |z_n| = r_n\}.$$

2) *Observe that* $(dd^c\psi_r)^n$ *is* $(S^1)^n$*-invariant and conclude that* $(dd^c\psi_r)^n$ *is the normalized Lebesgue measure on* $\mathbb{T}^n(r)$.

Exercise 3.10. *Set*

$$\varphi(z) := \max_{1\le j\le n} \log^+ |z_j|, \text{ where } \log^+ x := \max(\log x, 0),$$

and $\varphi_j(z) = \frac{1}{2^j}\max(0, \log|z_1^j + \cdots + z_n^j|)$.

1) *Show that* $\varphi_j \to \varphi$ *in* $L^1_{\rm loc}(\mathbb{C}^n)$.

2) *Show that* $(dd^c\varphi_j)^n = 0$ *when* $n \ge 2$, *while* $(dd^c\varphi)^n$ *is the normalized Lebesgue measure on the torus* $(S^1)^n \subset \mathbb{C}^n$. *Conclude that the complex Monge–Ampère operator is not continuous for the* L^1*-topology.*

Exercise 3.11. *Let* φ *be a plurisubharmonic function in* $\mathbb{C}^n$ *whose gradient is in* $L^2_{\rm loc}$. *Let* φ_j *be a sequence of plurisubharmonic functions decreasing to* φ. *Show that* $\nabla\varphi_j \in L^2_{\rm loc}$ *and that* $\varphi_j \to \varphi$ *in the Sobolev sense* $W^{1,2}_{\rm loc}$.

Exercise 3.12. *Let* $u_1, \ldots, u_n$ *be continuous non-negative plurisubharmonic functions in* $\mathbb{C}^n$ *such that for all* i, u_i *is pluriharmonic in* $\{u_i > 0\}$. *Show that*

$$(dd^c\max(u_1, \ldots, u_n))^n = dd^cu_1 \wedge \ldots \wedge dd^cu_n.$$

Exercise 3.13. *Let* u *be a plurisubharmonic function in* $\mathbb{C}^n$. *Show that if*

$$\limsup_{|z|\to+\infty} \frac{u(z)}{\log|z|} = 0,$$

then u *is constant.*

Exercise 3.14. *Show that if* u *belongs to the Lelong class* $\mathcal{L}^+(\mathbb{C}^n)$, *i.e., if there exists a constant* $C_u \in \mathbb{R}$ *such that for all* $z \in \mathbb{C}^n$,

$$\log^+\|z\| - C_u \le u(z) \le \log^+\|z\| + C_u,$$

then $\int_{\mathbb{C}^n}(dd^cu)^n = 1$.

Exercise 3.15. *Set* $u(z) := \log|z|$, $v(z) = \max_{1\le j\le n}\log|z_j|$, *and*

$$w(z) = \max(\log|z_1|, \log|z_2 - z_1^2|, \dots, \log|z_n - z_{n-1}^2|).$$

Show that

$$(dd^c u)^n = (dd^c v)^n = (dd^c w)^n = \delta_0$$

is the Dirac mass at the origin of $\mathbb{C}^n$. *Conclude that the comparison principle cannot hold for these functions.*

Exercise 3.16. *Set* $\varphi(z) = \log^+|z| = \max(\log|z|, 0)$.

1) *Let* $\chi : \mathbb{R} \to \mathbb{R}$ *be a smooth convex function and set* $\psi(z) = \chi(\log|z|)$. *Show that* ψ *is a plurisubharmonic function with compact singularities. Prove that* $(dd^c\psi)^n$ *is absolutely continuous with respect to Lebesgue measure if* ψ *has zero Lelong number at the origin.*

2) *Approximate* $\max(x, 0)$ *by a smooth decreasing family of convex functions* χ_ε, *use* 1), *and let* ε *decrease to zero to conclude that* $(dd^c\varphi)^n$ *is the normalized Lebesgue measure on the unit sphere.*

3) *Use the invariance properties of* φ *and Exercise* 3.14 *to give an alternative proof of this result.*

Exercise 3.17. *Let* u *be a smooth plurisubharmonic function in some domain* $\Omega \subset \mathbb{C}^n$.

1) *Assume that for all* $p \in \Omega$ *there exists a holomorphic disc* $\mathbb{D} \subset \Omega$ *centered at* p *such that* $u|_\Delta$ *is harmonic. Prove that* $(dd^c u)^n \equiv 0$.

2) *Show, conversely, that if* $(dd^c u)^n \equiv 0$ *and* $(dd^c u)^{n-1} \not\equiv 0$, *then there exists a holomorphic foliation of* Ω *by Riemann surfaces* L_a *such that* $u|_{L_a}$ *is harmonic for all* a (*see* [BK77] *for some help*).

Exercise 3.18. *Let* μ *be a probability measure in the unit ball* $\mathbb{B}$ *of* $\mathbb{C}^n$ *and set*

$$V_\mu(z) := \int_{w\in\mathbb{B}} \log|z - w| d\mu(w).$$

Check that V_μ *is plurisubharmonic. Give conditions on* μ *to ensure that* V_μ *is locally bounded, and check that in this case* $(dd^c V_\mu)^n$ *is absolutely continuous with respect to Lebesgue measure* (*see* [Carl99] *for more information*).

4 The Monge–Ampère capacity

As we saw in Chapter 3, the polar locus (i.e., the $-\infty$ locus) of a plurisubharmonic function is the null set of several Borel measures. These small sets cannot be characterized by a single measure, one has to introduce *capacities*, a non-linear generalization of them.

Capacities play an important role in Complex Analysis as they allow to characterize small sets. There are various capacities, depending on the problem under study. We introduce here a generalized capacity in the sense of Choquet, whose null sets are pluripolar sets (i.e., sets that are locally contained in the polar locus of a plurisubharmonic function).

We follow the seminal work of Bedford and Taylor [BT82] (with subsequent simplifications by Cegrell [Ceg88] and Demailly [Dem91]).

4.1 Choquet capacity theory

A systematic study of general(ized) capacities was performed by Choquet in his famous work [Cho53]. We present here just a few aspects of Choquet theory that we use in the sequel.

4.1.1 Choquet capacities Let Ω be a Hausdorff locally compact topological space which we assume is σ-compact. We denote by 2^Ω the set of all subsets of Ω.

Definition 4.1. *A set function $c : 2^\Omega \to \bar{\mathbb{R}}^+ := [0, +\infty]$ is called a capacity on Ω if it satisfies the following properties:*

(i) $c(\emptyset) = 0$;

(ii) *c is monotone, i.e., $A \subset B \subset \Omega \Longrightarrow 0 \leq c(A) \leq c(B)$;*

(iii) *if $(A_n)_{n\in\mathbb{N}}$ is a non-decreasing sequence of subsets of Ω, then*

$$c\left(\bigcup_n A_n\right) = \lim_{n\to+\infty} c(A_n) = \sup_n c(A_n);$$

(iv) *if (K_n) is a non-increasing sequence of compact subsets of Ω,*

$$c\left(\bigcap_n K_n\right) = \lim_n c(K_n) = \inf_n c(K_n).$$

A capacity c is called a Choquet capacity if it is subadditive, i.e., if it satisfies the additional condition

(v) *if* $(A_n)_{n\in\mathbb{N}}$ *is any sequence of subsets of* Ω*, then*

$$c\left(\bigcup_n A_n\right) \leq \sum_n c(A_n),$$

Capacities are usually first defined for Borel subsets and then extended to all sets by building the appropriate outer set function (see Example 4.5). This motivates the following definition, where we let $\mathcal{B}(\Omega)$ denote the σ-algebra of Borel subsets of Ω:

Definition 4.2. *A set function* $c : \mathcal{B}(\Omega) \to \bar{\mathbb{R}}^+$ *is a precapacity if it satisfies the properties* (i), (ii), (iii) *for all Borel subsets of* Ω.

A precapacity c *is subadditive if it satisfies* (v). *It is outer regular if for all Borel subsets* $B \subset \Omega$,

$$c(B) = \inf\{c(G)|\ G \text{ open and } B \subset G \subset \Omega\}.$$

A precapacity c *is inner regular if for all Borel sets* $B \subset \Omega$,

$$c(B) = \sup\{c(K)|\ K \text{ compact and } K \subset B\}.$$

Note that a Choquet capacity is a precapacity when restricted to Borel sets; the converse however does not hold (see Example 4.5 below). The relation beween these two notions is given by the following result.

Proposition 4.3. *Let* $c : \mathcal{B}(\Omega) \to \bar{\mathbb{R}}^+$ *be a precapacity on* Ω. *The associated outer capacity, defined for any subset* $E \subset \Omega$ *by*

$$c^*(E) := \inf\{c(G)|\ G \text{ open and } E \subset G \subset \Omega\} \tag{4.1.1}$$

is a capacity on Ω.

If c *is moreover subadditive, then* c^* *is a Choquet capacity.*

Proof. It is clear that c^* satisfies (i) and (ii). We prove the upper continuity property (iii). By definition, there exists a sequence of open sets (G_j) such that $A_j \subset G_j$ and

$$c^*(A_j) \leq c(G_j) \leq c^*(A_j) + 1/j$$

for all $j \geq 1$.

Since (A_j) is a non-decreasing sequence, we can arrange so that that the corresponding open sets (G_j) are non-decreasing (replacing G_j by $\bigcap_{k\geq j} G_k$). Since c satisfies (iii) for Borel sets and $A \subset G := \bigcup_j G_j$, we conclude that

$$c^*(A) \leq c(G) = \lim_j c(G_j) = \lim_j c^*(A_j).$$

Since the opposite inequality is trivial, we have proved that c^* satisfies condition (iii).

We now check that c^* satisfies condition (iv). If (K_j) is a decreasing sequence of compact subsets, any open set G containing $K := \bigcap_j K_j$ will contain K_j for any j large enough. Thus $c(G) \geq \lim_j c^*(K_j)$, hence $c^*(K) \geq \lim_j c^*(K_j)$. Since the opposite inequality is again trivial, it follows that c^* satisfies the lower continuity property (iv).

Assume that c is subadditive and let (A_j) be a sequence of subsets of Ω. To prove (v) for $c^* \nearrow$ we can assume that $c^*(A_j) < +\infty$ for all j. By definition, given $\varepsilon > 0$ there exists an open set $G_j \supset A_j$ such that $c(G_j) \leq c^*(A_j) + \varepsilon 2^{-j-1}$. Then $G := \bigcup_j G_j$ is an open set such that $\bigcup_j A_j \subset G$ and

$$c(G) \leq \sum_j c(G_j) \leq \sum_j c^*(A_j) + \varepsilon,$$

We infer $c^*(\bigcup_j A_j) \leq \sum_j c^*(A_j) + \varepsilon$ and (v) follows by letting $\varepsilon \to 0$. □

Example 4.4. *A Borel measure is a positive measure defined on all Borels sets which is locally finite, inner, and outer regular. Therefore, any Borel measure μ on Ω is an outer regular precapacity and its associated outer measure μ^* is a Choquet capacity on Ω by Proposition 4.3.*

Example 4.5. *Let $\mathcal{M}$ be a family of Borel measures on Ω. The set function defined on any Borel subsets $A \subset \Omega$ by the formula*

$$c_{\mathcal{M}}(A) := \sup\{\mu(A) |\ \mu \in \mathcal{M}\}$$

is a precapacity on Ω. It is called the upper envelope of $\mathcal{M}$.

Observe that this precapacity need not be additive. Indeed, let λ be a probability measure and A, B be disjoint Borel subsets of positive λ-measure. Set $\mu_1 := \mathbf{1}_A \lambda/\lambda(A)$ and $\mu_2 := \mathbf{1}_B \lambda/\lambda(B)$. Then the upper envelope c of $\{\mu_1, \mu_2\}$ satisfies

$$c(A) = c(B) = 1 \ \text{ and } \ c(A \cup B) = 1.$$

The precapacity $c_{\mathcal{M}}$ need not be outer regular either, unless $\mathcal{M}$ is a finite set (see Exercise 4.4). However, if $\mathcal{M}$ is a compact set for the weak-topology, then $c^*_{\mathcal{M}}$ is a Choquet capacity (see Exercise 4.5).*

The problem we address now is whether a Choquet capacity is inner and outer regular. We first need a definition.

Definition 4.6. *Let $c : \mathcal{B}(\Omega) \to \bar{\mathbb{R}}^+$ be a precapacity. A subset $E \subset \Omega$ is said to be capacitable if $c^*(E) = c_*(E)$, where*

$$c_*(E) := \sup\{c(K) |\ K \subset \Omega \text{ is compact}\}$$

is the inner capacity of E.

Theorem 4.7. *Let c be a Choquet capacity on Ω. Then every Borel subset $B \subset \Omega$ satisfies*

$$c(B) = \sup\{c(K) |\ K \subset \Omega \text{ is compact}\}.$$

This result is a special case of Choquet's capacitability theorem which we prove in the next section.

Corollary 4.8. *Let $c : \mathcal{B}(\Omega) \to \bar{\mathbb{R}}^+$ be a precapacity on Ω satisfying (iv). Then c^* is inner regular, i.e., for every Borel set $B \subset \Omega$*

$$c^*(B) = \sup\{c(K) |\ K \text{ compact},\ K \subset B\}.$$

Proof. Property (iv) implies that $c^*(K) = c(K)$ for any compact subset $K \subset \Omega$. Indeed, given a compact set K there exists a sequence of compact subsets $(K_j)_j$ such that $K \subset K_{j-1} \subset K_j^\circ$ for all $j \geq 1$. Then

$$c^*(K) \leq c(K_j^\circ) \leq c(K_j)$$

for all $j \geq 1$, hence $c^*(K) \leq \lim_j c(K_j) = c(K)$ since c satisfies property (iv), thus $c^*(K) = c(K)$.

It follows from Proposition 4.3 that c^* is a Choquet capacity on Ω. Choquet's theorem therefore yields

$$c^*(B) = \sup\{c^*(K) |\ K \text{ compact } K \subset B\},$$

for any Borel subset $B \subset \Omega$. Thus $c^*(B) = c_*(B)$. □

4.1.2 Choquet's capacitability theorem

We prove here a general version of Choquet's capacitability theorem. In what follows all topological spaces to be considered are Hausdorff and locally compact.

Definition 4.9.

(1) *A $\mathcal{F}_\sigma$-set is a countable union of closed subsets of X.*
(2) *A $\mathcal{G}_\delta$-set is a countable intersection of open subsets of X.*
(3) *A $\mathcal{F}_{\sigma\delta}$-subset is a countable intersection of $\mathcal{F}_\sigma$-subsets of X.*
(4) *A space X is a $\mathcal{K}_\sigma$ (resp. $\mathcal{K}_{\sigma\delta}$) space if it is homeomorphic to a $\mathcal{F}_\sigma$-subset (resp. $\mathcal{F}_{\sigma\delta}$-subset) of some compact space Y.*

Here are some elementary properties which we use later on:

Proposition 4.10.

(1) *Every closed subset of a $\mathcal{K}_{\sigma\delta}$ space X is a $\mathcal{K}_{\sigma\delta}$ space.*
(2) *Every countable cartesian product of $\mathcal{K}_{\sigma\delta}$ spaces is a $\mathcal{K}_{\sigma\delta}$ space.*
(3) *Every countable disjoint sum of $\mathcal{K}_{\sigma\delta}$ spaces is a $\mathcal{K}_{\sigma\delta}$ space.*

Proof. (1) Let F be a closed subset of $X = \bigcap_{j\geq 1} G_j$, where (G_j) is a sequence of $\mathcal{F}_\sigma$ subsets of some compact space Y and $G_j = \bigcup_\ell K_{j\ell}$ is a countable union of closed subsets $K_{j\ell}$ of Y. If $\bar{F}$ denotes the closure of F in Y, we have

$$F = X \cap \bar{F} = \bigcap_{j\geq 1} G_j \cap \bar{F}, \quad G_j \cap \bar{F} = \bigcup_{\ell\geq 1} \bar{F} \cap K_{j\ell},$$

thus F is a $\mathcal{F}_{\sigma\delta}$ subset of Y.

(2) Write $X = \prod_{j\in\mathbb{N}^*} X_j$ as a cartesian product of a sequence (X_j) of $\mathcal{K}_{\sigma\delta}$ spaces. Write for every $j \in \mathbb{N}$

$$X_j = \bigcap_{\ell\in\mathbb{N}^*} G_\ell^j, \quad G_\ell^j = \bigcup_{m\in\mathbb{N}^*} K_{\ell,m}^j,$$

where $K_{\ell,m}^j$ are closed subsets of some compact space Y_j. Then

$$X = \bigcap_{\ell\in\mathbb{N}^*} G_\ell, \text{ with } G_\ell := G_\ell^1 \times G_{\ell-1}^2 \times \cdots \times G_1^\ell \times \prod_{i\in\mathbb{N}^*} Y_{\ell+1},$$

and

$$G_\ell = \bigcup_{m_1,\ldots,m_\ell\geq 1} K_{\ell m_1}^1 \times K_{\ell-1 m_2}^2 \times \cdots \times K_{1 m_\ell}^\ell \times \prod_{i\in\mathbb{N}^*} Y_{\ell+1},$$

where each term in the countable union is a closed subset in the compact space $Y := \prod_{\ell\in\mathbb{N}^*} Y_\ell$.

(3) With the same notations as in 2, define $X := \coprod X_j$ and $Y := \coprod Y_j$. Then Y can be embedded in the compact space $\hat{Y} := \prod_j (Y_j \coprod \{a\})$, where $\{a\}$ is any single-point topological space, by sending a point $w \in Y_j$ to the point $(a, \ldots, a, w, a, \ldots)$, where w is in the jth position. Then $X = \coprod X_j$ can be written as

$$X = \bigcap_{\ell\geq 1} G_\ell, \quad G_\ell := \bigcup_{m\geq 2} \coprod_{j\geq 1} K_{\ell m}^j.$$

Since $K_{\ell m}^j$ can be embedded into a closed subset of the compact space $\hat{Y}$, it follows that X is a $\mathcal{K}_{\sigma\delta}$ space. □

Observe that a $\mathcal{F}_{\sigma\delta}$-subset in X is a Borel subset of X. It is well known that the continuous image of a Borel subset is not in general a Borel subset (see Exercise 4.1). This motivates the following definition:

Definition 4.11. *A subset A in a topological space X is a $\mathcal{K}$-analytic subset of X if it is the continuous image of a $\mathcal{K}_{\sigma\delta}$ space of some topological space Y, i.e., there exists a continuous map $f : E \to X$ from a $\mathcal{K}_{\sigma\delta}$ space E into X such that $f(E) = A$.*

Proposition 4.12. *Let (A_j) be a sequence of $\mathcal{K}$-analytic subsets of Ω. Then*

$$\bigcup A_j \quad \text{and} \quad \bigcap A_j$$

are $\mathcal{K}$-analytic subsets of Ω.

Proof. Let $f_j : X_j \to \Omega$ be a continuous map from a $\mathcal{K}_{\sigma\delta}$-space X_j onto $A_j = f_j(X_j)$. Let $X := \coprod X_j$ and $f := \coprod f_j : X \to \Omega$. It follows from Proposition 4.10 that X is a $\mathcal{K}_{\sigma\delta}$ space and $f(X) = \bigcup A_j$.

For the intersection we set

$$Y := \{x = (x_j) \in X |\ \forall j \in \mathbb{N}^*, f_j(x_j) = f_1(x_1)\}$$

and $g(x) = f_1(x_1) = f_j(x_j)$ for all $j \in \mathbb{N}^*$. Then Y is closed in X, hence it is a $\mathcal{K}_{\sigma\delta}$-space and $g(Y) = \bigcap_j A_j$, by Proposition 4.10 again. □

Corollary 4.13. *Let X be a Hausdorff locally compact space. Then any Borel subset of X is a $\mathcal{K}$-analytic subset of X.*

Proof. Observe that any open or closed subset of X is a countable union of compact subsets, hence $\mathcal{K}$-analytic. On the other hand, by Proposition 4.12, the family

$$\mathcal{T} := \{E \in 2^X |\ E \text{ and } \Omega \setminus E \text{ are } \mathcal{K}\text{-analytic}\}$$

is a σ-algebra on X. Since it contains all open subsets of X, it also contains all Borel subsets of X. □

We need the following structural lemma about $\mathcal{K}$-analytic sets:

Lemma 4.14. *Let A be a relatively compact $\mathcal{K}$-analytic subset of a topological space X. There exist a compact space H, a continuous map $h : H \to X$, and a $\mathcal{F}_{\sigma\delta}$ subset $Y \subset H$ such that $h(Y) = A$.*

Proof. There exist a compact space Y, a $\mathcal{F}_{\sigma\delta}$ subset $E \subset Y$, and a continuous map $g : E \to A$ such that $A = g(E)$. Let

$$Z := \{(y, g(y))|\ y \in E\} \subset E \times A$$

be the graph of the map g. Let $H := \bar{Z}$ be the closure of Z in the compact space $\bar{E} \times \bar{A}$. Then A is the image of Z under the second projection $h : H \to \bar{A}$, i.e., $A = h(Z)$.

As g is continuous, Z is closed in $E \times \bar{A}$, thus $Z = H \cap (E \times \bar{A})$. Now E is a $\mathcal{F}_{\sigma\delta}$ subset of $\bar{E}$, hence $\bar{E} \times \bar{A}$ is a $\mathcal{F}_{\sigma\delta}$ subset of $\bar{E} \times \bar{A}$ and Z is a $\mathcal{F}_{\sigma\delta}$ subset of $H = \bar{Z}$. The lemma follows since $A = h(Z)$. □

We are finally ready to prove Choquet's capacitability theorem:

Theorem 4.15. *Let X be a $\mathcal{K}_\sigma$ space and c a capacity on X. Then every $\mathcal{K}$-analytic subset $A \subset X$ satisfies*

$$c(A) = \sup\{c(K) |\ K \ \text{compact} \subset A\}.$$

Proof. Since X is the union of an increasing sequence of compact sets K_j, it follows from property (iii) of Definition 4.1 that

$$c(A) = \lim_j c(A \cap K_j).$$

We may thus assume that A is relatively compact.

It follows from Lemma 4.14 that there exist a $\mathcal{F}_{\sigma\delta}$-subset H in some compact space and a continuous map $h : H \to \Omega$ such that $h(H) = A$.

We let the reader check in Exercice 4.6 that the set function $c_h := h^* c$ defined on subsets of H by $c_h(E) = c(h(E))$ is a capacity on H. We are thus reduced to proving the theorem when X is a compact space and A is a $\mathcal{F}_{\sigma\delta}$-subset of X. We then write

$$A = \bigcap_{i\in\mathbb{N}} G_i, \text{ and } G_i = \bigcup_{j\in\mathbb{N}} K_{ij},$$

where $K_{i,j}$ are closed subsets of X (hence compact), increasing with j. Fix a real number $0 < \alpha < c(A)$. Decomposing

$$A = G_1 \cap \left(\bigcap_{\ell\geq 2} G_\ell\right) = \bigcup_{j\geq 1} \left(K_{1j} \cap \bigcap_{\ell\geq 2} G_\ell\right)$$

and using property (ii) of Definition 4.1, we can find a subset

$$A_1 := K_{1j_1} \cap \bigcap_{\ell\geq 2} G_\ell \ \text{ such that } \ c(A_1) > \alpha.$$

By induction we find a decreasing sequence of subsets $A \supset A_1 \supset \cdots \supset A_m$ such that

$$A_m = K_{1j_1} \cap \cdots \cap K_{mj_m} \cap \bigcap_{\ell\geq m+1} G_\ell, \ \text{ and } \ c(A_m) > \alpha.$$

Set $K := \bigcap K_{mj_m} = \bigcap A_m \subset A$. This is a compact subset of A. By property (iv) of Definition 4.1,

$$c(K) = \lim_m c(K_{1j_1} \cap \cdots \cap K_{mj_m}) \geq \lim_m c(A_m) \geq \alpha,$$

which implies that $c_*(A) \geq c(A)$, and the proof is complete. □

4.1.3 The Monge–Ampère capacity In classical potential theory, the capacity is defined as the maximal amount of charge supported on a compact set $K \subset \Omega \subset \mathbb{R}^m$ so that the difference of potentials in the condenser (K, Ω) is 1. This definition can be formalized by setting

$$\mathrm{Cap}(K) := \sup\{\mu(K)|\ \mu \in \Gamma(K)\},$$

where $\Gamma(K)$ is the set of Borel measures that are supported on K and whose potential U_μ is bounded between 0 and 1 in Ω. Here U_μ denotes the Green potential of μ, i.e., the solution of the Dirichlet problem $\Delta u = \mu$ in $\Omega \setminus K$ with boundary values $u = -1$ on ∂K and 0 on $\partial\Omega$.

There is no simple formula for U_μ in $\mathbb{C}^n$ when $n \geq 2$, as the corresponding complex Monge–Ampère operator is non-linear. We will nevertheless mimic the above definition, using the family of bounded plurisubharmonic functions with zero boundary values on $\partial\Omega$.

4.1.3.1 Definition Let $\Omega \Subset \mathbb{C}^n$ be a smoothly bounded pseudoconvex domain. The Monge–Ampère capacity is defined as follows:

Definition 4.16. *For any Borel subset $E \subset \Omega$ we set*

$$\mathrm{Cap}(E;\Omega) := \sup\left\{\int_E (dd^c u)^n |\ u \in PSH(\Omega), -1 \leq u \leq 0\right\}.$$

We shall also use the notation $\mathrm{Cap}_\Omega(E)$ *in the sequel.*

It follows from Chern–Levine–Nirenberg inequalities that the capacity of relatively compact subsets $E \Subset \Omega$ is finite, $\mathrm{Cap}(E;\Omega) < +\infty$.

The real number $\mathrm{Cap}(E;\Omega)$ is the (inner) Monge–Ampère capacity of the condenser (E, Ω). Since any Borel measure is inner regular, it follows that the set function $\mathrm{Cap}(\cdot;\Omega)$ is inner regular. The outer Monge–Ampère capacity is then:

Definition 4.17. *For a any subset $E \subset \Omega$, we set*

$$\mathrm{Cap}^*(E;\Omega) := \inf\{\mathrm{Cap}(G;\Omega),\ G \text{ open},\ E \subset G \subset \Omega\}.$$

We say that E is capacitable if $\mathrm{Cap}^*(E;\Omega) = \mathrm{Cap}(E;\Omega) < +\infty$.

Let us show, using Choquet's Theorem, that all Borel sets are capacitable. We start by establishing some elementary properties of this capacity:

Proposition 4.18.
(1) *If $\Omega \subset \mathbb{B}(a, R) \Subset \mathbb{C}^n$, then for any Borel subset $E \subset \Omega$,*

$$\lambda_{2n}(E) \leq \left(\frac{\pi R^2}{2}\right)^n \mathrm{Cap}^*(E;\Omega).$$

(2) *If* $E_1 \subset E_2 \subset \Omega_2 \subset \Omega_1$, *then* $\mathrm{Cap}^*(E_1;\Omega_1) \le \mathrm{Cap}^*(E_2;\Omega_2)$.

(3) *The set function* $\mathrm{Cap}^*(\cdot;\Omega)$ *is subadditive, i.e., if* $(E_j)_{j\in\mathbb{N}}$ *is any sequence of subsets of* Ω, *then*

$$\mathrm{Cap}^*(E;\Omega) \le \sum_j \mathrm{Cap}^*(E_j;\Omega).$$

(4) *Let* $\Omega' \Subset \Omega'' \subset \Omega \subset \mathbb{C}^n$ *be open subsets. Then there exists a constant* $A = A(\Omega,\Omega',\Omega'') > 0$ *such that for all* $E \subset \Omega'$,

$$\mathrm{Cap}^*(E;\Omega) \le \mathrm{Cap}(E;\Omega'') \le A\mathrm{Cap}^*(E;\Omega).$$

(5) *Let* $f : \Omega_1 \subset \mathbb{C}^n \to \Omega_2 \subset \mathbb{C}^n$ *be a proper holomorphic map. Then for any Borel subset* $E \subset \Omega_2$ *we have*

$$\mathrm{Cap}(E;\Omega_2) \le \mathrm{Cap}(f^{-1}(E);\Omega_1).$$

Proof. The function $\rho(z) := |z-a|^2/R^2 - 1$ is plurisubharmonic in Ω and $-1 \le \rho \le 0$ in $\Omega \subset \mathbb{B}(a,R)$. By definition, we thus get

$$\int_E (dd^c\rho)^n \le \mathrm{Cap}(E;\Omega),$$

which proves (1) since $dd^c\rho = (2/\pi R^2)\beta$.

We let the reader check the properties (2), (3) and prove property (4). Fix ρ a plurisubharmonic defining function for Ω and $c > 0$ such that $\Omega'' \subset \Omega_c := \{\rho < -c\}$, hence $\mathrm{Cap}(\cdot;\Omega'') \ge \mathrm{Cap}(\cdot;\Omega_c)$. It is sufficient to prove the inequality for Ω_c. Choose $A > 1$ so that $\psi := A(\rho + c)$ satisfies $\psi \le -1$ on Ω'. Fix $u \in PSH(\Omega_c)$ with $-1 \le u \le 0$ and define

$$\tilde{u}(z) := \begin{cases} \max(u(z),\psi(z)), & \text{if } z \in \Omega_c, \\ \psi(z), & \text{if } z \in \Omega \setminus \Omega_c, \end{cases}$$

so that $\tilde{u}$ is plurisubharmonic on Ω and satisfies $\tilde{u} = u$ on Ω'.

The function $v := (a+1)^{-1}(\tilde{u} - a)$, where $a := \sup_\Omega \tilde{u}$, is plurisubharmonic in Ω and $-1 \le v \le 0$. Thus $\int_E (dd^c v)^n \le \mathrm{Cap}(E;\Omega)$. Since $v = (a+1)^{-1}(u-a)$ in Ω', we infer

$$\int_E (dd^c u)^n \le (a+1)^n \mathrm{Cap}(E;\Omega),$$

hence $\mathrm{Cap}(E;\Omega_c) \le (a+1)^n \mathrm{Cap}(E;\Omega)$.

We finally prove (5). Fix $u \in PSH(\Omega_2) \cap L^\infty_{\mathrm{loc}}(\Omega_2)$. It follows from Exercise 3.5 in Chapter 3 that $f_*(dd^c u \circ f)^n = (dd^c u)^n$ in the sense of Borel measures in Ω_1. Thus, if $E \subset \Omega_2$ is a Borel subset,

$$\int_E (dd^c u)^n = \int_{f^{-1}(E)} (dd^c u \circ f)^n \le \mathrm{Cap}(f^{-1}(E);\Omega_1).$$

Taking the supremum over all such u yields the required inequality for the inner capacities. □

4.1.3.2 Polar sets are null sets We now show that the polar locus of a plurisubharmonic function has zero outer capacity.

Proposition 4.19. *Let $\Omega' \Subset \Omega \subset \mathbb{C}^n$ be two open sets and $K \subset \Omega'$ a compact set. There exists $A = A(K, \Omega') > 0$ such that for all plurisubharmonic functions $V \in PSH(\Omega)$ and for all $s > 0$,*

$$\mathrm{Cap}^*(\{z \in K |\ V(z) < -s\}; \Omega) \leq \frac{A}{s} \|V\|_{L^1(\Omega')}. \tag{4.1.2}$$

In particular, the polar locus $P := \{z \in \Omega |\ V(z) = -\infty\}$ satisfies

$$\mathrm{Cap}^*(P, \Omega) = 0.$$

Proof. Let $u \in PSH(\Omega)$ be so that $-1 \leq u \leq 0$. It follows from the Chern–Levine–Nirenberg inequalities that there exists $A = A(K, \Omega') > 0$ such that

$$\int_K |V| (dd^c u)^n \leq A \int_{\Omega'} |V| d\lambda_{2n}.$$

We infer that

$$\int_{\{z \in K; V(z) \leq -s\}} (dd^c u)^n \leq \frac{1}{s} \int_K |V| (dd^c u)^n \leq \frac{A}{s} \int_{\Omega'} |V| d\lambda_{2n}.$$

The desired estimate for the inner capacity follows.

Since for any open set $D \Subset \Omega$, the sublevel sets $\{z \in D |\ V < -s\}$ are open sets, it follows that the same inequality holds for the outer capacity. The last statement follows from the subadditivity of the outer capacity. □

We will later on show that conversely, if $\mathrm{Cap}^*(P; \Omega) = 0$, then P is (locally) contained in the polar set of a plurisubharmonic function.

4.2 Continuity of the complex Monge–Ampère operator

4.2.1 Quasi-continuity of plurisubharmonic functions

Theorem 4.20. *Let V be a plurisubharmonic function in Ω. For all $\varepsilon > 0$, there exists an open set $G \subset \Omega$ such that $\mathrm{Cap}(G; \Omega) < \varepsilon$ and $V|(\Omega \setminus G)$ is continuous.*

Proof. Let $D \Subset \Omega$ be an open subset. It follows from Proposition 4.19 that $\mathrm{Cap}(G_1; \Omega) < \varepsilon$ if $s > 1$ is large enough, where

$$G_1 := \{z \in D |\ V(z) < -s\}.$$

Set $v = v_s := \sup\{V, -s\}$. The function v is plurisubharmonic and bounded in a neighborhood of $\overline{D}$, and $v = V$ in $\Omega \setminus G_1$. Let (v_j) a decreasing sequence of smooth plurisubharmonic functions in a neighborhood $\Omega' \Subset \Omega$ of $\overline{D}$ which converges to v. We can assume that $v_j = v = A\rho$ in a neighbourhood of $\partial\Omega'$. It follows from Proposition 3.21 that for all $\delta > 0$,

$$\mathrm{Cap}\left(D \cap \{v_j - v > \delta\}; \Omega\right) \leq \frac{(n+1)!}{\delta^{n+1}} \int_{\Omega'} (v_j - v)(dd^c v)^n.$$

The monotone convergence theorem therefore yields

$$\lim_j \mathrm{Cap}\left(D \cap \{v_j - v > \delta\}; \Omega\right) = 0.$$

Thus, for all $k \in \mathbb{N}^*$ there exists $j_k \in \mathbb{N}$ large enough so that

$$\mathrm{Cap}\left(D \cap \{v_{j_k} - v > 1/k\}; \Omega\right) \leq \varepsilon 2^{-k}.$$

We can assume that the sequence (j_k) is increasing. Set

$$G_2 := \bigcup_{k \geq 1} \{v_{j_k} - v > 1/k\} \text{ and } G := G_1 \cup G_2.$$

The sequence (v_{j_k}) decreases uniformly to v on the compact set $\overline{D} \setminus G_2$ and $\mathrm{Cap}_\Omega(G_2) \leq \varepsilon$, thus the open set G satisfies the required properties: it has small capacity and $V = v$ is continuous in $D \setminus G$.

To complete the proof of the theorem we take an exhaustive sequence (D_j) of relatively compact domains such that $\bigcup_j D_j = \Omega$, and apply the first part of the proof to find a sequence (G_j) of open subsets of Ω such that $\mathrm{Cap}(G_j; \Omega) \leq \varepsilon 2^{-j}$ and $V|(D_j \setminus G_j)$ is continuous. We finally set $G := \bigcup_j G_j$ and use the subadditivity of the capacity to conclude that $\mathrm{Cap}(G; \Omega) \leq \varepsilon$ and $V|(\Omega \setminus G)$ is continuous. □

The following lemma has been used in Chapter 3:

Lemma 4.21. *Let $\mathcal{P}$ be a locally uniformly bounded family of plurisubharmonic functions in Ω. Let $\mathcal{T}$ be the set of currents of the form $T := \bigwedge_{1 \leq i \leq p} dd^c u_i$, where $u_1, \ldots, u_p \in \mathcal{P}$.*

Assume that $(T_j)_{j \in \mathbb{N}}$ is a sequence of currents in $\mathcal{T}$ converging weakly to $T \in \mathcal{T}$. Then for any locally bounded quasi-continuous function f in Ω, $f T_j \to f T$ in the weak sense of currents in Ω.

The notion of quasi-continuous function is defined as follows:

Definition 4.22. *A Borel function $f : \Omega \to \overline{\mathbb{R}}$ is quasi-continuous if for all $\varepsilon > 0$ and all compact subsets $K \Subset \Omega$ there exists an open subset $G \subset \Omega$ such that $\mathrm{Cap}(G; \Omega) \leq \varepsilon$ and $f|(K \setminus G)$ is continuous.*

It follows from Theorem 4.20 that any plurisubharmonic function is quasi-continuous. So are the differences of plurisubharmonic functions.

Proof. Let Θ be a positive continuous test form of bidegree $(n-p, n-p)$ in Ω, $K \subset \Omega$ its compact support, and fix $\varepsilon > 0$ small.

It follows from the quasi-continuity of f that there is an open subset $G \subset \Omega$ with $\mathrm{Cap}(G;\Omega) \leq \varepsilon$ such that $f|(K \setminus G)$ is continuous. Let g be a continuous function in Ω with compact support such that $g = f$ on $K \setminus G$. Then

$$\int_\Omega f(T_j \wedge \Theta - T \wedge \Theta) = \int_\Omega g(T_j \wedge \Theta - T \wedge \Theta) + \int_G (f-g)(T_j \wedge \Theta - T \wedge \Theta).$$

Observe that $\lim_{j\to+\infty} \int_\Omega g(T_j \wedge \Theta - T \wedge \Theta) = 0$ since T_j converges weakly towards T. We claim that the second term is $O(\varepsilon)$. Indeed, since $|f-g|$ is bounded by a constant $M > 0$ on the support of Θ,

$$\left| \int_G (f-g)(T_j \wedge \Theta - T \wedge \Theta) \right| \leq M \int_G (T_j \wedge \Theta + T \wedge \Theta).$$

Observe that $T_j \wedge \Theta \leq C_1 T_j \wedge \beta^{n-p}$ for some $C_1 > 0$, where $\beta = dd^c|z|^2$. Moreover, there exists a uniform constant $C_2 > 0$ such that for all Borel subsets $E \subset \Omega$,

$$\int_E \bigwedge_{1\leq i\leq p} (dd^c v_i) \wedge \beta^{n-p} \leq C_2 \mathrm{Cap}(E,\Omega)$$

as $v_1, \ldots, v_p \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$ with $|v_i| \leq A$ (see Exercise 4.7). Therefore,

$$\int_G T_j \wedge \Theta + T \wedge \Theta \leq 2C_1 C_2 \varepsilon,$$

and the proof is complete. □

4.2.2 Convergence in capacity We have shown that the complex Monge–Ampère operator is continuous along monotone sequences of plurisubharmonic functions. We introduce here, following [X96], a more general notion of convergence that contains the above continuity properties as a particular case.

Definition 4.23. *A sequence of Borel functions $(f_j)_{j\geq 0}$ converges in capacity to a Borel function f in Ω if for all $\delta > 0$ and all compact subsets $K \subset \Omega$,*

$$\lim_{j\to+\infty} \mathrm{Cap}^*(K \cap \{|f_j - f| \geq \delta\};\Omega)) = 0.$$

Convergence in capacity implies convergence in L^1_{loc} (but the converse is not true):

Lemma 4.24. *Let $(f_j)_{j\geq 0}$ be sequence of (locally) uniformly bounded Borel functions which converges in capacity to a Borel function f in Ω. Then $f_j \to f$ in $L^1_{\mathrm{loc}}(\Omega)$.*

Proof. Fix a compact set $K \subset \Omega$ and $\delta > 0$. Since the sequence $(f_j)_{j\in\mathbb{N}}$ and f are uniformly bounded in K (by some $M > 0$), we get

$$\int_K |f_j - f| d\lambda \leq 2M\lambda(K \cap \{|f_j - f| > \delta\}) + \delta\lambda(K).$$

By Proposition 4.18, the Lebesgue measure λ is dominated by capacity, hence the first term in the right-hand side converges to 0. The claim follows since $\delta > 0$ is arbitrarily small. □

We note that convergence of monotone sequences of plurisubharmonic functions implies convergence in capacity.

Proposition 4.25. *Let $(V_j)_{j\in\mathbb{N}} \subset PSH(\Omega)$ be a monotone sequence of plurisubharmonic functions which converges almost everywhere to $V \in PSH(\Omega)$. Then (V_j) converges to V in capacity.*

We treat here two rather different settings at once. If (v_j) is non-increasing, then $v_j(x)$ converges towards $v(x)$ at all points x (see Chapter 1). When (v_j) is non-decreasing, then $w = \sup_j v_j$ is usually not u.s.c. hence equality $w(x) = v(x)$ does not hold everywhere.

Proof. By the subadditivity and monotonicity of the capacity, it is enough to prove that for any Euclidean ball $\mathbb{B} \Subset \Omega$, any compact $K \subset \mathbb{B}$, and any $\delta > 0$ we have

$$\lim_{j\to+\infty} \mathrm{Cap}^*(K \cap \{|V_j - V| \geq \delta\}; \mathbb{B}) = 0.$$

We use here the fact that $\mathrm{Cap}^*(\cdot\,; \Omega) \leq \mathrm{Cap}^*(\cdot\,; \mathbb{B})$.

We first reduce to the case where the sequence (V_j) is locally uniformly bounded. Indeed, fix $s \in \mathbb{N}$ and define

$$V_j^s := \sup\{V_j, -s\}, \;\; V^s := \sup\{V, -s\}.$$

Then

$$\{|V_j - V| \geq \delta\} \subset \{|V_j^s - V^s| \geq \delta\} \cup \{V \leq -s\} \cup \{V_j \leq -s\}.$$

Now by (4.1.2), we have for any $s \geq 1$ and $j \geq 1$,

$$\mathrm{Cap}^*(K \cap \{V_j < -s\}; \mathbb{B}) \leq \frac{A}{s} \|V_j\|_{L^1(\mathbb{B})},$$

where $A > 0$ is a constant which does not depend on j. Therefore

$$\mathrm{Cap}^*(K \cap \{|V_j - V| \geq \delta\}; \mathbb{B}) \leq \mathrm{Cap}^*(K \cap \{|V_j^s - V^s| \geq \delta\}; \mathbb{B}) + \frac{A'}{s},$$

hence it suffices to treat the case of sequences of plurisubharmonic functions that are uniformly bounded.

Assume thus that $-M \leq V_j, V \leq +M$ in $\mathbb{B}$, for some $M > 0$. Using the localization principle (see Chapter 3), we can assume that all the V_j's are equal in a neighborhood of $\partial\mathbb{B}$.

Fix $\varepsilon > 0$. The quasi-continuity property of plurisubharmonic functions and the subadditivity of the capacity ensure that we can find an open set $G \subset \Omega$ such that $\mathrm{Cap}(G;\mathbb{B}) \leq \varepsilon$ and all V_j's and V are continuous in $\bar{\mathbb{B}} \setminus G$. Since the sequence (V_j) is monotone, it follows from Dini's lemma that the convergence is uniform on the compact set $\bar{\mathbb{B}} \setminus G$.

Assume first that $(V_j)_{j\in\mathbb{N}}$ is a non-decreasing sequence. It follows from the Chebyshev inequality and Proposition 3.21 that

$$\begin{aligned}
&\frac{1}{(n+1)!}\mathrm{Cap}(K \cap \{V - V_j \geq \delta\}, \mathbb{B}) \\
\leq\ & \delta^{-n-1} \int_{\mathbb{B}} (V - V_j)(dd^c V_j)^n \\
\leq\ & \delta^{-n-1} \int_{\mathbb{B}\setminus G} (V - V_j)(dd^c V_j)^n + \delta^{-n-1} \int_{\mathbb{B}\cap G} (V - V_j)(dd^c V_j)^n \\
\leq\ & \delta^{-n-1} \|V_j - V\|_{\bar{\mathbb{B}}\setminus G} \int_{\mathbb{B}} (dd^c V_j)^n + 2M\delta^{-n-1} \int_G (dd^c V_j)^n.
\end{aligned}$$

Now $\int_{\mathbb{B}} (dd^c V_j)^n$ is uniformly bounded by the Chern–Levine–Nirenberg inequalities and $\int_G (dd^c V_j)^n \leq M^n \,\mathrm{Cap}(G;\mathbb{B}) \leq \varepsilon M^n$. The conclusion follows since $\lim_j \|V_j - V\|_{\bar{\mathbb{B}}\setminus G} = 0$.

Assume now that $(V_j)_{j\in\mathbb{N}}$ is non-increasing. We proceed as above to obtain

$$\mathrm{Cap}_{\mathbb{B}}(K \cap \{V_j - V| \geq \delta\}; \mathbb{B}) \leq \delta^{-n-1}\,(n+1)! \int_{\mathbb{B}} (V_j - V)(dd^c V)^n.$$

The conclusion follows from the monotone convergence theorem and the capacitability of Borel sets (Corollary 4.36). □

4.2.3 Continuity of the Monge–Ampère operator

Our aim in this section is to show that the complex Monge–Ampère operator is continuous with respect to the convergence in capacity.

Theorem 4.26. *Let $(f_j)_{j\in\mathbb{N}}$ be positive and uniformly bounded quasi-continuous functions which converge in capacity to a quasi-continuous function f in Ω. Let $(u_j^1)_{j\in\mathbb{N}}, \ldots, (u_j^p)_{j\in\mathbb{N}}$ be uniformly bounded plurisubharmonic functions which converge in capacity in Ω to locally bounded plurisubharmonic functions $u^1, \ldots, u^p$ respectively. Then*

$$f_j \bigwedge_{1\leq j\leq p} dd^c u_j^i \longrightarrow f \bigwedge_{1\leq j\leq p} dd^c u^i$$

in the weak sense of currents in Ω.

Proof. The proof proceeds by induction. It follows from Lemma 4.24 that f_j (resp. u_j) converges in $L^1_{\mathrm{loc}}(\Omega)$ to f (resp. u). If $p = 1$, we know that $dd^c u_j^1 \to dd^c u^1$ weakly. Using induction on p and setting

$$T := \bigwedge_{1\le i\le p} dd^c u^i, \quad T_j := \bigwedge_{1\le i\le p} dd^c u_j^i,$$

it is enough to prove that if $T_j \to T$ weakly, then $f_j T_j \to fT$ weakly. The statement is local, so we can assume that Ω is the unit ball and all the functions are bounded between -1 and 0. Fix a test form Θ of bidegree $(n-p, n-p)$ and observe that

$$\int_\Omega f_j T_j \wedge \Theta - \int_\Omega fT \wedge \Theta = \int_\Omega (f_j - f)T_j + \int_\Omega f\left(T_j \wedge \Theta - T \wedge \Theta\right) = I_j + J_j.$$

It follows from Lemma 4.21 that $\lim_j J_j = 0$. It thus remains to prove that $\lim_j I_j = 0$. Fix $\delta > 0$ small, let K be the support of Θ and set $E_j := K \cap \{|f_j - f| \ge \delta\}$. Then

$$|I_j| \le \int_\Omega |f_j - f| T_j \wedge \Theta \le \int_{K\cap E_j} |f_j - f| T_j \wedge \Theta + \delta \int_K T_j \wedge \Theta.$$

It follows from previous estimates that

$$|I_j| \le C' \mathrm{Cap}^*(E_j, \Omega) + \delta M(K),$$

where $M(K) := C \sup_j \int_K T_j \wedge \beta^{n-p}$ is finite by the Chern–Levine–Nirenberg inequalities. Hence, $\lim_j I_j = 0$. □

Corollary 4.27. *Let $(h_j)_{j\ge 0}$ and $(u_j)_{j\ge 0}$ be monotone sequences of uniformly bounded negative plurisubharmonic functions which converge almost everywhere to plurisubharmonic functions h and u, respectively. Then for all $p \ge 0$,*

$$(-h_j)^p (dd^c u_j)^n \longrightarrow (-h)^p (dd^c u)^n.$$

Proof. It follows from Proposition 4.25 that the sequences (u_j) and (h_j) converge in capacity to u and h, respectively.

Observe that if $u \in PSH^-(\Omega)$, then $(-u)^p$ is quasicontinuous on Ω. Thus the sequence $(-h_j)^p$ converges in capacity to $(-h)^p$. The conclusion follows therefore from the previous theorem. □

Corollary 4.28. *Let $\mathcal{P}$ be a family of plurisubharmonic functions in a domain $\Omega \subset \mathbb{C}^n$ which is locally uniformly bounded from above, and set $U := \sup\{u |\, u \in \mathcal{P}\}$. Then the exceptional set*

$$E := \{U < U^*\}$$

is negligible with respect to any measure $dd^c u_1 \wedge \ldots \wedge dd^c u_n$, where $u_1, \ldots, u_n \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$. In particular

$$\mathrm{Cap}(E, \Omega) = 0.$$

Proof. By Choquet's lemma (see Lemma 4.31 below) we reduce to the case when $\mathcal{P}$ is an increasing sequence of plurisubharmonic functions. Fix a sequence (v_j) of locally uniformly bounded plurisubharmonic functions which increases almost everywhere to v and set

$$E := \{\sup_j v_j < v\}.$$

It follows from Corollary 4.27 that

$$v_j dd^c u_1 \wedge \ldots \wedge dd^c u_n \longrightarrow v dd^c u_1 \wedge \ldots \wedge dd^c u_n$$

weakly in Ω. Thus the positive currents $(v - v_j)dd^c u_1 \wedge \ldots \wedge dd^c u_n$ converge to 0, hence for any compact subset $K \subset \Omega$,

$$\lim_j \int_K (v - v_j)dd^c u_1 \wedge \ldots \wedge dd^c u_n = 0.$$

On the other hand, if we set $w := \lim_j v_j$ in Ω, then $w \leq v$ and the monotone convergence theorem yields

$$\lim_j \int_K (w - v_j)dd^c u_1 \wedge \ldots \wedge dd^c u_n = 0.$$

Therefore $\int_K (w - v)dd^c u_1 \wedge \ldots \wedge dd^c u_n = 0$, hence $w = v$ almost everywhere in K for the measure $dd^c u_1 \wedge \ldots \wedge dd^c u_n$. □

Theorem 4.29. *Let $(u_1^j)_{j\geq 0}, \ldots, (u_q^j)_{j\geq 0}$ be decreasing sequences of bounded plurisubharmonic functions in Ω which converge respectively to $u_1, \ldots, u_q \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$. Let (V_j) be a decreasing sequence of plurisubharmonic functions which converges to $V \in PSH(\Omega)$. Then*

$$V_j dd^c u_1^j \wedge \ldots \wedge dd^c u_q^j \longrightarrow V dd^c u_1 \wedge \ldots \wedge dd^c u_q.$$

Proof. We already know $dd^c u_1^j \wedge \ldots \wedge dd^c u_q^j \to dd^c u_1 \wedge \ldots \wedge dd^c u_q$. It follows from the Chern–Levine–Nirenberg inequalities that the currents $V_j dd^c u_1^j \wedge \ldots \wedge dd^c u_q^j$ have locally uniformly bounded masses.

Assume that $V_j \bigwedge_{1\leq i\leq q} dd^c {u_i}^j \wedge \beta^{n-p-q} \to \Theta$. The reader can check that $\Theta \leq V dd^c u_1 \wedge \ldots \wedge dd^c u_q$ (using an argument that we have already used in previous such proofs) and the problem is to show that $\int_{\mathbb{B}} V \bigwedge_{1\leq i\leq q} dd^c u_i \wedge \beta^{n-p-q} \leq \int_{\mathbb{B}} \Theta$, for an arbitrary ball $\mathbb{B}$.

We can assume all the $V_j's$ are negative and set $V_j^k := \sup\{V_j, k\rho\}$ for $k \in \mathbb{N}$, where ρ is a defining function for the ball $\mathbb{B}$. Since $V \leq V_j \leq V_j^k$ on $\mathbb{B}$ and $V_j^k = 0$

in $\partial\mathbb{B}$, we get

$$\begin{aligned}\int_{\overline{\mathbb{B}}} V \bigwedge_{1\le i\le q} dd^c u_i \wedge \beta^{n-q} &\le \int_{\overline{\mathbb{B}}} V_j^k \bigwedge_{1\le i\le q} dd^c u_i \wedge \beta^{n-q}\\ &= \int_{\overline{\mathbb{B}}} u_1 \wedge dd^c V_j^k \bigwedge_{2\le i\le q} dd^c u_i \wedge \beta^{n-q}\\ &\le \int_{\overline{\mathbb{B}}} {u_1}^j \wedge dd^c V_j^k \bigwedge_{2\le i\le q} dd^c u_i \wedge \beta^{n-q}\\ &\le \cdots \le \int_{\overline{\mathbb{B}}} V_j^k \bigwedge_{1\le i\le q} dd^c u_i^j \wedge \beta^{n-q},\end{aligned}$$

for all $j,k\in\mathbb{N}$. The monotone convergence theorem yields

$$\int_{\overline{\mathbb{B}}} V \bigwedge_{1\le i\le q} dd^c u_i \wedge \beta^{n-q} \le \int_{\overline{\mathbb{B}}} V_j \bigwedge_{1\le i\le q} dd^c {u_i}^j \wedge \beta^{n-q},$$

for all $j\in\mathbb{N}$. Letting now $j\to+\infty$, we obtain

$$\int_{\overline{\mathbb{B}}} V \bigwedge_{1\le i\le q} dd^c u_i \wedge \beta^{n-q} \le \limsup_{j\to+\infty} \int_{\overline{\mathbb{B}}} V_j \bigwedge_{1\le i\le q} dd^c {u_i}^j \wedge \beta^{n-q} \le \int_{\overline{\mathbb{B}}} \Theta,$$

and the proof is complete. □

4.3 The relative extremal function

4.3.1 Definition and Choquet's lemma

Definition 4.30. *Let $E\subset\Omega$ be a Borel subset. The relative extremal function associated to (E,Ω) is*

$$h_E(z) = h_{E,\Omega}(z) := \sup\{u(z)|\ u\in PSH(\Omega), u\le 0, u|E\le -1\}.$$

Observe that the upper semi-continuous regularization $h^*_{E,\Omega}$ is a plurisubharmonic function in Ω. This function was first defined by Siciak as a multidimensional analogue of the classical harmonic measure in one complex variable [Sic69].

We use on several occasions the following elementary topological lemma due to Choquet:

Lemma 4.31. *Let $\mathcal{U}$ be a family of upper semi-continuous functions and let $U := \sup\{u; u\in\mathcal{U}\}$ be its upper envelope. There exists a countable sub-family (u_j) in $\mathcal{U}$ such that $U^* = (\sup_j u_j)^*$ in Ω, where*

$$U^*(z) = \limsup_{z'\to z} U(z').$$

Proof. Assume first that $\mathcal{U}$ is locally uniformly bounded from above. Let $B(z_j, r_j)$ be a countable basis for the topology of Ω. For each j let $z_{j,k}$ a sequence in the ball $B(z_j, r_j)$ such that

$$\sup_{B(z_j,r_j)} U = \sup_k U(z_{j,k}).$$

For each (j,k) there exists a sequence $(u^\ell_{j,k})_{\ell\in\mathbb{N}}$ in $\mathcal{U}$ such that

$$U(z_{j,k}) = \sup_\ell u^\ell_{j,k}(z_{j,k}).$$

Set $V := \sup_{j,k,\ell} u^\ell_{j,k}$. Then $V \le U$, hence $V^* \le U^*$. Now

$$\sup_{B(z_j,r_j)} V \ge \sup_k V(z_{j,k}) \ge \sup_{k,\ell} u^\ell_{j,k}(z_{j,k}) = \sup_k U(z_{j,k}) = \sup_{B(z_j,r_j)} U,$$

hence $\sup_{B(z_j,r_j)} V = \sup_{B(z_j,r_j)} U$ for all j. Since any $B(z,\varepsilon)$ is a union of balls $B(z_j, r_j)$, we get $\sup_{B(z,\varepsilon)} V = \sup_{B(z,\varepsilon)} U$ hence $V^*(z) = U^*(z)$.

To treat the general case we define, for $u \in \mathcal{U}$, $\tilde{u} := \chi \circ u$, where $\chi : t \in \mathbb{R} \mapsto t/(1+|t|) \in]-1,+1[$ is an increasing homeomorphism. Observe that

$$\tilde{U} = \chi \circ U \text{ and } \tilde{U}^* = \chi \circ U^*,$$

hence the conclusion follows from the previous case applied to $\tilde{U}$. □

The first basic properties of the relative extremal functions are summarized in the following:

Theorem 4.32.

(1) *If $E_1 \subset E_2 \subset \Omega_2 \subset \Omega_1$, then $-1 \le h_{E_2,\Omega_2} \le h_{E_1,\Omega_1} \le 0$.*

(2) *If $E \Subset \Omega$, then $h^*_{E,\Omega}(z) \to 0$ as $z \to \partial\Omega$.*

(3) *Let (K_j) be a non-increasing sequence of compact subsets in Ω and $K := \bigcap_j K_j$. Then $h^*_{K_j}$ increases almost everywhere to h^*_K.*

(4) *For all $E \subset \Omega$,*

$$(dd^c h^*_{E,\Omega})^n = 0 \text{ in } \Omega \setminus \bar{E}. \tag{4.3.1}$$

Proof. (1) is clear. Let $\rho : \Omega \to [-1, 0[$ be a defining function for Ω. If $E \Subset \Omega$, there exists a constant $A > 1$ such that $A\rho \le -1$ on $\bar{E}$. Thus $A\rho \le h_E \le h^*_E \le 0$ in Ω and (2) follows.

We now prove (3). Since (K_j) is a non-increasing sequence of compact subsets in Ω, the sequence $(h^*_{K_j})$ increases almost everywhere to a plurisubharmonic function h in Ω which satisfies $-1 \le h \le h^*_K$ in Ω.

Let u be a plurisubharmonic function in Ω such that $u \le 0$ in Ω and $u \le -1$ in K. Fix $\varepsilon \in]0, 1[$. The open set

$$\{u < -1+\varepsilon\} = \left\{\frac{u}{1-\varepsilon} < -1\right\}$$

contains the compact set K, hence contains K_j for j large enough. We infer $u \leq (1-\varepsilon)h_{K_j}$ in Ω, hence $u \leq h$ in Ω. Therefore $h_K \leq h$ in Ω, hence $h_K \leq h \leq h_K^*$ in Ω.

It follows from Corollary 4.28 that the set $\{h_K < h_K^*\}$ has Lebesgue measure 0. Thus $h = h_K^*$ almost everywhere in Ω, hence everywhere by the submean value inequalities: the proof of (3) is complete.

We now prove (4). By Choquet's lemma, there exists a sequence (u_j) of plurisubharmonic functions such that $u_j = -1$ on E, $u_j \leq 0$ in Ω and u_j increases almost everywhere to h_E^*.

Fix a small ball B in $\Omega \setminus \overline{E}$. By Corollary 5.21, there exists a unique plurisubharmonic function $\hat{u}_j$ such that

- $u_j \leq \hat{u}_j \leq 0$ in Ω and $u_j = \hat{u}_j$ in $\Omega \setminus B$;
- $j \mapsto \hat{u}_j$ is increasing and $(dd^c\hat{u}_j)^n = 0$ in B.

We infer that $\hat{u}_j$ again increases a.e. to h_E^*, therefore $(dd^c h_E^*)^n = 0$ in B. Since B is an arbitrary ball in $\Omega \setminus \overline{E}$, this proves (4). □

The technique used in the proof of (4) is the classical *balayage*. The key technical tool here is the solution of the Dirichlet problem for the complex Monge–Ampère operator, a fundamental result of Bedford and Taylor [BT82] that we prove in Chapter 5.

Remark 4.33. *It is also true that $h_{E_j}^*$ decreases to h_E^* if E_j increases to E. The proof is however more involved and will be achieved in Corollary 4.44.*

4.3.2 Capacity and relative extremal function

We connect here the capacity of the condenser (E,Ω) to the Monge–Ampère mass of its extremal function, and use this to show that the Monge–Ampère capacity is a Choquet capacity.

Theorem 4.34.

(1) *For any relatively compact subset $E \Subset \Omega$ and any $p \geq 0$,*

$$\mathrm{Cap}^*(E;\Omega) = \int_\Omega (-h_E^*)^p (dd^c h_E^*)^n = \int_{\bar{E}} (-h_E^*)^p (dd^c h_E^*)^n. \tag{4.3.2}$$

The measure $(dd^c h_E^*)^n$ is carried by $\{z \in \partial E \,|\, h_{E,\Omega}^*(z) = -1\}$.

(2) *Let (K_j) be a decreasing sequence of compact subsets of Ω and $K := \bigcap_j K_j$, then*

$$\lim \mathrm{Cap}(K_j;\Omega) = \mathrm{Cap}(K;\Omega) \quad \textit{hence} \quad \mathrm{Cap}^*(K;\Omega) = \mathrm{Cap}(K;\Omega).$$

(3) *Let $E_j \subset \Omega$ be an increasing sequence and $E := \bigcup_j E_j$. Then*

$$\mathrm{Cap}^*(E;\Omega) = \lim_{j\to+\infty} \mathrm{Cap}^*(E_j;\Omega). \tag{4.3.3}$$

(4) *Let (E_j) be any sequence of subsets of Ω and $E := \bigcup_j E_j$. Then*

$$\mathrm{Cap}^*(E;\Omega) \le \sum_j \mathrm{Cap}^*(E_j,\Omega).$$

Proof. We first prove (4.3.2) when $E = K$ is a compact set. Observe that $\int_K (dd^c h_K^*)^n \le \mathrm{Cap}(K;\Omega)$. Let u be a plurisubharmonic function in Ω with $-1 < u < 0$. We can modify u in a neighborhood of $\partial\Omega$ and assume that it has boundary values 0. It follows from Corollary 4.28 that $N := \{h_K < h_K^*\}$ has zero $(dd^c u)^n$-measure. As $K \subset N \cup \{h_K^* < u\}$, the comparison principle yields

$$\int_K (dd^c u)^n \le \int_{\{h_K^*<u\}} (dd^c u)^n \le \int_{\{h_K^*<u\}} (dd^c h_K^*)^n,$$

hence $\mathrm{Cap}(K;\Omega) \le \int_\Omega (dd^c h_K^*)^n$. Now (4.3.1) yields

$$\int_\Omega (dd^c h_K^*)^n = \int_K (dd^c h_K^*)^n \le \mathrm{Cap}(K;\Omega),$$

thus

$$\mathrm{Cap}(K;\Omega) = \int_\Omega (dd^c h_{K,\Omega}^*)^n = \int_K (dd^c h_{K,\Omega}^*)^n.$$

The set $K \cap \{h_{K,\Omega}^* > -1\} \subset \{h_{K,\Omega} < h_{K,\Omega}^*\}$ is a null set for the measure $(dd^c h_{k,\Omega}^*)^n$ (again by Corollary 4.28). Therefore

$$\begin{aligned}\mathrm{Cap}(K;\Omega) &= \int_K (dd^c h_{K,\Omega}^*)^n \\ &= \int_K (-h_K^*)^p (dd^c h_{K,\Omega}^*)^n = \int_\Omega (-h_K^*)^p (dd^c h_{K,\Omega}^*)^n,\end{aligned}$$

using (4.3.1). This proves (1) for compact sets.

We now prove (2). It follows from Theorem 4.32 that the sequence $h_{K_j}^*$ increases to h_K^* almost everywhere in Ω. By Theorem 4.26 the measures $(-h_{K_j}^*)^p (dd^c h_{K_j}^*)^n$ converge to $(-h_K^*)^p (dd^c h_K^*)^n$ weakly. All these measures have support in the same compact set $K_0 \subset \Omega$, hence

$$\lim_{j\to+\infty} \int_\Omega (-h_{K_j}^*)^p (dd^c h_{K_j}^*)^n = \int_\Omega (-h_K^*)^p (dd^c h_K^*)^n$$

and $\lim \mathrm{Cap}(K_j;\Omega) = \mathrm{Cap}(K;\Omega)$, using (4.3.2).

This property ensures that $\mathrm{Cap}^*(K;\Omega) = \mathrm{Cap}(K;\Omega)$ for any compact set K. Indeed, we can find a decreasing sequence of compact sets (K_j) such that $K \subset K_j \subset K_{j+1}^\circ$ and $K := \bigcap_j K_j$. Thus

$$\mathrm{Cap}^*(K;\Omega) \le \mathrm{Cap}(K_{j+1}^\circ;\Omega) \le \mathrm{Cap}(K_{j+1};\Omega),$$

and the previous property yields

$$\mathrm{Cap}^*(K;\Omega) = \lim_j \mathrm{Cap}(K_{j+1};\Omega) = \mathrm{Cap}(K;\Omega).$$

We now establish (4.3.2) when $E = G \Subset \Omega$ is an open subset. Fix an exhaustive sequence (K_j) of compact subsets of G which increases to G. We let the reader check in Exercise 4.10 that $h^*_{K_j} \downarrow h_G$ on Ω, hence

$$(-h^*_{K_j})^p (dd^c h^*_{K_j,\Omega})^n \longrightarrow (-h^*_G)^p (dd^c h_{G,\Omega})^n.$$

Since these measures are all compactly supported in $\bar{G}$, we infer

$$\begin{aligned}\int_\Omega (-h^*_G)^p (dd^c h_{G,\Omega})^n &= \lim_{j\to+\infty} \int_\Omega (-h^*_{K_j})^p (dd^c h^*_{K_j,\Omega})^n \\ &= \lim_{j\to+\infty} \mathrm{Cap}(K_j;\Omega) = \mathrm{Cap}(G;\Omega).\end{aligned}$$

This proves (4.3.2) when $E \subset \Omega$ is an open subset.

We finally prove (4.3.2) for an arbitrary subset $E \Subset \Omega$. By definition, there is a sequence of open subsets $(O_j)_{j\geq 1} \subset \Omega$ containing E such that $\mathrm{Cap}^*(E;\Omega) = \lim_{j\to+\infty} \mathrm{Cap}(O_j;\Omega)$. Replacing O_j by $\bigcap_{1\leq k\leq j} O_k \subset O_j$, we can assume that $(O_j)_{j\geq 1}$ is decreasing.

It follows from Lemma 4.31 that there exists an increasing sequence $(u_j)_{j\geq 1}$ of negative plurisubharmonic functions in Ω such that $u_j = -1$ on E and $u_j \nearrow h^*_E$ almost everywhere on Ω. Set

$$G_j := O_j \cap \{u_j < -1 + 1/j\}.$$

These are decreasing open subsets of Ω such that $E \subset G_j \subset O_j$ and $u_j - 1/j \leq h_{G_j} \leq h_E$. Thus $h_{G_j} \uparrow h^*_E$ almost everywhere on Ω and

$$(-h_{G_j})^p (dd^c h_{G_j,\Omega})^n \longrightarrow (-h^*_E)^p (dd^c h^*_{E,\Omega})^n.$$

Since these measures are supported in a fixed compact set,

$$\int_\Omega (-h^*_E)^p (dd^c h_{E,\Omega})^n = \lim_{j\to+\infty} \int_\Omega (-h^*_{G_j})^p (dd^c h^*_{G_j,\Omega})^n.$$

On the other hand,

$$\mathrm{Cap}^*(E;\Omega) \leq \lim_{j\to+\infty} \mathrm{Cap}^*(G_j;\Omega) \leq \lim_{j\to+\infty} \mathrm{Cap}^*(O_j;\Omega) = \mathrm{Cap}^*(E;\Omega).$$

The formula (4.3.2) for open subsets therefore yields

$$\int_\Omega (-h^*_E)^p (dd^c h_{E,\Omega})^n = \mathrm{Cap}^*(E;\Omega).$$

Recall that the measure $(dd^c h_E^*)^n$ is supported on ∂E. Set

$$E^* = \{z \in \partial E |\ h_E^*(z) = -1\}.$$

Thus

$$\mathrm{Cap}(E;\Omega) = \int_\Omega (-h_E^*)^p (dd^c h_E^*)^n = \int_{E^*} (dd^c h_E^*)^n + \int_{\partial E \setminus E^*} (-h_E^*)^p (dd^c h_E^*)^n.$$

Since $0 \le -h_E^* < 1$ on $\bar{E} \setminus E^*$, we can let $p \to +\infty$ and use the monotone convergence theorem to obtain

$$\int_\Omega (dd^c h_E^*)^n = \mathrm{Cap}(E;\Omega) = \int_{E^*} (dd^c h_E^*).$$

The measure $(dd^c h_E^*)^n$ is therefore supported on E^*.

We now prove (3). Since the sequence (E_j) is non-decreasing, the plurisubharmonic functions $h_{E_j}^*$ decrease to a plurisubharmonic function h in Ω which satisfies $h_E^* \le h \le 0$ in Ω. Observe that $h = -1$ in $E \setminus N_1$, where $N_1 := \bigcup_j \{h_{E_j}) < h_{E_j}^*\}$. Set

$$N := \{h_E < h_E^*\} \cup N_1 = \{h_E < h_E^*\} \cup \bigcup_j \{h_{E_j}) < h_{E_j}^*\}.$$

Hence, $h = -1$ in $E \setminus N$, hence $h \le h_{E'}^*$ in Ω, where $E' := E \setminus N$. Therefore $h = h_{E'}^*$ and Theorem 4.26 yields

$$(-h_{E_j}^*)^p (dd^c h_{E_j}^*)^n \longrightarrow (-h_{E'}^*)^p (dd^c h_{E'}^*)^n.$$

By lower semi-continuity and (4.3.2),

$$\begin{aligned} \mathrm{Cap}^*(E';\Omega) &= \int_\Omega (-h_{E'}^*)^p (dd^c h_{E'}^*)^n \\ &\le \liminf_j \int_\Omega (-h_{E_j}^*)^p (dd^c h_{E_j}^*)^n = \lim_j \mathrm{Cap}^*(E_j;\Omega) \\ &\le \mathrm{Cap}^*(E;\Omega), \end{aligned}$$

since (E_j) in non-decreasing.

We conclude by checking that $\mathrm{Cap}^*(E';\Omega) = \mathrm{Cap}^*(E;\Omega)$. Observe that $E \subset E' \cup N$, hence by subaddivity

$$\mathrm{Cap}^*(E;\Omega) \le \mathrm{Cap}^*(E';\Omega) + \mathrm{Cap}^*(N;\Omega) = \mathrm{Cap}(E';\Omega),$$

since the set N has zero outer capacity in Ω. The proof of (4.3.3) is complete.

Finally, let us prove (4). We can assume that $\mathrm{Cap}^*(E_j;\Omega) < \infty$ for all j. Fix $\varepsilon > 0$. There exists an open set $G_j \subset \Omega$ such that $E_j \subset G_j$ and

$$\mathrm{Cap}(G_j;\Omega) \le \mathrm{Cap}^*(E_j;\Omega) + \varepsilon 2^{-j-1}.$$

Thus $G := \bigcup_j G_j$ is an open set which contains E and satisfies

$$\mathrm{Cap}^*(E;\Omega) \le \mathrm{Cap}(G;\Omega) \le \sum_j \mathrm{Cap}(G_j;\Omega) \le \sum_j \mathrm{Cap}^*(E_j;\Omega) + \varepsilon,$$

which proves the required inequality. □

Definition 4.35. *The measure $(dd^c h_E^*)^n$ is called the equilibrium measure of E.*

Corollary 4.36. *The set function $E \mapsto \mathrm{Cap}^*(E;\Omega)$ is a Choquet capacity on Ω, hence any Borel subset $B \subset \Omega$ is capacitable,*

$$\mathrm{Cap}^*(E;\Omega) = \sup\{\mathrm{Cap}(K;\Omega);\ K \text{ compact } \subset \Omega\}.$$

Proof. This is a straightforward consequence of Theorems 4.32 and 4.15. □

Example 4.37. *Let $\mathbb{B}(a,r) \subset \mathbb{C}^n$ be the Euclidean open ball centered at a, and of radius $r > 0$. For $0 < r < R$ we have*

$$h_{\bar{\mathbb{B}}(a,r),\mathbb{B}(a,R)}(z) = \max\left\{\frac{\log|z| - \log R}{\log R - \log r}, -1\right\},$$

and

$$\mathrm{Cap}\left(\bar{\mathbb{B}}(a,r);\mathbb{B}(a,R)\right) = \frac{1}{(\log R/r)^n}.$$

We let the reader check these facts in Exercise 4.13.

4.4 Small sets

4.4.1 Pluripolar sets and extremal functions Our purpose in this section is to show thatpluripolar sets are exactly the sets of zero outer capacity.

Definition 4.38. *A set $E \subset \Omega$ is pluripolar in Ω if for all $z_0 \in E$ there exist an open neighborhood U of z_0 in Ω and a plurisubharmonic function $\varphi \in PSH(U)$ such that $E \cap U \subset \{z \in U \,|\, \varphi(z) = -\infty\}$.*

The set E is called complete pluripolar (in Ω) if it coincides with the polar set of a plurisubharmonic function $V \in PSH(\Omega)$.

Examples 4.39.

1. *A proper complex analytic subset of Ω is complete pluripolar.*
2. *A countable union of pluripolar sets is pluripolar.*
3. *The image of a pluripolar subset of Ω under a biholomorphic map $f : \Omega \to \Omega'$ is pluripolar in Ω' (see Exercice 4.8).*

4. *The set* $\mathbb{R}^n \subset \mathbb{C}^n$ *(the real part of* $\mathbb{C}^n$*) is not pluripolar since the restriction of any* $u \in PSH(\Omega)$ *is locally integrable on* $\Omega \cap \mathbb{R}^n$.

5. *Any subset of positive n-dimentional Lebesgue measure in a totally real analytic submanifold of dimension n is non- pluripolar* [Sad76].

6. *There exists a smooth curve in* $\mathbb{C}^n$ *which is non- pluripolar* [DF82] (*see* [CLP05] *for recent developments*).

Lelong asked in [Lel69] wether a (local) pluripolar set E in $\mathbb{C}^n$ can be defined by a global plurisubharmonic function $\phi \in PSH(\mathbb{C}^n)$. This delicate problem has been solved by Josefson using techniques from Padé approximation [Jos78]. We provide in this section an alternative proof using the Monge–Ampère capacity, following [BT82].

Theorem 4.40. *The following properties are equivalent:*

(i) *There exists* $u \in PSH(\Omega)$ *such that* $E \subset \{u = -\infty\}$.
(ii) $\mathrm{Cap}^*(E;\Omega) = 0$.
(iii) $h^*_{E,\Omega} \equiv 0$ *in* Ω.

Proof. The implication (i) $\Longrightarrow$ (ii) follows from Proposition 4.19 and the σ-subadditivity of the capacity.

We now prove (iii) $\Longrightarrow$ (i). Assume that $h^*_{E,\Omega} \equiv 0$ in Ω. By Corollary 4.28 the set $\{h_{E,\Omega} < h^*_{E,\Omega}\}$ is negligible hence there exists $a \in \Omega$ such that $h_{E,\Omega}(a) = 0$. By definition we can find a sequence $(u_j)_{j\in\mathbb{N}} \subset PSH(\Omega)$ such that for all $j \in \mathbb{N}$, $u_j \leq 0$ in Ω, $u_j \leq -1$ on E and $u_j(a) \geq -2^{-j}$. Therefore

$$z \longmapsto u(z) := \sum_{j=0}^{+\infty} u_j$$

is a negative plurisubharmonic function in Ω, such that $u(a) \geq -1$ (hence $u \in PSH(\Omega)$) and $u \mid E \equiv -\infty$, as desired.

To prove (ii) $\Longrightarrow$ (iii) we first assume that $E \Subset \Omega$ is relatively compact. Assume $\mathrm{Cap}^*(E;\Omega) = 0$. It follows from Theorem 4.34.1 that $(dd^c h^*_{E,\Omega})^n = 0$ in Ω. Since $h^*_{E,\Omega} \to 0$ at $\partial\Omega$ (by Theorem 4.34.2), the comparison principle ensures that $h^*_{E,\Omega} \equiv 0$ in Ω.

To treat the general case we fix an increasing sequence $E_j \Subset \Omega$ such that $\bigcup_j E_j = E$. If $\mathrm{Cap}(E;\Omega) = 0$ then $\mathrm{Cap}(E_j;\Omega) = 0$, hence $h^*_{E_j,\Omega} \equiv 0$ for all j. Thus there exists $u_j \in PSH^-(\Omega)$ such that $u_j|E_j \equiv -\infty$. Fix $a \in \Omega$ such that $u_j(a) > -\infty$ for all $j \in \mathbb{N}$ and choose for each $j \in \mathbb{N}$ a constant $\varepsilon_j > 0$ such that $\varepsilon_j u_j(a) \geq -2^{-j}$. Then

$$u := \sum_{j=1}^{+\infty} \varepsilon_j u_j,\ z \in \Omega,$$

is a negative plurisubharmonic function on Ω such that $u(a) \geq -1$ and $u|_E = -\infty$. We infer $h^*_{E,\Omega} \equiv 0$ in Ω: indeed for all $\varepsilon > 0$, εu is a negative plurisubharmonic function in Ω such that $\varepsilon u|E = -\infty \leq -1$. Thus $\varepsilon u \leq h_{E,\Omega} \leq h^*_{E,\Omega}$ in Ω. Letting $\varepsilon \to 0$ we obtain $0 \leq h^*_{E,\Omega}$ almost everywhere in Ω, as the set $\{u = -\infty\}$ is negligible since $u \not\equiv -\infty$. □

We are now ready to prove Josefson's theorem:

Corollary 4.41. *Let $E \subset \mathbb{C}^n$ be a pluripolar set. There exists $V \in \mathcal{L}(\mathbb{C}^n)$ such that $E \subset \{V = -\infty\}$.*

Proof. Applying Theorem 4.40, we find a sequence of subsets E_j and bounded strictly pseudoconvex domains (Euclidean balls) Ω_j such that $E_j \Subset \Omega_j \subset \mathbb{C}^n$, $E = \bigcup_j E_j$ and $\mathrm{Cap}(E_j, \Omega_j) = 0$.

Let m_j be a labelling of the positive integers such that each integer appears infinitely many times. Choose an increasing sequence of positive numbers $R_j > 1$ such that $\Omega_{m_j} \Subset \mathbb{B}_j := \{|z| < R_j\}$ and $\log^+ |z| - \log R_j \leq -2^j$ in E_{m_j}. Observe that

$$0 \leq \mathrm{Cap}^*(E_{m_j}; \mathbb{B}_j) \leq \mathrm{Cap}(E_{m_j}; \Omega_{m_j}),$$

hence $\mathrm{Cap}^*(E_{m_j}; \mathbb{B}_j) = 0$.

It thus follows from Theorem 4.40 that $h^*_{E_{m_j}\mathbb{B}_j} \equiv 0$ in $\mathbb{B}_j$. Applying Lemma 4.31, we find for each j a plurisubharmonic function u_j in $\mathbb{B}_j$ such that $u_j \leq 0$ in $\mathbb{B}_j$, $u_j = -1$ in E_{m_j}, and $\int_{\mathbb{B}_j} (-u_j) d\lambda \leq 2^{-j}$. Set

$$V_j(z) := \max\{u_j(z), 2^{-j}(\log^+ |z| - \log R_j)\}, \quad \text{if } z \in \mathbb{B}_j$$

and $V_j(z) := 2^{-j}(\log^+ |z| - \log R_j)$, if $z \in \mathbb{C}^n \setminus \mathbb{B}_j$. The function V_j is plurisubharmonic in $\mathbb{C}^n$ and $V_j = -1$ in E_{m_j}. Therefore

$$V := \sum_j V_j \in PSH(\mathbb{C}^n).$$

Since $V_j(z) \leq -1$ for infinitely many integers $k = m_j$, we infer

$$E \subset \{V = -\infty\}.$$

Our growth estimates yield $V(z) \leq \log^+ |z|$ in $\mathbb{C}^n$, hence V belongs to the Lelong class. □

4.4.2 Negligible sets

We have shown earlier that the set where the upper envelope of a family of plurisubharmonic functions is not plurisubharmonic has zero inner capacity. We now show the following stronger property:

Theorem 4.42. *Let $\mathcal{P} \subset PSH(\Omega)$ be a family which is locally uniformly bounded from above and set $U := \sup\{u | \, u \in \mathcal{P}\}$. Then $N := \{U < U^*\}$ has zero outer capacity,* $\mathrm{Cap}^*(N; \Omega) = 0$.

Proof. Using Lemma 4.31 we can find an increasing sequence $(u_j)_{j\in\mathbb{N}}$ of plurisubharmonic functions such that and $U := \sup_j u_j$. Recall that psh functions are quasi-continuous. Given $\varepsilon > 0$ we can thus find an open set $G \subset \Omega$ such that $\mathrm{Cap}(G;\Omega) < \varepsilon$ and u and the u_j's are continuous on $F := \Omega \setminus G$.

Fix a compact set $K \subset \Omega$ and observe that for all rational numbers $\alpha < \beta$, the sets

$$K_{\alpha\beta} := K \cap F \cap \{U \leq \alpha < \beta \leq U^*\}$$

are compact, since U is lower semi-continuous in $K \cap F$ and U^* is upper semi-continuous. Observe also that

$$K \cap N \subset G \cup \bigcup_{\alpha<\beta} K_{\alpha\beta}.$$

By subadditivity,

$$\mathrm{Cap}^*(K \cap N;\Omega) \leq \mathrm{Cap}(G;\Omega) + \sum_{\alpha<\beta} \mathrm{Cap}^*(K_{\alpha\beta}).$$

Since U is lower semi-continuous on K, there exists $m \in \mathbb{R}$ such that $U \geq m$ in K. Set $U_m := \sup\{U, m\}$ and observe that

$$K_{\alpha\beta} \subset L_{\alpha\beta} := K \cap F \cap \{U_m \leq \alpha < \beta \leq U_m^*\},$$

and $L_{\alpha\beta}$ is a compact subset. It follows from Theorem 4.34 that $\mathrm{Cap}^*(L_{\alpha\beta}) = \mathrm{Cap}(L_{\alpha\beta}) = 0$ since the set $\{U_m < U_m^*\}$ has inner capacity 0 by Corollary 4.28. □

Corollary 4.43. *Let (u_j) be a sequence of plurisubharmonic functions in Ω which are locally uniformly bounded from above, and set $u := \limsup_j u_j$. Then $N := \{u < u^*\}$ has outer capacity zero,*

$$\mathrm{Cap}^*(N;\Omega) = 0.$$

Proof. Set $v_j := (\sup_{k\geq j} u_k)$ for $j \in \mathbb{N}$. Then v_j^* is plurisubharmonic and the set $N_j := \{v_j < v_j^*\}$ has outer capacity 0, $\mathrm{Cap}^*(N_j;\Omega) = 0$, by the previous result.

Moreover, (v_j) is a decreasing sequence of plurisubharmonic functions which converges to a plurisubharmonic function v satisfying $v \geq u^*$ in Ω. Since $u^* = u = v$ on $\Omega \setminus \bigcup_j N_j$, it follows that $v = u^*$ and $N := \{u < u^*\} \subset \cup_j N_j$, hence $\mathrm{Cap}^*(N;\Omega) = 0$ by subadditivity of the capacity Cap^*. □

Corollary 4.44. *Let $(E_j)_{j\in\mathbb{N}^*}$ be a non-decreasing sequence of subsets of Ω and $E := \bigcup_j E_j$. Then $(h^*_{E_j,\Omega})$ decrease to $h^*_{E,\Omega}$.*

Proof. Observe that $h_j := h^*_{E_j,\Omega}$ is a non-increasing sequence of plurisubharmonic functions which lie between -1 and 0. It converges to a plurisubharmonic function h, bounded between -1 and 0 and such that $h_{E,\Omega} \leq h \leq 0$. Set

$$N := \bigcup_j \{h_{E_j\Omega} < h^*_{E_j\Omega}\}$$

and observe that $h \leq -1$ in $E \setminus N$. It follows from Corollary 4.43 and Theorem 4.40 that there exists $u \in PSH(\Omega)$ such that $u < 0$ and $N \subset P := \{u = -\infty\}$.

Thus, for all $\varepsilon > 0$, the function $h + \varepsilon u$ is plurisubharmonic and negative in Ω and satifies $u \leq -1$ in E. Hence, $h + \varepsilon u \leq h_{E,\Omega}$ and

$$h \leq h_{E,\Omega} \leq h^*_{E,\Omega} \text{ in } \Omega \setminus P.$$

By plurisubharmonicity we infer that $h \leq h^*_{E,\Omega}$ in Ω since P has Lebesgue measure 0, hence $h = h^*_{E,\Omega}$. □

Remark 4.45. *Recall that the Lelong class $\mathcal{L}(\mathbb{C}^n)$ is the set of plurisubharmonic functions v in $\mathbb{C}^n$ with logarithmic growth, i.e., for which there exists $C_v > 0$ such that for all $z \in \mathbb{C}^n$,*

$$v(z) \leq \log^+ \|z\| + C_v.$$

Corollary 4.41 shows that any pluripolar set is contained in the polar locus of a function from the Lelong class $\mathcal{L}(\mathbb{C}^n)$. Given a bounded set $E \Subset \mathbb{C}^n$, its global extremal function is

$$V_E(z) := \sup\{v(z) \mid v \in \mathcal{L}(\mathbb{C}^n), v|E \leq 0\}.$$

This function was introduced by Siciak [Sic69] *(by using polynomials P of arbitrary degree d and functions $v = d^{-1} \log|P|$) and further studied by Zahariuta* [Za76]. *We let the reader establish some of its basic properties in Exercise 4.14. We develop in Chapter 9 a theory that contains the Siciak–Zahariuta functions as a particular case.*

4.5 Exercises

Exercise 4.1. *Let $T \subset \mathbb{R}$ be the set of real numbers having a continued fraction expansion $x = (a_0, a_1, \ldots, a_k, \ldots)$ such that the sequence $(a_0, a_1, \ldots)$ admits a subsequence which is non-decreasing with respect to the divisibility order.*

Show that T is an analytic subset (continuous image of a complete separable metric space) which is not a Borel subset of $\mathbb{R}$. We refer the reader to [Luz27] *for more details.*

Exercise 4.2. *Prove that if c is a capacity in Ω and $\chi : \bar{\mathbb{R}}^+ \to \bar{\mathbb{R}}^+$ is a continuous increasing function, then $\chi \circ c$ is a capacity in Ω.*

Exercise 4.3. *Prove that if c is a precapacity (resp. a Choquet capacity) in Ω, then the associated outer capacity c^* defined by*

$$c^*(E) := \inf\{c(G) |\ G \text{ open}, G \supset E\}$$

is also a precapacity (resp. a Choquet capacity) which is outer regular.

Exercise 4.4. *Let $\mathcal{M}$ be the family of measures*

$$d\mu_\varepsilon := \varepsilon^{-1}\rho(x/\varepsilon)dx$$

for $\varepsilon > 0$, where ρ is a continuous function in $\mathbb{R}$ with compact support in $[-1,+1]$ such that $\int \rho(x)dx = 1$. Prove that its upper envelope c satisfies $c(\{0\}) = 0$, while any neighborhood of 0 has capacity 1, i.e., $c^*(\{0\}) = 1$. Hence it is not outer regular.

Exercise 4.5. *Let $\mathcal{M}$ be a compact (for the weak*-topology) family of Radon measures on Ω. Define the associated precapacity on Borel sets $E \subset \Omega$ by the formula*

$$c_{\mathcal{M}}(E) := sup\{\mu(E)|\ \mu \in \mathcal{M}\}.$$

Show that the associated outer capacity $c^*_{\mathcal{M}}$ is a Choquet capacity on Ω which is outer and inner regular.

Exercise 4.6. *Let $h : X \to Y$ be a continuous map and c a capacity in Y.*

(1) *Prove hat the set function $c_h := h^*c$ defined on subsets of X by $c_h(E) = c(h(E))$ is a capacity in X.*

(2) *Show that c_h is not additive in general, even when c is the exterior measure associated to a Borel measure.*

Exercise 4.7. *Let $v_1, \ldots, v_n$ be plurisubharmonic functions in a domain $\Omega \subset \mathbb{C}^n$ such that $0 \le v_i \le A_i$. Set $\mu = dd^c v_1 \wedge \ldots \wedge dd^c v_n$. Show that*

$$\mu(\cdot) \le A_1 \cdots A_n \operatorname{Cap}(\cdot\,;\Omega).$$

Exercise 4.8. *Let $f : \Omega \to \Omega'$ be a proper holomorphic map from a domain in $\mathbb{C}^n$ into a domain in $\mathbb{C}^m$. Show that if E is a pluripolar subset of Ω, its image $f(E)$ is a pluripolar subset of Ω'.*

Exercise 4.9. *Let $\Omega \subset \mathbb{C}^n$ be an open set, $v \in PSH^-(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$, and $B \subset \Omega$ a Borel subset.*

(1) *Set $s := \sup_B |v|$ and $w := \sup\{v/s, -1\}$. Show that*

$$(dd^c w)^n \ge \mathbf{1}_{\{v/s \ge -1\}}(dd^c v/s)^n.$$

(2) *Deduce the inequality*

$$\int_B (dd^c v)^n \le \left(\sup_B |v|\right)^n \operatorname{Cap}(B;\Omega). \tag{4.5.1}$$

Exercise 4.10. *Let $G \subset \Omega$ be an open set. Let $K_j \subset K_{j+1} \subset G$ be an exhaustive sequence of compact subsets of G, i.e., $\bigcup_j K_j = G$. Show that the relative extremal functions $h^*_{K_j,\Omega}$ decrease pointwise to $h^*_{G,\Omega}$.*

Exercise 4.11. *Prove the following variational characterization of the complex Monge–Ampère capacity in terms of the complex Monge–Ampère energy: for all subsets* $E \Subset \Omega$

$$\mathrm{Cap}^*(E;\Omega) = \inf\left\{\int_\Omega (-u)(dd^c u)^n \,|\, u \in PSH(\Omega)\cap L^\infty(\Omega), u \le -\mathbf{1}_E\right\}.$$

Exercise 4.12. *Let* $K \subset [0,1]^2 \subset \mathbb{R}^2 \simeq \mathbb{C}$ *be the standard Cantor set, so that the set obtained in the* nth *step consists of* 2^{2n} *intervals with edges of length* ℓ_n. *Let* $\mathbb{B}$ *denote the ball of radius* 2 *centered at the origin of* $\mathbb{C}$. *Show that*

$$\mathrm{Cap}(K;\mathbb{B}) > 0 \iff \sum_{n\ge 1} 4^{-n}\log \ell_n > -\infty.$$

We refer the reader to [Carl67, Ra] *for details and more examples of this kind in complex dimension* 1.

Exercise 4.13. *Let* $B(a,r) \subset \mathbb{C}^n$ *be the Euclidean open ball of center* a *and radius* $r > 0$. *Show that for* $0 < r < R$,

$$h_{\bar B(a,r),B(a,R)}(z) = \max\left\{\frac{\log|z| - \log R}{\log R - \log r}, -1\right\},$$

and

$$\mathrm{Cap}(\bar B(a,r); B(a,R) = \frac{1}{(\log R/r)^n}.$$

Exercise 4.14. *Let* $E \Subset \mathbb{C}^n$ *be a bounded subset. Show that*

(1) E *is pluripolar if and only if* $V_E^* \equiv +\infty$ *in* $\mathbb{C}^n$;

(2) *if* E *in non-pluripolar and* $E \subset B_R := \{z \in \mathbb{C}^n; |z| < R\}$, *then* $V_E^* \in \mathcal{L}(\mathbb{C}^n)$ *and there exists a constant* $C > 0$ *such that*

$$\log^+(|z|/R) \le V_E^*(z) \le \log^+|z| + C, \ \text{for all } z \in \mathbb{C}^n;$$

(3) *if* E *in non-pluripolar, then* $V_E^* = 0$ *in* E°,

$$(dd^c V_E^*)^n = 0 \ \text{in}\ \mathbb{C}^n \setminus \bar E, \ \text{and} \ \int_{\mathbb{C}^n}(dd^c V_E)^n = 1.$$

Exercise 4.15.

(1) *Let* u *be a bounded subharmonic function in* $\mathbb{C}$. *Show that* u *can be written as a countable sum of continuous subharmonic functions* (*this result does not hold for plurisubharmonic functions, see* [Kol01] *for a counterexample*).

(2) *Let* μ *be a probability measure on* $\mathbb{C}$. *Use* (1) *to prove that* μ *does not charge polar sets if and only if it can be decomposed as*

$$\mu = \sum_{j\ge 1}\mu_j,$$

where $\mu_j = \Delta u_j$ *is the Laplacian of a continuous subharmonic function.*

5 The Dirichlet problem

Let $\Omega \Subset \mathbb{C}^n$ be a bounded strongly pseudoconvex domain in $\mathbb{C}^n$. Given $\varphi \in C^0(\partial\Omega)$ and $0 \leq f \in C^0(\bar{\Omega})$, we consider the Dirichlet problem for the complex Monge–Ampère operator:

$$\begin{cases} u \in PSH(\Omega) \cap C^0(\bar{\Omega}) \\ (dd^c u)^n = f\beta^n \quad \text{in } \Omega, \\ u = \varphi \qquad\qquad\quad \text{on } \partial\Omega. \end{cases} \tag{5.0.1}$$

When $n = 1$ this is the classical Dirichlet problem for the Laplace operator. In this case one can find an explicit formula for the solution, when the domain is sufficiently regular, generalizing the Poisson–Jensen formula in the unit disc proved in Chapter 1.

For more general domains, as well as for the higher dimensional setting, one uses the method of upper envelopes due to Perron and the comparison principle to build the solution and show it is unique. Namely, we consider

$$U(z) = U_{\Omega,\varphi,f}(z) := \sup\{u(z) |\, u \in \mathcal{B}(\Omega, \varphi, f)\}, \tag{5.0.2}$$

where $\mathcal{B} = \mathcal{B}(\Omega, \varphi, f)$ is the class of subsolutions,

$$\mathcal{B} := \{u \in PSH \cap L^\infty(\Omega) |\, (dd^c u)^n \geq f\beta^n \text{ in } \Omega \text{ and } u^* \leq \varphi \text{ on } \partial\Omega\}.$$

Our goal is to show that U is the unique solution to (5.0.1).

When $f = 0$, Bremermann [Bre59] has shown that the envelope U is plurisubharmonic and has the right boundary values, by constructing appropriate barriers (this is where the pseudoconvexity assumption is used). Later on Walsh [Wa68] proved that U is continuous up to the boundary. Then Bedford and Taylor showed in [BT76] that the complex Monge–Ampère measure $(dd^c U)^n$ is well defined, and U solves the Dirichlet problem (5.0.1).

We give in this chapter a complete proof of these results using simplifications due to Demailly [Dem91], as well as ideas from optimal control and viscosity theory developed recently in [EGZ11]. These methods allow a new description of the Perron–Bremermann envelope using Laplace operators associated to a family of constant Kähler metrics on Ω, following an observation by Gaveau [Gav77]. This avoids the general measure-theoretic construction of Goffman and Serrin [GS64] used in [BT76].

5.1 The Dirichlet problem for the Laplace operator

We have shown in Chapter 1 that the Poisson transform solves explicitly the Dirichlet problem for theLaplace equation in $\mathbb{R}^2$ (complex dimension $n = 1$). We study here the Laplace operator in $\mathbb{R}^N$ with $N \geq 3$. This will help us later on in getting some information on plurisubharmonic functions in domains of $\mathbb{C}^n$, with $N = 2n$.

5.1.1 The maximum principle We fix here an integer $N \geq 3$ and a domain $\Omega \subset \mathbb{R}^N$. Let $x = (x_1, \ldots, x_N)$ denote the canonical coordinates in $\mathbb{R}^N$. For $x = (x_1, \ldots, x_N) \in \mathbb{R}^N$ and $y = (y_1, \ldots, y_N) \in \mathbb{R}^N$ we set

$$x \cdot y := \sum_{j=1}^{N} x_j y_j \quad \text{and} \quad |x|^2 = x \cdot x = \sum_{j=1}^{N} x_j^2.$$

The Laplace operator in $\mathbb{R}^N$ is defined by

$$\Delta = \sum_{i=1}^{N} \frac{\partial^2}{\partial x_i^2}.$$

Harmonic functions in a domain $\Omega \subset \mathbb{R}^N$ are those which satisfy the Laplace equation $\Delta h = 0$ in Ω (in the smooth or weak sense of distributions). They can also be characterized as continuous functions which satisfy the spherical mean value property.

Definition 5.1. *A function $u : \Omega \subset\to [-\infty, +\infty[$ is subharmonic if it is upper semi-continuous, and for any ball $\mathbb{B}(a, r) \Subset \Omega$,*

$$u(a) \leq \frac{1}{\sigma_{N-1}} \int_{|\xi|=1} u(+a + r\xi) d\sigma(\xi),$$

where $d\sigma$ is the area measure on the unit sphere $\mathbb{S} := \{x \in \mathbb{R}^N \,\big|\, |x| = 1\}$.

It follows from Chapter 1 that plurisubharmonic functions in a domain $\Omega \subset \mathbb{C}^n$ are subharmonic as functions of $2n$-real variables.

We denote by $SH(\Omega)$ the positive cone of subharmonic functions in the domain Ω which are not identically $-\infty$. It follows from the submean value inequalities that

$$SH(\Omega) \subset L^1_{\text{loc}}(\Omega).$$

Proposition 5.2. *Let u be a subharmonic function in a bounded domain $\Omega \Subset \mathbb{R}^N$ such that $\limsup_{x \to \partial\Omega} u(x) \leq 0$. Then $u \leq 0$ in Ω. Moreover, for all $x \in \Omega$,*

$$u(x) < \sup_{\partial\Omega} u,$$

unless u is constant in Ω.

The proof of this maximum principle is similar to the one given in Chapter 1.

5.1.2 Green functions For $N \geq 3$, the Newton kernel

$$K_N(x) := \frac{-1}{(N-2)\sigma_{N-1}}|x|^{2-N}$$

is a locally integrable function in $\mathbb{R}^N$ which satisfies

$$\Delta K_N = \delta_0$$

in the sense of distributions in $\mathbb{R}^N$, where δ_0 is the Dirac measure at the origin. In particular, K_N is subharmonic in $\mathbb{R}^N$ and harmonic in $\mathbb{R}^N \setminus \{0\}$. It is the *fundamental solution* to the Laplace operator in $\mathbb{R}^N$.

Definition 5.3. *The Green function for the unit ball $\mathbb{B}$ is*

$$G(x, y) = G_{\mathbb{B}}(x, y) := K_N(x - y) - K_N(|y|x - y/|y|),$$

where $(x, y) \in \mathbb{B} \times \mathbb{B}$.

Observe that G is well defined in $\mathbb{B} \times \mathbb{B}$ and has the same singularities as $K_N(x - y)$, since the second term is smooth. It satisfies the following properties:

(1) For all $y \in \mathbb{B}$, $x \mapsto G(x, y)$ is subharmonic in $\mathbb{B}$;

(2) $G(x, y) = G(y, x)$ in $\mathbb{B} \times \mathbb{B}$;

(3) $G < 0$ in $\mathbb{B} \times \mathbb{B}$, and $G(x, y) = 0$ if $(x, y) \in \partial\mathbb{B} \times \mathbb{B}$;

(4) for all $y \in \mathbb{B}$, $\Delta_x G(x, y) = \delta_y$.

Definition 5.4. *The function $x \mapsto G(x, y)$ is called the Green function of the ball $\mathbb{B}$ with pole at y.*

It follows from the maximum principle for the Laplace operator that G is the unique function satisfying these properties.

The explicit formula for the Newton kernel K_N yields

$$G(x, y) = \frac{-1}{(N-2)\sigma_{N-1}} \left(\|x - y|^{2-N} - (1 - 2x \cdot y + |x|^2|y|^2)^{1-N/2} \right).$$

Recall that if u, v are smooth functions in $\mathbb{B}$, then

$$\int_{\mathbb{B}} (u\Delta v - v\Delta u) d\lambda = \int_{\partial\mathbb{B}} \left(u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{v \partial \nu} \right) d\sigma.$$

Here $\partial/\partial\nu$ is the derivative along the outward normal vector ν in $\partial\mathbb{B}$ and $d\sigma$ is the Euclidean area measure on $\partial\mathbb{B} = \mathbb{S}$. To get a representation formula for a subharmonic function u, we apply this formula with $v := G(x, \cdot)$ and $x \in \mathbb{B}$ fixed. We thus consider the Poisson kernel

$$P(x, y) := \partial G(x, y)/\partial\nu(y), \quad (x, y) \in \mathbb{B} \times \partial\mathbb{B}.$$

An easy computation shows that

$$P(x,y) := \frac{1}{(N-2)\sigma_{N-1}} \frac{1-|x|^2}{|x-y|^N}, \quad (x,y) \in \mathbb{B} \times \partial\mathbb{B}.$$

Proposition 5.5. *Let $u \in SH(\mathbb{B}) \cap C^0(\bar{\mathbb{B}})$. Then for all $x \in \mathbb{B}$,*

$$u(x) = \int_{\mathbb{S}} u(y) P(x,y) d\sigma(y) + \int_{\mathbb{B}} G(x,y) d\mu_u(y), \tag{5.1.1}$$

where $\mu_u := \Delta u$ is the Riesz measure of u.

In particular, if u is harmonic in $\mathbb{B}$ and continuous in $\bar{\mathbb{B}}$, then

$$u(x) = \int_{\mathbb{S}} u(y) P(x,y) d\sigma(y). \tag{5.1.2}$$

The proof is left as Exercise 5.2. We can thus solve the Dirichlet problem for the homogeneous Laplace equation with continuous boundary values:

Theorem 5.6. *Fix $\varphi \in \mathcal{C}^0(\partial\mathbb{B})$. The Poisson transform of φ,*

$$P_\varphi(x) := \int_{|y|=1} \varphi(y) P(x,y) d\sigma(y), \quad x \in \mathbb{B}, \tag{5.1.3}$$

is harmonic in $\mathbb{B}$, continuous in $\bar{\mathbb{B}}$ and satisfies $P_\varphi(x) = \varphi(x)$ for $x \in \mathbb{S}$.

The proof is identical to the one for the unit disc (see Chapter 1).

5.1.3 A characterization of subharmonic functions

The following characterization of subharmonic functions will be quite useful:

Proposition 5.7. *Let $u : \Omega \to \mathbb{R}^N$ be an upper semi-continuous function in a domain $\Omega \subset \mathbb{R}^N$. The following conditions are equivalent:*

(1) the function u is subharmonic in Ω;

(2) $\Delta q(x_0) \geq 0$ for all $x_0 \in \Omega$ and all C^2-smooth functions q in a small ball B of center x_0 such that $u \leq q$ in B and $u(z_0) = q(z_0)$;

(3) $u \leq h$ in B, for all balls $B \Subset \Omega$ and all functions $h : \bar{B} \to \mathbb{R}$ continuous in $\bar{B}$ and harmonic in B such that $u \leq h$ in ∂B.

A C^2-smooth function q in a ball $B = B(x_0, r)$ such that $u \leq q$ in B and $u(x_0) = q(x_0)$ is called an upper test function for u at the point x_0.

Proof. We first prove (1) $\Longrightarrow$ (2). Assume that u is subharmonic and fix $x_0 \in \Omega$. If there exists a C^2-smooth upper test function q at x_0 such that $\Delta q(x_0) < 0$, then for $\varepsilon > 0$ small enough the function $x \mapsto u(x) - q(x) - \varepsilon|x - x_0|^2$ is subharmonic in a

small ball $B(x_0, r)$, equal to 0 at x_0, and negative for $0 < |x - x_0| < r$ small enough. This contradicts the maximum principle.

We now prove (2) $\Longrightarrow$ (3). Let h be a harmonic function in some ball $B(a, r) \Subset \Omega$, continuous on $\bar{B}(a, r)$, and such that $u \leq h$ in $\partial B(a, r)$. Fix $\varepsilon > 0$ and observe that

$$u(x) \leq h_\varepsilon(x) := h(x) - \varepsilon(|x - a|^2 - r^2) \text{ on } \partial B(a, r).$$

By upper semi-continuity, there exists $x_0 \in \bar{B}(a, r)$ such that

$$u(x_0) - h_\varepsilon(x_0) = \max_{\bar{B}(a,r)} (u - h_\varepsilon).$$

If $x_0 \in B$, we can fix $B(z_0, s) \subset B(a, r)$ so that $q := h_\varepsilon - h_\varepsilon(x_0) + u(x_0)$ is a smooth upper test function for u at x_0 such that $\Delta q(x_0) = -2n\varepsilon < 0$, a contradiction. Thus $x_0 \in \partial B$ and $u \leq h_\varepsilon$ in $B(a, r)$ for all $\varepsilon > 0$. We infer that $u \leq h$ in $B(a, r)$ by letting $\varepsilon \to 0$.

We finally prove (3) $\Longrightarrow$ (1). Fix $a \in \Omega$ and $r > 0$ such that $\bar{B}(a, r) \subset \Omega$. Let (φ_j) be a sequence of continuous functions decreasing to u in ∂B. It follows from Theorem 5.6 that there exists a harmonic function h_j in B continuous in $\bar{B}$ such that $h_j = \varphi_j$ in ∂B. Property (3) yields $u \leq h_j$ in B for any $j \geq 1$. Thus

$$u(a) \leq h_j(a) = \frac{1}{\sigma_{N-1}} \int_{|y|=1} \varphi_j(a + ry) d\sigma(y).$$

We infer ($j \to +\infty$) that u satisfies the submean value inequalities. □

The implication (1) $\Longrightarrow$ (2) is a soft version of the maximum principle. Property (2) can be used to define plurisubharmonicity in the sense of viscosity as we explain in the sequel.

5.2 The Perron–Bremermann envelope

5.2.1 A characterization of plurisubharmonicity Let H_n^+ denote the set of all semi-positive Hermitian $n \times n$ matrices. We set

$$\dot{H}_n^+ := \{H \in H_n^+ |\ \det H = n^{-n}\}.$$

The following observation of Gaveau [Gav77] is quite useful:

Lemma 5.8. *Fix $Q \in H_n^+$. Then*

$$(\det Q)^{\frac{1}{n}} = \inf\{\mathrm{tr}(H\,Q) |\ H \in \dot{H}_n^+\}$$

Proof. Every matrix $H \in \dot{H}_n^+$ has a square root which we denote by $H^{1/2} \in H_n^+$. Thus $H^{1/2}QH^{1/2} \in H_n^+$. Diagonalizing the latter and using the arithmetic-geometric inequality, we get

$$(\det Q)^{\frac{1}{n}}(\det H)^{\frac{1}{n}} = (\det(H^{1/2}QH^{1/2}))^{\frac{1}{n}} \leq \frac{1}{n}\mathrm{tr}(H^{1/2}QH^{1/2}).$$

Therefore, $(\det Q)^{\frac{1}{n}}\,(\det H)^{\frac{1}{n}} \leq \frac{1}{n}\,\mathrm{tr}(QH)$, hence

$$(\det Q)^{\frac{1}{n}} \leq \inf\{\mathrm{tr}(HQ)|\ H \in \dot{H}_n^+\}.$$

Suppose now that $Q \in H_n^+$ is positive. There exist an invertible Hermitian matrix P and a diagonal matrix $A = (\lambda_i)$ with positive entries $\lambda_i > 0$ such that $Q = PAP^{-1}$. Set

$$\alpha_i = \frac{(\prod_i \lambda_i)^{\frac{1}{n}}}{n\lambda_i}.$$

Observe that $H = (\alpha_i) \in \dot{H}_n^+$ and $(\det A)^{\frac{1}{n}} = \mathrm{tr}(AH)$, hence

$$(\det Q)^{\frac{1}{n}} = (\det A)^{\frac{1}{n}} = \mathrm{tr}(HA) = \mathrm{tr}(HPAP^{-1}) = \mathrm{tr}(H'Q),$$

where $H' = PHP^{-1} \in \dot{H}_n^+$.

If Q is merely semi-positive we consider $Q_\varepsilon := Q + \varepsilon I_n$, $\varepsilon > 0$, and apply the previous argument to obtain

$$(\det Q_\varepsilon)^{\frac{1}{n}} = \inf\{\mathrm{tr}(HQ_\varepsilon)|\ H \in \dot{H}_n^+\} \geq \inf\{\mathrm{tr}(HQ)|\ \dot{H}_n^+\}.$$

We conclude by letting $\varepsilon \to 0$. □

For $H \in \dot{H}_n^+$, we consider

$$\Delta_H := \sum_{j,k=1}^{n} h_{k\bar{j}} \frac{\partial^2}{\partial z_j\, \partial \bar{z_k}}, \tag{5.2.1}$$

the Laplace operator associated to the (constant) Kähler metric defined by $\bar{H}^{-1}$ in $\mathbb{C}^n$. The previous lemma yields the following interesting characterization of plurisubharmonicity:

Proposition 5.9. *Let $u : \Omega \to [-\infty, +\infty[$ be an upper semi-continuous function. The following properties are equivalent:*

(i) *The function u is plurisubharmonic in Ω;*

(ii) *$dd^c q(z_0) \geq 0$ for all $z_0 \in \Omega$ and all functions that are $q\,C^2$ in a neighborhood B of z_0 such that $u \leq q$ in B and $u(z_0) = q(z_0)$;*

(iii) *u is subharmonic and for all $H \in \dot{H}_n^+$, $\Delta_H u \geq 0$ in the sense of distributions.*

Proof. We first prove (i) $\Longrightarrow$ (ii). Assume that is u is plurisubharmonic and fix $z_0 \in \Omega$. If there exists a C^2-smooth upper test q at z_0 such that $dd^c q(z_0)$ is not semi-positive, there is a direction $\xi \in \mathbb{C}^n$, $\xi \neq 0$, such that $\sum_{j,k} \xi_j \bar{\xi}_k \frac{\partial^2 q}{\partial z_j \partial \bar{z}_k}(z_0) < 0$. Then for $\varepsilon > 0$ small enough

$$\tau \longmapsto u(z_0 + \tau\xi) - q(z_0 + \tau\xi) - \varepsilon|\tau|^2$$

is subharmonic in a neighborhood of the origin, where it reaches a local maximum, contradicting the maximum principle.

We now prove (ii) $\Longrightarrow$ (iii). Let h be a function that is harmonic in a ball $B = B(a,r) \Subset \Omega$, continuous on $\bar{B}(a,r)$, and such that $u \leq h$ in $\partial B(a,r)$. We claim that $u \leq h$ in B. Fix $\varepsilon > 0$ and observe that

$$u(x) \leq h_\varepsilon(x) := h(x) - \varepsilon(|x-a|^2 - r^2) \quad \text{on } \partial B(a,r).$$

By upper semi-continuity, there exists $x_0 \in \bar{B}(a,r)$ such that

$$u(x_0) - h_\varepsilon(x_0) = \max_{\bar{B}(a,r)} (u - h_\varepsilon).$$

If $x_0 \in B$, we can choose a small ball $B(x_0, s) \subset B$ so that

$$q := h_\varepsilon - h_\varepsilon(x_0) + u(x_0)$$

is an upper test for u at x_0 which satifies $\Delta q(x_0) = -2n\varepsilon < 0$, a contradiction. Therefore, $x_0 \in \partial B$ and $u \leq h_\varepsilon$ in $B(a,r)$ for all $\varepsilon > 0$. We infer that $u \leq h$ in $B(a,r)$ as $\varepsilon \to 0$.

This proves that u is subharmonic in Ω by Proposition 5.7, hence $\Delta u \geq 0$ in Ω in the sense of distributions. To prove that $\Delta_H u \geq 0$ for all $H \in \dot{H}_n^+$, we use the following observations:

(a) Let $T : \mathbb{C}^n \to \mathbb{C}^n$ be a $\mathbb{C}$-linear isomorphism and $q : \mathbb{C}^n_\zeta \to \mathbb{C}^n_z$ be a C^2-smooth function in a neighborhood of a point z_0. Set $z = T(\zeta)$ and $q_T(\zeta) := q(z) = q(T(\zeta))$. Then q_T is C^2-smooth function in a neighborhood of $\zeta_0 := T^{-1}(z_0)$ and

$$\Delta q_T(\zeta) = \sum_{j=1}^n \frac{\partial^2 q_T(\zeta)}{\partial \zeta_j \partial \bar{\zeta}_k} = \operatorname{tr}(T^* Q(z) T),$$

where T^* denotes the complex conjugate transpose of $T := \left(\frac{\partial z_j}{\partial \zeta_k}\right)$ and $Q(z) := \left(\frac{\partial^2 q}{\partial z_j \partial \bar{z}_k}(z)\right)$ is the complex hessian of q at z. If $H \in \dot{H}_n^+$, we can find a Hermitian positive matrix T such that $T^* T = H$, and then one gets

$$\Delta q_T(\zeta) = \Delta_H q(z).$$

We leave the details as Exercise 5.4.

(b) Fix $u : \Omega \to [-\infty, +\infty[$ an upper semi-continuous function and $z_0 \in \Omega$. Then q is an upper test function for u at z_0 iff $\tilde{q} := q_T := q \circ T$ is an upper test function for u_T at the point $\zeta_0 := T^{-1}(z_0)$.

Observation (b) shows that the condition (ii) is invariant under complex linear changes of coordinates. Observation (a) and the proof preceding it show that (ii) implies (iii).

We finally show that (iii) $\Longrightarrow$ (i). Assume first that u is smooth. Condition (iii) means that the complex hessian matrix A of u at $z_0 \in \Omega$ is a Hermitian matrix that satisfies $\mathrm{tr}(HA) \geq 0$ for all $H \in H_n$. We infer, by diagonalizing A, that A is a semi-positive Hermitian matrix. Thus u is plurisubharmonic in Ω.

To treat the general case we regularize $u_\varepsilon = u \star \rho_\varepsilon$ (see Chapter 1) and obtain smooth functions satisfying (iii) since

$$\Delta_H u_\varepsilon = (\Delta_H u) \star \rho_\varepsilon \quad \text{in } \Omega_\varepsilon$$

by linearity. Thus, u_ε is plurisubharmonic in Ω_ε. Since u is subharmonic in Ω, it follows that u_ε decreases to u as ε decreases to 0, hence u is plurisubharmonic in Ω. □

5.2.2 Perron envelopes

5.2.2.1 The viscosity point of view We establish here some useful facts which are also the first steps towards a viscosity approach to solving complex Monge–Ampère equations. We shall develop these concepts further in Chapter 6.

Proposition 5.10. *Let $u \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$ and $0 \leq f \in C^0(\Omega)$. The following conditions are equivalent :*

(1) $(dd^c u)^n \geq f\beta^n$;

(2) $\Delta_H u \geq f^{1/n}$ *for all* $H \in \dot{H}_n^+$.

This equivalence has been observed by Blocki [Blo96, Theorem 3.10] when u continuous, by using a slightly different argument.

Proof. We first prove (2) $\Longrightarrow$ (1). Suppose that $u \in C^2(\Omega)$. It follows from Lemma 5.8 that

$$\Delta_H u \geq f^{1/n}, \quad \forall H \in \dot{H}_n^+$$

is equivalent to

$$\left(\det\left(\frac{\partial^2 u}{\partial z_j \partial \bar{z}_k}\right)\right)^{1/n} \geq f^{1/n},$$

which is itself equivalent to $(dd^c u)^n \geq f\beta^n$. All these inequalities hold pointwise in Ω.

When u is not smooth, we fix $H \in \dot{H}_n^+$ and let (χ_ϵ) be standard mollifiers. The functions $u_\epsilon := u \star \chi_\epsilon$ are plurisubharmonic in Ω_ε and decrease to u as ε

decreases to 0. Since the condition (2) is linear, we infer that $\Delta_H u_\epsilon \geq (f^{1/n})_\epsilon$ pointwise in Ω_ε. We can use the first case, since u_ϵ is smooth, obtaining $(dd^c u_\epsilon)^n \geq ((f^{1/n})_\epsilon)^n \beta^n$ pointwise in Ω_ε. Letting $\epsilon \searrow 0$ and applying the convergence theorem for the complex Monge–Ampère operator, we get that $(dd^c u)^n \geq f\beta^n$ weakly in Ω.

We now prove that (1) $\Longrightarrow$ (2). Fix $x_0 \in \Omega$ and a C^2-smooth function q in a neighborhood B of x_0 such that $u \leq q$ in B and $u(x_0) = q(x_0)$. We know from Proposition 5.9 that $dd^c q(x_0) \geq 0$. We claim that $(dd^c q(x_0))^n \geq f(x_0)\beta^n$. Suppose by contradiction that $(dd^c q)^n_{x_0} < f(x_0)\beta^n$, and set

$$q^\epsilon(x) = q(x) + \epsilon\left(\|x - x_0\|^2 - \frac{r^2}{2}\right).$$

If $0 < \epsilon < 1$ is small, then $0 < (dd^c q^\epsilon(x_0))^n < f(x_0)\beta^n$. Since f is lower semi-continuous at x_0, there exists $r > 0$ such that

$$(dd^c q^\epsilon(x))^n \leq f(x)\beta^n \quad \text{in } \mathbb{B}(x_0, r)$$

It follows that $(dd^c q^\epsilon)^n \leq (dd^c u)^n$ in $\mathbb{B}(x_0, r)$ and $q^\epsilon \geq q \geq u$ on $\partial\mathbb{B}(x_0, r)$. The comparison principle implies that $q^\epsilon \geq u$ on $\mathbb{B}(x_0, r)$. But $q^\epsilon(x_0) = q(x_0) - \epsilon\frac{r^2}{2} = u(x_0) - \epsilon\frac{r^2}{2} < u(x_0)$, a contradiction.

It follows from Proposition 5.9 that for any $x_0 \in \Omega$, and any upper test q for u at x_0, we have $\Delta_H q(x_0) \geq f^{1/n}(x_0)$ for all $H \in \dot{H}_n^+$.

If f is positive and smooth, there exists $g \in C^\infty(\bar{\Omega})$ such that $\Delta_H g = f^{1/n}$. Thus $h = u - g$ is Δ_H-subharmonic by Proposition 5.7, i.e., $\Delta_H h \geq 0$, hence $\Delta_H u \geq f^{1/n}$ for any $H \in \dot{H}_n^+$.

If f is positive and merely continuous, we observe that

$$f = \sup\{g |\ g \in C^\infty(\bar{\Omega}),\ f \geq g > 0\},$$

hence $(dd^c u)^n \geq f\beta^n \geq g\beta^n$ for any such g. The previous case yields $\Delta_H u \geq g^{1/n}$. We infer that $\Delta_H u \geq f^{1/n}$ for all $H \in \dot{H}_n^+$.

Assume finally that f is merely continuous and semipositive. Observe that $u^\epsilon(z) = u(z) + \epsilon\|z\|^2$ satisfies

$$(dd^c u^\epsilon)^n \geq (f + \epsilon^n)\beta^n.$$

It follows from the previous case that for all $H \in \dot{H}_n^+$,

$$\Delta_H u^\epsilon \geq (f + \epsilon^n)^{1/n}.$$

Letting ϵ decrease to 0 yields $\Delta_H u \geq f^{1/n}$ for all $H \in \dot{H}_n^+$. □

5.2.2.2 Perron–Bremermann envelopes The previous proposition allows us to reinterpret the Perron–Bremermann envelope by using the Laplace operators Δ_H. Consider

$$\mathcal{V} = \{v \in PSH \cap L^\infty(\Omega) |\ v|_{\partial\Omega} \leq \varphi \quad \text{and} \quad \Delta_H v \geq f^{1/n}\ \forall H \in \dot{H}_n^+\}.$$

Proposition 5.11. *The class $\mathcal{V}$ is non-empty, stable under maxima, and bounded from above in Ω. The Perron–Bremermann envelope*

$$U_{\Omega,\varphi,f}(z) = \sup\{v(z)|\ v \in \mathcal{V}\} \tag{5.2.2}$$

is plurisubharmonic in Ω.

Proof. Let ρ be a strictly plurisubharmonic defining function for Ω. Choose $A > 0$ big enough so that $Add^c\rho \geq M^{1/n}\beta$, where $M := \|f\|_{L^\infty(\Omega)}$. We use here that Ω is bounded and ρ is strictly plurisubharmonic near $\bar{\Omega}$. Fix $B > 0$ so large that $-B \leq \varphi \leq B$. Then $v_0 := A\rho - B \in \mathcal{V}$, since

$$\Delta_H v_0 \geq A\Delta_H \rho \geq M^{1/n} \geq f^{1/n},$$

for all $H \in \dot{H}_n^+$. Thus $\mathcal{V} \neq \emptyset$.

Since φ is bounded from above by B, the maximum principle shows that all functions in $\mathcal{V}$ are bounded from above by B. It follows that $U := U_{\Omega,\varphi,f}$ is well defined and given by

$$U(z) = \sup\{v(z)|\ v \in \mathcal{V}_0\},\ z \in \Omega,$$

where

$$\mathcal{V}_0 := \{v \in \mathcal{V}|\ v_0 \leq v \leq M\}.$$

We claim that $\mathcal{V}_0 \subset L^1(\Omega)$ is compact. Indeed, let (v_j) be a sequence in $\mathcal{V}_0$. This is a bounded sequence of plurisubharmonic functions in $L^1(\Omega)$. Thus there exists a subsequence (w_j) such that $w_j \to w$ in $L^1(\Omega)$, where w is plurisubharmonic and satisfies $w = (\limsup_j w_j)^*$ in Ω. We infer that $v_0 \leq w \leq B$ and for all $H \in \dot{H}_n^+$,

$$f^{1/n} \leq \Delta_H w_j \longrightarrow \Delta_H w,$$

hence $\Delta_H w \geq f^{1/n}$. Therefore, $w \in \mathcal{V}_0$, which proves the claim. It follows that U is plurisubharmonic in Ω.

We now prove that the class $\mathcal{V}$ is stable under maxima, that is, if $u, v \in \mathcal{V}$, then $\max\{u, v\} \in \mathcal{V}$. It suffices to show that for all $H \in \dot{H}_n^+$

$$\Delta_H \max\{u, v\} \geq \min\{\Delta_H u, \Delta_H v\} \tag{5.2.3}$$

Indeed, let $\mu := \min\{\Delta_H u, \Delta_H v\}$ in the sense of Radon measures in Ω and suppose that $\mu(\{z|\ u(z) = v(z)\}) = 0$. The local maximum principle shows that $\Delta_H \max\{u, v\} \geq \mu$ in the sense of Borel measures in the Borel set $\Omega' = \{u \neq v\}$. Since $\mu(\Omega \setminus \Omega') = 0$, we get

$$\Delta_H \max\{u, v\} \geq \mu.$$

When $\mu(\{z|\ u(z) = v(z)\}) \neq 0$ we replace v by $v + \epsilon$, and observe that $\mu(\{z|u(z) = v(z)+\epsilon\}) \neq 0$ for at most countably many ϵ's. The previous case yields $\Delta_H \max\{u, v + \epsilon\} \geq \min\{\Delta_H u, \Delta_H v\} = \mu$ for those ε's. Since $\Delta_H \max\{u, v + \epsilon\}$ converges to $\Delta_H \max\{u, v\}$, we obtain (5.2.3). □

5.2.3 Continuity of the envelope

Theorem 5.12. *Let $0 \le f \in C(\bar\Omega)$ be a continuous function in $\bar\Omega$ and $\varphi \in C(\partial\Omega)$. The Perron–Bremermann envelope*

$$U = \sup\{v |\ v \in \mathcal{V}(\Omega, \varphi, f)\}$$

is a continuous plurisubharmonic function which belongs to $\mathcal{V}(\Omega, \varphi, f)$ and satisfies $U = \varphi$ on $\partial\Omega$.

If $\varphi \in \mathcal{C}^{1,1}(\partial\Omega)$, then the modulus of continuity of U satisfies

$$\omega_U(\delta) \le C\delta\|\varphi\|_{\mathcal{C}^{1,1}(\partial\Omega)} + B\omega_{f^{1/n}}(\delta),$$

where B and C depend only on the geometry of the domain Ω. In particular U is Lipschitz on $\bar\Omega$ whenever $f^{1/n}$ is Lipschitz on $\bar\Omega$.

Proof. The proof proceeds in several steps.

Step 1. We first show that $U \in \mathcal{V}$. We have already shown in Proposition 5.11 that U is plurisubharmonic and bounded. It follows from Choquet's lemma that there exists a sequence (v_j) in $\mathcal{V}(\Omega, \varphi, f)$ such that

$$U = (\sup_j v_j)^* \quad \text{in } \Omega.$$

By Proposition 5.11 (stability of the family under max), we can further assume that (v_j) in non-decreasing. Fix $H \in \dot H_n^+$. Since $\Delta_H v_j \ge f^{1/n}$ for all j and $v_j \to u$ in $L^1(\Omega)$, we infer that $\Delta_H u \ge f^{1/n}$, hence $U \in \mathcal{V}$.

Step 2. Construction of barriers at boundary points. Let ρ be a strictly plurisubharmonic defining function for Ω, $dd^c\rho \ge c_0 dd^c|z|^2$, for some $c_0 > 0$. Fix $\varepsilon > 0$ and let ψ be a $C^{1,1}$-smooth function in $\bar\Omega$ such that $|\psi - \varphi| \le \varepsilon$ on $\partial\Omega$. For $K > 1$ large enough,

$$v_0 := K\rho + \psi - 2\varepsilon,$$

is a smooth function near $\bar\Omega$ such that $v_0 \le \varphi$ on $\partial\Omega$, and

$$dd^c v_0 = K dd^c\rho + dd^c\psi \ge M^{1/n}\beta,$$

where $M = \sup_\Omega f$. Observe that K depends on the $C^{1,1}$-bound of ψ. Fix $H \in \dot H_n^+$. Then

$$\Delta_H v_0 \ge f^{1/n}.$$

Therefore, v_0 belongs to the class $\mathcal{V}$ and $v_0 \le U$. It follows that

$$\liminf_{z\to\zeta} U(z) \ge \psi(\zeta) - 2\varepsilon,$$

for all $\zeta \in \partial\Omega$. Letting ψ converge to φ and then $\varepsilon \to 0$, we obtain

$$\liminf_{z\to\zeta} U(z) \ge \varphi(\zeta).$$

The same argument shows that

$$w_0 := K\rho - \psi - 2\varepsilon$$

is plurisubharmonic and continuous on $\bar{\Omega}$, with $-\varphi - \varepsilon \le w_0 \le -\varphi$. Observe that $U \le -w_0$. Indeed, if $v \in \mathcal{V}(\Omega, \varphi, f)$ then $v + w_0$ is a bounded plurisubharmonic function in Ω that satisfies $v^* + w_0 \le 0$ on $\partial\Omega$. The maximum principle yields $v + w_0 \le 0$, hence $U \le -w_0$ in Ω. We infer that $\limsup_{z\to\zeta} U(z) \le \varphi(\zeta)$, for all $\zeta \in \partial\Omega$. Thus

$$\lim_{z\to\zeta} U(z) = \varphi(\zeta). \tag{5.2.4}$$

Step 3. U is continuous on $\bar{\Omega}$. It follows from the above estimates that for $\zeta \in \partial\Omega$ and $z \in \Omega$,

$$U(z) - \varphi(\zeta) \le -w_0(z) - \varphi(\zeta) \le -K\rho(z) + \varepsilon. \tag{5.2.5}$$

Fix $C > 0$ such that $-\rho(z) \le C|z - \zeta|$ for $\zeta \in \partial\Omega$ and $z \in \Omega$. If $\delta > 0$ is small enough we infer, for $z \in \Omega$, $\zeta \in \partial\Omega$, that

$$|z - \zeta| \le \delta \Longrightarrow U(z) - \varphi(\zeta) \le KC\delta + \varepsilon.$$

Fix $a \in \mathbb{C}^n$, $|a| < \delta$, and define $\Omega_a := \Omega - a$. For $v \in \mathcal{V}$ we set

$$v_1 := \max\{v, v_0\}.$$

It follows from (5.2.3) that $v_1 \in \mathcal{V}$ and $\varphi - \varepsilon \le v_1 \le \varphi$ in $\partial\Omega$. Moreover if $\zeta \in \Omega_a \cap \partial\Omega$,

$$v_1(\zeta + a) \le U(\zeta + a) \le \varphi(\zeta) + KC\delta + \varepsilon,$$

while $v_1(z + a) \le \varphi(z + a) + KC\delta + \varepsilon$, if $z \in \Omega \cap \partial\Omega_a$. Therefore

$$w(z) = \begin{cases} v_1(z), & \text{if } z \in \Omega \setminus \Omega_a, \\ \max\{v_1(z), v_1(z + a) - \varepsilon - KC\delta\}, & \text{if } z \in \Omega \cap \Omega_a, \end{cases}$$

is a bounded plurisubharmonic function in Ω such that $w \le \varphi$ in $\partial\Omega$ and, by (5.2.3), for all $H \in \dot{H}_n^+$,

$$\Delta_H w(z) \ge \min\{f^{1/n}(z), f^{1/n}(z + a)\}$$

in $\Omega \cap \Omega_a$, while $\Delta_H w(z) \ge f^{1/n}(z)$ in $\Omega \setminus \bar{\Omega}_a$.

Let $\omega_{f^{1/n}}$ denote the modulus of continuity of $f^{1/n}$ in Ω. Since $|a| \le \delta$, it follows that $f^{1/n}(z + a) \ge f^{1/n}(z) - \omega_{f^{1/n}}(\delta)$, hence

$$\Delta_H w(z) \ge f^{1/n}(z) - \omega_{f^{1/n}}(\delta)$$

in Ω. The function

$$\tilde{w}(z) := \omega_{f^{1/n}}(\delta)\rho(z) + w(z), \quad z \in \Omega,$$

therefore satisfies $\tilde{w} \in \mathcal{V}$ and $w^* \leq \varphi$ in $\partial\Omega$, hence $\tilde{w} \leq U$ in Ω. Thus

$$v(z+a) \leq v_1(z+a) - \varepsilon - KC\delta \leq U(z) + B\omega_{f^{1/n}}(\delta),$$

if $z \in \Omega$ and $z + a \in \Omega$, where $B = -\inf_\Omega \rho$. Since v was arbitrary in $\mathcal{V}$, it follows that

$$U(z+a) - \varepsilon - KC\delta - B\omega_{f^{1/n}}(\delta) \leq U(z),$$

if $z \in \Omega$ and $z + a \in \Omega$.

This proves that U is continuous in Ω, hence on $\bar{\Omega}$ by (5.2.4). Therefore, $U \in \mathcal{V}$ satisfies the requirements of the first part of the theorem.

Step 4. Modulus of continuity of U. In the construction above the constants B, C do not depend on ε and δ, while the constant $K = K(\psi)$ depends only on the $C^{1,1}$-bound of an ε-approximation ψ of φ in $\partial\Omega$. When φ is $C^{1,1}$-smooth we can take $\psi = \varphi$ and thus get a precise control on the modulus of continuity of U: for $\delta > 0$ small enough,

$$\omega_U(\delta) \leq C\delta\|\varphi\|_{C^{1,1}} + B\omega_{f^{1/n}}(\delta),$$

as desired. □

5.3 The case of the unit ball

We are going to show that the Perron–Bremermann envelope U solves the Monge–Ampère equation $(dd^c U)^n = f\beta^n$ in Ω. Following [BT76] (and simplifications by [Dem91]), we first prove this statement when $\Omega = \mathbb{B}$ is the unit ball and f, φ are regular enough.

5.3.1 $\mathcal{C}^{1,1}$-regularity

Theorem 5.13. *Assume $\Omega = \mathbb{B}$ is the unit ball, $f^{1/n} \in C^{1,1}(\bar{\mathbb{B}})$, and $\varphi \in C^{1,1}(\partial\mathbb{B})$. Then the Perron–Bremermann envelope $U = U_{\mathbb{B},\varphi,f}$ admits second-order partial derivates almost everywhere in $\mathbb{B}$ which are locally bounded in $\mathbb{B}$, i.e., $U \in C^{1,1}_{\mathrm{loc}}(\mathbb{B})$.*

Here and in the sequel we set

$$C^{0,\alpha}(\overline{\Omega}) = \{v \in C(\overline{\Omega}) |\ \|v\|_\alpha < +\infty\}$$

for $0 < \alpha \leq 1$, where the α-Hölder norm is given by

$$\|v\|_\alpha = \|v\|_{\alpha,\overline{\Omega}} = \sup\{|v(z)|\ |z \in \overline{\Omega}\} + \sup\left\{\frac{|v(z)-v(y)|}{|z-y|^\alpha} |\ z, y \in \overline{\Omega}\right\}.$$

If $0 < \alpha \leq 1$ and $k \in \mathbb{N}^*$, then $C^{k,\alpha}(\overline{\Omega})$ denotes the class of functions which admit continuous partial derivatives up to order k, and whose k-th order partial derivatives are Hölder continuous of order α in $\overline{\Omega}$. We shall also consider the spaces $C^{k,\alpha}_{\mathrm{loc}}(\Omega)$ and $C^{k,\alpha}(\partial\Omega)$, with obvious notations.

Proof. The proof of Theorem 5.13 consists of several steps and occupies the rest of this section. Recall from Theorem 5.12 that $U \in C^{0,1}(\bar{\mathbb{B}})$. We are going to show that for any fixed compact $K \subset \mathbb{B}$, there exists $C = C(K) > 0$ such that for any $z \in K$ and $|h|$ small enough,

$$U(z+h) + U(z-h) - 2U(z) \leq C|h|^2. \tag{5.3.1}$$

This implies that U has second-order partial derivatives almost everywhere that are locally bounded.

Step 1. Using automorphisms of the ball $\mathbb{B}$ as translations. The main difficulty with the expression $U(z+h) + U(z-h) - 2U(z)$ is that it is not defined in $\mathbb{B}$, since translations do not preserve the ball. We use automorphisms of $\mathbb{B}$ instead and study the corresponding invariant symmetric differences of second order. For $a \in \mathbb{B}$, we set

$$T_a(z) = \frac{P_a(z) - a + \sqrt{1-|a|^2}(z - P_a(z))}{1 - \langle z, a\rangle}, \qquad P_a(z) = \frac{\langle z, a\rangle}{|a|^2}a,$$

where $\langle \cdot, \cdot \rangle$ denote the Hermitian inner product in $\mathbb{C}^n$. The reader will check in Exercise 5.7 that T_a is a holomorphic automorphism of the unit ball such that $T_a(a) = 0$. Note that T_0 is the identity. We set

$$h = h(a, z) := a - \langle z, a\rangle z. \tag{5.3.2}$$

Observe that $h(-a, z) = -h(a, z)$. If $|a| \leq 1/2$, then

$$T_a(z) = z - h + O(|a|^2),$$

where $O(|a|^2) \leq C|a|^2$, with C a uniform constant independent of $z \in \mathbb{B}$ when $|a| \leq 1/2$. Thus $T_{\pm a}$ is the translation by $\mp h$ up to small second-order terms, when $|a|$ is small enough.

Step 2. Estimating the invariant symmetric differences of U. The invariant symmetric differences of U in $\mathbb{B}$ are

$$z \in \mathbb{B} \longmapsto U \circ T_a(z) + U \circ T_{-a}(z) - 2U(z) \in \mathbb{R}.$$

We also consider the plurisubharmonic function

$$z \longmapsto V_a(z) := \frac{1}{2}(U \circ T_a(z) + U \circ T_{-a}(z)),$$

and try to compare it to U in $\mathbb{B}$. We claim that

$$V_a(z) \leq U(z) + C|a|^2,$$

for some constant $C > 0$ and for all $z, a \in \mathbb{B}$ with $|a| \leq 1/2$. We verify this by showing that V_a belongs to the Perron–Bremermann class $\mathcal{V}(\mathbb{B}, \varphi, f)$, i.e., V_a is a subsolution to the Dirichlet problem.

2.1. Boundary values of V_a. Observe that if $g \in C^{0,1}(\bar{\mathbb{B}})$, then

$$|g \circ T_a(z) - g(z-h)| \leq \|g\|_{C^{0,1}(\bar{\mathbb{B}})} \cdot |T_a(z) - z + h| \leq c_1 |a|^2 \|g\|_{C^{0,1}(\bar{\mathbb{B}})}, \quad (5.3.3)$$

where $c_1 > 0$ is a geometric constant. A Taylor yields

$$g \circ T_a(z) = g(z - h + O(|a|^2)) = g(z) + dg(z) \cdot h + O(|a|^2),$$

for $g \in C^{1,1}(\bar{\mathbb{B}})$, hence

$$g \circ T_a(z) + g \circ T_{-a}(z) \leq 2g(z) + 2C_2|a|, \quad (5.3.4)$$

where $C_2 = C_2(g)$ depends on the $C^{1,1}$-norm of g.

Extending φ as a function in $C^{1,1}(\bar{\mathbb{B}})$ and applying (5.3.4) yields

$$\varphi \circ T_a + \varphi \circ T_{-a} \leq 2\varphi + 2C_2|a|^2, \quad (5.3.5)$$

where $C_2 = C_2(\varphi)$ depends on the $C^{1,1}$-norm of φ. We infer that

$$V_a(z) \leq \varphi + C_2|a|^2, \ \zeta \in \partial\mathbb{B}.$$

2.2. Estimating the Monge–Ampère measure of V_a. We now estimate $\Delta_H V_a$ from below, for $H \in \dot{H}_n^+$ fixed. Observe that

$$\Delta_H(U \circ T_a) \geq (\det T_a')^{2/n} \ (f^{1/n} \circ T_a),$$

where $\det T_a'(z) = 1 + (n+1)\langle z, a\rangle + O(|a|^2)$. Hence,

$$\big(\det T_a'(z)\big)^{2/n} = 1 + \frac{2(n+1)}{n}\langle z, a\rangle + O(|a|^2).$$

Since $f^{1/n} \in C^{1,1}(\bar{\mathbb{B}})$, it follows from (5.3.4) that

$$f^{1/n} \circ T_a(z) = f^{1/n}(z - h + o(|a|^2)) = f^{1/n}(z) + \psi_1(z, a) + O(|a|^2),$$

setting $\psi_1(a, z) := df^{1/n}(z) \cdot h$. An elementary computation yields

$$|\det T_a(z)|^{2/n} \left(f^{1/n} \circ T_a(z)\right) + |\det T_{-a}(z)|^{2/n} \left(f^{1/n} \circ T_{-a}(z)\right) \geq 2f^{1/n}(z) - C_1|a|^2,$$

for $z \in \mathbb{B}$ and $|a| \leq 1/2$, where $C_1 = C_1(n) > 0$. Therefore

$$\Delta_H(U \circ T_a) + \Delta_H(U \circ T_{-a}) \geq 2f^{1/n} - C_2|a|^2,$$

hence

$$\Delta_H V_a \geq f^{1/n} - C_3|a|^2.$$

For $|a| \leq 1/2$ we consider the continuous plurisubharmonic function

$$z \mapsto v_a(z) = V_a(z) + C_3|a|^2(|z|^2 - 2).$$

Observe that $v_a \leq \varphi$ on $\partial\mathbb{B}$ and for every $H \in \dot{H}_n^+$,

$$\Delta_H v_a = \frac{1}{2}\Delta_H V_a + C|a|^2\Delta_H(|z|^2) \geq f^{1/n} - C_3|a|^2 + C_3|a|^2 \geq f^{1/n}.$$

Thus $v_a \in \mathcal{V}(\Omega, \varphi, f)$, hence $v_a \leq U$. It follows that

$$\frac{1}{2}V_a(z) - C_3|a|^2 \leq \frac{1}{2}V_a(z) + C_3|a|^2(|z|^2 - 2) \leq U(z)$$

for $z \in \mathbb{B}$, hence

$$U \circ T_a(z) + U \circ T_{-a}(z) - 2U(z) \leq 2C_3|a|^2,$$

as claimed.

Step 3. Comparing invariant and usual symmetric differences. We now compare $U \circ T_a(z) + U \circ T_{-a}(z)$ and $U(z-h) + U(z+h)$, where h is defined by (5.3.2). Fix a compact set $K \subset \mathbb{B}$ and $|h|$ small enough.

Applying (5.3.3) with $g = U$, $z \in K$ and $|h| < \operatorname{dist}(K, \partial\mathbb{B})$, we get

$$\begin{aligned} U(z-h) + U(z+h) - 2U(z) & \\ &\leq U \circ T_a(z) + U \circ T_{-a}(z) - 2U(z) + 2c_1\|U\|_{C^{0,1}(\bar{\mathbb{B}})}|a|^2 \\ &\leq (2c_1\|U\|_{C^{0,1}(\bar{\mathbb{B}})} + 2C_3)|a|^2. \end{aligned}$$

Observe that $a \mapsto h(a,z) = a - \langle z, a\rangle z$ is a non-singular endomorphism of $\mathbb{C}^n$ which depends smoothly on $z \in \mathbb{B}$. The inverse mapping $h \mapsto a(h,z)$ is a linear map with norm less than $\frac{1}{1-|z|^2}$, since

$$|h| \geq |a| - |\langle z, a\rangle||z| \geq |a| - |z|^2|a| \geq |a|(1 - |z|^2).$$

For $z \in K$ and $|h| \leq \operatorname{dist}(K, \partial\mathbb{B})/2$, we infer that

$$U(z+h) + U(z-h) - 2U(z) \leq \frac{C_4}{(1-|z|^2)^2}|h|^2,$$

where $C_4 := (2c_1\|U\|_{C^{0,1}(\bar{\mathbb{B}})} + 2C_3)$.

Consider now a convolution with a regularizing kernel χ_ε, $\varepsilon > 0$ small enough. For $z \in K$ and $|h|$ small we obtain

$$U_\varepsilon(z+h) + U_\varepsilon(z-h) - 2U_\varepsilon(z) \le \frac{C_4}{(1-(|z|+\varepsilon)^2)^2}|h|^2.$$

The Taylor expansion of order two of U_ϵ yields

$$D^2U_\varepsilon(z)h^2 \le \frac{C_4}{(1-(|z|+\varepsilon)^2)^2}|h|^2 \le A|h|^2,$$

for $z \in K, h \in \mathbb{C}^n$, where $A := C_4/\mathrm{dist}(K, \partial\mathbb{B})^2$. Since $U_\varepsilon \in PSH(\mathbb{B}_\varepsilon)$,

$$D^2U_\varepsilon(z)h^2 + D^2U_\varepsilon(z)(ih)^2 = 4\sum_{j,k}\frac{\partial^2 U_\varepsilon}{\partial z_j\,\partial\bar{z_k}}h_j\bar{h_k} \ge 0.$$

Hence for $z \in K$ and $|h|$ small enough,

$$D^2U_\varepsilon(z)h^2 \ge -D^2U_\varepsilon(z)(ih)^2 \ge -A|h|^2.$$

Therefore for $z \in K$ we have a uniform bound $|D^2U_\varepsilon(z)| \le A$.

The Alaoglu–Banach theorem ensures that there exists $g \in L^\infty(K)$ such that D^2U_ε converges weakly to g in $L^\infty(K)$. Since $D^2U_\varepsilon \to D^2U$ in the sense of distributions, $D^2U = g$ in the sense of distributions. The second order derivatives of U therefore exist almost everywhere and are locally bounded in $\mathbb{B}$, with

$$\|D^2U\|_{L^\infty(K)} \le A,$$

where $A := C_4/\mathrm{dist}(K, \partial\mathbb{B})^2$ and C_4 depends on the $C^{0,1}$ norm of U, the $C^{1,1}$ norm of φ and $f^{1/n}$. We have thus shown that $U \in C^{1,1}_{\mathrm{loc}}(\mathbb{B})$. □

In general, U does not belong to $C^{1,1}(\bar{\mathbb{B}})$, as Exercise 5.11 shows.

5.3.2 Solution to the Dirichlet problem We now show that the Perron–Bremermann envelope is the solution to the Dirichlet problem:

Theorem 5.14. *Assume* $0 \le f^{1/n} \in C^{1,1}(\bar{\mathbb{B}})$ *and* $\varphi \in C^{1,1}(\partial\mathbb{B})$. *Then* $U = U(\mathbb{B}, \varphi, f)$ *is the unique solution to the Dirichlet problem*

$$\begin{cases} u \in PSH(\mathbb{B}) \cap C(\overline{\mathbb{B}}) \\ (dd^c u)^n = f\beta^n \quad \text{in } \mathbb{B}, \\ u = \varphi \qquad\qquad \text{on } \partial\mathbb{B}. \end{cases} \tag{5.3.6}$$

Proof. We already know that $U \in C^{1,1}_{\mathrm{loc}}(\mathbb{B}) \cap \mathcal{V}(\mathbb{B}, \varphi, f)$ and has the correct boundary values. It remains to show that $(dd^c U)^n = f\beta^n$. Since $U \in C^{1,1}_{\mathrm{loc}}(\mathbb{B})$, it suffices to show that for almost every $z \in \mathbb{B}$

$$\det\left(\frac{\partial^2 U}{\partial z_j \partial \bar{z}_k}(z)\right) = f(z).$$

The inequality $\geq$ holds almost everywhere since U is a subsolution.

Suppose by contradiction that there exists a point $z_0 \in \mathbb{B}$ at which U twice differentiable and satisfies

$$\det\left(\frac{\partial^2 U}{\partial z_j \partial \bar{z}_k}(z_0)\right) > f(z_0) + 2\varepsilon,$$

for some $\varepsilon > 0$. Then for any $H \in \dot{H}_n^+$,

$$\Delta_H U(z_0) > (f(z_0) + 2\epsilon)^{1/n}. \tag{5.3.7}$$

Using the Taylor expansion of U of order two at the point z_0, we get

$$U(z) = U(z_0) + \Re P(z - z_0) + L(z - z_0) + o(|z - z_0|^2),$$

where P is a complex polynomial of degree 2 and

$$L(\zeta) = \sum_{j,k} \frac{\partial^2 U}{\partial z_j \partial \bar{z_k}}(z_0)\zeta_j \bar{\zeta}_k$$

is the Levi form of U at z_0.

Since L is positive, for any $0 < s < 1$ close to 1, there exist $\delta, r > 0$ small enough such that $B(z_0, r) \Subset \mathbb{B}$, and for $|z - z_0| = r$,

$$U(z) \geq U(z_0) + \Re P(z - z_0) + sL(z - z_0) + \delta.$$

Observe that the function defined by

$$w(z) := U(z_0) + \Re P(z - z_0) + sL(z - z_0) + \delta$$

is smooth and plurisubharmonic in $\mathbb{C}^n$.

Then the function

$$v(z) = \begin{cases} U(z), & \text{if } z \in \mathbb{B} \setminus \mathbb{B}(z_0, r), \\ \max\{U(z), w(z)\}, & \text{if } z \in \mathbb{B}(z_0, r), \end{cases} \tag{5.3.8}$$

is plurisubharmonic in $\mathbb{B}$, continuous near $\partial\mathbb{B}$ with $v = \varphi$ in $\partial\mathbb{B}$.

We claim that $\Delta_H v \geq f^{1/n}$ for all $H \in \dot{H}_n^+$. Indeed, if A is the complex hessian matrix associated to U at z_0, we have $\Delta_H w = s\,\mathrm{tr}(HA)$. Lemma 5.8 yields $\mathrm{tr}(HA) \geq (\det A)^{1/n}$, hence by (5.3.7) for $z \in \mathbb{B}(z_0, r)$,

$$\Delta_H w(z) \geq s(f(z_0) + 2\varepsilon)^{1/n} \geq (f(z_0) + \varepsilon)^{1/n},$$

if $s < 1$ is chosen close enough to 1.

Since $f^{1/n}$ is continuous in $\bar{\mathbb{B}}$, shrinking r if necessary, we can assume that $(f(z_0)+\varepsilon)^{1/n} \geq f(z)^{1/n}$ for $z \in \mathbb{B}(z_0, r)$, and so

$$\Delta_H w(z) \geq f(z)^{1/n}$$

pointwise in $\mathbb{B}(z_0, r)$. It now follows from (5.2.3) that $\Delta_H v \geq f^{1/n}$.

We infer $v \in \mathcal{V}(\mathbb{B}, \varphi, f)$, hence $v \leq U$ in $\mathbb{B}$. On the other hand, $v(z_0) = U(z_0) + \delta > U(z_0)$, a contradiction. The proof is complete. □

Using an approximation process, we can now solve the Dirichlet problem in the unit ball with continuous data:

Corollary 5.15. *Assume $\varphi \in C(\partial\mathbb{B})$ and $0 \leq f \in C(\bar{\mathbb{B}})$. Then $U = U(\mathbb{B}, \varphi, f)$ is the solution to Dirichlet problem* (5.3.6).

Proof. Let (f_j) be a sequence of smooth positive functions which decrease to f uniformly on $\bar{\mathbb{B}}$. Fix also C^∞-smooth functions φ_j in $\partial\mathbb{B}$ such that φ_j increase to φ uniformly on $\partial\mathbb{B}$.

The upper envelope $U_j := U(\mathbb{B}, \varphi_j, f_j)$ is the unique plurisubharmonic solution to the Dirichlet problem (5.3.6) with boundary data φ_j and right-hand side f_j. Observe that (U_j) is non-decreasing in $\mathbb{B}$.

Fix $\varepsilon > 0$ small and set $\rho(z) := |z|^2 - 1$. Note that for $k \geq j$,

$$(dd^c(U_k + \varepsilon\rho)^n \geq (dd^c U_k)^n + \varepsilon^n \beta^n = (f_k + \varepsilon^n)\beta^n.$$

Since (f_k) decreases to f uniformly in $\bar{\mathbb{B}}$, we can find $j_0 > 0$ such that $f_j \leq f_k + \varepsilon^n$ in $\bar{\mathbb{B}}$ for $k \geq j \geq j_0$, hence

$$(dd^c(U_k + \varepsilon\rho)^n \geq f_j \beta^n.$$

Since $U_k + \varepsilon(|z|^2 - 1) = f_k \leq f_j$ in $\partial\mathbb{B}$, the comparison principle shows that $U_k + \varepsilon\rho \leq U_j$ in $\bar{\mathbb{B}}$. Further, since $U_j \leq U_k$ we infer for $k \geq j \geq j_0$,

$$\max_{\bar{\mathbb{B}}} |U_k - U_j| \leq \varepsilon.$$

The sequence (U_j) therefore converges uniformly in $\bar{\mathbb{B}}$ to a function $u \in PSH(\mathbb{B}) \cap C^0(\bar{\mathbb{B}})$ such that $u = \varphi$ in $\partial\mathbb{B}$. The uniform convergence ensures that $(dd^c U_j)^n$ converge to $(dd^c u)^n$ weakly, hence $(dd^c u)^n = f\beta^n$.

The comparison principle guarantees that $u = U(\mathbb{B}, \varphi, f)$ is the unique solution to the Dirichlet problem (5.3.6). □

5.4 Strictly pseudo-convex domains

5.4.1 Continuous densities We generalize here Corollary 5.15 to the case when $\Omega \Subset \mathbb{C}^n$ is a strictly pseudo-convex domain in $\mathbb{C}^n$.

Theorem 5.16. *Assume $\varphi \in C(\partial\Omega)$ and $0 \leq f \in C(\bar{\Omega})$. The envelope $U = U(\Omega, \varphi, f)$ is the unique solution to the Dirichlet problem.*

Proof. We already know that $U \in PSH(\Omega) \cap C^0(\bar{\Omega})$ and satisfies $(dd^cU)^n \geq f\beta^n$ weakly. It remains then to check that $(dd^cU)^n \leq f\beta^n$.

We use the classical balayage technique. Let $B \Subset \Omega$ be an arbitrary Euclidean ball. By Corollary 5.15, we can solve the Dirichlet problem

$$(dd^cu)^n = f\beta^n \text{ in } B \text{ and } u = U \text{ on } \partial B.$$

The comparison principle ensures that $U \leq u$ in B. Hence, by Proposition 5.10,

$$z \longmapsto v(z) = \begin{cases} u(z), & \text{if } z \in B, \\ U(z), & \text{if } z \in \overline{\Omega} \setminus B, \end{cases}$$

belongs to the class $\mathcal{V}(\Omega, \varphi, f)$ and $v = U = \varphi$ on $\partial\Omega$. We infer $v \leq U$, hence $u = U$ in B, so that $(dd^cU)^n = (dd^cu)^n = f\beta^n$ in B. The equality holds in Ω since B was arbitrary. □

5.4.2 More general densities We start by extending Theorem 5.16 to the case when the density is merely bounded:

Theorem 5.17. *Assume $\varphi \in C(\partial\Omega)$ and $0 \leq f \in L^\infty(\Omega)$. Then the envelope $U(\Omega, \varphi, f)$ is a bounded plurisubharmonic function in Ω which is the unique solution to the Dirichlet problem*

$$\begin{cases} u \in PSH(\Omega) \cap L^\infty(\Omega) \\ (dd^cu)^n = f\beta^n \quad \text{in } \Omega, \\ \lim_{z\to\zeta} u(z) = \varphi(\zeta) \quad \text{for } \zeta \in \partial\Omega, \end{cases} \tag{5.4.1}$$

Proof. Let (f_j) be a sequence of continuous functions in $\bar{\Omega}$ which converge to f in $L^1(\Omega)$ and almost everywhere. Theorem 5.16 yields, for each $j \in \mathbb{N}$, a solution $U_j \in PSH(\Omega) \cap C^0(\bar{\Omega})$ such that $(dd^cU_j)^n = f_j\beta^n$ in Ω and $U_j = \varphi$ in $\partial\Omega$.

Set $V_0 := U(\Omega, \varphi, 0)$ and $V_1 := U(\Omega, \varphi, M)$, where where M is a uniform L^∞-bound for the f_j's. The comparison principle yields $V_1 \leq U_j \leq V_0$, so the U_j's are uniformly bounded.

Extracting and relabelling we can assume that (U_j) converges in $L^1(\Omega)$ and almost everywhere to a bounded plurisubharmonic function U such that $U = (\limsup_j U_j)^*$ in Ω.

We claim that (U_j) converges to U in capacity. Indeed fix a compact set $K \subset \Omega$ and $\delta, \varepsilon > 0$. There exists an open set $G \subset \Omega$ such that $\mathrm{Cap}_\Omega(G) < \varepsilon$ and all the function U_j, U are continuous in $\Omega \setminus G$, by quasicontinuity. Hartogs' lemma yields

$$\limsup_{j\to+\infty} \max_{K\setminus G}(U_j - U) \leq 0,$$

hence $\{U_j - U \geq 2\delta\} \subset G$ for $j > 1$ large enough. We infer that

$$\lim_{j\to+\infty} \mathrm{Cap}_\Omega(\{U_j - U \geq 2\delta\}) = 0.$$

On the other hand, Lemma 5.18 shows that for all $j \in \mathbb{N}$,

$$\begin{aligned} \mathrm{Cap}_\Omega(\{U - U_j \geq 2\delta\}) &\leq \delta^{-n} \int_{\{U-U_j \geq \delta\}} (dd^c U_j)^n \\ &\leq M\delta^{-n} \int_\Omega (U - U_j)_+ \beta^n. \end{aligned}$$

The right-hand side converges to 0, since $U_j \to U$ in L^1, hence

$$\lim_{j \to +\infty} \mathrm{Cap}_\Omega(\{U - U_j \geq 2\delta\}) = 0.$$

Our claim is proved.

We conclude that $(dd^c U_j)^n \to (dd^c U)^n$ and $(dd^c U)^n = f\beta^n$ weakly in Ω. Since $V_1 \leq U \leq V_0$, Theorem 5.16 shows that U tends to φ at the boundary of Ω.

The comparison principle ensures that $U = U(\Omega, \varphi, f)$ is the unique solution to the Dirichlet problem (5.4.1). □

We need to establish the following lemma which was used in the previous proof.

Lemma 5.18. *Assume u, v are bounded plurisubharmonic functions such that $\{u < v\} \Subset \Omega$. Then for all $s, t > 0$*

$$t^n \mathrm{Cap}_\Omega(\{u - v \leq -s - t\}) \leq \int_{\{u-v \leq -s\}} (dd^c u)^n.$$

Proof. Fix $w \in PSH(\Omega)$ such that $-1 \leq w \leq 0, s, t > 0$. Then

$$\{u \leq v - s - t\} \subset \{u \leq v - s + tw\} \subset \{u \leq v - s\} \Subset \Omega.$$

The comparison principle thus yields

$$\begin{aligned} t^n \int_{\{u<v-s-t\}} (dd^c w)^n &\leq \int_{\{u<v-s+tw\}} (dd^c(v - s + tw))^n \\ &\leq \int_{\{u<v-s+tw\}} (dd^c u)^n \\ &\leq \int_{\{u<v-s\}} (dd^c u)^n. \end{aligned}$$

The required estimate follows by taking the sup over all such w's. □

When the density f is merely in $L^1_{\mathrm{loc}}(\Omega)$, we can show that the existence of a subsolution implies the existence of a solution:

Corollary 5.19. *Fix $\varphi \in C^0(\partial\Omega)$ and $0 \leq f \in L^1_{\mathrm{loc}}(\Omega)$. Assume there exists $v \in PSH(\Omega) \cap L^\infty(\Omega)$ such that $v = \varphi$ in $\partial\Omega$ and $(dd^c v)^n \geq f\beta^n$ in the weak sense. Then the Perron–Bremermann envelope $U(\Omega, \varphi, f)$ is the unique solution to the Dirichlet problem* (5.4.1).

Proof. Set $f_j := \min\{f, j\}$. Then (f_j) is a sequence of bounded densities which increase to f everywhere and in $L^1_{\mathrm{loc}}(\Omega)$.

Theorem 5.17 guarantees the existence of a unique $U_j \in PSH(\Omega) \cap L^\infty(\Omega)$ such that $(dd^c U_j)^n = f_j \beta^n$ in Ω and $U_j = \varphi$ in $\partial\Omega$. Set

$$u_\varphi := U(\Omega, \varphi, 0).$$

The comparison principle yields $U_j \le U_{j+1} \le u_\varphi$, therefore (U_j) is uniformly bounded in Ω.

Since (U_j) is non-decreasing, it converges to a bounded plurisubharmonic function U in Ω such that $U_1 \le U \le u_\varphi$. The continuity of the complex Monge–Ampère operator for decreasing sequences ensures that $(dd^c U)^n = f\beta^n$. Now the comparison principle implies that $U = U(\Omega, \varphi, f)$ is the unique solution to the Dirichlet problem (5.4.1). □

Remark 5.20. *The result above is due to Cegrell and Sadullaev* [CS92]*, who also gave examples of densities $0 \le f \in L^1(\Omega)$ for which there is no bounded plurisubharmonic subsolution to the Dirichlet problem* (5.4.1) *(see Exercise 5.8).*

When $0 \le f \in L^p(\Omega)$, $p > 1$, Kolodziej has shown in [Kol98] *that the Dirichlet problem* (5.4.1) *has a unique continuous solution. The case $p = 2$ was proved earlier by Cegrell and Person* [CP92].

The balayage procedure used in the proof of Theorem 5.16 is quite classical in Potential Theory. It can be be generalized as follows.

Corollary 5.21. *Let $B \Subset \Omega$ be a ball and $0 \le f \in L^1(B)$. Fix $u \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$ such that $(dd^c u)^n \ge f\beta^n$ in B. There exists a unique $\hat{u} \in PSH(\Omega) \cap L^\infty_{\mathrm{loc}}(\Omega)$ such that $\hat{u} = u$ in $\Omega \setminus B$, $u \le \hat{u}$ in B, and $(dd^c \hat{u})^n = f\beta^n$ in B. If $f \in C^0(\bar{B})$ and $u \in C^0(\partial B)$, then $\hat{u} \in C^0(\bar{B})$.*

Proof. Let (φ_j) a decreasing sequence of continuous function converging to u in $\bar{B}$. Applying Theorem 5.17, we find $U_j \in PSH(B) \cap L^\infty(\bar{B})$ such that $U_j = \varphi_j$ in ∂B and $(dd^c U_j)^n = f\beta^n$ in B.

The comparison principle ensures that $u \le U_j \le U_{j+1}$ in B. Therefore, (U_j) converges to a plurisubharmonic function U in B such that $u \le U \le U_j$ and $(dd^c U)^n = f\beta^n$ in B. Moreover,

$$U^*(\zeta) := \limsup_{B \ni z \to \zeta} U(z) \le \limsup_{B \ni z \to \zeta} U_j(z) = \varphi_j(\zeta)$$

for any $\zeta \in \partial B$, hence $U^*(\zeta) \le u(\zeta)$ in ∂B.

The function $\hat{u}$ defined by $\hat{u} = u$ in $\Omega \setminus B$ and $\hat{u} = U$ in B is therefore plurisubharmonic in Ω and has all the required properties. □

5.4.3 Stability estimates We address here the following issue: if f_1 and f_2 (resp. φ_1 and φ_2) are close (in an appropriate sense), does it follow that so are $U(f_1, \varphi_2)$ and $U(f_2, \varphi_2)$?

Proposition 5.22. *Fix $\varphi_1, \varphi_2 \in C^0(\partial\Omega)$ and $f_1, f_2 \in C^0(\overline{\Omega})$. The solutions $U_1 = U(\Omega, \varphi_1, f_1)$, $U_2 = U(\Omega, \varphi_2, f_2)$ satisfy*

$$\|U_1 - U_2\|_{L^\infty(\overline{\Omega})} \leq R^2 \|f_1 - f_2\|^{1/n}_{L^\infty(\overline{\Omega})} + \|\varphi_1 - \varphi_2\|_{L^\infty(\partial\Omega)}, \tag{5.4.2}$$

where $R := \operatorname{diam}(\Omega)$. In particular, if $\varphi \in C^0(\partial\Omega)$ and $f \in C^0(\overline{\Omega})$, then

$$\|U(\Omega, \varphi, f)\|_{L^\infty(\Omega)} \leq R^2 \|f\|_{L^\infty(\overline{\Omega})} + \|\varphi\|_{L^\infty(\partial\Omega}. \tag{5.4.3}$$

Proof. For $z_0 \in \Omega$ and $R > 0$ such that $B(z_0, R) \subset \Omega$ we set

$$v_1(z) = \|f_1 - f_2\|^{1/n}_{L^\infty(\overline{\Omega})}(|z - z_0|^2 - R^2) + U_2(z)$$

and

$$v_2(z) = U_1(z) + \|\varphi_1 - \varphi_2\|_{L^\infty(\partial\Omega)}.$$

Observe that $v_1, v_2 \in PSH(\Omega) \cap C(\overline{\Omega})$, $v_1 \leq v_2$ in $\partial\Omega$, and $(dd^c v_1)^n \geq (dd^c v_2)^n$ in Ω. Therefore, by the comparison principle, $v_1 \leq v_2$ on $\overline{\Omega}$. Consequently,

$$U_2 - U_1 \leq R^2 \|f_1 - f_2\|^{1/n}_{L^\infty(\overline{\Omega})} + \|\varphi_1 - \varphi_2\|_{L^\infty(\partial\Omega)}.$$

The inequality (5.4.2) follows by reversing the roles of U_1 and U_2. □

These stability estimates show that the operator

$$\begin{array}{rcl} U_\Omega : C^0(\partial\Omega) \times L^\infty_+(\Omega) & \longrightarrow & PSH(\Omega) \cap L^\infty(\Omega), \\ (\varphi, f) & \longmapsto & U(\Omega, \varphi, f), \end{array}$$

is continuous in the corresponding topologies. Here $L^\infty_+(\Omega)$ denotes the set of all non-negative measurable and bounded densities in Ω. The reader will check in Exercise 5.13 that this stability property does not hold for arbitrary measures.

Remark 5.23. *Finer stability estimates have been established by Cegrell and Persson* [CP92] *when $f \in L^2(\Omega)$, and by Kolodziej* [Kol02] *when $f \in L^p(\Omega)$, $p > 1$ (see also* [GKZ08]*). We develop such stability estimates in Chapter 12, in the setting of compact Kähler manifolds.*

5.4.4 More general right-hand side We now allow the right-hand side to depend on the unknown function, following Cegrell in [Ceg84]. Specifically, we consider the Dirichlet problem

$$\begin{cases} u \in PSH(\Omega) \cap L^\infty(\Omega) \\ (dd^c u)^n = e^u f\beta^n \quad \text{in } \Omega, \\ \lim_{z\to\zeta} u(z) = \varphi(\zeta) \quad \text{on } \partial\Omega, \end{cases} \tag{5.4.4}$$

where $\varphi \in C(\partial\Omega)$ and $0 \le f \in L^\infty(\Omega)$.

The class $\mathcal{W}(\Omega, \varphi, f)$ of subsolutions to this Dirichlet problem is the set of all functions $w \in PSH(\Omega) \cap L^\infty(\Omega)$ such that $w^* \le \varphi$ in $\partial\Omega$ and $(dd^c w)^n \ge e^w f\beta^n$ in Ω. The corresponding upper envelope is

$$W(\Omega, \varphi, f) = W(z) := \sup\{w(z) |\ w \in \mathcal{W}(\Omega, \varphi, f)\}. \tag{5.4.5}$$

Theorem 5.24. *Fix $\varphi \in C(\partial\Omega)$ and $0 \le f \in L^\infty(\Omega)$. Then $W(\Omega, \varphi, f)$ is the unique solution to the Dirichlet problem* (5.4.4).

Proof. We can always assume that $\varphi \le 0$ in $\partial\Omega$. Let ρ be a defining function for $\bar\Omega$ defining Ω and $u_0 := U(\Omega, \varphi, 0)$. The function $v_0 := A\rho + u_0$ is a subsolution to the Dirichlet problem (5.4.4) if we choose $A > 1$ so large that $A^n(dd^c\rho)^n \ge f\beta^n$ in Ω. Thus $\mathcal{W}(\Omega, \varphi, f)$ is not empty.

We prove that the envelope $W(\Omega, \varphi, f)$ is *the* solution by using Theorem 5.17 and applying the Schauder fixed point theorem. Set

$$\mathcal{C} := \{w \in PSH(\Omega) \cap L^\infty(\Omega) |\ v_0 \le w \le u_0\}.$$

Observe that $\mathcal{C}$ is compact and convex in $L^1(\Omega)$. It follows from Theorem 5.17 that for each $w \in \mathcal{C}$ there exists a unique function $v = S(w) \in PSH(\Omega) \cap L^\infty(\Omega)$ such that $v = \varphi$ in $\partial\Omega$ and

$$(dd^c v)^n = e^w f\beta^n \quad \text{in } \Omega.$$

Observe that $(dd^c v)^n \le f\beta^n \le (dd^c v_0)^n$, as $w \le 0$. Since $v = v_0 = u_0$ in $\partial\Omega$, the comparison principle yields $v_0 \le v \le u_0$, hence $S(w) \in \mathcal{C}$.

We claim that the operator $S : \mathcal{C} \to \mathcal{C}$ is continuous in the L^1-topology. Indeed, assume $(w_j) \in \mathcal{C}^{\mathbb{N}}$ converges to $w \in \mathcal{C}$ in $L^1(\Omega)$ and set $v_j := S(w_j)$. Extracting and relabelling, we can assume that $v_j \to v$ in $\mathcal{C}$ and $w_j \to w$ almost everywhere.

The sequence (v_j) converges to v in capacity. Indeed, fix $\delta > 0$. Since $(dd^c v_j)^n = e^{w_j} f\beta^n$ and $w_j \le 0$, Lemma 5.18 yields for all $j \in \mathbb{N}$,

$$\begin{aligned} \mathrm{Cap}_\Omega(\{v - v_j \ge 2\delta\}) &\le \delta^{-n} \int_{\{v - v_j \ge \delta\}} (dd^c v_j)^n \\ &\le \delta^{-(n+1)} \|f\|_{L^\infty(\Omega)} \int_\Omega (v - v_j)_+ \beta^n. \end{aligned}$$

The right-hand side converges to 0 for $v_j \to v$ in $L^1(\Omega)$, hence the claim. We infer that $(dd^c v_j)^n \to (dd^c v)^n$ weakly in Ω.

On the other hand, (w_j) is uniformly bounded in Ω and $e^{w_j} f \to e^w f$ in $L^1(\Omega)$, hence $(dd^c v)^n = e^w f\beta^n$. Since $v_0 \le v \le u_0$ in Ω, it follows that $v = \varphi$ in $\partial\Omega$. The comparison principle now yields $v = S(w)$.

It follows that $S(w_j) \to S(w)$ in $L^1(\Omega)$, hence $S : \mathcal{C} \to \mathcal{C}$ is continuous. The Schauder fixed point theorem ensures that S admits a fixed point $w \in \mathcal{C}$, i.e., w is a solution to the Dirichlet problem (5.4.4).

The uniqueness of solutions to the Dirichlet problem (5.4.4) is a consequence of the next lemma. It guarantees that $w = W(\Omega, \varphi, f)$. □

Lemma 5.25. *Assume $0 \le f_1 \le f_2$ and $w_1, w_2 \in PSH(\Omega) \cap L^\infty(\Omega)$ are such that $(dd^c w_1)^n \le e^{w_1} f_1\beta$ and $(dd^c w_2)^n \ge e^{w_2} f_2$ in Ω.*

If $w_2 \le w_1$ on $\partial\Omega$, then $w_2 \le w_1$ in Ω.

Proof. We infer from $f_1 \le f_2$ that

$$\int_{\{w_1<w_2\}} e^{w_2} f_1\beta^n \le \int_{\{w_1<w_2\}} e^{w_2} f_2\beta^n = \int_{\{w_1<w_2\}} (dd^c w_2)^n.$$

The comparison principle thus yields

$$\int_{\{w_1<w_2\}} (dd^c w_2)^n \le \int_{\{w_1<w_2\}} (dd^c w_1)^n \le \int_{\{w_1<w_2\}} e^{w_1} f_1.$$

Therefore $\int_{\{w_1<w_2\}} e^{w_2} f_1\beta^n \le \int_{\{w_1<w_2\}} e^{w_1} f_1\beta^n$, whence

$$\int_{\{w_1<w_2\}} (e^{w_2} - e^{w_1}) f_1\beta^n = 0 \ \text{ and } \ \mathbf{1}_{\{w_1<w_2\}}(e^{w_2} - e^{w_1}) = 0,$$

almost everywhere with respect to $\mu_1 := f_1\beta^n$.

Since $(dd^c w_1)^n \le \mu_1$ we infer that $w_2 \le w_1$ almost everywhere for $(dd^c w_1^n)$. The domination principle now yields $w_2 \le w_1$ everywhere. □

5.4.5 Further results For the sake of completeness, we mention here without proof a few important results.

Theorem 5.26 (Krylov). *Let $\Omega \subset \mathbb{C}^n$ be a smoothly bounded strictly pseudoconvex domain and fix $\varphi \in \mathcal{C}^{3,1}(\partial\Omega)$. The unique plurisubharmonic solution $U = U_{\Omega,\varphi,0}$ of the homogeneous complex Monge–Ampère equation in Ω with boundary values φ is $\mathcal{C}^{1,1}$-smooth on $\overline{\Omega}$.*

This result (together with many other regularity results) was obtained by Krylov by probabilistic methods, in a series of articles (see notably [Kry89]). We refer the interested reader to the lecture notes by Delarue [Del12] for an overwiev of these techniques.

Theorem 5.27 (Caffarelli–Kohn–Nirenberg–Spruck). *Let $\Omega \subset \mathbb{C}^n$ be a smoothly bounded strictly pseudoconvex domain and fix $\varphi \in \mathcal{C}^\infty(\partial\Omega)$. If the density f is smooth and strictly positive on $\overline{\Omega}$, then the unique solution $U = U_{\Omega,\varphi,f}$ is smooth up to the boundary.*

This result (together with many others) was obtained by Caffarelli, Kohn, Nirenberg, and Spruck in [CKNS85]. We refer the interested reader to the lecture notes by Boucksom [Bou12] for an up-to-date presentation.

5.5 Exercises

Exercise 5.1. *Let μ be a probability measure with compact support in $\mathbb{R}^N$.*

1. *Show that the function*

$$\mathbb{R}^N \ni x \longmapsto U_\mu(x) := \int_{\mathbb{R}^n} K_N(x-y)d\mu(y) \in \mathbb{R},$$

is subharmonic and satisfies the Poisson equation $\Delta U_\mu = \mu$ in $\mathbb{R}^N$.

2. *Let $D \Subset \mathbb{R}^N$ be a domain and u a subharmonic function in a neighborhood of $\overline{D}$. Show that u can be written, in D, as*

$$u = U_\mu + h_D,$$

where μ is the Riesz measure of u and h_D is harmonic in D.

Exercise 5.2. *Let $u \in SH(\mathbb{B}) \cap C^0(\bar{\mathbb{B}})$. Show that for all $x \in \mathbb{B}$,*

$$u(x) = \int_{\mathbb{S}} u(y)P(x,y)d\sigma(y) + \int_{\mathbb{B}} G(x,y)d\mu_u(y),$$

where $\mu_u := \Delta u$ is the the Riesz measure of u.

Exercise 5.3. *Let φ be a continous function on $\partial\mathbb{B}$. Show that*

$$P_\varphi(x) := \int_{|y|=1} \varphi(y)P(x,y)d\sigma(y), \quad x \in \mathbb{B},$$

defines a harmonic function in the ball $\mathbb{B}$, which is continuous on $\bar{\mathbb{B}}$ and satisfies $P_\varphi(x) = \varphi(x)$ for $x \in \mathbb{S}$.

Exercise 5.4. *Let $T : \mathbb{C}^n \to \mathbb{C}^n$ be a $\mathbb{C}$-linear isomorphism and $q : \mathbb{C}^n_\zeta \to \mathbb{C}^n_z$ be a C^2-smooth function in a neighborhood of a point z_0. Set $z = T(\zeta)$ and $q_T(\zeta) := q(z) = q(T(\zeta)$.*

1. *Check that q_T is C^2-smooth near $\zeta_0 := T^{-1}(z_0)$ and*

$$\Delta q_T(\zeta) = \sum_{j=1}^{n} \frac{\partial^2 q_T(\zeta)}{\partial \zeta_j \partial \bar{\zeta}_k} = \operatorname{tr}(T^* Q(z) T),$$

where T^* denotes the complex conjugate transpose of $T := \left(\frac{\partial z_j}{\partial \zeta_k}\right)$ and $Q(z) := \left(\frac{\partial^2 q}{\partial z_j \partial \bar{z}_k}(z)\right)$ is the complex hessian of q at z.

2. *Fix $H \in \dot{H}_n$. Show that there exists a unitary complex matrix U such that $U^* H U = D$ is a diagonal matrix with positive entries. Set $T = D^{1/2} U$. Check that T is Hermitian, $T^* T = H$, and*

$$\operatorname{tr}(T^* A T) = \operatorname{tr}(H T^{-1} A T) = \operatorname{tr}(HA)$$

for all Hermitian matrix A.

3. *Apply* 2. *with $A(z)$ the complex hessian of q at $z = T(\zeta)$ to get*

$$\Delta q_T(\zeta) = \Delta_H q(z).$$

Exercise 5.5. *Let μ be a non-negative Borel measure in $\Omega \subset \mathbb{C}^n$ such that there exists $u \in PSH(\Omega) \cap L^\infty(\Omega)$ with $\mu \leq (dd^c u)^n$ in Ω.*

Show that that $PSH(\Omega) \subset L^1_{\mathrm{loc}}(\mu)$, and for any compact subsets $K \subset E \subset \Omega$ with $K \subset E^\circ$, there exists $C > 0$ such that

$$\int_K |V| d\mu \leq C \int_E |V| d\lambda,$$

for all $V \in PSH(\Omega)$, where λ is the Lebesgue measure on Ω.

Exercise 5.6. *Let u be a plurisubharmonic function in $\Omega \Subset \mathbb{C}^n$.*

1. *Assume that there exists constants $A, \delta > 0$ such that*

$$u(z+h) + u(z-h) - 2u(z) \leq A\|h\|^2,$$

for all $0 < \|h\| < \delta$ and for all $z \in \Omega$ such that $\operatorname{dist}(z, \partial\Omega) > \delta$.

Show that u is $\mathcal{C}^{1,1}$-smooth and its second derivatives, which exist almost everywhere, satisfy $\|D^2 u\|_{L^\infty(\Omega)} \leq A$.

2. *Show that the Monge–Ampère measure $(dd^c u)^n$ is absolutely continuous with respect to the Lebesgue measure dV in Ω, with*

$$(dd^c u)^n = c_n \det\left(\frac{\partial^2 u}{\partial z_j \partial \bar{z}_k}\right) dV,$$

for some constant $c_n > 0$. Check that this last result actually holds whenever u belongs to the Sobolev space $W^{2,n}_{\mathrm{loc}}$.

Exercise 5.7. *Let $\mathbb{B}$ denote the unit ball. For $a \in \mathbb{B}$, set*

$$T_a(z) = \frac{P_a(z) - a + \sqrt{1-|a|^2}(z - P_a(z))}{1 - \langle z, a\rangle}, \qquad P_a(z) = \frac{\langle z, a\rangle}{|a|^2} a,$$

where $\langle \cdot, \cdot \rangle$ denotes the Hermitian inner product in $\mathbb{C}^n$.

Check that T_a is a holomorphic automorphism of $\mathbb{B}$ such that $T_a(a) = 0$ and that T_0 is the identity.

Exercise 5.8. *Fix $\alpha > 1$ and set*

$$f_\alpha(z) := \frac{1}{|z|^{2n}(1 - \log|z|)^\alpha}.$$

1. *Check that $f_\alpha \in L^1(\mathbb{B}) \setminus L^p(\mathbb{B})$ for all $p > 1$ and show that there is no $u \in PSH(\mathbb{B}) \cap L^\infty(\mathbb{B})$ such that $(dd^c u)^n \geq f\beta^n$ in $\mathbb{B}$.*

2. *Show that there exists a unique radial plurisubharmonic function U in $\mathbb{B}$ which is smooth in $\mathbb{B} \setminus \{0\}$ and such that $U(z) = 0$ in $\partial\mathbb{B}$, $(dd^c U)^n = f\beta^n$ on $\mathbb{B}$, and $U(0) = -\infty$.*

Exercise 5.9. *Give an example of a non-negative Borel measure μ on Ω such that the class $\mathcal{B}(\Omega, \varphi, f)$ is non-empty but contains no element u such that $\lim u(z) = \varphi(\zeta)$ for all $\zeta \in \partial\Omega$ (see [CS92]).*

Exercise 5.10. *Let $h : \Omega \to \mathbb{R}$ be an upper semi-continuous function. The plurisubharmonic envelope of h is defined, for $z \in \Omega$, by*

$$P_\Omega h(z) := \sup\{u(z) |\, u \in PSH(\Omega), u \leq h \text{ in } \Omega\}.$$

1. *Show that $P_\Omega h$ is a plurisubharmonic function in Ω. Observe that $P_\Omega h = h$ if h is plurisubharmonic in Ω.*

2. *Show that if $h : \Omega \to \mathbb{R}$ is continuous, then $P_\Omega h$ is continuous in Ω and satisfies*

$$\mathbf{1}_{\{P_\Omega h < h\}}(dd^c P_\Omega h)^n = 0.$$

3. *Show that if $\Omega = \mathbb{B}$ and h is locally $C^{1,1}$, then $P_\mathbb{B} h$ is locally $C^{1,1}$ in $\mathbb{B}$ and satisfies the Monge–Ampère equation:*

$$(dd^c P_\mathbb{B} h)^n = \mathbf{1}_{\{P_\mathbb{B} h = h\}}(dd^c h)^n.$$

Exercise 5.11. *Let $\mathbb{B} \subset \mathbb{C}^2$ the unit ball. Fix $\alpha > 0$ and set*

$$\varphi(z, w) := (1 + \Re z)^\alpha \text{ for } (z, w) \in \partial\mathbb{B}.$$

Observe that the function φ is C^∞-smooth in $\partial\mathbb{B}$, except at the point $(-1, 0)$, and near this point we have $1 + \Re z = O(|w|^2 + (\Im z)^2)$.

1) *Assume that* $\alpha := 1 + \epsilon$ *with* $0 < \varepsilon \leq 1$. *Show that* $\varphi \in C^{2,2\epsilon}(\partial\mathbb{B})$ *if* $\varepsilon \leq 1/2$ *and* $\varphi \in C^{3,2\varepsilon-1}(\partial\mathbb{B})$ *if* $1/2 < \varepsilon \leq 1$.

2) *Observe that the function defined by* $U(z,w) := (1 + \Re z)^{1+\epsilon}$, $(z,w) \in \mathbb{B}$, *is plurisubharmonic and continuous in* $\bar{\mathbb{B}}$ *and that it is the unique solution to the homogeneous Dirichlet problem*

$$\begin{cases} u \in PSH(\mathbb{B}) \cap C(\overline{\mathbb{B}}) \\ (dd^c u)^2 = 0 \quad in\ \mathbb{B}, \\ u = \varphi \qquad\quad on\ \partial\mathbb{B}. \end{cases}$$

3) *Check that* $U \in C^{1,\epsilon}(\bar{\mathbb{B}}) \cap C^\infty(\mathbb{B})$. *When* $\varepsilon = 1/2$, *verify that* $\varphi \in C^{2,1}(\partial\mathbb{B})$ *and* U *is no better than* $C^{1,1/2}$.

For $1/2 < \alpha < 1$, *show that Theorem* 5.12 *is optimal: the solution is no better than Lipschitz on* $\bar{\mathbb{B}}$ *when* φ *in no better than* $C^{1,1}$ *on* $\partial\mathbb{B}$.

Exercise 5.12. *Let* $f \in L^p(\mathbb{B})$, *with* $p > 1$, *be a radial non-negative density and* $\varphi \equiv 0$ *on* $\partial\mathbb{B}$.

1) *Prove that the solution* U *of the Dirichlet problem* $(dd^c U)^n = f\beta^n$ *with* $U = 0$ *on* $\partial\mathbb{B}$ *is radial.*

2) *Show that* U *is given, for* $r := |z| < 1$, *by*

$$U(r) = -\int_r^1 \frac{2}{t}\left(\int_0^t \rho^{2n-1} f(\rho)d\rho\right)^{1/n} dt,$$

3) *Check that* $U \in C^{0,2-\frac{2}{p}}(\bar{\mathbb{B}})$ *for* $1 < p < 2$, *hence* $U \in C^{0,1}(\bar{\mathbb{B}})$ *for* $p \geq 2$ (*see* [Mon86] *for more details*).

Exercise 5.13. *Let* φ_j *be a sequence of uniformly bounded plurisubharmonic functions in the unit ball* $\mathbb{B}$ *of* $\mathbb{C}^n$ *such that* $\varphi_j \to \varphi$ *in* L^1, *but* $(dd^c\varphi_j)^n$ *does not converge to* $(dd^c\varphi)^n$. *By modifying* φ_j *near* $\partial\mathbb{B}$, *construct a sequence of probability measures* μ_j *in* $\mathbb{B}$ *and one of plurisubharmonic functions* ψ_j *in* $\mathbb{B}$ *such that:*

- *the* ψ_j*'s are uniformly bounded and continuous in* $\overline{\mathbb{B}}$;
- *the* ψ_j*'s are solutions of the Dirichlet problem* $\mathrm{Dir}(\mathbb{B}, 0, \mu_j)$;
- *the sequence* (μ_j) *converges weakly to a probability measure* μ;
- (ψ_j) *does not converge to the solution of* $\mathrm{Dir}(\mathbb{B}, 0, \mu)$.

This shows that the stability property obtained in Proposition 5.22 *does not hold in general. We refer the interested reader to* [CK94, CK06] *for more information.*

Exercise 5.14. *Let* K *be a compact subset of* $\mathbb{C}^n$. *Recall that the polynomial hull of* K *is*

$$\hat{K} := \{z \in \mathbb{C}^n \,|\, |P(z)| \leq \sup_K |P| \text{ for all polynomials } P\}.$$

Fix a bounded pseudoconvex domain $\Omega \subset \mathbb{C}^n$ *and* $\phi \in \mathcal{C}^\infty(\partial\Omega)$*, and set*

$$F := \{(z, w) \in \partial\Omega \times \mathbb{C} |\ |w| \leq \exp(-\phi(z))\}\,.$$

Show that

$$\hat{F} = \{(z, w) \in \partial\Omega \times \mathbb{C} |\ |w| \leq \exp(-u(z))\}\,,$$

where $u = U_{\mathbb{B},\phi,0}$ *is the unique maximal plurisubharmonic function in* $\mathbb{B}$ *with* ϕ *as boundary values.*

6 Viscosity solutions

A standard approach to non-linear PDEs is the method of viscosity solutions introduced in [Lio83]. This method is local in nature and has been used to prove existence and uniqueness of solutions in a generalized sense for first order non-linear equations of Hamilton–Jacobi type.

It turned out quickly that this method can be used to prove existence and uniqueness of generalized weak solutions to certain fully non-linear second-order degenerate elliptic PDEs [CIL92], especially for those equations for which the notions of classical solution (i.e., smooth solution), generalized solution (i.e., belonging to the Sobolev space $W^{2,\infty}$), or weak solution (i.e., in the sense of distributions) do not make sense.

The main tool in this approach is the *Comparison Principle*. It allows to compare subsolutions and supersolutions with appropriate boundary conditions. It implies uniqueness of viscosity solutions for the associated Dirichlet problem. Perron's method can then be applied to construct the unique solution as the upper envelope of all subsolutions.

Whereas the viscosity approach has been developed for real Monge–Ampère equations [IL90], the complex case has not been studied until recently (see [HL09, EGZ11, HL11, Wan12]).

In this chapter we present the basic ideas underlying the viscosity approach, freely using several useful survey articles [CIL92, CC95, Bar97, DI04]. The presentation is self-contained, except for the Jensen–Ishii maximum principle, whose proof can be found [CIL92]. We then treat degenerate complex Monge–Ampère equations, following [EGZ11].

6.1 The viscosity approach to degenerate non-linear PDEs

We first present a few examples where the standard notions of generalized or weak solution do not make sense.

6.1.1 Classical solutions A fully non-linear second order PDE can be written in the form

$$F(x, u, Du, D^2u) = 0, \tag{6.1.1}$$

where $\Omega \subset \mathbb{R}^N$ is an open set, $\mathcal{S}_N$ is the space of real symmetric matrices of order N, and $F : \Omega \times \mathbb{R} \times \mathbb{R}^N \times \mathcal{S}_N \to \mathbb{R}$ is a function satisfying some conditions that we list below. The first one is the *Degenerate Ellipticity Condition*: for all $x \in \Omega, s \in \mathbb{R}, p \in \mathbb{R}^N, Q_i \in \mathcal{S}_N$,

(DEC) $$Q_1 \leq Q_2 \Longrightarrow F(x, s, p, Q_2) \leq F(x, s, p, Q_1).$$

Here $Q \geq 0$ means that the symmetric matrix Q is semi-positive, i.e., all its eigenvalues are non-negative. Note that the condition (DEC) is usually not satisfied by $-F$.

We say that $u : \Omega \to \mathbb{R}$ is a *classical solution* of the equation (6.1.1) if u is C^2-smooth in Ω and satisfies the differential identity

$$F(x, u(x), Du(x), D^2u(x)) = 0,$$

for all $x \in \Omega$.

Definition 6.1. *If u satisfies the differential inequality*

$$F(x, u(x), Du(x), D^2u(x)) \leq 0$$

(*resp.* $F(x, u(x), Du(x), D^2u(x)) \geq 0$) *pointwise in Ω, we say that u is a classical subsolution* (*resp. supersolution*) *of the equation* (6.1.1).

A function u is a classical solution of (6.1.1) *if and only if it is a classical subsolution and a classical supersolution of* (6.1.1) *in Ω.*

6.1.2 Examples

6.1.2.1 Hamilton–Jacobi equations These equations are first-order equations

$$H(x, u, Du) = 0 \tag{6.1.2}$$

associated to a continuous Hamiltonian $H : \Omega \times \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}$, where $\Omega \subset \mathbb{R}^n$ is an open set.

A prototypical case is the eikonal equation corresponding to the Hamiltonian $H(u) := |Du(x)| - 1$. Consider the Dirichlet problem for the eikonal equation: for all $x \in (-1, 1)$,

$$|u'(x)| - 1 = 0, \text{ with } u(\pm 1) = 0. \tag{6.1.3}$$

A classical solution to this equation is a C^1-smooth function satisfying $|u'(x)| = 1$ pointwise in $]-1, 1[$ and the boundary condition $u(-1) = u(1) = 0$. Such a function cannot exist, since Rolle's lemma imposes the existence of at least one critical point in $]-1, 1[$.

On the other hand, equation (6.1.3) has plenty of generalized solutions, i.e., functions $u \in W^{1,\infty}(]-1, 1[, \mathbb{R})$ satisfying $|u'(x)| = 1$ almost everywhere in $]-1, 1[$. The function $u_0(x) := 1 - |x|$ is one such generalized solution and it is easy to cook up piecewise-affine functions that satisfy (6.1.3) in $[-1, +1]$, except in a given finite set.

If $u \in W^{1,\infty}(]-1, +1[)$ is a generalized solution to the equation associated to the Hamiltonian function $H(u)$, then $-u$ is a generalized solution of the equation associated to the Hamiltonian $\tilde{H}(u) := -H(-u)$. From the point of view of generalized solutions, the corresponding equations are the same and u and $-u$ are different solutions to the same Dirichlet problem. From the viscosity point of view they should however be considered as different, since they correspond to different Hamiltonian functions (see Example 6.12 below).

6.1.2.2 Quasi-linear elliptic equations of second order An important class of elliptic second-order PDE's are the quasi-linear ones,

$$-\sum_{j,k} a^{j,k}(x)\frac{\partial^2 u}{\partial x_j \partial x_k} + H(x,u,Du) = 0, \quad \text{in } \Omega, \tag{6.1.4}$$

where $a = (a^{j,k})$ is an $N \times N$ symmetric matrix-valued function with continuous entries on Ω satisfying the (uniform) ellipticity condition

$$\sum_{j,k} a^{j,k}(x)\xi_j\xi_k \geq \nu|\xi|^2, \quad \forall x \in \Omega,\ \forall \xi \in \mathbb{R}^N, \tag{6.1.5}$$

where $\nu > 0$. These equations are of the type (6.1.1), with

$$F(x,s,p,Q) := -\operatorname{tr}(A(x)Q) + H(x,s,p),$$

where $(x,s,p,Q) \in \Omega \times \mathbb{R} \times \mathbb{R}^N \times \mathcal{S}_N$.

The degenerate ellipticity condition for F is a generalization of the ellipticity condition (6.1.5) (see Exercise 6.3).

A typical example of non-linear but quasi-linear second-order elliptic equation is, for $\varepsilon > 0$ small,

$$-\varepsilon\Delta u + H(x,u,Du) = 0. \tag{6.1.6}$$

This equation can be considered as a small perturbation of the Hamilton–Jacobi equation $H(x,u,Du) = 0$. The small perturbation term $-\varepsilon\Delta u$ is called a *viscosity term* (in fluid mechanics).

This perturbation is uniformly elliptic, hence one can find a unique C^2-smooth solution u_ε with suitable boundary conditions. One can often get uniform L^∞-estimates of u_ε and ∇u_ε independent of $\varepsilon > 0$. Ascoli's theorem ensures that a subsequence converges uniformly to a continuous function u; however, the corresponding subsequence ∇u_ε only converges weakly in L^∞. This does not allow one to pass to the limit in (6.1.6) as $\varepsilon \searrow 0$ and does not produce a generalized/weak solution to the equation (6.1.2), as the eikonal equation shows.

On the other hand, one should expect that u is a "solution" of the equation (6.1.6). The *vanishing viscosity method* will soon allow us to justify this claim.

6.1.2.3 Degenerate Monge–Ampère equations We now consider

$$-\det(D^2u) + f(x,u,Du) = 0, \quad \text{in } \Omega, \tag{6.1.7}$$

where $\Omega \subset \mathbb{R}^N$ is a convex domain, $f : \Omega \times \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}^+$ is a continuous non-negative function in Ω, non-decreasing in u, and we assume that the unknown is a convex function $u : \Omega \to \mathbb{R}$.

This equation is degenerate elliptic when restricted to an appropriate convex subset of the space of symmetric matrices: set

$$F(x,s,p,Q) := \begin{cases} -\det(Q) + f(x,s,p), & \text{if } Q \geq 0, \\ +\infty, & \text{otherwise.} \end{cases}$$

Then F is lower semi-continuous on $\Omega \times \mathbb{R} \times \mathbb{R}^N \times \mathcal{S}_n$ and continuous on its domain $\{F < +\infty\}$, and $F(x, u, Du, D^2u) = 0$ is a *real* degenerate elliptic equation of Monge–Ampère type.

We are mainly interested in *complex* Monge–Ampère equations

$$-\det\left(\frac{\partial^2 u}{\partial z_j \partial \bar{z}_k}\right) + f(z, u, Du) = 0, \quad \text{in } \Omega, \tag{6.1.8}$$

where $\Omega \subset \mathbb{C}^n$ is an open set, and $f : \Omega \times \mathbb{R} \times \mathbb{R}^{2n} \to \mathbb{R}^+$ is non-negative, continuous, and increasing in u. The unknown function u is here a bounded plurisubharmonic function $u : \Omega \to \mathbb{R}$.

This equation can also be written $-(dd^c u)^n + f(z, u, Du)\beta^n = 0$. As in the real case, it is degenerate elliptic when restricted to an appropriate convex subset of the space $\mathcal{H}_n$ of Hermitian matrices: identifying $\mathbb{C}^n$ with $\mathbb{R}^{2n}$, we set

$$F(z, s, p, Q) := \begin{cases} -\det(Q^{1,1}) + f(z, s, p), & \text{if } Q^{1,1} \geq 0, \\ +\infty, & \text{otherwise,} \end{cases}$$

where $(z, s, p, Q) \in \Omega \times \mathbb{R} \times \mathbb{C}^n \times \mathcal{S}_{2n}$ and $Q^{1,1} \in \mathcal{H}_n$ is the Hermitian $(1, 1)$-part of $Q \in \mathcal{S}_{2n}$ considered as a real quadratic form Q on $\mathbb{C}^n$.

Again F is a lower semi-continuous degenerate elliptic Hamiltonian on $\Omega \times \mathbb{R} \times \mathbb{C}^n \times \mathcal{S}_{2n}$, continuous on its domain $\{F < +\infty\}$. In this chapter we focus on the special case

$$-(dd^c\varphi)^n + e^{\varepsilon\varphi}\mu = 0,$$

where $\varepsilon \geq 0$ and $\mu = f\beta^n$ is a volume form with continuous density.

6.1.3 Viscosity concepts

Definition 6.2. *A function F satisfying the degenerate ellipticity condition and the properness condition is called a Hamiltonian.*

6.1.3.1 Hypotheses on F We assume in the sequel that F satisfies the following conditions:

Degenerate ellipticity: $\forall x \in \Omega, (s, p) \in \mathbb{R} \times \mathbb{R}^N,\ Q_i \in \mathcal{S}_N,$

$$Q_1 \leq Q_2 \Longrightarrow F(x, s, p, Q_2) \leq F(x, s, p, Q_1). \tag{6.1.9}$$

Properness: $\forall x \in \Omega, \forall (s_1, s_2) \in \mathbb{R}^2, \forall p \in \mathbb{R}^n, \forall Q \in \mathcal{S}_N,$

$$s_1 \leq s_2 \Longrightarrow F(x, s_1, p, Q) \leq F(x, s_2, p, Q). \tag{6.1.10}$$

Observe that this last condition is satisfied when F does not depend on u.

If F is a Hamiltonian, then $-F$ is not, unless it is independent of u and D^2u. Even when F only depends on Du, as for the eikonal example, the theory will usually provide for F and for $-F$ two different solutions.

6.1.3.2 Viscosity sub- and supersolutions

Definition 6.3. *A function φ defined in a neighborhood of x_0 is called an upper test for u at x_0 if*

- *φ is $\mathcal{C}^2$-smooth;*
- *$u(x_0) = \varphi(x_0)$ and $u \leq \varphi$ near x_0.*

A function ψ defined near x_0 is called a lower test for v at x_0 if

- *ψ is $\mathcal{C}^2$-smooth;*
- *$v(x_0) = \psi(x_0)$ and $v \geq \psi$ near x_0.*

The idea of the viscosity approach is to test differential (in)equalities (or properties) of a non-differentiable function by using these differential tests:

Definition 6.4.

1. *A viscosity subsolution of the equation $F(x, u, Du, D^2u) = 0$ is an upper semi-continuous function $u : \Omega \to \mathbb{R}$ such that for any x_0, all upper tests φ for u at x_0 satisfy*

$$F(x_0, \varphi(x_0), D\varphi(x_0); D^2\varphi(x_0)) \leq 0.$$

2. *A viscosity supersolution of the equation $F(x, u, Du, D^2u) = 0$ is a lower semi-continuous function $v : \Omega \to \mathbb{R}$ such that for any x_0, all lower tests ψ for v at x_0 satisfy*

$$F(x_0, \psi(x_0), D\psi(x_0); D^2\psi(x_0)) \leq 0.$$

3. *A function $w : \Omega \to \mathbb{R}$ is a viscosity solution of the equation $F(x, w, Dw, D^2w) = 0$ if it is both a subsolution and a supersolution.*

Note that a viscosity solution is continuous in Ω. These definitions are consistent with the classical ones, as shown by the following result which emphasizes the role of the maximum principle:

Proposition 6.5. *Assume F is degenerate elliptic and $u \in \mathcal{C}^2(\Omega)$.*

1. *The function u is a classical subsolution of (6.1.1) if and only if it is a viscosity subsolution.*
2. *The function u is a classical supersolution of (6.1.1) if and only if it is a viscosity supersolution.*

Proof. We only prove the first item. Assume u is a viscosity subsolution. Since u is $\mathcal{C}^2$, it can be used as an upper test function at any point and has to satisfy the corresponding differential inequality. It is thus a subsolution in the classical sense.

Assume now u is a classical solution. Fix an upper-test φ for u at x_0. Then $u - \varphi$ is a C^2-smooth function in a neighborhood of x_0 which attains its local maximum at x_0. We infer $D(u - \varphi)(x_0) = 0$ and $D^2(u - \varphi)(x_0) \leq 0$. Since $u(x_0) = \varphi(x_0)$,

$D\varphi(x_0) = Du(x_0)$, $D^2u(x_0) \leq D^2\varphi(x_0)$, and u is a classical subsolution, it follows from the degenerate ellipticity condition that

$$F(x_0, \varphi(x_0), D\varphi(x_0), D^2\varphi(x_0)) \leq F(x_0, u(x_0), Du(x_0), D^2u(x_0)) \leq 0.$$

Thus u is a viscosity subsolution. □

A basic observation is that the class of subsolutions is stable under maximum:

Lemma 6.6. *If u_1, u_2 are subsolutions of $F(x, u, Du, D^2u) = 0$, so is $\sup\{u_1, u_2\}$.*

Similarly, the minimum of two supersolutions is again a supersolution. The proof are straightforward and left to the reader.

6.1.3.3 The classical comparison principle We now illustrate the importance of the Properness condition (PRC) with the following:

Proposition 6.7. *Let $\Omega \Subset \mathbb{R}^N$ be a bounded domain. Assume the Hamiltonian $F(x, s, p, Q)$ is strictly increasing in s. If $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ is a classical subsolution of (6.1.1) and v is a viscosity supersolution of (6.1.1) such that $u \leq v$ on $\partial\Omega$ then $u \leq v$ on Ω.*

In particular, equation (6.1.1) has at most one classical solution with the prescribed continuous boundary values.

The same conclusion holds if we rather assume that u is a viscosity subsolution and v is a classical supersolution.

Proof. Since $u - v$ is continuous on the compact set $\overline{\Omega}$, it attains its maximum at some point $x_0 \in \Omega$, $\max_{\overline{\Omega}}(u - v) = u(x_0) - v(x_0)$.

If $x_0 \in \partial\Omega$, then $u(x_0) \leq v(x_0)$ and we are done. Assume that $x_0 \in \Omega$. Then $u(x) - u(x_0) + v(x_0) \leq v(x)$ in a neighborhood of x_0, hence the C^2-function $q(x) := u(x) - u(x_0) + v(x_0)$ is a lower test function for v at x_0. Since v is a supersolution, it follows that

$$F(x_0, v(x_0), Dq(x_0), D^2q(x_0)) \geq 0.$$

On the other hand, u is a classical subsolution, so

$$F(x_0, u(x_0), Du(x_0), D^2u(x_0)) \leq 0.$$

Combining these two inequalities yields

$$F(x_0, u(x_0), Du(x_0), D^2u(x_0)) \leq F(x_0, v(x_0), Dq(x_0), D^2q(x_0)).$$

Since F is strictly increasing, we infer that $u(x_0) \leq v(x_0)$. □

When F does not depend on r (no strict monotonicity), the result still holds, but the proof is more involved and requires a refined Maximum Principle due to Alexandrov, Bakelman, and Pucci (see [CC95]).

6.1.4 Jet differentials As we have seen, the crucial information is contained in the jet (2-jet) of order 2 of the function u. We therefore develop a sub-differential calculus and study sub/super-jets of order 2.

Definition 6.8.

1. *Let $u : \Omega \to \mathbb{R}$ be an upper semi-continuous function and $x_0 \in \Omega$. The super-differential 2-jet of u at x_0 is the set $J^{2,+}u(x_0)$ of all $(p, Q) \in \mathbb{R}^N \times \mathcal{S}_N$ such that for all $\xi \in \mathbb{R}^N$ with $|\xi| \ll 1$,*

$$u(x_0 + \xi) \leq u(x_0) + \langle p, \xi \rangle + \frac{1}{2}\langle Q\xi, \xi \rangle + o(|\xi|^2).$$

2. *Let $u : \Omega \to \mathbb{R}$ a lower semi-continuous function and $x_0 \in \Omega$. The sub-differential 2-jet of u at x_0 is the set $J^{2,-}u(x_0)$ of all $(p, Q) \in \mathbb{R}^N \times \mathcal{S}_N$ such that for all $\xi \in \mathbb{R}^N$ with $|\xi| \ll 1$,*

$$u(x_0 + \xi) \geq u(x_0) + \langle p, \xi \rangle + \frac{1}{2}\langle Q\xi, \xi \rangle + o(|\xi|^2).$$

3. *If u is continuous we define the differential 2-jet as*

$$J^2u(x_0) := J^{2,+}u(x_0) \cap J^{2,-}u(x_0).$$

Remark 6.9. *If u is twice differentiable at x_0 then*

$$J^{2,+}u(x_0) = \{(Du(x_0), Q) |\ Q \geq D^2u(x_0)\},$$

and

$$J^{2,-}u(x_0) = \{(Du(x_0), Q) |\ Q \leq D^2u(x_0)\},$$

so that

$$J^2u(x_0) = \{(Du(x_0), D^2u(x_0)\}.$$

For an arbitrary upper semi-continuous function, the set $J^{2,+}u(x_0)$ may be empty. However, there are usually many points nearby with non-empty jets, as the convex function $u(x) := |x|$ shows at $x_0 = 0$. For an upper semi-continuous function u which is bounded from above, the set

$$\mathcal{D}^{2,+}_\Omega u := \{x \in \Omega |\ J^{2,+}u(x) \neq \emptyset\}$$

is dense in Ω (see Exercise 6.7).

It is useful to extend the previous definitions by introducing the notions of approximate super(sub)-differential jets:

Definition 6.10. *Let $u : \Omega \to \mathbb{R}$ be an upper semi-continuous function and $x_0 \in \Omega$ and $(p, Q) \in \mathbb{R}^N \times \mathcal{S}_N$. We say that $(p, Q) \in \bar{J}^{2,+}u(x_0)$ if there exist a sequence of points $y_j \to x_0$ in Ω and a sequence $(p_j, Q_j) \in J^{2,+}u(y_j)$ such that $(p_j, Q_j) \to (p, Q)$.*

We define similarly $\bar{J}^{2,-}u(x_0)$ for a lower semi-continuous function.

Proposition 6.11.

1. *Assume F is lower semi-continuous. An upper semi-continuous function $u : \Omega \to \mathbb{R}$ is a viscosity subsolution of $F(x, u, Du, D^2u) = 0$ if and only if for all $x_0 \in \Omega$, $(p, Q) \in \bar{J}^{2,+}u(x_0)$, $F(x_0, u(x_0), p, Q) \leq 0$.*

2. *Assume F is upper semi-continuous. A lower semi-continuous function v is a viscosity supersolution of $F(x, u, Du, D^2u) = 0$ if and only if for all $x_0 \in \Omega$, $(p, Q) \in \bar{J}^{2,-}u(x_0)$, $F(x_0, u(x_0), p, Q) \geq 0$.*

Proof. It suffices to prove the first claim. To prove that the condition is sufficient, it is enough to show that if ϕ is an upper test function for u at some point x_0 then $(D\phi(x_0), D^2\phi(x_0)) \in J^{2,+}u(x_0)$. This follows from Taylor's formula: if $|\xi| \ll 1$ and $x = x_0 + \xi \in \Omega$, then

$$\phi(x) = \phi(x_0) + \langle D\phi(x_0), \xi\rangle + \frac{1}{2}\langle D^2\phi(x_0) \cdot \xi, \xi\rangle + o(|\xi|^2).$$

Since $u \leq_{x_0} \phi$ with $u(x_0) = \phi(x_0)$, it follows that

$$u(x) \leq u(x_0) + \langle D\phi(x_0), \xi\rangle + \frac{1}{2}\langle D^2\phi(x_0)\xi, \xi\rangle + o(|\xi|^2),$$

hence $(D\phi(x_0), D^2\phi(x_0)) \in J^{2,+}u(x_0)$.

To prove the converse, we can assume $(p, Q) \in J^{2,+}u(x_0)$: by approximation, the result will then follow by lower semi-continuity of F. We let the reader check in Exercise 6.8 that any $(p, Q) \in J^{2,+}(u)$ is the 2-jet of a C^2-function at the point x_0. □

Example 6.12. *Consider*

$$F(x, u, Du, D^2u) = H(u') = |u'| - 1 = 0$$

on $[-1, +1]$, *with the the boundary condition* $u(\pm 1) = 0$.

The function $u_0(x) = 1 - |x|$ is a generalized solution, as it satisfies $|u_0'(x)| = 1$ except at the origin, where it is not differentiable.

The equation has infinitely many piecewise-affine generalized solutions with these boundary values. Among these, u_0 is the only one which is a viscosity solution: the only problem is at the origin. Namely, any upper test function q for u_0 at 0 satisfies the condition $|q'(0)| \leq 1$, while there is no lower test function for u at the origin.

Observe also that u_0 is not a subsolution for the equation $1 - |u'| = 0$, since any upper test function q at the origin should satisfy $1 \leq |q'(0)|$, while we have noticed that $|q'(0)| \leq 1$ hence $|q'(0)| = 1$, a contradiction. The same reasoning as above shows that $v_0(x) = |x| - 1$ is the viscosity solution of $1 - |u'| = 0$ with boundary values 0.

6.1.5 Jensen's maximum principle

Definition 6.13. *A function $\varphi : \Omega \to \mathbb{R}$ is semi-convex if there exists $k > 0$ such that $x \mapsto \varphi(x) + \frac{k}{2}|x|^2$ is convex in Ω. We also say that φ is a k-convex function in Ω.*

A function φ is k-concave in Ω if $-\varphi$ is k-convex in Ω.

Definition 6.14. *We say that $\varphi : \Omega \to \mathbb{R}$ is twice differentiable at the point x_0 (in the sense of Alexandrov) if there exists $p \in \mathbb{R}^N$ and $Q \in \mathcal{S}_N$ such that for $|\xi| \ll 1$,*

$$(A) \qquad w(x_0 + \xi) = w(x_0) + \langle p, \xi \rangle + \frac{1}{2}\langle Q \cdot \xi, \xi \rangle + o(|\xi|^2).$$

Remark 6.15. *The condition (A) means that $J^2 w(x_0) = \{(p, Q)\}$. This implies that w is differentiable at x_0 and $Dw(x_0) = p$, but in general it does not mean that u is twice differentiable in the usual sense at x_0: the function w is not necessarily differentiable near x_0, however the quadratic form Q satisfying (A) is unique, given, for $\xi \in \mathbb{R}^N$, by*

$$Q(\xi) = \lim_{t \to 0} \frac{w(x_0 + t\xi) + w(x_0 - t\xi) - 2w(x_0)}{t^2},$$

We will write $Q = D^2 w(x_0)$ and $J^2 w(x_0) = \{(Dw(x_0), D^2 w(x_0))\}$.

It follows from the definitions that if a k-convex function w is twice differentiable at $x_0 \in \Omega$, then $D^2 w(x_0) \geq -k I_N$ in the sense of quadratic forms on $\mathbb{R}^N$.

Recall the following fundamental result of A. D. Alexandrov:

Theorem 6.16. *Let φ be a k-convex function in Ω. There exists a Borel set $E \subset \Omega$ of Lebesgue measure 0 such that φ is twice differentiable at any point $x_0 \in \Omega \setminus E$, hence $D^2 \varphi(x_0) \geq -k I_N$.*

For a convex function, the set $\mathcal{D}^{2,+}_\Omega u$ is thus dense with full Lebesgue measure in Ω. Alexandrov's theorem is an important tool in the proof of the following maximum principle due to Jensen:

Theorem 6.17. *Let w be a semi-convex function in an open set $\Omega \subset \mathbb{R}^N$ which has a strict local maximum at $x_0 \in \Omega$.*

Fix $r, \delta > 0$. The set of points $y \in B(x_0, r)$ where w is twice differentiable and there exists $p \in \mathbb{R}^N$, $|p| < \delta$, such that the function $x \mapsto w(x) + \langle p, x \rangle$ achieves its maximum on $\bar{B}(x_0, r)$ at y has positive Lebesgue measure.

Thus there exist sequences $y_j \to x_0$ in Ω and $p_j \to 0$ in $\mathbb{R}^n$ such that w is twice differentiable at each y_j, the function $x \mapsto w(x) + \langle p_j, x \rangle$ has a local maximum at y_j, $Dw_j(y_j) \to 0$ and $D^2 w(y_j) \to Q^+$ in $\mathcal{S}_N$, with $Q^+ \leq 0$, hence $(0, Q^+) \in \bar{J}^{2,+} w(x_0)$.

We refer the reader to [CIL92, DI04] for a proof of this important result.

Let $u : \Omega \to \mathbb{R}$ be a bounded upper semi-continuous function. For $\varepsilon > 0$ and $x \in \Omega_\varepsilon$, we define the sup-convolutions of u as follows:

$$u^\varepsilon(x) := \sup_{y\in\Omega_\varepsilon} \left\{ u(y) - \frac{1}{2\varepsilon^2}|y-x|^2 \right\} = \sup_{|y-x|\le A\varepsilon} \left\{ u(y) - \frac{1}{2\varepsilon^2}|y-x|^2 \right\},$$

where $A > 0$ is large enough so that $A^2 > 2\,\mathrm{osc}_\Omega\, u$.

Similarly, if $v : \Omega \to \mathbb{R}$ is a bounded lower semi-continuous function, $\varepsilon > 0$ and $x \in \Omega_\varepsilon$, we define the inf-convolutions of v by

$$v_\varepsilon(x) := \inf_{y\in\Omega_\varepsilon} \left\{ v(y) + \frac{1}{2\varepsilon^2}|y-x|^2 \right\} = \inf_{|y-x|\le A\varepsilon} \left\{ v(y) + \frac{1}{2\varepsilon^2}|y-x|^2 \right\}.$$

We let the reader check in Exercise 6.5 the following properties:

Proposition 6.18.
1. *Let $u : \Omega \to \mathbb{R}$ be a bounded upper semi-continuous function. Then u_ε is ε^{-2}-convex in Ω_ε and decreases to u in Ω as $\varepsilon \searrow 0$. Moreover, u_ε is a subsolution of the equation $F_\varepsilon(x, w, Dw, D^2w) = 0$, where*

$$F_\varepsilon(x, s, p, Q) := \inf\{F(y, s, p, Q)|\ |y - x| \le A\varepsilon\}.$$

2. *Let $v : \Omega \to \mathbb{R}$ be a bounded lower semi-continuous function. Then v_ε is ε^{-2}-concave in Ω_ε and increases to v in Ω as $\varepsilon \searrow 0$. Moreover, v_ε is a supersolution of $F^\varepsilon(x, w, Dw, D^2w) = 0$, where*

$$F^\varepsilon(x, s, p, Q) := \sup\{F(y, s, p, Q)|\ |y - x| \le A\varepsilon\}.$$

Observe that F_ε (resp. F^ε) is a continuous Hamiltonian which increases (resp. decreases) to F in Ω as ε decreases to 0.

6.2 Degenerate complex Monge–Ampère equations

The purpose of this section is to make the connection between the pluripotential theory of complex Monge–Ampère operators from the previous chapters and the concept of viscosity solutions.

In this chapter (and only in this chapter), we denote by $(dd^c\varphi)^n_{\mathrm{BT}}$ the Monge–Ampère measure in the pluripotential sense, while the notation $(dd^c\varphi)^n$ is reserved for the viscosity sense.

6.2.1 Viscosity subsolutions of $(dd^c\varphi)^n = \mu$ Let Ω be a bounded domain in $\mathbb{C}^n$ and $\mu = f\beta^n$ a semi-positive volume form in Ω with a continuous density $f \ge 0$ in Ω.

Definition 6.19. *An upper semi-continuous function $\varphi : \Omega \to \mathbb{R} \cup \{-\infty\}$ is a viscosity subsolution of the Monge–Ampère equation*

$$(dd^c\varphi)^n = f\beta^n \tag{6.2.1}$$

if it satisfies the following conditions

1. $\varphi|_\Omega \not\equiv -\infty$.
2. *For every $x_0 \in \Omega$ and any $\mathcal{C}^2$-function q defined on a neighborhood of x_0 such that $\varphi - q$ has a local maximum at x_0,*

$$(dd^c q(x_0))^n \geq f(x_0)\beta^n(x_0).$$

We first observe that a function φ satisfies $(dd^c\varphi)^n \geq 0$ in the viscosity sense if and only if it is plurisubharmonic:

Proposition 6.20. *The viscosity subsolutions of the homogeneous complex Monge–Ampère equation $(dd^c\varphi)^n = 0$ are precisely the plurisubharmonic functions in Ω.*

Proof. The fact that plurisubharmonic functions are viscosity subsolutions of the complex Monge–Ampère equation $(dd^c\varphi)^n = 0$ follows from Proposition 5.9.

Conversely, assume φ is a viscosity subsolution of the complex Monge–Ampère equation $(dd^c\varphi)^n = 0$. Let $z_0 \in \Omega$ and $q \in \mathcal{C}^2(z_0)$ such that $\varphi \leq_{z_0} q$. Then for any semi-positive Hermitian matrix $H \in H_n^+$, the function

$$q_H(z) := q(z) + \langle H(z - z_0), z - z_0\rangle$$

is also an upper test function for u at z_0. By the definition of viscosity subsolution, we conclude that $(dd^c q_H(z_0))^n \geq 0$. This shows that $\det(Q + H) \geq 0$.

Hence, by Exercise 6.9, $dd^c q(z_0) \geq 0$. Proposition 5.9 finally shows that u is plurisubharmonic. □

When φ is plurisubharmonic and locally bounded, its Monge–Ampère measure $(dd^c\varphi)^n_{\mathrm{BT}}$ in Ω is well defined in the pluripotential sense of Bedford and Taylor, as we have explained in previous chapters.

Our next result establishes the basic connection between pluripotential and viscosity subsolutions:

Proposition 6.21. *Let φ be a locally bounded upper semi-continuous function in Ω and f a continuous density in Ω. The following are equivalent:*

1. *φ satisfies $(dd^c\varphi)^n \geq f\beta^n$ in the viscosity sense.*
2. *φ is plurisubharmonic and $(dd^c\varphi)^n_{\mathrm{BT}} \geq f\beta^n$.*

Proof. Assume $\varphi \in PSH \cap L^\infty(\Omega)$ satisfies $(dd^c\varphi)^n_{\mathrm{BT}} \geq \mu := f\beta^n$. Consider a $\mathcal{C}^2$ upper test function q for u at a point $z_0 \in \Omega$, i.e., $\varphi - q$ achieves a local maximum at z_0 and $\varphi(z_0) = q(z_0)$. Since φ satisfies $(dd^c\varphi)^n \geq 0$ in the viscosity sense, $dd^c q(z_0) \geq 0$ by Exercise 6.9 and $(dd^c q)^n_{z_0} \geq 0$. We want to prove that $(dd^c q)^n_{z_0} \geq \mu_{z_0}$. Assume, by contradiction, that $(dd^c q)^n_{z_0} < \mu_{z_0}$. Let $q^\varepsilon(z) := q(z) + \varepsilon\|z - z_0\|^2$. Choosing $\varepsilon > 0$ small enough, we also have $0 < (dd^c q^\varepsilon(z_0))^n < \mu(z_0)$. Since μ has a lower semi-continuous density, we can choose a small ball B centered at z_0 of radius $r > 0$ such that $\bar{q}^\varepsilon = q^\varepsilon - \varepsilon\frac{r^2}{2} \geq \varphi$ near ∂B and $(dd^c\bar{q}^\varepsilon)^n_{\mathrm{BT}} \leq (dd^c\varphi)^n_{\mathrm{BT}}$ in B. The pluripotential comparison principle yields $\bar{q}^\varepsilon \geq \varphi$ on B. But this fails at x_0. Hence, $(dd^c q)^n_{x_0} \geq \mu_{x_0}$ and φ is a viscosity subsolution.

Conversely, assume φ is a viscosity subsolution. Fix $z_0 \in B$ such that $\varphi(z_0) \neq -\infty$ and $q \in \mathcal{C}^2$ such that $\varphi - q$ has a local maximum at x_0. Then the Hermitian matrix $Q = dd^c q(z_0)$ satisfies $\det(Q) \geq \mu(z_0)$.

It follows from Lemma 5.8 that for every positive definite Hermitian matrix $(h_{i\bar{j}})$ with $\det(h) = n^{-n}$,

$$\Delta_H q(z_0) := \sum_{i,j} h_{i\bar{j}} \frac{\partial^2 q}{\partial z_i \partial \bar{z}_j}(z_0) \geq f^{1/n}(z_0),$$

i.e., φ is a viscosity subsolution of the linear equation $\Delta_H \varphi \geq f^{1/n}$.

This is a constant coefficient linear partial differential equation. Assume $f^{1/n}$ is C^∞ and choose a $\mathcal{C}^2$ solution of $\Delta_H \varphi = f^{1/n}$ in a neighborhood of z_0 (see [GT83]). Then $u = \varphi - f$ satisfies $\Delta_H u \geq 0$ in the viscosity sense. It follows from the proof of Proposition 5.9 that u is Δ_H-subharmonic, i.e., $\Delta_H u \geq 0$, hence $\Delta_H \varphi \geq f^{1/n}$ in the sense of distributions.

Using convolution to regularize φ and setting $\varphi_\varepsilon = \varphi * \rho_\varepsilon$ we see that $\Delta_H \varphi_\varepsilon \geq (f^{1/n})_\varepsilon$. Another application of the above reasoning yields

$$(dd^c\varphi_\varepsilon)^n \geq ((f^{1/n})_\varepsilon)^n.$$

Here $\tilde{\varphi}_k = \varphi_{1/k}$ is a decreasing sequence of smooth functions converging to φ. We infer that $(dd^c\varphi)^n_{\mathrm{BT}} \geq f\beta^n$.

This settles the case when $f > 0$ is smooth. In the case where $f > 0$ is merely continuous we observe that $f = \sup\{g | g \in \mathcal{C}^\infty,\ f \geq g > 0\}$ and take into account the fact that any subsolution of $(dd^c\varphi)^n = f\beta^n$ is a subsolution of $(dd^c\varphi)^n = g$ provided $f \geq g$, hence $(dd^c\varphi)^n_{\mathrm{BT}} \geq f\beta^n$.

To treat the general case, we observe that $\psi_\varepsilon(z) = \varphi(z) + \varepsilon\|z\|^2$ satisfies $(dd^c\psi_\varepsilon)^n \geq (f + \varepsilon^n)\beta^n$ in the viscosity sense. Hence,

$$(dd^c\psi_\varepsilon)^n_{\mathrm{BT}} \geq f,$$

which implies that $(dd^c\varphi)^n_{\mathrm{BT}} \geq f\beta^n$. □

We now relax the boundedness assumption and connect viscosity subsolutions to pluripotential subsolutions through the following:

Theorem 6.22. *Assume $f\beta^n \le (dd^c\rho)^n_{\mathrm{BT}}$ for some locally bounded plurisubharmonic function ρ in Ω. Let φ be an upper semi-continuous function such that $\varphi \not\equiv -\infty$ on any connected component. The following are equivalent:*

(i) *φ satisfies $(dd^c\varphi)^n \ge f\beta^n$ in the viscosity sense in Ω;*

(ii) *φ is plurisubharmonic and for all $c > 0$,*

$$(dd^c \sup\{\varphi, \rho - c\})^n_{\mathrm{BT}} \ge f\beta^n,$$

in the pluripotential sense in Ω.

Proof. Assume first that φ is a viscosity subsolution of $(dd^c\varphi)^n = f\beta^n$. Since $\rho - c$ is also a subsolution, it follows from Lemma 6.6 that $\sup\{\varphi, \rho - c\}$ is a subsolution, hence Proposition 6.21 yields

$$(dd^c \sup\{\varphi, \rho - c\})^n_{\mathrm{BT}} \ge v.$$

Conversely, fix $z_0 \in \Omega$ and assume (i) holds. If φ is locally bounded near z_0, Proposition 6.21 implies that φ is a viscosity subsolution near z_0. Assume $\varphi(z_0) \ne -\infty$, but φ is not locally bounded near z_0. Fix $q \in \mathcal{C}^2$ such that $q \ge \varphi$ near z_0 and $q(z_0) = \varphi(z_0)$. Then for $c > 0$ big enough we have $q \ge \varphi_c = \sup(\varphi, \rho - c)$ and $q(z_0) = \varphi_c(z_0)$, hence $(dd^c q(z_0))^n \ge f(z_0)\beta^n$, by Proposition 6.21 again.

Finally if $\varphi(z_0) = -\infty$ there are no q to be tested against the differential inequality, hence it holds for every test function q. □

Remark 6.23. *Condition* (ii) *might seem a bit cumbersome. The point is that the Monge–Ampère operator cannot be defined on the whole space of plurisubharmonic functions. The above arguments actually work in any class of plurisubharmonic functions in which the Monge–Ampère operator is continuous by decreasing limits of locally bounded functions and the*

comparison principle holds. These are precisely the finite energy classes studied in [Ceg98, GZ07]*: when φ belongs to one of these classes, condition* (ii) *is equivalent to $(dd^c\varphi)^n_{\mathrm{BT}} \ge f\beta^n$.*

6.2.2 Viscosity subsolutions of $(dd^c\varphi)^n = e^{\varepsilon\varphi}\mu$

Fix $\mu := f\beta^n$ a continuous volume form in Ω and $\varepsilon > 0$.

Definition 6.24. *An upper semi-continuous function φ is a viscosity subsolution of $(dd^c\varphi)^n = e^{\varepsilon\varphi}\mu$ if φ is not identically $-\infty$ and for all $z_0 \in \Omega$ and all $q \in \mathcal{C}^2(\Omega)$ such that $\varphi - q$ has a local maximum at z_0 and $\varphi(z_0) = q(z_0)$, one has $(dd^c q(z_0))^n \ge e^{\varepsilon q(z_0)}\mu(z_0)$.*

Although the definition of viscosity subsolution is here the same as the one considered previously when $\varepsilon = 0$, the proof that it coincides with the pluripotential concept is considerably more difficult:

Proposition 6.25. *Let $\varphi : \Omega \to \mathbb{R}$ be a bounded upper semi-continuous function. Then φ satisfies $(dd^c\varphi)^n \geq e^{\varepsilon\varphi} f\beta^n$ in the viscosity sense if and only if it is plurisubharmonic and it does so in the pluripotential sense.*

Proof. When φ is continuous, so is the density $\tilde{f} = e^{\varepsilon\varphi} f$ and the result follows from Proposition 6.21. When φ is not assumed to be continuous, the proof is slightly more delicate. Without loss of generality we only treat the case $\varepsilon = 1$.

Assume φ is a viscosity subsolution. It follows from Proposition 6.20 that φ is plurisubharmonic. We approximate φ by its sup-convolutions:

$$\varphi^\delta(z) := \sup_{\zeta\in\Omega} \left\{ \varphi(\zeta) - \frac{1}{2\delta^2}|z-\zeta|^2 \right\}, \quad z \in \Omega_\delta.$$

This is a Lipschitz function such that $z \mapsto \varphi^\delta(z) + \frac{1}{2\delta^2}|z|^2$ is convex.

Fix $A > 0$ so large that $A^2 > 2\operatorname{osc}_\Omega \varphi$. For $\delta > 0$ small enough define $\Omega_\delta := \{x \in \Omega \,|\, \operatorname{dist}(x, \partial\Omega) > A\delta\}$. Then for $z \in \Omega_\delta$, we have

$$\varphi^\delta(z) = \sup_{|\zeta|\leq A\delta} \left\{ \varphi(z-\zeta) - \frac{1}{2\delta^2}|\zeta|^2 \right\}.$$

The functions φ_δ are plurisubharmonic in Ω_δ and decrease towards φ as δ decreases to zero. Furthermore, φ^δ satisfies

$$(dd^c\varphi^\delta)^n \geq e^{\varphi^\delta} f_\delta \beta_n, \text{ with } f_\delta(x) = \inf\{f(y) \,|\, |y-x| \leq A\delta\}$$

in the sense of viscosity (see Lemma 6.26). Since φ^δ is *continuous*, we can invoke Proposition 6.21 and get

$$(dd^c\varphi^\delta)^n_{\mathrm{BT}} \geq e^{\varphi^\delta} f_\delta\, \beta_n \geq e^{\varphi} f_\delta\, \beta_n$$

in the pluripotential sense.

Since the complex Monge–Ampère operator is continuous along decreasing sequences of bounded plurisubharmonic functions, and since f_δ increases towards f, we finally obtain $(dd^c\varphi)^n_{\mathrm{BT}} \geq e^\varphi v$ in the pluripotential sense.

We now treat the other implication. Let φ be a plurisubharmonic function satisfying

$$(dd^c\varphi)^n_{\mathrm{BT}} \geq e^\varphi f\beta^n$$

in the pluripotential sense. We approximate φ using sup-convolution φ^δ as above. Lemma 6.26 shows that

$$(dd^c\varphi^\delta)^n_{\mathrm{BT}} \geq e^{\varphi^\delta} f_\delta \beta_n \tag{6.2.2}$$

in the sense of pluripotential theory in Ω_δ.

Since φ^δ is continuous, we can apply Proposition 6.21 to conclude that φ^δ is a viscosity subsolution of the equation $(dd^cu)^n = e^u f_\delta \beta_n$ in Ω_δ. The rest of the

proof explains how to deduce that φ is a viscosity subsolution of $(dd^c\varphi)^n = e^{\varphi} f\beta_n$, by letting δ decrease to 0.

Fix $z_0 \in \Omega$ and a quadratic polynomial q such that $\varphi(z_0) = q(z_0)$ and $\varphi \le q$ on a ball $2B$, where $B := B(z_0, r) \Subset \Omega$. Since φ is plurisubharmonic in Ω, it satisfies $(dd^c\varphi)^n \ge 0$ in the viscosity sense by Proposition 6.21, hence $dd^c q(z_0) \ge 0$ by (6.9). Replacing q by $q(z)+\varepsilon|z-z_0|^2$ and taking $r > 0$ small enough, we can assume that q is plurisubharmonic on the ball $2B$. We need to prove that $(dd^c q(z_0))^n \ge e^{\varphi(z_0)} f(z_0)\beta_n$. Fix $\varepsilon > 0$, $z \in B$ and set

$$q_\varepsilon(z) := q(x) + 2\varepsilon(|z - z_0|^2 - r^2) + \varepsilon r^2.$$

Since $\varphi \le q$ on $2B$, we deduce that

- if $z \in \partial B$, $\varphi^\delta(z) - q_\varepsilon(z) = \varphi^\delta(z) - q(z) - \varepsilon r^2 < 0$, if $0 < \delta \ll 1$;
- if $z = z_0$, then $\varphi_\delta(z_0) - q^\varepsilon(z_0) = \varphi_\delta(z_0) - q(z_0) + \varepsilon r^2$.

Since $\varphi_\delta(z_0) - q(z_0) \to \varphi(z_0) - q(z_0) + \varepsilon r^2 = \varepsilon r^2$ as $\delta \to 0$, it follows that for δ small, the function $\varphi^\delta(z) - q_\varepsilon(z)$ reaches it maximum on $\bar B$ at some interior point $z_\delta \in B$ with

$$\lim_{\delta\to 0} \max_{\bar B}(\varphi_\delta - q^\varepsilon) = \lim_{\delta\to 0}(\varphi_\delta(z_\delta) - q^\varepsilon(z_\delta)) \ge \varepsilon r^2. \tag{6.2.3}$$

We claim that $z_\delta \to x_0$ as $\delta \to 0$. Observe indeed that

$$\begin{aligned} \varphi^\delta(z_\delta) - q_\varepsilon(z_\delta) &= \varphi^\delta(z_\delta) - q(z_\delta) - 2\varepsilon(|z_\delta - z_0|^2 - r^2) - \varepsilon r^2 \\ &\le q^\delta(z_\delta) - q(z_\delta) - 2\varepsilon|z_\delta - z_0|^2 + \varepsilon r^2. \end{aligned}$$

Since $q^\delta(z_\delta) - q(z_\delta)$ converges to 0, any cluster point z_0' of the family (z_δ) is in $\bar B$, so that, using (6.2.3),

$$\varepsilon r^2 \le \limsup \max_{\bar B}(\varphi_\delta - q^\varepsilon) \le -2\varepsilon|z_0' - z_0|^2 + \varepsilon r^2.$$

Therefore, $-2\varepsilon|z_0' - z_0|^2 \ge 0$, and so $z_0' = z_0$, as claimed.

Since $\max_{\bar B}(\varphi^\delta - q_\varepsilon)$ is realized at $z_\delta \in B$ and φ^δ is a viscosity subsolution of $(dd^c u) = e^u f_\delta \beta_n$, we infer that

$$(dd^c q_\varepsilon(z_\delta))^n \ge e^{\varphi^\delta(z_\delta)} f_\delta(z_\delta)\beta_n = e^{\varphi^\delta(z_\delta) - q_\varepsilon(z_\delta)} e^{q_\varepsilon(z_\delta)} f_\delta(z_\delta)\beta_n.$$

Now $\limsup_{\delta\to 0} \max_{\bar B}(\varphi^\delta - q) = \max_{\bar B}(\varphi - q) = 0$, therefore

$$\limsup_{\delta\to 0}(\varphi^\delta(z_\delta) - q_\varepsilon(z_\delta)) \ge \min_{\bar B}(q - q_\varepsilon) = \min_{\bar B}(-2\varepsilon|z - z_0|^2 + \varepsilon r^2) = -\varepsilon r^2.$$

It follows that $(dd^c q_\varepsilon(z_0))^n \ge e^{q(z_0) - 2\varepsilon r^2} f(z_0)\beta_n$. Letting $\varepsilon \to 0$, we obtain the desired inequality

$$(dd^c q(z_0))^n \ge e^{\varphi(z_0)} f(z_0)\beta_n,$$

since $q(z_0) = \varphi(z_0)$. □

Lemma 6.26. *Let φ be a bounded plurisubharmonic function such that*

$$(dd^c\varphi)^n_{\mathrm{BT}} \geq e^{\varphi} f\beta_n.$$

Then the sup-convolutions (φ^δ) satisfy

$$(dd^c\varphi^\delta)^n_{\mathrm{BT}} \geq e^{\varphi^\delta} f_\delta \beta_n,$$

where $f_\delta(x) := \inf\{f(y); |y-x| \leq A\delta\}$.

Proof. Fix $\delta > 0$ small and set $\psi_y(x) := \varphi(x-y) - \frac{1}{2\delta^2}|y|^2$, where $y \in B(0, A\delta)$ and $x \in \Omega_\delta$. Observe that ψ_y is a bounded plurisubharmonic function in Ω_δ which satisfies

$$(dd^c\psi_y)^n_{\mathrm{BT}} \geq e^{\psi_y} \tilde{f}_\delta \beta_n \geq e^{\psi_y} f_\delta \beta_n,$$

in Ω_δ thanks to the invariance of β_n under translations, where

$$\tilde{f}_\delta(x) = \inf\{f(y)e^{|x-y|^2/2\delta^2} \,|\, |y-x| \leq A\delta\}.$$

Since φ is the upper envelope of the family $\{\psi_y \,|\, y \in B(0, A\delta)\}$, it follows from Lemma 4.31 that there exists $(y_j)_{j\in\mathbb{N}}$ in $B(0, A\delta)$ such that $\varphi^\delta = (\sup_j \psi_{y_j})^*$ in Ω_δ. Set $\theta_j := \sup_{0\leq k\leq j} \psi_{y_k}$. These are bounded plurisubharmonic functions which increase a.e. to φ^δ on Ω_δ. We claim that θ_j is also a pluripotential subsolution of the same equation, i.e.,

$$(dd^c\theta_j)^n_{\mathrm{BT}} \geq e^{\theta_j} f_\delta \beta_n. \tag{6.2.4}$$

This follows Chapter 4, but we recall the argument for convenience: if w_1 and w_2 are two bounded plurisubharmonic functions on Ω, then

$$(dd^c \sup\{w_1, w_2\})^n_{\mathrm{BT}} \geq \mathbf{1}_{\{w_1>w_2\}}(dd^c w_1)^n_{\mathrm{BT}} + \mathbf{1}_{\{w_1\leq w_2\}}(dd^c w_2)^n_{\mathrm{BT}}.$$

If moreover $(dd^c w_i)^n_{\mathrm{BT}} \geq e^{w_i} v$ for $i = 1, 2$, then $w := \sup\{w_1, w_2\}$ also satisfies $(dd^c w)^n_{\mathrm{BT}} \geq e^w v$. This justifies (6.2.4).

Since θ_j increase to φ^δ quasi everywere, the continuity property of the complex Monge–Ampère operator and the inequalities (6.2.4) yield $(dd^c\varphi^\delta)^n_{\mathrm{BT}} \geq e^{\varphi^\delta} f_\delta \beta_n$. □

Remark 6.27. *Let u be a bounded plurisubharmonic function in a domain $\Omega \subset \mathbb{C}^n$ and $\mu = f\beta_n$ a continuous volume form with continuous density $f \geq 0$. Assume that*

$$(dd^c\varphi)^n_{\mathrm{BT}} \geq e^{\varphi} f\beta_n,$$

in the pluripotential sense in Ω. Then for $\delta > 0$ small enough, the usual regularization by convolution $\tilde{\varphi}_\delta := \varphi \star \chi_\delta$ satisfies

$$(dd^c\tilde{\varphi}_\delta)^n_{\mathrm{BT}} \geq e^{\tilde{\varphi}_\delta} \tilde{f}_\delta \beta_n$$

pointwise in Ω_δ, where

$$\tilde{f}_\delta(x) := \inf\{f(y)e^{\frac{1}{2\delta^2}|x-y|^2} \,|\, |y-x| \leq A\delta\},$$

as the reader will check in Exercise 6.12.

6.2.3 Supersolutions of the equation $(dd^c\varphi)^n = e^{\varepsilon\varphi}\mu$ The definition of supersolutions is more delicate. In order to use the framework of the viscosity approach, we need to use a Hamiltonian which is upper semi-continuous. This is not the case for the Hamiltonian used for subsolutions, hence our notion of supersolution is not dual to that of subsolution. We consider, as above,

$$(dd^c\varphi)^n = e^{\varepsilon\varphi} f\beta^n,$$

where $\varepsilon \geq 0$ and f is a continuous function in Ω.

To fit in with the viscosity point of view, we identify $\mathbb{C}^n \simeq \mathbb{R}^{2n}$ and set, for $(z, s, p, Q) \in \Omega \times \mathbb{R} \times \mathbb{C}^n \times \mathcal{S}_{2n}$,

$$F(z, s, p, Q) = \begin{cases} -\det(Q^{1,1}) + e^{\varepsilon s} f(z), & \text{if } Q^{1,1} \geq 0, \\ +\infty, & \text{otherwise,} \end{cases}$$

where $Q^{1,1}$ is the Hermitian $(1,1)$-part of the (real) quadratic form Q on $\mathbb{C}^n \simeq \mathbb{R}^{2n}$. We identify here a Hermitian form with the real $(1,1)$-form associated to it.

This Hamiltonian F is lower semi-continuous in $(z, s, p, Q) \in \Omega \times \mathbb{R} \times \mathbb{C}^n \times \mathcal{S}_{2n}$, continuous in its domain, and satisfies the degenerate ellipticity and the properness conditions. We can thus use the notions of sub/super solutions from the previous sections. However, F is not upper semi-continuous, hence the concept of supersolution is not dual to that of subsolution: when v is a supersolution to $F = 0$, it is not clear if $-v$ is a subsolution.

We introduce the Hamiltonian

$$F_+(z, s, p, Q) = \begin{cases} -\det(Q^{1,1}) + e^{\varepsilon s} f(z), & \text{if } Q^{1,1} \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

If q is a lower test function for φ at x_0, i.e., $\varphi \geq_{x_0} q$, the condition $(dd^c q(x_0))_+^n \leq e^{\varepsilon q(x_0)} f\beta^n$ is always satisfied when $dd^c q(x_0)$ is not semi-positive; so is the condition $F_+(x_0, \varphi(x_0), Dq(x_0), dd^c q(x_0)) \geq 0$. Hence, $F_+(x, s, p, Q) \geq 0$ is consistent only when $dd^c Q \geq 0$. The notion of supersolution corresponding to the Hamiltonian F_+ is:

Definition 6.28. *A supersolution of $(dd^c\varphi)^n = e^{\varepsilon\varphi}\mu$ is a lower semi-continuous function $\psi : \Omega \to \mathbb{R} \cup \{+\infty\}$ such that $\psi \not\equiv +\infty$ and the following property is satisfied: if for any $z_0 \in \Omega$ and any $q \in \mathcal{C}^2(z_0)$, defined in a neighborhood of z_0, such that $\psi(z_0) = q(z_0)$ and $\psi - q$ has a local minimum at z_0, then*

$$(dd^c q(z_0))_+^n \leq e^{\varepsilon\psi(z_0)} f(z_0)\beta^n.$$

The Hamiltonian function F_+ is upper semi-continuous everywhere, degenerate elliptic, and proper, and coincides with F in its domain. It turns out that the notion of subsolution is the same for the two Hamiltonians, as we explain below.

Remark 6.29. *We will use this definition as follows: if φ is not a supersolution of the equation, then there exist a point z_0 and a lower test function q at z_0 such that*

$(dd^c q(z_0))_+^n > e^{\varepsilon q(z_0)} f(z_0)\beta^n \geq 0$. *Therefore,* $dd^c q(z_0) \geq 0$ *and* $(dd^c q(z_0))^n > 0$, *hence* $dd^c q(z_0) > 0$.

If h is an upper semi-continuous function in a ball B, its plurisubharmonic envelope is

$$P(h)(z) = P_B f(z) := (\sup\{\psi(x) |\ \psi \text{ is plurisubharmonic in } B \text{ and } \psi \leq h\})^*,$$

the greatest plurisubharmonic function that lies below h in B. We refer the reader to Exercise 5.10 for some basic properties of these envelopes.

Lemma 6.30.

(1) *Let* ψ *be a bounded plurisubharmonic function satisfying* $(dd^c\psi)_{\mathrm{BT}}^n \leq f(z_0)\beta^n$. *Then its lower semi-continuous regularization* ψ_* *is a viscosity supersolution of the equation* $(dd^c\varphi)^n = \mu$.

(2) *Let* φ *be a continuous viscosity supersolution of the equation* $(dd^c\varphi)^n = f\beta^n$. *Then for any small ball* B, $\psi := P_B(\varphi)$ *is a continuous plurisubharmonic viscosity supersolution of* $(dd^c\psi)^n = f\beta^n$.

(3) *Let* φ *be a* $C^{1,1}$*-smooth viscosity supersolution of* $(dd^c\varphi)^n = f\beta^n$. *Then for any small ball* B, *its plurisubharmonic envelope* $P_B(\varphi)$ *satisfies* $(dd^c P(\varphi))_{\mathrm{BT}}^n \leq f\beta^n$ *in the pluripotential sense in* B.

Proof. (1) Assume $\psi \in PSH \cap L^\infty(\Omega)$ satisfies $(dd^c\psi)_{\mathrm{BT}}^n \leq v$ in the pluripotential sense on Ω. Consider a $\mathcal{C}^2$ function q such that $\psi_*(x_0) = q(x_0)$ and $\psi_* - q$ has a local minimum at x_0. We want to prove that $(dd^c q(x_0))_+^n \leq v(x_0)$. Assume that $(dd^c q(x_0))_+^n > v_{x_0}$. Then $dd^c q(x_0) \geq 0$ and $(dd^c q(x_0))^n > v_{x_0} > 0$, which implies that $dd^c q(x_0) > 0$. Set

$$q^\varepsilon := q - 2\varepsilon(\|x - x_0\|^2 - r^2) - \varepsilon r^2.$$

Since v has continuous density, we can choose $\varepsilon > 0$ small enough and a small ball $B(x_0, r)$ containing x_0 of radius $r > 0$ such that $dd^c q^\varepsilon > 0$ in $B(x_0, r)$ and $(dd^c q^\varepsilon)^n > v$ on the ball $B(x_0, r)$. Thus we have $q^\varepsilon = q - \varepsilon r^2 < \psi_* \leq \psi$ near $\partial B(x_0, r)$, while $(dd^c q^\varepsilon)_{\mathrm{BT}}^n \geq v \geq (dd^c\psi)_{\mathrm{BT}}^n$ on $B(x_0, r)$. The comparison principle yields $q^\varepsilon \leq \psi$ on $B(x_0, r)$, hence

$$q^\varepsilon(x_0) = \liminf_{x \to x_0} q^\varepsilon(x) \leq \liminf_{x \to x_0} \psi(x) = \psi_*(x_0),$$

i.e., $q(x_0) + \varepsilon r^2 \leq \psi_*(x_0) = q(x_0)$, which is a contradiction.

(2) Set $\psi := P(\varphi)$, fix a point $x_0 \in \Omega$, and consider a super test function q for ψ at x_0, i.e., q is a C^2-function on a small ball $B(x_0, r) \subset \Omega$ such that $\psi(x_0) = q(x_0)$ and $\psi - q$ attains its minimum at x_0. We want to prove that $(dd^c q(x_0))_+^n \leq v(x_0)$. Since $\psi \leq \varphi$, there are two cases:

- if $\psi(x_0) = \varphi(x_0)$, then q is also a super test function for φ at x_0, and then $(dd^c q(x_0))^n_+ \leq v(x_0)$, since φ is a supersolution of the same equation;
- if $\psi(x_0) < \varphi(x_0)$, then by the continuity of φ there exists a ball $B(x_0, s)$, $0 < s < r$, such that $\psi = P(\varphi) < \varphi$ on the ball $B(x_0, s)$, and then $(dd^c\psi)^n = 0$ on $B(x_0, s)$ since $(dd^c P(\varphi))^n$ is supported on the set $\{P(\varphi) = \varphi\}$.

Therefore, ψ is a continuous plurisubharmonic function satisfying the inequality $(dd^c\psi)^n = 0 \leq v$ in the sense of pluripotential theory on $B(x_0, s)$. Assume that $(dd^c q(x_0))^n_+ > v(x_0)$. Then, by definition, $dd^c q(x_0) > 0$ and $(dd^c q(x_0))^n > v(x_0)$. Taking $s > 0$ small enough and $\varepsilon > 0$ small enough, we can assume that

$$q^\varepsilon := q - \varepsilon(|x - x_0|^2 - s^2)$$

is plurisubharmonic on $B(x_0, s)$ and $(dd^c q^\varepsilon)^n > v \geq (dd^c\psi)^n$ on the ball $B(x_0, s)$, while $q^\varepsilon = q \leq \psi$ on $\partial B(x_0, s)$. By the pluripotential

comparison principle for the complex Monge–Ampère operator, it follows that $q^\varepsilon \leq \psi$ on $B(x_0, s)$, so $q(x_0) + \varepsilon s^2 \leq \psi(x_0)$, which is a contradiction.

(3) It follows from the tools developed in Chapter 5 that $P(\varphi)$ is a $C^{1,1}$-smooth function and its Monge–Ampère measure $(dd^c P(\varphi))^n_{\mathrm{BT}}$ is concentrated on the set where $P(\varphi) = \varphi$, and satisfies

$$(dd^c P(\varphi))^n_{\mathrm{BT}} = \mathbf{1}_{\{P(\varphi)=\varphi\}}(dd^c\varphi)^n.$$

The conclusion follows from the definition of supersolutions. □

It would be useful to know whether assertion (3) of the lemma is valid for less regular supersolutions.

6.3 The viscosity comparison principle

We say that the viscosity comparison principle holds for the complex Monge–Ampère equation

$$(dd^c\varphi)^n = e^{\varepsilon\varphi} f\beta^n \tag{6.3.1}$$

if for all bounded subsolutions $\underline{u}$ (resp. supersolutions $\overline{u}$) of (6.3.1) in Ω such that $\limsup_{z\to\partial\Omega}[\underline{u}(z) - \overline{u}(z)] \leq 0$, one has $\underline{u} \leq \overline{u}$ in Ω.

The goal of this section is to establish the comparison principle:

Theorem 6.31. *The viscosity comparison principle holds for* (6.3.1) *if* $\varepsilon > 0$ *and* $f \geq 0$.

Proof. We can assume, without loss of generality, that $\varepsilon = 1$. We first assume that $f > 0$ in Ω. Let $\underline{u}$ be a bounded subsolution and $\overline{u}$ be a bounded supersolution of (6.3.1) in some bounded open set Ω, such that $\underline{u} \leq \overline{u}$ on $\partial\Omega$. Replacing first $\underline{u}, \overline{u}$

by $\underline{u} - \delta, \overline{u} + \delta$, we can assume that the inequality is strict and holds in a small neighborhood of $\partial\Omega$.

As in the proof of Proposition 6.25, we regularize $\underline{u}$ and $\overline{u}$ using their sup/inf convolutions. Since $\underline{u}$ and $\overline{u}$ are bounded, multiplying by a small constant we can assume that for $\alpha > 0$ small and $x \in \Omega_\alpha$,

$$\underline{u}^\alpha(x) := \sup_{y\in\Omega}\left\{\underline{u}(y) - \frac{1}{2\alpha^2}|y-x|^2\right\} = \sup_{|y-x|\le\alpha}\left\{\underline{u}(y) - \frac{1}{2\alpha^2}|y-x|^2\right\},$$

and

$$\overline{u}_\alpha(x) := \inf_{y\in\Omega_\alpha}\left\{\overline{u}(y) + \frac{1}{2\alpha^2}|y-x|^2\right\} = \inf_{|y-x|\le\alpha}\left\{\overline{u}(y) + \frac{1}{2\alpha^2}|y-x|^2\right\}.$$

For $\alpha > 0$ small enough we thus get $\underline{u}^\alpha(x) \le \overline{u}_\alpha(x)$ near the boundary of Ω_α. Observe that, if we set $M_\alpha := \sup_{\overline{\Omega}_\alpha}(\underline{u}^\alpha - \overline{u}_\alpha)$, then

$$\liminf_{\alpha\to 0^+} M_\alpha \ge \sup_\Omega(\underline{u} - \overline{u}).$$

Arguing by contradiction, assume that $\sup_\Omega(\underline{u} - \overline{u}) > 0$. Then for $\alpha > 0$ small enough, the supremum M_α is > 0 and then it is attained at some point $x_\alpha \in \Omega_\alpha$. The function $\underline{u}^\alpha$ is semi-convex and $\overline{u}_\alpha$ is semi-concave. In particular, they are twice differentiable almost everywhere in Ω_α by Alexandrov' Theorem 6.16.

We first reach a contradiction under the unrealistic assumption that they are twice differentiable at the point x_α. The classical maximum principle yields

$$D^2\underline{u}^\alpha(x_\alpha) \le D^2\overline{u}_\alpha(x_\alpha),$$

in the sense of quadratic forms on $\mathbb{R}^{2n}$. Applying this inequality for vectors of the form (Z, Z) and (iZ, iZ) and adding, we get the same inequality for Levi forms on $\mathbb{C}^n$, i.e.,

$$0 \le dd^c\underline{u}^\alpha(x_\alpha) \le dd^c\overline{u}_\alpha(x_\alpha),$$

where the first inequality follows from the fact that $\underline{u}^\alpha$ is plurisubharmonic on Ω_α since $\underline{u}$ is. From this inequality between non-negative Hermitian forms on $\mathbb{C}^n$ it follows that the same inequality holds between their determinants, i.e.,

$$(dd^c\underline{u}^\alpha)^n(x_\alpha) \le (dd^c\overline{u}_\alpha)^n(x_\alpha).$$

Now

$$(dd^c\overline{u}_\alpha)^n(x_\alpha) \le e^{\overline{u}_\alpha(x_\alpha)} f^\alpha(x_\alpha)\beta_n,$$

where f_α increases pointwise towards f, with $v = f\beta_n$, and

$$(dd^c\underline{u}^\alpha)^n(x_\alpha) \ge e^{\underline{u}^\alpha(x_\alpha)} f_\alpha(x_\alpha)\beta_n,$$

where f^α decreases pointwise towards f. For α small enough we infer

$$e^{\underline{u}^\alpha(x_\alpha)} f_\alpha(x_\alpha) \le e^{\overline{u}_\alpha(x_\alpha)} f^\alpha(x_\alpha), \tag{6.3.2}$$

hence

$$\sup_\Omega (\underline{u} - \overline{u}) \le \overline{\lim}_{\alpha\to 0} M_\alpha \le 0 = \lim_{\alpha\to 0} \log \frac{f^\alpha(x_\alpha)}{f_\alpha(x_\alpha)},$$

which contradicts our assumption that $\sup_\Omega (\underline{u} - \overline{u}) > 0$.

Assume now that $\underline{u}^\alpha$ and $\overline{u}_\alpha$ are not twice differentiable at x_α. We prove that the inequality (6.3.2) is still valid by approximating x_α by a sequence of points where the functions are twice differentiable and not far from attaining their local maximum at those points. More precisely, for $k \in \mathbb{N}^*$ the semiconvex function $\underline{u}^\alpha - \overline{u}_\alpha - (1/2k)|x - x_\alpha|^2$ attains a strict local maximum at x_α. By Theorem 6.17, there exist a sequence $(p_k)_{k\ge 1}$ of vectors converging to 0 in $\mathbb{R}^n$ and a sequence of points (y_k) converging to x_α in Ω_α such that the functions $\underline{u}^\alpha$ and $\overline{u}_\alpha$ are twice differentiable at y_k and, if we set $q_k(x) = (1/2k)|x - x_\alpha|^2 + \langle p_k, x\rangle$, the function $\underline{u}^\alpha - \overline{u}_\alpha - q_k$ attains its local maximum at the point y_k.

Applying the classical maximum principle at each point y_k, we get

$$D^2\underline{u}^\alpha(y_k) \le D^2\overline{u}_\alpha(y_k) + (1/k) I_n,$$

in the sense of quadratic forms on $\mathbb{R}^{2n}$. We infer as earlier that

$$0 \le dd^c\underline{u}^\alpha(y_k) \le dd^c\overline{u}_\alpha(y_k) + (1/k)\, dd^c|x|^2, \tag{6.3.3}$$

in the sense of positive Hermitian forms on $\mathbb{C}^n$, where the first inequality follows from the fact that $\underline{u}^\alpha$ is plurisubharmonic on Ω_α. The inequality (6.3.3) between positive Hermitian forms implies

$$(dd^c\underline{u}^\alpha(y_k))^n \le (dd^c\overline{u}_\alpha(y_k) + (1/k)\, dd^c|x|^2)^n \tag{6.3.4}$$

Recall that $\overline{u}_\alpha - (1/2\alpha^2)|x|^2$ is concave on Ω_α, hence

$$D^2\overline{u}_\alpha(y_k) \le (1/2\alpha^2) I_n,$$

in the sense of quadratic forms on $\mathbb{R}^{2n}$. Therefore:

$$dd^c\overline{u}_\alpha(y_k) \le (1/2\alpha^2)\, dd^c|x|^2. \tag{6.3.5}$$

It follows from (6.3.3) and (6.3.5) that

$$(dd^c\overline{u}_\alpha(y_k) + (1/k)\, dd^c|x|^2)^n = (dd^c\overline{u}_\alpha(y_k))^n + O(1/k). \tag{6.3.6}$$

By the definition of subsolutions and supersolutions,

$$(dd^c\underline{u}^\alpha(y_k))^n \ge e^{\underline{u}^\alpha(y_k)} f_\alpha(y_k)\beta_n, \quad (dd^c\overline{u}_\alpha(y_k))^n \le e^{\overline{u}_\alpha(y_k)} f^\alpha(y_k)\beta_n, \tag{6.3.7}$$

therefore (6.3.4), (6.3.6), and (6.3.7) yield, for all $k \geq 1$,

$$e^{\underline{u}^\alpha(y_k)} f_\alpha(y_k) \leq e^{\overline{u}_\alpha(y_k)} f^\alpha(y_k) + O(1/k),$$

which implies (6.3.2) as $k \to +\infty$. This yields, as above, a contradiction.

Assume finally that $f \geq 0$. Set $u_\varepsilon(x) := \underline{u}(x) + \varepsilon(|x|^2 - R^2)$ for $\varepsilon > 0$ and $R > 1$ large enough so that $\Omega \subset B(0, R)$. Then u_ε is a viscosity subsolution of the equation

$$(dd^c w)^n = e^w (f + \varepsilon^n e^{-C})\beta^n,$$

where $C = \sup_\Omega \underline{u}$. Note that $\overline{u}$ is a supersolution of this modified equation. Since $u_\varepsilon \leq \overline{u}$ in $\partial\Omega$, it follows from our previous analysis that $u_\varepsilon \leq \overline{u}$ in Ω, hence $\underline{u} \leq \overline{u}$ in Ω. □

6.4 Perron's method

Once the comparison principle holds, one can easily construct continuous solutions by Perron's method as we now explain. Consider

$$(dd^c\varphi)^n = e^{\varepsilon\varphi} f\beta^n, \tag{6.4.1}$$

where $0 \leq f \in C^0(\bar{\Omega})$ and $\varepsilon > 0$. This equation can be written

$$F(x, \varphi, D\varphi, D^2\varphi) = 0,$$

where $F : \Omega \times \mathbb{R} \times \mathbb{R}^{2n} \times \mathcal{S}_{2n} \to \mathbb{R}$ is defined by

$$F(x, r, p, Q) := -(dd^c Q^{1,1})^n + e^{\varepsilon\varphi} f\beta^n, \quad \text{if } dd^c Q^{1,1} \geq 0,$$

and $F(x, r, p, Q) := +\infty$ if $dd^c Q^{1,1}$ is not semi-positive.

Theorem 6.32. *Let γ be a continuous function in $\partial\Omega$. Assume that the equation (6.4.1) has a subsolution $\underline{u}$ and a supersolution $\overline{u}$ such that $\underline{u}_* = \gamma = \overline{u}^*$ on $\partial\Omega$.*

Then the upper envelope φ of subsolutions u satisfying $\underline{u} \leq u \leq \overline{u}$ is the unique viscosity solution of (6.4.1) such that $u = \gamma$ in $\partial\Omega$. It coincides with the unique pluripotential solution.

Proof. The upper semi-continuous regularization φ^* of the upper envelope φ is a subsolution of (6.4.1) by Exercice 6.1, since F is lower semi-continuous. Hence φ^* is again a subsolution, i.e., $\varphi = \varphi^*$ is the maximal subsolution of (6.4.1).

We claim that φ_* is a viscosity supersolution of (6.4.1). We argue by contradiction, using a bump construction. Assume we can find $z_0 \in \Omega$ and a lower test function q for φ_* at z_0 such that

$$F_+(z_0, \varphi_*(z_0), Dq(z_0), dd^c q(x_0)) < 0.$$

This implies $Q := dd^c q(z_0) \geq 0$ and $F(z_0, \varphi_*(z_0), Dq(z_0), Q) < 0$.

We set $q_\delta(z) := q(z) - \delta(|z - z_0|^2 - r^2)$, where $\delta > 0$, $r > 0$ are small and $|z - z_0| < 2r$. Then

$$\begin{aligned} q_\delta(z_0) &= \varphi_*(z_0) + \delta r^2, \\ Dq_\delta(z_0) &= Dq(z_0), \\ D^2 q_\delta(z_0) &= Q - 2\delta I_n. \end{aligned}$$

Since $F(z_0, \varphi_*(z_0), Dq(z_0), Q) < 0$, the continuity of F in its domain ensures that for $\delta > 0$ small, we can find a small $r > 0$ such that

$$F(z, q_\delta(z), Dq_\delta(z), Q_\delta) < 0,$$

for $|z - z_0| < 2r$. Thus q_δ is a subsolution of (6.4.1) in the ball $|z - z_0| < 2r$.

Now for $|z - z_0| = 2r$,

$$q_\delta(z) = q(z) - \delta r^2 \le \varphi_*(z) - 3\delta r^2 \le \varphi - 3\delta r^2,$$

therefore

$$\psi(z) := \begin{cases} \max\{\varphi, q_\delta\}, & \text{in } B_r(z_0), \\ \varphi, & \text{in } \Omega \setminus B_r(z_0), \end{cases}$$

is a subsolution of (6.4.1) in Ω, where $B_r(z_0) = \{|z - z_0| < 2r\}$ (see Exercise 6.2). Since φ is the maximal subsolution of the equation in Ω, we infer that $\psi \le \varphi$ in Ω, hence $q_\delta \le \varphi$ in the ball $B_r(z_0)$.

On the other hand, $q_\delta(x_0) - \varphi_*(x_0) = \delta r^2$, hence there exists (y_j) converging to z_0 such that $\lim_{j\to+\infty} q_\delta(y_j) - \varphi(y_j) = q_\delta(z_0) - \varphi_*(z_0)$. For $j > 1$ large enough, we have $y_j \in B_r(z_0)$ and

$$q_\delta(y_j) - \varphi(y_j) > \delta r^2/2 > 0,$$

which contradicts the inequality $q_\delta \le \varphi$ on the ball $B_r(x_0)$. This proves that φ_* is a supersolution of the equation (6.4.1).

Since $\underline{u} \le \varphi \le \overline{u}$, it follows that $\underline{u}_* \le \varphi_* \le \overline{u}$ in Ω. Now $\varphi \le \overline{u}^* = \gamma$ on $\partial\Omega$, while $\gamma = \underline{u}_* \le \varphi_*$ on $\partial\Omega$, thus $\varphi \le \varphi_*$ on $\partial\Omega$.

The comparison principle therefore yields $\varphi \le \varphi_*$ in Ω, hence $\varphi = \varphi_*$ is a viscosity solution of the equation (6.4.1). The fact that it is the pluripotential solution follows from Proposition 6.33. □

We end this section by comparing viscosity solutions and pluripotential solutions.

Proposition 6.33. *Let $f \ge 0$ be a continuous density in Ω. The viscosity solutions of the equation $(dd^c\varphi)^n = e^\varphi f\beta^n$ are precisely the continuous plurisubharmonic solutions φ of the equation $(dd^c\varphi)^n = e^\varphi f\beta^n$ in the pluripotential sense.*

Proof. It follows from Proposition 6.25 and Lemma 6.30 that every pluripotential solution is a viscosity solution.

Assume conversely that φ is a viscosity solution and fix a ball $B \subset \Omega$. It follows from Theorem 6.32 that φ is the unique pluripotential solution of the Dirichlet problem $(dd^c\varphi)^n = e^\varphi f\beta^n$ in B with boundary values φ. The conclusion follows as B is arbitrary. □

6.5 Exercises

Exercise 6.1. *Let $F : \Omega \times \mathbb{R} \times \mathbb{R}^{2n} \times \mathcal{S}_{2n} \to \mathbb{R} \cup \{+\infty\}$ be a lower semi-continuous function satisfying the degenerate ellipticity condition.*

1. *Let u_1, u_2 be subsolutions of $F = 0$. Show that so is* $\max\{u_1, u_2\}$.

2. *Let $\mathcal{U}$ a family of subsolutions to the equation $F = 0$. Assume that $\mathcal{U}$ is non-empty and locally bounded from above in Ω. Prove that*

$$U^* := (\sup\{u |\, u \in \mathcal{U}\})^*$$

is a subsolution of the equation $F(x, u, Du, D^2u) = 0$.

3. *Prove the dual statements for supersolutions.*

Exercise 6.2. *Let $\Omega_1, \Omega_2 \subset \mathbb{R}^N$ be two bounded domains such that $\bar{\Omega}_2 \subset \Omega_1$. Let u be a viscosity subsolution of $F(x, u, Du, D^2u) = 0$ in Ω_1 and v a subsolution of the same equation in Ω_2 such that $u \geq v$ in $\Omega_2 \cap \partial\Omega_1$. Prove that*

$$w := \begin{cases} \sup\{u, v\}, & \text{in } \Omega_2, \\ u, & \text{in } \Omega_1 \setminus \Omega_2, \end{cases}$$

is a subsolution of the equation $F(x, u, Du, D^2u) = 0$ in Ω_1.

Exercise 6.3. *Let $A \in \mathcal{S}_N$ be a real symmetric matrix such that* $\operatorname{tr}(AQ) \geq 0$*, for all $Q \in \mathcal{S}_N$ with $Q \geq 0$. Show that $A \geq 0$.*

Exercise 6.4. *Consider*

$$F(x, u, Du, D^2u) = |u'| - 1 = 0$$

in $(-1, +1)$ with the boundary conditions $u(\pm 1) = 0$.

1. *Show that the function*

$$u(x) := \begin{cases} x, & \text{if} \quad x \in [-1, -1/2] \cup [0, 1/2], \\ -x, & \text{if} \quad x \in [-1/2, 0] \cup [1/2, 1], \end{cases}$$

is a subsolution but not a supersolution of this Dirichlet problem.

2. *Show that $x \mapsto U(x) = 1 - |x|$ is the viscosity solution.*

3. *For $\varepsilon > 0$, let u_ε denote the unique solution of*

$$-\varepsilon \Delta u_\varepsilon + |u_\varepsilon'| - 1 = 0 \text{ in } (-1, +1),$$

with $u_\varepsilon(\pm 1) = 0$. Show that u_ε uniformly converges to U on $[-1, +1]$.

Exercise 6.5.

1. *Let $u : \Omega \to \mathbb{R}$ be a bounded upper semi-continuous function. Show that its sup-convolution u^ε is ε^{-2}-convex in Ω_ε and decreases to u in Ω as $\varepsilon \searrow 0$. Check that if u is a subsolution of the equation $F = 0$, then u^ε is a subsolution of the equation $F_\varepsilon(x, w, Dw, D^2w) = 0$, where*

$$F_\varepsilon(x, s, p, Q) := \inf\{F(y, s, p, Q)|\ |y - x| \le A\varepsilon\}.$$

2. *Let $v : \Omega \to \mathbb{R}$ be a bounded lower semi-continuous function. Show that its inf-convolution v_ε is ε^{-2}-concave in Ω_ε and increases to v in Ω as $\varepsilon \searrow 0$. Check that if v is a supersolution of the equation $F = 0$, then v_ε is a supersolution of $F^\varepsilon(x, w, Dw, D^2w) = 0$, where*

$$F^\varepsilon(x, s, p, Q) := \sup\{F(y, s, p, Q)|\ |y - x| \le A\varepsilon\}.$$

Exercise 6.6. *Fix $\Omega \subset \mathbb{C}^n$ a domain. Let $\varphi : \Omega \to \mathbb{R}$ be a bounded upper semi-continuous function that satisfies $(dd^c\varphi)^n \ge e^{\varepsilon\varphi} f\beta^n$ in the viscosity sense. Let φ^δ be the approximation by supconvolution. Prove that φ_δ satisfies*

$$(dd^c\varphi^\delta)^n \ge e^{\varepsilon\varphi^\delta} f_\delta \beta_n, \text{ with } f_\delta(x) = \inf\{f(y)|\ |y - x| \le A\delta\}.$$

in the sense of viscosity on Ω_δ.

Exercise 6.7. *Let $u : \Omega \to \mathbb{R}$ be an upper semi-continuous function which is locally bounded from above in Ω. Set*

$$\mathcal{D}^{2,+}_\Omega u := \{x \in \Omega|\ J^{2,+}u(x) \ne \emptyset\}.$$

Fix $x_0 \in \Omega$ and a ball $B = B(x_0, r) \Subset \Omega$ with $r > 0$.

1. *Show that $\exists A > 0$ such that $u(x) - A|x - x_0|^2 < u(x_0)$ for $|x - x_0| = r$.*

2. *Set $q(x) := A|x - x_0|^2$. Show that the function $u - q$ attains its maximum M in $\bar{B}$ at some point $\hat{x} \in \bar{B}$.*

3. *Show that $\hat{x} \in B$ and deduce that $(D\hat{q}(\hat{x}), D^2\hat{q}(\hat{x})) \in J^{2,+}u(\hat{x})$.*

4. *Conclude that $D^{2,+}u(x_0)$ is dense in Ω.*

Exercise 6.8. *Let u be an upper semi-continuous function in Ω. Fix $x_0 \in \Omega$ and $(p, Q) \in J^{2,+}u(x_0)$. Show that there exists a C^2-function q near x_0 such that $Dq(x_0) = p$, $D^2q(x_0) = Q$, and $u \le_{x_0} q$, i.e., $J^2q(x_0) = \{(p, Q)\}$.*

Exercise 6.9. *Let $Q \in H_n$ be a Hermitian matrix such that, for every semi-positive Hermitian matrix H, $\det(Q + H) \ge 0$. Show that $Q \ge 0$ is semi-positive.*

Exercise 6.10. *Let u be an upper semi-continuous function which is locally bounded from above in $\Omega \subset \mathbb{R}^N$. Let $H \in H_N^+$ be a positive Hermitian matrix and Δ_H the associated Laplace operator. Show that u is Δ_H-subharmonic iff u is a viscosity subsolution of $\Delta_H\varphi \ge 0$.*

Exercise 6.11. *Prove that if $F : \mathbb{R}^N \mapsto \mathbb{R}^N$ is C-Lipschitz on a compact set $K \subset \mathbb{R}^N$, then*

$$\lambda_N(F(K)) \leq C^N \lambda_N(K),$$

where λ_N is the Lebesgue measure on $\mathbb{R}^N$.

Exercise 6.12. *Let u be a bounded plurisubharmonic function in a domain $\Omega \subset \mathbb{C}^n$ and $\mu = f\beta_n$ a continuous volume form with continuous density $f \geq 0$. Assume that*

$$(dd^c\varphi)^n_{\mathrm{BT}} \geq e^{\varphi} f\beta_n.$$

Show that the usual regularization by convolution $\tilde{\varphi}_\delta := \varphi \star \chi_\delta$ satisfies

$$(dd^c\tilde{\varphi}_\delta)^n_{\mathrm{BT}} \geq e^{\tilde{\varphi}_\delta} \tilde{f}_\delta \beta_n$$

pointwise in Ω_δ, where

$$\tilde{f}_\delta(x) := \inf\{f(y)e^{\frac{1}{2\delta^2}|x-y|^2} \mid |y-x| \leq A\delta\}.$$

Exercise 6.13. *Let $h : \Omega \subset \mathbb{R}^N \to \mathbb{R}$ be a $C^{1,1}$-smooth function in some open set Ω. Assume that $h(x_0) > \max_{\partial B(x_0,r)} h$.*

1. *Show that there exists $k > 0$ such that $x \mapsto h(x) + (k/2)\|x\|^2$ is convex in $\bar{B}(x_0, r)$, hence $D^2h(x) \geq -kI_N$ a.e. in $\bar{B}(x_0, r)$.*

2. *Prove that for $\delta > 0$ small enough the function $x \mapsto h(x) + \langle p, x\rangle$ attains its maximum on $\bar{B}(x_0, r)$ at some interior point of $B(x_0, r)$.*

3. *Let $E := E(x_0, r, \delta)$ denote the set of points $y \in \bar{B}(x_0, r)$ such that there exists $p \in \mathbb{R}^N$ with $|p| < \delta$ and the function $x \mapsto h(x) + \langle p, x\rangle$ attains its maximum on $\bar{B}(x_0, r)$ at the point y. Prove that $B(0, \delta) \subset Dh(E)$ and for any $x \in E$ we have $-(k/2)I_N \leq D^2h(x) \leq 0$.*

4. *Deduce the following Jensen inequality:*

$$\lambda_N(E) \geq k^{-N}\kappa_N\delta^N > 0, \tag{6.5.1}$$

where κ_N is the volume of the unit ball in $\mathbb{R}^N$.

5. *Let h be a k-convex function in a neighborhood of $\bar{B}(x_0, r)$. Show that one can approximate h uniformly by smooth k-convex functions in a neighborhood of $\bar{B}(x_0, r)$ and deduce that the inequality (6.5.1) still holds using Exercise 6.11.*

Exercise 6.14. *Let φ a subsolution to $(dd^c\varphi)^n \geq f\beta^n$ in some open set Ω, where f is a continuous density. Assume $f(z_0) > 0$, $z_0 \in \Omega$.*

1. *Show that φ is strictly plurisubharmonic in the sense of viscosity near z_0, i.e., $dd^cq(z) > 0$ for all z close to z_0 and q an upper test function for φ at z.*

2. *Show that the function $\varphi(z) := |z|$ is strictly plurisubharmonic in the sense of viscosity in $\mathbb{C}^n$, and*

$$dd^c|z| \geq |z|^{-1/4}\beta \geq \beta,$$

when $|z| < 1$ in the sense of currents.

3. *Show that if* $(dd^c\varphi)^n > 0$ *in the sense of viscosity, then* φ *is strictly plurisubharmonic in the sense of viscosity.*

4. *Give an example of a continuous plurisubharmonic function* φ *in* $\mathbb{C}^n$ *such that* $(dd^c\varphi)^n = \beta$ *and* φ *is not strictly plurisubharmonic in the usual sense, i.e., there is no* $\varepsilon > 0$ *such that* $\varphi - \varepsilon|z|^2$ *is plurisubharmonic.*

Part II

Pluripotential theory on compact manifolds

7 Compact Kähler manifolds

We introduce in this chapter several notions and theorems without giving proofs. This is to fix the frame, notations and examples that we will use in the sequel. Although it is not necessary (and probably impossible) that you feel immediately familiar with these concepts, it is important that you read more about them in textbooks, such as [Beauville, GH, Wells, Dem, V, BHPV, Laz] to name a few.

7.1 Examples of compact complex manifolds

7.1.1 Tangent spaces Let X be a compact topological Hausdorff space, which is always assumed to be *connected* in the sequel.

Definition 7.1. *A system of local complex* (*analytic*) *coordinates of dimension* $n \in \mathbb{N}^*$ *on* X *is a collection* $z_\alpha : U_\alpha \to \mathbb{C}^n$ *of homeomorphisms* (*onto their images*), *where* $U = \{U_\alpha, \alpha \in A\}$ *is an open covering of* X *and the mappings*

$$f_{\alpha\beta} := z_\beta \circ z_\alpha^{-1} : z_\beta(U_\alpha \cap U_\beta) \longrightarrow z_\alpha(U_\alpha \cap U_\beta)$$

are biholomorphisms. Two systems $(z_\alpha), (w_i)$ *are equivalent if the maps* $z_\alpha(p) \mapsto w_i(p)$ *are biholomorphic wherever they are defined.*

We say that X *is a complex manifold of dimension* n, *if it can be endowed with a complex structure, i.e., an equivalence class of systems of local complex* (*analytic*) *coordinates of dimension* n.

Note that a complex manifold X of complex dimension n is in particular a differentiable manifold of real dimension $2n$. There are three different notions of tangent space to X at a point $p \in X$:

(1) the usual real tangent space $T_{\mathbb{R},p}(X)$: it can be realized as the space of real derivations on the ring of real-valued $\mathcal{C}^\infty$-functions near p;

(2) the complexified tangent space $T_{\mathbb{C},p}(X) := T_{\mathbb{R},p}(X) \otimes \mathbb{C}$;

(3) the holomorphic tangent space $T'_p(X) \subset T_{\mathbb{C},p}(X)$, consisting of the derivations that vanish on antiholomorphic functions.

Definition 7.2. *A map* $f : X \to Y$ *between two complex manifolds is called holomorphic if it is a smooth map whose induced linear map* $f_* : T_{\mathbb{C},p}(X) \to T_{\mathbb{C},f(p)}(Y)$ *sends* $T'_p(X)$ *to* $T'_{f(p)}(Y)$ *for all* $p \in X$.

The reader will check in Exercise 7.1 that a map $f : X \to Y$ is holomorphic if and only if the maps $w_i \circ f \circ z_\alpha^{-1}$ are holomorphic for any system of local complex coordinates z_α (resp. w_i) on X (resp. on Y).

Example 7.3. *The complex projective space $\mathbb{CP}^n$ is one of the most fundamental examples of compact complex manifolds. It can be constructed from $\mathbb{C}^{n+1} \setminus \{0\}$ by identifying $(x_0, \ldots, x_n) = x \sim w = (w_0, \ldots, w_n)$ whenever there exists $\lambda \in \mathbb{C}^*$ such that $x = \lambda w$. This is a compact Hausdorff space for the quotient topology. We let $[x]$ denote the equivalence class of x. Note that $\mathbb{CP}^n$ can be identified with the set of complex lines through the origin in $\mathbb{C}^{n+1}$.*

A system of complex coordinates can be obtained as follows. We let

$$U_\alpha := \{[x] \in \mathbb{CP}^n \mid x_\alpha \neq 0\}.$$

Then $\mathcal{U} := \{U_\alpha, 0 \leq \alpha \leq n\}$ is an open covering of $\mathbb{CP}^n$ and the map

$$z_\alpha : [x] \in U_\alpha \longmapsto (x_0/x_\alpha, \ldots, x_{\alpha-1}/x_\alpha, x_{\alpha+1}/x_\alpha, \ldots, x_n/x_\alpha) \in \mathbb{C}^n$$

is a homeomorphism onto $z_\alpha(U_\alpha) = \mathbb{C}^n$ such that $z_\beta \circ z_\alpha^{-1}$ is a biholomorphism from $\mathbb{C}^n \setminus \{z_\beta = 0\}$ onto $\mathbb{C}^n \setminus \{z_\alpha = 0\}$.

The coordinates $[x] = [x_0 : \ldots : x_n] \in \mathbb{CP}^n$ are called the homogeneous coordinates. If P is a homogeneous polynomial in x of degree $d \in \mathbb{N}^*$, then

$$H := \{[x] \in \mathbb{CP}^n \mid P(x) = 0\}$$

is a well-defined analytic subset of $\mathbb{CP}^n$ of complex codimension one which is a smooth hypersurface (of degree $d = \deg P$) if ∇P does not vanish on H.

Definition 7.4. *A subset $V \subset \mathbb{CP}^n$ is called a projective algebraic variety if there exists homogeneous polynomials $P_1, \ldots, P_s$ such that*

$$V = \{[x] \in \mathbb{CP}^n \mid P_1(x) = \cdots = P_s(x) = 0\}.$$

The variety V is a smooth submanifold of $\mathbb{CP}^n$ if the rank of the matrix $(\partial P_i / \partial z_\alpha)$ is the same at each point of V.

An important *theorem of Chow* (see [Cirka] for a proof) asserts that every closed analytic subvariety of $\mathbb{CP}^n$ is a projective algebraic variety.

Example 7.5. *Let $\mathrm{Gr}(r, n, \mathbb{C})$ denote the Grassmannian of complex r-planes in $\mathbb{C}^n$, i.e. the set of all r-dimensional complex linear subspaces of $\mathbb{C}^n$. If we fix a basis $e_1, \ldots, e_r \in \mathbb{C}^n$ of such a r-plane, then every other basis is obtained from this one by multiplication by a matrix $A \in \mathrm{GL}(r, \mathbb{C})$. Thus*

$$\mathrm{Gr}(r, n, \mathbb{C}) = M_{r,n}(\mathbb{C})/\mathrm{GL}(r, \mathbb{C}),$$

where $M_{r,n}(\mathbb{C})$ denotes the set of rank r-matrices with r rows and n columns.

We let the reader check in Exercise 7.2 that $\mathrm{Gr}(r, n, \mathbb{C})$ is a complex projective algebraic variety of complex dimension $r(n - r)$.

7.1.2 Quotients There are many recipes for constructing compact complex manifolds starting from known ones: one can consider products, submanifolds, quotients by groups of biholomorphisms acting properly discontinuously without fixed points, holomorphic coverings, etc. In this subsection we consider several examples of complex quotients.

7.1.2.1 Complex tori Let $\Lambda \subset \mathbb{C}^n$ be a lattice of maximal rank $2n$. The quotient $X = \mathbb{C}^n/\Lambda$ is homeomorphic to $\mathbb{R}^{2n}/\mathbb{Z}^{2n}$ and inherits the structure of a complex compact manifold of (complex) dimension n.

When $n = 1$, $X = \mathbb{C}/\Lambda$ is called an *elliptic curve*. Elliptic curves are biholomorphic to one of the curves $E_\tau = \mathbb{C}/\mathbb{Z} \otimes \mathbb{Z}[\tau]$, where $\tau \in \mathbb{C} \setminus \mathbb{R}$ is a complex number with positive imaginary part.

Elliptic curves can be realized as smooth cubics in $\mathbb{CP}^2$ (by using the Weierstrass $\mathcal{P}$-function and its derivative). In particular, they are projective algebraic. However, most complex tori in higher dimension $n \geq 2$ are not projective algebraic. Those which are, are called *abelian varieties*.

7.1.2.2 Compact Riemann surfaces Let S be a compact complex manifold of dimension $n = 1$; it is called a compact Riemann surface. It follows from the Riemann mapping theorem that S is biholomorphic

- either to the Riemann sphere $\mathbb{CP}^1$,
- or to an elliptic curve $E_\tau = \mathbb{C}/\mathbb{Z} \otimes \mathbb{Z}[\tau]$,
- or else to an hyperbolic surface Δ/Γ.

In the last case, Δ denotes the open unit disc in $\mathbb{C}$ and Γ is a discrete subgroup of $\mathrm{PSL}(2, \mathbb{R})$ which acts freely and properly on Δ.

7.1.2.3 Hopf surfaces Another interesting example is the quotient $X = \mathbb{C}^2 \setminus \{0\}/\Gamma$, where Γ is the group generated by the polynomial contraction

$$\gamma : (x, y) \in \mathbb{C}^2 \longmapsto (\lambda x, \mu y) \in \mathbb{C}^2, \text{ where } 0 < |\lambda| \leq |\mu| < 1.$$

We let the reader check that the image (under the projection map) of the x-axis is an elliptic curve. Can you see other elliptic curves inside X?

These compact complex surfaces are all diffeomorphic to $S^3 \times S^1$, in particular their first Betti number is one. It therefore follows from Hodge's theorem that these are not Kähler surfaces (see Corollary 7.30).

7.1.3 Blow-ups

7.1.3.1 Blow-up of a point Let $\mathbb{B}$ be a small ball centered at $0 \in \mathbb{C}^n$. The blow-up of $\mathbb{B}$ at 0 is the complex manifold $\tilde{\mathbb{B}}$ obtained by replacing 0 in $\mathbb{B}$ by the set of all complex lines through $0 \in \mathbb{C}^n$, more precisely

$$\tilde{\mathbb{B}} := \{(z, \ell) \in \mathbb{B} \times \mathbb{P}^{n-1} \mid z \in \ell\}.$$

This is a complex submanifold of $\mathbb{B} \times \mathbb{P}^{n-1}$ of complex dimension n.

Here and in the sequel we let $\mathbb{P}^k = \mathbb{CP}^k = \mathbb{P}^k(\mathbb{C})$ denote the *complex* projective space of complex dimension k.

Observe that the projection $\pi : (z, \ell) \in \tilde{\mathbb{B}} \mapsto z \in \mathbb{B}$ realizes a biholomorphism from $\tilde{\mathbb{B}} \setminus E$ onto $\mathbb{B} \setminus \{0\}$, where $E := \pi^{-1}(0) \simeq \mathbb{P}^{n-1}$ is called the exceptional divisor.

As this construction is local, we can define the blow-up $\pi : \tilde{X} \to X$ of a complex manifold X at any point $p \in X$. Set $E := \pi^{-1}(p) \simeq \mathbb{P}^{n-1}$. The manifold $\tilde{X}$ is n-dimensional and $\tilde{X} \setminus E$ is biholomorphic to $X \setminus \{p\}$.

7.1.3.2 Blow-up with smooth center Let X be a complex manifold and $Y \subset X$ a complex submanifold of codimension k. Locally along Y there exists holomorphic functions $f_1, \ldots, f_k$ with independent differentials such that $Y = \{z \in X \mid f_1(z) = \cdots = f_k(z) = 0\}$. These equations are not unique, but we have the following result:

Lemma 7.6. *Let $g_1, \ldots, g_k$ be another system of local equations for Y. Then there exists an invertible (k, k)-matrix A of holomorphic functions defined in a neighborhood of Y such that*

$$(g_1, \ldots, g_k) = A \cdot (f_1, \ldots, f_k).$$

Moreover, $A|_Y$ is uniquely determined.

We leave the proof of this lemma as Exercise 7.4. Fix an open subset U of X on which there exists a local system $\{f_1, \ldots, f_k\}$ of local equations for Y. We set

$$\tilde{U}_Y := \{([x], z) \in \mathbb{P}^{k-1} \times U \mid x_i f_j(z) = x_j f_i(z),\ \forall i, j \leq k\}$$

and

$$\pi : ([x], z) \in \tilde{U}_Y \longmapsto z \in U.$$

Set $E := \pi^{-1}(Y) \sim \mathbb{P}^{k-1}$. Observe that $\tilde{U}_Y \setminus E$ is biholomorphic to $U \setminus Y$.

If V is another open set and $\{g_1, \ldots, g_k\}$ another system of local equations for Y in V, we let the reader check, by using Lemma 7.6, that the pieces of $\tilde{U}_Y$ and $\tilde{V}_Y$ coming from $U \cap V$ are biholomorphic. This allows to glue together the manifolds $\tilde{U}_Y$ (obtained from an open cover $\{U\}$ of X) so as to obtain a manifold $\tilde{X}_Y$:

Definition 7.7. *The manifold $\tilde{X}_Y$ is called the blow-up of X along Y.*

7.1.4 Meromorphic maps

We have already defined the notion of holomorphic map $f : X \to Y$ between two complex manifolds. There are however many interesting maps that are merely *meromorphic*:

Definition 7.8. *Let X, Y be two complex manifolds. A map $f : X \dashrightarrow Y$ is meromorphic if it is holomorphic outside an analytic set I_f (the indeterminacy locus) of codimension ≥ 2, and if the closure of its holomorphic graph,*

$$\Gamma_f := \{(x, y) \in X \times Y \mid x \notin I_f \text{ and } y = f(x)\},$$

in $X \times Y$ is an analytic subset of $X \times Y$.

The map f is called bimeromorphic if it has topological degree one. This equivalently means that there exists a meromorphic map $g : Y \dashrightarrow X$ such that $f \circ g$ and $g \circ f$ are the identity mappings outside a complex hypersurface.

Meromorphic maps are called *rational* when the manifolds X and Y are projective algebraic. This is because they are defined by rational functions, as the following example illustrates:

Example 7.9. *Any meromorphic map $f : \mathbb{CP}^n \dashrightarrow \mathbb{CP}^n$ is given, in homogeneous coordinates, by a collection $P_0, \ldots, P_n$ of homogeneous polynomials of the same degree with no common factor:*

$$f[z] = [P_0(z) : \cdots : P_n(z)].$$

It is holomorphic if the P_j's have no common zero (except for the origin), it is meromorphic otherwise, with

$$I_f = \{[z] \in \mathbb{CP}^n \mid P_0(z) = \cdots = P_n(z) = 0\}.$$

We leave the proof of this fact as Exercise 7.11 for the reader.

In the blow-up construction, the projection map $\pi : \tilde{X}_Y \to X$ is holomorphic, while its "inverse", the contraction map $\pi^{-1} : X \dashrightarrow \tilde{X}_Y$ is meromorphic with indeterminacy locus equal to Y. Blow-ups allow one to understand the structure *any* birational map, as shown by the following deep theorem (see [Kollar]):

Theorem 7.10. *Let X, Y be projective algebraic complex manifolds and $f : X \dashrightarrow Y$ a bimeromorphic map. Then there exists complex manifolds X_j, $0 \le j \le 2s$, and blow-ups with smooth centers $f_i : X_{2i-1} \to X_{2i}$, $\pi_i : X_{2i-1} \to X_{2i-2}$, such that f is given by*

$$\begin{array}{ccccccccccc} & & X_1 & & & & & & X_{2s-1} & & \\ & \swarrow^{\pi_1} & & \searrow^{\varphi_1} & & & & \swarrow^{\pi_s} & & \searrow^{\varphi_s} & \\ X = X_0 & & & & X_2 & \cdots & X_{2s-2} & & & & X_{2s} = Y. \end{array}$$

The proof is much easier in dimension $n = 2$, and the result is also stronger: in this case one can first perform a sequence of blow-ups (of points) and then a sequence of contractions. Thus there exist $\pi_1 : W \to X$ and $\pi_2 : W \to Y$, compositions of blow-ups (of points), such that

$$\begin{array}{ccccc} & & W & & \\ & \swarrow^{\pi_1} & & \searrow^{\pi_2} & \\ X & & \overset{f}{\dashrightarrow} & & Y. \end{array}$$

7.2 Holomorphic vector bundles

7.2.1 Holomorphic line bundles and divisors

7.2.1.1 Definition Let X be complex manifold. A holomorphic vector bundle $E \xrightarrow{\pi} X$ of rank $r \in \mathbb{N}^*$ is a complex manifold such that

(1) for each $x \in X$, $E_x := \pi^{-1}(x)$ is a complex vector space of dimension r;

(2) the projection map $\pi : E \to X$ is holomorphic;

(3) for each $x \in X$, there exists an open neighborhood $U \subset X$ of x and

$$\varphi_U : \pi^{-1}(U) \longrightarrow U \times \mathbb{C}^r$$

a biholomorphism taking E_x isomorphically onto $\{x\} \times \mathbb{C}^r$.

The map φ_U is called a (local) trivialization of E. Note that for any pair of trivializations φ_U, φ_V, there is a holomorphic map $g_{UV} \in \mathcal{O}(U \cap V, \mathrm{GL}(r, \mathbb{C}))$ such that the map $\varphi_V \circ \varphi_U^{-1}$ induces an automorphism of $U \cap V \times \mathbb{C}^r$ by

$$\varphi_U \circ \varphi_V^{-1} : (x, \zeta) \in U \cap V \times \mathbb{C}^r \longmapsto (x, g_{UV}(x)(\zeta)) \in U \cap V \times \mathbb{C}^r.$$

The functions $\{g_{UV}\}$ are called transition functions. They satisfy

$$g_{UV} \cdot g_{VU} = \mathrm{Id} \quad \text{and} \quad g_{UV} \cdot g_{VW} \cdot g_{WU} = \mathrm{Id} \tag{7.2.1}$$

Conversely, given any open cover $\{U\}$ of X and holomorphic maps $g_{UV} \in \mathcal{O}(U \cap V, \mathrm{GL}(r, \mathbb{C}))$ satisfying (7.2.1), there exists a unique — up to isomorphism — holomorphic vector bundle $E \xrightarrow{\pi} X$ of rank r on X with the g_{UV}'s as its transition functions. Here a holomorphic map $E \to F$ between two holomorphic vector bundles is an isomorphism if it maps E_x isomorphically onto F_x.

7.2.1.2 Operations Given two holomorphic vector bundles E, F of rank r, s on X, we can define the following holomorphic vector bundles:

- the dual E^*, with fibers $E_x* = (E_x)^*$ and transition functions ${}^t g_{UV}^{-1}$;
- the sum $E \oplus F$, with fibers $E_x \oplus F_y$ and transitions $\begin{pmatrix} g_{UV} & 0 \\ 0 & h_{UV} \end{pmatrix}$;
- $E \otimes F$, with transition functions $g_{UV} \otimes h_{UV} \in \mathrm{GL}(k, \mathbb{C}^r \otimes \mathbb{C}^s)$;
- $\Lambda^j E$, with transition functions $\Lambda^j g_{UV} \in \mathrm{GL}(\Lambda^j \mathbb{C}^r)$.

In particular, $\Lambda^r E =: \det E$ is a holomorphic line bundle (rank 1), called the *determinant bundle* of E.

Example 7.11. *The holomorphic tangent space $T'(X) \xrightarrow{\pi} X$ is a holomorphic vector bundle of rank n whenever X is an n-dimensional complex manifold. The line bundle*

$$K_X := \Lambda^n (T'(X))^*$$

is called the canonical line bundle. It plays a central role in many problems of complex algebraic and differential geometry.

7.2.1.3 Line bundles A holomorphic vector bundle L of rank 1 is called a *holomorphic line bundle*. Let $\{U\}$ be an open covering of X which trivializes L, and let $\{g_{UV}\} \in \mathcal{O}^*(U \cap V) = \mathcal{O}(U \cap V, \mathbb{C}^*)$ denote its associated transition functions (note that these have values in $\mathbb{C}^*$, since the rank is 1). The relations

$$g_{UV} \cdot g_{VU} \equiv 1 \quad \text{and} \quad g_{UV} \cdot g_{VW} \cdot g_{WU} \equiv 1$$

show that $\{g_{UV}\}$ defines a Čech 1-cochain on X with values in $\mathcal{O}^*$ which is a cocycle, hence it defines a class in $H^1(X, \mathcal{O}^*)$.

If $L' \to X$ is a holomorphic line bundle isomorphic to L, then the corresponding trivializations φ'_U satisfy

$$\varphi'_U = f_U \varphi_U \text{ for some } f_U \in \mathcal{O}^*(U),$$

hence the associated transition functions g'_{UV} satisfy

$$g'_{UV} = \frac{f_U}{f_V} g_{UV}.$$

Therefore, $\{g_{UV}\}$ and $\{g'_{UV}\}$ represent the same class in $H^1(X, \mathcal{O}^*)$.

Conversely, any class in $H^1(X, \mathcal{O}^*)$ can be represented by a cocycle $\{g_{UV}\}$ which defines a unique holomorphic line bundle L, up to isomorphism. Thus the Picard group of X,

$$\text{Pic}(X) := \{\text{holomorphic line bundles on } X\}/\text{modulo isomorphism},$$

is isomorphic to $H^1(X, \mathcal{O}^*)$. This is a group for the law induced by the tensor product (check it!).

Example 7.12. *The Picard group of the complex projective space $\mathbb{P}^n$ is $\mathbb{Z}$, it is generated by the hyperplane bundle $\mathcal{O}(1)$ which is associated to a hyperplane H in $\mathbb{P}^n$ (see the correspondence divisors/line bundles below). It is the dual of the universal bundle $\mathcal{O}(-1)$ on $\mathbb{P}^n$ which is defined by*

$$\mathcal{O}(-1) = \{([z], \zeta) \in \mathbb{P}^n \times \mathbb{C}^{n+1} \mid \zeta \in [z]\}.$$

This bundle is trivialized by the covering

$$U_\alpha = \{[z] \in \mathbb{P}^n \mid z_\alpha \neq 0, \quad 0 \leq \alpha \leq n\},$$

with transition functions $g_{\alpha\beta} = z_\alpha / z_\beta$.

7.2.1.4 Divisors A *divisor* $D = \sum a_i V_i$ on X is a locally finite formal linear combination of irreducible analytic hypersurfaces V_i of X. The coefficients are assumed here to be integers $\in \mathbb{Z}$ (however rational and even real divisors may be useful to

consider, depending on the context). The divisor D is called *effective* if all coefficients $a_i \geq 0$ are non-negative. We can locally, say in a small open set U, find a holomorphic function f_U such that

$$D \cap U = \{f_U = 0\}.$$

If V is another open set intersecting V, then the holomorphic function

$$g_{UV} := \frac{f_U}{f_V} \text{ is non-vanishing,}$$

hence it is associated to a holomorphic line bundle L_D. We let the reader check that if $D = (f)$ is the divisor associated to a meromorphic function on X, i.e., $D = (\text{zeroes of } f) - (\text{poles of } f)$, then $L_D = \mathbb{C} \times X$ is the trivial line bundle. Two divisors D and D' are *linearly equivalent* if they differ by the divisor of a meromorphic function. We write $D \sim D'$ in this case.

We have thus defined an injective map that is a group morphism,

$$\mathrm{Div}(X)/\sim \longrightarrow \mathrm{Pic}(X).$$

This mapping is an isomorphism if X is projective. The correspondence between holomorphic line bundles and divisors is both important and very helpful in practice, and the reader is encouraged to read more about it.

7.2.1.5 Metric and first Chern class The exact sequence of sheaves

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \longrightarrow 0$$

induces a boundary map in cohomology

$$H^1(X, \mathcal{O}^*) \xrightarrow{\delta} H^2(X, \mathbb{Z}).$$

The *first Chern class* $c_1(L) \in H^2(X, \mathbb{Z})$ of a holomorphic line bundle $L \in \mathrm{Pic}(X)$ is $\delta(L)$, identifying L with an element in $H^1(X, \mathcal{O}^*)$.

From an analytic point of view, we will be mostly interested in the image of $c_1(L)$ in the de Rham cohomology vector space $H^2(X, \mathbb{R})$, where the map

$$H^2(X, \mathbb{Z}) \longrightarrow H^2(X, \mathbb{R})$$

is induced by the inclusion map $\mathbb{Z} \to \mathbb{R}$.

Definition 7.13. *Let $L \to X$ be a holomorphic line bundle with transition functions g_{UV}. A smooth metric $h = \{h_U\}$ of L on X is a collection of smooth non-vanishing functions $h_U = e^{-\varphi_U} \in \mathcal{C}^\infty(U, \mathbb{R}^+)$ such that*

$$\varphi_U = \varphi_V + \log |g_{UV}| \quad \text{in } U \cap V.$$

The curvature Θ_h of h is the differential form of bidegree $(1,1)$,

$$\Theta_h := dd^c\varphi_U = dd^c\varphi_V.$$

Here $d = \partial + \overline{\partial}$ and $d^c = \frac{1}{2i\pi}(\partial - \overline{\partial})$ are both real operators and $dd^c = \frac{i}{\pi}\partial\overline{\partial}$.

Note that the curvature makes sense globally, since the functions $\log|g_{UV}|$ are pluriharmonic in $U \cap V$. As it is closed, it represents a class in $H^2(X, \mathbb{R})$:

Lemma 7.14. *One has* $c_1(L) = \{\Theta_h\} \in H^2(X, \mathbb{R})$.

We leave (again) the interesting proof to the reader as Exercise 7.7. Metrics of L will allow to measure the size of its sections in the sequel.

Definition 7.15. *A holomorphic line bundle $L \to X$ is positive if it admits a smooth metric whose curvature is a Kähler form, i.e., can be locally written as*

$$\Theta_h = \sum_{\alpha,\beta} \omega_{\alpha\beta}\, i\, dz_\alpha \wedge d\overline{z}_\beta,$$

where $(\omega_{\alpha\beta})$ is Hermitian positive definite.

The existence of such a holomorphic line bundle has tremendous consequences, as we shall soon see (see Theorem 7.20).

7.2.2 Holomorphic sections

7.2.2.1 Asymptotic growth Let $E \xrightarrow{\pi} X$ be a holomorphic vector bundle on a complex manifold X.

Definition 7.16. *A holomorphic section s of E on X is a holomorphic map $s : X \to E$ such that $\pi \circ s = \mathrm{Id}_X$, i.e., such that $s(x) \in E_x$ for all $x \in X$.*

We let $\Gamma(X, E)$ or $H^0(X, E)$ denote the set of holomorphic sections of E on X. The set $H^0(X, E)$ is a complex vector space of finite dimension when X is compact. We let $h^0(X, E) := \dim_{\mathbb{C}} H^0(X, E)$ denote its dimension.

When $E = L$ is a line bundle, it follows from the *asymptotic Riemann–Roch theorem* that

- either $h^0(X, L^k) = 0$ for all $k \in \mathbb{N}^*$,
- or $h^0(X, L^k) = 0$ or 1 (but not always zero),
- or else there exists $C > 0$ and $\nu \in \{1, \dots, n = \dim_{\mathbb{C}} X\}$ such that

$$\frac{1}{C}k^\nu \le h^0(X, L^k) \le Ck^\nu \text{ for all } k \in \mathbb{N}^*.$$

By convention, we set $\nu = -\infty$ and $\nu = 0$, respectively, in the first two cases. The integer $\nu(L) = \nu$ is called the *Itaka dimension* of L.

If $\{U\}$ is an open covering of X trivializing L with transition functions $\{g_{UV}\}$, then a holomorphic section $s \in \Gamma(X, L)$ is a collection $s = \{s_U \in \mathcal{O}(U)\}$ of holomorphic functions in U such that

$$s_U = g_{UV} s_V \text{ in } U \cap V.$$

Example 7.17. *The hyperplane bundle $\mathcal{O}(1)$ on the projective space $\mathbb{P}^n$ has transition functions $g_{\alpha\beta} = z_\beta/z_\alpha$ associated to the trivializing covering $U_\alpha := \{[z] \in \mathbb{P}^n |, |\, z_\alpha \neq 0\} \sim \mathbb{C}^n$. A holomorphic section of $\mathcal{O}(1)$ is thus a set of holomorphic functions $P_\alpha \in \mathcal{O}(U_\alpha)$ such that*

$$z_\alpha P_\alpha(z_0/z_\alpha, \ldots, z_n/z_\alpha) = z_\beta P_\beta(z_0/z_\beta, \ldots, z_n/z_\beta) = P(z_0, \ldots, z_n),$$

where P is a homogeneous polynomial of degree 1 in $z = (z_0, \ldots, z_n)$. Similarly, holomorphic sections of $\mathcal{O}(k)$ are homogeneous polynomials of degree k in $n+1$ variables. Note that $h^0(\mathbb{P}^n, \mathcal{O}(k)) \sim C k^n$, hence $\nu(\mathcal{O}(1)) = n$.

7.2.2.2 Holomorphic vector fields and forms Let X be a complex compact manifold. A smooth section of the tangent bundle T_X is called a (smooth) vector field. A holomorphic section of the holomorphic tangent bundle T'_X is called a holomorphic vector field.

A (smooth) differential form of degree k is a (smooth) section of the vector bundle $\Lambda^k(T_X)^*$. The decomposition of the complexified tangent bundle $T_{\mathbb{C}}(X) = T'(X) \oplus T''(X)$ induces a decomposition of the complexified bundle of differential forms,

$$\Omega_{\mathbb{C}}(X) = T_{\mathbb{C}}(X)^* = \Omega^{1,0}(X) \oplus \Omega^{0,1}(X).$$

The bundle $\Omega^{1,0}(X)$ of complex differential forms of type $(1,0)$ is generated, in holomorphic local coordinates $z_1, \ldots, z_n$, by the dz_i's. A holomorphic section of $\Omega^{1,0}(X)$ is called a holomorphic 1-form.

The above decomposition also induces a decomposition of complex k-forms,

$$\Lambda^k \Omega_{\mathbb{C}}(X) = \sum_{p+q=k} \Omega^{p,q}(X),$$

where $\Omega^{p,q}(X) = \Lambda^p \Omega^{1,0}(X) \otimes \Lambda^q \Omega^{0,1}(X)$.

The canonical bundle (one of the most important holomorphic line bundles) is $K_X = \Omega^{n,0}(X) = \Lambda^n \Omega^{1,0}(X)$.

Example 7.18. *When $X = \mathbb{C}^n/\Lambda$ is a compact complex torus, the Euclidean holomorphic 1-forms $dz_1, \ldots, dz_n$ descend to the quotient and induce a basis of nowhere vanishing holomorphic sections of T'_X which is therefore the trivial vector bundle. Thus $K_X = X \times \mathbb{C}$ is trivial and $\nu(K_X) = 0$.*

7.2.3 Kodaira embedding theorem

7.2.3.1 Ample line bundle Let $L \to X$ be a holomorphic line bundle on a compact complex manifold. Fix a basis $\{s_0, \ldots, s_N\}$ of $H^0(X, L)$ and assume that for all x there is one section $s \in H^0(X, L)$ such that $s(x) \neq 0$ (one says in this case that L is *base point free*). Then the evaluation

$$\Phi_L : X \ni x \longmapsto [s_0(x) : \cdots : s_N(x)] \in \mathbb{P}(H^0(X, L))$$

is a well-defined holomorphic map.

Definition 7.19. *A holomorphic line bundle $L \to X$ is called very ample if the mapping $\Phi_L : X \to \mathbb{P}(H^0(X, L))$ is an embedding.*

L is called ample if there exists $k \in \mathbb{N}$ such that L^k is very ample.

Here L^k denotes $L \otimes \cdots \otimes L$ (k times). It is also very common to use the additive notation kL. The existence of an ample line bundle is thus equivalent to X being projective algebraic.

7.2.3.2 Positivity versus ampleness

Theorem 7.20 (Kodaira). *Let $L \to X$ be a holomorphic line bundle on a compact complex manifold. Then L is positive if and only if it is ample.*

One direction is clear: if L is ample, then L^k is very ample for some $k \in \mathbb{N}^*$. The evaluation map Φ_{L^k} allows then to identify L^k with the pull-back $\phi_{L^k}^* \mathcal{O}(1)$ of the (restriction to $\phi_{L^k}(X)$ of the) hyperplane bundle $\mathcal{O}(1)$. If h is a smooth positive metric of the latter, then $(\phi_{L^k}^* h)^{1/k}$ is a positive metric of L on X. The other direction is more involved and can be proved by using Hörmander's resolution of $\overline{\partial}$ with L^2-estimates (see [Dem, Bern10]).

Corollary 7.21. *A compact complex manifold X is projective algebraic if and only if it admits a Hodge form, i.e., a Kähler form ω with integer cohomology class $\{\omega\} \in H^2(X, \mathbb{Z})$.*

Proof. If X is projective algebraic, then the restriction of the Fubini–Study form does the job, in any projective embedding. Conversely, if ω is a Kähler form with cohomology class $\{\omega\} \in H^{1,1}(X, \mathbb{R}) \cap H^2(X, Z)$, then there exists a holomorphic line bundle $L \to X$ whose first Chern class $c_1(L)$ coincides with $\{\omega\}$. We can find a smooth metric h of L with curvature form equal to ω, thus L is positive and hence ample by Kodaira's theorem. □

This explains at least intuitively why most complex tori $X = \mathbb{C}^n/\Lambda$ are not abelian in dimension $n \geq 2$ (see [Deb99] for a comprehensive introduction to tori and abelian manifolds).

7.2.4 Canonical bundle and Kodaira dimension

It is natural to try and classify all compact complex manifolds, up to an equivalence relation. It turns out that the right equivalence notion is that of bimeromorphic conjugacy. Finding invariants is of course a fundamental problem. The positivity properties of the canonical bundle K_X furnish a first set of such invariants. The Kodaira dimension of X is the Itaka dimension of its canonical bundle,

$$\mathrm{kod}(X) = \nu(K_X) \in \{-\infty, 0, \ldots, n\}.$$

Example 7.22. *The canonical bundle of the complex projective space* $\mathbb{P}^n$ *is*

$$K_{\mathbb{P}^n} = \mathcal{O}(-n-1),$$

hence $\mathrm{kod}(\mathbb{P}^n) = -\infty$.

We have already noticed that the canonical bundle of a complex torus $T = \mathbb{C}^n/\Lambda$ is the trivial line bundle, hence $\mathrm{kod}(T) = 0$.

7.2.4.1 The adjunction formula Let Y be a smooth hypersurface of a complex manifold X. The reader is encouraged to prove (Exercise 7.15) the following *adjunction formula*:

Lemma 7.23. *The canonical bundle of* Y *is given by*

$$K_Y = (K_X + L_Y)\,|_Y.$$

In particular, if H is an hypersurface of degree $d \in \mathbb{N}^*$ in $\mathbb{P}^n$, then

$$K_H = \mathcal{O}(d-n-1)_{|H}.$$

It is thus ample if $d \geq n+2$ so that $\mathrm{kod}(H) = n-1$.

It is trivial if $d = n+1$, and negative if $d \leq n$. Manifolds with a negative canonical line bundle (i.e., with ample anti-canonical K_X^*) are called *Fano*. They will play an important role in the sequel.

7.2.4.2 Riemann surfaces Let S be a compact Riemann surface. Then

- $K_S = \mathcal{O}(-2)$ is negative and $\mathrm{kod}(S) = -\infty$ if $S = \mathbb{P}^1$;
- $K_S = S \times \mathbb{C}$ is trivial and $\mathrm{kod}(S) = 0$ if S is an elliptic curve,
- K_S is ample and $\mathrm{kod}(S) = 1$ if S is hyperbolic.

7.2.4.3 Higher dimension The classification of complex *algebraic* surfaces ($\dim_{\mathbb{C}} X = 2$) is rather well understood thanks to the works of the Italian school (late 19th/early 20th century) and was cleaned up in the 50's–60's by Kodaira (see [Beauville] for a gentle introduction):

- if $\mathrm{kod}(X) = -\infty$, then X is bimeromorphic to a geometrically ruled surface $X' \xrightarrow{\pi} \mathcal{C}$ over a curve $\mathcal{C}$ (with $\mathbb{P}^1$-fibers);
- if $\mathrm{kod}(X) \geq 0$, then one can contract finitely many curves and reach a unique *minimal model* $X \xrightarrow{\pi} X_{\min}$ such that $K_{X_{\min}}$ is nef.

A line bundle L is nef if for any ample line bundle $A \to X$ and $N \in \mathbb{N}$, the line bundle $L^N \otimes A$ is ample. We shall come back to this important notion later on in the book.

In dimension $n \geq 3$, it becomes necessary to work on varieties with mild singularities. We will thus have to study analytic objects in this context as well. There has been recent important progress in the classification of higher dimensional algebraic varieties, we refer the reader to [KM] for an introduction to the Minimal Model Program and to [BCHM10] for one of the most recent developments.

7.3 Hodge theory

7.3.1 Positive forms and currents We have reviewed the basic local properties of positive closed currents in Chapter 2. We briefly recall here the notions that we are going to use in the context of compact Kähler manifolds.

7.3.1.1 Kähler forms Recall that a current T of dimension q on an n-dimensional complex manifold X is a continuous linear form on the space of smooth differential forms of degree q with compact support. The current dT is the current of dimension $q-1$ defined by

$$\langle dT, \theta\rangle := (-1)^{q+1}\langle T, d\theta\rangle.$$

The current T is *closed* if $dT = 0$. We let $\{T\}$ denote the cohomology class defined by the current T. It is a theorem of de Rham that the corresponding cohomology vector space

$$H^{2n-q}(X,\mathbb{R}) = \frac{\{\text{closed currents of dimension } q\}}{\{dS \mid S \text{ current of dimension } q-1\}}$$

is isomorphic to the one defined by using closed smooth differential forms.

Since X is complex, differential forms can be decomposed accordingly to their bidegrees (in local coordinates with respect to the basis $dz_I \wedge d\overline{z}_J$). We say that T is of bidimension (p,q) (or bidegree $(n-p, n-q)$) if it acts on forms of bidegree (p,q). Such a current can be locally written as

$$T = \sum_{|I|=p,|J|=q} T_{I,J}\, dz_I \wedge d\overline{z}_J,$$

where the coefficients $T_{I,J}$'s are distributions. We let $H^{p,q}(X,\mathbb{R})$ denote the corresponding vector space of cohomology classes.

A current T of bidimension (p,p) is *positive* if for all 1-forms $\alpha_1, \ldots, \alpha_p$,

$$\langle T, i\alpha_1 \wedge \overline{\alpha}_1 \wedge \ldots \wedge i\alpha_p \wedge \overline{\alpha}_p\rangle \geq 0.$$

A positive current is of order zero, i.e., the coefficients $T_{I,J}$ are measures.

In particular a positive closed current of bidegree $(1,1)$ can be locally written as $T = dd^c\varphi$, for some real valued function φ which is plurisubharmonic. Two extreme examples of closed positive currents are

(1) currents of integration along analytic subsets;
(2) positive smooth closed differential forms.

Definition 7.24. *A smooth closed current T of bidegree $(1,1)$ is called a Kähler form if it is strictly positive, i.e., if it can be locally written as*

$$T = \sum_{\alpha,\beta} T_{\alpha\beta}\, i\, dz_\alpha \wedge d\overline{z}_\beta,$$

where the smooth matrix $(T_{\alpha\beta})$ is Hermitian positive definite.

A complex manifold X is called Kähler if it admits a Kähler metric.

While the existence of a *Hermitian form* (a strictly positive $(1,1)$-form) is easily shown by using a partition of unity, the existence of a *Kähler form* has strong consequences on the structure of the underlying manifold, as we shall soon indicate.

Example 7.25. *The complex projective space is a Kähler manifold since*

$$\omega = dd^c \log \left[1 + \sum_{i \neq \alpha} |z_i|^2 \right] \text{ in } U_\alpha$$

defines a Kähler form which is independent of α, hence defines a global Kähler form on $\mathbb{CP}^n$. The form $\omega_{\mathrm{FS}} := \omega/2$ is called the Fubini–Study Kähler form.

A complex torus $X = \mathbb{C}^n/\Lambda$ is a Kähler manifold since the Euclidean form $\omega = dd^c \sum_{i=1}^n |z_i|^2$ is invariant under translations, hence passes to the quotient and defines on X a Kähler form.

Observe that any complex submanifold of a Kähler manifold is a Kähler manifold, hence so is any projective algebraic manifold.

7.3.1.2 Extension and decomposition of closed positive currents Recall that a (locally) pluripolar set is a Borel set which is locally included in the $-\infty$ locus of a plurisubharmonic function. The following extension result is very useful (see [Dem, Chapter 3]):

Theorem 7.26. *Let X be a compact complex manifold, ω a smooth Hermitian form on X and $A \subset X$ a closed subset. Let T be a positive closed current of bidimension (q,q) on $X \setminus A$.*

i) *If A has zero $(2q-1)$-dimensional Hausdorff measure, then*

$$\int_{X \setminus A} T \wedge \omega^q < +\infty.$$

This is the case if A is an analytic subset of dimension $\leq q-1$.

ii) *If A is pluripolar and T has finite mass on $X \setminus A$, then the trivial extension $\tilde{T}$ of T through A is still closed.*

Here $\tilde{T}$ is obtained by extending the measures coefficients $T_{I,J}$ of T as zero on A. The following theorem of Siu is also fundamental:

Theorem 7.27. *Let T be a positive closed current of bidimension (q,q) on a complex manifold X.*

i) *For all $c > 0$, the (super)level sets*

$$E_c(T) := \{x \in X \mid \nu(T,x) \geq c\}$$

are closed analytic subsets of X.

ii) *There exist a countable family* (V_j) *of* q*-dimensional analytic subsets and* $(\varepsilon_j) \in \mathbb{R}_+^{\mathbb{N}}$ *such that*

$$T = \sum_{j \geq 0} \varepsilon_j [V_j] + R,$$

where R *is a positive closed current of bidimension* (q, q) *with the property that for all* $c > 0$*,*

$$\dim_{\mathbb{C}} E_c(R) \leq q - 1.$$

We shall use the following consequence of both theorems on several occasions:

Corollary 7.28. *Let* T *be a positive closed current of bidimension* (q, q) *on a complex manifold* X*,* $V \subset X$ *an irreducible analytic subset of dimension* q*. Then*

$$T = c[V] + R,$$

where R *is a positive closed current of bidimension* (q, q) *which does not charge* V *and* $c = \inf_{x \in V} \nu(T, x) \geq 0$ *is the Lelong number of* T *at a generic point of* V*.*

7.3.1.3 Mass of positive currents Let X be a compact Kähler manifold and fix a Kähler form ω. Let T be a positive current of bidimension (q, q) on X and decompose it locally as

$$T = \sum_{I,J} T_{I,J} i dz_I \wedge d\overline{z}_J.$$

The total variation of its measure coefficients is then controlled by the mass $\|T\| := \int_X T \wedge \omega^q$ of T with respect to ω, i.e.,

$$C^{-1} \int_X T \wedge \omega^q \leq \sum \|T_{I,J}\| \leq C \int_X T \wedge \omega^q,$$

where C is independent of T.

An important fact that we shall use over and over is that the set of positive currents with mass uniformly bounded by a fixed constant is compact in the weak topology.

7.3.1.4 Push-forward/Pull-back of positive currents Let $f : X \to Y$ be a holomorphic map between two compact complex manifolds. One can push forward a current T on X by duality, setting

$$\langle f_* T, \theta \rangle := \langle T, f^* \theta \rangle,$$

since the pull-back of a smooth differential form (with compact support) is again a smooth differential form (with compact support). The push-forward preserves positivity, closedness, and bidimension.

In general one cannot define the pull-back of a current S on Y by a holomorphic map (see [Meo96]). It is however possible to pull-back positive closed currents of

bidegree $(1,1)$: such a current can be locally written as $S = dd^c\varphi$ for some local plurisubharmonic function φ, and one can set

$$f^*S := dd^c(\varphi \circ f).$$

We let the reader check that this definition does not depend on the choice of φ and that f^*S is a globally well-defined current of bidegree $(1,1)$ on X, which is positive and closed. In particular, one can always pull-back the current of integration along an analytic hypersurface.

7.3.2 Hodge decomposition theorem Hodge theory (valid for any Riemannian manifold) identifies peculiar representatives (harmonic forms) of de Rham cohomology classes. When the manifold is complex Kähler, several differential relations (the Kähler identities) hold, having important consequences. We encourage the reader to consult [Wells, V], indicating here only three major results:

7.3.2.1 Decomposition of de Rham cohomology

Theorem 7.29. *Let X be a compact Kähler manifold. Then*

$$H^k(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X,\mathbb{C}).$$

The space $H^{p,q}(X,\mathbb{C})$ is isomorphic to $\overline{H^{q,p}}(X,\mathbb{C})$.

Corollary 7.30. *The odd Betti numbers $b_{2k+1} = \dim_{\mathbb{C}} H^{2k+1}(X,\mathbb{C})$ of a compact Kähler manifold are even.*

This shows for instance that the Hopf surfaces are not Kähler.

7.3.2.2 The Hodge index theorem

The wedge product of closed differential forms of bidegree $(1,1)$ induces a cup product at the level of cohomology,

$$H^{1,1}(X,\mathbb{R})^n \ni (\alpha_1,\dots,\alpha_n) \longmapsto \alpha_1 \cdot \dots \cdot \alpha_n \in \mathbb{R},$$

which coincides with the geometric intersection of divisors, in case the α_i's are the first Chern class of effective divisors D_i intersecting transversally,

$$c_1(D_1)\cdots c_1(D_n) = \sum_{x \in D_1 \cap \dots \cap D_n} \mathrm{mult}_x(D_1 \cap \dots \cap D_n).$$

When X is a compact Kähler surface, the cup-product form is non-degenerate with signature $(1, h^{1,1}(X) - 1)$. This is the *Hodge index theorem*.

7.3.3 The $\partial\overline{\partial}$-lemma Let T be a positive closed current of bidegree $(1,1)$ on a compact manifold X. It can be locally written as $T = dd^c\varphi$, where φ is a plurisubharmonic function. This cannot hold globally, since the only plurisubharmonic functions are the constants, by the maximum principle. On the other hand, if θ is a smooth closed form representing the cohomology class of T, one may wonder whether $T-\theta$ (which is d-exact) is furthermore dd^c-exact. This is true on a Kähler manifold:

Lemma 7.31. *Let X be a compact Kähler manifold. Let S be a current which is both ∂ and $\overline{\partial}$-closed. Then S is d-exact if and only if it is dd^c-exact.*

Note that a current of bidegree (p,p) is d-closed if and only if it is both ∂ and $\overline{\partial}$-closed. The following particular case will be used on several occasions in this book: fix a cohomology class $\alpha \in H^{1,1}(X,\mathbb{R})$ that can be represented by a positive closed current (such a class is called *pseudoeffective*). Fix a smooth representative θ of α. Then any closed positive current of bidegree $(1,1)$ in α can be written, if X is Kähler, as

$$T = \theta + dd^c\varphi$$

for some upper semi-continuous function $\varphi : X \to \mathbb{R}\cup\{-\infty\}$, which is uniquely determined up to an additive constant. Such functions are called θ-plurisubharmonic and will be the main object of study in Chapter 8.

7.4 Curvatures of metrics

7.4.1 Hermitian and Kähler metrics Let X be a n-dimensional complex manifold. It can be considered as a real manifold of real dimension $2n$, equipped with an endomorphism J of the real tangent bundle $T_{\mathbb{R}}(X)$ such that

$$J(\partial/\partial x_j) = \partial/\partial y_j \text{ and } J(\partial/\partial y_j) = -\partial/\partial x_j,$$

so that $J^2 = -\mathrm{Id}$. Note that the holomorphic tangent bundle $T'(X)$ is the i-eigenspace of J acting on $T_{\mathbb{C}}(X) = T_{\mathbb{R}}(X)\otimes\mathbb{C}$.

Recall that a Riemannian metric g smoothly associates to each point $p \in X$ a scalar product $\langle\cdot,\cdot\rangle_p$ on $T_p(X)$ (see [GHL]).

Definition 7.32. *A Riemannian metric g on X is Hermitian if it is J-invariant, i.e.,*

$$g(Ju, Jv) = g(u,v), \text{ for all } u, v \in T_{\mathbb{R}}(X).$$

We let the reader check that a Hermitian metric g is of the form

$$g = 2\Re\left(\sum_{\alpha,\beta} g_{\alpha\beta}\, dz_\alpha \otimes d\overline{z}_\beta\right),$$

where the matrix $(g_{\alpha\beta})$ is Hermitian.

Definition 7.33. *A Hermitian metric is Kähler if the* $(1,1)$*-form*

$$\omega := \sum_{\alpha,\beta} g_{\alpha\beta} i\, dz_\alpha \wedge d\overline{z}_\beta$$

is closed, $d\omega = 0$.

The following classical proposition will be used on several occasions. We leave the proof as an Exercise 7.18 for the reader:

Proposition 7.34. *Let* (X, ω) *be a complex Hermitian manifold. It is Kähler if and only if for each* $p \in X$ *we can find a holomorphic coordinate system centered at* p *such that*

$$\omega = \sum_{\alpha,\beta} g_{\alpha\beta} i\, dz_\alpha \wedge d\overline{z}_\beta, \text{ with } g_{\alpha\beta} = \delta_{\alpha\beta} + O(\|z\|^2).$$

Such coordinates are called *normal coordinates*.

7.4.2 The Chern–Levi-Civita connection

7.4.2.1 Levi-Civita connection Let (X, g) be a Riemannian manifold of real dimension m. Recall that a connection ∇ on the tangent bundle allows to differentiate vector fields as if they were functions. It is given by m^3 smooth functions $\{\gamma_{ij}^\ell\}$ (the Christoffel symbols) such that

$$\nabla_{\partial_{x_i}} \partial_{x_j} = \sum_\ell \gamma_{ij}^\ell \partial_{x_\ell},$$

with suggestive notations.

The Levi-Civita connection is the unique connection on the (real) tangent bundle of X that preserves the metric (under parallel transport) and is torsion free. More precisely, it is the unique connection ∇ such that for any vector fields V, Y, Z,

(1) $L_V\langle Y, Z\rangle = \langle \nabla_V Y, Z\rangle + \langle Y, \nabla_V Z\rangle$ (invariance),
(2) $\nabla_V Y - \nabla_Y V = [V, Y]$ (zero torsion),

where $L_V(\cdot)$ denotes the Lie derivative, i.e., the derivative in the direction of the vector field V (see [GHL, Chapter 2.3]).

7.4.2.2 Chern connection Let now (X, ω) be a complex Hermitian manifold. There is a unique connection, the Chern connection, defined on the holomophic tangent bundle $T'(X)$, with the following properties:

(1) it preserves the metric, i.e., $\nabla g = 0$,
(2) it preserves the complex structure, i.e., $\nabla J = 0$,
(3) the torsion is pure in its indices.

We refer the reader to [Wells, Dem, V] for a proof.

Remark 7.35. *While there is no naturally defined exterior derivative d on the space of smooth sections of a vector bundle, on a holomorphic vector bundle E, the $\overline{\partial}$-operator is always well defined, acting on the space of E-valued (p,q)-forms.*

It is natural to try and compare the Levi-Civita connection on $T_{\mathbb{C}}(X)$ and the Chern connection. They turn out to coincide when the manifold (X,ω) is Kähler. We let Γ_{ij}^{ℓ} denote the complex Christoffel symbols, defined by

$$\nabla_{\frac{\partial}{\partial z_i}} \frac{\partial}{\partial z_j} = \sum_{\ell} \Gamma_{ij}^{\ell} \frac{\partial}{\partial z_\ell} + \sum_{k} \Gamma_{ij}^{\overline{k}} \frac{\partial}{\partial z_{\overline{k}}}$$

and

$$\nabla_{\frac{\partial}{\partial z_i}} \frac{\partial}{\partial \overline{z}_j} = \sum_{\ell} \Gamma_{i\overline{j}}^{\ell} \frac{\partial}{\partial z_\ell} + \sum_{k} \Gamma_{i\overline{j}}^{\overline{k}} \frac{\partial}{\partial z_{\overline{k}}}.$$

The Kähler condition implies that the only possibly non-zero coefficients are of pure type, these are Γ_{ij}^{ℓ} and $\Gamma_{\overline{i}\,\overline{j}}^{\overline{k}}$.

7.4.3 Curvatures Let ∇ denote the Levi-Civita–Chern connection of a Kähler manifold (X,ω).

7.4.3.1 Riemannian curvature We extend in a $\mathbb{C}$-linear way to $T_{\mathbb{C}}(X)$ the map

$$R(u,v)w = \nabla_u \nabla_v w - \nabla_v \nabla_u w - \nabla_{[u,v]} w.$$

The Riemannian curvature tensor is the map $R = T(X)^4 \to \mathbb{R}$ defined by

$$R(u,v,w,x) := \omega(R(u,v)x, w).$$

In local coordinates,

$$R_{ijk\ell} := R\left(\frac{\partial}{\partial z_i}, \frac{\partial}{\partial z_j}, \frac{\partial}{\partial z_k}, \frac{\partial}{\partial z_\ell}\right),$$

with analogous notations $R_{\overline{i}jk\ell}$ if we replace $\frac{\partial}{\partial z_i}$ by $\frac{\partial}{\partial \overline{z}_i}$, etc.

Lemma 7.36. *The only non-zero terms are (essentially) the coefficients $R_{i\overline{j}k\overline{\ell}}$. These can be expressed by*

$$R_{i\overline{j}k\overline{\ell}} = -\frac{\partial^2 g_{ij}}{\partial z_k \partial \overline{z}_\ell} + \sum_{s,t} g^{st} \frac{\partial g_{sj}}{\partial z_k} \frac{\partial g_{it}}{\partial \overline{z}_\ell},$$

where $(g^{st}) = {}^{\mathrm{t}}(g_{st})^{-1}$ denotes the transposed inverse matrix.

We invite the reader to prove this in Exercise 7.20.

7.4.3.2 Holomorphic bisectional curvature Let σ be a plane in $T_p(X)$, i.e., a real two-dimensional subspace of $T_p(X)$. Choosing an orthonormal basis x, y for σ, we define the sectional curvature $K(\sigma)$ by

$$K(\sigma) = K(x, y) := R(x, y, x, y).$$

The restriction of the sectional curvature to the set of J-invariant planes is called the *holomorphic sectional curvature*.

If we now consider two J-invariant planes, we obtain the important concept of holomorphic bisectional curvature:

Definition 7.37. *The holomorphic bisectional curvature of* (X, ω) *is*

$$R(x, y, y, x) + R(x, Jy, Jy, x).$$

Although this is a sum of two sectional curvatures (and as such carries less information than the sectional curvature itself), prescribing the sign or the value of the holomorphic bisectional curvature imposes drastic conditions on the manifold X, as the following two instances indicate:

- the *uniformization theorem* states that a complete Kähler manifold of constant holomorphic bisectional curvature is covered either by the complex projective space, or by $\mathbb{C}^n$, or else by the unit ball $\mathbb{B} \subset \mathbb{C}^n$;
- if the holomorphic sectional curvature of a compact Kähler manifold (X, ω) is non-negative, then X is (up to an étale cover) the product of a symmetric space of compact type (e.g., a projective space, a Grassmannian, etc) and a compact complex torus.

We refer the reader to [Mok88] for the proofs of these results.

7.4.3.3 Scalar and Ricci curvature Let ω be a Kähler form on X, which writes locally $\omega = \sum_{\alpha,\beta} g_{\alpha\beta} \frac{i}{2} dz_\alpha \wedge d\overline{z}_\beta$. The *Ricci curvature tensor* is the trace of the Riemannian curvature tensor,

$$\mathrm{Ric}_{k\ell} = \sum_{i,j} g^{ij} R_{i\overline{j}k\overline{\ell}} = -\frac{\partial^2}{\partial z_k \partial \overline{z}_\ell} \log \det(g_{pq}).$$

The *Ricci curvature form* is, by definition,

$$\mathrm{Ric}(\omega) = -c \sum_{\alpha,\beta} \frac{\partial^2}{\partial z_\alpha \partial \overline{z}_\beta} \log \det(g_{pq}) \frac{i}{2} dz_\alpha \wedge d\overline{z}_\beta,$$

where $c > 0$ is a normalizing constant such that the following holds:

Lemma 7.38. *The Ricci form* $\mathrm{Ric}(\omega)$ *is a closed* $(1,1)$*-form representing the first Chern class* $c_1(X)$ *of* X.

The reader will check this in Exercise 7.21. In the sequel we will often forget c to simplify notations, with no dange of confusion.

The *scalar curvature* is the trace of the Ricci curvature: it is defined by

$$\mathrm{Scal}(\omega) := \frac{\mathrm{Ric}(\omega) \wedge \omega_{n-1}}{\omega_n} = n \frac{\mathrm{Ric}(\omega) \wedge \omega^{n-1}}{\omega^n}.$$

Here $\omega_j := \frac{1}{j!}\omega^j$. In local coordinates,

$$\mathrm{Scal}(\omega) = \sum_{\alpha,\beta} \omega^{\alpha\beta} R_{\alpha\beta}, \text{ where } (\omega^{\alpha\beta}) = {}^t(\omega_{\alpha\beta})^{-1}$$

and $\mathrm{Ric}(\omega) = \sum_{\alpha\beta} R_{\alpha\beta} i dz_\alpha \wedge d\overline{z}_\beta$.

Definition 7.39. *A Kähler metric ω has constant scalar curvature if the function* $\mathrm{Scal}(\omega)$ *is constant.*

Observe that if a Kähler metric ω has constant scalar curvature, then the constant $\mathrm{Scal}(\omega) \equiv \lambda_\alpha$ is uniquely determined by the cohomology class α of ω: integration yields

$$\lambda_\alpha \cdot \alpha^n = n\, c_1(X) \cdot \alpha^{n-1}.$$

7.5 Exercises

Exercise 7.1. *Let $f : X \to Y$ be a map between two complex manifolds X, Y. Show that f is holomorphic if and only if the maps $w_i \circ f \circ z_\alpha^{-1}$ are holomorphic for any system of local complex coordinates z_α (resp. w_i) on X (resp. on Y).*

Exercise 7.2. *Let $e_1, \ldots, e_n$ be the canonical basis of $\mathbb{C}^n$ and let*

$$e_I := e_{i_1} \wedge \cdots e_{i_r}, \quad I = (i_1, \ldots, i_r), \quad 1 \le i_1 < \cdots < i_r \le n,$$

be the standard basis of $\Lambda^r \mathbb{C}^n$. For $Z = [Z^i]_{1 \le i \le r} \in \mathbb{C}^{nr}$, we let Δ_I denote the homogeneous polynomials of degree r such that

$$Z^1 \wedge \cdots Z^r = \sum_I \Delta_I(Z) e_I.$$

i) *Check that for all $A \in \mathrm{GL}(r, \mathbb{C})$, $\Delta_I(A \cdot Z) = \det A \Delta_I(Z)$ and that*

$$\Phi : \mathrm{Gr}(r, n, \mathbb{C}) \ni [Z] \longmapsto [Z^1 \wedge \ldots \wedge Z^r] \in \mathbb{P}(\Lambda^r \mathbb{C})$$

defines an embedding (the so called Plücker embedding).

ii) *Show that the charts $U_I := \{[Z] \in \mathrm{Gr}(r, n, \mathbb{C}) \mid \Delta_I(Z) \neq 0\}$ allow to define a complex analytic system of coordinates on* $\mathrm{Gr}(r, n, \mathbb{C})$.

Exercise 7.3. *Let $\mathcal{C} \subset \mathbb{P}^2$ be a non-singular curve of degree d. Show that $\mathcal{C}$ is a Riemann surface of genus $g = \frac{d(d-3)}{2} + 1$.*

Exercise 7.4. *Prove Lemma 7.6.*

Exercise 7.5. *Let $E \xrightarrow{\pi} X$ be a holomorphic vector bundle of rank $r+1$ on a compact complex manifold X. Let $\mathbb{P}(E)$ denote the quotient of E minus the zero section, modulo the natural $\mathbb{C}^*$-action. Show that this projective bundle is a compact complex manifold, which is Kähler if X is so.*

Exercise 7.6. *Let $L \to X$ be a holomorphic line bundle on a complex manifold X. Show that L is trivial (i.e., isomorphic to the trivial line bundle $\mathbb{C} \times X$) if and only if it admits a nowhere vanishing holomorphic section $s \in \Gamma(X, L)$.*

Exercise 7.7. *Let $L \to X$ be a holomorphic line bundle on a compact complex manifold X. Let h be a smooth metric of L on X. Show that*

$$c_1(L) = \{\Theta_h\} \in H^2(X, \mathbb{R}).$$

Exercise 7.8. *Show that the Picard group of the complex projective space is $\mathbb{Z}$ and that of $\mathbb{P}^1 \times \mathbb{P}^1$ is $\mathbb{Z}^2$.*

Let $\tilde{X}_Y \xrightarrow{\pi} X$ denote the blow-up of a compact complex manifold X along a smooth submanifold Y of codimension ≥ 2. Show that

$$\mathrm{Pic}(\tilde{X}_Y) \simeq \pi^* \mathrm{Pic}(X) \otimes \mathbb{Z}.$$

Exercise 7.9. *Show that any holomorphic vector bundle on the Riemann sphere $\mathbb{P}^1$ decomposes as a sum of holomorphic line bundles. Show that this is not necessarily the case on an elliptic curve.*

Exercise 7.10. *Check that*

$$\sigma : [z_0 : z_1 : z_2] \longmapsto [z_1 z_2 : z_0 z_2 : z_0 z_1]$$

defines a birational self-map of $\mathbb{P}^2$. Compute I_f, $f(I_f)$, and $\sigma \circ \sigma$.

Show that σ induces and automorphism of $\mathbb{P}^2$ blown-up at the three points $[1:0:0], [0:1:0], [0:0:1]$.

Exercise 7.11. *Justify the assertions of Example 7.9: any meromorphic map $f : \mathbb{CP}^n \dashrightarrow \mathbb{CP}^n$ is given, in homogeneous coordinates, by a collection $P_0, \ldots, P_n$ of homogeneous polynomials of the same degree with no common factor:*

$$f[z] = [P_0(z) : \cdots : P_n(z)].$$

It is holomorphic if the P_j's have no common zero (except for the origin), and it is meromorphic otherwise, with

$$I_f = \{[z] \in \mathbb{CP}^n \mid P_0(z) = \cdots = P_n(z) = 0\}.$$

Exercise 7.12. *Let X denote the blow-up of $\mathbb{P}^2$ at some point p. Show that the holomorphic self-maps of X are defined by holomorphic self-maps g of $\mathbb{P}^2$ such that $g^{-1}(p) = p$. What can you say about holomorphic self-maps of $\mathbb{P}^2$ blown-up at several points?*

Exercise 7.13. *Show that the canonical bundle of the complex projective space is $\mathcal{O}(-n-1)$. Can you describe the canonical bundle of $\mathbb{P}^n \times \mathbb{P}^k$?*

Exercise 7.14. *Let $\tilde{X}_p \xrightarrow{\pi} X$ be the blow-up of a complex manifold at some point p. Show that*

$$K_{\tilde{X}_p} = \pi^* K_X + (n-1)L_E,$$

where $E = \pi^{-1}(p)$ denotes the exceptional divisor.

Exercise 7.15. *Let Y be a smooth complex hypersurface of a complex manifold X. Prove the adjunction formula (see Lemma 7.23)*

$$K_Y = (K_X + L_Y)|_Y.$$

Exercise 7.16. *Show that if $\pi : \tilde{X} \to X$ is the blow-up of a compact Kähler manifold X at a point p, then $\tilde{X}$ is also a compact Kähler manifold. Show that if S is a positive closed current of bidegree $(1,1)$ on $\tilde{X}$, then there exist a positive closed current T of bidegree $(1,1)$ on X and $\lambda \geq 0$ such that*

$$S = \pi^* T + (\lambda - \nu(T,p))[E],$$

where $E = \pi^{-1}(p) \subset \tilde{X}$ is the exceptional divisor.

Exercise 7.17. *Let $X = \mathbb{C}^2 \setminus \{0\}/\Gamma$ be a complex Hopf surface, where Γ is generated by $\gamma(x,y) = (\lambda x, \mu y)$, with $0 < |\lambda| \leq |\mu| < 1$. Describe all positive closed currents of bidegree $(1,1)$ on X and deduce, without using Hodge's theorem, that X is not Kähler (see [HL83] for more information).*

Exercise 7.18. *Prove that a Hermitian metric is Kähler if and only if it osculates the Euclidean metric up to order 2 (Proposition 7.34).*

Exercise 7.19. *Let (X,ω) be a complex Kähler manifold. Show that for each $p \in X$, one can find holomorphic coordinates centered at p such that $\omega = \sum \omega_{\alpha\beta} i\, dz_\alpha \wedge d\overline{z}_\beta$, with*

$$\omega_{\alpha\beta} = \delta_{\alpha\beta} - \sum_{p,q} c_{pq\alpha\beta} z_p \overline{z}_q + O(\|z\|^3),$$

where the $c_{pq\alpha\beta}$'s are the coefficients of the curvature tensor.

Exercise 7.20. *Prove the assertions of Lemma 7.36.*

Exercise 7.21. *Let (X, ω) be a Kähler manifold. Show that the Ricci form* $\mathrm{Ric}(\omega)$ *of ω (conveniently normalized) represents the first Chern class $c_1(X)$ of X.*

Exercise 7.22. *Verify that the following Kähler manifolds have constant scalar curvature:*

(i) *The complex projective space $\mathbb{P}^n$ with its Fubini–Study form.*
(ii) *$\mathbb{C}^n$ equipped with its Euclidean Kähler form.*
(iii) *The unit ball $\mathbb{B} = \{z \in \mathbb{C}^n \mid \|z\| < 1\}$ with*

$$\omega = -d\,d^{\mathrm{c}} \log(1 - \|z\|^2).$$

8 Quasi-plurisubharmonic functions

The purpose of this chapter is to present the first basic properties of quasi-plurisubharmonic functions. These are functions which are locally given as the sum of a smooth and a plurisubharmonic function.

It follows from the maximum principle that on a compact manifold X there are no global plurisubharmonic functions except for the constants; however, there are plenty of quasi-plurisubharmonic (qpsh for short) functions.

When (X, ω) is compact Kähler, any qpsh function φ is $A\omega$-psh for some $A > 0$ large enough, i.e., $A\omega + dd^c\varphi \geq 0$ in the weak sense of currents. Indeed, $dd^c\varphi$ is bounded below by a smooth form, which is itself bounded below by $-A\omega$ if $A > 0$ is large enough. By rescaling, one can assume $A = 1$.

In this chapter we restrict ourselves to studying the class $PSH(X, \omega)$ of ω-psh functions. It is very useful to study this class when ω is not necessarily positive. To simplify our exposition, we assume throughout the book that ω is *semi-positive* and *big*, i.e., such that the volume

$$V = V_\omega = \mathrm{Vol}_\omega(X) := \int_X \omega^n$$

is positive: $V > 0$. The case when ω is allowed to become negative is also of interest, we shall come back to it in the last part of the book.

8.1 Topology

8.1.1 Definition

Definition 8.1. *We let $PSH(X, \omega)$ denote the set of all ω-plurisubharmonic (ω-psh) functions, i.e., the set of functions $\varphi \in L^1(X, \mathbb{R} \cup \{-\infty\})$ which can be locally written as the sum of a smooth and a psh function, and such that*

$$\omega + dd^c\varphi \geq 0$$

in the weak sense of positive currents.

Note that ω-psh functions are in particular upper semi-continuous hence bounded from above on X, which is compact.

We endow $PSH(X, \omega)$ with the L^1-topology. Observe that $PSH(X, \omega)$ is a closed subset of $L^1(X)$ which is convex. More precisely:

Proposition 8.2.

(1) *If* $\omega_1 \leq \omega_2$, *then* $PSH(X, \omega_1) \subset PSH(X, \omega_2)$.

(2) $\forall A \in \mathbb{R}_+^*$, $PSH(X, A\omega) = A \cdot PSH(X, \omega)$.

(3) *If* ω' *is cohomologous to* ω, $\omega' = \omega + dd^c\chi$, *then*

$$PSH(X, \omega') = PSH(X, \omega) - \chi.$$

(4) *If* $\varphi, \psi \in PSH(X, \omega)$, *then*

$$\max(\varphi, \psi)\ ,\ \frac{\varphi + \psi}{2}\ ,\ \log[e^{\varphi} + e^{\psi}] \in PSH(X, \omega).$$

Proof. Assertions (1), (2), and (3) follow straightforwardly from the definition. Observe that (4) says that $PSH(X, \omega)$ is a convex set which is stable under taking the maximum and also under the operation $(\varphi, \psi) \mapsto \log[e^{\varphi} + e^{\psi}]$. These are all consequences of the corresponding local properties of psh functions. We nevertheless give a proof for the reader's convenience. That $(\varphi + \psi)/2 \in PSH(X, \omega)$ follows by linearity. The latter assertion is a consequence of the following computation:

$$dd^c \log[e^{\varphi} + e^{\psi}] = \frac{e^{\varphi} dd^c\varphi + e^{\psi} dd^c\psi}{e^{\varphi} + e^{\psi}} + \frac{e^{\varphi+\psi} d(\varphi - \psi) \wedge d^c(\varphi - \psi)}{[e^{\varphi} + e^{\psi}]^2},$$

using that $df \wedge d^c f \geq 0$. This computation makes sense if for instance φ and ψ are smooth. The general case follows then by (locally) regularizing φ and ψ. Finally, observe that $\max(\varphi, \psi) = \lim j^{-1} \log[e^{j\varphi} + e^{j\psi}] \in PSH(X, \omega)$. □

It follows from (3) above that $PSH(X, \omega)$ essentially depends on the cohomology class $\{\omega\} \in H^{1,1}(X, \mathbb{R})$. In the same vein we have the following:

Proposition 8.3. *Let* $\mathcal{T}_{\{\omega\}}(X)$ *denote the set of positive closed currents* ω' *of bidegree* $(1, 1)$ *on* X *which are cohomologous to* ω. *Then*

$$PSH(X, \omega) \simeq \mathcal{T}_{\{\omega\}}(X) \oplus \mathbb{R}.$$

Proof. The mapping

$$\Phi : PSH(X, \omega) \ni \varphi \longmapsto \omega_\varphi := \omega + dd^c\varphi \in \mathcal{T}_{\{\omega\}}(X)$$

is a continuous affine mapping whose kernel consists of constants: indeed, $\omega_\varphi = \omega_\psi$ implies that $\varphi - \psi$ is pluriharmonic, hence constant by the maximum principle. Moreover, Φ is surjective: if ω' is cohomologous to ω, then $\omega' = \omega + dd^c\varphi$ for some $\varphi \in L^1(X, \mathbb{R})$, as follows from the dd^c-lemma. Thus φ coincides almost everywhere with a function of $PSH(X, \omega)$ and $\omega' = \Phi(\varphi)$. □

8.1.2 Compactness properties It follows from the previous proposition that any set of "normalized" ω-psh functions is in 1-to-1 correspondence with $\mathcal{T}_{\{\omega\}}(X)$, which is compact for the weak topology of currents. This is the key to several results to follow: normalized ω-psh functions form a compact family in $L^1(X)$.

Proposition 8.4. *Let $(\varphi_j) \in PSH(X,\omega)^{\mathbb{N}}$.*

(1) If (φ_j) is uniformly bounded from above on X, then either φ_j converge uniformly to $-\infty$ on X or the sequence (φ_j) is relatively compact in $L^1(X)$.

(2) If $\varphi_j \to \varphi$ in $L^1(X)$, then φ coincides almost everywhere with a unique function $\varphi^* \in PSH(X,\omega)$. Moreover,

$$\limsup \varphi_j(x) \leq \varphi^*(x),$$

with equality outside a pluripolar set, and

$$\lim_{j\to+\infty} \sup_X \varphi_j = \sup_X \varphi^*.$$

(3) In particular, if φ_j is decreasing, then either $\varphi_j \to -\infty$ or $\varphi = \lim \varphi_j \in PSH(X,\omega)$. Similarly, if φ_j is increasing and uniformly bounded from above, then $\varphi := (\lim \varphi_j)^* \in PSH(X,\omega)$, where * denotes the upper semi-continuous regularization.

(4) If $\varphi_j \to \varphi$ in the weak sense of distributions, then φ coincides a.e. with a unique function $\varphi^* \in PSH(X,\omega)$ and $\varphi_j \to \varphi^*$ in $L^p(X)$ for any $p > 1$. Moreover, $\Phi_j := (\sup_{l\geq j} \varphi_l)^*$ decrease to φ^*.

The second item is often called "Hartogs lemma".

Proof. This is a straighforward consequence of the analogous local result for sequences of plurisubharmonic functions (see Chapter 1). □

Proposition 8.5. *The family*

$$\mathcal{F}_0 := \{\varphi \in PSH(X,\omega) |\ \sup_X \varphi = 0\}$$

is a compact subset of $PSH(X,\omega)$.

If μ is a probability measure such that $PSH(X,\omega) \subset L^1(\mu)$, then

$$\mathcal{F}_\mu := \{\varphi \in PSH(X,\omega) |\ \int_X \varphi d\mu = 0\}$$

is a relatively compact subset of $PSH(X,\omega)$. In particular there exists C_μ such that for any $\varphi \in PSH(X,\omega)$,

$$-C_\mu + \sup_X \varphi \leq \int_X \varphi d\mu \leq \sup_X \varphi.$$

Proof. It immediately follows from Proposition 8.4 that $\mathcal{F}_0$ is a relatively compact subset of $PSH(X,\omega)$. Moreover, $\mathcal{F}_0$ is closed by Hartogs' lemma.

Let $(\varphi_j) \in \mathcal{F}_\mu^{\mathbb{N}}$. Then $\psi_j := \varphi_j - \sup_X \varphi_j \in \mathcal{F}_0$, which is relatively compact. Assume first μ is smooth. Then $(\int_X \psi_j d\mu)$ is bounded: this is because if $\psi_{j_k} \to \psi$ in $L^1(X)$, then $\psi_{j_k}\mu \to \psi\mu$ in the weak sense of (negative) measures, hence $\int_X \psi_{j_k} d\mu \to \int_X \psi d\mu > -\infty$. Now $\int_X \psi_j d\mu = \int_X \varphi_j d\mu - \int_X \sup_X \varphi_j d\mu = -\sup_X \varphi_j$, thus $(\sup_X \varphi_j)$ is bounded and we can apply the previous proposition to conclude that (φ_j) is relatively compact (it cannot converge uniformly to $-\infty$ since $\int_X \varphi_j d\mu = 0$).

When μ is not smooth, it only remains to prove that $(\int_X \psi_j d\mu)$ is bounded. Assume on the contrary that $\int_X \psi_j d\mu \to -\infty$. Extracting a subsequence if necessary we can assume $\int_X \psi_j d\mu \leq -2^j$. Set $\psi = \sum_{j\geq 1} 2^{-j}\psi_j$. This is a decreasing sequence of ω-psh functions, hence $\psi \in PSH(X,\omega)$ or $\psi \equiv -\infty$. Now it follows from the previous discussion that $\int_X \psi_j dV \geq -C$ if dV denotes some smooth probability measure on X. Thus $\int_X \psi dV > -\infty$, hence $\psi \in PSH(X,\omega)$. We obtain a contradiction, since by the monotone convergence theorem, $\int_X \psi d\mu = \sum_{j\geq 1} 2^{-j} \int_X \psi_j d\mu = -\infty$. □

Remark 8.6. *It is a delicate question to determine whether $\mathcal{F}_\mu$ is closed* (*hence compact*). *This is obviously the case when μ is a smooth volume form. In dimension one, the reader will check in Exercise 12.1 that $\mathcal{F}_\mu$ is closed if and only if μ has continuous potentials.*

8.2 Examples

If ω is a Kähler form and $u : X \to \mathbb{R}$ is a $\mathcal{C}^2$-smooth function, then its curvature form $dd^c u$ is bounded below by $-A\omega$ for some $A > 0$ large enough (which linearly depends on $\|u\|_{\mathcal{C}^2}$). Up to rescaling, this shows that any smooth function is ω-psh.

The set $PSH(X,\omega)$ also contains singular functions. The local model of singular behavior is that of psh functions, the ω-psh condition however also encodes a global information which restricts the possible type of singularities. This is well illustrated in the global model projective case when $X = \mathbb{P}^n$ is the complex projective space and $\omega = \omega_{\mathrm{FS}}$ is the Fubini–Study Kähler form.

8.2.1 Positive metrics of holomorphic lines bundles

8.2.1.1 The Lelong class The most fundamental example which may serve as a guide to everything that follows is the case where $X = \mathbb{CP}^n$ is the complex projective space and $\omega = \omega_{\mathrm{FS}}$ is the Fubini–Study Kähler form. There is then a 1-to-1 correspondence between $PSH(\mathbb{CP}^n, \omega_{\mathrm{FS}})$ and the Lelong class

$$\mathcal{L}(\mathbb{C}^n) := \left\{ \psi \in PSH(\mathbb{C}^n) |\ \psi(z) \leq \frac{1}{2}\log[1+|z|^2] + C_\psi \right\}$$

which is given by the natural mapping

$$\mathcal{L}(\mathbb{C}^n) \ni \psi \longmapsto \varphi(x) = \begin{cases} \psi(x) - \frac{1}{2}\log[1+|x|^2], & \text{if } x \in \mathbb{C}^n, \\ \overline{\lim}_{y\in\mathbb{C}^n \to x}(\psi(y) - \frac{1}{2}\log[1+|y|^2]), & \text{if } x \in H_\infty, \end{cases}$$

where H_∞ denotes the hyperplane at infinity. One can easily show that this mapping is bicontinuous for the L^1_{loc}-topology.

8.2.1.2 More general line bundles Recall that a holomorphic line bundle L on X is a family of complex lines $\{L_x\}_{x\in X}$ together with a structure of complex manifold of dimension $1 + \dim_{\mathbb{C}} X$ such that the projection map $\pi : L \to X$ taking L_x to x is holomorphic. Moreover, one can always locally trivialize L: there exist an open covering $\{\mathcal{U}_\alpha\}$ of X and biholomorphisms $\Phi_\alpha : \pi^{-1}(\mathcal{U}_\alpha) \to \mathcal{U}_\alpha \times \mathbb{C}$ which map $L_x = \pi^{-1}(x)$ isomorphically onto $\{x\} \times \mathbb{C}$. The line bundle L is then uniquely (i.e., up to isomorphism) determined by its transition functions $g_{\alpha\beta} \in \mathcal{O}^*(\mathcal{U}_{\alpha\beta})$, $\mathcal{U}_{\alpha\beta} := \mathcal{U}_\alpha \cap \mathcal{U}_\beta$, where

$$g_{\alpha\beta} := (\Phi_\alpha \circ \Phi_\beta^{-1})|_{\{x\}\times\mathbb{C}}.$$

Note that the $g_{\alpha\beta}$'s satisfy the cocycle condition $g_{\alpha\beta} \cdot g_{\beta\gamma} \cdot g_{\gamma\alpha} \equiv 1$, hence define a class $[\{g_{\alpha\beta}\}] \in H^1(X, \mathcal{O}^*)$. The first Chern class of L is the image $c_1(L) \in H^2(X, \mathbb{Z})$ of $[\{g_{\alpha\beta}\}]$ under the mapping $c_1 : H^1(X, \mathcal{O}^*) \to H^2(X, \mathbb{Z})$ induced by the exponential short exact sequence $0 \to \mathbb{Z} \to \mathcal{O} \to \mathcal{O}^* \to 0$.

We let $\Gamma(X, L)$ denote the set of holomorphic sections of L on X: $s \in \Gamma(X, L)$ is a collection $s = \{s_\alpha\}$ of holomorphic functions s_α on $\mathcal{U}_\alpha$ satisfying the compatibility condition $s_\alpha = g_{\alpha\beta} s_\beta$ on $\mathcal{U}_{\alpha\beta}$. Similarly:

Definition 8.7. *A (singular) metric ψ of L on X is a collection $\psi = \{\psi_\alpha\}$ of functions $\psi_\alpha \in L^1(\mathcal{U}_\alpha)$ satisfying $\psi_\alpha = \psi_\beta + \log|g_{\alpha\beta}|$ in $\mathcal{U}_{\alpha\beta}$. The metric is said to be smooth if the ψ'_αs are $\mathcal{C}^\infty$-smooth functions. A smooth metric always exists. The metric ψ is said to be positive*[1] *if the ψ_α's are psh functions.*

In particular, if $s = \{s_\alpha\}$ is a holomorphic section of L on X, then $\psi = \{\psi_\alpha := \log|s_\alpha|\}$ is a positive (singular) metric of L on X. Note that we make here a slight abuse of terminology: differential geometers usually call "metric" the non-negative (usually smooth and non-vanishing) quantities $e^{-\psi} = \{e^{-\psi_\alpha}\}$.

Given a (singular) metric $\psi = \{\psi_\alpha\}$ of L on X, we consider its curvature $\Theta_\psi := dd^c\psi_\alpha$ in $\mathcal{U}_\alpha$. This yields a globally well-defined real closed current on X, since $dd^c \log|g_{\alpha\beta}| = 0$ in $\mathcal{U}_{\alpha\beta}$. It is a standard consequence of de Rham's isomorphism that this current represents the image of the first Chern class of L under the mapping $i : H^2(X, \mathbb{Z}) \to H^2(X, \mathbb{R})$ (induced by the inclusion $i : \mathbb{Z} \to \mathbb{R}$). The line bundle L is said to be *positive* if it admits a smooth metric whose curvature is a Kähler form.

[1] In the french weak sense: the metric is here said to be positive if its curvature current $dd^c\psi$ is (weakly) positive. One should perhaps say non-negative.

Assume L is *positive*, fix a smooth positive metric $h = \{h_\alpha\}$ of L on X, and set $\omega := \Theta_h$. Then $PSH(X,\omega)$ is in 1-to-1 correspondence with the set of positive singular metrics of L on X. Indeed, if ψ is such a metric, then $\varphi := \psi - h$ is globally well defined on X and such that $dd^c\varphi \geq -\omega$. Conversely, if $\varphi \in PSH(X,\omega)$, then $\psi = \{\psi_\alpha := \varphi + h_\alpha\}$ defines a positive singular metric of L on X.

Since L is an ample line bundle (by Kodaira's theorem), a multiple L^k defines embeddings

$$\Phi_k : X \ni x \longmapsto [s_0(x) : \ldots : s_{N_k}(x)] \in \mathbb{P}^{N_k},$$

where $\{s_0, \ldots, s_{N_k}\}$ denotes a basis of $H^0(X, L^k)$. In this setting, L^k can be identified with the restriction to $\Phi_k(X)$ of the hyperplane bundle $\mathcal{O}_{\mathbb{P}^{N_k}}(1)$. Any ω_{FS}-psh function on $\mathbb{P}^{N_k}$ gives thus rise, by restriction and pullback, to an ω-psh function on X. In the next section we address the converse problem: is any ω-psh function obtained that way?

8.2.2 Extension property In this section we assume X is an analytic submanifold of a compact Kähler manifold (V,ω). Every ω-psh function φ on V yields, by restriction, an $\omega|_X$-psh function $\varphi|_X$ on X, provided $\varphi|_X \not\equiv -\infty$. The question we address here is whether this restriction operator is surjective. In other words, is there equality

$$PSH(X,\omega|_X) \stackrel{?}{=} PSH(V,\omega)|_X.$$

We make here the elementary observation that smooth strictly ω-psh functions can easily be extended.

Proposition 8.8. *Let V be a compact Kähler manifold equipped with a Kähler form ω, and let X be a complex submanifold of V. Then*

$$PSH^+(X,\omega|_X) \cap \mathcal{C}^\infty(X,\mathbb{R}) = \big(PSH^+(V,\omega) \cap \mathcal{C}^\infty(V,\mathbb{R})\big)|_X.$$

We let here $PSH^+(V,\omega)$ denote the set of strictly ω-psh functions φ, i.e.,

$$PSH^+(V,\omega) := \bigcup_{0<\varepsilon<1} PSH^+(V,(1-\varepsilon)\omega).$$

Proof. Let $\varphi \in \mathcal{C}^\infty(X,\mathbb{R})$ be such that $(1-\varepsilon)\omega|_X + dd^c\varphi \geq 0$ on X, for some $\varepsilon > 0$. We first choose $\tilde{\varphi}$ to be any smooth extension of φ to V. Consider

$$\psi := \tilde{\varphi} + A\chi\,\mathrm{dist}(\cdot,X)^2,$$

where χ is a test function supported in a small neighborhood of X and such that $\chi \equiv 1$ near X. Here dist is any Riemannian distance on V, for instance the distance associated to the Kähler metric ω. Then ψ is yet another smooth extension of φ to V, which now satisfies $(1-\varepsilon/2)\omega + dd^c\psi \geq 0$ near X, if A is chosen large enough.

The function $\log(\mathrm{dist}(\cdot,X)^2)$ is well defined and qpsh in a neighborhood of X. We can assume that χ is supported in this neighborhood. The function $u = \chi \log(\mathrm{dist}(\cdot,X)^2)$ is $N\omega$-psh on V for a large integer N. Moreover, $\exp(u)$ is smooth and

$X = \{u = -\infty\}$. Replacing ω by $N\omega$ and φ by $N\varphi$, we may assume that $N = 1$. Now set

$$\psi_C := \frac{1}{2}\log\left[e^{2\psi} + e^{u+C}\right].$$

This again is a smooth extension of φ, and a straightforward computation yields

$$dd^c\psi_C \geq \frac{2e^{2\psi}\, dd^c\psi + e^{u+C}\, dd^c u}{2(e^{2\psi} + e^{u+C})}.$$

Hence,

$$\left(1 - \frac{\varepsilon}{2}\right)\omega + dd^c\psi_C \geq \frac{2e^{2\psi}\left[\left(1 - \frac{\varepsilon}{2}\right)\omega + dd^c\psi\right] + (1-\varepsilon)e^{u+C}\omega}{2(e^{2\psi} + e^{u+C})} \geq 0,$$

if C is chosen large enough. □

This proof breaks down when φ is singular and hence a different approach is needed. The general case is actually an open question, the interested reader will find in [CGZ13] an extension result in the particular case when ω is a Hodge form, i.e., (strictly) positive (singular) metrics of holomorphic line bundles do extend, and in [CT14] an extension result for Kähler currents with analytic singularities.

8.2.3 Approximation by holomorphic sections Let $L \to X$ be a *positive* holomorphic line bundle on a compact projective manifold X. If $s \in \Gamma(X, L^N)$ is a holomorphic section, $N \in \mathbb{N}^*$, then $N^{-1}\log|s|$ defines a singular positive metric of L on X. The purpose of this section is to show that this set of special metrics is actually dense in the set of all positive singular metrics, for the L^1-topology:

Theorem 8.9. *Let $L \to X$ be a positive holomorphic line bundle on a compact projective manifold X, equipped with a singular positive metric ψ. Then there exist $N_j \in \mathbb{N}^{\mathbb{N}}$ and $S_j \in \Gamma(X, L^{N_j})$ such that, as $j \to +\infty$,*

$$\frac{1}{N_j}\log|S_j| \xrightarrow{L^1} \psi.$$

Proof. Let h be a smooth positive metric of L on X with Kähler curvature form ω. Let $\varphi \in PSH(X,\omega)$. We can assume w.l.o.g. that $\varphi \leq 0$ on X. We let $\psi = \{\psi_\alpha := \varphi + h_\alpha \in PSH(\mathcal{U}_\alpha)\}$ denote the associated (singular) positive metric of L on X, where $\{\mathcal{U}_\alpha\}$ denotes an open cover of X trivializing L.

We consider the Bergman spaces

$$\mathcal{H}_{j,j_0} := \left\{s \in \Gamma(X, L^j) \,|\, \int_X |s|^2 e^{-2h_{j,j_0}}\, dV_\omega < +\infty\right\},$$

where $h_{j,j_0} = (j - j_0)\psi + j_0 h$, j_0 a fixed large integer (to be specified later). Let $\sigma_1^{(j,j_0)}, \ldots, \sigma_{s_j}^{(j,j_0)}$ be an orthonormal basis of $\mathcal{H}_{j,j_0}$ and set

$$\psi_{j,j_0} := \frac{1}{2j} \log \left[\sum_{l=1}^{s_j} \left| \sigma_l^{(j,j_0)} \right|^2 \right] = \frac{1}{2j} \sup_{s \in B_{j,j_0}} \log |s|^2,$$

where B_{j,j_0} denotes the unit ball of radius 1 centered at 0 in $\mathcal{H}_{j,j_0}$. Clearly, ψ_{j,j_0} defines a positive (singular) metric of L on X, equivalently $\varphi_{j,j_0} := \psi_{j,j_0} - h \in PSH(X, \omega)$. If $x \in \mathcal{U}_\alpha$ and $s = \{s_\alpha\} \in \mathcal{H}_{j,j_0}$, then $|s_\alpha|^2$ is subharmonic in $\mathcal{U}_\alpha$, hence

$$|s_\alpha(x)|^2 \le \frac{C_1}{r^{2n}} \int_{B(x,r)} |s_\alpha(x)|^2 \le \frac{C_2}{r^{2n}} e^{2 \sup_{B(x,r)} h_{j,j_0}} \int_X |s|^2 e^{-2h_{j,j_0}} \, dV_\omega,$$

where $r > 0$ is so small that $B(x, r) \subset \mathcal{U}_\alpha$. We infer

$$\varphi_{j,j_0}(x) \le (1 - j_0/j) \sup_{B(x,r)} \varphi + \frac{C_3 - n \log r}{j}. \tag{8.2.1}$$

There is also a reverse inequality which uses a deep extension result of Ohsawa and Takegoshi (see [Bern10]): there exist $j_0 \in \mathbb{N}$ and $C_4 > 0$ large enough so that $\forall x \in X$, $\forall j \in \mathbb{N}$, there exists $s \in \Gamma(X, L^j)$ with

$$\int_X |s|^2 e^{-2h_{j,j_0}} \, dV_\omega \le C_4 |s(x)|^2 e^{-2h_{j,j_0}(x)}.$$

Choose s so that the right-hand side is equal to 1, hence $s \in B_{j,j_0}$. Then

$$\psi_{j,j_0}(x) \ge \frac{1}{2j} \log |s(x)|^2 = \left(1 - \frac{j_0}{j}\right) \psi(x) + \frac{j_0}{j} h(x) - \frac{\log C_4}{2j}.$$

It follows that

$$\varphi_{j,j_0}(x) \ge \left(1 - \frac{j_0}{j}\right) \varphi(x) - \frac{\log C_4}{2j} \ge \varphi(x) - \frac{\log C_4}{2j}, \tag{8.2.2}$$

since $\varphi \le 0$ on X.

It follows from previous inequalities that $\varphi_j \to \varphi$ in $L^1(X)$. We can now approximate φ_j by $\tilde{\varphi}_j = j^{-1} \log |S_{\theta,j}|$, where $S_{\theta,j} \in \Gamma(X, L^j)$ is defined by

$$S_{\theta,j} = \sum_{l=1}^{s_j} e^{i\theta_l} \sigma_l^{(j,j_0)},$$

by choosing $\theta = (\theta_1, \ldots, \theta_{s_j}) \in \mathbb{R}^{s_j}$ conveniently. Observe indeed that

$$\frac{1}{(2\pi)^{s_j}} \int_{[0,2\pi]^{s_j}} |S_{\theta,j}|^2 \, d\theta = \sum_{l=1}^{s_j} \left| \sigma_l^{(j,j_0)} \right|^2. \qquad \square$$

8.3 Integrability properties

Observe that qpsh functions are in $L^p(X)$ for every $p \geq 1$, since their local psh counterparts are in L^p_{loc}. Similarly, qpsh functions have gradient in L^r for every $r < 2$.

We have established in Chapter 2 a uniform version of Skoda's integrability theorem. We apply it to families of qpsh functions and discuss precise weighted estimates.

8.3.1 Skoda's integrability theorem

8.3.1.1 Lelong numbers of qpsh functions Recall that if u is a plurisubharmonic function defined in a neighborhood of the origin in $\mathbb{C}^n$, its Lelong number at the origin can be defined by

$$\nu(u,0) := \lim_{r\to 0^+} \frac{\sup_{\mathbb{B}(r)} u}{\log r} = \sup\{\gamma \geq 0 \,|\, u(z) \leq \gamma \log \|z\| + O(1)\}.$$

An important result of Siu ensures that this notion is invariant under biholomorphic changes of coordinates (see Chapter 2). Note that $\nu(u,0) = 0$ if u is smooth, hence one can define the Lelong number of a qpsh function on any complex manifold. In the sequel $\nu(\varphi, x)$ will denote the Lelong number of a qpsh function φ at a given point $x \in X$.

Lemma 8.10. *The number*

$$\nu(\{\omega\}) := \sup\{\nu(\varphi, x) \,|\, x \in X \ \text{ and } \ \varphi \in PSH(X,\omega)\} < +\infty$$

is finite and only depends on the cohomology class of ω.

We let the reader check that $\nu(\{\omega_{\mathrm{FS}}\}) = 1$ when $X = \mathbb{CP}^n$ and $\omega = \omega_{\mathrm{FS}}$ is the Fubini–Study Kähler form.

Proof. Assume first that ω is a Kähler form and let d denote the Riemannian distance associated to ω. Given $a \in X$, we can find a cut-off function $\chi \in \mathcal{C}^\infty(X, \mathbb{R})$ with support near a and which is identical to 1 in a smaller neighborhood of a such that

$$g_a(x) := \chi(x) \log d(x,a)$$

is $\mathcal{C}^\infty$-smooth on $X \setminus \{a\}$. Observe that g_a is psh in a neighborhood of a, thus g_a is qpsh and even $B\omega$-psh for some $B > 0$.

An easy compactness argument allows to find an absolute $B = B(\omega)$ independent of point a, and quasi-smooth $B\omega$-psh functions g_a as above.

Given $\varphi \in PSH(X,\omega)$, it follows from the local theory of complex Monge–Ampère operators that the measures $\omega_\varphi \wedge (B\omega + dd^c g_a)^{n-1}$ are well-defined for all $a \in X$. They moreover have a Dirac mass with multiplicity $\nu(\varphi, a)$ at the point a. Since the total mass can be computed in cohomology, we infer that

$$\nu(\varphi, a) \leq B^{n-1} \mathrm{Vol}_\omega(X).$$

When ω is not Kähler, we can find a Kähler form ω' such that $\omega \leq \omega'$ and observe that $PSH(X,\omega) \subset PSH(X,\omega')$, $\nu(\{\omega\}) \leq \nu(\{\omega'\})$ to conclude. □

8.3.1.2 Uniform integrability We now recall a powerful integrability result which goes back to Skoda [Sko72].

Theorem 8.11. *Fix $\varphi \in PSH(X,\omega)$ and $A < 2[\sup_{x\in X} \nu(\varphi,x)]^{-1}$. Then $\exp(-A\varphi) \in L^1(X)$. Moreover, if $A < 2\nu(\{\omega\})^{-1}$, then*

$$\sup\left\{\int_X e^{-A\varphi} dV \,|\, \varphi \in PSH(X,\omega) \text{ and } \sup_X \varphi = 0\right\} < +\infty.$$

This is a consequence of Theorem 2.50. This result has been reproved and used in the context of Kähler–Einstein geometry in [Tian87] (Tian only considers smooth ω-psh functions, but it follows from Demailly's regularization result and lower semi-continuous properties of these integrability exponents that the two definitions agree).

On the other hand it easily follows from the convexity of the exponential and the above definition that $e^{-A\varphi}$ is not locally integrable near a point x such that $\nu(\varphi,x) \geq 2n/A$ (see Exercise 8.6). If φ has some positive Lelong number, it therefore follows that

$$\alpha(\varphi) := \sup\{A > 0 \,|\, e^{-A\varphi} \in L^1(X)\}$$

is both finite and positive.

The map $\varphi \mapsto \alpha(\varphi)$ is lower semi-continuous, as follows from the work of Phong and Sturm [PS00] and Demailly and Kollar [DK01]. It turns out that $e^{-\alpha(\varphi)\varphi}$ is not integrable: this is a delicate problem (known for a decade as the openness conjecture) which has been solved recently (see [FJ05, Bern13, GZhou15b] and Exercise 8.7).

Remark 8.12. *As the proof of Theorem 2.50 shows, the uniform control holds for any compact subfamily of qpsh functions, as long as we have a uniform control on the Lelong numbers.*

8.3.1.3 Log canonical thresholds

Definition 8.13. *We set*

$$\alpha(\omega) := \inf\{\alpha(\varphi) \,|\, \varphi \in PSH(X,\omega)\}.$$

The number $\alpha(\omega)$ is often called "Tian's α-invariant" in the literature. It only depends on the cohomology class of the form ω. It follows from Theorem 8.11 that

$$\alpha(\omega) = \sup\left\{A > 0 \,|\, \exists C > 0,\ \forall \varphi \in PSH_0(X,\omega), \int_X e^{-A\varphi} dV \leq C\right\},$$

where $PSH_0(X,\omega)$ is the set of ω-psh functions that are normalized by $\sup_X \varphi = 0$ and dV is an arbitrary smooth volume form.

Assume now that ω belongs to $c_1(L)$, the first Chern class of a semi-ample line bundle. and h is a smooth metric of L with curvature ω. We set $\alpha(L) := \alpha(\omega)$ (as noted above, the definition of α only depends on the cohomology class of ω, i.e., $c_1(L)$).

If $D = (s = 0)$ is the zero set of a holomorphic section of L^k, $k \in \mathbb{N}^*$, then $\varphi := k^{-1} \log |s|_{kh} \in PSH(X, \omega)$ and we can consider the corresponding integrability index

$$c(k^{-1} \log |s|_{kh}) =: \mathrm{lct}\left(\frac{1}{k} D\right).$$

This number, the *log canonical threshold* of the $\mathbb{Q}$-divisor $\frac{1}{k} D$, can also be defined in a purely algebraic way using multiplier ideals (see [Laz, Section 9.3.B]) and plays an important role in algebraic geometry (see e.g., [Chel08, CS08]).

The corresponding uniform integrability indices

$$\alpha_k(L) = \inf\{c(k^{-1} \log |s|_{kh}) \mid s \in H^0(X, L^k)\}$$

satisfy

$$\alpha(L) = \inf_{k \in \mathbb{N}^*} \alpha_k(L),$$

as follows from the Ohsawa–Takegoshi extension theorem (see the Appendix by Demailly in [CS08]).

When $L = K_X^{-1}$ is the anticanonical bundle of a Fano manifold, the invariant is denoted by $\alpha(X)$ and its values plays an important role in the Kähler–Einstein problem as we shall see in the sequel.

8.3.2 Gradient of a qpsh function We assume here for simplicity that ω is a Kähler form. The reader will easily extend the following a priori estimates to a slightly more general context.

Proposition 8.14. *Fix $0 < \alpha < 1$ and $\varphi \in PSH(X, \omega)$ such that $\varphi \leq -1$. Then*

$$\int_X \frac{d\varphi \wedge d^c\varphi}{(-\varphi)^{2-\alpha}} \wedge \omega^{n-1} \leq \frac{\mathrm{Vol}_\omega(X)}{\alpha(1-\alpha)}.$$

In particular, $\nabla\varphi \in L^r(X)$ for every $r < 2$ with

$$\|\nabla\varphi\|_{L^r(X)} \leq \left(\frac{n\mathrm{Vol}_\omega(X)}{\alpha(1-\alpha)}\right)^{\frac{1}{2}} \cdot \left(\int_X |\varphi|^{\frac{r(2-\alpha)}{2-r}} \omega^n\right)^{\frac{2-r}{2r}}.$$

Here $\nabla\varphi = \nabla_\omega\varphi$ denotes the gradient of φ with respect to the Kähler metric ω; it satisfies

$$|\nabla_\omega\varphi|^2 = n\frac{d\varphi \wedge d^c\varphi \wedge \omega^{n-1}}{\omega^n}.$$

Proof. Set $\varphi_\alpha := -(-\varphi)^\alpha$. We first check that φ_α belongs to $PSH(X,\omega)$, by computing

$$dd^c\varphi_\alpha = \alpha(1-\alpha)(-\varphi)^{\alpha-2}d\varphi\wedge d^c\varphi + \alpha(-\varphi)^{\alpha-1}dd^c\varphi,$$

thus

$$\omega_{\varphi_\alpha} = \alpha(1-\alpha)(-\varphi)^{\alpha-2}d\varphi\wedge d^c\varphi + \alpha(-\varphi)^{\alpha-1}\omega_\varphi + \left[1-\frac{\alpha}{(-\varphi)^{1-\alpha}}\right]\omega \geq 0.$$

We note moreover that $\alpha(1-\alpha)(-\varphi)^{\alpha-2}d\varphi\wedge d^c\varphi \leq \omega_{\varphi_\alpha}$, hence

$$\alpha(1-\alpha)\int_X \frac{d\varphi\wedge d^c\varphi}{(-\varphi)^{2-\alpha}}\wedge\omega^{n-1} \leq \int_X(\omega+dd^c\varphi_\alpha)\wedge\omega^{n-1} = \mathrm{Vol}_\omega(X),$$

as claimed.

The last inequality follows from Hölder's inequality. All these computations can be done first for smooth functions and then by approximation. □

Observe that bounded functions have gradient in L^2: if φ is a bounded ω-psh function, we can assume, up to translating and rescaling, that φ takes values in $[0,1/2]$, so that

$$\omega + dd^c(\varphi^2) = 2d\varphi\wedge d^c\varphi + 2\varphi\omega_\varphi + (1-2\varphi)\omega \geq 2d\varphi\wedge d^c\varphi.$$

This shows that φ^2 is ω-psh and moreover $\|\nabla\varphi\|^2_{L^2} \leq \mathrm{Vol}_\omega(X)/2$.

We observe, more generally, that ω-psh functions with confined singularities automatically have gradient in L^2:

Proposition 8.15. *Let $D\subset X$ be an ample divisor. If a qpsh function φ is bounded near D, then $\nabla\varphi\in L^2(X)$.*

Proof. Let ω_D be a Kähler form cohomologous to $[D]$, the current of integration along D. Replacing D by a multiple of it and ω by ω_D, we can assume φ is ω-psh, $\varphi\leq 0$, and $[D]=\omega+dd^c\psi_D$, where the ω-psh function ψ_D has poles along D and e^{ψ_D} is smooth.

Fix a neighborhood V of D so that φ is bounded near $\overline{V}$. Let $\chi:\mathbb{R}\to\mathbb{R}$ be a smooth convex increasing function such that $\chi\equiv -1$ near $-\infty$ and $\chi(t)\equiv t$ on $\mathbb{R}^+$. Observe that the smooth form

$$\omega_C := \omega + dd^c\chi\circ(\psi_D + C) \geq 0$$

is semi-positive and satisfies $\omega_C\equiv 0$ in $X\setminus V$ if C is large enough.

Assume now that φ is smooth. Writing $\omega_\varphi=\omega+dd^c\varphi$, we obtain

$$\begin{aligned}\int_X d\varphi\wedge d^c\varphi\wedge\omega^{n-1} &\leq \int_X(-\varphi)\omega_\varphi\wedge\omega^{n-1}\\ &= \int_V(-\varphi)\omega_\varphi\wedge\omega_C\wedge\omega^{n-2} - \int_X(-\varphi)\omega_\varphi\wedge dd^c\chi\circ(\psi_D+C)\wedge\omega^{n-2}\end{aligned}$$

The last but one integral is finite since φ is bounded in $\overline{V}$. The last one is also under control since $u = \chi \circ (\psi_D + C) - M$ is negative for M large enough (and bounded from below by $-M - 1$), hence

$$\int_X \varphi\, \omega_\varphi \wedge d\,d^c u \wedge \omega^{n-2} = \int_X (-u)\,(-d\,d^c\varphi) \wedge \omega_\varphi \wedge \omega^{n-2} \leq (M+1)\mathrm{Vol}_\omega(X).$$

We have thus obtained the uniform bound

$$\int_X d\varphi \wedge d^c\varphi \wedge \omega^{n-1} \leq \left[\|\varphi\|_{L^\infty(V)} + M + 1\right] \mathrm{Vol}_\omega(X).$$

When φ is not smooth, we can approximate it by a decreasing sequence (φ_j) of smooth ω-psh functions. These functions belong to a fixed ball of the Sobolev space

$$W^{1,2} := \{u \in L^2(X) |\ \nabla u \in L^2(X)\}.$$

Any cluster point in $W^{1,2}$ is also a cluster point in the weak sense of distributions, hence it has to be φ, so that $\nabla\varphi \in L^2(X)$. □

8.4 Regularization

It is easy to approximate a given ω-psh function φ by a decreasing sequence of less singular ω-psh functions. We have used above attenuation of singularities, such as

$$\varphi_\alpha := -(-\varphi)^\alpha,$$

where $0 < \alpha < 1$ and φ is assumed to be normalized so that $\varphi \leq -1$. Letting α increase to 1 yields a decreasing family of ω-psh functions which are less singular (for example, these have zero Lelong numbers) but still have the same polar set as that of φ,

$$\{\varphi_\alpha = -\infty\} = \{\varphi = -\infty\}.$$

One can actually also approximate φ by the following sequence of *bounded* ω-psh functions:

$$\varphi_j := \max(\varphi, -j).$$

It is more delicate to find a decreasing sequence of *continuous* ω-psh approximants. Chapter 13 is devoted to this issue.

Passing from *continuous* to *smooth* approximants can be done using an old observation of Richberg [Rich68].

8.4.1 Elementary smoothing The purpose of this section is to explain how one can approximate a given ω-psh function by a decreasing sequence of $\mathcal{C}^\infty$-smooth ω-psh functions, following the elementary approach in [BK07].

We assume here and in the next section that ω is a Kähler form.

Proposition 8.16. *Fix $\varphi \in PSH(X,\omega)$. Then there exists smooth ω-psh functions $\varphi_j \in PSH(X,\omega) \cap \mathcal{C}^\infty(X)$ which decrease towards φ.*

Proof. We can assume without loss of generality that φ is bounded (replacing φ by $\max(\varphi, -j)$) and $\varphi \leq 0$ (translating if necessary). Replacing then φ by $\varphi/(1+j)$ (which decrease to φ since $\varphi \leq 0$), we can further assume that φ is actually $(1-\varepsilon)\omega$-psh for some $\varepsilon > 0$ (i.e., belongs to $PSH^+(X,\omega)$).

Fix two coverings V_α, U_α by biholomorphic images of balls in $\mathbb{C}^n$, such that $V_\alpha \subset U_\alpha$. Let h_α denote a local potential of ω in U_α, $\omega = dd^c h_\alpha$. Then the function $u_\alpha := \varphi + (1-\varepsilon)h_\alpha$ is psh in U_α and we can regularize it by convolution (looking first at the image under a biholomorphic map F to $\mathbb{C}^n$, then using convolution, then pulling back by F^{-1} to U_α). We let $u_{\alpha,\delta}$ denote the resulting functions, obtained by using convolutions in balls of radius δ.

We let the reader check in Exercise 8.16 that, as $\delta \to 0$,

$$u_{\alpha,\delta} - u_{\beta,\delta} - (1-\varepsilon)(h_\alpha - h_\beta) \longrightarrow 0 \quad \text{locally uniformly in} \quad U_\alpha \cap U_\beta. \tag{8.4.1}$$

Let χ_α be a smooth cut-off function such that $\chi_\alpha \equiv 0$ in V_α and $\chi_\alpha \equiv -1$ near ∂U_α. Observe that $dd^c\chi_\alpha \geq -C\omega$ for some constant $C > 0$ which only depends on the size of the coverings, since we have assumed that ω is Kähler. Set

$$\varphi_\delta := \max_\alpha \left\{ u_{\alpha,\delta} - (1-\varepsilon)h_\alpha + \varepsilon\chi_\alpha/C \right\}.$$

This is a maximum of ω-psh functions in each set U_α, hence it is ω-psh there. If $\delta > 0$ is small enough, it follows from (8.4.1) that $u_{\alpha,\delta} - (1-\varepsilon)h_\alpha + \varepsilon\chi_\alpha/C$ does not contribute to the maximum when $\chi_\alpha \equiv -1$. This ensures that φ_δ is globally ω-psh and continuous.

Observe that $u_{\alpha,\delta}$ decreases to u_α, hence φ_δ decreases to φ, as δ decreases to zero. To obtain a smooth approximation, it suffices to replace the max by a regularized maximum. □

8.4.2 Demailly estimates Recall that if u is a psh function in a domain in $\mathbb{C}^n$, then a convolution with a radial smoothing kernel preserves positivity of $dd^c u$. For a non-flat Kähler metric ω on a compact manifold X, this may not be the case any longer.

However, an approximation technique due to Demailly allows to control the negative part of the smooth form. It is described in detail in [Dem82, Dem94]. Here we shall briefly highlight its main features. Consider the exponential mapping from the tangent space of a given point $z \in X$,

$$\exp_z : T_z X \ni \zeta \longmapsto \exp_z(\zeta) \in X,$$

which is defined by $\exp_z(\zeta) = \gamma(1)$ with γ being the geodesic[2] starting from z with initial velocity $\gamma'(0) = \zeta$. Given any function $u \in L^1(X)$, we define its δ-regularization $\rho_\delta u$ to be

$$\rho_\delta u(z) = \frac{1}{\delta^{2n}} \int_{\zeta \in T_z X} u(\exp_z(\zeta)) \rho\left(\frac{|\zeta|^2_\omega}{\delta^2}\right) dV_\omega(\zeta), \ \delta > 0, \tag{8.4.2}$$

according to [Dem82]. Here ρ is a smoothing kernel,

$$|\zeta|^2_\omega = \sum_{i,j=1}^{n} g_{i\bar{j}}(z)\zeta_i \bar{\zeta}_j,$$

and $dV_\omega(\zeta)$ is the induced measure $\frac{1}{2^n n!}(dd^c|\zeta|^2_\omega)^n$.

This may be formally extended as a function on $X \times \mathbb{C}$ by putting

$$U(z,w) := \rho_\delta u(z) \text{ for } w \in \mathbb{C}, \ |w| = \delta.$$

The introduction of the variable w is convenient for an application of Kiselman's minimum principle (see Theorem 2.44) to that function.

The estimates obtained in [Dem82] show that all results of [Dem94] and [BD12] are still valid with the unmodified definition of $\rho_\delta u$, at least when (X, ω) is Kähler. By [Dem82, Lemma 8.2], the exponential function

$$\exp : TX \longrightarrow X, \quad TX \ni (z,\zeta) \longmapsto \exp_z(\zeta) \in X, \ \zeta \in T_z X$$

satisfies the following properties:

(1) exp is a $\mathcal{C}^\infty$ smooth mapping;
(2) $\forall z \in X$, $\exp_z(0) = z$ and $d_\zeta \exp(0) = \mathrm{Id}_{T_z X}$;
(3) $\forall z \in X$, the mapping $\zeta \longrightarrow \exp_z \zeta$ has a third-order Taylor expansion at $\zeta = 0$ of the form

$$\left| \exp_z(\zeta)_m - z_m - \zeta_m - \frac{1}{2} \sum_{j,k,l} R_{j\bar{k}l\bar{m}}\left(\bar{z}_k + \tfrac{1}{3}\bar{\zeta}_k\right)\zeta_j \zeta_l \right| \leq C(|\zeta|^2(|z| + |\zeta|)^2), \tag{8.4.3}$$

$|\zeta| < r$, for small enough $r > 0$. The expansion is valid in holomorphic normal coordinates with respect to the Kähler metric.

It is convenient to select a particular smoothing kernel, namely

$$\rho(t) = \begin{cases} \frac{\eta}{(1-t)^2} \exp(\frac{1}{t-1}), & \text{if } 0 \leq t \leq 1, \\ 0, & \text{if } t > 1, \end{cases}$$

with a suitable constant η, such that $\int_{\mathbb{C}^n} \rho(|z|^2)\, dV(z) = 1$ (dV denotes the Lebesgue measure in $\mathbb{C}^n$).

[2] corresponding to the Riemannian metric associated to the Kähler form ω.

The crucial estimate of the Hessian of $U(z,w)$ given in [Dem94, Proposition 3.8] (see also [Dem82, Proposition 8.5]), coupled with Kiselman's theorem (see Theorem 2.44) provide a lemma stated in this form in [BD12, Lemma 1.12]:

Lemma 8.17. *Fix $u \in PSH(X,\omega)$ and let ρ_δ denote its regularization as defined above. Define the Kiselman–Legendre transform with level c by*

$$u_{c,\delta} := \inf_{0\le t\le\delta}\Big[\rho_t u(z) + Kt^2 - K\delta^2 - c\log\Big(\frac{t}{\delta}\Big)\Big]. \tag{8.4.4}$$

Then for some positive constant K depending on the curvature, the function $\rho_t u(z) + Kt^2$ is increasing in t and one has

$$\omega + dd^c u_{c,\delta} \ge -(A\min\{c,\lambda(z,\delta)\} + K\delta^2)\,\omega, \tag{8.4.5}$$

where A is a lower bound of the negative part of the bisectional curvature of ω, while

$$\lambda(z,t) := \frac{\partial}{\partial\log t}(\rho_t u(z) + Kt^2) \overset{t\to 0^+}{\longrightarrow} \nu(u,z).$$

These estimates are extremely useful. We encourage the reader to spend some efforts in checking them. They allow to regularize qpsh functions in a very broad context and have several applications. One is given hereafter.

8.4.3 Regularity of upper envelopes As an application of Demailly's estimates, we briefly sketch the proof of the following regularity result due to Berman and Demailly [BD12]:

Theorem 8.18. *For $h \in \mathcal{C}^2(X,\mathbb{R})$ we consider*

$$P(h)(x) := \sup\{\varphi(x) \,|\, \varphi \in PSH(X,\omega) \text{ and } \varphi \le h\}.$$

Then $P(h)$ is ω-plurisubharmonic and has bounded Laplacian. In particular, $P(h) \in \mathcal{C}^{1,\alpha}(X)$ for any $\alpha < 1$ and

$$(\omega + dd^c P(h))^n = \mathbf{1}_{\{P(h)=h\}}(\omega + dd^c h)^n.$$

Proof. We can assume without loss of generality that $h \ge 0$. Set $u := P(h)$. Note that the upper semi-continuity is lost in general when taking upper envelopes. Now u^* (the upper semi-continuous regularization of u) is ω-psh and satisfies $u^* \le h^* = h$, thus $u^* \le u$ and $u = u^*$, i.e., u is actually upper semi-continuous, hence ω-psh.

Our goal is to show that $\Delta u \in L^\infty(X)$ by establishing uniform bounds

$$-K' \le \frac{u(z) - \rho_t u(z)}{t^2} \le K, \tag{8.4.6}$$

as $t \to 0$. We use the Kiselman–Legendre transform

$$u_{c,\delta} := \inf_{0\le t\le\delta} \Big[\rho_t u(z) + Kt^2 - K\delta^2 - c\log(t/\delta)\Big],$$

with $c = K\delta^2/A$. Since $u = P(h) \le h$, we obtain

$$u_{c,\delta}(z) \le \rho_\delta u(z) \le \rho_\delta h(z) \le h(z) + B\delta^2,$$

where $B = \|\Delta h\|_{L^\infty(X)}$. It follows therefore that

$$V_\delta := \frac{u_{c,\delta} - B\delta^2}{1 + 2K\delta^2} \le u,$$

since $V_\delta \le h$ and V_δ is ω-psh by Lemma 8.17.

Let $t = t(z) \in]0, \delta]$ denote the time realizing the infimum in the definition of $u_{c,\delta}$. Using that $\rho_t u + Kt^2 \ge u$ and $(1 + 2K\delta^2)u \le u \le 0$, we infer

$$u(z) - K\delta^2 - \frac{K\delta^2}{A}\log\left(\frac{t}{\delta}\right) - B\delta^2 \le u(z),$$

hence

$$\delta\exp\left(-\frac{A(K+B)}{K}\right) \le t(z) \le \delta.$$

Therefore,

$$\frac{\rho_t u(z) + Kt^2 - K\delta^2 - c\log(t/\delta) - B\delta^2}{1 + 2K\delta^2} \le u \le \rho_t u + Kt^2,$$

which yields (8.4.6).

It follows that the complex Monge–Ampère measure $(\omega + dd^c P(h))^n$ is absolutely continuous with respect to Lebesgue measure and concentrated on the contact set $\{P(h) = h\}$. We refer the reader to Exercise 5.10 for more details on this last assertion. □

Remark 8.19. *When ω is merely semi-positive (or when the cohomology class of ω is even merely big), a similar result holds on the ample locus of $\{\omega\}$.*

8.5 Exercises

Exercise 8.1. *Consider, on $X = \mathbb{P}^n$, the $j\omega_{\mathrm{FS}}$-psh functions*

$$\varphi_j[z_0 : \cdots : z_n] = j\log\|(z_1, \ldots, z_n)\| - j\log\|z\|.$$

Check that they satisfy $\sup_X \varphi_j = 0$, but do not form a relatively compact sequence. Is this in contradiction with Proposition 8.5?

Exercise 8.2. *Let φ be a plurisubharmonic function in $\mathbb{C}^n$.*

(1) *Let $u : \mathbb{R} \to \mathbb{R}$ be a non-constant convex increasing function. Show that* $\limsup_{x\to+\infty} u(x)/x > 0$.

(2) *Show that if* $\limsup_{z\to\infty} \varphi(z)/\log\|z\| \leq 0$ *then φ is constant.*

(3) *Assume now that* $\limsup_{z\to\infty} \varphi(z)/\log\|z\| = \gamma \in \mathbb{R}^+$. *Show that φ defines canonically a $\gamma\omega_{\mathrm{FS}}$-psh function on the complex projective space $\mathbb{P}^n$.*

(4) *Conversely, show that a $\gamma\omega_{\mathrm{FS}}$-psh function ψ on $\mathbb{P}^n$ induces a psh function φ on $\mathbb{C}^n$ such that* $\limsup_{z\to\infty} \varphi(z)/\log\|z\| \leq \gamma$. *Characterize, in term of the curvature current $\gamma\omega_{\mathrm{FS}} + dd^c\psi$, the equality*

$$\limsup_{z\to\infty} \varphi(z)/\log\|z\| = \gamma.$$

Exercise 8.3. *Let (X, ω_X) and (Y, ω_Y) be compact Kähler manifolds and assume $f : X \to Y$ is a holomorphic map. Show that*

$$PSH(X, f^*\omega_Y) = f^*PSH(Y, \omega_Y).$$

Exercise 8.4. *Let μ be a probability measure on X such that $\mu = \Theta + dd^c(S)$, where Θ is smooth and S is a positive current of bidimension $(1,1)$ on X.*

Show that $PSH(X, \omega) \subset L^1(\mu)$.

Exercise 8.5. *Show that if u is psh and locally bounded near the unit ball $\mathbb{B}$ of $\mathbb{C}^n$, it can be extended to $\mathbb{C}^n$ as a global psh function with logarithmic growth by considering*

$$U(z) := \begin{cases} u(z), & \text{if } z \in B, \\ \max(u(z), A\log^+|z| - \sup_B|u| - 1), & \text{if } z \in (1+\varepsilon)B \setminus B, \\ A\log^+|z| - \sup_B|u| - 1, & \text{if } z \in \mathbb{C}^n \setminus (1+\varepsilon)B, \end{cases}$$

where $\log^+|z| := \max(\log|z|, 0)$ *and with A large enough.*

Exercise 8.6. *Let φ be a qpsh function on a compact Kähler manifold (X, ω) such that $\nu(\varphi, x) \geq 2n$ for some point $x \in X$. Show that $e^{-\varphi} \notin L^1(X)$.*

Exercise 8.7. *Let φ be a quasi-subharmonic function on a compact Riemann surface X. Set*

$$c = c(\varphi) := \sup\{\gamma > 0 |\ \exp(-\gamma\varphi) \in L^1(X)\}.$$

Show that $\exp(-c\varphi) \notin L^1(X)$.

Show a similar result in higher dimension when φ has analytic singularities, i.e., assuming that φ can be locally decomposed as

$$\varphi = c\log\left[\sum_{i=1}^{s}|f_i|^2\right] + u,$$

where $c \geq 0$, u is smooth and the f_i's are (local) holomorphic functions.

Exercise 8.8. *Let $(a_j) \in \mathbb{C}^{\mathbb{N}}$ be a sequence converging to the origin and set*

$$\varphi(z) := \frac{3}{\pi^2} \sum_{j \geq 1} j^{-2} \log \left[|z - a_j|^2 + e^{-j^3} \right].$$

(1) *Show that φ belongs to the Lelong class $\mathcal{L}(\mathbb{C})$ and that it is locally bounded in $\mathbb{C}^*$, but not locally bounded near 0.*

(2) *Show that one can choose (a_j) so that $\varphi(0)$ is finite.*

(3) *Explain how to similarly construct an ω-psh function on a compact Kähler manifold (X, ω) which is finite at all points, but not locally bounded on a "large" set S. How large can you make S?*

Exercise 8.9. *Let (X, ω) be a compact Kähler manifold. Explain how to construct an ω-psh function φ which has positive Lelong numbers on a countable and dense set of points. Using a Baire category argument, show that the set $\{\varphi = -\infty\}$ is uncountable.*

Exercise 8.10. *Consider $X = \mathbb{P}^1 \times \mathbb{P}^1$ equipped with the Kähler form $\omega(x, y) = \omega_{\mathrm{FS}}(x) + \omega_{\mathrm{FS}}(y)$, where ω_{FS} denotes the Fubini–Study Kähler form on the Riemann sphere. Compute*

$$\nu(\omega) := \sup\{\nu(\varphi, p) \mid p \in X \text{ and } \varphi \in PSH(X, \omega)\},$$

the maximal Lelong number of ω-psh functions.

Exercise 8.11. *Let (X, ω) be a compact Kähler manifold. We set*

$$\alpha(\{\omega\}) := \sup \left\{ \alpha > 0 \,\Big|\, \sup_{\varphi \in \mathcal{F}_0} \int_X e^{-\alpha\varphi} \omega^n < +\infty \right\},$$

where $\mathcal{F}_0 = \{\varphi \in PSH(X, \omega) |\ \sup_X \varphi = 0\}$. Show that $0 < \alpha(\{\omega\}) < +\infty$ and $\alpha(\{\lambda\omega\}) = \lambda\alpha(\{\omega\})$ for all $\lambda > 0$. Compute α when (X, ω) is

(i) $(\mathbb{P}^n, \omega_{\mathrm{FS}})$;

(ii) $(\mathbb{P}^1_x \times \mathbb{P}^1_y, a\omega_{\mathrm{FS}}(x) + b\omega_{\mathrm{FS}}(y))$, $a, b > 0$;

(iii) $(X_1, \pi^*\{\omega_{\mathrm{FS}}\} - \varepsilon[E])$, *where $\pi = X_1 \to \mathbb{P}^2$ is the blow-up of $\mathbb{P}^2$ at one point.*

Exercise 8.12. *Let X be the blow-up of the complex projective plane $\mathbb{P}^2$ at eight points in general position.*

(1) *Check that X is a Fano manifold, i.e., $-K_X$ is ample ($c_1(X) > 0$).*

(2) *Fix a Kähler form ω in $c_1(X)$ and $\varphi \in PSH(X, \omega)$. Assume that the current ω_φ does not charge the strict transform of any cubic passing through these eight points. Show that $c(\varphi) \geq 1$.*

(3) *Analyze the singular behavior of the pencil of cubics going through these eight points in $\mathbb{P}^2$ and show that $\alpha(c_1(X)) \geq 5/6$ (see [Dolgachev] for some help).*

Exercise 8.13. *Let (X,ω) be a compact Kähler manifold. Let $\chi : \mathbb{R}^- \to \mathbb{R}$ be a convex increasing function of class $\mathcal{C}^2$ and φ be a qpsh function on X. Translating and rescaling, we can assume without loss of generality that $\chi'(-1) \leq 1$ and φ is ω-psh with $\varphi \leq -1$. Show that $\chi \circ \varphi \in PSH(X,\omega)$ and*

$$\int_X \chi'' \circ \varphi \, d\varphi \wedge d^c\varphi \wedge \omega^{n-1} \leq \mathrm{Vol}_\omega(X).$$

Choosing $\chi(t) = t/\log(8-t)$, deduce that

$$\int_X \frac{|\nabla\varphi|^2}{|\varphi|\,(\log[1+|\varphi|])^2}\omega^n < +\infty.$$

Considering the case when $\omega_\varphi = [D]$ is the current of integration along a divisor D, show that the exponent of the log-term cannot be lowered to 1.

Exercise 8.14. *Let (X,ω) be a compact Kähler manifold. Fix an ω-psh function φ with $\nabla\varphi \in L^2(X)$. Let φ_j be a sequence of ω-psh functions which decrease to φ. Show that $\nabla\varphi_j \in L^2(X)$ and φ_j converges to φ in the Sobolev norm $W^{1,2}(X)$.*

Exercise 8.15. *Let φ be an ω-psh function on a compact Kähler manifold (X,ω) such that $\varphi \leq -1$. For $0 < \alpha < 1$, we set*

$$\varphi_\alpha := -(-\varphi)^\alpha.$$

(i) *Show that $\varphi_\alpha \in PSH(X,\omega)$ has zero Lelong number at all points.*
(ii) *Generalize this to $\varphi_\chi := \chi \circ \varphi$ under appropriate conditions on χ, φ.*
(iii) *Give an example such that $\nabla\varphi_\alpha \notin L^2$ if $\alpha \geq 1/2$.*

Exercise 8.16. *Let u be a plurisubharmonic function in a neighborhood of the origin in $\mathbb{C}^n$. Using standard radial mollifiers $\rho_\delta \geq 0$, consider*

$$u_\delta(z) := u * \rho_\delta(z) = \int u(z-\delta w)\rho(w)dV(w).$$

Also set $\hat{u}_\delta(z) := \sup_{\mathbb{B}(z,\delta)} u$. Fix $A > 1$. Show that

$$\hat{u}_{A\delta} - \hat{u}_\delta \longrightarrow 0 \ \textit{locally uniformly}$$

if and only if u has zero Lelong numbers. Show similarly that

$$\hat{u}_\delta - u_\delta \longrightarrow 0 \ \textit{locally uniformly}$$

if and only if u has zero Lelong numbers. Use these results to complete the proof of Proposition 8.16.

9 Envelopes and capacities

In this chapter we fix a compact Kähler manifold X of complex dimension n and a semi-positive closed $(1,1)$ form ω such that

$$V = V_\omega = \mathrm{Vol}_\omega(X) := \int_X \omega^n > 0.$$

9.1 Complex Monge–Ampère operators

9.1.1 Dirichlet problem It follows from the local theory of Bedford and Taylor [BT82] that if φ is a *bounded* ω-plurisubharmonic function, then

$$MA(\varphi) := V^{-1}(\omega + dd^c\varphi)^n$$

is a well-defined probability measure. The operator

$$\varphi \longmapsto MA(\varphi)$$

is continuous along *monotone* sequences, but it is in general discontinuous for the L^1-topology (see Exercise 9.2).

As in the local theory, it is actually necessary to study intermediate Monge–Ampère type operators which are defined as follows: let T be a positive closed current of bidegree (p,p) on X, $0 \le p \le n = \dim_{\mathbb{C}} X$. It can be thought of as a closed differential form of bidegree (p,p) with measure coefficients whose total variation is controlled by

$$\|T\| := \int_X T \wedge \omega^{n-p}.$$

We refer the reader to Chapter 2 and [Dem] for basic properties of positive currents.

Given $\varphi \in PSH(X,\omega)$, we write $\varphi \in L^1(T)$ if φ is integrable with respect to each (measure) coefficient of T. This is equivalent to φ being integrable with respect to the trace measure $T \wedge \omega^{n-p}$. In this case the current φT is well defined, hence so is

$$\omega_\varphi \wedge T := \omega \wedge T + dd^c(\varphi T).$$

This is again a *positive* closed current on X, of bidegree $(p+1,p+1)$. Indeed positivity is a local property which is stable under taking limits. One can locally regularize φ and approximate $\omega_\varphi \wedge T$ by the currents $\omega_{\varphi_\varepsilon} \wedge T$ which are positive, since $\omega_{\varphi_\varepsilon}$ are smooth positive forms.

When $\varphi \in PSH(X,\omega) \cap L^\infty(X)$, then $\varphi \in L^1(T)$ for any positive closed current T of bidegree (p,p). One can thus inductively define $\omega_\varphi^j \wedge T$, $1 \le j \le n-p$, for $\varphi \in PSH(X,\omega) \cap L^\infty(X)$.

For $T = 0$ and $j = n$ one obtains the (unormalized) complex Monge–Ampère operator

$$\omega_\varphi^n = (\omega + dd^c\varphi)^n.$$

In the sequel we shall need the following immediate consequence of the local theory:

Theorem 9.1. *Let B be a small "ball" in X and $\varphi \in PSH(X,\omega) \cap L^\infty(X)$. There exists a unique $\psi \in PSH(X,\omega) \cap L^\infty(X)$ such that*

$$\psi \equiv \varphi \text{ in } X \setminus B,\ \psi \geq \varphi \text{ on } X, \text{ and } MA(\psi) \equiv 0 \text{ in } B.$$

By "ball" we mean here any inverse image of a ball in some coordinate chart.

9.1.2 Maximum principles At the heart of Bedford–Taylor's theory lies the following *maximum principle*: if u, v are bounded ω-psh functions, then

$$\mathbf{1}_{\{v<u\}} MA(\max(u,v)) = \mathbf{1}_{\{v<u\}} MA(u). \tag{MP}$$

This equality is elementary when u is *continuous*, as $\{u > v\}$ is then a Borel open subset of X. When u is merely *bounded*, this set is only open in the plurifine topology. Since Monge–Ampère measures of bounded qpsh functions do not charge pluripolar sets (by the Chern–Levine–Nirenberg inequalities), and since u is nevertheless *quasi-continuous*, this gives a heuristic justification of (MP). We refer the reader to Theorem 3.27 in Chapter 3 for a detailed proof of this maximum principle.

We now show how (MP) implies the comparison principle:

Proposition 9.2. *Let u, v be bounded ω-plurisubharmonic functions. Then*

$$\int_{\{v<u\}} MA(u) \leq \int_{\{v<u\}} MA(v).$$

Proof. It follows from (MP) that

$$\begin{aligned}\int_{\{v<u\}} MA(u) &= \int_{\{v<u\}} MA(\max(u,v)) = 1 - \int_{\{v\geq u\}} MA(\max(u,v)) \\ &\leq \int_X MA(v) - \int_{\{v>u\}} MA(\max(u,v)) = \int_{\{v\leq u\}} MA(v).\end{aligned}$$

Replacing u by $u-\varepsilon$ we obtain

$$\int_{\{v<u-\varepsilon\}} MA(u) \leq \int_{\{v\leq u-\varepsilon\}} MA(v) \leq \int_{\{v<u\}} MA(v).$$

The conclusion follows by observing that the sets $\{v < u-\varepsilon\}$ increase to $\{v < u\}$ when ε decreases to 0. □

9.1.3 Choquet's lemma Recall that the upper-envelope of a family of upper semi-continuous functions need not be upper semi-continuous. We will use on several occasions the following result due to Choquet (see Chapter 4):

Lemma 9.3. *Every family $(u_\alpha)_{\alpha\in A}$ of upper semi-continuous functions $u_\alpha : X \to \mathbb{R}\cup\{-\infty\}$ which are locally uniformly bounded from above admits a countable subfamily $(u_{\alpha_j})_{j\in\mathbb{N}}$ whose upper envelope satisfies*

$$\left(\sup_{\alpha\in A} u_\alpha\right)^* = \left(\sup_{j\in\mathbb{N}} u_{\alpha_j}\right)^*.$$

9.2 The Monge–Ampère capacity

9.2.1 Chern–Levine–Nirenberg inequalities We establish in this section global uniform versions of the Chern–Levine–Nirenberg inequalities.

Proposition 9.4. *Let T be a positive closed current of bidegree (p,p) on X and $\varphi \in PSH(X,\omega)\cap L^\infty(X)$. Then $\|\omega_\varphi\wedge T\| = \|T\|$. Moreover, if $\psi \in PSH(X,\omega)\cap L^1(T)$, then $\psi \in L^1(T\wedge\omega_\varphi)$ and*

$$\|\psi\|_{L^1(T\wedge\omega_\varphi)} \le \|\psi\|_{L^1(T)} + \left(2\sup_X \psi + \sup_X \varphi - \inf_X \varphi\right)\|T\|.$$

Proof. By the Stokes theorem, $\int_X dd^c\varphi\wedge T\wedge\omega^{n-p-1} = 0$, hence

$$\|\omega_\varphi\wedge T\| := \int_X \omega_\varphi\wedge T\wedge\omega^{n-p-1} = \int_X T\wedge\omega^{n-p} := \|T\|.$$

Consider now $\psi \in PSH(X,\omega)\cap L^1(T)$. Since T has measure coefficients, this simply means that ψ is integrable with respect to the total variation of these measures. Assume first $\psi\le 0$, $\varphi\ge 0$. Then

$$\begin{aligned}\|\psi\|_{L^1(T\wedge\omega_\varphi)} &:= \int_X (-\psi)T\wedge\omega_\varphi\wedge\omega^{n-p-1}\\ &= \|\psi\|_{L^1(T)} + \int_X (-\psi)T\wedge dd^c\varphi\wedge\omega^{n-p-1}.\end{aligned}$$

Now it follows from the Stokes theorem that

$$\int_X (-\psi)T\wedge dd^c\varphi\wedge\omega^{n-p-1} = \int_X \varphi T\wedge(-dd^c\psi)\wedge\omega^{n-p-1}$$

$$\le \int_X \varphi T\wedge\omega^{n-p} \le \sup_X\varphi\int_X T\wedge\omega^{n-p},$$

where the next to last inequality follows from $\varphi T\wedge\omega^{n-p}\ge 0$ and $-dd^c\psi\le\omega$.

This yields

$$\|\psi\|_{L^1(T\wedge\omega_\varphi)} \leq \|\psi\|_{L^1(T)} + \sup_X \varphi \, \|T\|.$$

The general case follows by replacing φ by $\varphi' = \varphi - \inf_X \varphi \geq 0$, and decomposing $\psi = \psi' + \sup_X \psi$ with $\psi' = \psi - \sup_X \psi \leq 0$. □

A straightforward induction yields the following a priori estimates, which are often called *Chern–Levine–Nirenberg inequalities*:

Corollary 9.5. *Let $\psi, \varphi \in PSH(X,\omega)$ with $0 \leq \varphi \leq 1$. Then*

$$0 \leq \int_X |\psi|\omega_\varphi^n \leq \int_X |\psi|\omega^n + n\left(1 + 2\sup_X \psi\right)\int_X \omega^n.$$

Following Bedford and Taylor [BT82] and Kolodziej [Kol05] we introduce the following Monge–Ampère capacity:

Definition 9.6. *Let K be a Borel subset of X. We set*

$$\mathrm{Cap}_\omega(K) := \sup\left\{\int_K MA(\varphi) \;\middle|\; \varphi \in PSH(X,\omega), 0 \leq \varphi \leq 1\right\}.$$

We list a few elementary properties of this capacity whose proof is left to the reader:

Proposition 9.7.

(1) *If $K \subset K' \subset X$ are Borel subsets then $\mathrm{Vol}_\omega \leq \mathrm{Cap}_\omega$ and*

$$0 \leq \mathrm{Cap}_\omega(K) \leq \mathrm{Cap}_\omega(K') \leq \mathrm{Cap}_\omega(X) = 1.$$

(2) *If K_j are Borel subsets of X then $\mathrm{Cap}_\omega\left(\bigcup K_j\right) \leq \sum \mathrm{Cap}_\omega(K_j)$. Moreover, $\mathrm{Cap}_\omega\left(\bigcup K_j\right) = \lim \mathrm{Cap}_\omega(K_j)$ if $K_j \subset K_{j+1}$.*

(3) *If $\omega_1 \leq \omega_2$ then $V_{\omega_1}\mathrm{Cap}_{\omega_1}(\cdot) \leq V_{\omega_2}\mathrm{Cap}_{\omega_2}(\cdot)$. For all $A \geq 1$,*

$$\mathrm{Cap}_\omega(\cdot) \leq \mathrm{Cap}_{A\omega}(\cdot) \leq A^n \mathrm{Cap}_\omega(\cdot).$$

In particular, if ω, ω' are two Kähler forms, there exists $C \geq 1$ such that

$$C^{-1}\mathrm{Cap}_\omega(\cdot) \leq \mathrm{Cap}_{\omega'}(\cdot) \leq C \cdot \mathrm{Cap}_\omega(\cdot).$$

The definition of Cap_ω mimics the definition of the relative Monge–Ampère capacity introduced by Bedford and Taylor in [BT82]. Fix a finite covering $\mathcal{U} = \{\mathcal{U}_\alpha\}$ of X by strictly pseudoconvex open subsets of X, $\mathcal{U}_\alpha = \{x \in X \mid \varrho_\alpha(x) < 0\}$, where ϱ_α is a strictly psh smooth function defined in a neighborhood of $\overline{\mathcal{U}_\alpha}$. Fix $\delta > 0$ such

that $\mathcal{U}^\delta = \{\mathcal{U}_\alpha^\delta\}$ is still a covering of X, where $\mathcal{U}_\alpha^\delta = \{x \in X \mid \varrho_\alpha(x) < -\delta\}$. For a Borel subset K of X, we set

$$\mathrm{Cap}_{\mathrm{BT}}(K) := \sum_\alpha \mathrm{Cap}_{\mathrm{BT}}(K \cap \mathcal{U}_\alpha^\delta, \mathcal{U}_\alpha),$$

where

$$\mathrm{Cap}_{\mathrm{BT}}(E, \Omega) := \sup \left\{ \int_E (dd^c u)^n \mid u \in PSH(\Omega),\ 0 \le u \le 1 \right\}$$

is the capacity studied by Bedford and Taylor.

Proposition 9.8. *There exists $C \ge 1$ such that*

$$C^{-1}\mathrm{Cap}_\omega(\cdot) \le \mathrm{Cap}_{\mathrm{BT}}(\cdot).$$

When ω is Kähler, there exists $C' \ge 1$ such that

$$\mathrm{Cap}_{\mathrm{BT}}(\cdot) \le C' \cdot \mathrm{Cap}_\omega(\cdot).$$

Proof. Let E be a Borel subset of X. Since $\mathrm{Cap}_\omega(E \cap \mathcal{U}_\alpha^\delta) \le \mathrm{Cap}_\omega(E) \le \sum_\alpha \mathrm{Cap}_\omega(E \cap \mathcal{U}_\alpha^\delta)$, it is sufficient to show that if $\Omega = \{x \in X \mid \varrho(x) < 0\}$ is a smooth hyperconvex subset of X, then there exists $C \ge 1$ such that for all $E \subset \Omega_\delta$,

$$C^{-1}\mathrm{Cap}_\omega(E) \le \mathrm{Cap}_{\mathrm{BT}}(E, \Omega),$$

where $\Omega_\delta = \{x \in X \mid \varrho(x) < -\delta\}$.

It is an easy fact in the local theory that the capacities $\mathrm{Cap}(\cdot, \Omega)$ and $\mathrm{Cap}(\cdot, \Omega')$ are comparable when $\Omega' \subset \Omega$. Therefore, we can assume (passing to a finer covering if necessary) that $\omega = dd^c\psi$ near $\overline{\Omega}$. Fix $C_1 > 0$ such that $-C_1 \le \psi \le C_1$ on Ω. Fix $\varphi \in PSH(X, \omega)$ such that $0 \le \varphi \le 1$ on X and set $u = (1 + 2C_1)^{-1}(\varphi + \psi + C_1)$. Then $u \in PSH(\Omega)$ and $0 \le u \le 1$, hence

$$\int_E \omega_\varphi^n = (1 + 2C_1)^n \int_E (dd^c u)^n \le (1 + 2C_1)^n \mathrm{Cap}_{\mathrm{BT}}(E, \Omega),$$

which yields $\mathrm{Cap}_\omega(E) \le C \cdot \mathrm{Cap}_{\mathrm{BT}}(E, \Omega)$.

For the reverse inequality we consider $\chi \in \mathcal{C}^\infty(X)$ such that $\chi \equiv 0$ in $X \setminus \Omega$ and $\chi < 0$ in Ω. Replacing χ by $\varepsilon\chi$ if necessary, we can assume $\chi \in PSH(X, \omega)$. This is the only place where we use the assumption that ω is positive.

Fix $\varepsilon > 0$ so small that $\chi \le -\varepsilon$ on Ω_δ. Let now $u \in PSH(\Omega)$ be such that $0 \le u \le 1$ on Ω. Consider

$$\varphi(x) = \begin{cases} \frac{u - \psi + C_1}{2 + 2C_1}, & \text{in } \Omega_\delta, \\ \max\left(\frac{u - \psi + C_1}{2 + 2C_1}, \frac{2}{\varepsilon}\chi(x) + 1\right), & \text{in } \Omega \setminus \Omega_\delta, \\ 1, & \text{in } X \setminus \Omega, \end{cases}$$

Observe that

$$0 \leq u' := \frac{u - \psi + C_1}{2 + 2C_1} \leq \frac{1 + 2C_1}{2 + 2C_1} < 1$$

in Ω. Therefore, $\varphi \in PSH(X, \frac{2}{\varepsilon}\omega)$ since $\frac{2}{\varepsilon}\chi(x) + 1 \leq -1 < u'$ in Ω_δ, while $\frac{2}{\varepsilon}\chi(x) + 1 \equiv 1 > u'$ on $\partial\Omega$. Note also that $0 \leq \varphi \leq 1$, thus for $E \subset \Omega_\delta$,

$$\frac{1}{(2 + 2C_1)^n} \int_E (dd^c u)^n = \int_E \left(\frac{\omega}{2 + 2C_1} + dd^c\varphi\right)^n \leq \int_E \left(\frac{2}{\varepsilon}\omega + dd^c\varphi\right)^n$$
$$\leq \mathrm{Cap}_{2\omega/\varepsilon}(E) \leq \frac{2^n}{\varepsilon^n}\mathrm{Cap}_\omega(E).$$

Therefore, $\mathrm{Cap}_{\mathrm{BT}}(E, \Omega) \leq 4^n(1 + C_1)^n\varepsilon^{-n}\mathrm{Cap}_\omega(E)$. □

Since locally pluripolar sets are precisely the sets of zero relative capacity (see Chapter 4), we obtain the following

Corollary 9.9. *Assume ω is Kähler. Then*

$$\mathrm{Cap}^*_\omega(P) = 0 \Longleftrightarrow \mathrm{Cap}^*_{\mathrm{BT}}(P) = 0 \Longleftrightarrow P \text{ is locally pluripolar.}$$

When the form ω is merely semi-positive and big, the same result holds, but the proof requires the notion of generalized capacities (see Chapter 12).

We shall show later on that locally pluripolar sets are $PSH(X, \omega)$-polar (see Josefson's theorem). A set is said to be $PSH(X, \omega)$-*polar* if it is included in the $-\infty$ locus of some function $\psi \in PSH(X, \omega)$, $\psi \not\equiv -\infty$.

9.2.2 Quasicontinuity

Let $PSH^-(X, \omega)$ denote the set of negative ω-psh functions.

Proposition 9.10. *If P is a $PSH(X, \omega)$-polar set, then* $\mathrm{Cap}_\omega(P) = 0$. *More precisely, if $\psi \in PSH^-(X, \omega)$ then*

$$\mathrm{Cap}_\omega(\psi < -t) \leq \frac{1}{t}\left[\int_X (-\psi)\frac{\omega^n}{V_\omega} + n\right], \ \forall t > 0.$$

Proof. Fix $\varphi \in PSH(X, \omega)$ such that $0 \leq \varphi \leq 1$. Fix $t > 0$ and set $K_t = \{x \in X \mid \psi(x) < -t\}$. By Chebyshev's inequality,

$$\int_{K_t} \omega_\varphi^n \leq \int_X (-\psi/t)\omega_\varphi^n \leq \frac{1}{t}\left[\int_X (-\psi)\omega^n + n\mathrm{Vol}_\omega(X)\right],$$

where the last inequality follows from Corollary 9.5. Taking the supremum over all φ's yields the claim. □

Observe that the previous proposition says that $\mathrm{Cap}^*_\omega(P) = 0$, where

$$\mathrm{Cap}^*_\omega(E) := \inf\{\mathrm{Cap}_\omega(G) \,|G \text{ open with } E \subset G\},$$

is the outer capacity associated to Cap_ω.

Our aim is now to show that ω-psh functions are quasicontinuous with respect to Cap_ω. We first need to show that decreasing sequences of ω-psh functions converge "in capacity".

Proposition 9.11. *Let $\psi, \psi_j \in PSH(X,\omega) \cap L^\infty(X)$ such that (ψ_j) decreases to ψ. Then for each $\delta > 0$,*

$$\mathrm{Cap}_\omega(\{\psi_j > \psi + \delta\}) \longrightarrow 0.$$

Proof. We can assume w.l.o.g. that $\mathrm{Vol}_\omega(X) = 1$ and $0 \le \psi_j - \psi \le 1$. Fix $\delta > 0$ and $\varphi \in PSH(X,\omega)$, $0 \le \varphi \le 1$. By the Chebyshev inequality, it suffices to control $\int_X (\psi_j - \psi)\omega_\varphi^n$ uniformly in φ. It follows from the Stokes theorem that

$$\int_X (\psi_j - \psi)\omega_\varphi^n = \int_X (\psi_j - \psi)\omega \wedge \omega_\varphi^{n-1} - \int_X d(\psi_j - \psi) \wedge d^c\varphi \wedge \omega_\varphi^{n-1}.$$

Now by the Cauchy–Schwartz inequality,

$$\left|\int_X df_j \wedge d^c\varphi \wedge \omega_\varphi^{n-1}\right| \le \left(\int_X df_j \wedge d^c f_j \wedge \omega_\varphi^{n-1}\right)^{1/2} \cdot \left(\int_X d\varphi \wedge d^c\varphi \wedge \omega_\varphi^{n-1}\right)^{1/2},$$

where we set $f_j := \psi_j - \psi \ge 0$. Moreover,

$$\int_X d\varphi \wedge d^c\varphi \wedge \omega_\varphi^{n-1} = \int_X \varphi(-dd^c\varphi) \wedge \omega_\varphi^{n-1} \le \int_X \varphi\omega \wedge \omega_\varphi^{n-1} \le 1,$$

since $\varphi\omega_\varphi^{n-1} \ge 0$, $-dd^c\varphi \le \omega$ and $\varphi \le 1$. Similarly,

$$\int_X df_j \wedge d^c f_j \wedge \omega_\varphi^{n-1} = \int_X -f_j dd^c f_j \wedge \omega_\varphi^{n-1} \le \int_X f_j\omega_\psi \wedge \omega_\varphi^{n-1}.$$

Altogether this yields

$$\begin{aligned}\int_X (\psi_j - \psi)\omega_\varphi^n &\le \int_X (\psi_j - \psi)\omega \wedge \omega_\varphi^{n-1} + \left(\int_X (\psi_j - \psi)\omega_\psi \wedge \omega_\varphi^{n-1}\right)^{1/2} \\ &\le \sqrt{2}\left(\int_X (\psi_j - \psi)(\omega + \omega_\psi) \wedge \omega_\varphi^{n-1}\right)^{1/2},\end{aligned}$$

where the last inequality follows from the elementary inequalities $0 \le a \le \sqrt{a} \le 1$ and $\sqrt{a} + \sqrt{b} \le \sqrt{2}\sqrt{a+b}$.

Going on replacing at each step a term ω_φ by $\omega + \omega_\psi$, we end up with

$$\int_X (\psi_j - \psi)\omega_\varphi^n \leq 2\left(\int_X (\psi_j - \psi)(\omega + \omega_\psi)^n\right)^{1/2^n}.$$

The majorant being independent of φ and converging to 0 as $j \to +\infty$ (by the dominated convergence theorem), this completes the proof. □

Corollary 9.12 (Quasicontinuity). *Let $\varphi \in PSH(X,\omega)$. For each $\varepsilon > 0$ there exists an open subset O_ε of X such that $\mathrm{Cap}_\omega(O_\varepsilon) < \varepsilon$ and φ is continuous on $X \setminus O_\varepsilon$.*

Proof. For $t > 0$ large enough, the set $O_1 = \{\varphi < -t\}$ has capacity $< \varepsilon/2$ by Proposition 9.10. Working in $X \setminus O_1$ we can thus replace φ by $\varphi_t = \max(\varphi, -t)$ which is bounded on X. Regularizing φ, we can find a sequence ψ_j of smooth $A\omega$-psh functions which decrease to φ_t on X, for some $A \geq 1$. By Proposition 9.11, the set $O_j = \{\psi_{k_j} > \varphi_t + 1/j\}$ has capacity $< \varepsilon 2^{-j-1}$ if k_j is large enough. Now ψ_{k_j} converges uniformly to $\varphi = \varphi_t$ on $X \setminus O_\varepsilon$, $O_\varepsilon = \bigcup_{j\geq 1} O_j$, so φ is continuous on $X \setminus O_\varepsilon$ and $\mathrm{Cap}_\omega(O_\varepsilon) \leq \varepsilon$. □

9.3 Outer regularity

9.3.1 The relative extremal function We now introduce a substitute for the relative extremal function which has revealed so useful in the local theory.

Definition 9.13. *If E is a Borel subset of X, we set*

$$h_{E,\omega}(x) := \sup\{\varphi(x) \,|\, \varphi \in PSH(X,\omega),\ \varphi \leq 0 \text{ and } \varphi|_E \leq -1\}.$$

We let $h^*_{E,\omega}$ denote its upper semi-continuous regularization, which we call the relative ω-plurisubharmonic extremal function of the subset E.

This extremal function enjoys several natural properties; we list some of them below. The proofs follow from standard arguments:

- The function $h^*_{E,\omega}$ is ω-psh. It satisfies $-1 \leq h^*_{E,\omega} \leq 0$ on X and $h^*_{E,\omega} = -1$ on $E \setminus P$, where P is pluripolar.
- If $E \subset X$ and $P \subset X$ is pluripolar, then $h^*_{E\setminus P} \equiv h^*_E$.
- If (E_j) increases towards E, then $h^*_{E_j}$ decreases towards h^*_E.
- If (K_j) is a sequence of compact subsets decreasing towards K, then $h^*_{K_j}$ increases (a.e.) towards h^*_K.

As in the local theory, the complex Monge–Ampère measure of the relative extremal function of a subset $E \subset X$ vanishes outside $\overline{E}$, except perhaps on the set $\{h^*_E = 0\}$ which, in the local theory, lies in the boundary of the domain.

Proposition 9.14. *If the open set* $\Omega_E := \{x \in X \mid h^*_{E,\omega}(x) < 0\}$ *is non-empty, then*

$$MA\left(h^*_{E,\omega}\right) = 0 \text{ in } \Omega_E \setminus \overline{E}.$$

Proof. Assume that $\Omega_E := \{h^*_{E,\omega} < 0\}$ is non-empty. It follows from Choquet's lemma (Lemma 4.31) that there exists an increasing sequence φ_j of ω-psh functions such that $\varphi_j = -1$ on E, $\varphi_j \leq 0$ on X, and $h^*_{E,\omega} = (\lim \varphi_j)^*$.

Let $a \in \Omega_E \setminus \overline{E}$ and fix a small ball $B \subset \Omega_E$ centered at a point a. Let $\widehat{\varphi_j} = \widehat{(\varphi_j)_B}$ denote the functions obtained by applying Theorem 9.1, so that $MA(\widehat{\varphi_j}) = 0$ in B. If B is chosen small enough, then $\widehat{\varphi_j} < 0$ in B, hence $\widehat{\varphi_j} \leq 0$ on X, while $\widehat{\varphi_j} = \varphi_j = -1$ on E. This can be seen by showing that $\widehat{\varphi_j}_B \to 0$ as the radius of the ball B shrinks to 0. Therefore, $\lim \nearrow \widehat{\varphi_j} = h^*_{E,\omega}$, hence $MA(h^*_{E,\omega}) \equiv 0$ in a neighborhood of a, which proves our claim. □

9.3.2 A twisted formula We now establish an important result which expresses the capacity in terms of the relative extremal function for any subset. It will show, in particular, that the set function Cap^*_ω is a capacity in the sense of Choquet which is outer regular. For simplicity, we write h_E for $h_{E,\omega}$.

Theorem 9.15. *Let* $E \subset X$ *be any Borel subset, then*

$$\mathrm{Cap}^*_\omega(E) = \int_X (-h^*_E) MA\left(h^*_E\right). \tag{†}$$

The Monge–Ampère capacity enjoys the following continuity properties:

(1) *If* $(E_j)_{j\geq 0}$ *is an increasing sequence and* $E := \bigcup_{j\geq 0} E_j$, *then*

$$\mathrm{Cap}^*_\omega(E) = \lim_{j\to+\infty} \mathrm{Cap}^*_\omega(E_j).$$

(2) *If* $(K_j)_{j\geq 0}$ *are decreasing compact subsets and* $K := \bigcap_{j\geq 0} K_j$, *then*

$$\mathrm{Cap}_\omega(K) = \mathrm{Cap}^*_\omega(K) = \lim_{j\to+\infty} \mathrm{Cap}_\omega(K_j).$$

In particular, $\mathrm{Cap}^*_\omega(\cdot)$ *is an outer regular Choquet capacity on* X.

Proof. We first establish (†) when $E = K \subset X$ is compact. Observe that in the definition of the capacity, it is enough to restrict ourselves to ω-psh functions φ such that $-1 < \varphi < 0$. Let φ be such a function. The set $N := \{h_K < h^*_K\}$ is of measure 0 for the measure $MA(\varphi)$. Since $-1 < \varphi$, we have $K \subset N \cup \{h^*_K < \varphi\}$, hence the comparison principle yields

$$\int_K MA(\varphi) \leq \int_{\{h^*_K<\varphi\}} MA(\varphi) \leq \int_{\{h^*_K<\varphi\}} MA(h^*_K).$$

This shows $\mathrm{Cap}_\omega(K) \leq \int_{\Omega_K} MA(h_K^*)$ since $\{h_K^* < \varphi\} \subset \Omega_K$. It follows from the Chern–Levine–Nirenberg inequalities that

$$\mathrm{Cap}_\omega(K) \leq \int_{\Omega_K} MA(h_K^*) = \int_K MA(h_K^*),$$

whence equality. Recall that $K \cap \{h_K^* > -1\} \subset \{h_K < h_K^*\}$ is of measure 0 for $MA(h_K^*)$, so

$$\mathrm{Cap}_\omega(K) = \int_K MA(h_K^*) = \int_K (-h_K^*) MA(h_K^*) = \int_X (-h_K^*) MA(h_K^*),$$

where the last equality follows from the fact that the equilibrium measure $MA(h_K^*)$ is supported on $K \cup \{h_K^* = 0\}$.

We assume now that $E = G \subset X$ is an *open* subset. Let (K_j) be an exhaustive sequence of compact subsets of G which increases to G. Since $h_{K_j}^* \downarrow h_G$ on X, we infer that $(-h_{K_j}^*) MA(h_{K_j}^*) \to (-h_G^*) MA(h_G)$ in the weak sense of measures on X, therefore

$$\begin{aligned} \int_X (-h_G) MA(h_G) &= \lim_{j \to +\infty} \int_X (-h_{K_j}^*) MA(h_{K_j}^*) \\ &= \lim_{j \to +\infty} \mathrm{Cap}_\omega(K_j) = \mathrm{Cap}_\omega(G). \end{aligned}$$

This proves (†) when $E \subset X$ is an open subset.

Finally, let $E \subset X$ be any subset. By the definition of the outer capacity, there is a sequence of open subsets $(O_j)_{j \geq 1}$ of X containing E such that $\mathrm{Cap}_\omega^*(E) = \lim_{j \to +\infty} \mathrm{Cap}_\omega(O_j)$. We can assume w.l.o.g. that the sequence $(O_j)_{j \geq 1}$ is decreasing.

By Choquet's lemma (Lemma 4.31), there exists an increasing sequence $(u_j)_{j \geq 1}$ of negative ω-psh functions on X s.t. $u_j = -1$ on E with $u_j \uparrow h_E^*$ almost everywhere on X. We set for each $j \in \mathbb{N}$,

$$G_j := O_j \cap \{u_j < -1 + 1/j\}.$$

Then (G_j) is a decreasing sequence of open subsets of X such that $E \subset G_j \subset O_j$ and $u_j - 1/j \leq h_{G_j} \leq h_E$, so $h_{G_j} \uparrow h_E^*$ almost everywhere on X. We infer that $(-h_{G_j}) MA(h_{G_j}) \to (-h_E^*) MA(h_E)$ in the weak sense of measures on X, thus

$$\int_X (-h_E^*) MA(h_E^*) = \lim_{j \to +\infty} \int_X (-h_{G_j}^*) MA(h_{G_j}).$$

On the other hand we have by construction

$$\mathrm{Cap}_\omega^*(E) \leq \lim_{j \to +\infty} \mathrm{Cap}_\omega^*(G_j) \leq \lim_{j \to +\infty} \mathrm{Cap}_\omega^*(O_j) = \mathrm{Cap}_\omega^*(E).$$

Therefore, using (†) for open subsets, we get

$$\int_X (-h_E^*) MA(h_E^*) = \mathrm{Cap}_\omega^*(E).$$

Now (1) follows from this formula: if (E_j) increases towards E, then $h_{E_j}^*$ decreases towards h_E^*, hence $(-h_{E_j}^*)MA(h_{E_j}^*) \to (-h_E^*)MA(h_E^*)$, so that

$$\begin{aligned}\mathrm{Cap}_\omega^*(E) &= \int_X (-h_E^*) MA(h_E^*) \\ &= \lim \int_X (-h_{E_j}^*) MA(h_{E_j}^*) = \lim \mathrm{Cap}_\omega^*(E_j).\end{aligned}$$

It remains to prove (2). Let (K_j) be a decreasing sequence of compact subsets of X which converges to K. We claim that $h_{K_j}^* \uparrow h_K^*$ almost everywhere on X. Indeed, the extremal function $h_{K_j}^*$ increases almost everywhere to an ω-psh function h such that $h \le h_K^*$ on X. We want to prove that $h_K \le h$ on X. Let $u \in PSH(X,\omega)$ be such that $u \le 0$ on X and $u|_K \le -1$. Fix $\varepsilon > 0$ and consider the open subset $G_\varepsilon := \{u < -1 + \varepsilon\}$. Then $K \subset G_\varepsilon$, thus $K_j \subset G_\varepsilon$ for j large enough. This yields $u - \varepsilon \le h_{K_j}$ for j large enough, hence $u \le h$ on X. Therefore $h_K \le h$ on X as claimed.

Since $(-h_{K_j})^* MA(h_{K_j}^*)$ converges weakly to $(-h_K^*)MA(h_K^*)$ on X, we infer that

$$\begin{aligned}\mathrm{Cap}_\omega(K) &= \int_X (-h_K^*) MA(h_K^*) \\ &= \lim_{j\to+\infty} \int_X (-h_{K_j}^*) MA(h_{K_j}^*) = \lim_{j\to+\infty} \mathrm{Cap}_\omega(K_j).\end{aligned}$$

From this last property, taking a decreasing sequence $(K_j)_{j\ge 0}$ of compact subsets such that $K = \bigcap_{j\ge 0} K_j$ and $K_{j+1} \subset K_j^\circ$ for any $j \in \mathbb{N}$ we obtain $\mathrm{Cap}_\omega^*(K) = \lim_{j\to+\infty} \mathrm{Cap}_\omega(K_j) = \mathrm{Cap}_\omega(K)$. □

9.4 Alexander–Taylor capacity

We now introduce another family of capacities which are defined by means of global extremal functions.

9.4.1 Global extremal functions

Definition 9.16. *Let K be a Borel subset of X. We set*

$$V_{K,\omega} := \sup\{\varphi(x) \,|\, \varphi \in PSH(X,\omega),\ \varphi \le 0 \text{ on } K\}.$$

This definition mimics the definition of the so-called "Siciak's extremal function", which is defined for bounded Borel subsets of $\mathbb{C}^n$.

Theorem 9.17. *Let K be a Borel subset of X.*

(1) *K is $PSH(X,\omega)$-polar iff $\sup_X V^*_{K,\omega} = +\infty$ iff $V^*_{K,\omega} \equiv +\infty$.*

(2) *If K is not $PSH(X,\omega)$-polar, then $V^*_{K,\omega} \in PSH(X,\omega)$ and satisfies $V^*_{K,\omega} \equiv 0$ in the interior of K, $MA(V^*_{K,\omega}) = 0$ in $X \setminus \overline{K}$, and*

$$\int_{\overline{K}} MA(V^*_{K,\omega}) = 1.$$

Proof. Assume $\sup_X V^*_{K,\omega} = +\infty$. By Choquet's lemma (Lemma 4.31), we can find an increasing sequence of functions $\varphi_j \in PSH(X,\omega)$ such that $\varphi_j = 0$ on K and $V^*_{K,\omega} = (\lim \nearrow \varphi_j)^*$. Extracting a subsequence if necessary, we can assume that $\sup_X \varphi_j \geq 2^j$. Set

$$\psi_j = \varphi_j - \sup_X \varphi_j.$$

These functions belong to $\mathcal{F}_0$, which is a compact subfamily of $PSH(X,\omega)$. Recall that if μ is a smooth volume form on X, then there exists C_μ such that $\int \psi_j d\mu \geq -C_\mu$ for all j. Set $\psi := \sum_{j\geq 1} 2^{-j}\psi_j$. Then $\psi \in PSH(X,\omega)$ as a decreasing limit of functions in $PSH(X,\omega)$ with $\int_X \psi d\mu \geq -C_\mu > -\infty$. Now for every $x \in K$ we get

$$\psi(x) = -\sum_{j\geq 1} 2^{-j} \sup_X \varphi_j = -\infty,$$

hence $K \subset \{\psi = -\infty\}$, i.e., K is $PSH(X,\omega)$-polar.

Conversely assume K is $PSH(X,\omega)$-polar, $K \subset \{\psi = -\infty\}$ for some $\psi \in PSH(X,\omega)$. Then for all $c \in \mathbb{R}$, $\psi + c \in PSH(X,\omega)$ and $\psi + c \leq 0$ on K. Therefore, $V_{K,\omega} \geq \psi + c$, $\forall c \in \mathbb{R}$. This yields $V_{K,\omega} = +\infty$ on $X \setminus \{\psi = -\infty\}$, hence $V^*_{K,\omega} \equiv +\infty$ on X since $\{\psi = -\infty\}$ has zero volume. We have thus shown the following circle of implications:

$$\begin{aligned} K \text{ is } PSH(X,\omega)\text{-polar} &\Rightarrow V^*_{K,\omega} \equiv +\infty \\ &\Rightarrow \sup_X V^*_{K,\omega} = +\infty \\ &\Rightarrow K \text{ is } PSH(X,\omega)\text{-polar}. \end{aligned}$$

Assume now that K is not $PSH(X,\omega)$-polar. Then the function $V^*_{K,\omega} \in PSH(X,\omega)$ and clearly satisfies $V^*_{K,\omega} = 0$ in the interior of K. If we show that $MA(V^*_{K,\omega}) = 0$ in $X \setminus \overline{K}$, then

$$\int_{\overline{K}} MA(V^*_{K,\omega}) = \int_X MA(V^*_{K,\omega}) = 1.$$

Let $\varphi_j \in PSH(X,\omega)$ be an increasing sequence such that $\varphi_j = 0$ on K and $V^*_{K,\omega} = (\lim \nearrow \varphi_j)^*$. Fix a small ball B in $X \setminus \overline{K}$. Let $\hat{\varphi_j}$ be the solution of the Dirichlet problem with boundary values φ_j. Then $\hat{\varphi_j} \in PSH(X,\omega)$, $\hat{\varphi_j} = \varphi_j$ in $X \setminus B$ (in particular $\hat{\varphi_j} = 0$ on K hence $\hat{\varphi_j} \leq V_{K,\omega}$), and the sequence $(\hat{\varphi_j})$ is again increasing (Theorem 9.1). Since $MA(\hat{\varphi_j}) = 0$ in B and $(\lim \nearrow \hat{\varphi_j}) = V^*_{K,\omega}$, it

follows from the continuity of the complex Monge–Ampère operator along increasing sequences that $MA(V^*_{K,\omega}) = 0$ in B. As B was an arbitrarily small ball in $X \setminus \overline{K}$ we conclude that $MA(V^*_{K,\omega}) = 0$ in $X \setminus \overline{K}$. □

Corollary 9.18. *Let K be a Borel subset of X and set*

$$\mathcal{F}_K := \{\varphi \in PSH(X,\omega) \mid \sup_K \varphi = 0\}.$$

Then $\mathcal{F}_K$ is relatively compact iff K is not $PSH(X,\omega)$-polar.

Proof. Observe that $V_{K,\omega}(x) = \sup\{\varphi(x) \mid \varphi \in \mathcal{F}_K\}$. Thus if $\mathcal{F}_K$ is relatively compact, then it is uniformly bounded from above, hence $\sup_X V_{K,\omega} < +\infty$, i.e., K is not $PSH(X,\omega)$-polar.

Assume conversely that K is not $PSH(X,\omega)$-polar. Let $(\varphi_j) \in \mathcal{F}_K^{\mathbb{N}}$. Then $\varphi_j \leq V_{K,\omega} \leq \sup_X V_{K,\omega} < +\infty$, hence (φ_j) is uniformly bounded from above. We infer that (φ_j) is relatively compact. Indeed it cannot converge uniformly to $-\infty$, since $\sup_K \varphi_j = 0$. □

Proposition 9.19.

(1) *If E is an open subset, then $V_E = V^*_E$.*

(2) *If $E \subset X$ and P is a $PSH(X,\omega)$-polar set, then $V^*_{E\cup P} \equiv V^*_E$.*

(3) *If (E_j) is an increasing sequence and $E := \bigcup E_j$, then*

$$V^*_{E,\omega} = \lim \searrow V^*_{E_j,\omega}.$$

(4) *Let K_j be a decreasing sequence of compact subsets of X and set $K = \bigcap K_j$. Then $V_{K_j,\omega} \nearrow V_{K,\omega}$, hence $V^*_{K_j,\omega} \nearrow V^*_{K,\omega}$ a.e.*

(5) *Fix a non-pluripolar set $E \subset X$. Then there exists a decreasing sequence of open subsets G_j, $E \subset G_j$, such that $V^*_E = \lim V^*_{G_j}$.*

Proof. We write here V_E for $V_{E,\omega}$ since ω is fixed and no confusion can arise.

Let E be an open subset of X. Observe that $V_E \leq 0$ on E, hence $V^*_E \leq 0$ on E, which is open. Therefore, $V^*_E \leq V_E$, whence the equality. This proves (1).

Let $w \in PSH(X,\omega)$, $w \leq 0$, and fix $P \subset \{w = -\infty\}$. Fix a Borel subset E of X. Clearly $V_{E\cup P} \leq V_E$, hence $V^*_{E\cup P} \leq V^*_E$. Conversely, let $\varphi \in PSH(X,\omega)$ be such that $\varphi \leq 0$ on E. Then $\forall \varepsilon > 0$, $\psi_\varepsilon := (1-\varepsilon)\varphi + \varepsilon w \in PSH(X,\omega)$ satisfies $\psi_\varepsilon \leq 0$ on $E \cup P$, hence $\psi_\varepsilon \leq V_{E\cup P}$. Letting $\varepsilon \to 0$ we infer $\varphi \leq V_{E\cup P}$ on $X \setminus P$, hence $\varphi \leq V^*_{E\cup P}$ on X. Thus $V^*_E \leq V^*_{E\cup P}$.

Let E_j be an increasing sequence of subsets of X and set $E = \bigcup_{j\geq 1} E_j$. Let $v := \lim \searrow V^*_{E_j}$ (the limit is obviously decreasing). If E is $PSH(X,\omega)$-polar, then so are all the E_j's, hence $V^*_E \equiv +\infty = \lim V^*_{E_j}$. So let us assume E is not $PSH(X,\omega)$-polar. Then $v \in PSH(X,\omega)$ since $v \geq V^*_{E,\omega} \not\equiv -\infty$. Observe that $v = 0$ on the set $E \setminus N$, where $N = \bigcup_{j\geq 1}\{V_{E_j} < V^*_{E_j}\}$. The latter is called a

negligible set. It follows from the local theory together with Josefson's theorem (see Chapter 12) that N is $PSH(X,\omega)$-polar. Therefore

$$V_E^* \le v \le V_{E\setminus N}^* = V_E^*$$

by (2).

Let K_j be a decreasing sequence of compact subsets and set $K = \bigcap_j K_j$. Clearly $\lim \nearrow V_{K_j} \le V_K$. Fix $\varepsilon > 0$ and let $\varphi \in PSH(X,\omega)$ be such that $\varphi \le 0$ on K. Then $\{\varphi < \varepsilon\}$ is an open set which contains all K_j's with $j \ge j_\varepsilon$ large enough. Thus $\varphi - \varepsilon \le 0$ on K_j, hence $\varphi - \varepsilon \le \lim \nearrow V_{K_j}$. Taking the supremum over all such φ's and letting $\varepsilon \to 0$ yields the reverse inequality $V_K \le \lim \nearrow V_{K_j}$. The conclusion on the convergence of the upper semi-continuous regularizations follows now from the properties of sequences of qpsh functions that we have established in the previous chapter.

It remains to prove (5). By Choquet's lemma, there exists an increasing sequence $\varphi_j \in PSH(X,\omega)$ such that $\varphi_j \le 0$ on E and $V_E^* = (\sup_j \varphi_j)^*$. Set $G_j := \{\varphi_j < 1/j\}$. This defines a decreasing sequence of open subsets containing E. Observe that $\varphi_j - 1/j \le V_{G_j} \le V_E$, hence $\lim \varphi_j \le \lim V_{G_j} \le V_E$. Therefore $V_E^* = \lim V_{G_j}^*$. □

9.4.2 Alexander–Taylor capacity

Definition 9.20. *Let K be a Borel subset of X. We set*

$$T_\omega(K) := \exp\left(-\sup_X V_{K,\omega}^*\right).$$

This capacity characterizes again $PSH(X,\omega)$-polar sets:

Proposition 9.21. *Let P be a Borel subset. Then $T_\omega(P) = 0$ iff P is $PSH(X,\omega)$-polar. Moreover, if $\varphi \in PSH(X,\omega)$, then*

$$T_\omega(\varphi < -t) \le C_\varphi \exp(-t), \ \forall t \in \mathbb{R},$$

where $C_\varphi = \exp(-\sup_X \varphi)$.

Proof. The first assertion follows from Theorem 9.17. Fix $\varphi \in PSH(X,\omega)$, $t \in \mathbb{R}$ and set $K_t = \{\varphi < -t\}$. Then $\varphi + t \le 0$ on K_t, hence $\varphi + t \le V_{K_t,\omega}^*$. We infer that $\sup_X \varphi + t \le \sup_X V_{K_t,\omega}^*$, hence $T_\omega(K_t) \le \exp(-\sup_X \varphi)\exp(-t)$. □

It follows from Proposition 9.19 that T_ω is a generalized capacity in the sense of Choquet which is outer regular.

Example 9.22. *Assume $X = \mathbb{CP}^n$, ω is the Fubini–Study Kähler form, and B_R is the Euclidean ball centered at the origin and of radius R in an affine chart $\mathbb{C}^n \subset \mathbb{CP}^n$.*

The reader will explicitly compute the extremal function in this case (Exercise 9.8). This yields

$$T_\omega(B_R) = \frac{R}{\sqrt{1+R^2}}.$$

Observe that $T_\omega(B_R) \overset{0}{\sim} R$, *showing the optimality of the rate of decrease in the previous proposition.*

The capacity $T_{\omega_{\mathrm{FS}}} = T_\omega$ has to be related to the capacity $T_{\mathbb{B}^n}$ which measures compact subsets of $\mathbb{C}^n$. It is defined as follows: given a Borel subset K of $\mathbb{C}^n$, $T_{\mathbb{B}^n}(K) := \exp(-\sup_{\mathbb{B}^n} L_K)$, where

$$L_K(z) = \sup\{v(z) \mid v \in \mathcal{L}(\mathbb{C}^n),\ \sup_K v \le 0\}$$

is the Siciak's extremal function of K and $\mathcal{L}(\mathbb{C}^n)$ denotes the Lelong class of psh functions with logarithmic growth in $\mathbb{C}^n$. One easily checks that

$$V_{K,\omega} - \log\sqrt{2} \le L_K - \frac{1}{2}\log[1+|z|^2] \le V_{K,\omega} \text{ in } \mathbb{C}^n.$$

We infer $\sup_{\mathbb{B}^n} L_K \le \log\sqrt{2} + \sup_{\mathbb{CP}^n} V_{K,\omega}$ hence $T_{\mathbb{B}^n}(K) \ge 2^{-1/2} T_\omega(K)$.

We also have a reverse inequality. Indeed $\forall \varphi \in PSH(\mathbb{CP}^n, \omega)$, $\sup_{\mathbb{CP}^n} \varphi \le \sup_{\mathbb{B}^n} \varphi + C_1$, where $C_1 = \sup_{\mathbb{CP}^n} V_{\mathbb{B}^n,\omega} = \log\sqrt{2}$. Therefore

$$\sup_{\mathbb{CP}^n} V_{K,\omega} \le \sup_{\mathbb{B}^n} V_{K,\omega} + \log\sqrt{2} \le \sup_{\mathbb{B}^n} L_K + \log 2,$$

which yields

$$\frac{1}{\sqrt{2}} T_\omega(K) \le T_{\mathbb{B}^n}(K) \le 2T_\omega(K).$$

Example 9.23. *Assume* $X = \mathbb{CP}^n$, $\omega = \omega_{\mathrm{FS}}$ *and consider the totally real subspace* $\mathbb{RP}^n$ *of points with real coordinates* (*closure of* $\mathbb{R}^n \subset \mathbb{C}^n \subset \mathbb{CP}^n$). *Then*

$$\frac{1}{2(1+\sqrt{2})} \le T_\omega(\mathbb{RP}^n) \le 1.$$

Indeed, set $B_{\mathbb{R}^n} := \mathbb{R}^n \cap \mathbb{B}^n$. *It follows from the discussion above that*

$$T_\omega(\mathbb{RP}^n) \ge \frac{1}{2} T_{\mathbb{B}^n}(\mathbb{B}_{\mathbb{R}^n}).$$

Now there is an explicit formula for $L^*_{B_{\mathbb{R}^n}}$ (*Lundin's formula, see* [Klim]),

$$L^*_{B_{\mathbb{R}^n}}(z) = \sup\{\log^+ |h(\langle z, \xi\rangle)| \mid \|\xi\| = 1\},\ z \in \mathbb{C}^n,$$

where $h(\zeta) = \zeta + \sqrt{\zeta^2 - 1}$. *A simple computation yields* $|h(\zeta)| \leq \log\left(|z| + \sqrt{|z|^2 + 1}\right)$ *for* $\zeta = \langle z, \xi\rangle$ *with* $|\xi| = 1$. *We infer*

$$L_{\mathbb{B}_{\mathbb{R}^n}}(z) \leq \log\left(|z| + \sqrt{|z|^2 + 1}\right) \leq \log\left(1 + \sqrt{2}\right) \quad \text{in } \mathbb{B}^n,$$

which yields the desired inequality.

Observe that the minorant is independent of the dimension n.

9.4.3 Chebyshev constants In this section we consider the case where $\{\omega\} = c_1(L)$ is the first Chern class of a positive holomorphic line bundle L on X. We let h denote a smooth metric of L on X whose curvature is ω. For $s \in \Gamma(X, L^N)$, we let $\|s\|_{Nh}$ denote the norm of s computed with respect to the metric Nh.

For a given Borel subset K of X, we define its *Chebyshev constants*

$$M_{N\omega}(K) := \inf\left\{\sup_K \|s\|_{Nh}\,;\, s \in \Gamma(X, L^N),\ \sup_X \|s\|_{Nh} = 1\right\}.$$

Note that a simple rescaling argument shows that $M_{N\omega}$ remains unchanged if we replace h by $h + C$ so that it only depends on $\omega = \Theta_h$. Consider

$$T'_\omega(K) := \inf_{N \geq 1} [M_{N\omega}(K)]^{1/N}.$$

Theorem 9.24. *Let* K *be a compact subset of* X. *Then*

$$T_\omega(K) = T'_\omega(K).$$

Proof. The core of the proof consists in showing that

$$V_{K,\omega}(x) = \sup\left\{\frac{1}{N}\log\|s\|_{Nh}(x)|\ N \geq 1, s \in \Gamma\left(X, L^N\right),\ \sup_K \|s\|_{Nh} \leq 1\right\}.$$

Note that for any s involved in the supremum, $\varphi := N^{-1}\log\|s\|_{Nh}$ belongs to $PSH(X, \omega)$ and satisfies $\varphi \leq 0$ on K. Therefore $\varphi \leq V_{K,\omega}$.

Conversely, fix $x_0 \in X$ and $a < V_{K,\omega}(x_0)$. Fix $\varphi \in PSH(X, \omega)$ such that $\sup_K \varphi \leq 0$ and $\varphi(x_0) > a$. Regularizing φ and translating, we can assume $\varphi \in PSH(X, \omega) \cap \mathcal{C}^\infty(X)$, $\sup_K \varphi < 0$, and $\varphi(x_0) > a$. Fix $\varepsilon > 0$. Let $B = B(x_0, r)$ be a small ball on which $\varphi > a$. We choose B so small that the oscillation of h is smaller than ε on B. Let χ be a test function with compact support in B and such that $\chi \equiv 1$ in $B(x_0, r/2)$. We can assume w.l.o.g. that $B \subset \mathcal{U}_{\alpha_0}$ for some α_0, but $B \cap \mathcal{U}_\beta = \emptyset$ for all $\beta \neq \alpha_0$. This ensures that χ is a smooth section of L^N for all $N \geq 1$.

Let ψ_1 be a smooth positive metric of $L^{N_1} \otimes K_X^*$ on X (this is possible if N_1 is chosen large enough since L is positive). Let ψ_2 be a positive metric of L^{N_2} on X which is smooth in $X \setminus \{x_0\}$ and with Lelong number $\nu(\psi_2, x_0) \geq n = \dim_{\mathbb{C}} X$

(this is again possible if N_2 is large enough, since L is ample). Observe that $\overline{\partial}\chi$ is a smooth $\overline{\partial}$-closed $(0,1)$-form with values in L^N (for all $N \geq 1$). Alternatively, it is a smooth $\overline{\partial}$-closed $(n,1)$-form with values in $L^N \otimes K_X^*$. Applying Hörmander's L^2-estimates (see [Dem, Chapter VIII]) with weight

$$\psi_N := (N - N_1 - N_2)(\varphi + h) + \psi_1 + \psi_2,$$

we find a smooth section f of L^N such that $\overline{\partial} f = \overline{\partial}\chi$ and

$$\int_X |f|^2 e^{-2(N-N_1-N_2)(\varphi+h)-2\psi_1-2\psi_2} dV_\omega \leq C_1 \int_X |\overline{\partial}\chi|^2 e^{-2\psi_N} dV_\omega.$$

Note that $\overline{\partial}\chi$ has support in $B \setminus B(x_0, r/2)$, where ψ_N is smooth so that both integrals are finite. Since $\nu(\psi_2, x_0) \geq n$, this forces $f(x_0) = 0$. The second integral is actually bounded from above by $C_2 e^{-2N(a-\varepsilon)}$, where C_2 is independent of N, since $-\varphi < -a$ on B and the oscillation of h is smaller than ε on B. Therefore, $s := \chi - f \in \Gamma(X, L^N)$ satisfies $s(x_0) = 1$ and

$$\int_X |s|^2 e^{-2N(\varphi+h)} dV_\omega \leq C_3 e^{-2N(a-\varepsilon)},$$

where C_3 is independent of N. Now $\varphi < 0$ in a neighborhood of K, so the mean value inequality applied to the subharmonic functions $|s_\alpha|^2$ yields, for all x in K,

$$\begin{aligned} |s|^2 e^{-2Nh}(x) &\leq C_\delta \int_{B(x,\delta)} |s|^2(y) e^{-2N[\varphi+h](y)} e^{2N[h(y)-h(x)+\varphi(y)]} d\lambda(y) \\ &\leq C_4 e^{-2N(a-\varepsilon)}, \end{aligned}$$

if δ is so small that $|\sup_{B(x,\delta)} \varphi| > 0$ is bigger than the oscillation of h on $B(x,\delta)$. Therefore, $S := C_4^{-1/2} e^{N(a-\varepsilon)} s \in \Gamma(X, L^N)$ satisfies $\sup_K \|S\|_{Nh} \leq 1$, and $N^{-1} \log \|S\|_{Nh}(x_0) \geq a - \varepsilon - \frac{\log C_4}{2N}$. Letting $N \to +\infty$, $\varepsilon \to 0$ and $a \to V_{K,\omega}(x_0)$ completes the proof of the equality.

To conclude observe that by rescaling one gets

$$\begin{aligned} -\log T_\omega(K) &= \sup_X V_{K,\omega} \\ &= \sup \left\{ \frac{1}{N} \sup_X \log \|S\|_{Nh} ;\ N \geq 1, S \in \Gamma(X, L^N) \text{ and } \sup_K \|S\|_{Nh} = 1 \right\} \\ &= \sup \left\{ -\frac{1}{N} \sup_K \log \|S\|_{Nh} ;\ N \geq 1, S \in \Gamma(X, L^N) \text{ and } \sup_X \|S\|_{Nh} = 1 \right\} \\ &= -\log T'_\omega(K). \end{aligned}$$

□

9.4.4 The weighted theory For applications, it is important to study how previous capacities depend on the cohomology class $\{\omega\} \in H^{1,1}(X, \mathbb{R})$ and its representatives. To do so, it is very convenient to generalize previous constructions and

add a weight $h : X \to \mathbb{R}$. Our purpose here is to study a very special case, by using Berman–Demailly's regularizing result: recall that

$$P(h)(x) := \sup\{\varphi(x) \,|\, \varphi \in PSH(X,\omega) \text{ and } \varphi \leq h\}$$

denotes the greatest ω-psh function that lies below h. It has bounded Laplacian, whenever h is $\mathcal{C}^2$-smooth. For $h \equiv 0$, $P(h) \equiv 0$ equals $V_{X,\omega}$. In general, the function $P(h)$ should be interpreted as a global extremal function with respect to the weight h. One can compute its Monge–Ampère measure as follows:

Proposition 9.25. *Assume $h : X \to \mathbb{R}$ is $\mathcal{C}^2$-smooth. Then*

$$(\omega + dd^c P(h))^n = \mathbf{1}_{\{h=P(h)\}}(\omega + dd^c h)^n.$$

Proof. Set $u = P(h) \in PSH(X,\omega)$. It follows again from a *balayage* argument that $MA(u)$ vanishes on the open set $\{P(h) < h\}$. Indeed, Choquet's lemma ensures that the envelope u is achieved along an increasing sequence (u_j) of ω-psh functions. If B is a small "ball" inside $\{P(h) < h\}$, we can consider $v_j \in PSH(X,\omega)$ such that $v_j \equiv u_j$ in $X \setminus B$ and

$$MA(v_j) = 0 \text{ in } B.$$

Then $v_j \geq u_j$ and we still have $v_j \leq h$ by continuity if B is small enough. Therefore $u = \lim \nearrow v_j$, hence $MA(u) = 0$ in B by continuity of the Monge–Ampère operator along increasing sequences. Since B was arbitrary, this shows that

$$MA(u) = \mathbf{1}_D MA(u), \text{ where } D := \{h = P(h)\}.$$

It follows from Theorem 8.18 that u has bounded Laplacian, hence the mixed derivatives $\partial^2 u/\partial z_j \partial \overline{z}_k$ are (locally) bounded, hence in L^p for all $p > 1$. Using the theory of singular integral operators (as presented e.g., in [Stein]), we infer that the full real Hessian $D^2\varphi$ (i.e., all second derivatives $\partial^2 u/\partial x_j \partial x_k$, $1 \leq j,k \leq 2n$) is in L^p. Thus they exist almost everywhere and it follows from a double use of Lemma 9.26 below that they coincide with $\partial^2 h/\partial x_j \partial x_k$ on the contact set $D := \{P(h) = h\}$.

Since $\partial^2 u/\partial z_j \partial \overline{z}_k \in L^n_{\text{loc}}$, its Monge–Ampère measure is absolutely continuous with respect to the Lebesgue measure and can be computed (almost everywhere) by taking the pointwise wedge product. Since $\omega + dd^c u = \omega + dd^c h$ on D, we infer that

$$MA(u) = \mathbf{1}_D(\omega + dd^c h)^n,$$

as claimed. □

We let the reader prove the following classical lemma from Lebesgue theory:

Lemma 9.26. *Let v be a real valued function defined in an open set $\Omega \subset \mathbb{R}^k$ such that the directional derivatives $\partial v/\partial x_i$ exist almost everywhere. If v vanishes on a Borel set $D \subset \Omega$, then $\partial v/\partial x_i = 0$ almost everywhere on D.*

9.5 Exercises

Exercise 9.1. *Let μ be a probability measure on $X = \mathbb{P}^2$ and let ω denote the Fubini–Study Kähler form. Show that*

$$\varphi_\mu(x) := \int_{\mathbb{P}^2} \log\left(\frac{\|x \wedge y\|}{\|x\| \cdot \|y\|}\right) d\mu(y)$$

defines a ω-psh function on $\mathbb{P}^2$ whose gradient is in L^2.

Verify that $MA(\varphi_\mu)$ has a Dirac mass at point a if so has μ. Show that $MA(\varphi_\mu)$ is absolutely continuous with respect to the Lebesgue measure if φ_μ is bounded on $\mathbb{P}^2$.

Exercise 9.2. *We assume here $X = \mathbb{CP}^2$ is the complex projective space, and $\omega = \omega_{\mathrm{FS}}$ is the Fubini–Study Kähler form. Let $\varepsilon_j > 0$ be a sequence of real numbers which decrease (fast enough) towards 0. Check that the functions*

$$\varphi_j[z_0 : z_1 : z_2] = \frac{1}{2j} \log\left[\varepsilon_j |z_0|^{2j} + |z_0^j + z_1^j + z_2^j|^2\right] - \log\|z\|$$

are ω-psh and bounded, and converge in $L^1(X)$ to

$$\varphi[z_0 : z_1 : z_2] = \log \max_{0 \le j \le 2} |z_j| - \log\|z\|.$$

Verify that $MA(\varphi)$ is the (normalized) Lebesgue measure on the 2-dimensional torus

$$\mathbf{T}^2 := \{[z] \in \mathbb{CP}^2 \,|\, |z_0| = |z_1| = |z_2|\},$$

while the Monge–Ampère measures $MA(\varphi_j)$ converge towards the (normalized) Lebesgue measure on the 1-dimensional torus

$$S^1 := \{[z] \in \mathbb{CP}^2 \,|\, z_0 = 0 \text{ and } |z_1| = |z_2|\}.$$

Exercise 9.3. *Let (X, ω) be a compact Kähler manifold and D an ample divisor such that $[D] = \omega + dd^c\varphi$. Estimate precisely the rate of decreasing, as $t \to +\infty$, of*

$$\mathrm{Cap}(\varphi < -t) \quad \text{and} \quad T_\omega(\varphi < -t).$$

Exercise 9.4. *Let (X, ω) be a compact Kähler manifold, $p \in X$ and let $\mathbb{B}(p, r)$ denote the ball centered at p of radius $r > 0$ with respect to the distance d_ω. Estimate the rate of decreasing, as $r \to 0$, of*

$$\mathrm{Cap}(\mathbb{B}(p, r)) \quad \text{and} \quad T_\omega(\mathbb{B}(p, r)).$$

Exercise 9.5. *Show that if $\varphi_j \in PSH(X, \omega) \cap L^\infty(X)$ decrease to $\varphi \in PSH(X, \omega) \cap L^\infty(X)$, then for all $u \in PSH(X, \omega) \cap L^\infty(X)$,*

$$\int_X u MA(\varphi_j) \longrightarrow \int_X u MA(\varphi).$$

Exercise 9.6. *Let D be an ample divisor on a compact projective manifold X. Fix a Hodge form ω and $\varphi \in PSH(X,\omega)$ such that $dd^c\varphi = [D] - \omega$, with $\sup_X \varphi = 0$. Consider $\varphi_c = \max(\varphi, -c) \in PSH(X,\omega) \cap \mathcal{C}^0(X)$. Then $\varphi_c \equiv \varphi$ outside some neighborhood $V_c = \{\varphi < -c\}$ of D. Show that*

$$\mathrm{Cap}_\omega(V_c) = \mathrm{Cap}_\omega(X)$$

for all $c \leq 1$.

Exercise 9.7. *Assume $X = \mathbb{CP}^n$, $\omega = \omega_{\mathrm{FS}}$, and*

$$\varphi[z:t] = \log|t| - \log\|(z,t)\|,$$

where z denotes the Euclidean coordinates in $\mathbb{C}^n = \mathbb{CP}^n \setminus \{t = 0\}$ and $D = \{t = 0\}$ is the hyperplane at infinity. Check that $\sup_{\mathbb{CP}^n} \varphi = 0$ and, following the previous exercise, that the Monge–Ampère capacity of the complement of a Euclidean ball of radius $\sqrt{e^2 - 1}$ equals 1.

Exercise 9.8. *Assume $X = \mathbb{CP}^n$, ω is the Fubini–Study Kähler form, and let B_R denote the Euclidean ball centered at the origin and of radius R in $\mathbb{C}^n \subset \mathbb{CP}^n$. Show that for $x \in \mathbb{C}^n$,*

$$V_{B_R,\omega}(x) = \max\left(\log\frac{\|x\|}{R} + \frac{1}{2}\log[1 + R^2] - \frac{1}{2}\log[1 + \|x\|^2]; 0\right).$$

Exercise 9.9. *Let K be a Borel subset of a compact Kähler manifold (X,ω). Establish the following properties:*

(1) *If $K' \subset K$ then, $V_{X,\omega} \leq V_{K,\omega} \leq V_{K',\omega}$. Moreover, $V_{X,\omega} \equiv 0$.*

(2) *If $\omega_1 \leq \omega_2$, then $V_{K,\omega_1} \leq V_{K,\omega_2}$.*

(3) *For all $A > 0$, $V_{K,A\omega} = A \cdot V_{K,\omega}$.*

(4) *If $\omega' = \omega + dd^c\chi$, then $\inf_X \chi \leq V_{K,\omega'} - V_{K,\omega} + \chi \leq \sup_X \chi$.*

(5) *If $f : X \to X$ is holomorphic, then $V_{f(K),\omega} \circ f \leq V_{K,f^*\omega}$.*

In particular, if f is a ω-isometry, then $V_{f(K),\omega} = V_{K,\omega}$.

Exercise 9.10. *Establish the following properties:*

(1) *For all Borel subsets $K' \subset K \subset X$,*

$$T_\omega(K') \leq T_\omega(K) \leq T_\omega(X) = 1.$$

(2) *If $\omega_1 \leq \omega_2$, then $T_{\omega_1}(\cdot) \geq T_{\omega_2}(\cdot)$.*

(3) *Forall $A > 0$, $T_{A\omega}(\cdot) = [T_\omega(\cdot)]^A$.*

(4) *If ω and ω' are Kähler forms, then there exists $C \geq 1$ such that*

$$[T_\omega(\cdot)]^C \leq T_{\omega'}(\cdot) \leq [T_\omega(\cdot)]^{1/C}.$$

(5) *If* $\omega' = \omega + dd^c\chi$ *then*

$$C^{-1}T_\omega(\cdot) \le T_{\omega'}(\cdot) \le C \cdot T_\omega(\cdot),$$

where $C = \exp(\sup_X \chi - \inf_X \chi) \ge 1$.

(6) *If* $f : X \to X$ *is a holomorphic map, then* $T_{f^*\omega}(\cdot) \le T_\omega \circ f(\cdot)$. *In particular, if* f *is a* ω*-isometry, then* $T_\omega \circ f = T_\omega$.

Exercise 9.11. *Assume* $(X, \omega) = (\mathbb{P}^1, \omega_{\mathrm{FS}})$. *Give an example of a smooth function* $h : \mathbb{P}^1 \to \mathbb{R}$ *whose envelope* $P(h)$ *is not* $\mathcal{C}^2$*-smooth.*

Exercise 9.12. *Let* v *be a real valued function defined in an open set* $\Omega \subset \mathbb{R}^k$ *such that the directional derivatives* $\partial v/\partial x_i$ *exist almost everywhere. Show that if* v *vanishes on a Borel set* $D \subset \Omega$, *then* $\partial v/\partial x_i = 0$ *almost everywhere on* D.

10 Finite energy classes

Our goal in this chapter is to extend the definition of the complex Monge–Ampère operator to a class of functions $\mathcal{E}(X,\omega)$ which contains ω-psh functions which are not necessarily bounded, in such a way that the comparison principle is still valid.

Once again we assume that ω is a semi-positive form which is *big*, i.e., such that

$$V := \int_X \omega^n > 0.$$

10.1 The class $\mathcal{E}(X,\omega)$

10.1.1 Definition Given $\varphi \in PSH(X,\omega)$, we consider its canonical approximants

$$\varphi_j := \max(\varphi, -j) \in PSH(X,\omega) \cap L^\infty(X).$$

It follows from Bedford–Taylor theory that the measures

$$MA(\varphi_j) := \frac{1}{V}(\omega + dd^c\varphi_j)^n$$

are well defined probability measures. Since the φ_j's are decreasing, it is natural to expect that these measures converge (in the weak sense). The following strong monotonicity property holds:

Lemma 10.1. *The sequence $\mu_j := \mathbf{1}_{\{\varphi>-j\}}MA(\varphi_j)$ is an increasing sequence of Borel measures.*

Proof. It follows from the maximum principle that

$$\mathbf{1}_{\{\varphi_j>-k\}}[\omega + dd^c\varphi_j]^n = \mathbf{1}_{\{\varphi_j>-k\}}[\omega + dd^c\max(\varphi_j,-k)]^n.$$

Now if $j \geq k$, then $(\varphi_j > -k) = (\varphi > -k)$ and $\max(\varphi_j,-k) = \varphi_k$, thus

$$\mathbf{1}_{\{\varphi>-k\}}[\omega + dd^c\varphi_j]^n = \mathbf{1}_{\{\varphi>-k\}}[\omega + dd^c\varphi_k]^n. \tag{10.1.1}$$

Observe also that $\{\varphi > -k\} \subset \{\varphi > -j\}$, therefore

$$j \geq k \Longrightarrow \mathbf{1}_{\{\varphi>-j\}}[\omega + dd^c\varphi_j]^n \geq \mathbf{1}_{\{\varphi>-k\}}[\omega + dd^c\varphi_k]^n,$$

in the weak sense of Borel measures. □

Since the μ_j's all have total mass bounded from above by 1 (the total mass of the measure $MA(\varphi_j)$), we can consider

$$\mu_\varphi := \lim_{j\to+\infty} \mu_j,$$

which is a positive Radon measure on X, with total mass ≤ 1.

Definition 10.2. *The finite energy class is*

$$\mathcal{E}(X,\omega) := \left\{\varphi \in PSH(X,\omega) \mid \mu_\varphi(X) = 1\right\}.$$

Of course, every bounded ω-psh function belongs to $\mathcal{E}(X,\omega)$ since in this case $\{\varphi > -j\} = X$ for j large enough, hence

$$\mu_\varphi \equiv \mu_j = MA(\varphi_j) = MA(\varphi).$$

The class $\mathcal{E}(X,\omega)$ also contains many ω-psh functions which are unbounded, as we shall see later on (see Exercise 10.9). When X is a compact Riemann surface ($n = \dim_{\mathbb{C}} X = 1$), the set $\mathcal{E}(X,\omega)$ is the set of ω-subharmonic functions whose Laplacian does not charge polar sets (see Exercise 10.1).

Remark 10.3. *The functions which belong to the class $\mathcal{E}(X,\omega)$, although usually unbounded, have relatively mild singularities. In particular, they have zero Lelong numbers when ω is a Kähler form, as the reader will check in Exercice 10.7.*

It is an interesting open problem to prove such is the case when ω is merely semi-positive and big.

Let us stress that the convergence of μ_j towards μ_φ holds in the (strong) sense of Borel measures, i.e., for all Borel sets B,

$$\mu_\varphi(B) := \lim_{j\to+\infty} \mu_j(B).$$

In particular, when $B = P$ is a pluripolar set, we obtain $\mu_j(P) = 0$, hence

$$\mu_\varphi(P) = 0 \text{ for all pluripolar sets } P.$$

We shall see in the next chapter that conversely, a probability measure μ equals μ_φ for some $\varphi \in \mathcal{E}(X,\omega)$ whenever μ does not charge pluripolar sets (one then says that μ is non-pluripolar).

Notation 10.4. *For $\varphi \in \mathcal{E}(X,\omega)$, we set $MA(\varphi) := \mu_\varphi$.*

As we shall see in the sequel, this notation is justified since functions φ in $\mathcal{E}(X,\omega)$ have a well-defined Monge–Ampère measure which turns out to be μ_φ! For the time being, we observe that the Monge–Ampère measures $MA(\varphi_j)$ of the canonical approximants $\varphi_j = \max(\varphi, -j)$ converge to μ_φ in a rather strong sense:

Proposition 10.5. *Let $\varphi \in \mathcal{E}(X,\omega)$. Then for all bounded Borel function b,*

$$\langle \mu_\varphi, b\rangle = \lim_{j\to+\infty} \langle MA(\varphi_j), b\rangle, \tag{10.1.2}$$

where $\varphi_j := \max(\varphi, -j)$. In particular,

$$\mathbf{1}_{\{\varphi>-j\}} MA(\varphi)(B) = \mathbf{1}_{\{\varphi>-j\}} MA(\varphi_j)(B) \tag{10.1.3}$$

for all Borel subsets $B \subset X$.

Proof. Observe that $\varphi \in \mathcal{E}(X,\omega)$ if and only if

$$MA(\varphi_j)\{\varphi \le -j\} \longrightarrow 0, \quad \text{when } j \to +\infty$$

where $\varphi_j := \max(\varphi, -j)$. We infer that for all Borel subsets $B \subset X$,

$$MA(\varphi)(B) := \lim_{j\to+\infty} \int_{B\cap\{\varphi>-j\}} MA(\varphi_j) = \lim_{j\to+\infty} \int_B MA(\varphi_j).$$

This yields (10.1.2), by using the Lebesgue dominated convergence theorem.

Since $MA(\varphi_j)$ does not charge any pluripolar set B, the same property holds for $MA(\varphi)$. Equality (10.1.3) now follows from (10.1.2) and the corresponding equality on bounded approximants. □

10.1.2 Comparison principles Our goal in this section is to establish various types of maximum principles that are going to be quite useful in the sequel.

We start by extending the *comparison principle* to the class $\mathcal{E}(X,\omega)$.

Proposition 10.6. *Let $\varphi, \psi \in \mathcal{E}(X,\omega)$, then*

$$\int_{\{\varphi<\psi\}} MA(\psi) \le \int_{\{\varphi<\psi\}} MA(\varphi).$$

Proof. Set $\varphi_j = \max(\varphi, -j)$ and $\psi_k = \max(\psi, -k)$. The comparison principle for bounded ω-psh functions yields

$$\int_{\{\varphi_j<\psi_k\}} MA(\psi_k) \le \int_{\{\varphi_j<\psi_k\}} MA(\varphi_j).$$

Observe that $\{\varphi_j < \psi\} \subset \{\varphi_j < \psi_k\} \subset \{\varphi < \psi_k\}$ hence

$$\int_{\{\varphi_j<\psi\}} MA(\psi_k) \le \int_{\{\varphi<\psi_k\}} MA(\varphi_j).$$

Letting $k \to +\infty$ yields, by Proposition 10.5,

$$\int_{\{\varphi_j<\psi\}} MA(\psi) \le \int_{\{\varphi\le\psi\}} MA(\varphi_j).$$

Letting now $j \to +\infty$, we infer $\int_{\{\varphi<\psi\}} MA(\psi) \le \int_{\{\varphi\le\psi\}} MA(\varphi)$. The desired inequality follows by replacing ψ by $\psi - \varepsilon$, $\varepsilon > 0$, and letting $\varepsilon \to 0^+$. □

Proposition 10.7. *The set $\mathcal{E}(X,\omega)$ is convex. It is an extremal face of the convex set $PSH(X,\omega)$ which is stable under the max operation: if $\varphi,\psi\in PSH(X,\omega)$ are such that $\varphi\le\psi$ and $\varphi\in\mathcal{E}(X,\omega)$, then $\psi\in\mathcal{E}(X,\omega)$.*

Recall that an extremal face F of a convex set C is a convex subset of C such that for all $\varphi,\psi\in C$, if $t\varphi+(1-t)\psi\in F$ for some $0<t<1$, then φ and ψ are in F.

Proof. The proof follows from the comparison principle together with the following elementary observation: *If $\varphi\in PSH(X,\omega)$ and $\varphi/2\in\mathcal{E}(X,\omega)$, then $\varphi\in\mathcal{E}(X,\omega)$.* Indeed set $u=\varphi/2$, $u_j:=\max(u,-j)$, and $\varphi_j:=\max(\varphi,-j)$. Observe that $u_j=\varphi_{2j}/2$ and

$$\omega_{u_j}:=\omega+dd^c(\varphi_{2j}/2)=\frac{1}{2}(\omega+\omega_{\varphi_{2j}})\ge\frac{1}{2}\omega_{\varphi_{2j}},$$

therefore

$$\begin{aligned}\int_{\{\varphi\le-2j\}}(\omega+dd^c\varphi_{2j})^n&=\int_{\{u\le-j\}}(\omega+dd^c\varphi_{2j})^n\\&\le 2^n\int_{\{u\le-j\}}(\omega+dd^cu_j)^n\longrightarrow 0.\end{aligned}$$

We now use the comparison principle to show that if $\varphi\in\mathcal{E}(X,\omega)$, $\psi\in PSH(X,\omega)$ are such that $\varphi\le\psi$, then $\psi/2\in\mathcal{E}(X,\omega)$, so that $\psi\in\mathcal{E}(X,\omega)$ by the previous observation. Set $v:=\psi/2$ and $v_j:=\max(v,-j)$, $\varphi_j:=\max(\varphi,-j)$. We can assume without loss of generality that $\varphi\le\psi\le-2$, hence $v\le-1$. It follows from $\varphi\le\psi$ that

$$\{v\le-j\}\subset\left\{\varphi_{2j}<v_j-j+1\right\}\subset\{\varphi\le-j\},$$

where the last inclusion simply uses $v_j\le-1$. We infer that

$$\begin{aligned}\omega_{v_j}^n\{v\le-j\}&\le\int_{\{\varphi_{2j}<v_j-j+1\}}\omega_{v_j}^n\le\int_{\{\varphi_{2j}<v_j-j+1\}}\omega_{\varphi_{2j}}^n\\&\le\omega_{\varphi_{2j}}^n\{\varphi\le-j\}\to 0.\end{aligned}$$

Thus, $v=\psi/2\in\mathcal{E}(X,\omega)$.

Using the comparison principle again, we now show that if $\varphi,\psi\in\mathcal{E}(X,\omega)$, then $(\varphi+\psi)/4\in\mathcal{E}(X,\omega)$. It follows then from our first observation that $(\varphi+\psi)/2\in\mathcal{E}(X,\omega)$, thus $\mathcal{E}(X,\omega)$ is convex. Set $w:=(\varphi+\psi)/4$, $w_j:=\max(w,-j)$, $\varphi_j:=\max(\varphi,-j)$, and $\psi_j:=\max(\psi,-j)$. Observe that

$$\{w\le-j\}\subset\{\varphi\le-2j\}\cup\{\psi\le-2j\},$$

thus it suffices to show that $\omega_{w_j}^n\{\varphi\le-2j\}\to 0$. Assuming, as above, that $\varphi,\psi\le-2$ yields $\{\varphi\le-2j\}\subset\left\{\varphi_{2j}<w_j-j+1\right\}\subset\{\varphi\le-j\}$, hence

$$\omega_{w_j}^n\{\varphi\le-2j\}\le\omega_{\varphi_{2j}}^n\{\varphi\le-j\}\longrightarrow 0.$$

We let the reader check, along similar lines, that $\mathcal{E}(X,\omega)$ is an extremal face of $PSH(X,\omega)$. □

We now extend the *maximum principle* to the class $\mathcal{E}(X,\omega)$.

Corollary 10.8. *If $\varphi \in \mathcal{E}(X,\omega)$, $\psi \in PSH(X,\omega)$, then* $\max(\varphi,\psi) \in \mathcal{E}(X,\omega)$ *and*

$$\mathbf{1}_{\{\varphi>\psi\}} MA(\varphi) = \mathbf{1}_{\{\varphi>\psi\}} MA(\max(\varphi,\psi)).$$

Proof. By the previous proposition that $u := \max(\varphi,\psi)$ belongs to $\mathcal{E}(X,\omega)$. Set $\varphi_j = \max(\varphi,-j)$, $\psi_{j+1} = \max(\psi,-j-1)$ and $u_j := \max(u,-j)$. Observe that $\max(\varphi_j,\psi_{j+1}) = \max(\varphi,\psi,-j) = u_j$. The maximum principle for bounded function yields

$$\mathbf{1}_{\{\varphi_j>\psi_{j+1}\}}(\omega + dd^c\varphi_j)^n = \mathbf{1}_{\{\varphi_j>\psi_{j+1}\}}(\omega + dd^c u_j)^n.$$

Recall that $\mathbf{1}_{\{\varphi>\psi\}}\omega^n_{\varphi_j} \to \mathbf{1}_{\{\varphi>\psi\}}\omega^n_{\varphi}$. Now

$$\{\varphi>\psi\} \subset \{\varphi_j > \psi_{j+1}\} \text{ and } \{\varphi_j>\psi_{j+1}\} \setminus \{\varphi>\psi\} \subset \{\varphi \le -j\},$$

hence

$$0 \le \left[\mathbf{1}_{\{\varphi_j>\psi_{j+1}\}} - \mathbf{1}_{\{\varphi>\psi\}}\right]\omega^n_{\varphi_j} \le \mathbf{1}_{\{\varphi\le -j\}}\omega^n_{\varphi_j} \to 0,$$

since $\varphi \in \mathcal{E}(X,\omega)$. This shows that

$$\mathbf{1}_{\{\varphi_j>\psi_{j+1}\}}(\omega + dd^c\varphi_j)^n \longrightarrow \mathbf{1}_{\{\varphi>\psi\}}(\omega + dd^c\varphi)^n.$$

One proves similarly that $\mathbf{1}_{\{\varphi_j>\psi_{j+1}\}}(\omega+dd^c u_j)^n \to \mathbf{1}_{\{\varphi>\psi\}}(\omega+dd^c u)^n$, observing that $\{\varphi_j > \psi_{j+1}\} \setminus \{\varphi>\psi\} \subset \{\max[\varphi,\psi] \le -j\}$. □

The following result will be used on several occasions, it is due to Bedford and Taylor [BT82] when φ,ψ are bounded:

Proposition 10.9. *Assume $\varphi,\psi \in \mathcal{E}(X,\omega)$ are such that $MA(\varphi) \ge \mu$ and $MA(\psi) \ge \mu$ for some positive Borel measure μ on X. Then*

$$MA(\max(\varphi,\psi)) \ge \mu.$$

Observe that the measure μ is *non-pluripolar* (i.e., μ does not charge pluripolar sets), since so are $MA(\varphi)$, $MA(\psi)$.

Proof. It follows from previous corollary that $\max(\varphi,\psi) \in \mathcal{E}(X,\omega)$ and

$$MA(\max(\varphi,\psi)) \ge \mathbf{1}_{\{\varphi>\psi\}} MA(\varphi) + \mathbf{1}_{\{\varphi<\psi\}} MA(\psi) \ge \mathbf{1}_{\{\varphi\ne\psi\}}\mu.$$

Thus we are done if $\mu\{\varphi=\psi\} = 0$.

Assume now that $\mu\{\varphi=\psi\} > 0$. We show hereafter that $\mu(\{\varphi = \psi + t_0\}) = 0$ for all $t_0 \in \mathbb{R} \setminus I_\mu$ where I_μ is at most countable. Assuming this, we can find a

decreasing sequence $\varepsilon_j \searrow 0$ such that $\mu\left(\{\varphi = \psi + \varepsilon_j\}\right) = 0$. Replacing ψ by $\psi + \varepsilon_j$ above yields

$$[\omega + dd^c \max(\varphi, \psi + \varepsilon_j)]^n \geq \mu.$$

The desired inequality therefore follows from the continuity of the complex Monge–Ampère operator along decreasing sequences (see Theorem 10.18).

It remains to show that $I_\mu := \{t_0 \in \mathbb{R} \,|\, \mu(\{\varphi = \psi + t_0\}) > 0\}$ is at most countable. Consider $f : t \in \mathbb{R} \mapsto \mu(\{\varphi < \psi + t\}) \in \mathbb{R}^+$. This is an increasing function which is left continuous since μ is a Borel measure. Moreover

$$\lim_{\substack{t > t_0 \\ t \to t_0}} f(t) = \mu(\{\varphi \leq \psi + t_0\} \setminus \{\psi = -\infty\}) = \mu(\{\varphi \leq \psi + t_0\})$$

since $\mu(\{\psi = -\infty\}) = 0$. Therefore f is continuous at t_0 unless $\mu(\{\varphi = \psi + t_0\}) > 0$. Thus the set I_μ coincides with the set of discontinuity of f which is at most countable. □

Remark 10.10. *The same proof shows the following useful inequality which goes back to Demailly* [Dem91]*:*

$$MA(\max(\varphi, \psi)) \geq \mathbf{1}_{\{\varphi > \psi\}} MA(\varphi) + \mathbf{1}_{\{\varphi \leq \psi\}} MA(\psi),$$

whenever $\varphi, \psi \in \mathcal{E}(X, \omega)$.

We finally consider the *domination principle*.

Proposition 10.11. *Assume* $\varphi, \psi \in \mathcal{E}(X, \omega)$ *are such that*

$$\psi(x) \leq \varphi(x) \quad \textit{for } MA(\varphi)\textit{-almost every } x \in X.$$

Then $\psi(x) \leq \varphi(x)$ *for all* $x \in X$.

Proof. We only treat here the case when φ and ψ are bounded. We can assume in this case that $\psi \geq 0$ (shifting φ and ψ by a constant).

Fix $0 < \delta < 1$. It follows from Lemma 12.9 that for all $t > 0$,

$$\delta^n \mathrm{Cap}(\{\varphi - \psi < -t - \delta\}) \leq \int_{\{\varphi - \psi < -t - \delta\, \psi\}} MA(\varphi) \leq \int_{\{\varphi - \psi < -t\}} MA(\varphi).$$

Our assumption ensures that the right-hand side vanishes, hence

$$\varphi(x) - \psi(x) \geq -t - \delta \quad \text{for all } x \in X \setminus P,$$

where P has zero capacity, hence zero Lebesgue measure. The mean value property of qpsh functions ensures then that the above inequality holds everywhere on X. Letting t, δ decrease to zero yields the desired result.

The proof for finite energy functions requires slightly more sophisticated arguments and will be given in Proposition 12.30. □

10.2 Weighted energies

10.2.1 The classes $\mathcal{E}_\chi(X,\omega)$ Fix $\varphi \in \mathcal{E}(X,\omega)$. Since $MA(\varphi) := \mu_\varphi$ does not charge the pluripolar set $\{\varphi = -\infty\}$, one can construct a continuous increasing function $h : \mathbb{R}^+ \to \mathbb{R}^+$ such that $h(+\infty) = +\infty$ and $h \circ |\varphi| \in L^1(MA(\varphi))$. This integrability property can actually be checked on the canonical approximants:

Lemma 10.12. *Fix $\varphi \in \mathcal{E}(X,\omega)$ and let $h : \mathbb{R}^+ \to \mathbb{R}^+$ be a continuous increasing function such that $h(+\infty) = +\infty$. Then*

$$\int_X h \circ |\varphi| MA(\varphi) < +\infty \Longleftrightarrow \sup_{j\geq 0} \int_X h \circ |\varphi_j| MA(\varphi_j) < +\infty,$$

where $\varphi_j := \max(\varphi, -j)$. Moreover, if this holds then for all Borel subsets B, then

$$\int_B h \circ |\varphi_j| MA(\varphi_j) \longrightarrow \int_B h \circ |\varphi| MA(\varphi). \tag{10.2.1}$$

Proof. We can assume without loss of generality that $\varphi, \varphi_j \leq 0$. Assume first that $\sup_{j\geq 0} \int_X h \circ |\varphi_j| MA(\varphi_j) < +\infty$. Since the Borel measures $h(-\varphi_j)MA(\varphi_j)$ have uniformly bounded masses, they form a weakly compact sequence. Let ν be a cluster point. Since $h(-\varphi_j)$ increase towards $h(-\varphi)$ and $MA(\varphi_j)$ converges towards $MA(\varphi)$, it follows from semi-continuity that $h(-\varphi)MA(\varphi) \leq \nu$, hence

$$\int_X h(-\varphi) MA(\varphi) \leq \nu(X) < +\infty.$$

Conversely, assume that $h(-\varphi) \in L^1(MA(\varphi))$. It follows from (10.1.3) that

$$\int_{\{\varphi\leq -j\}} MA(\varphi_j) = \int_X MA(\varphi_j) - \int_{\{\varphi > -j\}} MA(\varphi_j) = \int_{\{\varphi\leq -j\}} MA(\varphi).$$

Thus by (10.1.3) again,

$$\begin{aligned}\int_X h(-\varphi_j) MA(\varphi_j) &= h(j) \int_{\{\varphi\leq -j\}} MA(\varphi_j) + \int_{\{\varphi > -j\}} h(-\varphi) MA(\varphi_j)\\ &= \int_{\{\varphi\leq -j\}} h(j) MA(\varphi) + \int_{\{\varphi > -j\}} h(-\varphi) MA(\varphi)\\ &\leq \int_X h(-\varphi) MA(\varphi).\end{aligned}$$

Moreover, if $B \subset X$ is a Borel subset, then

$$\begin{aligned}&\left| \int_B h(-\varphi_j) MA(\varphi_j) - \int_B h(-\varphi) MA(\varphi) \right|\\ &\qquad \leq \int_{B\cap\{\varphi\leq -j\}} h(-\varphi_j) MA(\varphi_j) + \int_{B\cap\{\varphi\leq -j\}} h(-\varphi) MA(\varphi)\\ &\qquad \leq 2 \int_{\{\varphi\leq -j\}} h(-\varphi) MA(\varphi) \longrightarrow 0.\end{aligned}$$

□

Observe that the stronger the growth of h at $+\infty$, the better the control on $MA(\varphi)$. We can assume without loss of generality that h is smooth. It is actually more convenient to consider $\chi(t) := -h(-t)$:

Definition 10.13. *A weight is a smooth increasing function $\chi : \mathbb{R} \to \mathbb{R}$ such that $\chi(-\infty) = -\infty$. We let $\mathcal{W}$ denote the set of weights.*

We let $\mathcal{W}_+$ denote the set of weights that are convex and normalized so that $\chi \leq 0$ on $\mathbb{R}^-$ and $\chi' \leq 1$. For $\varphi \in PSH(X,\omega) \cap L^\infty(X)$, we set

$$E_\chi(\varphi) := \frac{1}{(n+1)V} \sum_{j=0}^{n} \int_X \chi \circ \varphi \, (\omega + dd^c\varphi)^j \wedge \omega^{n-j}.$$

Observe that convex weights have smaller growth at $-\infty$. Note also that if $\varphi \in PSH(X,\omega)$, then $\chi \circ \varphi \in PSH(X,\omega)$ when $\chi \in \mathcal{W}_+$, since

$$dd^c \chi \circ \varphi = \chi'' \circ \varphi \, d\varphi \wedge d^c\varphi + \chi' \circ \varphi \, dd^c\varphi \geq -\chi' \circ \varphi \, \omega \geq -\omega.$$

Proposition 10.14. *Fix $\chi \in \mathcal{W}_+$ and $\varphi, \psi \in PSH(X,\omega) \cap L^\infty(X)$. Then*

(i) $\varphi \leq \psi \Rightarrow E_\chi(\varphi) \leq E_\chi(\psi)$;

(ii) $j \geq k \Rightarrow \int_X \chi \circ \varphi \, (\omega + dd^c\varphi)^j \wedge \omega^{n-j} \leq \int_X \chi \circ \varphi \, (\omega + dd^c\varphi)^k \wedge \omega^{n-k}$.

In particular,

$$\int_X \chi \circ \varphi \, MA(\varphi) \leq E_\chi(\varphi) \leq \frac{1}{n+1} \int_X \chi \circ \varphi \, MA(\varphi),$$

where the latter inequality only holds when $\varphi \leq 0$.

Proof. To prove (i) we set $v = \psi - \varphi \geq 0$ and $\varphi_t = \varphi + tv = (1-t)\varphi + t\psi$ (which is ω-psh and bounded). It suffices to show that $t \mapsto E_\chi(\varphi_t)$ is increasing on $[0,1]$. Now

$$\begin{aligned}(n+1)V \frac{dE_\chi(\varphi_t)}{dt} &= \sum_{j=0}^{n} \int_X v\chi' \circ \varphi_t (\omega + dd^c\varphi_t)^j \wedge \omega^{n-j} \\ &\quad + \sum_{j=0}^{n} j \int_X \chi \circ \varphi_t \, dd^c v \wedge (\omega + dd^c\varphi_t)^{j-1} \wedge \omega^{n-j}.\end{aligned}$$

Integrating by part in the last integral yields

$$\begin{aligned}&\int_X \chi \circ \varphi_t \, dd^c v \wedge (\omega + dd^c\varphi_t)^{j-1} \wedge \omega^{n-j} \\ &\quad \geq \int_X v\chi' \circ \varphi_t \, (\omega_{\varphi_t} - \omega) \wedge (\omega + dd^c\varphi_t)^{j-1} \wedge \omega^{n-j}\end{aligned}$$

since $v \geq 0$ and $dd^c \chi \circ \varphi_t \geq \chi' \circ \varphi_t (\omega_{\varphi_t} - \omega)$.

Rearranging the second sum yields a telescopic series and we end up with

$$(n+1)V\frac{dE_\chi(\varphi_t)}{dt} \geq n\int_X v\chi'\circ\varphi_t(\omega+dd^c\varphi_t)^n \geq 0.$$

We now prove (ii) when $j = k+1$ (the general case follows by induction). It suffices to observe that if T is a "sufficiently regular" closed positive current of bidimension $(1,1)$, then by Stokes' theorem,

$$\int_X \chi\circ\varphi\omega_\varphi\wedge T = \int_X \chi\circ\varphi\omega\wedge T - \int_X \chi'\circ\varphi d\varphi\wedge d^c\varphi\wedge T \leq \int_X \chi\circ\varphi\omega\wedge T.$$

Using this inequality when $T = \omega_\varphi^j\wedge\omega^{n-j-1}$ yields the case $k = j$. □

The monotonicity property (i) motivates the following definition:

Definition 10.15. *Fix $\chi\in\mathcal{W}$. For $\varphi\in PSH(X,\omega)$ we set*

$$E_\chi(\varphi) := \inf\{E_\chi(\psi)\mid \psi\ \omega\text{-psh and bounded with } \varphi\leq\psi\}$$

and

$$\mathcal{E}_\chi(X,\omega) := \{\varphi\in PSH(X,\omega)\mid E_\chi(\varphi) > -\infty\}.$$

Proposition 10.16. *We have*

$$\mathcal{E}(X,\omega) = \bigcup_{\chi\in\mathcal{W}}\mathcal{E}_\chi(X,\omega) = \bigcup_{\chi\in\mathcal{W}_+}\mathcal{E}_\chi(X,\omega).$$

Observe that these unions are increasing in the sense that $\mathcal{E}_\chi(X,\omega)\subset\mathcal{E}_{\tilde\chi}(X,\omega)$ whenever $\tilde\chi = O(\chi)$ is dominated by χ at $-\infty$.

Proof. We first observe that each class $\mathcal{E}_\chi(X,\omega)$ is included in $\mathcal{E}(X,\omega)$. Indeed if $\varphi\in\mathcal{E}_\chi(X,\omega)$ and $\varphi_j := \max(\varphi,-j)$, it follows from Proposition 10.14 that

$$\int_X \chi\circ\varphi_j\, MA(\varphi_j) \geq (n+1)E_\chi(\varphi_j) \geq (n+1)E_\chi(\varphi)$$

is uniformly bounded below. Therefore

$$\int_{(\varphi\leq -j)} MA(\varphi_j) \leq \frac{C}{|\chi(-j)|}\longrightarrow 0,$$

as $j\to+\infty$, so that $\varphi\in\mathcal{E}(X,\omega)$.

Conversely assume that $\varphi\in\mathcal{E}(X,\omega)$ and set again $\varphi_j = \max(\varphi,-j)$. Since the probability measure $MA(\varphi) := \mu_\varphi$ does not charge the pluripolar set $\{\varphi=-\infty\}$, we can construct an increasing function $\chi:\mathbb{R}\to\mathbb{R}$ such that $\chi(-\infty) = -\infty$ and $\chi\circ\varphi\in L^1(\mu_\varphi)$.

We can further assume that $\chi\in\mathcal{W}_+$ (see Exercise 10.4). It follows now from Lemma 10.12 and Proposition 10.14 that $E_\chi(\varphi_j)$ is uniformly bounded below. By Proposition 10.14 again, this yields a uniform bound from below for $E_\chi(\psi)$, where $\psi\geq\varphi$ is ω-psh and bounded. Thus $\varphi\in\mathcal{E}_\chi(X,\omega)$. □

Remark 10.17. *Classes of high energies (when the weight χ is concave) are less useful than classes of low energies (when $\chi \in \mathcal{W}_+$). They nevertheless allow to interpolate nicely between the class $\mathcal{E}(X,\omega)$ and the class $PSH(X,\omega) \cap L^\infty(X)$ as the reader will check in Exercise 10.2.*

10.2.2 Continuity of twisted Monge–Ampère operators We now show that the complex Monge–Ampère operator is well defined on the class $\mathcal{E}(X,\omega)$, by establishing several continuity properties. Recall that we have set $MA(\varphi) := \mu_\varphi$ when $\varphi \in \mathcal{E}(X,\omega)$.

Theorem 10.18. *Fix $\varphi \in \mathcal{E}(X,\omega)$ and let $\varphi_j \in \mathcal{E}(X,\omega)$ be a monotone sequence converging[1] to φ. Then $MA(\varphi_j)$ weakly converge to $MA(\varphi)$. More precisely, if $\varphi, \varphi_j \in \mathcal{E}_\chi(X,\omega)$ for some $\chi \in \mathcal{W}_+$, then*

(i) *the measures $\chi\circ\varphi_j\, MA(\varphi_j)$ weakly converge to $\chi\circ\varphi\, MA(\varphi)$;*
(ii) *for any bounded ω-psh function u, $\int_X u MA(\varphi_j) \to \int_X u MA(\varphi)$.*

Proof. The idea behind the proof of this result is quite simple: since these results are known to hold when the φ_j's are uniformly bounded, we'll try to reduce to that case approximating φ, φ_j by

$$\varphi^{(K)} := \max(\varphi, -K), \quad \varphi_j^{(K)} := \max(\varphi_j, -K).$$

Of course this necessitates to establish uniform bounds, which are going to follow from uniform controls on the energies.

We first prove (i). We can assume $\varphi_j, \varphi \le 0$. We fix $\tilde\chi \in \mathcal{W}$ a weight such that $\chi = o(\tilde\chi)$ at $-\infty$ and still $\varphi, \varphi_j \in \mathcal{E}_{\tilde\chi}(X,\omega)$[2]. Let θ be a test function. Then

$$\begin{aligned}
&\left|\int \theta\left[\chi\circ\varphi_j^{(K)} MA\left(\varphi_j^{(K)}\right) - \chi\circ\varphi_j MA(\varphi_j)\right]\right| \\
&\quad\le \|\theta\|_{L^\infty}\left|\frac{\chi(-K)}{\tilde\chi(-K)}\right| \int_{(\varphi_j\le -K)}\left[|\tilde\chi|\circ\varphi_j^{(K)} MA\left(\varphi_j^{(K)}\right) + |\tilde\chi|\circ\varphi_j MA\left(\varphi_j\right)\right] \\
&\quad\le \|\theta\|_{L^\infty}\left|\frac{\chi(-K)}{\tilde\chi(-K)}\right|\left[\left|E_{\tilde\chi}\left(\varphi_j^{(K)}\right)\right| + \left|E_{\tilde\chi}\left(\varphi_j\right)\right|\right](n+1)V \\
&\quad\le C\left|\frac{\chi(-K)}{\tilde\chi(-K)}\right|.
\end{aligned}$$

Since $\chi(-K)/\tilde\chi(-K) \to 0$ and for each fixed K

$$\chi\circ\varphi_j^{(K)} MA\left(\varphi_j^{(K)}\right) \overset{j\to+\infty}{\longrightarrow} \chi\circ\varphi^{(K)} MA\left(\varphi^{(K)}\right),$$

this yields the desired result (i).

The proof of (ii) is completely similar. □

[1] The convergence necessary holds at every point when the sequence is decreasing, it holds almost everywhere with respect to Lebesgue measure when the sequence is increasing.

[2] We can always find such a weight, why?

The following semi-continuity result will also be useful:

Proposition 10.19. *Fix $\chi \in \mathcal{W}_+$. The energy map $\varphi \mapsto E_\chi(\varphi)$ is upper semi-continuous for the L^1-topology. It is continuous along decreasing sequences.*

Proof. Let (φ_j) be a sequence of ω-psh functions which converge in $L^1(X)$ to $\varphi \in PSH(X,\omega)$. We wish to show that

$$\limsup E_\chi(\varphi_j) \leq E_\chi(\varphi).$$

If $\limsup E_\chi(\varphi_j) = -\infty$ there is nothing to prove, so we can assume without loss of generality that $\varphi_j \in \mathcal{E}_\chi(X,\omega)$ and $E_\chi(\varphi_j) \geq -C$ is uniformly bounded below. Set

$$\Phi_j := \left(\sup_{l\geq j} \varphi_l\right)^*,$$

so that $\Phi_j \geq \varphi_j$ and Φ_j decreases pointwise to φ.

Fix ψ a bounded ω-psh such that $\psi \geq \varphi$. Then $\max(\psi,\Phi_j) \geq \psi$ decrease to $\max(\psi,\varphi) = \psi$. It follows from the continuity properties of complex Monge–Ampère operators acting on *bounded* ω-psh functions and the monotonicity property of E_χ that

$$E_\chi(\psi) = \lim E_\chi\left(\max(\psi,\Phi_j)\right) \geq \limsup E_\chi(\Phi_j) \geq \limsup E_\chi(\varphi_j).$$

Taking the infimum over all ψ's dominating φ yields, by definition,

$$E_\chi(\varphi) \geq \limsup E_\chi(\varphi_j).$$

When (φ_j) is decreasing, the monotonicity property of E_χ yields the reverse inequality $E_\chi(\varphi) \leq \liminf E_\chi(\varphi_j)$, whence continuity. □

10.2.3 Mixed Monge–Ampère measures Let $\omega_1,\dots,\omega_n$ be Kähler forms and fix $\varphi_i \in PSH(X,\omega_i)$. The maximum principle shows (as above when $\varphi_1 = \cdots = \varphi_n$) that the measures $\mu_j(\varphi_1,\dots,\varphi_n) =$

$$\mathbf{1}_{\{\varphi_1>-j\}\cap\cdots\cap\{\varphi_n>-j\}}(\omega_1 + dd^c\max(\varphi_1,-j)) \wedge \ldots \wedge (\omega_n + dd^c\max(\varphi_n,-j))$$

form an increasing sequence of Borel measures, whose mass is uniformly bounded from above by $\{\omega_1\}\cdots\{\omega_n\}$.

Definition 10.20. *We set*

$$\mu(\varphi_1,\dots,\varphi_n) := \lim_{j\to+\infty} \mu_j(\varphi_1,\dots,\varphi_n).$$

We let the reader check in Exercise 10.8 the following assertions:

Lemma 10.21.
1. *For every permutation* σ, $\mu(\varphi_1, \ldots, \varphi_n) = \mu(\varphi_{\sigma(1)}, \ldots, \varphi_{\sigma(n)})$.
2. *The mapping* $(\varphi_1, \ldots, \varphi_n) \mapsto \mu(\varphi_1, \ldots, \varphi_n)$ *is multilinear.*

It follows from these observations and the convexity of $\mathcal{E}(X, \omega)$ that:

Proposition 10.22. *When* $\varphi_1, \ldots, \varphi_n \in \mathcal{E}(X, \omega)$, *the mixed Monge–Ampère operators*

$$(\varphi_1, \ldots, \varphi_n) \longmapsto V^{-1}(\omega + dd^c\varphi_1) \wedge \ldots \wedge (\omega + dd^c\varphi_n)$$

are well-defined probability measures which have the same continuity properties as $\varphi \mapsto MA(\varphi)$.

Proof. Consider indeed, for any $t = (t_1, \ldots, t_n) \in \mathbb{R}^n_+$,

$$P(t_1, \ldots, t_n) := \mu\left(\sum_{i=1}^n t_i\varphi_i\right).$$

The multilinearity of μ ensures that P is a homogeneous polynomial of degree n. Since $\mathcal{E}(X, \omega)$ is convex, we moreover see that

$$P(t_1, \ldots, t_n) = (t_1 + \cdots + t_n)^n$$

when the t_i's lie on the simplex

$$\Delta := \{t \in \mathbb{R}^n_+ \ , \ t_1 + \cdots + t_n = 1\}.$$

By homogeneity, the equality therefore holds in $\mathbb{R}^n_+$, i.e.,

$$P(t_1, \ldots, t_n) = (t_1 + \cdots + t_n)^n$$

for all t. Identifying the coefficients of the monomial $t_1 \cdots t_n$ on both sides shows that $\mu(\varphi_1, \ldots, \varphi_n)$ is a probability measure.

The continuity properties follow similarly from the corresponding results for $\varphi \mapsto \mu(\varphi, \ldots, \varphi)$. □

10.2.4 Compactness properties

Proposition 10.23. *Fix* $\chi \in \mathcal{W}_+$ *and* $C > 0$. *The set*

$$\mathcal{E}_{\chi,C}(X, \omega) := \left\{\varphi \in \mathcal{E}_\chi(X, \omega) \mid E_\chi(\varphi) \geq -C \text{ and } \varphi \leq 0\right\}$$

is compact for the L^1*-topology.*

Proof. We observe that, since χ is increasing,

$$\chi(\varphi) \leq \chi\left(\sup_X \varphi\right) \quad \text{hence} \quad E_\chi(\varphi) \leq \chi\left(\sup_X \varphi\right).$$

Recall that $\chi(t) \to -\infty$ as $t \to -\infty$, therefore

$$\mathcal{E}_{\chi,C}(X,\omega) \subset \{\varphi \in PSH(X,\omega) \mid -C_\chi \leq \sup_X \varphi \leq 0\},$$

the latter set being a compact subset of $PSH(X,\omega)$ by Hartogs' lemma.

Since $\varphi \mapsto E_\chi(\varphi)$ is upper semi-continuous, the set $\mathcal{E}_{\chi,C}(X,\omega)$ is also closed, hence compact for the L^1-topology. □

Example 10.24. *Assume $n = 1$, i.e., X is a compact Riemann surface and $\chi(t) \equiv t$. In this case, the class $\mathcal{E}_\chi(X,\omega)$ is denoted by $\mathcal{E}^1(X,\omega)$ and*

$$\mathcal{E}^1(X,\omega) = \{\varphi \in PSH(X,\omega) \mid \nabla\varphi \in L^2\},$$

since

$$E_\chi(\varphi) = \frac{1}{2V}\left\{\int_X \varphi\omega_\varphi + \int_X \varphi\omega\right\} = -\frac{1}{2V}\int_X d\varphi \wedge d^c\varphi + \frac{1}{V}\int_X \varphi\omega.$$

The set $\mathcal{E}_{\chi,C}(X,\omega)$ in this case is (contained in) the intersection of $PSH(X,\omega)$ with a finite ball in the Sobolev space $W^{1,2}(X)$, the latter is indeed compact for the weak topology.

10.2.5 Integrability properties

Proposition 10.25. *Fix $\chi \in \mathcal{W}_+$ and $0 \geq \varphi, \psi \in \mathcal{E}_\chi(X,\omega)$. Then*

$$2(n+1)\left\{E_\chi(\varphi) + E_\chi(\psi)\right\} \leq \int_X \chi\circ\varphi\, MA(\psi) \leq 0.$$

In particular, $\chi \circ \mathcal{E}_\chi(X,\omega) \subset L^1(MA(\mathcal{E}_\chi(X,\omega)))$.

Proof. Recall that

$$(n+1)E_\chi(\varphi) \leq \int_X \chi\circ\varphi\, MA(\varphi).$$

Thus it suffices to show that

$$\int_X \chi\circ\varphi MA(\psi) \geq 2\left\{\int_X \chi\circ\varphi\, MA(\varphi) + \int_X \chi\circ\psi\, MA(\psi)\right\}.$$

Observe that $\chi'(2t) \leq \chi'(t)$ for all $t < 0$ since χ is convex, hence

$$\int_X (-\chi)\circ\varphi\,\omega_\psi^n = \int_{-\infty}^0 \chi'(t)\omega_\psi^n\{\varphi < t\}\,dt \leq 2\int_{-\infty}^0 \chi'(t)\omega_\psi^n\{\varphi < 2t\}\,dt.$$

Now $\{\varphi < 2t\} \subset \{\varphi < \psi + t\} \cup \{\psi < t\}$, hence

$$\int_X (-\chi)\circ\varphi\,\omega_\psi^n \le 2\int_{-\infty}^0 \chi'(t)\omega_\psi^n\{\varphi < \psi + t\}\,dt + 2\int_X (-\chi)\circ\psi\,\omega_\psi^n.$$

The comparison principle yields $\omega_\psi^n\{\varphi < \psi + t\} \le \omega_\varphi^n\{\varphi < \psi + t\}$. The desired inequality follows by observing that $\{\varphi < \psi + t\} \subset \{\varphi < t\}$. □

When the weight $\chi \in \mathcal{W}$ is rather concave, functions in $\mathcal{E}_\chi(X,\omega)$ are less singular, as the following property indicates:

Proposition 10.26. *Let $\chi \in \mathcal{W}$ be a concave weight. Then all functions in $\mathcal{E}_\chi(X,\omega)$ have gradient in L^2.*

Proof. It follows from the concavity of χ that $\mathrm{Id} = O(\chi)$ at $-\infty$, hence

$$\mathcal{E}_\chi(X,\omega) \subset \mathcal{E}^1(X,\omega),$$

where $\mathcal{E}^1(X,\omega)$ is the class that corresponds to the identity weight $\chi(t) \equiv t$ (we shall investigate its properties further in the next section).

Thus it suffices to prove the result for $\chi(t) \equiv t$. Fix $\varphi \in \mathcal{E}^1(X,\omega)$ so that $E_{\mathrm{Id}}(\varphi) > -\infty$. We can assume without loss of generality that φ is normalized so that $\int_X \varphi\omega^n = 0$. It follows then from the Stokes theorem that

$$\begin{aligned} +\infty > -E_{\mathrm{Id}}(\varphi) &= \frac{1}{(n+1)V}\sum_{j=0}^n \int_X \varphi(\omega^j - \omega_\varphi^j)\wedge\omega^{n-j} \\ &= \frac{1}{(n+1)V}\sum_{j=1}^n\sum_{\ell=0}^{j-1}\int_X d\varphi\wedge d^c\varphi\wedge\omega_\varphi^\ell\wedge\omega^{n-\ell+1} \\ &\ge \frac{1}{(n+1)V}\int_X d\varphi\wedge d^c\varphi\wedge\omega^{n-1}, \end{aligned}$$

which yields the desired result. □

One can even show in this case that functions in $\mathcal{E}_\chi(X,\omega)$ belong to the local domain of definition of the complex Monge–Ampère operator (see Section 10.4).

10.3 The class $\mathcal{E}^1(X,\omega)$

The following class will play a central role in the sequel:

Definition 10.27. *We let $\mathcal{E}^1(X,\omega)$ denote the class $\mathcal{E}_\chi(X,\omega)$ for $\chi(t) \equiv t$.*

Note that the weight $\chi = \mathrm{Id}$ is both convex and concave, hence for the class $\mathcal{E}^1(X,\omega)$ all results from both categories hold. This class even enjoys further properties as we now explain.

10.3.1 Primitive of the Monge–Ampère operator

We simply denote by

$$E(\varphi) = \frac{1}{(n+1)V}\sum_{j=0}^{n}\int_X \varphi(\omega + dd^c\varphi)^j \wedge \omega^{n-j}.$$

the corresponding functional E_{Id}.

Proposition 10.28. *The functional E is a primitive of the complex Monge–Ampère operator, namely whenever $\varphi + tv$ belongs to $\mathcal{E}^1(X,\omega)$,*

$$\left.\frac{dE(\varphi+tv)}{dt}\right|_{t=0} = \int_X v MA(\varphi).$$

The functional E is concave increasing, satisfies $E(\varphi + c) = E(\varphi) + c$ for all $c \in \mathbb{R}$, $\varphi \in \mathcal{E}^1(X,\omega)$, and the cocycle condition

$$E(\varphi) - E(\psi) = \frac{1}{(n+1)V}\sum_{j=0}^{n}\int_X (\varphi-\psi)(\omega + dd^c\varphi)^j \wedge (\omega + dd^c\psi)^{n-j}.$$

for all $\varphi, \psi \in \mathcal{E}^1(X,\omega)$.

Note that the concavity of E implies, in particular, that the set

$$\mathcal{E}^1_C(X,\omega) := \{\varphi \in \mathcal{E}^1(X,\omega) \mid E(\varphi) \geq -C \text{ and } \varphi \leq 0\}$$

is *convex*. It was previously known to be compact (Proposition 10.23).

Proof. We have already computed the derivative of $E_\chi(\varphi + tv)$ for a general weight $\chi \in \mathcal{W}_+$ in Proposition 10.14. The estimate we have obtained there by bounding from below $dd^c\chi \circ \varphi_t$ by $\chi' \circ \varphi_t(\omega_{\varphi_t} - \omega)$ becomes an equality here. Thus the same use of the Stokes theorem yields the desired result (we leave the details as an exercise for the reader), which we formally write as $E' = MA$.

If we formally replace ω by ω_ψ and φ by $\varphi - \psi$ in all these computations (this amounts to changing the representative of the cohomology class $\{\omega\}$), we end up similarly with

$$g'(t) = \left.\frac{dG(\varphi+tv)}{dt}\right|_{t=0} = \int_X v\frac{(\omega_\psi + dd^c[\varphi-\psi])^n}{V} = \int_X v MA(\varphi),$$

where

$$G(\varphi) := \frac{1}{(n+1)V}\sum_{j=0}^{n}\int_X (\varphi-\psi)(\omega + dd^c\varphi)^j \wedge (\omega + dd^c\psi)^{n-j}.$$

For $v = \psi - \varphi$ and $e(t) = E(\varphi + tv)$, we therefore obtain $e' \equiv g'$ and $e(0) = E(\varphi), e(1) = E(\psi)$, while $g(0) = G(\varphi)$ and $g(1) = G(\psi) = 0$. This yields the cocycle condition $G(\varphi) = E(\varphi) - E(\psi)$.

The invariance under translations $E(\varphi + c) = E(\varphi) + c$ is immediate: it remains to show that E is concave (it is increasing as $E' = MA$ is a non-negative measure). Set again, for $v = \psi - \varphi$,

$$e(t) := E(\varphi + tv) = E(t\psi + (1-t)\varphi).$$

We have already shown that

$$e'(t) = \int_X vMA(t\psi + (1-t)\varphi) = V^{-1}\int_X v\left(t\omega_\psi + (1-t)\omega_\varphi\right)^n.$$

Taking derivative again yields

$$\begin{aligned} e''(t) &= \frac{n}{V}\int_X vdd^c v \wedge \left(t\omega_\psi + (1-t)\omega_\varphi\right)^{n-1} \\ &= -\frac{n}{V}\int_X dv \wedge d^c v \wedge \left(t\omega_\psi + (1-t)\omega_\varphi\right)^{n-1} \leq 0, \end{aligned}$$

as follows from the Stokes theorem. □

10.3.2 Capacity of sublevel sets

Lemma 10.29. *Fix $\varphi \in PSH(X,\omega)$. If*

$$\int_0^{+\infty} t^n \mathrm{Cap}_\omega\{\varphi < -t\}\, dt < +\infty,$$

then $\varphi \in \mathcal{E}^1(X,\omega)$. Conversely for each $C > 0$,

$$\sup\left\{\int_0^{+\infty} t\,\mathrm{Cap}_\omega\{\varphi < -t\}\, dt \;\middle|\; \varphi \in \mathcal{E}^1_C(X,\omega)\right\} < +\infty.$$

Proof. Fix $\varphi \in PSH(X,\omega)$. We can assume without loss of generality that $\varphi \leq -1$. Observe that for each $t \geq 1$, the function $1 + t^{-1}\max(\varphi, -t)$ is ω-psh with values in $[0,1]$, hence

$$MA(\max(\varphi,-t)) \leq t^n \mathrm{Cap}_\omega.$$

Thus $\int_1^{+\infty} MA(\max\{\varphi,-t\})\{\varphi < -t\}\, dt < +\infty$ if $\int_1^{+\infty} t^n \mathrm{Cap}_\omega\{\varphi < -t\}\, dt < +\infty$. We infer in particular that

$$MA(\max\{\varphi,-t\})\{\varphi < -t\} \longrightarrow 0,$$

so that $\varphi \in \mathcal{E}(X,\omega)$.

Observe that $\varphi \in \mathcal{E}^1(X,\omega)$ iff $\varphi \in \mathcal{E}(X,\omega)$ and $\varphi \in L^1(MA(\varphi))$. We assume without loss of generality that $\varphi \leq -1$. Now

$$\begin{aligned} -\int_X \varphi MA(\varphi) &= \int_1^{+\infty} MA(\varphi)(\varphi \leq -t)dt + \int_X \omega^n \\ &= \int_1^{+\infty} MA(\max\{\varphi,-t\})\{\varphi \leq -t\}\, dt + \int_X \omega^n \end{aligned}$$

is finite, since $MA(\varphi)$ coincides with $MA(\max(\varphi,-t))$ on the Borel set $\{\varphi > -t\}$, by the maximum principle. Thus $\varphi \in \mathcal{E}^1(X,\omega)$, as claimed.

The second assertion follows from the comparison principle. Fix $u \in PSH(X,\omega)$ with values in $[0,1]$. Observe that

$$\{\varphi < -2t\} \subset \left\{t^{-1}\varphi < u - 1\right\} \subset \{\varphi < -t\}\,.$$

It follows therefore from the comparison principle that

$$\int_{\{\varphi<-2t\}} MA(u) \leq \int_{\{\varphi<-t\}} MA(t^{-1}\varphi).$$

Expanding $MA(t^{-1}\varphi) \leq V^{-1}(t^{-1}\omega_\varphi + \omega)^n$ yields

$$\begin{aligned}\int_2^{+\infty} t\,\mathrm{Cap}\,\{\varphi < -t\}\,dt &= 4\int_1^{+\infty} t\,\mathrm{Cap}\,\{\varphi < -2t\}\,dt \\ &\leq 2\int_X \varphi^2\frac{\omega^n}{V} + \frac{4}{V}\sum_{j=1}^n \binom{n}{j}\int_X(-\varphi)\omega_\varphi^j\wedge\omega^{n-j}.\end{aligned}$$

The latter sum is uniformly bounded when $\varphi \in \mathcal{E}^1_C(X,\omega)$ and so is $\int_X \varphi^2\omega^n$, since $-C \leq E(\varphi) \leq \sup_X \varphi \leq 0$. □

10.3.3 Convergence in energy We let $I(\varphi,\psi)$ denote the symmetric (bi)functional

$$I(\varphi,\psi) := \int_X (\psi-\varphi)(MA(\varphi) - MA(\psi)) = -\left(E'(\varphi) - E'(\psi)\right)\cdot(\varphi-\psi),$$

which is non-negative as follows from the concavity of E.

10.3.3.1 A quasi-triangle inequality for I When X has dimension 1,

$$I(\varphi,\psi) = \int_X d(\varphi-\psi)\wedge d^c(\varphi-\psi)$$

coincides with the squared L^2-norm of $d(\varphi-\psi)$ for all $\varphi,\psi \in \mathcal{E}^1(X,\omega_0)$. By Cauchy–Schwarz, $I^{1/2}$ satisfies the triangle inequality, and the convexity inequality $(x+y)^2 \leq 2(x^2+y^2)$ for $x,y\in\mathbb{R}$ yields

$$\tfrac{1}{2}I(\varphi_1,\varphi_2) \leq I(\varphi_1,\varphi_3) + I(\varphi_3,\varphi_2)$$

for any $\varphi_1,\varphi_2,\varphi_3 \in \mathcal{E}^1(X,\omega_0)$. We establish here the following higher dimensional version of this inequality.

Theorem 10.30. *There exists a constant $c_n > 0$, only depending on the dimension n, such that*

$$c_n I(\varphi_1, \varphi_2) \le I(\varphi_1, \varphi_3) + I(\varphi_3, \varphi_1).$$

for all $\varphi_1, \varphi_2, \varphi_3 \in \mathcal{E}^1(X, \omega_0)$.

For any $\varphi_1, \varphi_2, \psi \in \mathcal{E}^1(X, \omega)$ we set

$$\|d(\varphi_1 - \varphi_2)\|_\psi := \left(\int_X d(\varphi_1 - \varphi_2) \wedge d^c(\varphi_1 - \varphi_2) \wedge (\omega + dd^c\psi)^{n-1} \right)^{1/2}, \tag{10.3.1}$$

which is the L^2-norm of $d(\varphi_1 - \varphi_2)$ with respect to ω_ψ when the latter is a Kähler form. It is easy to see that

$$\|d(\varphi_1 - \varphi_2)\|^2_{\frac{\varphi_1+\varphi_2}{2}} \le I(\varphi_1, \varphi_2) \le 2^{n-1} \|d(\varphi_1 - \varphi_2)\|^2_{\frac{\varphi_1+\varphi_2}{2}}. \tag{10.3.2}$$

Lemma 10.31. *There exists $c_n > 0$ only depending on n such that for all $\varphi_1, \varphi_2, \psi \in \mathcal{E}(X, \omega_0)$,*

$$c_n \|d(\varphi_1 - \varphi_2)\|^2_\psi \le I(\varphi_1, \varphi_2)^{1/2^{n-1}} \left(I(\varphi_1, \psi)^{1-1/2^{n-1}} + I(\varphi_2, \psi)^{1-1/2^{n-1}} \right).$$

Proof. Set $u := \varphi_1 - \varphi_2$, $v := (\varphi_1 + \varphi_2)/2$, and for each p,

$$b_p := \int_X du \wedge d^c u \wedge \omega_\psi^p \wedge \omega_v^{n-p-1}.$$

By (10.3.2), we have $b_0 \le I(\varphi_1, \varphi_2)$, while

$$b_{n-1} = \|d(\varphi_1 - \varphi_2)\|^2_\psi$$

is the quantity we are trying to bound. Let us first check that

$$b_{p+1} \le b_p + 4\sqrt{b_p I(\psi, v)} \tag{10.3.3}$$

for $p = 0, \dots, n-2$. Using integration by parts and the identity $dd^c u = \omega_{\varphi_1} - \omega_{\varphi_2}$, we compute

$$\begin{aligned} b_{p+1} - b_p &= \int_X du \wedge d^c u \wedge dd^c(\psi - v) \wedge \omega_\psi^p \wedge \omega_v^{n-p-2} \\ &= -\int_X du \wedge d^c(\psi - v) \wedge dd^c u \wedge \omega_\psi^p \wedge \omega_v^{n-p-2} \\ &= -\int_X du \wedge d^c(\psi - v) \wedge \omega_{\varphi_1} \wedge \omega_\psi^p \wedge \omega_v^{n-p-2} \\ &\quad + \int_X du \wedge d^c(\psi - v) \wedge \omega_{\varphi_2} \wedge \omega_\psi^p \wedge \omega_v^{n-p-2}. \end{aligned}$$

For $i = 1, 2$ the Cauchy–Schwarz inequality yields

$$\begin{aligned}
&\left| \int_X du \wedge d^c(\psi - v) \wedge \omega_{\varphi_i} \wedge \omega_\psi^p \wedge \omega_v^{n-p-2} \right| \\
&\quad \leq \left(\int_X du \wedge d^c u \wedge \wedge \omega_{\varphi_i} \wedge \omega_\psi^p \wedge \omega_v^{n-p-2} \right)^{1/2} \\
&\quad \times \left(\int_X d(\psi - v) \wedge d^c(\psi - v) \wedge \omega_{\varphi_i} \wedge \omega_\psi^p \wedge \omega_v^{n-p-2} \right)^{1/2} \leq 2 b_p^{1/2} I(\psi, v)^{1/2},
\end{aligned}$$

noting that $\omega_{\varphi_i} \leq 2\omega_v$. This proves (10.3.3).

Next we use the convexity of $\varphi \mapsto J_\psi(\varphi)$ to get

$$2I(v, \psi) \leq (n+1) \left(I(\varphi_1, \psi) + I(\varphi_2, \psi) \right).$$

Setting

$$H_n := 8(n+1) \left(I(\varphi_1, \psi) + I(\varphi_2, \psi) \right) \tag{10.3.4}$$

and

$$h(t) := t + \sqrt{H_n t} \tag{10.3.5}$$

we thus have $b_{p+1} \leq h(b_p)$ for $p = 0, \ldots, n-1$ by (10.3.3). Since $b_0 \leq I(\varphi_1, \varphi_2)$ and h is non-decreasing, it follows that

$$\|d(\varphi_1 - \varphi_2)\|_\psi^2 \leq h^{n-1} \left(I(\varphi_1, \varphi_2) \right), \tag{10.3.6}$$

where $h^{n-1} := h \circ \cdots \circ h$ denotes the $(n-1)$-st iterate of $h : \mathbb{R}_+ \to \mathbb{R}_+$.

It is easy to check by induction on $p \in \mathbb{N}$ that

$$h^p(t) \leq 4 H_n^{1-1/2^p} t^{1/2^p} \text{ for } 0 \leq t \leq 2^{-2^{p+1}} H_n \tag{10.3.7}$$

There are two cases. If $I(\varphi_1, \varphi_2) \leq 2^{-2^n} H_n$, we can apply (10.3.7) with $p = n-1$, which combines with (10.3.6) to yield

$$\|d(\varphi_1 - \varphi_2)\|_\psi^2 \leq 4 H_n^{1-1/2^{n-1}} I(\varphi_1, \varphi_2)^{1/2^{n-1}}.$$

We conclude by using the subadditivity of $t \mapsto t^{1/2^{n-1}}$.

Assume now that $I(\varphi_1, \varphi_2) \geq 2^{-2^n} H_n$. We have

$$\begin{aligned}
\|d(\varphi_1 - \varphi_2)\|_\psi &\leq \|d(\varphi_1 - \psi)\|_\psi + \|d(\psi - \varphi_2)\|_\psi, \\
&\leq I(\varphi_1, \psi)^{1/2} + I(\varphi_2, \psi)^{1/2}
\end{aligned}$$

hence

$$\begin{aligned}
\|d(\varphi_1 - \varphi_2)\|_\psi^2 &\leq 2(I(\varphi_1, \psi) + I(\varphi_2, \psi)) = \frac{1}{4(n+1)} H_n \\
&= \frac{1}{4(n+1)} H_n^{1-1/2^{n-1}} H_n^{1/2^{n-1}} \leq \frac{1}{n+1} H_n^{1-1/2^{n-1}} I(\varphi_1, \varphi_2)^{1/2^{n-1}},
\end{aligned}$$

and we conclude as before. □

Proof of Theorem 10.30. Set $\psi := \frac{\varphi_1+\varphi_2}{2}$. The triangle inequality

$$\|d(\varphi_1-\varphi_2)\|_\psi \le \|d(\varphi_1-\varphi_3)\|_\psi + \|d(\varphi_2-\varphi_3)\|_\psi$$

and (10.3.2) yield

$$I(\varphi_1,\varphi_2) \le 2^n \left(\|d(\varphi_1-\varphi_3)\|_\psi^2 + \|d(\varphi_2-\varphi_3)\|_\psi^2\right)$$

Applying Lemma 10.31 we obtain

$$\begin{aligned} c_n I(\varphi_1,\varphi_2) \le\ & I(\varphi_1,\varphi_3)^{1/2^{n-1}} \left(I(\varphi_1,\psi)^{1-1/2^{n-1}} + I(\varphi_3,\psi)^{1-1/2^{n-1}}\right) \\ & + I(\varphi_2,\varphi_3)^{1/2^{n-1}} \left(I(\varphi_2,\psi)^{1-1/2^{n-1}} + I(\varphi_3,\psi)^{1-1/2^{n-1}}\right). \end{aligned}$$

As before, the convexity of J_ψ implies that

$$I(\varphi_1,\psi) \le \frac{n+1}{2} I(\varphi_1,\varphi_2),\ \ I(\varphi_2,\psi) \le \frac{n+1}{2} I(\varphi_1,\varphi_2)$$

and

$$I(\varphi_3,\psi) \le \frac{n+1}{2} \left(I(\varphi_1,\varphi_3) + I(\varphi_2,\varphi_3)\right).$$

Plugging this in the above inequality, we obtain after changing c_n,

$$\begin{aligned} c_n I(\varphi_1,\varphi_2) \le\ & \left(I(\varphi_1,\varphi_3)^{1/2^{n-1}} + I(\varphi_2,\varphi_3)^{1/2^{n-1}}\right) \times \\ & \times \left(I(\varphi_1,\varphi_2)^{1-1/2^{n-1}} + I(\varphi_1,\varphi_3)^{1-1/2^{n-1}} + I(\varphi_2,\varphi_3)^{1-1/2^{n-1}}\right). \end{aligned}$$

Note that we can assume $I(\varphi_1,\varphi_2) \ge \max\{I(\varphi_1,\varphi_3), I(\varphi_2,\varphi_3)\}$, otherwise the usual triangle inequality holds and we are done. Therefore, our last inequality implies that

$$c_n I(\varphi_1,\varphi_2) \le 3 \left(I(\varphi_1,\varphi_3)^{1/2^{n-1}} + I(\varphi_2,\varphi_3)^{1/2^{n-1}}\right) I(\varphi_1,\varphi_2)^{1-1/2^{n-1}},$$

and the result follows thanks to the convexity inequality $(x+y)^{2^{n-1}} \le 2^{2^{n-1}-1}\left(x^{2^{n-1}} + y^{2^{n-1}}\right)$ for $x, y \in \mathbb{R}$. □

10.3.3.2 The strong topology In order to circumvent the discontinuity of the Monge–Ampère operator with respect to the weak topology, we introduce, following [BEGZ10, BBEGZ11], a strong topology that makes it continuous.

Definition 10.32. *The* strong topology *on $\mathcal{E}^1(X,\omega_0)$ is defined as the coarsest refinement of the weak topology such that E become continuous.*

Lemma 10.33. *Every strongly convergent sequence in $\mathcal{E}^1(X,\omega_0)$ is contained in $\mathcal{E}^1_C(X,\omega_0)$ for some $C > 0$.*

Proof. If $\varphi_j \to \varphi$ is a strongly convergent sequence in $\mathcal{E}^1(X,\omega_0)$, then both $\sup_X \varphi_j$ and $E(\varphi_j)$ are convergent sequences, hence are bounded. □

The strong topology corresponds to the *convergence in energy*:

Definition 10.34. *Let $\varphi_j \in \mathcal{E}^1(X,\omega)$ be a sequence normalized by $\int_X \varphi_j dV = 0$. We say that φ_j converges to $\varphi \in \mathcal{E}^1(X,\omega)$ in energy if $I(\varphi_j,\varphi) \to 0$.*

Proposition 10.35. *If $\varphi_j, \varphi \in \mathcal{E}^1_{\rm norm}(X,\omega_0)$ are* normalized *ω_0-psh functions, then $\varphi_j \to \varphi$ in the strong topology iff $I(\varphi_j,\varphi) \to 0$.*

In the sequel we set

$$J_\varphi(\psi) = E(\varphi) - E(\psi) + \int_X (\psi - \varphi) MA(\varphi).$$

Proof. We have

$$\frac{1}{n+1} I(\varphi_j,\varphi) \le E(\varphi) - E(\varphi_j) + \int_X (\varphi_j - \varphi) MA(\varphi) = J_\varphi(\varphi_j) \le I(\varphi_j,\varphi). \tag{10.3.8}$$

If $\varphi_j \to \varphi$ strongly, then all φ_j belong to $\mathcal{E}^1_C(X,\omega_0)$ for some $C > 0$ by Lemma 10.33. It follows that $\int_X (\varphi_j - \varphi) MA(\varphi) \to 0$, hence $I(\varphi_j,\varphi) \to 0$ by (10.3.8).

Assume conversely that $I(\varphi_j,\varphi) \to 0$. Then $\varphi_j \to \varphi$ weakly and it remains to show that $E(\varphi_j) \to E(\varphi)$. Now $\varphi_j \in \mathcal{E}^1_C(X,\omega_0)$ for some fixed $C > 0$, hence $\int_X (\varphi_j - \varphi) MA(\varphi) \to 0$ and we get $E(\varphi_j) \to E(\varphi)$ using again (10.3.8). □

10.3.3.3 $(\mathcal{E}^1(X,\omega), I)$ is a complete metric space We are now going to show that I defines a *complete metrizable uniform structure* on $\mathcal{E}^1_{\rm norm}(X,\omega)$, whose underlying topology is the strong topology.

When $n = \dim_{\mathbb{C}} X = 1$, $I(u,v)^{1/2}$ coincides with the L^2-norm of the gradient of $u - v$. As a consequence, $I^{1/2}$ defines a complete metric space structure on $\mathcal{E}^1_{\rm norm}(X,\omega)$. Since the unit ball of the Sobolev space $W^{1,2}$ is not compact, it is easy to see that the sets $\mathcal{E}^1_C(X,\omega_0)$, even though they are weakly compact, are *not* compact in the strong topology already for $n = 1$.

In higher dimension, the quasi-triangle inequality of Theorem 10.30 implies that I defines a uniform structure, which is furthermore metrizable for general reasons. Let us now show that it is complete.

Proposition 10.36. *Every sequence (φ_j) in $\mathcal{E}^1_{\rm norm}(X,\omega_0)$ such that $\lim_{j,k\to+\infty} I(\varphi_j,\varphi_k) = 0$ converges in the strong topology of $\mathcal{E}^1_{\rm norm}(X,\omega_0)$.*

Proof. Pick j_0 such that $I(\varphi_j,\varphi_k) \le 1$ for all $j,k \ge j_0$. It follows that $J_{\varphi_{j_0}}(\varphi_j)$ is bounded, hence so is $J(\varphi_j)$.

We may thus find $C > 0$ such that $\varphi_j \in \mathcal{E}^1_C(X,\omega_0)$ for all j. Using the Cauchy-like assumption, it is as usual enough to show that some subsequence of

(φ_j) is strongly convergent. By the weak compactness of $\mathcal{E}^1_C(X,\omega_0)$, we may thus assume after perhaps passing to a subsequence that φ_j converges weakly to some $\varphi \in \mathcal{E}^1_C(X,\omega_0)$. Let us show that $\varphi_j \to \varphi$ strongly as well. Let $\varepsilon > 0$, and pick $N \in \mathbb{N}$ such that $I(\varphi_j,\varphi_k) \leq \varepsilon$ for all $j,k \geq N$. We get

$$E(\varphi_j) - E(\varphi_k) + \int_X (\varphi_k - \varphi_j) MA(\varphi_j) = J_{\varphi_j}(\varphi_k) \leq I(\varphi_j,\varphi_k) \leq \varepsilon.$$

Since E is usc in the weak topology and $MA(\varphi_j)$ is weakly continuous on $\mathcal{E}^1_C(X,\omega_0)$, it follows that by letting $k \to \infty$ with j fixed that

$$J_{\varphi_j}(\varphi) = E(\varphi_j) - E(\varphi) + \int_X (\varphi - \varphi_j) MA(\varphi_j) \leq \varepsilon$$

for all $j \geq N$. It follows that $I(\varphi_j,\varphi) \to 0$. □

The convergence in energy is well adapted to the complex Monge–Ampère operator acting on $\mathcal{E}^1(X,\omega)$, as the following result indicates:

Theorem 10.37. *If $\varphi_j \in \mathcal{E}^1(X,\omega)$ converges to $\varphi \in \mathcal{E}^1(X,\omega)$ in energy, then*

$$\int_X \psi MA(\varphi_j) \to \int_X \psi MA(\varphi)$$

and

$$\int_X \varphi_j MA(\psi) \to \int_X \varphi MA(\psi)$$

for all $\psi \in \mathcal{E}^1(X,\omega)$. These convergences are uniform with respect to ψ if the latter stays in a compact set $\mathcal{E}^1_C(X,\omega)$. In particular $\varphi_j \to \varphi$ in capacity.

We leave the proof as an Exercise 10.20 to the reader.

Remark 10.38. *The reader will check that if a sequence $(\varphi_j) \subset \mathcal{E}^1(X,\omega)$ is monotone and converges to $\varphi \in \mathcal{E}^1(X,\omega)$, then the convergence holds in energy.*

10.4 Defining the Monge–Ampère operator

We analyze now the domain of definition of the complex Monge–Ampère operator, i.e., the maximal subclass of ω-psh functions for which $MA(\varphi)$ is a well-defined probability measure, in a sense to be made precise below.

10.4.1 The local domain of definition By comparison with the real Monge–Ampère operator, which is well defined for *all* convex functions, it is not always possible to make sense of $(dd^c\cdot)^n$ for all psh functions, say in the unit ball $\mathbb{B}$ of $\mathbb{C}^n$.

The following example is due to Kiselman:

Example 10.39. *Consider*

$$u : \mathbb{B} \ni (z, w) \longmapsto (|w|^2 - 4)\sqrt{-\log|z|} \in \mathbb{R} \cup \{-\infty\}.$$

The reader will check (Exercise 10.13) that u is a well-defined psh function in the unit ball $\mathbb{B}$ of $\mathbb{C}^2$, which is smooth, except on $\{z = 0\} = \{u = -\infty\}$, and such that

$$\int_{\mathbb{B}(\frac{1}{2})\setminus\{z=0\}} (dd^c u)^2 = +\infty.$$

This example shows that one cannot expect to define the complex Monge–Ampère operator for all psh functions. It was known since the fifties (work of Chern, Levine, and Nirenberg) that one can make sense of $(dd^c\cdot)^n$ for *continuous* psh functions. Bedford and Taylor have shown [BT82] how to extend this definition so as to cover the case of all *bounded* psh functions. Following their work, it is tempting to set the following:

Definition 10.40. *We say that a psh function φ belongs to the domain of definition* $\mathrm{DMA}_{\mathrm{loc}}(\mathbb{B})$ *of the complex Monge–Ampère operator $(dd^c\cdot)^n$ if there exists a positive Radon measure μ_φ such that, for all balls $B' \Subset B$, for all sequences (φ_j) of bounded psh functions decreasing to φ in B', the measures $(dd^c\varphi_j)^n$ weakly converge to μ_φ in B'.*

In this case, one sets $(dd^c\varphi)^n := \mu_\varphi$.

Bedford and Taylor had already observed in [BT78] that when $n = 2$, one can make sense of $(dd^c\varphi)^2$ as soon as $\nabla\varphi \in L^2_{\mathrm{loc}}$: in this case $\varphi \in L^1_{\mathrm{loc}}(dd^c\varphi \wedge dd^c|z|^2)$ as can be seen by integration by parts, since

$$\int_{\mathbb{B}} -\varphi dd^c\varphi \wedge \theta = \int_{\mathbb{B}} d\varphi \wedge d^c\varphi \wedge \theta,$$

for all test forms θ. The reader will check that

$$PSH(\mathbb{B}) \cap W^{1,2}_{\mathrm{loc}}(\mathbb{B}) \subset \mathrm{DMA}_{\mathrm{loc}}(\mathbb{B})$$

when $n = 2$, by showing that any decreasing sequence actually converges in the Sobolev sense (see Exercise 8.14 in the compact case).

The domain $\mathrm{DMA}_{\mathrm{loc}}(\mathbb{B})$ was studied by several authors, notably Demailly [Dem93] and Fornaess and Sibony [FS95], who noticed that it contains functions which are bounded near the boundary $\partial\mathbb{B}$, and Cegrell [Ceg98, Ceg04] and Blocki [Blo04, Blo06], who have described it very accurately:

Theorem 10.41. *If $n = 2$, then* $\mathrm{DMA}_{\mathrm{loc}}(\mathbb{B}) = PSH(\mathbb{B}) \cap W^{1,2}_{\mathrm{loc}}(\mathbb{B})$.

This is done by showing that if a psh function φ does not have gradient locally in L^2, then one can construct a sequence φ_j of bounded psh functions decreasing to φ such that the Monge–Ampère measures $(dd^c\varphi_j)^2$ have locally unbounded masses. The reader will work out an example in Exercise 10.15.

A more involved description is also available in higher dimensions [Ceg04, Blo06]. Note that Monge–Ampère masses are always uniformly bounded in the compact setting: the story is somewhat different as we now explain.

10.4.2 The global domain of definition By analogy with the local setting, it is natural to say that a function $\varphi \in PSH(X,\omega)$ has a well defined Monge–Ampère measure $MA(\varphi)$, if the latter is the weak limit of the probability measures $MA(\varphi_j) := V^{-1}(\omega + dd^c\varphi_j)^n$, whenever the φ_j's are bounded ω-psh functions decreasing to φ. It turns out however that this notion is slightly to weak in the compact setting. We rather propose the following definition, following observations made in [CGZ08]:

Definition 10.42. *We say that φ belongs to* $\mathrm{DMA}(X,\omega)$ *if $\varphi \in PSH(X,\omega)$ and there exists a probability measure $MA(\varphi)$ such that for all sequences φ_j of bounded ω-psh functions decreasing to φ and for all test functions $u \in PSH(X,\omega) \cap L^\infty(X)$,*

$$\int_X uMA(\varphi_j) \longrightarrow \int_X uMA(\varphi).$$

In other words, we use a slightly stronger notion of convergence than the standard weak Radon sense: test functions are allowed to be as discontinuous as ω-psh functions can be. It follows from Theorem 10.18 that $\mathcal{E}(X,\omega) \subset \mathrm{DMA}(X,\omega)$. The connection between these sets is made by the following:

Proposition 10.43. *If $\varphi \in \mathrm{DMA}(X,\omega)$, then*

$$\mu_\varphi = \mathbf{1}_{\{\varphi > -\infty\}} MA(\varphi).$$

In particular, $\varphi \in \mathrm{DMA}(X,\omega)$ belongs to $\mathcal{E}(X,\omega)$ if and only if $MA(\varphi)(\{\varphi = -\infty\}) = 0$.

Proof. We can assume w.l.o.g. that $\varphi \leq 0$. Set

$$u_s := \max(\varphi/s + 1, 0).$$

Note that u_s are bounded ω-psh functions which increase towards $\mathbf{1}_{\{\varphi>-\infty\}}$. Moreover $\{u_s > 0\} = \{\varphi > -s\}$ and $u_s = 0$ elsewhere. We infer, when $j > s$, that

$$u_s MA(\varphi_j) = u_s \mathbf{1}_{\{\varphi > -j\}} MA(\varphi_j), \tag{10.4.1}$$

where $\varphi_j = \max(\varphi, -j)$ are the canonical approximants. Letting $j \to +\infty$, we infer $u_s \mu_\varphi \leq u_s MA(\varphi)$, since u_s is upper semi-continuous. This yields

$$\mu_\varphi = \mathbf{1}_{\{\varphi > -\infty\}} \mu(\varphi) \leq \mathbf{1}_{\{\varphi > -\infty\}} MA(\varphi).$$

Now these two measures have the same total mass, as can be seen by integrating (10.4.1), which gives

$$\int_X u_s MA(\varphi) = \int_X u_s d\mu_\varphi.$$

Letting $s \to +\infty$ yields $\int_X \mathbf{1}_{\{\varphi > -\infty\}} MA(\varphi) = \int_X \mu_\varphi$. □

A complete description of $\mathrm{DMA}(X, \omega)$ is not yet available (see [CGZ08] for some more information and open problems).

10.4.3 A characterization of $\mathcal{E}(X, \omega)$ It follows from previous sections that the complex Monge–Ampère operator $MA(\varphi)$ is well defined for functions $\varphi \in \mathcal{E}(X, \omega)$, although these functions need not have gradient in $L^2(X)$ (see Exercise 10.3). This is in contrast with the local theory.

We have the following interesting characterization of $\mathcal{E}(X, \omega)$:

Theorem 10.44. *The class $\mathcal{E}(X, \omega)$ is the largest class of ω-psh functions on which the operator $MA(\varphi)$ is well defined and the comparison principle holds.*

Proof. Let $\mathcal{F}$ be the largest class with these properties, so that $\mathcal{E}(X, \omega) \subset \mathcal{F} \subset PSH(X, \omega)$ and

$$\int_{(u \leq \psi)} MA(\psi) \leq \int_{(u \leq \psi)} MA(u),$$

for all $u, \psi \in \mathcal{F}$. Note that this inequality is equivalent to the comparison principle since both $MA(u)$ and $MA(\psi)$ are probability measures.

Fix $\varphi \in \mathcal{E}(X, \omega)$ and $\psi \in \mathcal{F}$ and apply the previous inequality with $u = \varphi + c$, $c \in \mathbb{R}$, to obtain

$$\begin{aligned} MA(\psi)(\varphi = -\infty) &= \lim_{c \to +\infty} \int_{\{\varphi + c \leq \psi\}} MA(\psi) \\ &\leq MA(\varphi)\{\varphi = -\infty\} = 0. \end{aligned}$$

Since $\mathcal{E}(X, \omega)$ characterizes pluripolar sets, we infer $MA(\psi)\{\psi = -\infty\} = 0$, hence $\psi \in \mathcal{E}(X, \omega)$, i.e., $\mathcal{F} = \mathcal{E}(X, \omega)$. □

10.5 Exercises

Exercise 10.1. *Let (X,ω) be a compact Riemann surface equipped with a Kähler form ω normalized so that $\mathrm{Vol}_\omega(X)=1$.*

(i) *Fix $a\in X$. Explain why there exists a unique function $g_a\in PSH(X,\omega)$ such that $\int_X g_a\,\omega=0$ and $\omega+dd^c g_a=\delta_a$ (Dirac mass at point a). Give an explicit formula for g_a in case $(X,\omega)=(\mathbb{P}^1,\omega_{\mathrm{FS}})$.*

(ii) *Show that $\varphi=\int_X g_a\,d\mu(a)\in PSH(X,\omega)$ is the unique ω-psh function such that $\mu=\omega+dd^c\varphi$ and $\int_X\varphi\,d\mu=0$.*

(iii) *Show that for $\varphi\in PSH(X,\omega)$, one has*

$$\mu_\varphi=\mathbf{1}_{\{\varphi>-\infty\}}(\omega+dd^c\varphi)$$

and deduce that $\varphi\in\mathcal{E}(X,\omega)$ iff $\omega+dd^c\varphi$ does not charge polar sets.

Exercise 10.2. *Show that*

$$\bigcap_{\chi\in\mathcal{W}}\mathcal{E}_\chi(X,\omega)=PSH(X,\omega)\cap L^\infty(X).$$

Exercise 10.3. *Let (X,ω) be a compact Kähler manifold and fix $0<\alpha<1$. For $\varphi\in PSH(X,\omega)$ such that $\varphi\le -1$, show that $\varphi_\alpha:=-(-\varphi)^\alpha$ belongs to $\mathcal{E}(X,\omega)$. Give an example such that φ_α does not have gradient in L^2 when $\alpha\ge 1/2$.*

Exercise 10.4. *Let (X,μ) be a probability space and $f\ge 0$ a measurable function such that $\mu\{f=+\infty\}=0$. Show that there exists a concave increasing function $h:\mathbb{R}^+\to\mathbb{R}^+$ such that $h(+\infty)=+\infty$ and*

$$\int_X h\circ f\,d\mu<+\infty.$$

Exercise 10.5. *Fix $\chi\in\mathcal{W}$ and $\varphi\in\mathcal{E}_\chi(X,\omega)$. Show that there exists $\tilde\chi\in\mathcal{W}$ such that $\chi=o(\tilde\chi)$ at $-\infty$ and still $\varphi\in\mathcal{E}_{\tilde\chi}(X,\omega)$.*

Exercise 10.6. *Show that the class $\mathcal{E}(X,\omega)$ is an extremal face of $PSH(X,\omega)$.*

Exercise 10.7. *Assume ω is a Kähler form. Show that a function $\varphi\in\mathcal{E}(X,\omega)$ has zero Lelong number at every point $x\in X$ by justifying the following assertions:*

(i) *for any $x\in X$, one can construct $\varrho\in PSH(X,\omega)$ which is smooth except at the point x, and such that $\varrho\sim c\log dist(\cdot,x)$ near x, for some $c>0$. Such a function ϱ has a well-defined Monge–Ampère measure $MA(\varphi)$ and the latter has a Dirac mass at x. Therefore, $\rho\notin\mathcal{E}(X,\omega)$.*

(ii) *if $\varphi\in PSH(X,\omega)$ has a positive Lelong number at point x, then $\varphi\le\gamma\varrho+C$ on X, for some constants $\gamma,C>0$, so $\varphi\notin\mathcal{E}(X,\omega)$.*

Exercise 10.8. *Fix Kähler forms $\omega_1, \dots, \omega_n$ and $\varphi_i \in PSH(X, \omega_i)$.*

(1) *Show that the measures $\mu_j(\varphi_1, \dots, \varphi_n) =$*

$$\mathbf{1}_{\{\varphi_1 > -j\} \cap \dots \cap \{\varphi_n > -j\}} (\omega_1 + dd^c \max[\varphi_1, -j]) \wedge \dots \wedge (\omega_n + dd^c \max[\varphi_n, -j])$$

form an increasing sequence of Borel measures with bounded masses.

(2) *We set $\mu(\varphi_1, \dots, \varphi_n) := \lim_{j \to +\infty} \mu_j(\varphi_1, \dots, \varphi_n)$. Check that the mapping*

$$(\varphi_1, \dots, \varphi_n) \longmapsto \mu(\varphi_1, \dots, \varphi_n)$$

is multilinear and symmetrical.

Exercise 10.9. *Let $\chi : \mathbb{R} \to \mathbb{R}$ be a smooth convex increasing function such that $\chi'(-\infty) = 0$.*

(i) *For $\varphi \in PSH(X, \omega)$ conveniently normalized, show that*

$$\chi \circ \varphi \in \mathcal{E}(X, \omega).$$

(ii) *Show that if ω_φ is supported on $\{\varphi = -\infty\}$, then $MA(\chi \circ \varphi)$ is absolutely continuous with respect to the Lebesgue measure.*

Exercise 10.10. *Let $\chi : \mathbb{R} \to \mathbb{R}$ be a convex increasing weight.*

(i) *Show that if $\varphi \in \mathcal{E}_\chi(X, \omega)$, then there exists $C_\varphi > 0$ such that*

$$\forall t > 1,\ \mathrm{Cap}_\omega \{\varphi < -t\} \leq C_\varphi |t\ \chi(-t)|^{-1}.$$

(ii) *Show that $\varphi \in \mathcal{E}_\chi(X, \omega)$ whenever there exists $C_\varphi, \varepsilon > 0$ such that*

$$\forall t > 1,\ \mathrm{Cap}_\omega \{\varphi < -t\} \leq C_\varphi |t^{n+\varepsilon}\ \chi(-t)|^{-1}.$$

Exercise 10.11. *Give an example of a probability measure μ absolutely continuous with respect to Lebesgue measure such that*

$$\mathcal{E}^1(X, \omega) \not\subset L^1(\mu).$$

Exercise 10.12. *Let $\chi \in \mathcal{W}^+$ be convex weight.*

(i) *Show that if $\chi(0) = 0$, then for all $a, b \leq 0$,*

$$\chi(a + b) \geq \chi(a) + \chi(b).$$

(ii) *Use* (i) *and Proposition* 10.25 *to show that $\mathcal{E}_\chi(X, \omega)$ is convex.*

Exercise 10.13. *Verify the claims made in Example 10.39. Namely, check that*

$$u : \mathbb{B} \ni (z,w) \mapsto (|w|^2 - 4)\sqrt{-\log|z|} \in \mathbb{R} \cup \{-\infty\}.$$

is a well-defined plurisubharmonic function in the unit ball $\mathbb{B}$ of $\mathbb{C}^2$, which is smooth except on $\{z =\} = \{u = -\infty\}$, and such that

$$\int_{\mathbb{B}(\frac{1}{2})\setminus\{z=0\}} (dd^c u)^2 = +\infty.$$

Exercise 10.14. *Let X be a compact Kähler surface* ($\dim_{\mathbb{C}} X = 2$) *equipped with a Kähler form ω.*

(i) *Find an example of $\varphi \in \mathcal{E}^1(X,\omega)$ such that $\varphi \notin L^2(\omega_\varphi \wedge \omega)$.*

(ii) *Show that if $\varphi \in \mathcal{E}^1(X,\omega)$ and φ is locally bounded near an ample divisor, then $\varphi \in L^2(\omega_\varphi \wedge \omega)$.*

Exercise 10.15. *Consider $u(z,w) := -\sqrt{-\log|z|}$. Check that $\nabla u \notin L^2_{\rm loc}$ and construct a sequence (u_j) of locally bounded psh functions in $\mathbb{B}$ which decrease to u and are such that the Monge–Ampère measures $(dd^c u_j)^2$ have locally unbounded masses in $\mathbb{B}$.*

Exercise 10.16. *Fix $\chi \in \mathcal{W}$ and let $P \subset X$ be a $PSH(X,\omega)$-polar subset. Show that there exists $\varphi \in \mathcal{E}_\chi(X,\omega)$ such that*

$$P \subset \{\varphi = -\infty\}.$$

Exercise 10.17. *Let $\varphi \in PSH(X,\omega)$ and set, for $t \in \mathbb{R}$, $\varphi_t = \max(\varphi, -t)$. Show that $t \mapsto E(\varphi_t)$ is convex and that $E(\varphi_t)/t$ admits a limit ℓ_φ as $t \to +\infty$. Check that $\varphi \in \mathcal{E}(X,\omega)$ if and only if $\ell_\varphi = 0$.*

Exercise 10.18. *Fix $\chi \in \mathcal{W}_+$. Show that $\varphi \in \mathcal{E}_\chi(X,\omega)$ if and only if $\varphi \in \mathrm{DMA}(X,\omega)$ and $\chi \circ \varphi \in L^1(MA(\varphi))$.*

Exercise 10.19. *Show that $\varphi \in \mathcal{E}^1(X,\omega)$ if and only if*

$$\int_0^{+\infty} MA((\max(\varphi,-t))\,\{\varphi \le -t\}\,dt < +\infty.$$

Exercise 10.20. *Prove Theorem 10.37. Namely, show that if $\varphi_j \in \mathcal{E}^1(X,\omega)$ converges to $\varphi \in \mathcal{E}^1(X,\omega)$ in energy, then*

$$\int_X \psi MA(\varphi_j) \longrightarrow \int_X \psi MA(\varphi)$$

and

$$\int_X \varphi_j MA(\psi) \longrightarrow \int_X \varphi MA(\psi)$$

for all $\psi \in \mathcal{E}^1(X,\omega)$, and that these convergences are uniform with respect to ψ if ψ stays in a compact set $\mathcal{E}^1_C(X,\omega)$.

Verify that this implies that $\varphi_j \to \varphi$ in capacity.

Exercise 10.21. *Let $\varphi_j, \varphi \in \mathcal{E}^1(X,\omega)$ be normalized by $\int_X \varphi dV = \int_X \varphi_j dV = 0$. Show that the following properties are equivalent:*

(i) *φ_j converges to φ in energy;*

(ii) *$\varphi_j \to \varphi$ in L^1 and $E(\varphi_j) \to E(\varphi)$;*

(iii) *$\int_X (\varphi_j - \varphi) MA(\varphi_j) \to 0$.*

Part III

Solving complex Monge–Ampère equations

11 The variational approach

Our goal in this chapter is to explain a variational approach developed in [BBGZ13] to solve the following degenerate complex Monge–Ampère equations

$$(\mathrm{MA})_\lambda \qquad\qquad MA(\varphi) = e^{-\lambda\varphi}\mu,$$

where $\varphi \in \mathcal{E}(X,\omega)$, $\lambda \in \mathbb{R}$, and μ is a non-pluripolar probability measure. Here again X is a compact Kähler manifold and ω is semi-positive and big.

11.1 The $\mathcal{F}_\lambda$-functionals

11.1.1 Euler–Lagrange equations Given $\lambda \in \mathbb{R}$, we consider the functional $\mathcal{F}_\lambda : \mathcal{E}^1(X,\omega) \to \mathbb{R}$ defined by

$$\mathcal{F}_\lambda(\varphi) = E(\varphi) + \frac{1}{\lambda}\log\left[\int_X e^{-\lambda\varphi}d\mu\right].$$

The limit case $\lambda = 0$ corresponds to

$$\mathcal{F}_0(\varphi) = E(\varphi) - \int_X \varphi\, d\mu = \lim_{\lambda\to 0}\mathcal{F}_\lambda(\varphi).$$

Note that $E(\varphi)$ is well defined for all functions $\varphi \in \mathcal{E}^1(X,\omega)$. The second term is always well defined (i.e., finite) when $\lambda < 0$ (since ω-psh functions are bounded from above on X), while its finiteness depends on properties of the measure μ, when $\lambda \geq 0$.

Definition 11.1. *We say that* (μ,λ) **satisfies condition (I)** *if*

- μ *does not charge pluripolar sets when* $\lambda < 0$*;*
- $\mathcal{E}^1(X,\omega) \subset L^1(\mu)$ *when* $\lambda = 0$*;*
- $\mu = fdV$ *with* $f \in L^p(dV)$ *for some* $p > 1$*, when* $\lambda > 0$*, for some volume form* dV*.*

Note that the second integrability condition is stronger than the first, since pluripolar sets can be characterized by functions in $\mathcal{E}^1(X,\omega)$. It follows from the Hölder inequality that the third condition implies the second one. Therefore, when $\mu = fdV$, $p > 1$, the couple (μ,λ) satisfies condition (I) for all values of $\lambda \in \mathbb{R}$.

It follows from the Hölder inequality and Skoda's uniform integrability theorem that $\mathcal{F}_\lambda$ is well defined on $\mathcal{E}^1(X,\omega)$ when $\lambda > 0$ and $\mu = fdV$, $f \in L^p(dV)$, $p > 1$.

Our first basic observation is that $(\mathrm{MA})_\lambda$ is the Euler–Lagrange equation of the functional $\mathcal{F}_\lambda$. To simplify our task, we assume here that the functions under consideration are smooth:

Lemma 11.2. *Let $\varphi, v : X \to \mathbb{R}$ be smooth functions. Then*

$$\left.\frac{d\mathcal{F}_\lambda(\varphi + tv)}{dt}\right|_{t=0} = \int_X v MA(\varphi) - \int_X v \frac{e^{-\lambda\varphi} d\mu}{\int_X e^{-\lambda\varphi} d\mu}.$$

Therefore, if φ is a ω-plurisubharmonic critical point of $\mathcal{F}_\lambda$, then $\varphi + c$ is a solution of $(\mathrm{MA})_\lambda$ for an appropriate choice of $c \in \mathbb{R}$.

It is understood here that $\frac{e^{-\lambda\varphi} d\mu}{\int_X e^{-\lambda\varphi} d\mu} = d\mu$ when $\lambda = 0$ (there is no need to change φ to $\varphi + c$ in this case).

Proof. We have already shown that

$$\left.\frac{dE(\varphi + tv)}{dt}\right|_{t=0} = \int_X v MA(\varphi)$$

while the second term comes from the derivative of the log term.

If φ is a ω-plurisubharmonic critical point of $\mathcal{F}_\lambda$, the above derivative vanishes for all function $v \in \mathcal{C}^\infty(X, \mathbb{R})$, therefore

$$MA(\varphi) = \frac{e^{-\lambda\varphi}\mu}{\int_X e^{-\lambda\varphi} d\mu}.$$

When $\lambda = 0$, this is thus a solution to $(\mathrm{MA})_0$. When $\lambda \neq 0$, we note that

$$\tilde{\varphi} := \varphi + \frac{1}{\lambda} \log \left[\int_X e^{-\lambda\varphi} d\mu \right]$$

is a solution to $(\mathrm{MA})_\lambda$. □

11.1.2 Upper semi-continuity

Say that μ satisfies $(\mathrm{DC})_A$ if there exists $A > 0$ such that

$$\forall K \subset X, \quad \mu(K) \leq A \, \mathrm{Cap}_\omega(K).$$

We shall see in Chapter 12 that $\mu = f dV$ satisfies $(\mathrm{DC})_A$ when $f \in L^p(dV)$, $p > 1$.

Proposition 11.3. *Let μ be a probability measure such that (μ, λ) satisfies condition* (I). *Then $\mathcal{F}_\lambda : \mathcal{E}^1(X, \omega) \to \mathbb{R}$ is invariant under translations and it is*

(i) *upper semi-continuous when $\lambda < 0$;*

(ii) *upper semi-continuous on each compact subset $\mathcal{E}^1_C(X, \omega)$ when $\lambda = 0$ and there exists $A > 0$ such that μ further satisfies $(\mathrm{DC})_A$;*

(iii) *upper semi-continuous on each subset $\mathcal{E}^1_C(X, \omega)$ when $\lambda > 0$.*

Proof. The invariance under translations is straightforward. Recall that $\varphi \mapsto E(\varphi)$ is upper semi-continuous (the class $\mathcal{E}^1(X,\omega)$ is endowed with the induced L^1-topology). We are going to show the semi-continuity property separately for each case $\lambda > 0$, $\lambda = 0, \lambda > 0$.

Case $\lambda < 0$. We can assume without loss of generality that $\lambda = -1$ (up to rescaling ω). We are to show that

$$\varphi \in \mathcal{E}^1(X,\omega) \longmapsto \int_X e^{\varphi} d\mu \in \mathbb{R}$$

is lower semi-continuous. Here μ is a probability measure which does not charge pluripolar sets. Let φ_j be a sequence of functions in $\mathcal{E}^1(X,\omega)$ which converge in $L^1(X)$ to $\varphi \in \mathcal{E}^1(X,\omega)$. We can assume without loss of generality that $\varphi, \varphi_j \leq 0$, so that

$$\psi = e^{\varphi}, \quad \psi_j = e^{\varphi_j} \in PSH(X,\omega)$$

are uniformly bounded (with values in $[0,1]$). It then follows from Lemma 11.5 below that $\int_X \psi_j \, d\mu \to \int_X \psi \, d\mu$.

Case $\lambda = 0$. Our aim here is to show that

$$\varphi \in \mathcal{E}^1_C(X,\omega) \longmapsto \int_X \varphi d\mu \in \mathbb{R}$$

is lower semi-continuous. Here μ is such that $\mathcal{E}^1(X,\omega) \subset L^1(\mu)$.

Let φ_j be a sequence of functions in $\mathcal{E}^1(X,\omega)$ which converge towards $\varphi \in \mathcal{E}^1(X,\omega)$. Since the φ_j's are uniformly bounded from above, we can assume without loss of generality that $\varphi_j \leq 0$. By Fatou's lemma,

$$\limsup \int_X \varphi_j d\mu \leq \int_X \limsup \varphi_j d\mu \leq \int_X \varphi d\mu,$$

where the last inequality follows from the ω-plurisubharmonicity property. This shows that $\varphi \mapsto \int_X \varphi \, d\mu$ is upper semi-continuous, but we are looking for lower semi-continuity (hence continuity)! We now use an additional assumption on μ: since $\mu \leq A \operatorname{Cap}_\omega$, we infer that

$$\begin{aligned}\int_X \varphi_j^2 \, d\mu &= 2 \int_0^{+\infty} t \, \mu(\varphi_j < -t) dt \\ &\leq 2A \int_0^{+\infty} t \operatorname{Cap}_\omega \{\varphi_j < -t\} \, dt \leq C',\end{aligned}$$

as follows from Lemma 10.29 and our restriction that φ_j stays in $\mathcal{E}^1_C(X,\omega)$. It follows therefore from Lemma 11.5 that $\int_X \varphi_j \, d\mu \to \int_X \varphi \, d\mu$.

Case $\lambda > 0$. We can assume, without loss of generality, that $\lambda = +1$ (up to rescaling ω). We need to check that

$$\varphi \in \mathcal{E}^1_C(X,\omega) \longmapsto \int_X e^{-\varphi} d\mu \in \mathbb{R}$$

is upper semi-continuous. Here $\mu = f\omega^n$ is absolutely continuous with respect to the Lebesgue measure, with density $f \in L^p(\omega^n)$ for some $p > 1$. It follows again from Fatou's lemma that $\varphi \in \mathcal{E}^1(X,\omega) \mapsto \int_X e^{-\varphi} d\mu \in \mathbb{R}$ is lower semi-continuous.

To establish an upper semi-continuity property we once again restrict ourselves to the compact convex subset $\mathcal{E}^1_C(X,\omega)$: let φ_j be a sequence in $\mathcal{E}^1_C(X,\omega)$ converging to φ; since all these functions have zero Lelong number, it follows from Skoda's uniform integrability theorem and Hölder inequality that

$$\int_X e^{-2(\varphi_j+\varphi)} d\mu = \int_X e^{-2(\varphi_j+\varphi)} f dV \leq \|f\|_{L^p} \left(\int_X e^{-2q(\varphi_j+\varphi)} dV \right)^{1/q} \leq C'.$$

Now

$$\left| \int_X e^{-\varphi_j} d\mu - \int_X e^{-\varphi} d\mu \right| \leq \int_X |\varphi_j - \varphi| e^{-(\varphi_j+\varphi)} d\mu \leq C' ||\varphi_j - \varphi||_{L^2(\mu)},$$

as follows from the Cauchy–Schwarz inequality and the elementary inequality

$$\left| e^a - e^b \right| \leq |a-b|\, e^{a+b}, \text{ for all } a, b \geq 0.$$

The conclusion follows since (φ_j) converges to φ in $L^2(\mu)$. □

Remark 11.4. *It will be true a posteriori that*

$$\mathcal{E}^1_C(X,\omega) \ni \varphi \longmapsto \int_X \varphi \, d\mu \in \mathbb{R}$$

is continuous when $\mathcal{E}^1(X,\omega) \subset L^1(\mu)$ without additional hypotheses on μ: this will be a consequence of the fact that there exists then $\psi \in \mathcal{E}^1(X,\omega)$ such that $\mu = MA(\psi)$ (see Exercise 11.14). As the reader will have noticed, the continuity of $\varphi \mapsto \int_X \varphi \, d\mu$ is a subtle question, whose answer depends on the subclass to which φ belongs.

Lemma 11.5. *Let μ be a probability measure which does not charge pluripolar sets. Let $\psi_j \in PSH(X,\omega)$ be a sequence which converges in $L^1(X)$ to $\psi \in PSH(X,\omega)$. If the ψ_j's are uniformly bounded, then*

$$\int_X \psi_j \, d\mu \longrightarrow \int_X \psi \, d\mu.$$

The same conclusion holds if we merely assume that there exists $p > 1$ such that $\sup_j \int_X |\psi_j|^p \, d\mu < +\infty$.

Proof. Recall that $\limsup \psi_j(x) \leq \psi(x) \leq C$, hence by Fatou's lemma

$$\limsup \int_X \psi_j d\mu \leq \int_X (\limsup \psi_j) \, d\mu \leq \int_X \psi \, d\mu.$$

To show there is continuity, it clearly suffices to prove that

$$\liminf \int \mathrm{Comb}(\psi_j)\, d\mu \geq \int \psi\, d\mu,$$

where $\mathrm{Comb}(\psi_j)$ denotes an arbitrary convex combination of the ψ_j's.

Assume first $p = 2$. Then (ψ_j) is weakly compact in $L^2(\mu)$. It follows from the Banach–Saks theorem that we can find a subsequence such that

$$\tilde{\psi}_N = N^{-1} \sum_{k=1}^{N} \psi_{j_k}$$

converges in $L^2(\mu)$ to some function $g \in L^2(\mu)$. Extracting again, we can further assume that $\tilde{\psi}_N$ converges μ-almost everywhere to $g(x)$. Set now

$$\varphi_N(x) := \left(\sup_{j \geq N} \tilde{\psi}_j(x) \right)^* .$$

Then $\varphi_N(x) = \sup_{j \geq N} \tilde{\psi}_j(x)$ μ-almost everywhere, since μ does not charge the pluripolar set

$$P := \left\{ x \in X \;\middle|\; \sup_{j \geq N} \psi_j(x) < \left(\sup_{j \geq N} \psi_j(x) \right)^* \right\}.$$

Now $\varphi_N(x)$ decreases to $\psi(x)$ everywhere, hence μ-almost everywhere, but it also converges to $g(x)$ μ-almost everywhere. Thus $\psi \equiv g$, in particular

$$\lim \int_X N^{-1} \sum_{k=1}^{N} \psi_{j_k}\, d\mu = \int_X \psi\, d\mu.$$

For $p > 1$, we use a similar reasoning using the reflexivity of $L^p(\mu)$ and replacing Cesàro means by more general convex combinations (convex sets of a Banach space are closed if and only if they are weakly closed). □

11.2 Solutions in $\mathcal{E}^1(X,\omega)$

11.2.1 Maximizers are critical points

Definition 11.6. *We say that the functional $\mathcal{F}_\lambda$ is proper if whenever $\varphi_j \in \mathcal{E}^1(X,\omega)$ is a sequence of functions such that $E(\varphi_j) \to -\infty$ and $\int_X \varphi_j \omega^n = 0$, then $\mathcal{F}_\lambda(\varphi_j) \to -\infty$.*

In many articles, this condition is expressed by saying that $\mathcal{F}_\lambda(\varphi_j) \to -\infty$ whenever $J(\varphi_j) \to -\infty$, where $J(\varphi) = E(\varphi) - \frac{1}{V}\int_X \varphi\omega^n$ is the translation invariant functional associated to E.

The main result of this section is the following:

Theorem 11.7. *Let μ be a probability measure such that (μ, λ) satisfies condition* (I) *with the additional assumption that $\mu \leq A\,\mathrm{Cap}_\omega$ when $\lambda = 0$. If $\mathcal{F}_\lambda$ is proper, then there exists $\varphi \in \mathcal{E}^1(X,\omega)$ which solves* $(\mathrm{MA})_\lambda$ *and such that*

$$\mathcal{F}_\lambda(\varphi) = \sup_{\mathcal{E}^1(X,\omega)} \mathcal{F}_\lambda.$$

Proof. Since $\mathcal{F}_\lambda$ is invariant under translations and proper, we can find $C > 0$ so large that

$$\sup_{\mathcal{E}^1(X,\omega)} \mathcal{F}_\lambda = \sup_{\mathcal{E}^1_C(X,\omega)} \mathcal{F}_\lambda,$$

where we recall that

$$\mathcal{E}^1_C(X,\omega) := \{\varphi \in \mathcal{E}^1(X,\omega) \,|\, E(\varphi) \geq -C \text{ and } \varphi \leq 0\}$$

is a compact convex subset of $\mathcal{E}^1(X,\omega)$. Now $\mathcal{F}_\lambda$ is upper semi-continuous on $\mathcal{E}^1_C(X,\omega)$, thus we can find $\varphi \in \mathcal{E}^1_C(X,\omega) \subset \mathcal{E}^1(X,\omega)$ which maximizes the functional $\mathcal{F}_\lambda$ on $\mathcal{E}^1(X,\omega)$.

If φ were known to be smooth and strictly ω-psh, we could consider $\varphi + tv$ for $t > 0$ small and v an arbitrary smooth function and use Lemma 11.2 to conclude that φ is a critical point of $\mathcal{F}_\lambda$, hence $\varphi + c$ solves $(\mathrm{MA})_\lambda$. This turns out to be the case when μ is a smooth volume form, however this requires a lot of extra work and our measures μ are here allowed to vanish, hence we cannot expect φ to be strictly ω-psh.

We therefore use the following detour. Fix an arbitrary continuous function $v \in \mathcal{C}^0(X,\mathbb{R})$ and consider

$$g(t) := E \circ P(\varphi + tv) + \frac{1}{\lambda}\log\left[\int_X e^{-\lambda(\varphi+tv)}d\mu\right],$$

where, given an upper semi-continuous function h on X, we let

$$P(h)(x) := \sup\{\psi(x) \in \mathbb{R} \,|\, \psi \in PSH(X,\omega) \text{ and } \psi \leq h\}$$

denote the ω-psh envelope of h. We are going to show in Theorem 11.11 that the function g is differentiable at the origin, with

$$g'(0) = \langle MA(\varphi), v\rangle - \left\langle \frac{e^{-\lambda\varphi}\mu}{\int_X e^{-\lambda\varphi}d\mu}, v\right\rangle.$$

Since $\varphi \in \mathcal{E}^1(X,\omega)$ and tv is bounded below, we see that $P(\varphi + tv) \in \mathcal{E}^1(X,\omega)$. Now $P(\varphi + tv) \leq \varphi + tv$, hence

$$g(t) \leq \mathcal{F}_\lambda \circ P(\varphi + tv) \leq g(0),$$

as φ is a maximizer. We infer $g'(0) = 0$, which finishes the proof. □

11.2.2 Properness of the functionals An important issue is thus to determine whether $\mathcal{F}_\lambda$ is proper. This turns out to be always the case when $\lambda \leq 0$. This is not necessarily so when $\lambda > 0$, and indeed one cannot always solve $(\mathrm{MA})_\lambda$ when $\lambda > 0$.

Theorem 11.8. *Assume (μ,λ) satisfies condition* (I). *Then*

(i) $\mathcal{F}_\lambda(\varphi) \leq E(\varphi) - \sup_X \varphi + C_\mu$ *is always proper when* $\lambda < 0$;

(ii) $\mathcal{F}_0$ *is proper: there exists* $C > 0$ *such that for all* $\varphi \in \mathcal{E}^1(X,\omega)$

$$\mathcal{F}_0(\varphi) \leq E(\varphi) - \sup_X \varphi + C \left| E(\varphi) - \sup_X \varphi \right|^{1/2}.$$

(iii) *There exists* $\alpha(\{\omega\}) > 0$ *such that for all* $0 < \lambda < \alpha(\{\omega\})$, *there exists* $C_\lambda > 0$ *such that for all* $\varphi \in \mathcal{E}^1(X,\omega)$,

$$\mathcal{F}_\lambda(\varphi) \leq E(\varphi) - \sup_X \varphi + C_\lambda,$$

hence $\mathcal{F}_\lambda$ *is proper for* $0 < \lambda < \alpha(\{\omega\})$.

The reader should keep in mind that the more positive λ, the more difficult it is to solve $(\mathrm{MA})_\lambda$. The upper bound in (ii) is not as good as that in (iii), because we are allowing for more singular measures in (ii): if $\mu = f dV$ has density in L^p, $p > 1$, it easily follows from Hölder's inequality that a linear upper bound holds in (ii) as well (see Exercise 11.3).

Proof. (i) is straightforward: it suffices to check, setting $\tau = -\lambda > 0$, that there exists $\delta > 0$ such that

$$\int_X e^{\tau(\varphi - \sup_X \varphi)} d\mu \geq \delta > 0$$

for all $\varphi \in PSH(X,\omega)$. This follows easily from Lemma 11.5 and the fact that the set of normalized $\tau\omega$-psh functions is compact (see Exercise 11.4).

To prove (ii), we assume that $\lambda = 0$ and μ is a probability measure such that $\mathcal{E}^1(X,\omega) \subset L^1(\mu)$. We let the reader check in Exercise 11.1 that for each $C > 0$, there exists $M_C > 0$ such that

$$\forall \psi \in \mathcal{E}^1_C(X,\omega), \quad \int_X \psi d\mu \geq -M_C.$$

Fix $\varphi \in \mathcal{E}^1(X,\omega)$ such that $\sup_X \varphi = 0$ and $E(\varphi) \leq -1$. We set $\varepsilon := |E(\varphi)|^{-1/2} \in]0,1]$ and consider

$$\psi := \varepsilon\varphi = \frac{\varphi}{|E(\varphi)|^{1/2}} \in PSH(X,\omega).$$

Observe that

$$E(\psi) = \frac{\varepsilon}{(n+1)V} \sum_{j=0}^{n} \int_X \varphi\, \omega_\psi^j \wedge \omega^{n-j}.$$

We let the reader check that there exists $N \in \mathbb{N}$ such that for all $j \geq 1$,

$$\omega_\psi^j \wedge \omega^{n-j} \leq \omega^n + N\varepsilon \sum_{l=1}^{n} \omega_\varphi^l \wedge \omega^{n-l}.$$

We infer, since $\varphi, \psi \leq 0$, that

$$E(\psi) \geq \frac{1}{V} \int_X \varphi \omega^n + \varepsilon^2 N' E(\varphi) \geq -C.$$

Since $I(\psi) := \int_X \psi d\mu$ is uniformly bounded on $\mathcal{E}_C^1(X, \omega)$, this yields

$$\int_X \varphi \, d\mu = |E(\varphi)|^{1/2} \int_X \psi \, d\mu \geq -C' |E(\varphi)|^{1/2},$$

so the proof of (ii) is complete.

We finally treat the case $\lambda > 0$. Here $\mu = fdV$ is assumed to be absolutely continuous with respect to Lebesgue measure, with density $f \in L^p$ for some $p > 1$. The upper bound (iii) is a consequence of the uniform upper bound

$$\int_X e^{-\lambda(\varphi - \sup_X \varphi)} f \omega^n \leq C,$$

which in turn follows from Hölder inequality and Skoda's uniform integrability theorem (Theorem 8.11), which states that for all $\varphi \in PSH(X, \omega)$,

$$\int_X e^{-q\lambda(\varphi - \sup_X \varphi)} \omega^n \leq C(q\lambda)$$

is uniformly bounded from above since the set of ω-psh functions $\psi = \varphi - \sup_X \varphi$ is compact, hence these integrals are uniformly bounded from above as soon as λ is so small that the Lelong numbers of the functions $q\lambda\psi$ are smaller than 2. □

11.2.3 Solutions in $\mathcal{E}^1(X, \omega)$

Corollary 11.9. *Let μ be a probability measure which does not charge pluripolar sets. Then for all $\tau > 0$, there exists a unique $\varphi \in \mathcal{E}^1(X, \omega)$ such that*

$$MA(\varphi) = e^{\tau\varphi}\mu.$$

The function φ is the unique maximizer of the functional $\mathcal{F}_{-\tau}$.

Proof. The existence follows from previous results since $(\mu, -\tau)$ satisfies condition (I) and $\mathcal{F}_{-\tau}$ is proper. The uniqueness follows from the *comparison principle*. Indeed if ψ is another solutions of $(\mathrm{MA})_{-\tau}$, then

$$MA(\psi) = e^{\tau\psi}\mu = e^{\tau(\psi-\varphi)} MA(\varphi)$$

and the comparison principle shows that

$$\int_{(\varphi<\psi)} e^{\tau(\psi-\varphi)} MA(\varphi) = \int_{(\varphi<\psi)} MA(\psi) \leq \int_{(\varphi<\psi)} MA(\varphi),$$

hence $\mu(\varphi < \psi) = 0$. We infer that $\psi \leq \varphi$ almost everywhere with respect to the measure $MA(\varphi)$, and so, by the *domination principle* (Proposition 10.11), $\psi \leq \varphi$ everywhere on X. By symmetry we conclude that $\varphi \equiv \psi$. □

Corollary 11.10. *Let μ be a probability measure such that $\mathcal{E}^1(X,\omega) \subset L^1(\mu)$. There exists a function $\varphi \in \mathcal{E}^1(X,\omega)$ such that*

$$MA(\varphi) = \mu,$$

which is moreover unique up to an additive constant.

The functions $\varphi + c$ are the unique maximizers of the functional $\mathcal{F}_0$.

Proof. The existence follows from previous results, since $\mathcal{F}_0$ is proper, when $\mu \leq A\,\mathrm{Cap}_\omega$ for some constant $A > 0$ (this condition was used to establish the upper semi-continuity of $\mathcal{F}_0$ on the sets $\mathcal{E}^1_C(X,\omega)$).

To show the existence under the sole assumption that $\mathcal{E}^1(X,\omega) \subset L^1(\mu)$, we project μ onto the convex compact set

$$\mathcal{C} := \{\nu \text{ probability measure} \mid \nu \leq \mathrm{Cap}_\omega\}.$$

A generalized Radon–Nikodým decomposition ensures that $\mu = f\nu + \sigma$, where $\nu \in \mathcal{C}$, $0 \leq f \in L^1(\nu)$ and $\sigma \perp \mathcal{C}$. One necessarily has $\sigma \equiv 0$, since μ vanishes on pluripolar sets.

Set now $\mu_j := c_j \min(f, j)\nu$, where $c_j \geq 1$ decreases to 1 is a normalization constant such that μ_j is again a probability measure. Since $\mu_j \leq jc_j\mathrm{Cap}_\omega$, it follows from the previous analysis that $\mu_j = MA(\varphi_j)$ for some function $\varphi_j \in \mathcal{E}^1(X,\omega)$ which we can assume is normalized so that $\sup_X \varphi_j = 0$. Extracting, if necessary, we can further assume that $\varphi_j \to \varphi$ in L^1. Now

$$|E(\varphi_j)| \leq \int_X (-\varphi_j) MA(\varphi_j) \leq c_j \int_X (-\varphi_j) d\mu \leq C|E(\varphi_j)|^{1/2}$$

by Theorem 11.8, hence $E(\varphi_j)$ is uniformly bounded and $\varphi \in \mathcal{E}^1(X,\omega)$ (by semi-continuity of E). Now $\Phi_j := (\sup_{\ell \geq j} \varphi_\ell)^*$ decreases to φ and

$$MA(\Phi_j) \geq \min(f, j)\nu,$$

so that $MA(\varphi) \geq \mu$ by Proposition 10.9. There is thus equality as these are both probability measures.

The uniqueness is a more delicate issue here, we refer the reader to Section 11.4. □

When $\lambda > 0$, solutions to $(\mathrm{MA})_\lambda$ do not always exist. When they do, the reader will check in Exercise 11.13 that they are usually far from being unique. When the cohomology class of ω is the first Chern class of X, an important result of Bando and Mabuchi [BM87] asserts that any two solutions to $(\mathrm{MA})_\lambda$, $\lambda > 0$, are connected by the flow of a holomorphic vector field. We will come back to this in the last part of the book.

11.2.4 The projection theorem Recall that the ω-psh envelope $P(u)$ of an upper semi-continuous function $u : X \to \mathbb{R} \cup \{-\infty\}$ is

$$P(u)(x) := \sup\{\psi(x) \,|\, \psi \in PSH(X,\omega) \text{ with } \psi \leq u\}.$$

The goal of this section is to prove the following result of Berman and Boucksom [BeBo10], on which the variational approach we've just presented relies:

Theorem 11.11. *Fix $\varphi \in \mathcal{E}^1(X,\omega)$ and $v \in \mathcal{C}^0(X,\mathbb{R})$. Then $t \mapsto E \circ P(\varphi + tv)$ is differentiable at zero, with*

$$\left.\frac{dE \circ P(\varphi + tv)}{dt}\right|_{t=0} = \int_X v MA(\varphi).$$

The proof we propose is a simplification of the original proof of Berman and Boucksom due to Lu and Nguyen [LN15].

Proof. We want to show that

$$\frac{E \circ P(\varphi + tv) - E(\varphi)}{t} \xrightarrow{t\to 0} \int_X v\, MA(\varphi).$$

Changing v to $-v$, it suffices to consider $t > 0$.

It follows from the concavity of E together with $E' = MA$ that

$$E \circ P(\varphi + tv) \leq E(\varphi) + E'(\varphi) \cdot [P(\varphi + tv) - \varphi]\,.$$

Since $t > 0$ and $P(\varphi + tv) \leq \varphi + tv$, we infer that

$$\begin{aligned}\frac{E \circ P(\varphi + tv) - E(\varphi)}{t} &\leq \int_X \frac{P(\varphi + tv) - \varphi}{t}\, MA(\varphi) \\ &\leq \int_X v\, MA(\varphi).\end{aligned}$$

By the concavity of E we also obtain, for $t > 0$,

$$\frac{E \circ P(\varphi + tv) - E(\varphi)}{t} \geq \int_X \frac{P(\varphi + tv) - \varphi}{t}\, MA(P(\varphi + tv)).$$

The measure $MA(P(\varphi + tv))$ is concentrated on the coincidence set $\{\varphi + tv = P(\varphi + tv)\}$ thus

$$\int_X \frac{P(\varphi + tv) - \varphi}{t} MA(P(\varphi + tv)) = \int_X v\, MA(P(\varphi + tv)).$$

Observe now that the projection P is uniformly Lipschitz, thus

$$\sup_X |P(\varphi + tv) - P(\varphi)| = \sup_X |P(\varphi + tv) - \varphi| \leq t \sup_X |v| \longrightarrow 0.$$

Since the complex Monge–Ampère operator is continuous under uniform convergence, we conclude that

$$\int_X v\, MA(P(\varphi + tv)) \longrightarrow \int_X v\, MA(\varphi)$$

and the proof is complete. □

11.3 More solutions

11.3.1 A properness criterion when $\lambda > 0$ When X is a Fano manifold (i.e., when the first Chern class $c_1(X)$ of X is positive), Tian [Tian87] has established a useful analytic criterion which ensures that one can solve $(MA)_1$ when ω is a Kähler form in $c_1(X)$. We generalize this criterion here following [BBEGZ11].

The following uniform integrability index plays an important role when the reference cohomology class is the canonical class:

Definition 11.12. *Let μ be a probability measure. We set*

$$\alpha_\mu(\{\omega\}) := \sup\left\{\alpha > 0 \,\middle|\, \sup_{\varphi \in \mathcal{F}_0} \int_X e^{-\alpha\varphi} d\mu < +\infty\right\},$$

where $\mathcal{F}_0 = \{\varphi \in PSH(X, \omega) \mid \sup_X \varphi = 0\}$.

When $\mu = f dV$ is a probability measure with density $f \in L^p$, $p > 1$, it follows from Hölder's inequality and Skoda's uniform integrability theorem that

$$\alpha_\mu(\{\omega\}) > 0.$$

When X is a Fano manifold and μ is a volume form, we set

$$\alpha(X) := \alpha_\mu(c_1(X)).$$

This analytically defined exponent $\alpha(X)$ coincides with the algebraic *log-canonical threshold* of X, which plays an important role in birational geometry (see Section 8.3.1.3 in Chapter 8).

Theorem 11.13. *Fix $\lambda > 0$. If $\alpha_\mu(X) > \lambda n/(n+1)$, then there exists $\varphi \in \mathcal{E}^1(X,\omega)$ solving*

$$MA(\varphi) = e^{-\lambda\varphi}\mu.$$

We infer that one can always solve $(\mathrm{MA})_\lambda$ for small enough values of $\lambda > 0$ when $\mu = f dV$ is a probability measure with density $f \in L^p$, $p > 1$.

Proof. By rescaling it suffices to consider the case $\lambda = 1$. Recall that

$$\mathcal{F}_1(\varphi) = E(\varphi) + \log\left[\int_X e^{-\varphi} d\mu\right].$$

We set

$$\mathcal{M}_\mu(\varphi) = E(\varphi) - \int_X \varphi MA(\varphi) - H_\mu(MA(\varphi)),$$

where

$$H_\mu(\nu) := \int_X \left(\frac{\nu}{\mu}\right) \log\left(\frac{\nu}{\mu}\right) d\mu,$$

is the relative entropy of ν with respect to μ (which equals by convention $+\infty$ when ν is not absolutely continuous with respect to μ).

By Lemma 11.16, $\mathcal{M}_\mu$ is proper when $\alpha_\mu > n/(n+1)$. Therefore

$$\sup_{\mathcal{E}^1(X,\omega)} \mathcal{M}_\mu = \sup_{\mathcal{E}^1_C(X,\omega)} \mathcal{M}_\mu$$

for $C > 0$ large enough and it follows from Lemmas 11.15 and 11.17 that

$$\sup_{\mathcal{E}^1_C(X,\omega)} \mathcal{M}_\mu = \sup_{\mathcal{E}^1_C(X,\omega)} \mathcal{F}_1 = \sup_{\mathcal{E}^1(X,\omega)} \mathcal{F}_1.$$

We can now proceed as in previous proofs, finding a maximizer $\varphi \in \mathcal{E}^1(X,\omega)$ of the functional $\mathcal{F}_1$ which is a critical point, hence a desired solution. □

Remark 11.14. *When μ is a volume form, it will follow from Chapter 14 that the solution is actually smooth. We recover this way the main result of* [Tian87]*, which deals with the case of Fano manifolds. In this context the functional $\mathcal{M}_\mu$ is often called the K-energy of Mabuchi; we shall come back to this in the last part of the book.*

It follows from Exercise 8.12 that the blow-up of $\mathbb{P}^2$ at eight points in general position is a Fano surface (a Del Pezzo surface of degree 1) whose α-invariant is larger than $2/3$.

In the sequel we set

$$\mu_\varphi := \frac{e^{-\varphi}\mu}{\int_X e^{-\varphi} d\mu}.$$

Lemma 11.15. *For all* $\varphi \in \mathcal{E}^1(X,\omega)$,

$$\mathcal{F}_1(\varphi) = \mathcal{M}_\mu(\varphi) + H_{\mu_\varphi}(MA(\varphi)) \geq \mathcal{M}_\mu(\varphi),$$

with equality if and only if $MA(\varphi) = \mu_\varphi$.

Proof. The equality follows from the definitions and observing that

$$H_{\mu_\varphi}(MA(\varphi)) = H_\mu(MA(\varphi)) + \int_X \varphi MA(\varphi) + \log\left[\int_X e^{-\varphi} d\mu\right].$$

The inequality follows from the concavity of the logarithm: a relative entropy $H_\mu(\nu)$ between two probability measures is always non-negative, it vanishes if and only if these are equal. □

We now show that the functional $\mathcal{M}_\mu$ is proper if $\alpha_\mu > n/(n+1)$:

Lemma 11.16. *Fix* $0 < \alpha < \alpha_\mu$. *There exists* $C_\alpha > 0$ *such that*

$$\mathcal{M}_\mu(\varphi) \leq [1 - (1-\alpha)(n+1)]\, E(\varphi) + C_\alpha,$$

for all $\varphi \in \mathcal{E}^1(X,\omega)$ *such that* $\sup_X \varphi = 0$. *Therefore,*

$$\mathcal{M}_\mu(\varphi) \leq \varepsilon E(\varphi) + C_\varepsilon \text{ with } \varepsilon > 0,$$

if $\alpha_\mu > n/(n+1)$.

Proof. Set $\psi = \varphi - \sup_X \varphi$. By assumption,

$$\int_X e^{-\alpha\psi - \log(MA(\psi)/\mu)} MA(\psi) = \int_X e^{-\alpha\psi} d\mu \leq e^{C_\alpha},$$

for some uniform constant $C_\alpha > 0$. It follows from the concavity of the logarithm that

$$-\alpha \int_X \psi MA(\psi) - C_\alpha \leq \int_X \log\left(\frac{MA(\psi)}{\mu}\right) MA(\psi) = H_\mu((MA(\psi)).$$

We infer that

$$\mathcal{M}_\mu(\psi) \leq E(\psi) - (1-\alpha)\int_X \psi MA(\psi) + C_\alpha.$$

The conclusion follows, since $-\int_X \psi MA(\psi) \leq -(n+1)E(\psi)$. □

It follows from our resolution of $(\mathrm{MA})_0$ in $\mathcal{E}^1(X,\omega)$ that for each $\varphi \in \mathcal{E}^1(X,\omega)$ normalized by $\sup_X \varphi = 0$, there exists a unique $\psi \in \mathcal{E}^1(X,\omega)$ also normalized by $\sup_X \psi = 0$ such that

$$MA(\psi) = \frac{e^{-\varphi}\mu}{\int_X e^{-\varphi}\, d\mu}.$$

We denote ψ by $\mathrm{Ric}^{-1}(\varphi) := \psi$.

Lemma 11.17. *For all $\varphi \in \mathcal{E}^1(X,\omega)$, $\sup_X \varphi = 0$, one has*

$$\mathcal{F}_1(\mathrm{Ric}^{-1}(\varphi)) \geq \mathcal{F}_1(\varphi) \quad \textit{and} \quad \mathcal{M}_\mu(\mathrm{Ric}^{-1}(\varphi)) \geq \mathcal{F}_1(\varphi).$$

Proof. Observe that all quantities are invariant under translations. We can thus assume without loss of generality that φ and $\psi = \mathrm{Ric}^{-1}(\varphi)$ are actually normalized so that

$$\int_X e^{-\varphi} d\mu = \int_X e^{-\psi} d\mu = 1.$$

Thus $\mu_\varphi = e^{-\varphi}\mu = MA(\psi)$ and $\mu_\psi = e^{-\psi}\mu$ are both probability measures and we have to show that $\mathcal{F}_1(\psi) = E(\psi) \geq E(\varphi) = \mathcal{F}_1(\varphi)$.

It follows from the cocycle property of E that

$$E(\varphi) - E(\psi) = \frac{1}{(n+1)V} \sum_{j=0}^{n} \int_X (\varphi - \psi)\omega_\varphi^j \wedge \omega_\psi^{n-j} \leq \int_X (\varphi - \psi) MA(\psi).$$

The latter integral equals

$$\int_X (\varphi - \psi) d\mu_\varphi = \int_X (\varphi - \psi) e^{-(\varphi-\psi)} d\mu_\psi = -H_{\mu_\psi}(\mu_\varphi) \leq 0,$$

hence $\mathcal{F}_1(\psi) \geq \mathcal{F}_1(\varphi)$. Now

$$\mathcal{M}_\mu(\psi) = E(\psi) + \int_X (\varphi - \psi) MA(\psi) \geq E(\varphi) = \mathcal{F}_1(\varphi),$$

and the proof is complete. □

11.3.2 Non-pluripolar measures

Theorem 11.18. *Let μ be a probability measure on X. There exists $\varphi \in \mathcal{E}(X,\omega)$ such that $\mu = MA(\varphi)$ iff μ does not charge pluripolar sets.*

The solution is moreover unique up to an additive constant.

It follows from the very definition of $\mathcal{E}(X,\omega)$ that the Monge–Ampère measures $MA(\varphi)$ do not charge pluripolar sets whenever $\varphi \in \mathcal{E}(X,\omega)$, thus the important contents of the theorem is that, conversely, any non-pluripolar probability measure is the Monge–Ampère measure of a (unique up to an additive constant) function in $\mathcal{E}(X,\omega)$.

Proof. Assume that μ does not charge pluripolar sets. By using a projection argument as earlier, we can find $u \in \mathcal{E}^1(X,\omega)$ and $0 \leq f \in L^1(MA(u))$ such that $\mu = f\, MA(u)$.

We can assume without loss of generality that $u \leq 0$ and even assume that u is bounded by arguing as follows: set $v = \exp(u)$ and observe that it is a bounded ω-psh function:

$$\omega + dd^c e^u = (1 - e^u)\omega + e^u \omega_u + e^u du \wedge d^c u \geq 0,$$

such that

$$e^{nu} MA(u) \leq MA(v).$$

Thus μ is also absolutely continuous with respect to $MA(v)$, hence up to replacing u by v, we assume that u is bounded in the sequel. Set

$$\mu_j := c_j \min(f, j) MA(u),$$

where $c_j \searrow 1$ is such that $\mu_j(X) = \mu(X) = 1$. We can assume without loss of generality that $1 \leq c_j \leq 2$. Since $\mu_j \leq 2jMA(u)$, our previous results ensure the existence of $\varphi_j \in \mathcal{E}^1(X, \omega)$ such that

$$\mu_j = MA(\varphi_j) \text{ and } \sup_X \varphi_j = 0.$$

Passing to a subsequence if necessary, we can assume $\varphi_j \to \varphi$ in $L^1(X)$, for some function $\varphi \in PSH(X, \omega)$ such that $\sup_X \varphi = 0$.

We claim that $\varphi \in \mathcal{E}_\chi(X, \omega)$ for some $\chi \in \mathcal{W}_+$, and that $\mu = MA(\varphi)$. The function χ is defined as follows. Let $\gamma : \mathbb{R}^+ \to \mathbb{R}^+$ be a convex increasing function with $\lim_{z \to +\infty} \gamma(z)/z = +\infty$ such that $\gamma \circ f$ still belongs to $L^1(MA(u))$ (see [RR02] for the construction of γ). Let γ^* be Legendre transform of γ,

$$\gamma^* : \mathbb{R}^+ \ni z \longmapsto \sup\{zy - \gamma(y) \mid y \in \mathbb{R}^+\} \in \mathbb{R}^+,$$

and set $\chi(t) := -(\gamma^*)^{-1}(-t)$: this is a convex increasing function such that $\chi(-\infty) = -\infty$, which satisfies

$$(-\chi)(-t) \cdot f(x) \leq -t + \gamma \circ f(x), \quad \text{for all } (t, x) \in \mathbb{R}^- \times X, \tag{11.3.1}$$

since $zy \leq \gamma^*(z) + \gamma(y)$ for all $z, y \in \mathbb{R}^+$.

Note that we can assume $\chi(0) = 0$ without loss of generality, by imposing $\gamma(0) = 0$. Applying (11.3.1) with $t = \varphi_j(x)$ and averaging against $MA(u)$, we infer that

$$\begin{aligned} 0 \leq \int_X (-\chi) \circ \varphi_j \, MA(\varphi_j) &\leq 2 \int_X (-\chi) \circ \varphi_j \, f MA(u) \\ &\leq 2 \int_X (-\varphi_j) MA(u) + 2 \int_X \gamma \circ f \, MA(u). \end{aligned}$$

It follows from the Chern–Levine–Nirenberg inequalities that the integral $\int_X (-\varphi_j) MA(u)$ is uniformly bounded from above, because u and $\int_X (-\varphi_j)\omega^n$ are bounded. Thus the sequence of integrals $(\int_X (-\chi) \circ \varphi_j \, MA(\varphi_j))$ is bounded.

Set $\phi_j := (\sup_{l\geq j}\varphi_l)^* \in \mathcal{E}^1(X,\omega)$. Observe that

$$MA(\phi_j) \geq \inf_{\ell\geq j} MA(\varphi_\ell) \geq \min(f,j)MA(u).$$

The sequence (ϕ_j) decreases towards φ and satisfies $\varphi_j \leq \phi_j \leq 0$, hence

$$0 \geq E_\chi(\phi_j) \geq E_\chi(\varphi_j) \geq \int_X \chi\circ\varphi_j\, MA(\varphi_j) \geq -C.$$

Therefore $\varphi \in \mathcal{E}_\chi(X,\omega) \subset \mathcal{E}(X,\omega)$ and

$$MA(\varphi) = \lim_{j\to+\infty} MA(\phi_j) \geq \lim_{j\to+\infty} \min(f,j)MA(u) = \mu.$$

Since μ and $MA(\varphi)$ are both probability measures, this actually yields equality, hence the proof of the existence is complete.

The uniqueness will be treated in the last section of this chapter. □

11.3.3 The classes $\mathcal{E}^p(X,\omega)$

Definition 11.19. *Fix $p > 0$. We let $\mathcal{E}^p(X,\omega)$ denote the class $\mathcal{E}_\chi(X,\omega)$ for the weight $\chi : t \mapsto -(-t)^p$ lies in $\mathcal{W}$.*

Note that $\chi(t) = -(-t)^p$ is only convex when $0 < p \leq 1$ (and $t \leq 0$), however these classes share several properties thanks to the homogeneity of the weight. We let

$$E_p(\varphi) := \frac{1}{(n+1)V}\sum_{j=0}^{n}\int_X -(-\varphi)^p\,\omega_\varphi^j\wedge\omega^{n-j}$$

denote the corresponding energy E_χ.

We have a perfect understanding of the range of the complex Monge–Ampère operator acting on the classes $\mathcal{E}^p(X,\omega)$ as the following result shows:

Theorem 11.20. *Let μ be a probability measure. There exists $\varphi \in \mathcal{E}^p(X,\omega)$ such that $\mu = MA(\varphi)$ if and only if $\mathcal{E}^p(X,\omega) \subset L^p(\mu)$.*

Proof. Assume first that $\mu = MA(\varphi)$ for some $\varphi \in \mathcal{E}^p(X,\omega)$. When $0 < p \leq 1$, it follows from Proposition 10.25 that $\mathcal{E}^p(X,\omega) \subset L^p(\mu)$. One can adapt the proof of Proposition 10.25 so as to cover as well the case of non-convex homogeneous weights $\chi(t) = -(-t)^p$, $p > 1$. We let the reader verify this in Exercise 11.11.

It follows from Exercise 10.18 that the class $\mathcal{E}^p(X,\omega)$ characterizes pluripolar sets. Therefore, μ does not charge pluripolar sets. Theorem 11.18 thus yields the existence of $\varphi \in \mathcal{E}(X,\omega)$ such that $\sup_X \varphi = -1$ and $\mu = MA(\varphi)$. Set $\varphi_j := \max(\varphi, -j)$. When $0 < p \leq 1$, we obtain

$$0 \leq |E_p(\varphi_j)| \leq \int_X (-\varphi_j)^p\, MA(\varphi_j) = \int_X (-\varphi_j)^p\, d\mu \leq C\,|E_p(\varphi_j)|^{\frac{p}{p+1}}$$

by Lemma 11.21 below, showing that $E_p(\varphi_j)$ is bounded, hence $\varphi \in \mathcal{E}^p(X,\omega)$.

When $p > 1$, the first inequality has to be weakened and replaced by

$$|E_p(\varphi_j)| \leq C_p \int_X (-\varphi_j)^p \, MA(\varphi_j),$$

as the reader will figure out (see the proof of [GZ07, Proposition 2.10]). □

Lemma 11.21. *Let μ be a probability measure such that $\mathcal{E}^p(X,\omega) \subset L^p(\mu)$. Then there exists $C > 0$ such that for all $\varphi \in \mathcal{E}^p(X,\omega)$ normalized by $\sup_X \varphi = -1$, one has*

$$\int_X |\varphi|^p d\mu \leq C\,|E_p(\varphi)|^{\frac{p}{p+1}}.$$

We leave the proof of this lemma as Exercise 11.12.

11.4 Uniqueness

We have shown above that the uniqueness of solutions to $(\mathrm{MA})_\lambda$ is a consequence of the domination principle when $\lambda < 0$. We study here the uniqueness of solutions to $(\mathrm{MA})_0$. The case $\lambda > 0$ is more delicate and will be adressed in the last part of the book.

11.4.1 Calabi's trick We start by explaining an argument due to Calabi which shows the uniqueness of solutions φ to $(\mathrm{MA})_0$ when ω_φ is further assumed to be a Kähler form.

Let φ, ψ be two solutions of $(\mathrm{MA})_0$ such that both $\omega_\varphi := \omega + dd^c\varphi$ and $\omega_\psi := \omega + dd^c\psi$ are Kähler forms. Observe that

$$0 = (\omega + dd^c\varphi)^n - (\omega + dd^c\psi)^n = dd^c(\varphi - \psi) \wedge S,$$

where

$$S := \sum_{j=0}^{n-1} \omega_\varphi^j \wedge \omega_\psi^{n-1-j}$$

is a closed positive $(n-1, n-1)$-form. Multiplying the previous equality by $\psi - \varphi$ and integrating by parts, we obtain

$$\int_X d(\varphi - \psi) \wedge d^c(\varphi - \psi) \wedge \omega_\varphi^{n-1} \leq \int_X d(\varphi - \psi) \wedge d^c(\varphi - \psi) \wedge S = 0,$$

so that $\nabla_{\omega_\varphi}(\varphi - \psi) = 0$, hence $\varphi - \psi$ is constant.

11.4.2 Uniqueness in $\mathcal{E}^1(X,\omega)$ The argument of Calabi has been adapted and generalized by Blocki [Blo03] to the case when φ,ψ are bounded ω-psh functions. We have further generalized this approach in [GZ07] as we now explain.

Theorem 11.22. *Assume* $(\omega + dd^c\varphi)^n \equiv (\omega + dd^c\psi)^n$, *where* $\varphi,\psi \in \mathcal{E}^1(X,\omega)$. *Then* $\varphi - \psi$ *is constant.*

Proof. Set $f = (\varphi-\psi)/2$ and $h = (\varphi+\psi)/2$. It follows from the convexity property of $\mathcal{E}^1(X,\omega)$ that $h \in \mathcal{E}^1(X,\omega)$. We can assume, without loss of generality, that $\varphi,\psi \le -C_\omega$, where $C_\omega > 0$ is chosen so that $\int_X (-h)\omega_h^n \ge 1$. We are going to prove that $\nabla_\omega f = 0$ by showing that

$$\int_X df \wedge d^c f \wedge \omega^{n-1} = 0.$$

This will be done by establishing an upper bound involving the energy of h and by using the following formal computation,

$$\begin{aligned} 0 \le \int_X df \wedge d^c f \wedge \omega_h^{n-1} &\le \int_X df \wedge d^c f \wedge \sum_{l=0}^{n-1} \omega_\varphi^l \wedge \omega_\psi^{n-1-l} \\ &= \int_X \frac{-f}{2}(\omega_\varphi^n - \omega_\psi^n) = 0. \end{aligned}$$

Of course we should (and could) justify these integrations by parts and make sense of all terms involved in this computation. Rather, we are going to establish the following a priori bound, when φ,ψ are bounded,

$$(\dagger) \qquad \int_X df \wedge d^c f \wedge \omega^{n-1} \le 3^{n-1} \left(\int_X df \wedge d^c f \wedge \omega_h^{n-1} \right)^{1/2^{n-1}} \int_X (-h)\omega_h^n.$$

Approximating φ,ψ by $\varphi_j = \max(\varphi,-j), \psi_j = \max(\psi,-j)$, it will then follow from the continuity properties and $(\dagger)$ that $2f = \varphi - \psi$ is constant, if $\omega_\varphi^n \equiv \omega_\psi^n$.

The a priori bound $(\dagger)$ follows by applying inductively the following inequality, where $T = \omega_h^l \wedge \omega^{n-2-l}$ is a closed positive current of bidimension $(2,2)$ and $l = n-2,\dots,0$,

$$(\dagger)_T \quad \int_X df \wedge d^c f \wedge \omega \wedge T \le 3 \left(\int_X df \wedge d^c f \wedge \omega_h \wedge T \right)^{\frac{1}{2}} \left(\int_X (-h)\omega_h^2 \wedge T \right)^{\frac{1}{2}}.$$

Indeed, observe that for $T = \omega_h^l \wedge \omega^{n-2-l}$,

$$\int_X (-h)\omega_h^{2+l} \wedge \omega^{n-2-l} \le \int_X (-h)\omega_h^n, \text{ for all } l = n-2,\dots,0$$

and use that $\int_X (-h)\omega_h^n \ge 1$ since $h \le -C_\omega$, to derive $(\dagger)$ from $(\dagger)_T$.

We now establish $(\dagger)_T$. Note that

$$df \wedge d^c f \wedge \omega = df \wedge d^c f \wedge \omega_h - df \wedge d^c f \wedge dd^c h,$$

hence integrating by parts yields

$$\begin{aligned}\int_X df \wedge d^c f \wedge \omega \wedge T &= \int_X df \wedge d^c f \wedge \omega_h \wedge T \\ &+ \int_X df \wedge d^c h \wedge \frac{(\omega_\varphi - \omega_\psi)}{2} \wedge T.\end{aligned}$$

It follows from the Cauchy–Schwarz inequality that

$$\begin{aligned}\left|\int_X df \wedge d^c h \wedge \omega_\varphi \wedge T\right| &\leq 2\left(\int_X df \wedge d^c f \wedge \omega_h \wedge T\right)^{\frac{1}{2}} \\ &\times \left(\int_X dh \wedge d^c h \wedge \omega_h \wedge T\right)^{\frac{1}{2}}.\end{aligned}$$

Note that we can get a similar control on $\int_X df \wedge d^c f \wedge \omega_\psi \wedge T$. Thus $(\dagger)_T$ follows upon observing that

$$\int_X df \wedge d^c f \wedge S = \frac{1}{4}\int_X (\psi - \varphi)(\omega_\varphi - \omega_\psi) \wedge S \leq \int_X (-h)\omega_h \wedge S,$$

where S is any positive closed current of bidimension $(1,1)$. □

11.4.3 Uniqueness in $\mathcal{E}(X,\omega)$

In this section we show that solutions to $(\mathrm{MA})_0$ in $\mathcal{E}(X,\omega)$ are unique (up to an additive constant):

Theorem 11.23. *Assume $(\omega + dd^c\varphi)^n \equiv (\omega + dd^c\psi)^n$, where $\varphi, \psi \in \mathcal{E}(X,\omega)$. Then $\varphi - \psi$ is constant.*

One cannot try and estimate the L^2-norm of the gradient of $\varphi - \psi$, since in general functions in $\mathcal{E}(X,\omega)$ do not have gradient in L^2. We follow instead a different path, due to Dinew [Din09].

11.4.3.1 Proof of the uniqueness

Set $T_1 = \omega + dd^c\varphi$, $T_2 = \omega + dd^c\psi$ and

$$\mu := T_1^n = T_2^n.$$

We first remark that the log-concavity property of mixed Monge–Ampère measures (Proposition 11.24) implies that

$$T_1^k \wedge T_2^{n-k} \geq \mu,$$

hence

$$T_1^k \wedge T_2^{n-k} = \mu \tag{11.4.1}$$

for $k = 0, \dots, n$ since both measures have same total mass V.

Since μ puts no mass on pluripolar subsets, the set of $t \in \mathbb{R}$ such that $\mu\{\varphi = \psi + t\} > 0$ coincides with the discontinuity locus of the non-decreasing function

$$t \longmapsto \mu\{\varphi < \psi + t\},$$

and is thus at most countable. We are going to show by contradiction that there exists $t \in \mathbb{R}$ such that $\varphi = \psi + t$ μ-almost everywhere.

Assume by contradiction that $\mu\{\varphi = \psi + t\} < V$ for all t, so that we can find $t \in \mathbb{R}$ such that

$$0 < \mu\{\varphi < \psi + t\} < V.$$

By monotone convergence, upon slightly pertubing t we can furthermore arrange that

$$\mu\{\varphi = \psi + t\} = 0.$$

Replacing ψ by $\psi + t$, we can finally assume that $t = 0$, so that both $\mu\{\varphi < \psi\}$ and $\mu\{\varphi < \psi\}$ are less than V while $\mu\{\varphi = \psi\} = 0$. We can thus pick $\varepsilon > 0$ small enough such that the mass of $(1+\varepsilon)^n\mu$ on both $\{\varphi < \psi\}$ and $\{\psi < \varphi\}$ is still less than V.

Fix $\eta > 0$ such that the non-pluripolar measure

$$\mathbf{1}_{\{\varphi<\psi\}}(1+\varepsilon)^n\mu + \mathbf{1}_{\{\varphi>\psi\}}\eta\mu$$

has total mass V. It follows from Theorem 11.18 (existence part) that there exists $0 \geq u \in \mathcal{E}(X, \omega)$ such that $T = \omega + dd^c u$ satisfies

$$T^n = \mathbf{1}_{\{\varphi<\psi\}}(1+\varepsilon)^n\mu + \mathbf{1}_{\{\varphi>\psi\}}\eta\mu.$$

It follows from Proposition 11.24 that for $j = 1, 2$,

$$T \wedge T_j^{n-1} \geq (1+\varepsilon)\mu$$

on $\{\varphi < \psi\}$. Consider

$$O_\delta := \{(1-\delta)\varphi < (1-\delta)\psi + \delta u\} \subset \{\varphi < \psi\}.$$

Since u, φ, ψ have finite energy the comparison principle yields

$$\int_{O_\delta} ((1-\delta)T_2 + \delta T) \wedge T_j^{n-1} \leq \int_{O_\delta} ((1-\delta)T_1 + \delta\omega) \wedge T_j^{n-1}$$

for $j = 1, 2$. By (11.4.1) it follows that

$$\delta(1+\varepsilon)\int_{O_\delta} \mu \leq \delta \int_{O_\delta} T \wedge T_j^{n-1} \leq \delta \int_{\{\varphi<\psi\}} \omega \wedge T_j^{n-1}.$$

Dividing by $\delta > 0$ and letting δ tend to 0, we infer that

$$(1+\varepsilon)\int_{\{\varphi<\psi\}}\mu \leq \int_{\{\varphi<\psi\}}\omega\wedge T_j^{n-1}$$

for $j = 1, 2$, by monotone convergence.

We can now play the same game with $\{\varphi < \psi\}$ (and a possibly different T) to conclude that

$$(1+\varepsilon)\int_{\{\varphi>\psi\}}\mu \leq \int_{\{\varphi>\psi\}}\omega\wedge T_j^{n-1}.$$

Summing these two inequalities and using $\mu\{\varphi = \psi\} = 0$ we obtain

$$(1+\varepsilon)V \leq \int_X \omega\wedge T_j^{n-1} \leq V,$$

a contradiction.

At this point we have proved that $\varphi = \psi$ a.e. with respect to $MA(\varphi) = MA(\psi)$. One can now use the domination principle (Proposition 12.30) to conclude that $\varphi \leq \psi$, hence $\varphi = \psi$ by symmetry. The proof is complete.

11.4.3.2 Log concavity of mixed Monge–Ampère measures

Proposition 11.24. *Let μ be a non-pluripolar measure and fix $0 \leq f_1, \ldots, f_n \in L^1(\mu)$. Assume $\varphi_i \in \mathcal{E}(X,\omega)$ are such that*

$$T_i^n = (\omega + dd^c\varphi_i)^n \geq f_i\,\mu$$

for all $1 \leq i \leq n$. Then

$$T_1\wedge\ldots\wedge T_n \geq (f_1\cdots f_n)^{1/n}\mu.$$

This inequality is a well-known variation of the Brunn–Minkowski inequality when the whole data are smooth. We will reduce to this case by a smart regularization process due to [Din09b, DL15].

Proof. We first show how to reduce to the case when all the functions and densities involved are *bounded*.

Recall that one can write $\mu = g(\omega + dd^c u)^n$ with u ω-psh and bounded and $0 \leq g \in L^1(\mu)$. Replacing f_i by $f_i g$, one can thus assume that

$$\mu = (\omega + dd^c u)^n, \quad \text{with } u \in PSH(X)\cap L^\infty(X).$$

Replacing f_i by $\min(f_i, k)$ we can further assume that f_i is bounded and eventually let k increase to $+\infty$ to conclude (using the Monotone convergence theorem).

We now show that one can also assume that the Monge–Ampère potentials φ_i are bounded. Consider indeed $\varphi_i^{(k)} = \max(\varphi_i, -k)$. It follows from the maximum principle that

$$(\omega + dd^c\varphi_i^{(k)})^n \geq f_i^{(k)}\,\mu,$$

where $f_i^{(k)} = \mathbf{1}_{\{\varphi_i < -k\}} f_i \in L^\infty(X)$. If we can establish the log concavity inequality when all data are bounded, the conclusion will thus follow by letting k increase to $+\infty$, observing that μ does not charge the pluripolar sets ($\varphi_i = -\infty$).

From now on we assume that $\mu = (\omega + dd^c u)^n$ with u, φ_i ω-psh and bounded, and that f_i are bounded densities. Fix $\delta > 0$. We let $\varphi_{i,p}$ denote the unique ω-psh function that solves the complex Monge–Ampère equation

$$(\omega + dd^c\varphi_{i,p})^n = e^{p(\varphi_{i,p}-\varphi_i)}\,\{(1-\delta)\,f_i\mu + \delta(\omega + dd^c\varphi_i)^n\}\,.$$

It follows from the comparison principle that

$$\varphi_i \leq \varphi_{i,p} \leq \varphi_i + \frac{1}{p}\log\delta^{-1}.$$

Thus the $\varphi_{i,p}$'s are uniformly bounded and converge to φ_i as $p \to +\infty$. It follows from Lemma 11.25 that

$$(\omega + dd^c\varphi_{1,p}) \wedge \ldots \wedge (\omega + dd^c\varphi_{n,p}) \geq (1-\delta)(f_1 \cdots f_n)^{1/n}\mu.$$

The conclusion follows by letting $p \to +\infty$ and then $\delta \to 0^+$. □

We now establish the desired log-concavity inequality in a particular case that can be well reduced to the smooth setting. We follow here [DL15, Lemma 3.6] and need to use smooth resolutions of complex Monge–Ampère equations (Yau's theorem) that will be available in Chapter 14.

Lemma 11.25. *Let u, v_i, φ_i be bounded ω-psh functions and $0 \leq f_i, g_i$ be bounded densities such that*

$$(\omega + dd^c\varphi_i)^n = f_i(\omega + dd^c u)^n + g_i(\omega + dd^c v_i)^n$$

for all $1 \leq i \leq n$. Then

$$(\omega + dd^c\varphi_1) \wedge \ldots \wedge (\omega + dd^c\varphi_n) \geq (f_1 \cdots f_n)^{1/n}(\omega + dd^c u)^n.$$

Proof. As mentioned, the inequality is known to hold when the whole data are smooth so we reduce to this case step by step.

Assume first that u and the v_j's are smooth. We approximate in L^1 the densities f_i and g_i by uniformly bounded sequences of smooth densities $f_{i,\varepsilon}, g_{i,\varepsilon}$ normalized so that one can find unique normalized smooth solutions $\varphi_{i,\varepsilon}$ of the Monge–Ampère equations

$$(\omega + dd^c\varphi_{i,\varepsilon})^n = f_{i,\varepsilon}(\omega + dd^c u)^n + g_{i,\varepsilon}(\omega + dd^c v_i)^n.$$

We use here Yau's celebrated result (see Chapter 14). It follows from the stability properties in Chapter 12 that the solutions $\varphi_{i,\varepsilon}$ converge in capacity to φ_i as ε decrease to zero. The inequality in this case is therefore a consequence of the smooth one.

We now relax the smoothness assumption on u, v_j, but assume that ω is Kähler and the densities $f_i, g_i \geq \delta > 0$ are quasi-continuous and bounded away from zero. We approximate u, v_j by decreasing sequences of smooth ω-psh functions $u_\varepsilon, v_{j,\varepsilon}$ and solve

$$(\omega + dd^c\varphi_{i,\varepsilon})^n = e^{\varphi_{i,\varepsilon}-\varphi_i} f_i(\omega + dd^c u_\varepsilon)^n + g_i(\omega + dd^c v_{i,\varepsilon})^n.$$

It follows from the comparison principle that the $\varphi_{i,\varepsilon}$'s are uniformly bounded (use that the densities are bounded away from zero and infinity) and it follows from stability properties that $\varphi_{i,\varepsilon}$ converges in capacity to φ_i (the stability properties in Chapter 12 are not sufficient here, we need to add one more ingredient, for example [DL15, Lemma 3.2]). Using the previous step we get

$$(\omega + dd^c\varphi_{1,\varepsilon}) \wedge \ldots \wedge (\omega + dd^c\varphi_{n,\varepsilon}) \geq e^{\sum \frac{\varphi_{i,\varepsilon}-\varphi_i}{n}} (f_1 \cdots f_n)^{1/n} (\omega + dd^c u_\varepsilon)^n.$$

The left-hand side converge to $(\omega + dd^c\varphi_1) \wedge \ldots \wedge (\omega + dd^c\varphi_n)$ since the sequences converge in capacity, while the right-hand side converge to $(f_1 \cdots f_n)^{1/n}(\omega + dd^c u)^n$ by using that the densities are quasi-continuous.

We now get rid of the extra assumptions made on f_i, g_i. We can approximate f_i, g_i by uniformly bounded sequences of continuous densities $f_{i,\varepsilon}, g_{i,\varepsilon} \geq \varepsilon > 0$ which converge in L^1, and solve

$$(\omega + dd^c\varphi_{i,\varepsilon})^n = e^{\varphi_{i,\varepsilon}-\varphi} \{f_{i,\varepsilon}(\omega + dd^c u)^n + g_{i,\varepsilon}(\omega + dd^c v_i)^n\}.$$

The conclusion follows by using the previous step together with similar arguments to pass to the limit.

The proof is thus complete when ω is Kähler. One can now localize and obtain this result for (local) psh function that are bounded in a domain of $\mathbb{C}^n$. This inequality is thus essentially local (but the local regularization process is actually more delicate!) and we can apply it in the ample locus of $\{\omega\}$ when the latter is merely semi-positive and big. The conclusion follows since the non-pluripolar product puts no mass to the complement of the ample locus. □

11.5 Exercises

Exercise 11.1. *Let $I : PSH(X, \omega) \to \mathbb{R} \cup \{-\infty\}$ be a functional which is convex, non-decreasing, and satisfies the scaling property*

$$I(\varphi + c) = I(\varphi) + c \text{ for all } \varphi \in PSH(X, \omega) \text{ and } c \in \mathbb{R}.$$

Let $\mathcal{K} \subset PSH(X, \omega)$ be a compact convex subset. Show that if I is finite on $\mathcal{K}$, then it is uniformly bounded on $\mathcal{K}$.

Exercise 11.2. *Let (X,ω) be a compact Kähler manifold and μ a probability measure on X. Fix a smooth function $\varphi \in PSH(X,\omega)$. Check that*

$$\lambda^{-1}\log\left[\int_X e^{-\lambda\varphi}d\mu\right] \longrightarrow -\int_X \varphi d\mu$$

when $\lambda \to 0$. When φ is less regular, under what condition on φ (resp. μ) does this remain true?

Exercise 11.3. *Let dV be a volume form and $\mu = f dV$ be a probability measure with $f \in L^p(dV)$, $p > 1$. Show that there exists $C > 0$ such that for all $\varphi \in \mathcal{E}^1(X,\omega)$,*

$$\mathcal{F}_0(\varphi) \leq E(\varphi) - \sup_X \varphi + C.$$

Exercise 11.4. *Let μ be a non-pluripolar probability measure. Show that there exists $\delta > 0$ such that for all $\varphi \in PSH(X,\omega)$,*

$$\delta \leq \int_X e^{\varphi - \sup_X \varphi} d\mu.$$

Exercise 11.5. *Let μ be a probability measure on a compact Kähler manifold (X,ω) such that $\mathcal{E}^1(X,\omega) \subset L^1(\mu)$. For $\tau > 0$, we let φ_τ denote the unique function $\varphi_\tau \in \mathcal{E}^1(X,\omega)$ such that*

$$MA(\varphi_\tau) = e^{\tau\varphi_\tau}\mu.$$

Show that φ_τ converges, as τ decreases to 0^+, to the unique ω-psh function $\varphi_0 \in \mathcal{E}^1(X,\omega)$ such that

$$MA(\varphi_0) = \mu \quad \textit{and} \quad \int_X \varphi_0\, d\mu = 0.$$

Exercise 11.6. *Let (X,ω) be a compact Kähler manifold and fix $p \in X$. Explain why there can be no solution $\varphi \in \mathcal{E}(X,\omega)$ to the Monge–Ampère equation*

$$MA(\varphi) = e^{\varphi}\delta_p,$$

where δ_p denotes the Dirac mass at the point p.

Exercise 11.7. *Let μ be a non-pluripolar probability measure on a compact Kähler manifold (X,ω). Let $\varphi \in \mathcal{E}^1(X,\omega)$ denote the unique solution to*

$$MA(\varphi) = e^{\varphi}\mu.$$

Show that φ is the envelope of subsolutions,

$$\varphi(x) = \sup\{\psi(x) \mid \psi \in \mathcal{E}^1(X,\omega) \textit{ such that } MA(\psi) \geq e^{\psi}\mu\}.$$

Exercise 11.8. *We assume here $(X,\omega) = (\mathbb{P}^1, \omega_{\mathrm{FS}})$ is the Riemann sphere. We fix a probability measure $\mu = \omega + dd^c\psi$ which is smooth except at one point (the origin in some affine chart), where it has a radial singularity of the form*

$$\mu = \Delta\chi \circ \log|z|, \text{ with } \chi(t) = -(-t)^a,\ 0 < \alpha < 1/2.$$

(i) *Show that $\mathcal{E}^1(X,\omega) \subset L^1(\mu)$.*

(ii) *Show that one can choose ε_j (resp. δ_j) which decreases slowly (resp. fast) towards 0^+ such that $\varphi_j = \varepsilon_j \log\left(|z|^2 + \delta_j^2\right) \to 0$ in L^1, while $\int \varphi_j d\mu \to -\infty$.*

Exercise 11.9. *Let $\varphi \le 0$ be a ω-plurisubharmonic function. Show that*

$$\psi := -\log(1-\varphi) = \chi \circ \varphi \in \bigcap_{p\ge 1} \mathcal{E}^p(X,\omega).$$

Exercise 11.10. *Let μ be a probability measure on (X,ω).*

(i) *Show that if there exist $\alpha > p/(p+1)$ and $A > 0$ such that*

$$\mu(E) \le A\,\mathrm{Cap}_\omega(E)^\alpha, \quad \text{for all Borel sets } E \subset X, \tag{11.5.1}$$

then $\mathcal{E}^p(X,\omega) \subset L^p(\mu)$.

(ii) *Conversely, show that if $\mathcal{E}^p(X,\omega) \subset L^p(\mu)$, $p > 1$, then there exists $0 < \alpha < 1$ and $A > 0$ such that (11.5.1) is satisfied.*

Exercise 11.11. *Let $\mu = MA(\varphi)$ be a probability measure such that $\varphi \in \mathcal{E}^p(X,\omega)$ for some $p \ge 1$. Show that*

$$\mathcal{E}^p(X,\omega) \subset L^p(\mu).$$

Exercise 11.12. *Complete the proof of Lemma 11.21. Namely, show that if μ is a probability measure such that $\mathcal{E}^p(X,\omega) \subset L^p(\mu)$, then there exists $C > 0$ such that for all $\varphi \in \mathcal{E}^p(X,\omega)$ normalized by $\sup_X \varphi = -1$,*

$$\int_X |\varphi|^p d\mu \le C\,|E_p(\varphi)|^{\frac{p}{p+1}}.$$

Exercise 11.13. *We assume here $(X,\omega) = (\mathbb{P}^n, \omega_{\mathrm{FS}})$. Let $z \in \mathbb{C}^n$ denote affine coordinates in a chart $\mathbb{C}^n \subset X$. Set $u(z) := \log[1+|z|^2]$ and check that*

$$(dd^c u)^n = e^{-(n+1)u(z)} dV(z).$$

Use this to produce several solutions $\varphi \in PSH(X,(n+1)\omega)$ of the Monge–Ampère equation

$$(\omega + dd^c\varphi)^n = e^{-(n+1)\varphi}\omega^n.$$

Exercise 11.14. *Fix $\psi \in \mathcal{E}^1(X,\omega)$ and set $\mu = MA(\psi)$. Show that*

$$\varphi \in \mathcal{E}^1_C(X,\omega) \longmapsto \int_X \varphi \, d\mu$$

is continuous for each $C > 0$.

Exercise 11.15. *Let $\pi : X \to \mathbb{P}^n$ be the blow-up of the complex projective space at one point p. Fix a hyperplane $H \sim \mathbb{P}^{n-1}$ such that $p \notin H$ and let $g : \mathbb{P}^n \to H$ denote the (meromorphic) projection from p onto H: any point $q \in \mathbb{P}^n \setminus \{p\}$ is sent to the intersection of H with the line (pq). Let $f : X \to \mathbb{P}^{n-1}$ be the holomorphic projection obtained by desingularizing g.*

Fix a Kähler form ω on H cohomologous to the Fubini–Study form of $\mathbb{P}^{n-1}$ and consider

$$T := \pi_* f^* \omega.$$

(i) *Show that the current T is cohomologous to the Fubini–Study form ω_{FS} of $\mathbb{P}^n$, that it is smooth in $\mathbb{P}^n \setminus \{p\}$ with Lelong number one at p.*

(ii) *Show that $T = \omega_{\mathrm{FS}} + dd^c\varphi$ with $1\varphi \in \mathrm{DMA}(\mathbb{P}^n, \omega_{\mathrm{FS}}) \setminus \mathcal{E}(\mathbb{P}^n, \omega_{\mathrm{FS}})$ and*

$$(\omega_{\mathrm{FS}} + dd^c\varphi)^n = \delta_p = \textit{Dirac mass at } p.$$

(iii) *Produce many other solutions $\psi \in \mathrm{DMA}(\mathbb{P}^n, \omega_{\mathrm{FS}})$ of the equation $(\omega_{\mathrm{FS}} + dd^c\psi)^n = \delta_p$.*

12 Uniform a priori estimates

Our main goal in this chapter is to establish the following uniform a priori estimate due to Kolodziej [Kol98]:

Theorem 12.1. *Let $\mu = MA(\varphi) = f dV$ be a probability Monge–Ampère measure with density $f \in L^p$, $p > 1$. Then*

$$\mathrm{Osc}_X \varphi \leq C$$

where C only depends on $\omega, dV, \|f\|_{L^p}$.

This result generalizes the celebrated uniform a priori estimate of Yau [Yau78], which dealt with the case when μ is a smooth volume form. While Yau's approach relies on Moser's iterative process, Kolodiej's idea is to show that the Monge–Ampère capacity of sublevel sets $(\varphi < -t)$ vanishes if $t > 0$ is large enough, by a clever use of the comparison principle.

We are going to follow Kolodziej's approach and expose some extensions of this result as well. We encourage the interested reader to read Yau's original article, together with some more recent lectures notes like [Blo12] for a detailed presentation of Moser's iterative process.

As before, the manifold X is assumed to be compact Kähler and ω is a semi-positive form such that $\int_X \omega^n > 0$.

12.1 Bounded solutions

12.1.1 Comparison of capacities Recall that the *Alexander–Taylor capacity* of K is defined by considering

$$M_\omega(K) := \sup_X V^*_{K,\omega} \in [0, +\infty], \tag{12.1.1}$$

so that $M_\omega(K) = +\infty$ if and only if K is pluripolar. The Monge–Ampère capacity

$$\mathrm{Cap}_\omega(K) = \sup \left\{ \int_K MA(u) |\ u \in PSH(X, \omega) \text{ with } 0 \leq u \leq 1 \right\}$$

and the Alexander–Taylor capacity compare through the following:

Lemma 12.2. *For every non-pluripolar compact subset K of X,*

$$1 \leq \mathrm{Cap}_\omega(K)^{-1/n} \leq \max(1, M_\omega(K)).$$

Proof. The left-hand inequality is trivial. In order to prove the right-hand inequality we consider two cases. If $M_\omega(K) \leq 1$, then $V^*_{K,\omega}$ is a candidate in the definition of $\mathrm{Cap}_\omega(K)$. Since $MA(V^*_{K,\omega})$ is supported on K, we thus have

$$\mathrm{Cap}_\omega(K) \geq \int_K MA(V^*_{K,\omega}) = \int_X MA(V^*_{K,\omega}) = 1$$

and the desired inequality holds in that case.

On the other hand if $M := M_\omega(K) \geq 1$ we have

$$0 \leq M^{-1} V^*_{K,\omega} \leq 1$$

and it follows by definition of the capacity again that

$$\mathrm{Cap}_\omega(K) \geq \int_K MA(M^{-1} V^*_{K,\omega}).$$

But since

$$MA(M^{-1} V^*_{K,\omega}) \geq M^{-n} MA(V^*_{K,\omega}),$$

we deduce that

$$\int_K MA(M^{-1} V^*_{K,\omega}) \geq M^{-n} \int_X MA(V^*_{K,\omega}) = M^{-n},$$

and the result follows. □

We also have a reverse control due to Alexander and Taylor [AT84]:

Lemma 12.3. *There exists $A > 0$ such that for every non-pluripolar compact subset K of X,*

$$\exp\left[-\frac{A}{\mathrm{Cap}_\omega(K)}\right] \leq T_\omega(K).$$

Proof. We can assume $M_K \geq 1$, otherwise it is sufficient to adjust the value of A. Let $\varphi \in PSH(X,\omega)$ be such that $\varphi \leq 0$ on K. Then $\varphi \leq M_K$ on X, hence $w := M_K^{-1}(\varphi - M_K) \in PSH(X,\omega)$ satisfies $\sup_X w \leq 0$ and $w \leq -1$ on K. We infer that $w \leq h^*_{K,\omega}$, hence

$$w_K := \frac{V^*_{K,\omega} - M_K}{M_K} \leq h^*_{K,\omega} \leq 0.$$

Now $\sup_X(V^*_{K,\omega} - M_K) = 0$, hence $\int_X |V^*_{K,\omega} - M_K|\omega^n \leq C_1$ for some constant $C_1 > 0$ independent of K. We infer, using that $h^*_{K,\omega} = -1$ on K, except perhaps on a pluripolar set which has zero $(\omega_{h^*_{K,\omega}})^n$-measure,

$$\begin{aligned}\mathrm{Cap}_\omega(K) = \int_K (\omega_{h^*_{K,\omega}})^n &\leq \int_X [-h^*_{K,\omega}](\omega_{h^*_{K,\omega}})^n \\ &\leq \frac{1}{M_K}\int_X -(V^*_{K,\omega} - M_K)(\omega_{h^*_{K,\omega}})^n \leq \frac{C_2}{M_K},\end{aligned}$$

where the latter upper bound follows from Chern–Levine–Nirenberg inequalities. The proof is complete. □

We can now compare Monge–Ampère capacities with respect to different semi-positive and big forms, following an observation of DiNezza [DiN15, Theorem 4.6]:

Corollary 12.4. *Let ω_1, ω_2 be two semi-positive and big forms. There exists $A \geq 1$ such that*

$$A^{-1}\mathrm{Cap}^n_{\omega_1} \leq \mathrm{Cap}_{\omega_2} \leq A\,\mathrm{Cap}^{1/n}_{\omega_1}.$$

In particular, the capacities Cap_{ω_i} characterize pluripolar sets.

Proof. We first assume that $\omega_1 \leq \omega_2$. In this case $PSH(X,\omega_1) \subset PSH(X,\omega_2)$, hence

$$\mathrm{Cap}_{\omega_1} \leq \mathrm{Cap}_{\omega_2}, \quad \text{while } T_{\omega_1} \geq T_{\omega_2}.$$

Using Lemmas 12.2 and 12.3, we obtain

$$\begin{aligned}\mathrm{Cap}_{\omega_1} \leq \mathrm{Cap}_{\omega_2} &\leq \frac{A}{-\ln T_{\omega_1}}\\ &\leq \frac{A}{-\ln T_{\omega_2}} \leq A'\mathrm{Cap}^{1/n}_{\omega_1}.\end{aligned}$$

To treat the general case, we dominate ω_1 and ω_2 by $\omega = \omega_1 + \omega_2$ and use previous estimates to conclude.

Note that the capacity Cap_ω characterizes pluripolar sets whenever ω is big and semi-positive: we have established this earlier for those capacities associated to a Kähler form, but previous estimates show that these compare well to the ones associated to forms that are merely semi-positive and big. □

12.1.2 Josefson's theorem Let $\chi \in \mathcal{W}$ be an arbitrary weight. We have shown earlier that any $PSH(X,\omega)$-polar set can actually be defined by a function $\varphi \in \mathcal{E}_\chi(X,\omega)$. This turns out to be the case of any locally pluripolar set:

Theorem 12.5. *Locally pluripolar sets are $PSH(X,\omega)$-polar.*

This result is originally due to Josefson [Jos78]. We follow the simpler approach of El Mir [ElM80] and Alexander and Taylor [AT84].

Proof. Let P be a locally pluripolar set. Using finite convex sums we can assume P is included in an open subset $\Omega \subset X$ and

$$P \subset \{v = -\infty\},$$

where $v \in PSH^-(\Omega)$.

Fix $0 < \varepsilon < 1/n$ and $V_t := V_{G_t,\omega}$ where $G_t = \{x \in \Omega \,|\, v(x) < -t\}$. We are going to show that

$$\varphi_\varepsilon(x) := \frac{1}{\varepsilon}\int_1^{+\infty} \frac{1}{t^{1+\varepsilon}}\left(V_t(x) - \sup_X V_t\right) dt$$

is a ω-psh function such that $P \subset \{\varphi_\varepsilon = -\infty\}$.

Indeed since G_t is open, we have $V_t \in PSH(X,\omega)$ and $V_t = 0$ on G_t. Observe that φ_ε is a sum of negative ω-psh functions hence it is either identically $-\infty$ or a well-defined $A\omega$-psh function with $A = \varepsilon^{-1}\int_1^{+\infty} t^{-(1+\varepsilon)}dt = 1$. Recall that $-C + \sup_X V_t \leq \int_X V_t\omega^n \leq \sup_X V_t$. Therefore $\int_X \varphi_\varepsilon\omega^n \geq -C$, hence $\varphi_\varepsilon \in PSH(X,\omega)$.

Fix $x \in \Omega$ such that $v(x) < -1$. Observe that $V_t - \sup_X V_t \leq 0$, with $V_t(x) = 0$ if $x \in G_t$, i.e., when $|v(x)| > t$. Therefore

$$\varphi_\varepsilon(x) \leq -\frac{1}{\varepsilon}\int_1^{|v(x)|} \frac{\sup_X V_t}{t^{1+\varepsilon}} dt.$$

Recall now that Cap_ω is always dominated by $\mathrm{Cap}_{\mathrm{BT}}$, hence $\mathrm{Cap}_\omega(G_t) \leq C_1/t < 1$ if t is large enough. We infer from the previous lemma that

$$-\sup_X V_t \leq -\left[\mathrm{Cap}_\omega(G_t)\right]^{-1/n} \leq -C_2 t^{1/n},$$

which yields

$$\varphi_\varepsilon(x) \leq \frac{-C_2}{\varepsilon}\int_1^{|v(x)|} \frac{dt}{t^{1+\varepsilon-1/n}} \leq -C_3|v(x)|^{1/n-\varepsilon} + C_4.$$

Note that $\varphi_\varepsilon(x) = -\infty$ whenever $v(x) = -\infty$, hence

$$P \subset \{\varphi_\varepsilon = -\infty\}. \qquad \square$$

12.1.3 Domination by capacity

Proposition 12.6. *Let $\mu = f\,dV$ be a positive measure with L^p density with respect to the Lebesgue measure, with $p > 1$. Then there exists $C > 0$ such that*

$$\mu(B) \leq C \cdot \mathrm{Cap}_\omega(B)^2$$

for all Borel subsets B of X. We can take

$$C := (p-1)^{-2n} A \|f\|_{L^p(dV)},$$

for a constant A depending only on ω and dV.

Proof. By the inner regularity of μ, it is enough to consider the case where $B = K$ is compact. We can also assume that K is non-pluripolar since $\mu(K) = 0$ otherwise and the inequality is then trivial.

Now set

$$\nu_\alpha := \sup_{T,x} \nu(T, x), \tag{12.1.2}$$

where the supremum ranges over all positive currents $T \in \alpha := \{\omega\}$ and all $x \in X$ and $\nu(T, x)$ denotes the Lelong number of T at x.

Since all Lelong numbers of $\nu_\alpha^{-1} T$ are < 2 for each positive current $T \in \alpha$, the uniform version of Skoda's integrability theorem together with the compactness of normalized ω-psh functions yields $C_\omega > 0$ only depending on dV and ω such that

$$\int_X \exp(-\nu_\alpha^{-1}\psi) dV \le C_\omega$$

for all ω-psh functions ψ normalized by $\sup_X \psi = 0$.

Applying this to $\psi = V^*_{K,\omega} - M_\omega(K)$ (which has the right normalization by (12.1.1)) we get

$$\int_X \exp(-\nu_\alpha^{-1} V^*_{K,\omega}) dV \le C_\omega \exp(-\nu_\alpha^{-1} M_\omega(K)).$$

Now $V^*_{K,\omega} \le 0$ on K a.e. with respect to the Lebesgue measure, hence

$$\mathrm{Vol}(K) \le C_\omega \exp(-\nu_\alpha^{-1} M_\omega(K)). \tag{12.1.3}$$

On the other hand, Hölder's inequality yields

$$\mu(K) \le \|f\|_{L^p(dV)} \mathrm{Vol}(K)^{1/q}, \tag{12.1.4}$$

where q denotes the conjugate exponent.

We may also assume that $M_\omega(K) \ge 1$. Otherwise Lemma 12.2 implies $\mathrm{Cap}_\omega(K) = 1$, and the result is thus clear in that case. By Lemma 12.2, (12.1.3), and (12.1.4) together we thus get

$$\mu(K) \le C_\omega^{1/q} \|f\|_{L^p(dV)} \exp\left(-\frac{1}{q\nu_\alpha} \mathrm{Cap}_\omega(K)^{-1/n}\right),$$

and the result follows since $\exp(-t^{-1/n}) = O(t^2)$ when $t \to 0_+$. □

12.1.4 Proof of Theorem 12.1

Lemma 12.7. *Fix $\varphi \in \mathcal{E}(X, \omega)$. Then for all $t > 0$ and $0 < \delta < 1$ we have*

$$\mathrm{Cap}_\omega\{\varphi < -t - \delta\} \le \delta^{-n} \int_{\{\varphi < -t\}} MA(\varphi).$$

Proof. If ψ is a ω-psh function such that $0 \leq \psi \leq 1$, then

$$\{\varphi < -t - \delta\} \subset \{\varphi < \delta\psi - t - \delta\} \subset \{\varphi < -t\}.$$

Since $\delta^n MA(\psi) \leq MA(\delta\psi)$ and $\varphi \in \mathcal{E}(X, \omega)$, it follows from the comparison principle that

$$\begin{aligned}\delta^n \int_{\{\varphi<-t-\delta\}} MA(\psi) &\leq \int_{\{\varphi<\delta\psi-t-\delta\}} MA(\delta\psi) \\ &\leq \int_{\{\varphi<\delta\psi-t-\delta\}} MA(\varphi) \leq \int_{\{\varphi<-t\}} MA(\varphi)\end{aligned}$$

and the proof is complete. □

Consider now

$$g(t) := (\mathrm{Cap}_\omega\{\varphi < -t\})^{1/n}.$$

Our goal is to show that $g(M) = 0$ for some M under control. Indeed, we will then have $\varphi \geq -M$ on $X \setminus P$ for some Borel subset P such that $\mathrm{Cap}_\omega(P) = 0$. But it then follows in particular from Proposition 12.6 (applied to the Lebesgue measure itself) that P has Lebesgue measure zero, hence $\varphi \geq -M$ will hold everywhere.

Since $MA(\varphi) = \mu$, it follows from Proposition 12.6 and Lemma 12.7 that

$$g(t + \delta) \leq \frac{C^{1/n}}{\delta} g(t)^2 \quad \text{for all } t > 0 \text{ and } 0 < \delta < 1.$$

We can thus apply Lemma 12.8 below, which yields $g(M) = 0$ for $M := t_0 + 5C^{1/n}$. Here $t_0 > 0$ has to be chosen so that

$$g(t_0) < \frac{1}{2C^{1/n}}.$$

Lemma 12.7 (with $\delta = 1$) implies that

$$\begin{aligned}g(t)^n \leq \mu\{\varphi < -t + 1\} &\leq \frac{1}{t-1} \int_X |\varphi| f dV \\ &\leq \frac{1}{t-1} \|f\|_{L^p(dV)} \|\varphi\|_{L^q(dV)},\end{aligned}$$

by Hölder's inequality.

Since φ belongs to the compact set of ω-psh functions normalized by $\sup_X \varphi = 0$, its $L^q(dV)$-norms is bounded by a constant C_2 that depends only on ω, dV and p. It is thus enough to take

$$t_0 > 1 + 2^{n-1} C_2 C \|f\|_{L^p(dV)}.$$

Lemma 12.8. *Let $g : \mathbb{R}^+ \to [0,1]$ be a decreasing function such that $g(+\infty) = 0$ and*

$$g(t+\delta) \leq \frac{C^{1/n}}{\delta} g(t)^2 \quad \text{for all } t > 0 \text{ and } 0 < \delta < 1.$$

Then $g(t) = 0$ for all $t \geq t_0 + 5C^{1/n}$, where

$$t_0 = \inf\{s > 0 \mid g(s) \leq e^{-1} C^{-1/n}\}.$$

Proof. Set $f(t) = -\log g(t)$, so that $f : \mathbb{R}^+ \to \mathbb{R}^+$ is increasing with

$$f(t+\delta) \geq 2f(t) + \log\left(\delta / C^{1/n}\right).$$

We define by induction an increasing sequence t_j such that

$$t_{j+1} = t_j + \delta_j, \text{ with } \delta_j = eC^{1/n} \exp(-f(t_j)) = eC^{1/n} g(t_j).$$

Observe that $0 < \delta_0$ is smaller than 1 if we choose t_0 as indicated. Since (t_j) is increasing and g is decreasing, this ensures that δ_j is smaller than 1 for all $j \in \mathbb{N}$. We can thus use the growth estimate and obtain

$$f(t_{j+1}) = f(t_j + \delta_j) \geq f(t_j) + 1.$$

Since $f \geq 0$, we infer $f(t_j) \geq j$ for all j. Now

$$t_\infty := t_0 + \sum_{j \geq 0} (t_{j+1} - t_j) \leq t_0 + C^{1/n} e \sum_{j \geq 0} \exp(-j) \leq t_0 + 5C^{1/n}.$$

The proof is thus complete, since $f(t) \geq f(t_\infty) = +\infty$ for all $t \geq t_\infty$. □

12.2 Continuity of the solutions

12.2.1 Refinements of previous results

We need the following refinement of Lemma 12.7:

Lemma 12.9. *Let $\varphi, \psi \in \mathcal{E}(X, \omega)$ be two negative functions. Then for all $t > 0$ and $0 \leq \delta \leq 1$,*

$$\delta^n \mathrm{Cap}_\omega(\varphi - \psi < -\delta - t) \leq \int_{\{\varphi - \psi < -t - \delta\psi\}} MA(\varphi).$$

Proof. Fix $u \in PSH(X, \omega)$ with $0 \leq u \leq 1$. For $s > 0$ we set $\delta = s/(1+s)$. Observe that $0 \leq \delta \leq 1$ and

$$\{\varphi - \psi < -\delta - t\} \subset \left\{\varphi < \frac{\psi + su}{1+s} - \delta - t\right\} \subset \{\varphi - \psi < -t - \delta\psi\}.$$

Set $\tilde{\varphi} := (\psi + su)/(1+s) - \delta - t \in PSH(X,\omega)$. Observe that

$$\delta^n \int_{\{\varphi-\psi<-\delta-t\}} \omega_u^n \leq \int_{\{\varphi-\psi<-\delta-t\}} \left[\frac{1}{1+s}\omega_\psi + \frac{s}{1+s}\omega_u\right]^n \leq \int_{\{\varphi<\tilde{\varphi}\}} \omega_{\tilde{\varphi}}^n.$$

It follows from the comparison principle that that

$$\int_{\{\varphi<\tilde{\varphi}\}} [\omega + dd^c\tilde{\varphi}]^n \leq \int_{\{\varphi<\tilde{\varphi}\}} [\omega + dd^c\varphi]^n \leq \int_{\{\varphi-\psi<-t-\delta\psi\}} [\omega + dd^c\varphi]^n.$$

Taking the supremum over all u's yields the desired result. □

Proposition 12.10. *Fix two negative functions $\varphi, \psi \in \mathcal{E}(X,\omega)$ and $\varepsilon > 0$. Assume $\mu = MA(\varphi) \leq A\,\mathrm{Cap}_\omega^{1+\alpha}$, $\alpha > 0$, and ψ is bounded. There exists $C = C(\alpha, A, \omega, \|\psi\|_{L^\infty(X)}) > 0$ such that*

$$\sup_X(\psi - \varphi) \leq \varepsilon + C\,[\mathrm{Cap}_\omega(\varphi - \psi < -\varepsilon)]^{\alpha/n}.$$

This inequality can be interpreted as follows. Assume ψ is also a solution to a Monge–Ampère equation $\omega_\psi^n = \mu'$, where μ' is similarly dominated by the Monge–Ampère capacity. Then we can interchange the roles of φ and ψ and get an upper-bound on $\|\varphi - \psi\|_{L^\infty(X)}$. The proposition then tells us that if φ and ψ are close in capacity, they are close in $\mathcal{C}^0$-norm.

Proof. Set $M := \|\psi\|_{L^\infty(X)}$. Observe that when $\delta \geq 0$,

$$\{\varphi - \psi < -t - \delta\psi\} \subset \{\varphi - \psi < -t + \delta M\},$$

hence it follows from Lemma 12.9 that for all $t > 0$, $0 \leq \delta \leq 1$,

$$\delta^n \mathrm{Cap}_\omega(\{\varphi - \psi < -\delta - t\}) \leq (1+M)^n \int_{\{\varphi-\psi<-t\}} MA(\varphi), \tag{12.2.1}$$

using the obvious substitutions $t \mapsto t - M\delta$, $\delta \mapsto (1+M)\delta$.

Since $\mu = MA(\varphi) \leq A\,\mathrm{Cap}_\omega^{1+\alpha}$, we infer

$$\delta^n \mathrm{Cap}_\omega\{\varphi - \psi < -\delta - t\} \leq A(1+M)^n \mathrm{Cap}_\omega\{\varphi - \psi < -t\}^{1+\alpha}.$$

Consider

$$g(t) := [\mathrm{Cap}_\omega\{\varphi - \psi < -t - \varepsilon\}]^{1/n},\ t > 0.$$

Then

$$g(t+\delta) \leq \frac{B}{\delta} g(t)^{1+\alpha},\ \text{for all } 0 < t,\ 0 < \delta < 1,$$

where $B = A^{1/n}(1+M)$. A slight refinement of Lemma 12.8 shows then that $g(t) \equiv 0$ for $t \geq T_\infty$, where

$$T_\infty = t_0 + \frac{eB}{1-e^{-\alpha}} g(t_0)^\alpha \text{ and } t_0 = \inf\{s > 0 \mid g(s)^\alpha \leq e^{-1}B^{-1}\}.$$

This shows that $\varphi \geq \psi - T_\infty - \varepsilon$, hence

$$\sup_X(\psi - \varphi) \leq \varepsilon + T_\infty.$$

The conclusion follows now by adjusting the value of C. Indeed, either $t_0 = 0$ and we are done since

$$T_\infty \leq C\,[\mathrm{Cap}_\omega\,\{\varphi - \psi < -\varepsilon\}]^{\alpha/n}\,,$$

or

$$g(0)^\alpha = \mathrm{Cap}_\omega\,\{\varphi - \psi < -\varepsilon\}^{\alpha/n} > 1/eB,$$

in which case $\sup_X(\psi - \varphi)$ is bounded from above by our previous uniform estimate, while $\mathrm{Cap}_\omega\,\{\varphi - \psi < -\varepsilon\}$ is bounded below away from zero. □

12.2.2 Continuity We now prove that the Monge–Ampère potentials we are considering are actually continuous when ω is a Kähler form:

Theorem 12.11. *Assume ω is a Kähler form. Let $\mu = f\omega^n$ be a probability Monge–Ampère measure with $f \in L^p$, $p > 1$. Then $\mu = MA(\varphi)$, where $\varphi \in PSH(X,\omega)$ is continuous.*

Proof. We already know that $\mu = MA(\varphi)$, where $\varphi \in PSH(X,\omega)$ is bounded. We an assume without loss of generality that $\varphi < 0$.

Let (φ_j) be a sequence of smooth ω-psh functions decreasing to φ. We can assume $\varphi_j \leq 0$. Since φ is bounded and $\varphi_j \geq \varphi$, the functions φ_j are uniformly bounded. Recall that φ_j converges towards φ in capacity, hence

$$\lim \mathrm{Cap}_\omega\left(\{\varphi - \varphi_j < -\varepsilon\}\right) = 0, \text{ for all } \varepsilon > 0.$$

It follows from previous proposition (applied with $\psi = \varphi_j$) that for all $\varepsilon > 0$,

$$\lim_{j\to+\infty} \|\varphi - \varphi_j\|_{L^\infty(X)} = \lim_{j\to+\infty} \sup_X(\varphi_j - \varphi) \leq \varepsilon.$$

Thus (φ_j) converges uniformly to φ, hence φ is continuous. □

When ω is merely semi-positive the proof breaks down, as we cannot ensure that there exists a sequence (φ_j) of *continuous* ω-psh functions which decrease towards φ.

When $\pi : X \to Y$ is a log resolution of a projective variety Y, $L \to Y$ is an ample line bundle and $\omega \in c_1(\pi^*L)$ represents the first Chern class of the semi-positive line bundle $\pi^*L \to X$, such a sequence can be found by using the extension result of [CGZ13].

We develop in Chapter 13 an alternative viscosity approach.

12.2.3 Relative bounds We present here a recent improvement of this technique due to Di Nezza and Lu [DiNL14].

12.2.3.1 The ψ-capacity

Definition 12.12. *Let $\psi \in PSH(X,\omega)$. We define the ψ-capacity of a Borel subset $E \subset X$ by*

$$\mathrm{Cap}_\psi(E) := \sup\left\{\int_E MA(u) \mid u \in PSH(X,\omega),\ \psi - 1 \le u \le \psi\right\}.$$

The Monge–Ampère capacity that we have previously introduced corresponds to $\psi \equiv$ constant.

We list below some basic properties of the ψ-capacity.

Proposition 12.13.

(i) *If $E_1 \subset E_2 \subset X$, then $\mathrm{Cap}_\psi(E_1) \le \mathrm{Cap}_\psi(E_2)$.*

(ii) *If $E_1, E_2, \ldots$ are Borel subsets of X, then*

$$\mathrm{Cap}_\psi\left(\bigcup_{j=1}^{\infty} E_j\right) \le \sum_{j=1}^{+\infty} \mathrm{Cap}_\psi(E_j).$$

(iii) *If $E_1 \subset E_2 \subset \ldots$ are Borel subsets of X, then*

$$\mathrm{Cap}_\psi\left(\bigcup_{j=1}^{\infty} E_j\right) = \lim_{j\to+\infty} \mathrm{Cap}_\psi(E_j).$$

The following results are elementary and important for the sequel.

Lemma 12.14. *Let $\psi \in PSH(X,\omega)$ and $\varphi \in \mathcal{E}(X,\omega)$. Then the function*

$$H(t) := \mathrm{Cap}_\psi(\{\varphi < \psi - t\}),\ t \in \mathbb{R},$$

is right-continuous and $H(t) \to 0$ as $t \to +\infty$.

Proof. The right-continuity of H follows from (iii) of Proposition 12.13. Let us prove the second statement. We can assume that $\psi \le 0$ on X. Fix $v \in PSH(X,\omega)$ such that $\psi - 1 \le v \le \psi$. We apply the comparison principle to obtain

$$\int_{\{\varphi<\psi-t\}} MA(v) \le \int_{\{\varphi<v-t+1\}} MA(v) \le \int_{\{\varphi<-t+1\}} MA(\varphi).$$

The last term goes to zero as t goes to $+\infty$ since $\varphi \in \mathcal{E}(X,\omega)$. □

Lemma 12.15. *Let (X,ω) be a compact Kähler manifold and $\psi \in PSH(X,\omega/2)$. Then we have*

$$\mathrm{Cap}_{\omega/2}(E) \leq \mathrm{Cap}_{\psi}(E).$$

The above result ensures $\mathrm{Cap}_{\psi}(E) > 0$ for any Borel subset E which is not pluripolar.

Proof. Let $u \in PSH(X,\omega/2)$ be such that $-1 \leq u \leq 0$. Then $\varphi := \psi + u$ is a candidate defining Cap_{ψ}. Using the definition of the Monge–Ampère measure it is not difficult to see that

$$\int_E (\omega/2 + dd^c u)^n \leq \int_E (\omega + dd^c\varphi)^n \leq \mathrm{Cap}_{\psi}(E),$$

and taking the supremum over all u we get the result. □

12.2.3.2 Uniform estimate Let $\varphi \in \mathcal{E}(X,\omega)$ be the unique normalized solution to

$$(\omega + dd^c\varphi)^n = f\omega^n, \tag{12.2.2}$$

where $0 \leq f \in L^1(X)$ is such that $\int_X f\omega^n = \int_X \omega^n$ and φ is normalized by $\sup_X \varphi = 0$.

The following a priori bound has been established by Di Nezza and Lu in [DiNL14]:

Theorem 12.16. *Assume that $f \leq e^{-\phi}$ for some qpsh function ϕ. Let $\varphi \in \mathcal{E}(X,\omega)$ be the unique normalized solution to (12.2.2). Then for any $a > 0$ such that $a\phi \in PSH(X,\omega/2)$, there exists $A > 0$ depending only on $\int_X e^{-2\varphi/a}\omega^n$ such that*

$$\varphi \geq a\phi - A.$$

If ϕ is bounded on a compact set K, then φ is continuous on K.

This bound allows in particular to study complex Monge–Ampère equations on quasi-projective varieties, where the solution φ is not necessarily bounded near a given divisor D (e.g., metrics with Poincaré type singularities), see [DiNL14] and the references therein.

Proof. We can assume that $\phi \leq 0$. Fix $a > 0$ such that $\psi := a\phi$ belongs to $PSH(X,\omega/2)$. It follows from Lemma 12.15 that $\mathrm{Cap}_{\omega} \leq 2^n \mathrm{Cap}_{\omega/2} \leq 2^n \mathrm{Cap}_{\psi}$.

Fix $s \in [0,1]$, $t > 0$ we obtain

$$s^n \mathrm{Cap}_{\psi}\{\varphi < \psi - t - s\} \leq \int_{\{\varphi<\psi-t\}} MA(\varphi). \tag{12.2.3}$$

By the assumption on f, we have

$$\int_{\{\varphi<\psi-t\}} MA(\varphi) \leq \int_{\{\varphi<\psi-t\}} e^{-\varphi/a}e^{\psi/a}MA(\varphi) \leq \int_{\{\varphi<\psi-t\}} e^{-\varphi/a}\omega^n. \tag{12.2.4}$$

Recall that

$$\mathrm{Vol}_\omega \leq \exp\left(\frac{-C_1}{\mathrm{Cap}_\omega^{1/n}}\right).$$

Thus using Hölder inequality we get from (12.2.3) and (12.2.4) that

$$s^n \mathrm{Cap}_\psi\{\varphi < \psi - t - s\} \leq C_2 \left(\mathrm{Cap}_\omega\{\varphi < \psi - t\}\right)^2 \leq C_3 \left(\mathrm{Cap}_\psi\{\varphi < \psi - t\}\right)^2,$$

where C_3 depends only on $\int_X e^{-2\varphi/a}\omega^n$. Now consider the function

$$H(t) = \left[\mathrm{Cap}_\psi(\{\varphi < \psi - t\})\right]^{1/n},\ t > 0.$$

By the arguments above we get

$$sH(t+s) \leq C_4 H(t)^2, \quad \forall t > 0, \forall s \in [0,1],$$

where $C_4 > 0$ depends only on $\int_X e^{-2\varphi/a}\omega^n$.

It follows from Lemma 12.14 that H is right-continuous and $H(+\infty) = 0$. Arguing as above we get $\varphi \geq \psi - C_5$, where C_5 depends only on $\int_X e^{-2\varphi/a}\omega^n$. Indeed, there exists t_∞ such that $H(t) = 0$ if $t \geq t_\infty$. Here, we can take

$$t_\infty = 2 + s_0,$$

where $s_0 > 0$ is big enough such that

$$H(s_0) \leq \frac{1}{2C_4}.$$

By using Hölder's inequality it follows from (12.2.3) (take $s = 1$) and (12.2.4) that

$$\begin{aligned} H(t+1)^n &\leq \left(\int_X e^{-2\varphi/a}\omega^n\right)^{1/2}\left(\int_{\{\varphi<\psi-t\}}\omega^n\right)^{1/2} \\ &\leq \left(\int_X e^{-2\varphi/a}\omega^n\right)^{1/2}\left(\frac{1}{t}\int_X(-\varphi)\omega^n\right)^{1/2}. \end{aligned}$$

The last integral is bounded by a uniform constant since φ is normalized by $\sup_X \varphi = 0$. From this we can choose $s_0 > 0$ *depending only* on an upper bound of $\int_X e^{-2\varphi/a}\omega^n$.

Now, assume that ϕ is bounded on a compact subset $K \subset X$. Set $\psi := a\phi$ as above. Let us prove that φ is continuous on K. For convenience, we normalize φ so that $\sup_X \varphi = -1$. Let $0 \geq \varphi_j$ be a sequence of continuous ω-psh functions on X decreasing to φ. Fix $\lambda \in (0,1)$. For each $j \in \mathbb{N}$ set

$$\psi_j := \lambda\varphi_j + (1-\lambda)\psi - (1-\lambda)A - 2(1-\lambda).$$

Then ψ_j belongs to $PSH(X, \frac{1+\lambda}{2}\omega)$ and $\psi_j \leq \varphi_j - 2(1-\lambda)$. Set

$$H_j(t) := \left[\mathrm{Cap}_{\psi_j}\left(\{\varphi < \psi_j - t\}\right)\right]^{1/n}, \quad t > 0.$$

We can argue as above to get

$$sH_j(t+s) \le C_1 H_j(t)^2, \quad \forall t > 0,\ \forall s \in [0,1],$$

where $C_1 > 0$ depends on $\int_X e^{-2\varphi/(1-\lambda)a}$. Let $\chi : \mathbb{R}^- \to \mathbb{R}^-$ be an increasing convex weight such that $\chi(0) = 0$, $\chi(-\infty) = -\infty$, and $\varphi \in \mathcal{E}_\chi(X,\omega)$. By the comparison principle we also get

$$\begin{aligned}(1-\lambda)^n \mathrm{Cap}_{\psi_j}\{\varphi < \psi_j\} &\le \int_{\{\varphi<\psi_j+1-\lambda\}} MA(\varphi) \le \int_{\{\varphi<\varphi_j-(1-\lambda)\}} MA(\varphi)\\ &\le \frac{1}{-\chi(-1+\lambda)} \int_X (-\chi\circ(\varphi-\varphi_j)) f\omega^n.\end{aligned}$$

The latter converges to 0 as $j \to +\infty$, since φ_j decreases to φ. Thus for j big enough we have $H_j(0) \le 1/(2C_1)$. It then follows that $H_j(t) = 0$ if $t \ge t_\infty$, where $t_\infty \le C_2 H_j(0)$ and C_2 depends on C_1. We then get

$$\varphi \ge \lambda\varphi_j + (1-\lambda)\psi - (1-\lambda)(A+2) - C_2 H_j(0),$$

whence, letting $j \to +\infty$,

$$\lim_{j\to+\infty} \inf_K(\varphi - \varphi_j) \ge (\lambda-1)(\sup_K|\psi| + A + 2).$$

Finally, letting $\lambda \to 1$ we get the continuity of φ on K. □

12.3 Stability of the solutions

Let μ be a non-pluripolar probability measure. It follows from previous chapters that there exists a unique function $\varphi_\mu \in \mathcal{E}(X,\omega)$ such that $MA(\varphi_\mu) = \mu$ and normalized by $\sup_X \varphi_\mu = 0$.

The purpose of this section is to study the stability properties of the mapping $\mu \mapsto \varphi_\mu$. We first establish a qualitative stability (for the weakest topologies) and then move on to show a stronger stability property, by making stronger assumptions on the measure μ.

12.3.1 Qualitative stability Let $\mu_j = MA(\varphi_{\mu_j})$ and $\mu = MA(\varphi_\mu)$ be non-pluripolar probability measures on X, where

$$\varphi_{\mu_j}, \varphi_\mu \in \mathcal{E}(X,\omega) \text{ with } \sup_X \varphi_{\mu_j} = \sup_X \varphi_\mu = 0.$$

One cannot expect stability (i.e., $\varphi_{\mu_j} \to \varphi_\mu$) just by assuming that μ_j converges weakly to μ (see Exercise 5.13). On the other hand, we have the following:

Proposition 12.17. *Let $\mu_j = MA(\varphi_{\mu_j}), \mu = MA(\varphi_\mu)$ be non-pluripolar probability measures, with $\sup_X \varphi_{\mu_j} = \sup_X \varphi_\mu = 0$. If $\|\mu_j - \mu\| \to 0$ then*

$$\varphi_{\mu_j} \longrightarrow \varphi_\mu \text{ in } L^1(X).$$

Here $\|\mu_j - \mu\|$ denotes the total variation of the signed measure $\mu_j - \mu$.

Proof. We first observe that this result is equivalent to the following:

Claim. *If the measures $\mu_j = f_j \nu$ are all absolutely continuous with respect to a fixed non-pluripolar measure ν and $f_j \to f$ in $L^1(\nu)$, then*

$$\varphi_{\mu_j} \longrightarrow \varphi_\mu \text{ in } L^1(X), \text{ where } \mu = f\nu.$$

Indeed if $\mu_j = f_j \nu$ and $\mu = f\nu$, then by definition

$$\|\mu_j - \mu\| = \|f_j - f\|_{L^1(\nu)}$$

so that the claim is a particular case of the proposition.

Conversely, if μ_j, μ are non-pluripolar probability measures such that $\|\mu_j - \mu\| \to 0$, then

$$\nu := 2^{-1}\mu + \sum_{j \geq 2} 2^{-j}\mu_j$$

is a well-defined non-pluripolar probability measure with respect to which μ_j, μ are absolutely continuous, thus the hypotheses of the claim are satisfied.

It therefore suffices to prove the claim: We assume that $\mu_j = f_j \nu$, $\mu = f\nu$, where ν vanishes on pluripolar sets and $f_j \to f$ in $L^1(\nu)$, and we are going to show that in this case (φ_j) converges in $L^1(X)$ to φ.

By weak compactness, we can assume — extracting and relabelling — that $\varphi_j := \varphi_{\mu_j} \to \psi \in PSH(X, \omega)$, with $\sup_X \psi = 0$. Extracting again, we can also assume that there exists $g \in L^1(\nu)$ such that

$$f_j \leq g \text{ for all } j \in \mathbb{N}.$$

Since the measure $g\nu$ does not charge pluripolar sets, it follows from Proposition 12.18 below that we can construct a convex weight $\chi \in \mathcal{W}_+$ and $C > 0$ such that for all $j \in \mathbb{N}$,

$$\int -\chi \circ \varphi_j \, g d\nu \leq C.$$

This shows that $\int -\chi \circ \varphi_j \, MA(\varphi_j) \leq C$, hence $\psi \in \mathcal{E}_\chi(X, \omega)$.

The functions $\psi_j := (\sup_{l \geq j} \varphi_l)^* \in PSH(X, \omega)$ decrease to ψ and satisfy

$$MA(\psi_j) \geq (\inf_{l \geq j} f_l)\nu.$$

We infer $MA(\psi) \geq \mu = f\nu$, whence equality, since these are both probability measures. Now $MA(\psi) = MA(\varphi)$, hence these normalized potentials have to be equal, by the uniqueness of normalized solutions. □

Proposition 12.18. *Let μ be a non-pluripolar measure and $\mathcal{K}$ a compact set of qpsh functions on X. Then there exist a convex weight χ and $C > 0$ such that $\int_X (-\chi)(\varphi) d\mu \leq C$ for all $\varphi \in \mathcal{K}$.*

Proof. Functions in $\mathcal{K}$ are uniformly bounded from above by compactness, and we can thus assume that they are all non-positive. Now let ω be a given Kähler form, and let $\mathcal{M}$ be the (weakly) closed convex set of positive measures generated by all Monge–Ampère measures $(\omega + dd^c\psi)^n$ with $0 \leq \psi \leq 1$ is ω-psh. By the Chern–Levine–Nirenberg inequality, there exists a constant $C > 0$ such that

$$\int_X (-\varphi)(\omega + dd^c\psi)^n \leq C$$

for all $\varphi \in \mathcal{K}$ and all $0 \leq \psi \leq 1$ as above, and this remains true on $\mathcal{M}$ by convexity and lower semi-continuity of $\nu \mapsto \int_X (-\varphi) d\nu$ (since $-\varphi$ is lower semi-continuous — just write it as the supremum of all continuous functions below it).

Now recall that pluripolar subsets A are characterized by the condition that $\nu(A) = 0$ for all Monge–Ampère measures ν as above. By an extension of Radon–Nikodým's theorem proved in [Rai69], there exists measures $\nu \in \mathcal{M}$ and $\nu' \perp \mathcal{M}$ such that our given measure μ writes

$$\mu = g\nu + \nu',$$

with g a non-negative element of $L^1(\nu)$. Since $\mu(A) = 0$ for all pluripolar subsets by assumption, we infer that $\nu' = 0$, and we have thus proved that $\mu \ll \nu$ for some $\nu \in \mathcal{M}$. We now conclude the proof by the following standard result in the theory of Orlicz spaces. □

Lemma 12.19. *Let $\mu \ll \nu$ be two positive measures with finite mass on a measured space X. Then there exist a concave function $h : \mathbb{R}^+ \to \mathbb{R}^+$ and $C > 0$ such that*

$$\int_X h(f) d\mu \leq C + \int_X f d\nu$$

for all measurable functions $f \geq 0$.

Proof. Let $g \geq 0$ be the function in $L^1(\nu)$ such that $d\mu = g d\nu$. By an easy measure-theoretic result that we've already used, there exists a convex increasing function τ on $\mathbb{R}^+$ such that $\tau(x) \gg x$ for $x \to +\infty$, but still $\int_X \tau(g) d\nu < +\infty$. Now let

$$\tau^*(y) = \sup_{x \geq 0} (xy - \tau(x))$$

be the Legendre transform of τ and let h be the reciprocal function of τ^*. By definition, we thus have the following generalized form of Young's inequality:

$$h(y)x \leq \tau(x) + y.$$

If follows that

$$\int_X h(f)d\mu = \int_X h(f)g d\nu \leq \int_X \tau(g)d\nu + \int_X f d\nu,$$

which yields the desired result with $C := \int_X \tau(g)d\nu$. □

Remark 12.20. *When $\mathcal{K}$ consists of a single function, Proposition 12.18 follows from standard measure-theoretic considerations since $\mu\{\varphi = -\infty\} = 0$. The general case would thus follow by compactness provided that $\varphi \mapsto \int_X \varphi d\mu$ were known to be lower semi-continuous on θ-psh functions in the $L^1(X)$ topology (it is always* upper *semi-continuous by Fatou's lemma). But already when $n = 1$ the usc condition is not automatic: it holds iff μ has* continuous *local potentials.*

12.3.2 Strong stability Let $\mu_j = f_j dV = MA(\varphi_j), \mu = f dV = MA(\varphi)$ be probability measures. It is natural to expect that if the densities (f_j) converge in L^p to f, then the corresponding normalized Monge–Ampère potentials converge uniformly.

This is indeed the case. For simplicity, we only state and prove this stability result when $p = 2$:

Theorem 12.21. *Let $MA(\varphi) = f dV$ and $MA(\psi) = g dV$ be two probability measures with L^2-densities, where φ and ψ are bounded ω-psh functions normalized so that $\int_X (\varphi - \psi)dV = 0$. Then there exists $C > 0$, depending only on $\|f\|_{L^2}, \|g\|_{L^2}$, such that*

$$\|\varphi - \psi\|_{L^\infty(X)} \leq C\|f - g\|^{\gamma}_{L^2(X)},$$

for some uniform exponent $\gamma > 0$.

Proof. The proof splits into two main steps. We first claim that

$$\|\varphi - \psi\|_{L^2(X)} \leq C\|f - g\|^{\frac{1}{2^n-1}}_{L^2(X)}, \tag{12.3.1}$$

for some appropriate $C > 0$. Indeed, we are going to show that

$$\int_X d(\varphi - \psi) \wedge d^c(\varphi - \psi) \wedge \omega^{n-1} \leq C_1 I(\varphi, \psi)^{2^{-(n-1)}}, \tag{12.3.2}$$

where

$$I(\varphi, \psi) := \int_X (\varphi - \psi)\{(\omega + dd^c\psi)^n - (\omega + dd^c\varphi)^n\} \geq 0$$

is non-negative, as the reader can check that an alternative writing is

$$I(\varphi, \psi) = \sum_{j=0}^{n-1} \int_X d(\varphi - \psi) \wedge d^c(\varphi - \psi) \wedge \omega_\varphi^j \wedge \omega_\psi^{n-1-j}.$$

The Cauchy–Schwarz inequality yields

$$I(\varphi,\psi)=\int_X(\varphi-\psi)(g-f)\omega^n\le\|\varphi-\psi\|_{L^2}\|f-g\|_{L^2},$$

therefore (12.3.1) is a consequence of (12.3.2) and Poincaré's inequality.

To prove (12.3.2), we write $\omega=\omega_\varphi-dd^c\varphi$ and integrate by parts to obtain

$$\begin{aligned}\int d(\varphi-\psi)\wedge d^c(\varphi-\psi)\wedge\omega^{n-1}&\\ =\int d(\varphi-\psi)\wedge d^c(\varphi-\psi)\wedge\omega_\varphi\wedge\omega^{n-2}&\\ -\int d(\varphi-\psi)\wedge d^c(\varphi-\psi)\wedge dd^c\varphi\wedge\omega^{n-2}&\\ =\int d(\varphi-\psi)\wedge d^c(\varphi-\psi)\wedge\omega_{\varphi_1}\wedge\omega^{n-2}&\\ +\int d(\varphi-\psi)\wedge d^c\varphi\wedge(\omega_\varphi-\omega_\psi)\wedge\omega^{n-2}.&\end{aligned}$$

We take care of the last term by using the Cauchy–Schwarz inequality, which yields

$$\int d(\varphi-\psi)\wedge d^c\varphi\wedge\omega_\varphi\wedge\omega^{n-2}\le A\left(\int d(\varphi-\psi)\wedge d^c(\varphi-\psi)\wedge\omega_\varphi\wedge\omega^{n-2}\right)^{1/2},$$

where

$$A^2=\int d\varphi\wedge d^c\varphi\wedge\omega_\varphi\wedge\omega^{n-2}$$

is uniformly bounded from above, since φ is uniformly bounded in terms of $\|f\|_{L^2(X)}$ by the Chern–Levine–Nirenberg inequalities and Proposition 12.6. Similarly,

$$-\int d(\varphi-\psi)\wedge d^c\varphi\wedge\omega_\psi\wedge\omega^{n-2}\le B\left(\int d(\varphi-\psi)\wedge d^c(\varphi-\psi)\wedge\omega_\psi\wedge\omega^{n-2}\right)^{1/2},$$

where

$$B^2=\int d\varphi\wedge d^c\varphi\wedge\omega_\psi\wedge\omega^{n-2}$$

is uniformly bounded from above. Note that both terms can be further bounded from above by the same quantity by bounding from above ω_φ (resp. ω_ψ) by $\omega_\varphi+\omega_\psi$.

Proceeding this way by induction, replacing at each step ω by $\omega_\varphi+\omega_\psi$, we end up with a control from above of $\int d(\varphi-\psi)\wedge d^c(\varphi-\psi)\wedge\omega^{n-1}$ by a quantity that is bounded from above by $CI(\varphi,\psi)^{2^{-(n-1)}}$ (there are $(n-1)$-induction steps), for some uniform constant $C>0$. This finishes the proof of the first step.

The second step consists in showing that

$$\|\varphi-\psi\|_{L^\infty(X)}\le C_2\|\varphi-\psi\|^\gamma_{L^2(X)}$$

for some constants $C_2,\gamma>0$. This is the contents of our next proposition. □

Proposition 12.22. *Assume $\omega_\varphi^n = f\omega^n$, $\omega_\psi = g\omega^n$, where $\varphi, \psi \in PSH(X,\omega)$ are bounded and normalized, and $f, g \in L^2(\omega^n)$. Then for all $0 < \gamma < 1/(1+n)$,*

$$\|\varphi - \psi\|_{L^\infty(X)} \leq C \|\varphi - \psi\|^\gamma_{L^2(\omega^n)}.$$

Proof. Fix $\varepsilon > 0$ and $\alpha > 0$ to be chosen later. It follows from Proposition 12.10 that

$$\|\varphi - \psi\|_{L^\infty(X)} \leq \varepsilon + C_1 \left[\mathrm{Cap}_\omega(|\varphi - \psi| > \varepsilon)\right]^{\alpha/n}.$$

Now Lemma 12.9 yields

$$\mathrm{Cap}_\omega(|\varphi - \psi| > \varepsilon) \leq \frac{C_2}{\varepsilon^{n+1}} \int_X |\varphi - \psi|(f+g)\omega^n.$$

It follows thus from Hölder's inequality that

$$\mathrm{Cap}_\omega(|\varphi - \psi| > \varepsilon) \leq \frac{C_3 \|f+g\|_{L^2}}{\varepsilon^{n+1}} \|\varphi - \psi\|_{L^2(\omega^n)}.$$

Choose now $\varepsilon := \|\varphi - \psi\|^\gamma_{L^2}$ where $0 < \gamma < 1/(1+n)$. Then

$$\mathrm{Cap}_\omega(|\varphi - \psi| > \varepsilon) \leq C_4 \left[\|\varphi - \psi\|_{L^2}\right]^{1-\gamma(n+1)}.$$

We infer that

$$\|\varphi - \psi\|_{L^\infty(X)} \leq \|\varphi - \psi\|^\gamma_{L^2} + C_5 \|\varphi - \psi\|^{\gamma'}_{L^2},$$

where

$$\gamma' = \frac{\alpha}{n}\left[1 - \gamma(n+1)\right].$$

We finally choose $\alpha > 0$ so large that $\gamma \leq \gamma'$ and adjust the value of the constant C: this yields the desired estimate. □

12.4 Further results

12.4.1 Hölder continuous solutions We push our analysis one step further and show that the solutions to the previous complex Monge–Ampère equations are actually Hölder continuous:

Theorem 12.23. *Let $\mu = f\omega^n$ be a probability Monge–Ampère measure with $f \in L^p$, $p > 1$, and let $\varphi \in PSH(X,\omega) \cap L^\infty(X)$ be a solution to*

$$\mu = MA(\varphi).$$

Then φ is locally Hölder continuous on the ample locus of $\{\omega\}$ with exponent α arbitrarily close to $2/(1+nq)$, where $1/p + 1/q = 1$.

We recall the definition of the ample locus:

Definition 12.24. *The ample locus of a big cohomology class $\{\omega\}$ is the largest Zariski open subset of X consisting of points $x \in X$ such that there exists a positive current ω_x cohomologous to ω which is smooth and strictly positive in a neighborhood of x.*

Corollary 12.25. *The Monge–Ampère potential φ is Hölder continuous on X when ω is a Kähler form.*

This result is due to Kolodziej [Ko08]. The proof presented here follows the approach of [EGZ09, DDGHKZ14].

Showing that φ is globally Hölder continuous when ω is merely semi-positive and big is an interesting open problem. The solutions corresponding to Kähler–Einstein metrics on varieties with orbifold singularities show that one cannot expect more than Hölder continuity.

Proof. We let φ be a bounded ω-psh function solution of

$$V^{-1}(\omega + dd^c\varphi)^n = \mu.$$

The solution is unique up to an additive constant. We can thus assume, without loss of generality, that $-C_0 \leq \varphi \leq 0$. We let

$$\varphi \longmapsto \rho_\delta \varphi$$

denote the regularization operator of Demailly studied in Chapter 8. Recall that $t \mapsto \rho_t\varphi + Kt^2$ is increasing for $0 \leq t \leq \delta_0$ and some constant K.

We consider the Kiselman–Legendre transform,

$$\psi_{c,\delta}(z) := \inf_{t\in]0,\delta]} \left\{\rho_t\varphi(z) + Kt^2 - c\log(t/\delta)\right\},$$

where $0 \leq \delta \leq \delta_0$ and $c > 0$ will be chosen below. Observe that

$$\varphi \leq \psi_{c,\delta} \leq \rho_\delta\varphi + K\delta^2.$$

Fix a Kähler form ω_X on X. The Demailly curvature estimate yields

$$\omega + dd^c\psi_{c,\delta} \geq -(Ac + K\delta^2)\omega_X$$

for some constant $A > 0$.

Since the coholomogy class $\{\omega\}$ is big, there exists an ω-psh function ψ_0 on X such that $\omega + dd^c\psi_0 \geq \varepsilon_0\omega_X$, for some small constant $\varepsilon_0 > 0$. Subtracting a large constant, we can always assume that $\psi_0 \leq 0$. We can moreover choose ψ_0 such that it is smooth in the ample locus, with analytic singularities.

It follows that the function

$$\varphi_{c,\delta} := \frac{Ac + K\delta^2}{\varepsilon_0}\psi_0 + \left(1 - \frac{Ac + K\delta^2}{\varepsilon_0}\right)\psi_{c,\delta}$$

is ω-plurisubharmonic on X. Fix $0 < \delta < \delta_0$ and choose $c > 0$ such that

$$Ac + K\delta^2 = \varepsilon_0\delta^\alpha, \quad \text{where } \alpha := 2\gamma,$$

and observe that $c = \varepsilon_0 A^{-1}\delta^\alpha - KA^{-1}\delta^2 = O(\delta^\alpha)$. In the sequel we set

$$\varphi_\delta := \varphi_{c,\delta}.$$

Since $\psi_0 \le 0 \le \varphi + C_0$, we see from the definition that on the ample locus,

$$\begin{aligned}\varphi_\delta - \varphi &= \delta^\alpha(\psi_0 - \varphi) + (1 - \delta^\alpha)(\psi_{c,\delta} - \varphi)\\ &\le C_0\delta^\alpha + (1 - \delta^\alpha)(\rho_\delta\varphi - \varphi + K\delta^2).\end{aligned}$$

Furthermore, since $\varphi \le 0$, we get $\varrho_\delta\varphi \le 0$, thus

$$\psi_{c,\delta} \le K\delta^2 \le C_0\delta^\alpha$$

if $\delta \le \delta_0$ small enough, and so $\varphi_\delta \le C_0\delta^\alpha$.

This implies $\psi := \varphi_\delta - C_0\delta^\alpha \le 0$. It follows that

$$\begin{aligned}\sup_X(\varphi_\delta - \varphi) &\le B_0\|\max(\varphi_\delta - \varphi - C_0\delta^\alpha, 0)\|^\gamma_{L^1(X)} + C_0\delta^\alpha\\ &\le B_0\|\rho_\delta\varphi + K\delta^2 - \varphi\|^\gamma_{L^1(X)} + C_0\delta^\alpha\end{aligned}$$

for some constant $B_0 > 0$ which depends only on γ and the uniform norm of φ. Thus

$$\sup_X(\varphi_\delta - \varphi) \le C_1\delta^\alpha,$$

where $C_1 := B_0C_\omega + K^\gamma + C_0$.

This inequality $\varphi_\delta \le \varphi + C_1\delta^\alpha$ yields a uniform lower bound on the parameter $t = t(z)$ which realizes the infimum in the definition of $\varphi_\delta(z)$ for a fixed $z \in \Omega$. Namely, the last inequality gives

$$\begin{aligned}\varphi_\delta(z) - \varphi(z) &= \delta^\alpha(\psi_0(z) - \varphi(z)) + (1 - \delta^\alpha)(\rho_t\varphi(z) + Kt^2 - \varphi(z) - c\log(t/\delta))\\ &\le C_1\delta^\alpha.\end{aligned}$$

Since $-\varphi \ge 0$ and $\rho_t\varphi(z) + Kt^2 - \varphi(z) \ge 0$, it follows that

$$c(1 - \delta^\alpha)\log[t(z)/\delta] \ge \delta^\alpha(\psi_0(z) - C_1).$$

Since $c = \varepsilon_0 A^{-1}\delta^\alpha - KA^{-1}\delta^2$, the choice

$$\delta \le \delta_1 := \min\{\delta_0, (\varepsilon_0/2K)^{1/(2-\alpha)}\}$$

yields $c \geq \frac{1}{2}\varepsilon_0 A^{-1}\delta^\alpha$, and therefore

$$t(z) \geq \delta\kappa(z),$$

where

$$\kappa(z) := \exp\left(C_2(\psi_0(z) - C_1)\right), \tag{12.4.1}$$

$$C_2 := \frac{2A}{\varepsilon_0(1-\delta_0^\alpha)}. \tag{12.4.2}$$

We are now in position to conclude. Fix $z \in \mathrm{Amp}(\{\omega\})$. Since $t(z) \geq \kappa(z)\delta$ and $t \mapsto \rho_t\varphi + Kt^2$ is increasing, we get

$$\begin{aligned}\rho_{\kappa(z)\delta}\varphi(z) - \varphi(z) &\leq \rho_{t(z)}\varphi(z) + Kt(z)^2 - \varphi(z)\\ &= \psi_{c,\delta}(z) - \varphi(z)\\ &= \frac{1}{1-\delta^\alpha}(\varphi_\delta(z) - \delta^\alpha\psi_0(z)),\end{aligned}$$

and by the above and the assumption $\varphi \leq 0$ we find that

$$\begin{aligned}\varphi_\delta - \delta^\alpha\psi_0 &\leq \varphi + C_1\delta^\alpha - \delta^\alpha\psi_0\\ &\leq C_1\delta^\alpha + \delta^\alpha(-\psi_0),\end{aligned}$$

and

$$\rho_{\kappa(z)\delta}\varphi(z) - \varphi(z) \leq (1-\delta_0^\alpha)^{-1})\delta^\alpha(C_1 - \psi_0(z)).$$

Replacing δ by $\kappa(z)^{-1}\delta$ and using (12.4.1), we obtain for $\delta \leq \delta_0\kappa(z)$,

$$\begin{aligned}\rho_\delta\varphi(z) - \varphi(z) &\leq (1-\delta_0^\alpha)^{-1}\delta^\alpha(C_1 - \psi_0(z)) \qquad (12.4.3)\\ &\quad \cdot \exp\left(\alpha C_2(C_1 - \psi_0(z))\right)\\ &\leq C_3\exp\left(2\alpha C_2(C_1 - \psi_0(z))\right), \qquad (12.4.4)\end{aligned}$$

where

$$C_3 := (\alpha C_2)^{-1}(1-\delta_0^\alpha)^{-1}. \tag{12.4.5}$$

The proof is complete, since $\psi_0(z)$ is locally bounded from below as well as $\kappa(z)$ given by (12.4.1). □

12.4.2 More capacities We introduce here yet another family of Monge–Ampère capacities, following Di Nezza and Lu [DiNL15]. They allow to give a simple proof of the domination principle in finite energy classes.

12.4.2.1 The (φ, ψ)-capacity We fix $\varphi, \psi \in PSH(X,\omega)$ such that $\varphi < \psi$ quasi-everywhere on X.

Definition 12.26. *The (φ,ψ)-capacity of a Borel subset $E \subset X$ is defined by*

$$\mathrm{Cap}_{\varphi,\psi}(E) := \sup\left\{\int_E MA(u)\ ;\ \ u \in PSH(X,\omega),\ \varphi \leq u \leq \psi\right\}.$$

When $\varphi \equiv \psi - 1$, we recover the capacity $\mathrm{Cap}_\psi := \mathrm{Cap}_{\psi-1,\psi}$. previously introduced. If moreover $\psi \equiv C$ is constant, we recover the Monge–Ampère capacity Cap_ω.

The *outer* (φ,ψ)*-capacity* of E is defined by

$$\mathrm{Cap}^*_{\varphi,\psi}(E) := \inf\left\{\mathrm{Cap}_{\varphi,\psi}(U)\ ;\ U \text{ is an open subset of } X,\ E \subset U\right\}.$$

Theorem 12.27. *If $\varphi \in \mathcal{E}(X,\omega)$ and $E \subset X$ is pluripolar, then*

$$\mathrm{Cap}^*_{\varphi,\psi}(E) = 0.$$

Proof. Assume that $\varphi \in \mathcal{E}(X,\omega)$ and fix a pluripolar set $E \subset X$. By translating ψ and φ by a constant we can assume that $\psi \leq 0$. Recall that $\varphi \in \mathcal{E}_\chi(X,\omega)$ for some convex increasing function $\chi : \mathbb{R}^- \to \mathbb{R}^-$. We can find $u \in \mathcal{E}_\chi(X,\omega), u \leq 0$, such that $E \subset \{u = -\infty\}$. We claim that

$$\mathrm{Cap}_{\varphi,\psi}(\{u < -t\}) \leq \frac{-2}{\chi(-t)}\left(E_\chi(u) + 2^n E_\chi(\varphi)\right),\ \forall t > 0.$$

Indeed, let $v \in PSH(X,\omega)$ such that $\varphi \leq v \leq \psi$. Chebyshev's inequality yields

$$\int_{\{u<-t\}} MA(v) \leq \frac{1}{-\chi(-t)} \int_{\{u<-t\}} (-\chi \circ u) MA(v).$$

Therefore

$$\int_{\{u<-t\}} MA(v) \leq \frac{-2}{\chi(-t)}\left(E_\chi(u) + E_\chi(v)\right).$$

which yields the claim. Now for any $t > 0$, $E \subset \{u < -t\}$ we obtain

$$\mathrm{Cap}^*_{\varphi,\psi}(E) \leq \mathrm{Cap}_{\varphi,\psi}(u < -t) \longrightarrow 0 \text{ as } t \longrightarrow +\infty. \qquad \square$$

We say that the (φ,ψ)-capacity *characterizes pluripolar sets* on X if for any subset $E \subset X$, the following holds:

$$\mathrm{Cap}^*_{\varphi,\psi}(E) = 0 \iff \text{E is a pluripolar subset of } X.$$

This turns out to be the case, as shown in [DiNL15]. The proof requires to analyze appropriate notions of extremal functions, e.g., the following:

Definition 12.28. *If $E \subset X$ is a Borel subset we set*

$$h_{\varphi,\psi,E} := \sup\left\{u \in PSH(X,\omega),\ u \leq \varphi \text{ q.e. on } E, u \leq \psi \text{ on } X\right\},$$

where "quasi-everywhere" (q.e.) means outside a pluripolar set. The upper semi-continuous regularization of $h_{\varphi,\psi,E}$ is called the relative (φ,ψ)-extremal function of E.

We let the reader check the following properties in Exercise 12.13:

Proposition 12.29. *Let $E \subset X$.*

(i) *The function $h^*_{\varphi,\psi,E}$ is ω-psh. It satisfies $\varphi \leq h^*_{\varphi,\psi,E} \leq \psi$ on X and $h^*_{\varphi,\psi,E} = \varphi$ quasi-everywhere on E.*

(ii) *If $P \subset E$ is pluripolar, then $h^*_{\varphi,\psi,E\setminus P} \equiv h^*_{\varphi,\psi,E}$; in particular, $h^*_{\varphi,\psi,P} \equiv \psi$.*

(iii) *If (E_j) are subsets of X increasing towards $E \subset X$, then h^*_{φ,ψ,E_j} decreases towards $h^*_{\varphi,\psi,E}$.*

(iv) *If $h^*_{\varphi,\psi,E} \equiv \psi$, then E is pluripolar.*

We refer the interested reader to [DiNL15] for much more information on these generalized Monge–Ampère capacities.

12.4.2.2 The Domination Principle The domination principle for finite energy classes was proved by Dinew (see [BL12]). The (φ, ψ)-capacity can be used to give an alternative and simplified proof of this important principle:

Proposition 12.30. *If $u, v \in \mathcal{E}(X, \omega)$ such that $u \leq v$ $MA(v)$-almost everywhere, then $u \leq v$ on X.*

Proof. We first claim that for every $\varphi \in \mathcal{E}(X, \omega)$ such that $0 \leq \varphi - u \leq C$ for some constant $C > 0$ and for any $s > 0$ one has

$$\int_{\{v<u-s\}} MA(\varphi) = 0.$$

Indeed, fix $s > 0$ and let φ be such a function. Let $C > 0$ be a constant such that $\varphi - u \leq C$ on X. Choose $\delta \in (0, 1)$ such that $\delta C < s$. Now, by using the comparison principle and the fact that $0 \leq \varphi - u \leq C$, we get

$$\begin{aligned}
\delta^n \int_{\{v<u-s\}} MA(\varphi) &= \int_{\{v<u-s\}} (\delta\omega + dd^c\delta\varphi)^n \\
&\leq \int_{\{v<\delta\varphi+(1-\delta)u-s\}} MA\left(\delta\varphi + (1-\delta)u\right) \\
&\leq \int_{\{v<\delta\varphi+(1-\delta)u-s\}} MA(v) \\
&\leq \int_{\{v<u\}} MA(v) = 0.
\end{aligned}$$

Thus, the claim is proved.

Now for each $t > 0$ let h_t denote the $(u,0)$-extremal function of the open set $G_t := \{u < -t\}$. It is clear that for every $t > 0$, $h_t \in \mathcal{E}(X,\omega)$ and $\sup_X(h_t - u) < +\infty$. The previous step yields

$$\int_{\{v<u-s\}} MA(h_t) = 0, \ \forall s > 0.$$

Fix $\varepsilon > 0$. Since u is quasi-continuous, we can fix a continuous function $\tilde{u}$ on X such that $\mathrm{Cap}_\omega(\{u \neq \tilde{u}\}) < \varepsilon$. Since h_t increases to 0 (see Lemma 12.31 below), we infer that

$$\int_{\{v<\tilde{u}-s\}} \omega^n \leq \liminf_{t\to+\infty} \int_{\{v<u-s\}} MA(h_t) + \mathrm{Cap}_{u,0}(\{u \neq \tilde{u}\}).$$

Repeating this argument we get

$$\int_{\{v<u-s\}} \omega^n \leq \varepsilon + \mathrm{Cap}_{u,0}(\{u \neq \tilde{u}\}).$$

Letting $\varepsilon \to 0$ and using Theorem 12.27 we get $\mathrm{Vol}(\{v < u - s\}) = 0$, for any $s > 0$ which implies that $u \leq v$ on X as desired. □

Lemma 12.31. *Let $v \in PSH(X,\omega)$. For each $t > 0$, set $G_t := \{v < -t\}$. Denote by h_t the $(\varphi,0)$-extremal function of G_t. Then h_t increases quasi-everywhere on X to 0 when t increases to $+\infty$.*

Proof. We can assume that $v \leq 0$ and $v \in \mathcal{E}(X,\omega)$ (up to replacing v by $-(-v)^p$ for some $p \in (0,1)$). We know that h_t increases quasi-everywhere to $h \in \mathcal{E}(X,\omega)$ and that $h \leq 0$. The same proof as in Theorem 12.27 gives

$$\lim_{t\to+\infty} \mathrm{Cap}_{\varphi,0}(G_t) = 0.$$

It follows that for each $t > 0$,

$$\int_{\{h<0\}} MA(h_t) \leq \int_{\{h_t<0\}} MA(h_t) = \mathrm{Cap}_{\varphi,0}(G_t).$$

We thus get

$$\int_{\{h<0\}} MA(h) \leq \liminf_{t\to+\infty} \int_{\{h<0\}} MA(h_t) = 0.$$

The comparison principle yields $\mathrm{Vol}(\{h < 0\}) = 0$ which completes the proof. □

12.5 Exercises

Exercise 12.1. *Let (X,ω) be a compact Riemann surface equipped with a Kähler form ω. Let $\mu = \omega + dd^c u \geq 0$ be a positive Radon measure, where $u \in SH(X,\omega)$. We endow $SH(X,\omega)$ with the L^1-topology.*

(1) *Show that* $SH(X,\omega) \subset L^1(\mu)$ *if and only if* u *is bounded.*

(2) *Show that* $\varphi \in SH(X,\omega) \mapsto \int_X \varphi\, d\mu \in \mathbb{R}$ *is continuous if and only if* u *is continuous.*

Exercise 12.2. *Let* (X,ω) *be a compact Kähler manifold. Give an example of a probability measure* $\mu = f dV$ *absolutely continuous with respect to Lebesgue measure such that*

- *its density* $0 \leq f$ *is smooth except at one point;*
- $\mu = MA(\varphi)$ *with* $\varphi \in \mathcal{E}(X,\omega)$ *unbounded.*

Exercise 12.3. *Let* $\mu = f\omega^n$ *be a probability measure on a compact Kähler manifold* (X,ω) *such that* $f[\log(1+f)]^n \in L^1(\omega^n)$. *Show that there exists* $A > 0$ *such that*

$$\mu(K) \leq A\,\mathrm{Cap}_\omega(K), \quad \text{for all Borel subsets } K \subset X.$$

Exercise 12.4. *Let* (X,ω) *be a compact Kähler manifold and* μ *be a probability measure on* X. *Let* $\varepsilon : \mathbb{R} \to \mathbb{R}^+$ *be a continuous decreasing function such that* $\int^{+\infty} \varepsilon < +\infty$. *Set*

$$F_\varepsilon : x \in \mathbb{R}^+ \longmapsto x\,[\varepsilon(-\ln x/n)]^n \in \mathbb{R}^+$$

and assume that for all Borel subsets $K \subset X$,

$$\mu(K) \leq F_\varepsilon(\mathrm{Cap}_\omega(K)).$$

Show that $\mu = MA(\varphi)$ *for some continuous* ω*-psh function* φ (*see* [BGZ08] *for more details*).

Exercise 12.5. *Let* (X,ω) *be a compact Kähler manifold. Let* $\mu = f dV$ *be a probability measure which is absolutely continuous with respect to Lebesgue measure. Let* $\chi : \mathbb{R}^+ \to \mathbb{R}^+$ *be a convex increasing weight such that* $\chi(+\infty) = +\infty$. *One says that the density* f *belongs to the Orlicz class* L_χ *if*

$$\int_0^{+\infty} \chi \circ f(t)dt < +\infty.$$

By looking carefully at the proof of Theorem 12.1, *find optimal growth conditions on* χ *which ensure that* μ *has bounded Monge–Ampère potentials* (*see* [Kol98] *for more details*).

Exercise 12.6. *Consider* $\mu = f\omega$ *on* $X = \mathbb{P}^1(\mathbb{C})$, *where* $\omega = \omega_{\mathrm{FS}}$ *is the Fubini–Study volume form,* $f \geq 0$ *is continuous on* $\mathbb{P}^1 \setminus \{p\}$, *and*

$$f(z) \simeq \frac{\varepsilon(\log(-\log|z|))}{|z|^2(\log|z|)^2}$$

in local coordinates near $p = 0$. *Here* $\varepsilon : \mathbb{R} \to \mathbb{R}^+$ *decreases towards* 0 *at* $+\infty$. *We claim that there exists* $A > 0$ *such that*

$$\mu(K) \leq A\,\mathrm{Cap}_\omega(K)\varepsilon(-\log \mathrm{Cap}_\omega(K)), \text{ for all } K \subset X.$$

(i) *Explain why it suffices to establish this estimate when K is included in a local chart near $p = 0$.*

(ii) *Consider $\tilde{K} := \{r \in [0, R]\,;\ K \cap \{|z| = r\} \neq \emptyset\}$. Show that the logarithmic capacity $c(K)$ of K can be estimated from below by the length of $\tilde{K}$, namely*

$$\frac{l(\tilde{K})}{4} \leq c(\tilde{K}) \leq c(K).$$

Details can be found in [Ra].

(iii) *Using that ε is decreasing, deduce that*

$$\mu(K) \leq 2\pi \int_0^{l(\tilde{K})} f(r) r dr \leq 2\pi \frac{\varepsilon(\log(-\log 4c(K)))}{-\log 4c(K)}.$$

(iv) *Show that $-\log 4c(K) \simeq -\log T_\omega(K) \simeq 1/\mathrm{Cap}_\omega(K)$, and finally*

$$\mu(K) \leq A\,\mathrm{Cap}_\omega(K)\varepsilon(-\log \mathrm{Cap}_\omega(K)).$$

Exercise 12.7. *Let $\mu = \omega + dd^c\varphi$ be a probability measure on a compact Riemann surface (X, ω), where $\varphi \in PSH(X, \omega)$. Show that the following properties are equivalent:*

(i) *the function φ is Hölder continuous ;*

(ii) *there exist $\alpha, C > 0$ such that $\mu(\mathbb{B}(a, r)) \leq C r^\alpha$, for all $(a, r) \in X \times [0, 1]$;*

(iii) *there exists $\varepsilon > 0$ such that $\exp(-\varepsilon PSH(X, \omega)) \subset L^1(\mu)$.*

Here $\mathbb{B}(a, r)$ denotes the ball with respect to the metric ω centered at a and of radius r.

Exercise 12.8. *Let $F : \mathbb{R} \to \mathbb{R}$ be a non-decreasing function such that $F(+\infty) = +\infty$ and $F(-\infty) = -\infty$. Let $\mu = f dV$ be a Radon measure with bounded density $0 \leq f \in L^\infty(X)$. Show that there exists a unique bounded ω-psh function φ such that*

$$MA(\varphi) = e^{F(\varphi)}\mu.$$

Verify that φ is the upper envelope of subsolutions to this equation.

Exercise 12.9. *Generalize the previous exercise to the case where the density merely belongs to $L^p(X)$, for some $p > 1$.*

What can you say about the regularity of φ?

Exercise 12.10. *Let u_Φ denote the unique harmonic function in the unit disk $\Delta = \{\zeta \in \mathbb{C}\,|\,|\zeta| < 1\}$ which is continuous up to the boundary, with boundary values $\Phi \in \mathcal{C}(\partial\Delta, \mathbb{C})$.*

(1) *Show that $\Phi \in \mathrm{Lip}_\alpha(\partial\Delta) \Rightarrow u_\Phi \in \mathrm{Lip}_\alpha(\overline{\Delta})$ when $0 < \alpha < 1$.*

(2) *Using $\Phi(e^{2i\pi\theta}) = |\sin\theta|$, show that the result is false when $\alpha = 1$.*

Exercise 12.11. *Let (X,ω) be a compact Kähler manifold and consider $\mu = MA(\varphi)$, where $\varphi \in PSH(X,\omega)$ is Hölder continuous. Show that there exists $\alpha > 0$ such that for all $\psi \in PSH(X,\omega)$,*

$$\int_X \exp(-\alpha\psi)\, d\mu < +\infty.$$

More information is to be found in [DNS10].

Exercise 12.12. *Show the following properties of the generalized Monge–Ampère capacities:*

(i) *If $E_1 \subset E_2 \subset X$, then* $\mathrm{Cap}_{\varphi,\psi}(E_1) \leq \mathrm{Cap}_{\varphi,\psi}(E_2)$.

(ii) *If $E_1, E_2, \ldots$ are Borel subsets of X, then*

$$\mathrm{Cap}_{\varphi,\psi}\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{+\infty} \mathrm{Cap}_{\varphi,\psi}(E_j).$$

(iii) *If $E_1 \subset E_2 \subset \ldots$ are Borel subsets of X, then*

$$\mathrm{Cap}_{\varphi,\psi}\left(\bigcup_{j=1}^{\infty} E_j\right) = \lim_{j\to+\infty} \mathrm{Cap}_{\varphi,\psi}(E_j).$$

Exercise 12.13. *Let $E \subset X$. Show that*

(i) *The function $h^*_{\varphi,\psi,E}$ is ω-psh. It satisfies $\varphi \leq h^*_{\varphi,\psi,E} \leq \psi$ on X and $h^*_{\varphi,\psi,E} = \varphi$ quasi-everywhere on E.*

(ii) *If $P \subset E$ is pluripolar, then $h^*_{\varphi,\psi,E\setminus P} \equiv h^*_{\varphi,\psi,E}$; in particular, $h^*_{\varphi,\psi,P} \equiv \psi$.*

(iii) *If (E_j) are subsets of X increasing towards $E \subset X$, then h^*_{φ,ψ,E_j} decreases towards $h^*_{\varphi,\psi,E}$.*

(iv) *If $h^*_{\varphi,\psi,E} \equiv \psi$, then E is pluripolar.*

Exercise 12.14. *Let (X,ω) be a compact Kähler manifold of dimension n and fix a divisor $D \subset X$. Consider a Radon measure $\mu = f\omega^n$ with density $f \geq 0$ which is locally bounded outside D.*

(i) *Show that if there exists $\psi \in \mathcal{E}^1(X,\omega)$ such that*

$$(\omega + dd^c\psi)^n \geq e^{\psi}\mu,$$

then $(\omega + dd^c\varphi)^n = e^{\varphi}\mu$ admits a unique solution in $\mathcal{E}^1(X,\omega)$.

(ii) *Analyze the case when f is of Poincaré type,*

$$f \simeq \frac{1}{|s_D|^2 |\log |s_D||^2}.$$

Show that there exists a subsolution and that the unique solution to the corresponding equation is locally bounded in $X \setminus D$.

(iii) *Same question for a log canonical density,*

$$f \simeq \frac{1}{|s_D|^2}.$$

See [DiNL14] *for more examples and details.*

13 The viscosity approach

We develop in this chapter the viscosity approach to the equation

$$(\mathrm{DMA}_v^\varepsilon) \qquad (\omega + dd^c\varphi)^n = e^{\varepsilon\varphi} v,$$

where ω is a closed smooth semipositive $(1,1)$-form on a n-dimensional connected compact Kähler manifold X and v is a volume form with non-negative continuous density and $\varepsilon \in \mathbb{R}_+$.

We adapt to this compact setting the basic definitions from Chapter 6. The key technical tool is the global comparison principle, which lies at the heart of the viscosity approach. Once it is established, Perron's method can be applied to produce viscosity solutions.

Our main goal here is thus to establish the global comparison principle for $(\mathrm{DMA}_v^\varepsilon)$. We then use it to solve the above degenerate complex Monge–Ampère equation.

The structure of $(\mathrm{DMA}_v^\varepsilon)$ allows us to avoid any restrictive curvature assumptions on X (unlike e.g., in [AFS08]).

We refer the reader to the work of Harvey and Lawson [HL11] for an interesting alternative approach which allowed theses authors to work on almost complex Hermitian manifolds.

13.1 Definitions

In the whole chapter X is a compact Kähler manifold and ω is a smooth semi-positive closed $(1,1)$-form representing a big cohomology class α. We let $\mathrm{Amp}(\alpha)$ denote the ample locus of α (see Definition 12.24). To fit in with the viscosity point of view, we express the Monge–Ampère equation as

$$(\mathrm{DMA}_v^\varepsilon) \qquad e^{\varepsilon\varphi} v - (\omega + dd^c\varphi)^n = 0.$$

Let $x \in X$. If $\kappa \in \Lambda^{1,1} T_x X$, we define κ_+^n to be κ^n if $\kappa \geq 0$ and 0 otherwise. For a technical reason, we will also consider a slight variant of $(\mathrm{DMA}_v^\varepsilon)$,

$$(\mathrm{DMA}_v^\varepsilon)_+ \qquad e^{\varepsilon\varphi} v - (\omega + dd^c\varphi)_+^n = 0.$$

Lemma 13.1. *Let $\Omega \subset X$ be an open subset and $z : \Omega \to \mathbb{C}^n$ be a holomorphic coordinate chart. Let h be a smooth local potential for ω defined on Ω. Then* $(\mathrm{DMA}_v^\varepsilon)$ *reduces in these z-coordinates to the scalar equation*

$$(\mathrm{DMA}_{v|z}^\varepsilon) \qquad e^{\varepsilon u} W - \det(u_{z\bar{z}}) = 0,$$

where $u = (\varphi + h)|_\Omega \circ z^{-1}$, $z_* v = e^{\varepsilon h_{|\Omega} \circ z^{-1}} W d\lambda$ and λ is the Lebesgue measure on $z(\Omega)$.

On the other hand, $(\mathrm{DMA}_v^\varepsilon)_+$ *reduces to the scalar equation:*

$(\mathrm{DMA}^\varepsilon_{v|z})_+$ $\qquad e^{\varepsilon u} W - \det(u_{z\bar z})_+ = 0$

The proof is left to the reader (Exercise 13.1). Note in particular that conditions (1.2) p. 27 (i.e., degenerate ellipticity) (2.11) p. 32 (properness), (2.18) p. 34 in [IL90] are satisfied by $(\mathrm{DMA}^\varepsilon_{v|z})_+$. If $v > 0$ and $\varepsilon > 0$, (2.17) p. 33 is also satisfied, so that we can apply the tools exposed in [IL90, CIL92].

13.1.1 Subsolutions If $\varphi_x^{(2)}$ is the 2-jet at $x \in X$ of a $\mathcal{C}^2$ real valued function φ we set

$$F_+(\varphi_x^{(2)}) = F^\varepsilon_{+,v}(\varphi_x) = e^{\varepsilon\varphi(x)} v_x - (\omega_x + dd^c\varphi_x)^n_+.$$

Recall the following definition from [IL90].

Definition 13.2. *A subsolution of* $(\mathrm{DMA}_v^\varepsilon)_+$ *is a function* $\varphi : X \to \mathbb{R} \cup \{-\infty\}$ *which is upper semi-continuous in* $\mathrm{Amp}(\alpha)$*, bounded from above on* X*, such that* $\varphi \not\equiv -\infty$ *and the following property is satisfied: if* $x_0 \in \mathrm{Amp}(\alpha)$ *and* $q \in \mathcal{C}^2$*, defined in a neighborhood of* x_0*, is such that* $\varphi(x_0) = q(x_0)$ *and*

$$\varphi - q \ \text{ has a local maximum at } x_0,$$

then $F_+(q^{(2)}_{x_0}) \le 0$.

Actually, this concept of a subsolution seems to be a bit too weak. It is not the same concept for $\varepsilon = 0$ as in the local setting and does not behave well if $v = 0$ since any usc function is then a viscosity subsolution of $(dd^c\varphi)^n_+ = 0$. It behaves well however if $v > 0$. We introduce it nevertheless in order to be able to use the reference [IL90].

We now introduce what we believe to be the right definition, which leads to a slightly stronger statement. If $\varphi_x^{(2)}$ is the 2-jet at $x \in X$ of a $\mathcal{C}^2$ real valued function φ, we set

$$F(\varphi_x^{(2)}) = F^\varepsilon_v(\varphi_x) = \begin{cases} e^{\varepsilon\varphi(x)} v_x - (\omega_x + dd^c\varphi_x)^n, & \text{if } \omega + dd^c\varphi_x \ge 0, \\ +\infty, & \text{otherwise.} \end{cases}$$

Recall the following definition from [CIL92]:

Definition 13.3. *A subsolution of* $(\mathrm{DMA}_v^\varepsilon)$ *is a function* $\varphi : X \to \mathbb{R} \cup \{-\infty\}$ *which is upper semi-continuous in* $\mathrm{Amp}(\alpha)$*, bounded from above on* X*, such that* $\varphi \not\equiv -\infty$ *and the following property is satisfied: if* $x_0 \in \mathrm{Amp}(\alpha)$ *and* $q \in \mathcal{C}^2$*, defined in a neighborhood of* x_0*, is such that* $\varphi(x_0) = q(x_0)$ *and*

$$\varphi - q \ \text{ has a local maximum at } x_0,$$

then $F(q^{(2)}_{x_0}) \le 0$.

Remark 13.4. *The function* F^ε_v *is lower semi-continuous and satisfies conditions* (0.1) *and* (0.2) *in* [CIL92].

Note that it is easy to compare subsolutions of $(\mathrm{DMA}_v^\varepsilon)$ and $(\mathrm{DMA}_v^\varepsilon)_+$:

Lemma 13.5. *Every subsolution φ of $(\mathrm{DMA}_v^\varepsilon)$ is a subsolution of $(\mathrm{DMA}_v^\varepsilon)_+$, it is ω-plurisubharmonic.*

A locally bounded upper semi-continuous function is ω-psh and satisfies $(\omega + dd^c\varphi)^n_{\mathrm{BT}} \geq e^{\epsilon\varphi} v$ iff it is a (viscosity) subsolution of $(\mathrm{DMA}_v^\varepsilon)$.

If $v > 0$, subsolutions of $(\mathrm{DMA}_v^\varepsilon)_+$ are subsolutions of $(\mathrm{DMA}_v^\varepsilon)$.

Proof. Immediate consequence of the definitions and the results obtained in the local setting. One just has to choose a local potential ρ such that $dd^c\rho = \omega$ and set $\varphi' = \varphi + \rho$, $v' = e^{-\varepsilon\rho} v$ to apply the local results. □

Corollary 13.6. *Let φ be any ω-plurisubharmonic function. Then φ satisfies $(\omega + dd^c\varphi)^n \geq e^{\varepsilon\varphi} v$ in the vicosity sense iff $\langle(\omega + dd^c\varphi)^n\rangle \geq e^{\varepsilon\varphi} v$, where $\langle(\omega + dd^c\varphi)^n\rangle$ is the non-pluripolar (pluripotential) Monge–Ampère measure of φ.*

The non-pluripolar product $\langle(\omega + dd^c\varphi)^n\rangle$ is the measure μ_φ introduced in Chapter 10.

13.1.2 (Super)solutions

Definition 13.7. *A supersolution of $(\mathrm{DMA}_v^\varepsilon)$ is a supersolution of $(\mathrm{DMA}_v^\varepsilon)_+$, that is, a function $\varphi : X \to \mathbb{R} \cup \{-\infty\}$ which is lower semi-continuous in $\mathrm{Amp}(\alpha)$, bounded from below on X, such that $\varphi \not\equiv +\infty$ and the following property is satisfied: if $x_0 \in \mathrm{Amp}(\alpha)$ and $q \in \mathcal{C}^2$, defined in a neighborhood of x_0, is such that $\varphi(x_0) = q(x_0)$ and $\varphi - q$ has a local minimum at x_0, then $F_+(q^{(2)}_{x_0}) \geq 0$.*

Definition 13.8. *A viscosity solution of $(\mathrm{DMA}_v^\varepsilon)$ is a function that is both a sub- and a supersolution. In particular, viscosity solutions are bounded on X and continuous in $\mathrm{Amp}(\alpha)$.*

A pluripotential solution of $(\mathrm{DMA}_v^\varepsilon)$ is an upper semi-continuous function $\varphi \in L^\infty \cap PSH(X, \omega)$ such that $(\omega + dd^c\varphi)^n_{\mathrm{BT}} = e^{\varepsilon\varphi} v$.

Classical sub/supersolutions are $\mathcal{C}^2$ viscosity sub/supersolutions.

13.2 The global viscosity comparison principle

We establish here the global comparison principle. This is the heart of the viscosity approach.

13.2.1 A warm up The global comparison principle can be deduced from [AFS08] when X carries a Kähler metric with positive sectional curvature. This global curvature assumption is very strong (see [Mok88]). In the general case [AFS08] does not allow to establish a global comparison principle even in the simplest case that we now consider:

Proposition 13.9. *The global comparison principle holds when the cohomology class of ω is Kähler and v is continuous and positive.*

Proof. We can assume without loss of generality that $\varepsilon = 1$.

Assume first $v > 0$ is positive and smooth. By [Aub78, Yau78], there exist a classical solution $\varphi_Y \in C^2(M)$ of (DMA_v^1). If $\underline{u}$ is a subsolution then $\underline{u} \leq \varphi_Y$, as follows from Lemma 13.10 below. Similarly, if $\overline{u}$ is a supersolution $\overline{u} \geq \varphi$. If $v > 0$ is merely continuous, fix $\delta > 0$. Then, if $\underline{u}$ is a subsolution of (DMA_v^1), $\underline{u} - \delta$ is a subsolution of $(\mathrm{DMA}_{e^{\delta} v}^1)$. Similarly, if $\overline{u}$ is a supersolution of (DMA_v^1), $\overline{u} + \delta$ is a supersolution of $(\mathrm{DMA}_{e^{-\delta} v}^1)$. Choose a smooth volume form v^* such that $e^{-\delta} v < v^* < e^{\delta} v$. Then $\underline{u} - \delta$ is a subsolution of $(\mathrm{DMA}_{v^*}^1)$ and $\overline{u} + \delta$ a supersolution. Hence, $\underline{u} - \delta \leq \overline{u} + \delta$. Letting $\delta \to 0$, we conclude the proof. □

Lemma 13.10. *Assume $v > 0$. Let $\underline{u}$ be a bounded subsolution of $(\mathrm{DMA}_v^{\varepsilon})$ on X. If $\overline{u}$ is a $\mathcal{C}^2$ supersolution on X, then $\underline{u} \leq \overline{u}$.*

Note in particular that if $(\mathrm{DMA}_v^{\varepsilon})$ has a classical solution, then the latter dominates (resp. minorates) every subsolution (resp. supersolution), hence the global viscosity principle holds.

Proof. If $\overline{u}$ is classical, the fact that $\underline{u} \leq \overline{u}$ is a trivial consequence of the definition of subsolution at a maximum of $\underline{u} - \overline{u}$. Indeed let $x_0 \in X$ such that $\underline{u}(x_0) - \overline{u}(x_0) = \max_X (\underline{u} - \overline{u}) = m$. Use $q = \overline{u} + m$ as a test function in the definition of $\underline{u}$ being a viscosity subsolution to deduce:

$$(\omega + d d^c \overline{u})^n_{x_0} \geq e^{\overline{u}(x_0)+m} v.$$

On the other hand, $\overline{u}$ being a classical supersolution, we have

$$(\omega + d d^c \overline{u})^n_{x_0} \leq e^{\overline{u}(x_0)} v.$$

Hence $m \leq 0$. □

13.2.2 The global maximum principle The above remarks have only academic interest since we use existence of a classical solution to deduce a comparison principle whose main consequence is existence of a viscosity solution (see below). We need to establish a honest global comparison principle that will allow us to produce solutions without invoking [Aub78, Yau78]. We now come to this result:

Theorem 13.11. *The global viscosity comparison principle holds for $(\mathrm{DMA}_v^{\varepsilon})$, provided $\int_X \omega^n > 0$, $\int_X v > 0$ and $\varepsilon > 0$.*

Since $v > 0$ is positive, the subsolutions of $(\mathrm{DMA}_v^{\varepsilon})_+$ are those of $(\mathrm{DMA}_v^{\varepsilon})$ hence this could be stated as the global viscosity comparison principle for $(\mathrm{DMA}_v^{\varepsilon})_+$.

We first establish a useful lemma for the local equation

$$e^u \mu - (dd^c u)^n = 0, \tag{13.2.1}$$

Lemma 13.12. *Let $\mu(z) \geq 0, \nu(z) \geq 0$ be continuous volume forms on some domain $D \Subset \mathbb{C}^n$. Let u be a subsolution to (13.2.1) associated to μ and let w be a bounded supersolution to (13.2.1) associated to ν in D. Assume that*

(i) *the function $u - w$ achieves a local maximum at some $x_0 \in D$;*

(ii) *$\exists c > 0$ s.t. $z \longmapsto u(z) - 2c|z|^2$ is plurisubharmonic near x_0.*

Then $\nu(x_0) > 0$ and

$$e^{u(x_0)} \mu(x_0) \leq e^{w(x_0)} \nu(x_0). \tag{13.2.2}$$

Proof. We can assume that $u - w$ achieves a local maximum at 0,

$$M := \sup_{x \in \bar{\mathbb{B}}} (u(x) - w(x)) = u(0) - w(0),$$

where $\mathbb{B}$ is the unit ball in $\mathbb{C}^n$. The hypothesis (ii) insures that $u(z) - 2c|z|^2$ is plurisubharmonic in a neighborhood of $\bar{\mathbb{B}}$. Thus for any fixed $\alpha \in]0, 1[$, the function

$$u_\alpha(z) := u(z) - c\alpha|z|^2$$

is strictly psh in $\mathbb{B}$ and $u_\alpha - w$ achieves a strict maximum at 0 in $\bar{\mathbb{B}}$.

Observe that $u_\alpha - (1-\alpha)u - \alpha c|z|^2 = \alpha(u - 2c|z|^2)$ is psh in $\mathbb{B}$ hence

$$dd^c u_\alpha \geq (1-\alpha) dd^c u + \alpha\, c\, dd^c |z|^2.$$

It follows that, setting $dV_{\text{eucl}} := (dd^c|z|^2)^n$,

$$\begin{aligned}(dd^c u_\alpha)^n &\geq (1-\alpha)^n (dd^c u)^n + \alpha^n c^n dV_{\text{eucl}} \\ &\geq e^{u_\alpha(z) + n\log(1-\alpha)} \mu + \alpha^n c^n dV_{\text{eucl}},\end{aligned}$$

noticing that $u \geq u_\alpha$.

Since $u_\alpha(0) = u(0)$ we replace in the sequel u by u_α. We have thus reduced the situation to the case where $u(z) - c|z|^2$ is psh in $\mathbb{B}$, $u - w$ achieves a strict maximum in $\bar{\mathbb{B}}$ at 0 and u is a subsolution of the equation

$$e^{u(z) + n\log(1-\alpha)} \mu + \alpha^n c^n dV_{\text{eucl}} - (dd^c u)^n = 0.$$

We want to apply Jensen–Ishii's maximum principle to u and w by using the penalty method as in [CIL92]. Fixing $\varepsilon > 0$, we want to maximize on $\bar{\mathbb{B}} \times \bar{\mathbb{B}}$ the upper semi-continuous function

$$w_\varepsilon(x, y) := u(x) - w(y) - (1/2\varepsilon)|x - y|^2.$$

The penalty function forces the maximum of w_ε to be asymptotically attained along the diagonal. Since w_ε is upper semicontinuous on the compact set $\bar{\mathbb{B}} \times \bar{\mathbb{B}}$, there exists $(x_\varepsilon, y_\varepsilon) \in \bar{\mathbb{B}} \times \bar{\mathbb{B}}$ such that

$$\begin{aligned} M_\varepsilon &:= \sup_{(x,y)\in\bar{\mathbb{B}}^2} \left\{ u(x) - w(y) - \frac{1}{2\varepsilon}|x-y|^2 \right\} \\ &= u(x_\varepsilon) - w(y_\varepsilon) - \frac{1}{2\varepsilon}|x_\varepsilon - y_\varepsilon|^2. \end{aligned}$$

The following result is classical [CIL92, Proposition 3.7]:

Lemma 13.13. *We have* $|x_\varepsilon - y_\varepsilon|^2 = o(\varepsilon)$. *Every limit point* $(\hat{x}, \hat{y})$ *of* $(x_\varepsilon, y_\varepsilon)$ *satisfies* $\hat{x} = \hat{y}$, $(\hat{x}, \hat{x}) \in \bar{\mathbb{B}} \times \bar{\mathbb{B}}$ *and*

$$\lim_{\varepsilon\to 0} M_\varepsilon = \lim_{\varepsilon\to 0} (u(x_\varepsilon) - w(y_\varepsilon)) = u(\hat{x}) - w(\hat{x}) = M.$$

Our hypothesis of strict local maximum guarantees that $\hat{x} = 0$. Therefore $(x_\varepsilon, y_\varepsilon) \to (0,0)$ as $\varepsilon \to 0$. Take any sequence $(x_j, y_j) = (x_{\varepsilon_j}, y_{\varepsilon_j}) \to (0,0)$ and $(x_j, y_j) \in \mathbb{B}^2$ for any $j > 1$ so that the conditions in the lemma above are satisfied with $\varepsilon = \varepsilon_j \to 0$.

Set $\phi_j(x,y) := \frac{1}{2\varepsilon_j}|x-y|^2$. The function $(x,y) \mapsto u(x) - w(y) - \phi_j(x,y)$ achieves its maximum in $\bar{\mathbb{B}}^2$ at the interior point $(x_j, y_j) \in \mathbb{B}^2$. We can thus apply Jensen–Ishii's maximum principle and obtain the following estimates:

Lemma 13.14. *For any* $\gamma > 0$, *we can find* $(p^+, Q^+), (p_-, Q_-) \in \mathbb{C}^n \times \mathrm{Sym}^2_{\mathbb{R}}(\mathbb{C}^n)$ *such that*

(1) $(p^+, Q^+) \in \overline{\mathcal{J}}^{2+} u(x_j)$, $(p_-, Q_-) \in \overline{\mathcal{J}}^{2-} w(y_j)$, *where*

$$p^+ = \frac{(x_j - y_j)}{2\varepsilon_j} = -p_-.$$

(2) *The block diagonal matrix with entries* (Q^+, Q_-) *satisfies:*

$$-(\gamma^{-1} + \|A\|) I \le \begin{pmatrix} Q^+ & 0 \\ 0 & -Q_- \end{pmatrix} \le A + \gamma A^2,$$

where $A = D^2\phi_j(x_j, y_j)$, *i.e.*

$$A = \gamma^{-1} \begin{pmatrix} I & -I \\ -I & I \end{pmatrix},$$

and $\|A\|$ *is the spectral radius of* A.

We choose $\gamma = \varepsilon_j$. Thus

$$-(2\varepsilon_j^{-1})I \leq \begin{pmatrix} Q^+ & 0 \\ 0 & -Q_- \end{pmatrix} \leq \frac{3}{\varepsilon_j}\begin{pmatrix} I & -I \\ -I & I \end{pmatrix}.$$

Looking at the upper and lower diagonal terms we deduce that the eigenvalues of Q^+, Q_- are $O(\varepsilon_j^{-1})$. Evaluating the inequality on vectors of the form (Z, Z) we deduce that $Q^+ \leq Q_-$ in the sense of quadratic forms.

For a fixed $Q \in \mathrm{Sym}^2_{\mathbb{R}}(\mathbb{C}^n)$, denote by $H = Q^{1,1}$ its $(1,1)$-part. It is a hermitian matrix. Since $(p^+, Q^+) \in \overline{\mathcal{J}}^{2+}u(x_j)$, we deduce from the viscosity differential inequality satisfied by u that H^+ is positive definite. The inequality $H^+ \leq H_-$ forces $H_- > 0$, thus $0 \leq H^+ \leq H_-$ and $\det H^+ \leq \det H_-$.

The viscosity differential inequalities satisfied by u and w yield

$$e^{u(x_j)+n\log(1-\alpha)}\mu(x_j) + \alpha^n c^n d V_{\mathrm{eucl}}(x_j) \leq \det H^+ \leq \det H_- \leq e^{w(y_j)}\nu(y_j),$$

hence

$$e^{u(x_j)+n\log(1-\alpha)}\mu(x_j) + \alpha^n c^n d V_{\mathrm{eucl}}(x_j) \leq e^{w(y_j)}\nu(y_j). \tag{13.2.3}$$

Recall that $\lim_{j\to+\infty}(u(x_j) - w(y_j)) = u(0) - w(0)$. On the other hand the semi-continuity properties of u, w yield (taking a subsequence if necessary), $\lim_{j\to+\infty} u(x_j) \leq u(0)$, $\lim_{j\to+\infty} w(y_j) \geq w(0)$. Hence

$$\lim_{j\to+\infty} u(x_j) = u(0), \quad \lim_{j\to+\infty} w(y_j) = w(0).$$

Letting $j \to \infty$ in the inequality (13.2.3) we infer

$$e^{u(0)+n\log(1-\alpha)}\mu(0) + \alpha^n c^n d V_{\mathrm{eucl}}(0) \leq e^{w(0))}\nu(0).$$

Since $d V_{\mathrm{eucl}}(0) > 0$, we infer $\nu(0) > 0$. Letting $\alpha \to 0$ yields the desired inequality. □

We are now ready for the proof of Theorem 13.11:

Proof. We can assume without loss of generality that $\varepsilon = 1$. Assume that φ is a subsolution (resp. ψ is a supersolution) for (DMA^1_v).

Fix $\delta \in]0, 1[$ and set

$$\tilde{\varphi}(x) := (1-\delta)\varphi(x) + \delta\rho(x),$$

where $\rho \leq \varphi$ is a ω-psh function satisfying $\omega + dd^c\rho \geq \beta$, where β is a Kähler form on X. Such a function exists since the cohomology class of ω is big. One can moreover impose ρ to be smooth in the ample locus Ω with analytic singularities, and such that $\rho(x) \to -\infty$ as $x \to \partial\Omega = X \setminus \Omega$.

Since $\tilde{\varphi} - \psi$ is upper semicontinuous in X and tends to $-\infty$ when $x \to \partial\Omega$, the maximum of $\tilde{\varphi} - \psi$ is achieved at some point $x_0 \in \Omega$,

$$M := \sup_{x \in X} (\tilde{\varphi}(x) - \psi(x)) = \tilde{\varphi}(x_0) - \psi(x_0).$$

Observe that $\tilde{\varphi}$ satisfies $(\omega + dd^c\tilde{\varphi})^n \geq (1-\delta)^n(\omega + dd^c\varphi)^n + \delta^n\beta^n$, in the viscosity sense in Ω. Now $\tilde{\varphi} \leq \varphi$ since $\rho \leq \varphi$ hence

$$(\omega + dd^c\tilde{\varphi})^n \geq e^{\tilde{\varphi}} \left\{(1-\delta)^n v(x) + e^{-C}\delta^n\beta^n\right\},$$

where $\tilde{\varphi} \leq \varphi \leq C$.

The idea is to localize near x_0 and use Lemma 13.12. Choose complex coordinates $z = (z^1, \ldots, z^n)$ near x_0 defining a biholomorphism identifying a closed neighborhood $\bar{B}$ of x_0 to the closed complex ball $\bar{B}_2 := B(0,2) \subset \mathbb{C}^n$ of radius 2, sending x_0 to the origin in $\mathbb{C}^n$.

We let h_ω be a smooth local potential for ω in $z^{-1}(B_2)$, i.e. $dd^c h_\omega = \omega$. Setting $u := \tilde{\varphi} \circ z^{-1} + h_\omega \circ z^{-1}$ in B_2, we obtain

$$e^u \mu \leq (dd^c u)^n \quad \text{in } B_2, \tag{13.2.4}$$

where

$$\mu = e^{-h_\omega \circ z^{-1}} z_* \left\{(1-\delta)^n v(x) + e^{-C}\delta^n\beta^n\right\} > 0$$

is a continuous volume form on B_2.

Similarly the lower semi-continuous function $w := \psi \circ z^{-1} + h_\omega \circ z^{-1}$ satisfies the viscosity differential inequality

$$e^w \nu \geq (dd^c w)^n, \quad \text{in } B_2, \tag{13.2.5}$$

where $\nu := e^{-h_\omega \circ z^{-1}} z_*(v) > 0$ is a continuous volume form on B_2.

Our hypothesis guarantees

$$M = \sup_X\{\tilde{\varphi} - \psi\} = \max_{\bar{\mathbb{B}}}\{u(\zeta) - w(\zeta)\} = u(0) - w(0), \tag{13.2.6}$$

i.e. $u - w$ achieves its maximum at the interior point $0 \in \mathbb{B}$. Moreover $dd^c u = \omega + dd^c\varphi \geq \delta\beta$, i.e. u is $2c$-strictly psh in B_2 for some $c = c(\delta) > 0$. We apply Lemma 13.12 and conclude that $v(0) > 0$ and

$$e^{u(0)}\mu(0) \leq e^{w(0)}\nu(0).$$

Going back to φ and ψ we obtain $v(x_0) > 0$ and

$$(1-\delta)^n e^{\tilde{\varphi}(x_0)} v(x_0) \leq e^{\psi(x_0)} v(x_0).$$

hence

$$(1-\delta)\varphi(x) + \delta\rho(x) \leq \psi(x) - n\log(1-\delta),$$

for all $x \in X$ and $0 < \delta < 1$. We infer $\varphi \leq \psi$ in $X \setminus \{\rho = -\infty\} = \Omega$. □

Remark 13.15. *The miracle with the complex Monge–Ampère equation we are studying is that the equation does not depend on the gradient in complex coordinates. In fact, it takes the form $F(X) - f(x) = 0$. The localisation technique would fail without this structural feature.*

13.2.3 Perron's method Once we know that the global comparison principle holds, one easily constructs continuous (viscosity=pluripotential) solutions by Perron's method, as we now explain.

Theorem 13.16. *Assume the global comparison principle holds for* (DMA_v^1) *and that* (DMA_v^1) *has a bounded subsolution* $\underline{u}$ *and a bounded supersolution* $\overline{u}$*. Then,*

$$\varphi = \sup\{w \,|\, \underline{u} \leq w \leq \overline{u} \text{ and } w \text{ is a viscosity subsolution of } (\mathrm{DMA}_v^1)\}$$

is the unique viscosity solution of (DMA_v^1).

In particular, φ is a bounded ω-plurisubharmonic function which is continuous in $\mathrm{Amp}(\alpha)$. *Moreover, φ is also a solution of* (DMA_v^1) *in the pluripotential sense.*

Proof. The upper envelope φ of the subsolutions of (DMA_v^1) is a subsolution of (DMA_v^1) since F is lower semi-continuous (Exercise 13.3). Hence φ is a subsolution of $(\mathrm{DMA}_v^1)_+$.

The trick is now to consider its lower semi-continuous envelope φ_*. We claim that it is a supersolution of (DMA_v^1): otherwise, we could find $x_0 \in \Omega$ and q a $\mathcal{C}^2$ function such that $\varphi_* - q$ has zero as a local minimum at x_0 and $F_+(q_x^{(2)}) < 0$. This forces $v_{x_0} > 0$. We can then construct a subsolution U such that $U(x_1) > \varphi(x_1)$ for some $x_1 \in X$, as follows: let $(z^1, \ldots, z^n)$ be a coordinate system centered at x_0 giving a local isomorphism with the complex unit ball and assume $v > 0$ on this complex ball neighborhood. Then, for $\gamma, \delta, r > 0$ small enough $q_{\gamma,\delta} = q + \delta - \gamma\|z\|^2$ satisfies $F_+(q_{\gamma,\delta}^{(2)}) < 0$ for $\|z(x)\| \leq r$. Choose $\delta = (\gamma r^2)/8$, $r > 0$ small enough. Since $\varphi_*(x) - q(x) \geq 0$ for $\|z(x)\| \leq r$, we have

$$\varphi(x) \geq \varphi_*(x) > q_{\gamma,\delta}(x) \text{ if } r/2 \leq \|z(x)\| \leq r.$$

It follows that U, defined by

$$U(x) = \max(\varphi(x), q_{\delta,\gamma}(x))$$

if $\|z(x)\| \leq r$ and $U(x) = \varphi(x)$, otherwise is a subsolution of $(\mathrm{DMA}_v^1)_+$ and in fact of (DMA_v^1) since we may assume that $v > 0$ on the relevant part of X. Choose a sequence (x_n) converging to x_0 so that $\varphi(x_n) \to \varphi_*(x_0)$. Then $q_{\gamma,\delta}(x_n) \to \varphi_*(x_0) + \delta$. Hence, for $n \gg 0$, $U(x_n) = q_{\gamma,\delta}(x_n) > \varphi(x_n)$.

This contradiction leads to the conclusion that φ_* is a supersolution. It follows thus from the viscosity comparison principle that $\varphi_* \geq \varphi$ in $\mathrm{Amp}(\alpha)$. Since $\varphi = \varphi^* \geq \varphi_*$, we infer that

$$\varphi = \varphi_* = \varphi^*$$

is a continuous viscosity solution in $\mathrm{Amp}(\alpha)$.

It remains to check that φ is also a solution of (DMA_v^1) in the pluripotential sense. This follows from the previous argument in pluripotential theory. Since φ is a viscosity subsolution, we know that

$$(\omega + dd^c\varphi)^n_{\mathrm{BT}} \geq e^{\epsilon\varphi} v.$$

To prove the reverse inequality we can proceed as follows. Choose a "ball" $B \subset \mathrm{Amp}(\alpha)$ and solve a Dirichlet problem to get a continuous ω-psh function ψ on $\bar{B}$ with $(\omega + dd^c\psi)^n_{\mathrm{BT}} = e^{\epsilon\psi} v$ and $\psi = \varphi$ on ∂B. Set $\psi \equiv \varphi$ in $X \setminus B$ so that ψ is a bounded ω-psh function on X.

Thus, ψ is a pluripotential subsolution of the equation on X, hence $\psi \leq \varphi$, while the Bedford–Taylor comparison principle yields $\psi \geq \varphi$ on B. Therefore $\psi = \varphi$ and the function φ solves the equation on B (hence on X, as B was arbitrary). □

In situations where the global comparison principle is not available, one can employ the following substitute (this natural idea is used in the recent work [HL09]).

Proposition 13.17. *Assume that* (DMA_v^1) *has a bounded subsolution* $\underline{u}$ *and a classical supersolution* $\overline{u}$*. Then,*

$$\varphi = \sup\{w \,|\, \underline{u} \leq w \leq \overline{u} \text{ and } w \text{ is a subsolution of } (\mathrm{DMA}_v^1)\}$$

is the unique maximal viscosity subsolution of (DMA_v^1).

In particular, it is a bounded ω*-plurisubharmonic function.*

In the global case when $\varepsilon = 0$, i.e., for $(\omega + dd^c\varphi)^n = v$ on a compact Kähler manifold, classical strict sub/supersolutions do not exist and one cannot use directly the Perron method. A possibility is to perturb the equation to $(\mathrm{DMA}_v^\varepsilon)$ and let ε decrease to zero, if appropriate stability estimates have been established (see Chapter 12).

13.3 Regularity of potentials of singular K-E metrics

In this section we apply the viscosity approach to show that the bounded ω-psh solutions φ of

$$MA(\varphi) = e^{-\lambda\varphi} f dV$$

constructed in Chapters 11 and 12 are actually continuous in $\mathrm{Amp}(\alpha)$.

13.3.1 The case $\lambda = -\varepsilon < 0$ Assume X is compact Kähler and v is a continuous volume form with semi-positive density. We consider

$$MA(\varphi) = e^{\varepsilon\varphi} f dV.$$

Fix a Kähler form β on X. We impose the following condition on (the cohomology class of) ω:

$$(\dagger) \qquad \exists \eta > 0 \, \exists \psi \in L^\infty \cap PSH(X,\omega), \ (\omega + dd^c\psi)^n \geq \eta\beta^n.$$

When ω is a semipositive $(1,1)$-form with $\int_X \omega^n > 0$, then (X,ω) satisfies $(\dagger)$, as follows from Chapters 11 and 12.

Note that the inequality can be interpreted in the pluripotential or viscosity sense.

Lemma 13.18. *Assume* $(\dagger)$ *is satisfied and* v *has positive density.*

If $c \gg 0$ *is large enough, then* $\psi - c$ *is a subsolution of* (DMA_v^1) *and the constant function* $\varphi = c$ *is a supersolution of* (DMA_v^1).

Proof. The existence of ψ is needed for the subsolution, whereas the supersolution exists under the condition that there exists $C > 0$ such that $\omega^n \leq C \cdot v$, which follows here from our assumption that v has positive density. □

Corollary 13.19. *When* $(\dagger)$ *is satisfied and* v *is positive,* (DMA_v^1) *has a unique viscosity solution* φ*, which is also the unique solution in the pluripotential sense.*

Proof. Indeed, the global comparison principle holds and Theorem 13.16 enable us to conclude. □

We are now ready to establish that the (pluripotential) solutions of some Monge–Ampère equations constructed earlier are continuous:

Theorem 13.20. *Assume* X *is a compact Kähler manifold,* ω *is a semipositive* $(1,1)$*-form with* $\int_X \omega^n > 0$*, and* v *is a semi-positive continuous probability measure on* X*. Then* $(\dagger)$ *is satisfied and there exists a unique bounded* ω*-plurisubharmonic function* φ *which is the viscosity* (*equivalently, pluripotential*) *solution to the degenerate complex Monge–Ampère equation*

$$(\omega + dd^c\varphi)^n = e^\varphi v.$$

Corollary 13.21. *The function* $\varphi_P \in L^\infty \cap PSH(X,\omega)$ *such that*

$$(\omega + dd^c\varphi_P)^n = e^{\varphi_P} v$$

in the pluripotential sense constructed in Chapter 11 *is a viscosity solution, hence it is continuous in* $\mathrm{Amp}(\alpha)$.

Proof. Observe that $(\dagger)$ is obviously satisfied when the cohomology class of ω is Kähler. If moreover v has positive density, the result is an immediate consequence of Corollary 13.19 together with the uniqueness of solutions.

We treat the general case by approximation. We first still assume that v is positive, but the cohomology class $\{\omega\}$ is now merely semi-positive and big (i.e., $\int_X \omega^n > 0$).

By the above there exists, for each $0 < \varepsilon \leq 1$, a unique continuous $(\omega + \varepsilon\beta)$-psh function u_ε such that

$$(\omega + \varepsilon\beta + dd^c u_\varepsilon)^n = e^{u_\varepsilon} v.$$

We first observe that (u_ε) is relatively compact in $L^1(X)$. This is equivalent to checking that $\sup_X u_\varepsilon$ is bounded, as $\varepsilon \searrow 0^+$. Note that

$$e^{\sup_X u_\varepsilon} \geq \frac{\int_X \omega^n}{v(X)} = \int_X \omega^n,$$

hence $\sup_X u_\varepsilon$ is uniformly bounded from below.

Set $w_\varepsilon := u_\varepsilon - \sup_X u_\varepsilon$. This is a relatively compact family of $(\omega + \beta)$-psh functions, hence there exists $C > 0$ such that for all $0 < \varepsilon \leq 1$, $\int_X w_\varepsilon \, dv \geq -C$. It follows from the concavity of the logarithm that

$$\log \int_X (\omega + \beta)^n \geq \sup_X u_\varepsilon + \log \int_X (e^{w_\varepsilon} \, dv) \geq \sup_X u_\varepsilon - C.$$

Thus $(\sup_X u_\varepsilon)$ is bounded, as claimed.

We now assert that (u_ε) is decreasing as ε decreases to 0^+. Indeed, assume that $0 < \varepsilon' \leq \varepsilon$ and fix $\delta > 0$. Note that $u_{\varepsilon'}, u_\varepsilon$ are both $(\omega + \varepsilon\beta)$-plurisubharmonic. It follows from the (pluripotential) comparison principle that

$$\int_{\{u_{\varepsilon'} \geq u_\varepsilon + \delta\}} (\omega + \varepsilon\beta + dd^c u_{\varepsilon'})^n \leq \int_{\{u_{\varepsilon'} \geq u_\varepsilon + \delta\}} (\omega + \varepsilon\beta + dd^c u_\varepsilon)^n.$$

Since

$$(\omega + \varepsilon\beta + dd^c u_{\varepsilon'})^n \geq (\omega + \varepsilon'\beta + dd^c u_{\varepsilon'})^n \geq e^\delta (\omega + \varepsilon\beta + dd^c u_\varepsilon)^n$$

on the set $\{u_{\varepsilon'} \geq u_\varepsilon + \delta\}$, this shows that the latter set has zero Lebesgue measure. As $\delta > 0$ was arbitrary, we infer that $u_{\varepsilon'} \leq u_\varepsilon$.

We let $u = \lim_{\varepsilon \to 0} u_\varepsilon$ denote the decreasing limit of the functions u_ε. By construction this is an ω-psh function. It follows from Chapter 12 that u is a bounded and (pluripotential) solution of the Monge–Ampère equation

$$(\omega + dd^c u)^n = e^u \, v.$$

This shows that (†) is satisfied, hence we can use Corollary 13.19 to conclude that u is actually continuous in $\mathrm{Amp}(\alpha)$ and that it is a viscosity solution.

It remains to relax the positivity assumption made on v. From now on $\{\omega\}$ is semi-positive and big and v is a probability measure with semi-positive continuous density. We can solve

$$(\omega + dd^c \varphi_\varepsilon)^n = e^{\varphi_\varepsilon} [v + \varepsilon\beta^n],$$

where φ_ε are bounded ω-psh functions and $0 < \varepsilon \leq 1$. Observe that

$$e^{\sup_X \varphi_\varepsilon} \geq \frac{\int_X \omega^n}{1 + \int_X \beta^n},$$

hence $\sup_X \varphi_\varepsilon$ is bounded below.

It follows again from the concavity of the logarithm that $M_\varepsilon := \sup_X \varphi_\varepsilon$ is also bounded from above. Indeed, set $\psi_\varepsilon := \varphi_\varepsilon - M_\varepsilon$. This is a relatively compact family of non-positive ω-psh functions, thus there exists $C > 0$ such that $\int_X \psi_\varepsilon(v + \beta^n) \geq -C$. Now

$$\log\left(\int e^{\psi_\varepsilon} \frac{v + \varepsilon\beta^n}{\int_X v + \varepsilon\beta^n}\right) \geq \int \psi_\varepsilon \frac{v + \varepsilon\beta^n}{\int_X v + \varepsilon\beta^n} \geq \int \psi_\varepsilon(v + \beta^n) \geq -C$$

yields

$$\log \int_X \omega^n \geq M_\varepsilon + \log\left[1 + \varepsilon \int_X \beta^n\right] - C,$$

so that (M_ε) is uniformly bounded.

We infer that (φ_ε) is relatively compact in $L^1(X)$. It follows that (φ_ε) is actually uniformly bounded, as ε decreases to zero. We infer that for any $0 < \delta \ll 1$,

$$\begin{aligned}
\mathrm{Cap}_\omega\left(\{\varphi_\varepsilon - \varphi_{\varepsilon'} < -2\delta\}\right) &\leq \frac{C}{\delta^n} \int_{\{\varphi_\varepsilon - \varphi_{\varepsilon'} < -\delta\}} (\omega + dd^c\varphi_\varepsilon)^n \\
&\leq \frac{C}{\delta^{n+1}} \int_X |\varphi_\varepsilon - \varphi_{\varepsilon'}| (\omega + dd^c\varphi_\varepsilon)^n \\
&\leq \frac{C'}{\delta^{n+1}} \int_X |\varphi_\varepsilon - \varphi_{\varepsilon'}| (v + \beta^n).
\end{aligned}$$

Using Proposition 12.10 and optimizing the value of δ yields

$$\|\varphi_\varepsilon - \varphi_{\varepsilon'}\|_{L^\infty} \leq C\left(\|\varphi_\varepsilon - \varphi_{\varepsilon'}\|_{L^1}\right)^{\frac{1}{n+2}}.$$

Thus, if (ε_n) is a sequence decreasing to zero as n goes to $+\infty$ such that $(\varphi_{\varepsilon_n})_n$ converges in L^1, $(\varphi_{\varepsilon_n})$ is actually a Cauchy sequence of bounded functions, hence it uniformly converges, to the unique bounded pluripotential solution φ of (DMA^1_v). From this, it follows that (φ_ε) has a unique cluster value in L^1 when ε decreases to 0, hence converges in L^1. The preceding argument yields uniform convergence.

Theorem 13.16 ensures that φ is also a viscosity subsolution. Remark 6.3 p. 35 in [CIL92] actually enables one to conclude that φ is indeed a viscosity solution. □

13.3.2 The case $\lambda = \varepsilon = 0$ We now turn to the study of the degenerate equations (DMA^0_v)

$$(\omega + dd^c\varphi)^n = v$$

on a given compact Kähler manifold X. Here v is a continuous volume form with semipositive density and ω is a smooth semipositive closed real $(1,1)$ form on X. We

assume that v is normalized so that

$$v(X) = \int_X \omega^n.$$

This is an obvious necessary condition for solving the equation

$$(\omega + dd^c\varphi)^n = v$$

on X. Bounded solutions to such equations have been provided in Chapter 12 when v has L^p-density, $p > 1$. We also have shown that these solutions are continuous on a Zariski open subset of X, the ample locus of the cohomology class of ω.

Our aim here is to provide an alternative proof of this result. We treat here the case of continuous densities, as this is required in the viscosity context, and refer the reader to Section 13.4.1 for more general cases.

Theorem 13.22. *The pluripotential solutions to* (DMA_v^0) *are viscosity solutions, hence they are continuous in* $\mathrm{Amp}(\alpha)$.

The plan is to combine the viscosity approach for the family of equations $(\omega + dd^c\varphi)^n = e^{\varepsilon\varphi}v$ with the pluripotential tools developed in previous chapters.

Proof. For $\varepsilon > 0$ we let φ_ε denote the unique viscosity (or equivalently pluripotential) ω-psh bounded solution of the equation

$$(\omega + dd^c\varphi_\varepsilon)^n = e^{\varepsilon\varphi_\varepsilon}v.$$

Set $M_\varepsilon := \sup_X \varphi_\varepsilon$ and $\psi_\varepsilon := \varphi_\varepsilon - M_\varepsilon$. The latter form a relatively compact family of ω-psh functions, hence there exists $C > 0$ such that

$$\int_X \psi_\varepsilon dv = \int_X (\varphi_\varepsilon - M_\varepsilon)dv \geq -C, \text{ for all } \varepsilon > 0.$$

Observe that $M_\varepsilon \geq 0$ since $v(X) = \int_X \omega^n =: V$. The concavity of the logarithm yields

$$0 = \log\left(\int_X e^{\varepsilon\varphi_\varepsilon}\frac{dv}{V}\right) \geq \frac{1}{V}\int \varepsilon\varphi_\varepsilon dv,$$

therefore

$$0 \geq \int \varphi_\varepsilon dv \geq -C + VM_\varepsilon,$$

i.e., (M_ε) is uniformly bounded. We infer that (φ_ε) is relatively compact in L^1 and the Monge–Ampère measures $(\omega + dd^c\varphi_\varepsilon)^n$ have uniformly bounded densities in L^∞.

One shows as above that this family of bounded ω-psh functions is uniformly Cauchy, hence converges to a *bounded* pluripotential solution of (DMA_v^0) which is continuous in $\mathrm{Amp}(\alpha)$.

This pluripotential solution is also a viscosity solution by [CIL92] Remark 6.3.

Recall that the solutions of (DMA_v^0) are unique, up to an additive constant. It is natural to wonder which solution is reached by the the family φ_ε. Observe that $\int_X e^{\varepsilon\varphi_\varepsilon} dv = \int_X dv = \int_X \omega^n$, thus

$$0 = \int_X \frac{e^{\varepsilon\varphi_\varepsilon} - 1}{\varepsilon} dv = \int_X \varphi_\varepsilon dv + o(1).$$

Hence, the limit φ of φ_ε as ε decreases to zero is the unique solution of (DMA_v^0) that is normalized by $\int_X \varphi\, dv = 0$. □

13.3.3 The case $\lambda > 0$ We have established in previous chapters some criteria which allow us to solve:

$$(\mathrm{MA})_\lambda \qquad MA(\varphi) = e^{-\lambda\varphi} f dV,$$

for some small enough values of $\lambda > 0$. Here f is assumed to be a continuous density.

Treating $e^{-\varphi} f = F$ as a density (independent of φ), it follows from Chapter 12 that φ is bounded. We will see in the next section that φ is continuous in $\mathrm{Amp}(\alpha)$, once we will be able to treat densities which are in L^p, $p > 1$, rather than continuous.

13.4 Concluding remarks

13.4.1 More continuous solutions Let X be a compact Kähler manifold, $v = f dV$ a non-negative measure which is absolutely continuous with respect to some volume form dV on X, and ω a smooth semi-positive closed real $(1,1)$-form on X with positive volume. We assume v is normalized, so that

$$v(X) = \int_X \omega^n,$$

where $n = \dim_{\mathbb{C}} X$.

When ω is Kähler, we have shown in Chapter 12 that there exists a unique continuous ω-plurisubharmonic function φ such that

$$(\omega + dd^c\varphi)^n = v \text{ and } \int_X \varphi\, dV = 0,$$

as soon as the density f is "good enough" (i.e., belongs to some Orlicz class, e.g., L^p, $p > 1$, is good enough).

This result can now be extended to the case where ω is merely semi-positive: it suffices to approximate the density f by smooth positive densities f_ε (using normalized convolutions) and to show, as in the proofs of Theorems 13.20 and 13.22 that the corresponding solutions form a Cauchy sequence. We leave the details to the reader.

13.4.2 More comparison principles Let again $B \subset \mathbb{C}^n$ denote the open unit ball and let $B' = (1+\eta)B$ with $\eta > 0$ be a slightly larger open ball. Let $u, u' \in PSH(B')$ be plurisubharmonic functions.

By convolution with an adequate non-negative kernel of the form $\rho_\epsilon(z) = \epsilon^{-2n}\rho_1(\frac{z}{\epsilon})$ we construct $(u_\epsilon)_{\eta>\epsilon>0}$ a family of smooth plurisubharmonic functions decreasing to u as ϵ decreases to 0.

Lemma 13.23. *For all $z \in B$,*

$$u(z) + u'(z) = \lim_{n\to\infty} \sup\{u'(x) + u_{1/j}(x) | j \geq n,\ |x - z| \leq 1/n\}.$$

Proof. Indeed, we have, if $2/n < \eta$:

$$\begin{aligned} u(z) + u'(z) &\leq \sup\{u'(z) + u_{1/j}(z) | j \geq n\} \\ &\leq \sup\{u'(x) + u_{1/j}(x) | j \geq n,\ |x - z| \leq 1/n\} \\ &\leq \sup\{u'(x) + u(x) |\ |x - z| \leq 2/n\}. \end{aligned}$$

Since $u + u'$ is upper semi-continuous, we have:

$$u(z) + u'(z) = (u + u')^*(z) = \lim_{n\to\infty} \sup\{u + u'(x) |\ |x - z| \leq 2/n\}. \qquad \square$$

Lemma 13.24. *Let φ a bounded plurisubharmonic function on B and v a continuous non-negative volume form, such that $e^{-\varphi}(dd^c\varphi)^n \geq v$ in the viscosity sense.*

Let ψ be a bounded psh function and w a continuous positive volume form, both defined on B' such that $(dd^c\psi)^n \geq w$.

Then there exist $C, c > 0$ depending only on $\|\psi\|_{L^\infty}, \|\varphi\|_{L^\infty}$ such that for every $\epsilon \in [0,1]$ $\Phi = \varphi + \epsilon\psi$ satisfies:

$$e^{-\Phi}(dd^c\Phi)^n \geq (1-\epsilon)^n e^{-C\epsilon} v + c\epsilon^n w$$

in the viscosity sense in B.

Proof. We may assume $\epsilon > 0$ and w to be smooth. Let us begin by the case when ψ is of class C^2. Let $x_0 \in B$ and $q \in C^2$ such that $q(x_0) = \Phi(x_0)$ and $\Phi - q$ has a local maximum at x_0. Then, $\varphi - (q - \epsilon\psi)$ has a local maximum at x_0.

We infer that

$$\begin{aligned} dd^c(q - \epsilon\psi)_{x_0} &\geq 0, \\ e^{-q(x_0)+\epsilon\psi(x_0)}(dd^c(q - \epsilon\psi))_{x_0})^n &\geq v_{x_0}. \end{aligned}$$

We conclude by using the inequality

$$(dd^c q)^n_{x_0} \geq (dd^c(q - \epsilon\psi))_{x_0})^n + \epsilon^n (dd^c\psi)^n.$$

We now treat the general case. Since ψ is defined on B', we can construct by the above classical mollification a sequence of C^2 psh functions $(\psi_{1/k})$ converging to ψ as k goes to $+\infty$.

It follows from the local theory that $(dd^c\psi_k)^n \geq ((w^{1/n})_{1/k})^n = w_k$ in both the pluripotential and viscosity sense.

We conclude from the previous case that $\Phi_k = \varphi + \epsilon\psi_k$ satisfies

$$c\epsilon^n w_k + (1-\epsilon)^n e^{-C\epsilon} v \leq e^{-\Phi_k}(dd^c\Phi_k)^n$$

in the viscosity sense. Since $w_k > 0$, there is no difference between subsolution of (DMA) and of $(\mathrm{DMA})_+$, hence we have

$$c\epsilon^n w_k + (1-\epsilon)^n e^{-C\epsilon} v - e^{-\Phi_k}(dd^c\Phi_k)_+^n \leq 0$$

in the viscosity sense.

By Lemma 6.1 p. 34 and Remark 6.3 p. 35 in [CIL92], we conclude that

$$\bar{\Phi} = \limsup_{n\to\infty} \sup\{\Phi_j(x) | j \geq n,\ |x-z| \leq 1/n\}$$

satisfies the limit inequation

$$e^{-\bar{\Phi}}(dd^c\bar{\Phi})_+^n \geq (1-\epsilon)^n e^{-C\epsilon} v + c\epsilon^n w$$

in the viscosity sense[1]. Now Lemma 13.23 implies that $\bar{\Phi} = \Phi$. Since $w > 0$, the proof is complete. □

Theorem 13.25. *Let X be a compact Kähler manifold and $\omega \geq 0$ be a semi-Kähler smooth form.*

Then, the global viscosity comparison principle holds for $\left(\mathrm{DMA}^1_v\right)$ for any non-negative continuous probability measure v.

Proof. This is a variant of the argument sketched in [IL90] Sect. V.3 p. 56.

Let $\overline{u}$ be a supersolution and $\underline{u}$ be a subsolution. Perturb the supersolution $\overline{u}$ setting $\overline{u}_\delta = \overline{u} + \delta$. This $\overline{u}_\delta$ is a supersolution to $\left(\mathrm{DMA}^1_{\tilde{w}}\right)$ for every continuous volume form $\tilde{w}$ such that $\tilde{w} \geq e^{-\delta} v$.

Choose a continuous positive probability measure $w > 0$. Assuming without loss of generality that $\int_X \omega^n = 1$, we can construct ψ a bounded ω-psh function such that, in the viscosity sense

$$(\omega + dd^c\psi)^n = w.$$

Perturb the subsolution $\underline{u}$ setting

$$\underline{u}_\epsilon = (1-\epsilon)\underline{u} + \epsilon\psi.$$

By Lemma 13.24, $\underline{u}_\epsilon$ satisfies, in the viscosity sense,

$$e^{-(1+\epsilon)u}(\omega + dd^c u)^n \geq \left(\frac{1-\epsilon}{1+\epsilon}\right)^n e^{-C\epsilon} v + c\left(\frac{\epsilon}{1+\epsilon}\right)^n w\,.$$

[1] Here we use the fact that $(\mathrm{DMA})_+$ is a continuous equation.

This in turn implies that $\underline{u}_\epsilon$ satisfies, in the viscosity sense,

$$e^{-u}(\omega + dd^c u)^n \geq e^{-\epsilon\|u\|_\infty}\left[\left(\frac{1-\epsilon}{1+\epsilon}\right)^n e^{-C\epsilon} v + c\left(\frac{\epsilon}{1+\epsilon}\right)^n w\right].$$

Hence $\underline{u}_\epsilon$ satisifies, in the viscosity sense,

$$e^{-u}(\omega + dd^c u)^n \geq \tilde{w}$$

whenever $\tilde{w} \leq e^{-\epsilon\|u\|_\infty}((\frac{1-\epsilon}{1+\epsilon})^n e^{-C\epsilon} v + c(\frac{\epsilon}{1+\epsilon})^n w)$.

Choosing $1 \gg \delta \gg \epsilon > 0$, we find a continuous volume form $\tilde{w} > 0$ such that $\overline{u}_\delta$ is a supersolution and $\underline{u}_\epsilon$ is a viscosity subsolution of $e^{-u}(\omega + dd^c u)^n = \tilde{w}$. Using the viscosity comparison principle for $\tilde{w}$, we conclude that $\overline{u}_\delta \geq \underline{u}_\epsilon$ in $\mathrm{Amp}(\alpha)$. Letting $\delta \to 0$, we infer that $\overline{u} \geq \underline{u}$ in $\mathrm{Amp}(\alpha)$. □

This comparison principle could have been used instead of the pluripotential-theoretic arguments to establish existence of a viscosity solution in the case $v \geq 0$ of Theorem 13.20. This could be useful in dealing with similar problems where pluripotential tools are less efficient.

13.5 Exercises

Exercise 13.1. *Prove Lemma 13.1, i.e., that the global complex Monge–Ampère equation can be conveniently localized and satisfies the degenerate ellipticity and properness conditions.*

Exercise 13.2. *Let (X, ω) be a compact n-dimensional Kähler manifold. Show that there exists no smooth function $v : X \to \mathbb{R}$ such that $(\omega + dd^c v)^n_+ \leq 0$.*

Exercise 13.3. *Let (X, ω) be a compact n-dimensional Kähler manifold. Fix a continuous density $f \geq 0$. Let $(u_i)_{i\in I}$ be a family of viscosity subsolutions of the equation*

$$F = -(\omega + dd^c\varphi)^n + e^\varphi f\omega^n = 0.$$

Assume the family is locally uniformly bounded from above. Show that

$$u := (\sup_{i\in I} u_i)^*$$

is again a subsolution of $F = 0$.

Exercise 13.4. *Let (X, ω) be a compact Kähler manifold of complex dimension n.*

1) *Show that for all $A > 0$, there exists a unique viscosity solution of the equation*

$$(\omega + dd^c\varphi_A)^n = 2e^{A\varphi_A}\omega^n.$$

2) *Show that φ_A converges uniformly to zero as $A \to +\infty$. What happens if you replace ω^n by $f\omega^n$ for some continuous density $f \geq 0$?*

Exercise 13.5. *Let (X, ω) be a compact manifold. We let d denote the Riemannian distance associated to ω. Let $u : X \to \mathbb{R}$ (resp. $v : X \to \mathbb{R}$) be an upper semi-continuous (resp. lower semi-continuous) function. Set*

$$M_\alpha := \sup\{u(x) - v(y) - \alpha d^2(x, y);\ (x, y) \in X^2\},$$

where $\alpha > 0$. Show that when $\alpha \to +\infty$, one has

$$\alpha d^2(x_\alpha, y_\alpha) \longrightarrow 0 \text{ and } M_\alpha \longrightarrow \sup_X (u - v).$$

Exercise 13.6. *Let (X, ω) be a compact Kähler manifold of dimension n. Let $f \geq 0$ be a continuous density and $\varepsilon > 0$. Show that the viscosity solution φ_ε of*

$$(\omega + dd^c\varphi)^n = e^{\varepsilon\varphi} f\omega^n$$

converges uniformly, as ε decreases to zero, to the unique viscosity solution ψ of

$$(\omega + dd^c\psi)^n = f\omega^n,$$

normalized by $\int_X \psi\ f\omega^n = 0$.

Exercise 13.7. *Let (X, ω) be a compact Kähler manifold of dimension n. Give an example of a continuous ω-psh function φ such that $(\omega + dd^c\varphi)^n > 0$ in the viscosity sense, but φ is not strictly ω-psh, i.e., there is no $\varepsilon > 0$ such that φ is $(1 - \varepsilon)\omega$-psh.*

14 Smooth solutions

Our goal in this chapter is to establish higher order regularity of the Monge–Ampère potentials constructed in the previous chapters. A prototypical result is the following:

Theorem 14.1. *Let (X, ω) be a compact Kähler manifold of dimension n. Let $F : \mathbb{R} \to \mathbb{R}$ and $h : X \to \mathbb{R}$ be smooth functions. If there exists a bounded solution $\varphi \in PSH(X, \omega)$ to the complex Monge–Ampère equation*

$$(\omega + dd^c\varphi)^n = e^{F(\varphi)+h}\omega^n,$$

then φ is $\mathcal{C}^\infty$-smooth.

We have determined in previous chapters when these equations do admit bounded solutions, the point here is to discuss the regularity of the solutions when they exist.

In order to construct singular Kähler–Einstein metrics, we will also need to establish smoothness of the solutions to such equations when the reference form ω is merely semi-positive and big. In this case the solutions will be smooth in the ample locus of $\{\omega\}$.

The form ω is assumed to be Kähler in the first three sections. Semi-positive forms are then considered in Section 14.4.

14.1 The continuity method

We first consider the particular case when $F \equiv 0$:

$$\text{(CY)} \qquad (\omega + dd^c\varphi)^n = e^h\omega^n.$$

It is understood here that the function h is so normalized that

$$\int_X e^h\omega^n = \int_X \omega^n.$$

It follows from the previous chapter that such an equation admits a unique continuous solution $\varphi \in PSH(X, \omega)$, normalized by $\int_X \varphi\, \omega^n = 0$.

We are going to study the regularity of the solutions by using the so called continuity method, i.e., by considering a family of equations, which deform a known equation to the one we would like to solve:

$$\text{(CY)}_t \qquad (\omega + dd^c\varphi)^n = [te^h + (1-t)]\omega^n.$$

Note that $(\mathrm{CY})_0$ admits the trivial solution $\varphi_0 \equiv 0$, while $(\mathrm{CY})_1$ is the equation we are interested in. We fix $k \in \mathbb{N}$, $0 < \alpha < 1$ and let

$$S := \{s \in [0,1] \mid (\mathrm{CY})_s \text{ admits a solution } \varphi_s \in PSH(X,\omega) \cap \mathcal{C}^{k+2,\alpha}(X)\}.$$

We normalize φ_s so that $\int_X \varphi_s\, \omega^n = 0$.

Our goal is to show that 1 belongs to S. This will be achieved by showing that S is non-empty (because $0 \in S$) and both open and closed (thus $S = [0,1]$, since $[0,1]$ is connected). We first show the openness.

14.1.1 Openness

Proposition 14.2. *The set S is open.*

Proof. Assume $s \in S$. We must solve $(\mathrm{CY})_t$ for t close to s. Set $\omega_s = \omega + dd^c\varphi_s$ and $g_s = \log[se^h + (1-s)]$, so that

$$\omega_s^n = [se^h + (1-s)]\omega^n = e^{g_s}\omega^n.$$

We are looking for $\omega_t = \omega + dd^c\varphi_t$ such that $\omega_t^n = e^{g_t}\omega^n$, thus

$$(\omega_s + dd^c(\varphi_t - \varphi_s))^n = e^{g_t - g_s}\omega_s^n.$$

Set $\varepsilon = t - s$ and observe that

$$k_\varepsilon := g_t - g_s = \log\left[\frac{te^h + (1-t)}{se^h + (1-s)}\right] = \log\left[1 + \varepsilon\frac{e^h - 1}{se^h + (1-s)}\right] = O(\varepsilon).$$

Setting $\psi = \varphi_t - \varphi_s$, we have to show that we can solve

$$(\omega_s + dd^c\psi)^n = e^{k_\varepsilon}\omega_s^n$$

when ε is small enough. This is a perturbation problem and our plan is to apply the inverse function theorem in appropriate Banach spaces. In the sequel we write $\omega = \omega_s$ to simplify notations. Set

$$E := \{u \in \mathcal{C}^{k+2,\alpha}(X,\mathbb{R}) \mid \int_X u\,\omega^n = 0\}$$

and

$$F := \left\{v \in \mathcal{C}^{k,\alpha}(X,\mathbb{R}) \mid \int_X v\,\omega^n = 0\right\}.$$

The complex Monge–Ampère operator $\mathcal{M}$ maps the open set

$$A := \{u \in E \mid \omega + dd^c u > 0\} \subset E$$

smoothly into

$$B := \left\{f \in \mathcal{C}^{k,\alpha}(X,\mathbb{R}) \mid \int_X f\,\omega^n = \int_X \omega^n\right\},$$

which is a hyperplane of the Banach space $\mathcal{C}^{k,\alpha}(X,\mathbb{R})$, whose tangent space can be identified with F.

We want to show that for every $\varphi \in A$, the differential $D\mathcal{M}(\varphi) : E \to F$ is an isomorphism. Observe that for $u \in E$,

$$D\mathcal{M}(\varphi) \cdot u = \frac{d}{dt}\mathcal{M}(\varphi + tu)_{|t=0} = \Delta_\varphi u,$$

where

$$\Delta_\varphi u := n \frac{dd^c u \wedge \omega_\varphi^{n-1}}{\omega_\varphi^n}$$

is the Laplace operator associated to the Kähler form ω_φ.

It follows that $D\mathcal{M}(\varphi)$ is injective. The standard theory of Laplace–Beltrami operators (see [GT83]) ensures then that $D\mathcal{M}(\varphi)$ is actually also surjective, hence an isomorphism. Thus S is open. □

14.1.2 The strategy To show that the set S is closed, we are going to establish several a priori estimates. It suffices to show that we can find $\beta > \alpha$ and $C > 0$ such that for all $s \in S$,

$$\|\varphi_s\|_{\mathcal{C}^{k+2,\beta}} \leq C. \tag{14.1.1}$$

Indeed, in this case the family $\{\varphi_s, s \in S\}$ is relatively compact in $\mathcal{C}^{k+2,\alpha}$ and actually compact since each cluster point $\psi = \lim \varphi_{s_i}$, $s_i \to s$, is still normalized by $\int_X \psi\, \omega^n = 0$ and solves $(\mathrm{CY})_s$, hence $\psi = \varphi_s$.

Since the initial choice of $\alpha > 0$ is arbitrary, so is that of β. Since we are going to establish (14.1.1) for an arbitrary $k \in \mathbb{N}$, at the end of the day we obtain a smooth solution to (CY).

To establish (14.1.1) we proceed step by step:

- *Step 1.* It follows from Chapter 12 that the φ_s's are uniformly bounded,

$$\|\varphi_s\|_{L^\infty(X)} \leq C,$$

since the densities $f_s = se^h + (1-s) \leq e^{\sup_X h}$ are uniformly bounded.

- *Step 2.* Using the uniform bound from Step 1, we then move on to show that there exists $C_2 > 0$ such that for all $s \in S$,

$$\sup_X |\Delta_\omega \varphi_s| \leq C_2,$$

where

$$\Delta_\omega \varphi_s := n \frac{dd^c \varphi_s \wedge \omega^{n-1}}{\omega^n}$$

denotes the Laplace operator associated to the Kähler form ω.

- *Step 3.* We then explain how a complex version of the Evans–Krylov theory can be derived to produce (14.1.1) for $k = 0$ and some $\beta > 0$.
- *Step 4.* At this stage one can invoke the classical Schauder estimates and a standard bootstrapping argument to establish (14.1.1) inductively for each $k \in \mathbb{N}$.

A few comments are in order:

- the first two steps use global arguments, the last two are local;
- the first step is valid in a much greater context, as was explained in Chapter 12, this is also the case of the second one as we shall later on indicate;
- the original third step of Yau consisted in establishing directly a third-order a priori estimate; here we follow a simplification proposed by Siu [Siu87].

14.2 A priori $\mathcal{C}^2$ estimates

The goal of this section is to complete Step 2 by proving the following rather general a priori estimate:

Theorem 14.3. *Let dV be smooth volume form. Assume that $\varphi \in PSH(X,\omega)$ is smooth and satisfies*

$$(\omega + dd^c\varphi)^n = e^{\psi_+ - \psi_-} dV,$$

with $\int_X \varphi\omega^n = 0$. Assume we are given $C > 0$ and $p > 1$ such that

(i) $dd^c\psi^+ \geq -C\,\omega$ *and* $\sup_X \psi^+ \leq C$.

(ii) $dd^c\psi^- \geq -C\,\omega$ *and* $\|e^{-\psi^-}\|_{L^p} \leq C$.

Then there exists $A > 0$ depending only on ω, p, C, such that

$$0 \leq \omega + dd^c\varphi \leq A\, e^{-\psi^-}\,\omega.$$

This result recovers in particular [Yau78, Theorem 7, p. 398], and it is due to Păun [Pau08]. We follow the formulation and proof of [BBEGZ11].

14.2.1 Preliminaries

We shall need two auxiliary results.

Lemma 14.4. *Let α, β be positive $(1,1)$-forms. Then*

$$n\left(\frac{\alpha^n}{\beta^n}\right)^{\frac{1}{n}} \leq \mathrm{tr}_\beta(\alpha) \leq \left(\frac{\alpha^n}{\beta^n}\right)\cdot(\mathrm{tr}_\alpha(\beta))^{n-1}.$$

Proof. At a given point $x \in X$, we can assume that the quadratic forms defined by $\alpha(x)$ and $\beta(x)$ are co-reduced so that $\beta(x)$ is the identity matrix while $\alpha(x)$ is diagonal with entries $\lambda_1, \ldots, \lambda_n$. It follows then from the concavity of the logarithm that

$$\frac{1}{n}\log\prod_{j=1}^n \lambda_j = \frac{1}{n}\sum_{j=1}^n \log\lambda_j \leq \log\left(\frac{1}{n}\sum_{j=1}^n \lambda_j\right),$$

which yields the first inequality. The second one follows by observing that

$$\sum_j \lambda_j \leq (n-1)\sum_j \lambda_j \leq \left(\prod_j \lambda_j\right)\left(\sum_j \frac{1}{\lambda_j}\right)^{n-1},$$

when $n \geq 2$. There is actually equality when $n = 2$. □

The second result we need is the following estimate, which goes back to the work of Aubin [Aub78] and Yau [Yau78]; in the form presented here it is due to Siu [Siu87].

Lemma 14.5. *Let ω, ω' be arbitrary Kähler forms. Let $-B \in \mathbb{R}^-$ be a lower bound on the holomorphic bisectional curvature of (X, ω). Then*

$$\Delta_{\omega'} \log \mathrm{tr}_\omega(\omega') \geq -\frac{\mathrm{tr}_\omega(\mathrm{Ric}(\omega'))}{\mathrm{tr}_\omega(\omega')} - B\ \mathrm{tr}_{\omega'}(\omega).$$

Recall that the Ricci curvature of a Kähler form ω which is locally written as

$$\omega = \sum_{\alpha,\beta} \omega_{\alpha\beta}\, i\, dz_\alpha \wedge d\overline{z}_\beta$$

is given by

$$\mathrm{Ric}(\omega) = -c \sum_{\alpha,\beta} \frac{\partial^2 \log \det(\omega_{pq})}{\partial z_\alpha \partial \overline{z}_\beta} i\, dz_\alpha \wedge d\overline{z}_\beta.$$

The Ricci form $\mathrm{Ric}(\omega)$ represents the first Chern class of X (if the constant $c > 0$ is conveniently chosen), and is globally related to the Ricci curvature $\mathrm{Ric}(\omega')$ of another Kähler form ω' through

$$\mathrm{Ric}(\omega') = \mathrm{Ric}(\omega) - d\,d^c \log(\omega')^n/\omega^n.$$

When $(\omega')^n = e^{\psi_+ - \psi_-}\omega^n$, we thus obtain

$$\mathrm{Ric}(\omega') = \mathrm{Ric}(\omega) - d\,d^c(\psi_+ - \psi_-).$$

Proof. Since this is a local differential inequality, we can use a local system of normal coordinates so that near the origin (which is the point to be considered), $\omega = \sum \omega_{pq}\sqrt{-1}dz_p \wedge d\overline{z}_q$, with

$$\omega_{pq} = \delta_{pq} - \sum_{i,j} R_{i\bar{j}p\bar{q}} z_i \overline{z}_j + O(\|z\|^3).$$

Here $R_{i\bar{j}p\bar{q}}$ denotes the curvature tensor of ω and δ_{pq} stands for the Kronecker symbol. We can also impose that

$$\omega'_{pq} = \lambda_p \delta_{pq} + O(\|z\|), \text{ with } \lambda_1 \leq \ldots \leq \lambda_n.$$

Observe that the inverse matrix $(\omega^{pq}) = (\omega_{pq})^{-1}$ satisfies

$$\omega^{pq} = \delta_{pq} + \sum_{i,j} R_{i\bar{j}p\bar{q}} z_i \overline{z}_j + O(\|z\|^3).$$

We also recall that the curvature tensor of ω' is given by

$$R'_{i\bar{j}p\bar{q}} = \sum_k \frac{1}{\lambda_k} \partial_i \omega_{pk} \overline{\partial}_j \omega'_{pq} - \partial_i \overline{\partial}_j \omega'_{pq}.$$

Set $u := \mathrm{tr}_\omega(\omega')$ and observe that

$$d d^{\mathrm{c}} \log u = \frac{d d^{\mathrm{c}} u}{u} - \frac{du \wedge d^{\mathrm{c}} u}{u^2},$$

thus

$$\Delta_{\omega'} \log u = \frac{\Delta_{\omega'} u}{u} - \frac{\mathrm{tr}_{\omega'}(du \wedge d^{\mathrm{c}} u)}{u^2}.$$

Now $\Delta_{\omega'} u = \sum_{ik} \lambda_i^{-1} \partial_i \overline{\partial}_i (\omega^{kk} \omega'_{kk})$, and

$$\partial_i \overline{\partial}_i (\omega^{kk} \omega'_{kk}) = \lambda_k R_{i\bar{i}k\bar{k}} + \partial_i \overline{\partial}_i \omega'_{kk},$$

hence

$$\Delta_{\omega'} u = \sum_{ik} \lambda_k \lambda_i^{-1} R_{i\bar{i}k\bar{k}} + \sum_{i,k} \lambda_i^{-1} \partial_i \overline{\partial}_i \omega'_{kk}.$$

Our assumption on ω is $R_{i\bar{i}k\bar{k}} \geq -B$. Therefore the first term above is bounded below by

$$\sum_{ik} \lambda_k \lambda_i^{-1} R_{i\bar{i}k\bar{k}} \geq -B \left(\sum_i \lambda_i^{-1} \right) \left(\sum_k \lambda_k \right) \geq -B \, \mathrm{tr}_{\omega'}(\omega) \, \mathrm{tr}_\omega(\omega'),$$

whence

$$\frac{\sum_{ik} \lambda_k \lambda_i^{-1} R_{i\bar{i}k\bar{k}}}{u} \geq -B \, \mathrm{tr}_{\omega'}(\omega).$$

We now take care of the remaining terms. By the definition of R'_{ijpq},

$$\sum_{i,k} \lambda_i^{-1} \partial_i \overline{\partial}_i \omega'_{kk} = \sum_{i,k,p} \lambda_i^{-1} \lambda_p^{-1} |\partial_i \omega'_{kp}|^2 - \sum_{i,k} \lambda_i^{-1} R'_{i\bar{i}k\bar{k}}.$$

Note that $\sum_{i,k} \lambda_i^{-1} R'_{i\bar{i}k\bar{k}} = \mathrm{tr}_\omega(\mathrm{Ric}(\omega'))$, while

$$\begin{aligned} \sum_{i,k,p} \lambda_i^{-1} \lambda_p^{-1} |\partial_i \omega'_{kp}|^2 &\geq \sum_{i,k} \lambda_i^{-1} \lambda_k^{-1} |\partial_i \omega'_{kk}|^2 \\ &\geq \frac{\sum_{i,k,l} \lambda_i^{-1} \partial_i \omega'_{kk} \partial_i \omega'_{ll}}{\sum_i \lambda_i} = \frac{\mathrm{tr}_{\omega'}(du \wedge d^u)}{u}, \end{aligned}$$

where the last estimate follows from the Cauchy–Schwarz inequality. Altogether we obtain

$$\frac{\Delta_{\omega'} u}{u} \geq -\frac{\mathrm{tr}_\omega(\mathrm{Ric}(\omega'))}{\mathrm{tr}_\omega(\omega')} - B\ \mathrm{tr}_{\omega'}(\omega) + \frac{\mathrm{tr}_{\omega'}(du \wedge d^c u)}{u^2},$$

which yields the claim. □

14.2.2 A warm up To explain the main idea in a slightly simpler context, we first consider the $(\mathrm{CY})_s$ setting, letting $\varphi = \varphi_s$ be the normalized solution of

$$(\omega + dd^c\varphi)^n = e^{g_s}\omega^n, \text{ with } g_s = \log[se^h + (1-s)]$$

and $0 \leq s \leq 1$. Fix $C > 0$ such that $dd^c h \geq -C\omega$ Then

$$dd^c g_s = \frac{se^h dd^c h + se^h dh \wedge d^c h}{se^h + (1-s)} + \frac{s^2 e^{2h} dh \wedge d^c h}{[se^h + (1-s)]^2} \geq -C\omega, \tag{14.2.1}$$

This setting can therefore be considered as a particular case of Theorem 14.3 with $\psi_- = 0$; however, the situation is much simpler here and our aim is to show how one can establish an a priori estimate

$$\Delta_\omega \varphi_s \leq C_2, \tag{14.2.2}$$

which is uniform in s, thus completing Step 2.

We are going to use Lemma 14.5: we let $-B < 0$ denote a lower bound on the bisectional curvature of ω. We also fix $C' > 0$ such that

$$\mathrm{Ric}(\omega) \leq C'\omega \text{ and } A = C + C' + B + 1.$$

We then have the following estimate:

Lemma 14.6. *Set $\omega_s := \omega + dd^c\varphi_s$. Then*

$$\Delta_{\omega_s}(\log \mathrm{tr}_\omega(\omega_s) - (A-1)\varphi_s) \geq \mathrm{tr}_{\omega_s}(\omega) - An.$$

Proof. It follows from Lemma 14.5 that

$$\Delta_{\omega_s} \log \mathrm{tr}_\omega(\omega_s) \geq -\frac{\mathrm{tr}_\omega(\mathrm{Ric}(\omega_s))}{\mathrm{tr}_\omega(\omega_s)} - B\ \mathrm{tr}_{\omega_s}(\omega).$$

Now

$$-\mathrm{Ric}(\omega_s) = -\mathrm{Ric}(\omega) + dd^c g_s \geq -(C' + C)\omega,$$

as follows from (14.2.1). This yields $-\mathrm{tr}_\omega(\mathrm{Ric}(\omega_s)) \geq -(C'+C)n$. We can use the trivial inequality $\mathrm{tr}_\omega(\omega_s)\,\mathrm{tr}_{\omega_s}(\omega) \geq n$ to infer

$$-\frac{\mathrm{tr}_\omega(\mathrm{Ric}(\omega_s))}{\mathrm{tr}_\omega(\omega_s)} \geq -(C + C')\ \mathrm{tr}_{\omega_s}(\omega).$$

The conclusion follows by observing that $\mathrm{tr}_{\omega_s}(\omega) = n - \Delta_{\omega_s}(\varphi_s)$. □

We can now prove (14.2.2) as follows. The function

$$H = \log \mathrm{tr}_\omega(\omega_s) - (A-1)\varphi_s$$

attains its maximum at some point $x_s \in X$. At this point $\Delta_{\omega_s} H \leq 0$ hence

$$\mathrm{tr}_{\omega_s}(\omega)(x_s) \leq An.$$

It follows then from Lemma 14.4 that

$$\mathrm{tr}_\omega(\omega_s)(x_s) \leq n e^{\sup_X h} [\mathrm{tr}_{\omega_s}(\omega)(x_s)]^{n-1} \leq A_1.$$

Now $H(x) \leq H(x_s) \leq \log A_1 - (A-1)\inf_X \varphi_s$ yields

$$\mathrm{tr}_\omega(\omega_s)(x) \leq A_1 \exp\left((A-1)\mathrm{Osc}_X \varphi_s\right) \leq C_2,$$

where the last inequality uses the uniform bound on φ_s that was previously established. Since $\mathrm{tr}_\omega(\omega_s) = n + \Delta_\omega(\varphi_s)$, this yields (14.2.2).

14.2.3 The refined estimate We now prove Theorem 14.3. We set $\omega_\varphi := \omega + dd^c\varphi > 0$, so that

$$-\mathrm{Ric}(\omega_\varphi) = -\mathrm{Ric}(\omega) + dd^c(\psi_+ - \psi_-).$$

As above, the key is to establish the following:

Lemma 14.7. *Set $A = B + C + C' + 1 > 0$ where $C' > 0$ is chosen so that $\mathrm{Ric}(\omega) \leq C'\omega$. Then*

$$\Delta_{\omega_\varphi}\left(\log \mathrm{tr}_\omega(\omega_\varphi) + \psi_- - (A-1)\varphi\right) \geq \mathrm{tr}_{\omega_\varphi}(\omega) - An.$$

Proof. Observe that

$$-\mathrm{Ric}(\omega_\varphi) \geq -\mathrm{Ric}(\omega) - C\omega - dd^c\psi_- \geq -C'\omega - dd^c\psi_-,$$

thus

$$-\frac{\mathrm{tr}_\omega(\mathrm{Ric}(\omega_\varphi))}{\mathrm{tr}_\omega(\omega_\varphi)} \geq -C'\,\mathrm{tr}_{\omega_\varphi}(\omega) - \frac{\Delta_\omega(\psi_-)}{\mathrm{tr}_\omega(\omega_\varphi)},$$

using the trivial inequality $\mathrm{tr}_\omega(\omega_\varphi)\,\mathrm{tr}_{\omega_\varphi}(\omega) \geq n$. It follows therefore from Lemma 14.5 that

$$\Delta_{\omega_\varphi} \log \mathrm{tr}_\omega(\omega_\varphi) \geq -(B + C')\,\mathrm{tr}_{\omega_\varphi}(\omega) - \frac{\Delta_\omega(\psi_-)}{\mathrm{tr}_\omega(\omega_\varphi)}.$$

Observe now that $0 \leq C\omega + dd^c\psi_- \leq \mathrm{tr}_{\omega_\varphi}(C\omega + dd^c\psi_-)\omega_\varphi$, thus

$$0 \leq nC + \Delta_\omega(\psi_-) \leq \mathrm{tr}_{\omega_\varphi}(C\omega + dd^c\psi_-)\,\mathrm{tr}_\omega(\omega_\varphi),$$

which yields

$$-\frac{\Delta_\omega(\psi_-)}{\mathrm{tr}_\omega(\omega_\varphi)} \geq -C\,\mathrm{tr}_{\omega_\varphi}(\omega) - \Delta_{\omega_\varphi}(\psi_-).$$

The conclusion follows since $\mathrm{tr}_{\omega_\varphi}(\omega) = n - \Delta_{\omega_\varphi}(\varphi)$. □

We now apply the maximum principle to the function

$$H := \log \mathrm{tr}_\omega(\omega_\varphi) + \psi_- - (A-1)\varphi.$$

If x_0 is a point at which H realizes its maximum, then $\Delta_{\omega_\varphi} H(x_0) \leq 0$, hence

$$\mathrm{tr}_{\omega_\varphi}(\omega)(x_0) \leq An.$$

It follows from Lemma 14.4 that at each point in X,

$$\log \mathrm{tr}_\omega(\omega_\varphi) + \psi_- \leq (n-1)\log \mathrm{tr}_\omega(\omega_\varphi) + C + \log n;$$

in particular,

$$H(x_0) \leq -(A-1)\inf_X \varphi + (n-1)\log(An) + \log n.$$

Therefore, for all $x \in X$,

$$\log \mathrm{tr}_\omega(\omega_\varphi) \leq -\psi_-(x) + (A-1)\mathrm{Osc}_X(\varphi) + (n-1)\log(An) + \log n.$$

The proof is now complete if we use the uniform a priori estimate from Chapter 12, which gives a bound on $\mathrm{Osc}_X(\varphi)$ in terms of the L^p-norm of the density $e^{\psi_+ - \psi_-}$. The latter is equivalently controlled by the L^p-norm of $e^{-\psi_-}$, since we assume $\psi_+ \leq C$.

Remark 14.8. *We have implicitly assumed above that φ, ψ^-, ψ^+ are smooth. Theorem 14.3 can be extended to non-smooth functions by approximation.*

14.3 Higher-order estimates

In this section we follow Blocki's presentation [Blo12] of Siu's approach to higher-order estimates.

14.3.1 Complex Evans–Krylov method Aubin [Aub78] and Yau [Yau78] have established a priori estimates for third-order derivatives of φ. Since then a general theory of non-linear elliptic equations of second order has been developed which allows one to obtain interior $C^{2,\alpha}$-estimates, once an estimate for second-order derivatives is known. This was done by Evans and independently by Krylov; the method was subsequently simplified by Trudinger (see [GT83] for references).

The complex Monge–Ampère operator is uniformly elliptic in the real sense (see Lemma 14.11). We cannot however apply the estimates from the real theory directly: the latter allow to control the mixed complex derivatives $\varphi_{i\bar{j}}$, but not the full real Hessian $D^2\varphi$, which is required in the real estimate.

Siu nevertheless observed in [Siu87] that one can follow (and adapt) step by step the real method in the complex case. The goal of this section is thus to establish the following a priori estimate:

Theorem 14.9. *Let u be a $\mathcal{C}^4$-smooth plurisubharmonic function in an open set $\Omega \subset \mathbb{C}^n$ such that* $\det(u_{i\bar{j}}) = e^g > 0$. *Then for any $\Omega' \Subset \Omega$ there exists*

$$0 < \alpha = \alpha\left(n, \|u\|_{\mathcal{C}^{0,1}(\Omega)}, sup_\Omega \Delta u, \|g\|_{\mathcal{C}^{0,1}(\Omega)}, \inf_\Omega g\right) < 1,$$

and $C > 0$, depending moreover on a lower bound for $\operatorname{dist}(\Omega', \partial\Omega)$, *such that*

$$\|u\|_{\mathcal{C}^{2,\alpha}(\Omega')} \leq C.$$

This results, together with Steps 1 and 2, yields Step 3 of the strategy exposed in Section 14.1.2.

We note that an alternative (and optimal) method has been proposed recently in [Wan12, TWWY15] based on Caffarelli's theory of interior estimates for fully non-linear elliptic equations [CC95].

14.3.1.1 Preliminaries Let $\mathcal{H}$ denote the set of all $n \times n$ Hermitian matrices and set

$$\mathcal{H}_+ := \{A \in \mathcal{H} |\ A > 0\}.$$

A crucial property for the Evans–Trudinger method to work is that

$$\mathcal{H}_+ \ni A \longmapsto (\det A)^{1/n} \in \mathbb{R}_+ \text{ is concave.} \tag{14.3.1}$$

This can be deduced from the observation [Gav77] that the complex Monge–Ampère equation can be treated as a Bellman equation: for all $A \in \mathcal{H}_+$,

$$(\det A)^{1/n} = \frac{1}{n} \inf\{\operatorname{tr}(AB) |\ B \in \mathcal{H}_+,\ \det B = 1\}. \tag{14.3.2}$$

We refer the reader to Lemma 5.8 for a proof.

We shall need the following result from linear algebra:

Lemma 14.10. *Let $0 < \lambda < \Lambda < \infty$ and let $S(\lambda, \Lambda)$ denote the set of Hermitian matrices whose eigenvalues lie in the interval $[\lambda, \Lambda]$.*

One can find unit vectors $\zeta_1, \ldots, \zeta_N \in \mathbb{C}^n$ and $0 < \lambda_* < \Lambda_* < \infty$, depending only on n, λ, and Λ, such that every $A \in S(\lambda, \Lambda)$ can be written as

$$A = \sum_{k=1}^{N} \beta_k \zeta_k \otimes \overline{\zeta}_k, \quad \textit{i.e.,} \quad a_{i\bar{j}} = \sum_k \beta_k \zeta_{ki} \overline{\zeta}_{kj},$$

where $\beta_k \in [\lambda_*, \Lambda_*]$, $k = 1, \ldots, N$. The vectors $\zeta_1, \ldots, \zeta_N$ can be chosen so that they contain a given orthonormal basis of $\mathbb{C}^n$.

Proof. [Siu87, p. 103] The space $\mathcal{H}$ of all Hermitian matrices is of real dimension n^2. Every $A \in \mathcal{H}$ can be written as

$$A = \sum_{k=1}^{n} \lambda_k w_k \otimes \overline{w}_k,$$

where $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$ are the eigenvalues of A and $w_1, \ldots, w_n \in \mathbb{C}^n$ the corresponding unit eigenvectors. It follows that there exist unit vectors $\zeta_1, \ldots, \zeta_{n^2} \in \mathbb{C}^n$ such that the matrices $\zeta_k \otimes \overline{\zeta}_k$, $k = 1, \ldots, n^2$, span $\mathcal{H}$ over $\mathbb{R}$. For such sets of vectors we consider the sets of matrices

$$U(\zeta_1, \ldots, \zeta_{n^2}) = \left\{ \sum_k \beta_k \zeta_k \otimes \overline{\zeta}_k \mid 0 < \beta_k < 2\Lambda \right\}.$$

They form an open covering of $S(\lambda/2, \Lambda)$, a compact subset of $\mathcal{H}$. Choosing a finite subcovering we get unit vectors $\zeta_1, \ldots, \zeta_N \in \mathbb{C}^n$ such that

$$S(\lambda/2, \Lambda) \subset \left\{ \sum_{k=1}^{N} \beta_k \zeta_k \otimes \overline{\zeta}_k \mid 0 < \beta_k < 2\Lambda \right\}.$$

For $A \in S(\lambda, \Lambda)$ we have

$$A - \frac{\lambda}{2N} \sum_{k=1}^{N} \zeta_k \otimes \overline{\zeta}_k \in S(\lambda/2, \Lambda)$$

and the lemma follows. Observe that we can take arbitrary $\lambda_* < \lambda/N$ and $\Lambda_* > \Lambda$. □

14.3.1.2 The a priori $\mathcal{C}^{2,\alpha}$ estimate We are now ready for the proof of Theorem 14.9. Fix $\zeta \in \mathbb{C}^n$, $|\zeta| = 1$. Differentiating

$$\log \det(u_{i\bar{j}}) = \log f = g,$$

we let the reader verify in Exercise 14.13 that

$$u^{i\bar{j}} u_{\zeta\bar{\zeta} i\bar{j}} = (\log f)_{\zeta\bar{\zeta}} + u^{i\bar{l}} u^{k\bar{j}} u_{\zeta i\bar{j}} u_{\bar{\zeta} k\bar{l}} \geq (\log f)_{\zeta\bar{\zeta}}. \tag{14.3.3}$$

We use here standard Einstein summation conventions and the notation

$$\left(u^{i\bar{j}}\right) = {}^t\left(u_{i\bar{j}}\right)^{-1}.$$

The inequality $u^{i\bar{l}} u^{k\bar{j}} u_{\zeta i\bar{j}} u_{\bar{\zeta} k\bar{l}} \geq 0$ corresponds to the concavity of the mapping

$$\mathcal{H}_+ \ni A \longmapsto \log \det A \in \mathbb{R}$$

(see Exercise 14.7). The latter follows from the concavity of the mapping $A \mapsto (\det A)^{1/n}$.

It is convenient to write (14.3.3) in divergence form so as to use the tools developed in [GT83]. Set $a^{i\bar{j}} := f u^{i\bar{j}}$. Then for any fixed i

$$(a^{i\bar{j}})_{\bar{j}} = f(u^{i\bar{j}}u^{k\bar{l}} - u^{i\bar{l}}u^{k\bar{j}})u_{k\bar{l}\bar{j}} = 0$$

and, by (14.3.3),

$$(a^{i\bar{j}}u_{\xi\bar{\xi}i})_{\bar{j}} \geq f_{\xi\bar{\xi}} - \frac{|f_\xi|^2}{f} \geq f_{\xi\bar{\xi}} - C_1.$$

We observe that, since $f \in C^{0,1}$,

$$f_{\xi\bar{\xi}} = \sum_{i,j} c_{i,j}\frac{\partial^2 f}{\partial x_i \partial x_j} = \sum_{i,j} c_{i,j}\frac{\partial f^j}{\partial x_i},$$

where $(x_1, \ldots, x_{2n})$ denote real coordinates in $\mathbb{R}^{2n} \simeq \mathbb{C}^n$.

These notations are chosen so as to fit in with the ones used in [GT83, Chapter 8]. In particular, $u_{\xi\bar{\xi}}$ is thus a subsolution of the equation

$$Lv = g + \sum_j D_j f^j,$$

where $g \equiv -C_1$, $D_j = \sum_i c_{i,j}\frac{\partial}{\partial x_i}$, f^j is bounded since $f \in C^{0,1}$, and

$$Lv = \sum_{i,j}\left(a^{i\bar{j}}v_i\right)_{\bar{j}}.$$

The assumptions on u and Lemma 14.11 ensure that the operator L is uniformly elliptic (in the real sense). The weak Harnack inequality [GT83, Theorem 8.18] therefore yields

$$r^{-2n}\int_{B_r}\left(\sup_{B_{4r}} u_{\xi\bar{\xi}} - u_{\xi\bar{\xi}}\right) \leq C_3\left(\sup_{B_{4r}} u_{\xi\bar{\xi}} - \sup_{B_r} u_{\xi\bar{\xi}} + r\right), \tag{14.3.4}$$

where $B_{4r} = B(z_0, 4r) \subset \Omega$ and $z_0 \in \Omega'$.

Set $U := (u_{i,\bar{j}})$. For $x, y \in B_{4r} \subset \Omega$, we obtain

$$a^{i\bar{j}}(y)u_{i\bar{j}}(x) = f(y)u^{i\bar{j}}(y)u_{i\bar{j}}(x) = f(y)\operatorname{tr}\left(U(y)^{-1}U(x)\right).$$

In particular,

$$a^{i\bar{j}}(y)u_{i\bar{j}}(y) = nf(y).$$

On the other hand, $\det\left(f(y)^{1/n}U(y)^{-1}\right) = 1$, hence (14.3.2) yields

$$a^{i\bar{j}}(y)u_{i\bar{j}}(x) = f(y)^{1-1/n}\operatorname{tr}\left(f(y)^{1/n}U(y)^{-1}U(x)\right) \geq nf(y)^{1-1/n}f(x)^{1/n}.$$

Therefore, for $x, y \in B_{4r}$,

$$\begin{aligned} a^{i\bar{j}}(y)(u_{i\bar{j}}(y) - u_{i\bar{j}}(x)) &\le nf(y) - nf(y)^{1-1/n} f(x)^{1/n} \\ &= nf(y)^{1-1/n}(f(y)^{1/n} - f(x)^{1/n}) \\ &\le C_4|x-y|, \end{aligned} \tag{14.3.5}$$

where $C_4 > 0$ depends on $\sup f$ and the Lipschitz constant of $f^{1/n}$.

We now wish to combine (14.3.4) with (14.3.5). The eigenvalues of $(a^{i\bar{j}}(y))$ lie in $[\lambda, \Lambda]$, where $\lambda, \Lambda > 0$ are under control. By Lemma 14.10, we can find unit vectors $\zeta_1, \dots, \zeta_N \in \mathbb{C}^n$ such that for $x, y \in \Omega$

$$a^{i\bar{j}}(y)\big(u_{i\bar{j}}(y) - u_{i\bar{j}}(x)\big) = \sum_{k=1}^{N} \beta_k(y)\big(u_{\zeta_k\bar{\zeta}_k}(y) - u_{\zeta_k\bar{\zeta}_k}(x)\big),$$

where $\beta_k(y) \in [\lambda_*, \Lambda_*]$ and $\lambda_*, \Lambda_* > 0$ are under control. Set

$$M_{k,r} := \sup_{B_r} u_{\zeta_k\bar{\zeta}_k}, \quad m_{k,r} := \inf_{B_r} u_{\zeta_k\bar{\zeta}_k},$$

and

$$\eta(r) := \sum_{k=1}^{N} (M_{k,r} - m_{k,r}).$$

We need to show that $\eta(r) \le Cr^{\alpha}$. Since $\zeta_1, \dots, \zeta_N$ can be chosen so that they contain the coordinate vectors, it will then follow that $\|\Delta u\|_{C^{\alpha}(\Omega')}$ is under control and by the Schauder estimates for the Poisson equation [GT83, Theorem 4.6] that $\|D^2 u\|_{C^{\alpha}(\Omega')}$ is also under control. To establish the Hölder condition $\eta(r) \le Cr^{\alpha}$ it suffices to show that

$$\eta(r) \le \delta\eta(4r) + r, \quad 0 < r < r_0, \tag{14.3.6}$$

where $\delta \in (0,1)$ and $r_0 > 0$ are under control (see Exercise 14.14).

It follows from (14.3.5) that

$$\sum_{k=1}^{N} \beta_k(y)\big(u_{\zeta_k\bar{\zeta}_k}(y) - u_{\zeta_k\bar{\zeta}_k}(x)\big) \le C_4|x-y|. \tag{14.3.7}$$

Summing (14.3.4) over $l \ne k$, where k is fixed, we obtain

$$r^{-2n} \int_{B_r} \sum_{l \ne k} \big(M_{l,4r} - u_{\zeta_l\bar{\zeta}_l}\big) \le C_3(\eta(4r) - \eta(r) + r). \tag{14.3.8}$$

By (14.3.7), for $x \in B_{4r}$, $y \in B_r$ we have

$$\begin{aligned}\beta_k(y)\big(u_{\zeta_k\bar\zeta_k}(y) &- u_{\zeta_k\bar\zeta_k}(x)\big)\\ &\le C_4|x-y| + \sum_{l\neq k}\beta_l(y)\big(u_{\zeta_l\bar\zeta_l}(x) - u_{\zeta_l\bar\zeta_l}(y)\big)\\ &\le C_5 r + \Lambda^* \sum_{l\neq k}(M_{l,4r} - u_{\zeta_l\bar\zeta_l}(y)).\end{aligned}$$

Thus for all $y \in B_r$,

$$u_{\zeta_k\bar\zeta_k}(y) - m_{k,4r} \le \frac{1}{\lambda^*}\left(C_5 r + \Lambda^* \sum_{l\neq k}\big(M_{l,4r} - u_{\zeta_l\bar\zeta_l}(y)\big)\right)$$

and (14.3.8) gives

$$r^{-2n}\int_{B_r}\big(u_{\zeta_k\bar\zeta_k} - m_{k,4r}\big) \le C_6(\eta(4r) - \eta(r) + r).$$

This coupled with (14.3.4) implies $\eta(r) \le C_7(\eta(4r) - \eta(r) + r)$. The desired inequality (14.3.6) follows. □

14.3.2 Schauder estimates

The complex Monge–Ampère operator

$$F(D^2u) = \det(u_{i\bar j})$$

is elliptic if the $2n\times 2n$ real symmetric matrix $A := (\partial F/\partial u_{pq})$ is positive (we denote here by u_{pq} the elements of the real Hessian D^2u). The matrix A is determined by

$$\frac{d}{dt}F(D^2u + tB)|_{t=0} = \operatorname{tr}(A^tB).$$

The reader will find in [Blo99] a general discussion on how to infer that a complex operator induces a uniformly elliptic operator in the real sense. It follows here from the following observation, where we let $\lambda_{\min}A$ and $\lambda_{\max}A$ denote respectively the minimal and the maximal eigenvalues of A:

Lemma 14.11. *One has*

$$\lambda_{\min}(\partial F/\partial u_{pq}) = \frac{\det(u_{i\bar j})}{4\lambda_{\max}(u_{i\bar j})},\quad \lambda_{\max}(\partial F/\partial u_{pq}) = \frac{\det(u_{i\bar j})}{4\lambda_{\min}(u_{i\bar j})}.$$

We leave the proof of this lemma as an Exercise 14.8.

Thus the operator F is elliptic (in the real sense) for smooth strongly psh functions. When $\|u\|_{2,\alpha} \leq C$ (Δu is thus under control and so are the complex mixed derivatives $u_{i\bar{j}}$) it is even uniformly elliptic, i.e.,

$$|\zeta|^2/C \leq \sum_{p,q=1}^{2n} \partial F/\partial u_{pq}\zeta_p\zeta_q \leq C|\zeta|^2, \quad \zeta \in \mathbb{C}^n = \mathbb{R}^{2n}$$

for some uniform constant C.

For such a uniformly elliptic F, we can apply the standard elliptic theory (see [GT83, Lemma 17.16] for details) to the equation

$$F(D^2u) = f.$$

For a fixed unit vector ζ and small $h > 0$ we consider

$$u^h(x) = \frac{u(x+h\zeta) - u(x)}{h}$$

and

$$a_h^{pq} = \int_0^1 \frac{\partial F}{\partial u_{pq}}\big(tD^2u(x+h\zeta) + (1-t)D^2u(x)\big)dt.$$

Thus

$$a_h^{pq}(x)u_{pq}^h(x) = \frac{1}{h}\int_0^1 \frac{d}{dt}F\big(tD^2u(x+h\zeta) + (1-t)D^2u(x)\big)dt = f^h(x).$$

Schauder theory for linear elliptic equations with variable coefficients yields the a priori estimates

$$u \in C^{2,\alpha} \implies a_h^{pq} \in C^{0,\alpha} \overset{\text{Schauder}}{\implies} u^h \in C^{2,\alpha}.$$

Since these estimates are uniform in h we infer $u \in C^{3,\alpha}$. Applying the Schauder estimtes again yields

$$u \in C^{3,\alpha} \implies a_h^{pq} \in C^{1,\alpha} \overset{\text{Schauder}}{\implies} u^h \in C^{3,\alpha} \implies u \in C^{4,\alpha} \implies \dots$$

For the complex Monge–Ampère equation (CY),

$$(\omega + dd^c\varphi)^n = f\omega^n,$$

for $k \geq 1$ we thus get the following:

Proposition 14.12. *Assume $(\omega + dd^c\varphi)^n = f\omega^n$, where $f > 0$ and φ is ω-plurisubharmonic and $C^{2,\alpha}$-smooth. Then*

$$\varphi \in C^{2,\alpha},\ f \in C^{k,\alpha} \implies \varphi \in C^{k+2,\alpha}$$

and

$$\|\varphi\|_{k+2,\alpha} \leq C,$$

where $C > 0$ depends only on upper bounds for $\|\varphi\|_{2,\alpha}$, $\|f\|_{k,\alpha}$.

At this stage Step 4 is complete and we have finished the proof of Theorem 14.1 when $F \equiv 0$: since $f = e^h$ is $\mathcal{C}^\infty$-smooth, we can apply the previous proposition with an arbitrary $k \in \mathbb{N}$.

14.3.3 Proof of Theorem 14.1 It follows from Chapter 12 that the bounded solution $\varphi \in PSH(X,\omega)$ to

$$(\omega + dd^c\varphi)^n = e^{F(\varphi)+h}\omega^n$$

is actually continuous. We let $\varphi_j \in PSH(X,\omega) \cap \mathcal{C}^\infty(X)$ denote a sequence of smooth ω-psh functions uniformly converging to φ.

We let ψ_j denote the unique smooth ω-psh functions such that $\int_X \psi_j\, \omega^n = 0$ and

$$(\omega + dd^c\psi_j)^n = e^{F(\varphi_j)+h+c_j}\omega^n,$$

where $c_j \in \mathbb{R}$ are normalizing constants such that the measures on the right-hand side have constant mass V_ω.

It follows from the stability results established in Chapter 12 that the sequence (ψ_j) converges uniformly to

$$\psi := \varphi - \int_X \varphi\, \omega^n.$$

Since the φ_j's are uniformly bounded, we can decompose $F = F_+ - F_-$ as a difference of two convex increasing functions F_+, F_- near the range K of the φ_j's. Let $C > 0$ be such that $F'_\pm \leq C$ on K. Note that

$$\begin{aligned} dd^c F_\pm \circ \varphi_j &= F'_\pm \circ \varphi_j\, dd^c\varphi_j + F''_\pm \circ \varphi_j d\varphi_j \wedge d^c\varphi_j \\ &\geq F'_\pm \circ \varphi_j\, dd^c\varphi_j \geq -C\omega. \end{aligned}$$

Enlarging $C > 0$ if necessary, we can further assume that

$$\sup_X F_+ \circ \varphi_j \leq C \quad \text{and} \quad \|e^{-F_- \circ \varphi_j}\|_{L^\infty(X)} \leq C.$$

We are thus in position to apply Theorem 14.3 with $\psi_\pm = F_\pm \circ \varphi_j$. It yields a uniform bound

$$\Delta_\omega \psi_j \leq C_2,$$

since $\psi_- := F_- \circ \varphi_j$ is uniformly bounded below.

Theorem 14.9 further yields a uniform bound

$$\|\psi_j\|_{\mathcal{C}^{2,\alpha}} \leq C_3,$$

for some $\alpha > 0$. A boostrapping argument, using inductively Proposition 14.12, yields uniform bounds

$$\|\psi_j\|_{\mathcal{C}^{k,\alpha}} \leq C_{k+1}$$

for each $k \in \mathbb{N}$. This shows that the sequence (ψ_j) is relatively compact in $\mathcal{C}^\infty$, hence its limit ψ is $\mathcal{C}^\infty$-smooth and so is $\varphi = \psi + \int_X \varphi\, \omega^n$. □

14.3.4 More general equations We have followed Yau's original solution to the Calabi conjecture and some of its recent generalizations, in which the $\mathcal{C}^2$-estimates are shown to follow directly from the $\mathcal{C}^0$-estimate. The $\mathcal{C}^1$-estimates then follow from the $\mathcal{C}^0$ and $\mathcal{C}^2$-estimates by general elliptic theory.

For the more general complex Monge–Ampère equation

$$(\omega + dd^c\varphi)^n = e^{F(\varphi,x)}\omega^n$$

where the dependence of F in (φ, x) is smooth and more general than what we have been considering so far, the $\mathcal{C}^1$-estimates cannot be bypassed. It has been established by Blocki [Blo09] and extended by Phong and Sturm in [PS10b, PS12]:

Theorem 14.13. *Let φ be a $\mathcal{C}^4$-smooth solution to the above equation. Then there exists $A > 0$ such that*

$$|\nabla\varphi(z)| \leq A\exp(A\varphi(z)),$$

where A depends only on upper bounds for $\inf_X \varphi$, $\sup_{X\times[\inf\varphi,+\infty)} F$, $\sup_{X\times[\inf\varphi,+\infty)} \big(|\nabla F| + |\partial_\varphi F|\big)$ and a lower bound for the holomorphic bisectional curvature of (X,ω).

We refer the reader to the original articles for a proof which uses a key differential inequality discovered by Blocki [Blo09]. Once this estimate is obtained, the higher-order estimates for the above equation follow the same lines as above.

14.4 The case of semi-positive forms

Let (X,ω) be a compact Kähler manifold of dimension n and D an arbitrary divisor on X. Let $E = \sum_{j=1}^{M} a_j E_j$ be an effective divisor with simple normal crossings on X.

Let θ be a smooth semipositive form on X such that $\int_X \theta^n > 0$ and $\{\theta\} - c_1(E)$ is Kähler. We consider the following degenerate complex Monge–Ampère equation:

$$(\theta + dd^c\varphi)^n = e^{\lambda\varphi} f\omega^n, \tag{14.4.1}$$

where $\lambda \geq 0$, $0 \leq f \in L^1(X,\omega^n)$ satisfies $\int_X f\omega^n = \int_X \theta^n$, and

$$f = e^{\psi^+ - \psi^-},\ \ dd^c\psi^\pm \geq -C\omega,\ \sup_X \psi^+ \leq C,\ \ \psi^- \in L^\infty_{\mathrm{loc}}(X\setminus D), \tag{14.4.2}$$

for some constant $C > 0$.

Our goal in this section is to prove the following regularity result:

Theorem 14.14. *Assume that $0 < f \in \mathcal{C}^\infty(X\setminus D)$ satisfies (14.4.2). Then any solution to (14.4.1) is smooth in $X\setminus(D\cup E)$.*

This result was obtained in [EGZ09, BEGZ10, BBEGZ11] when the density is in L^p, $p > 1$. The general version given here is due to Di Nezza and Lu [DiNL14]; it allows to deal with situations where the solution φ is not necessarily bounded near D. This covers the case of Poincaré type singularities, as dealt (with a very different method) by Auvray in [Auv14].

14.4.1 Uniform estimate For each $j = 1, \dots, M$ let K_j be the holomorphic line bundle defined by E_j. Let σ_j be a holomorphic section of K_j that vanishes on E_j. We fix a Hermitian metric h_j on K_j such that $|\sigma_j| \leq 1/e$. Since $\{\theta\} - c_1(E)$ is Kähler, we can assume that

$$\theta + dd^c\phi = \omega_0 + [E],$$

where ω_0 is a Kähler form on X and

$$\phi := \sum_{j=1}^{M} a_j \log |\sigma_j|.$$

By rescaling ω we can also assume that $\omega_0 \geq \omega$.

The following lower bound is an adaptation of Theorem 12.16:

Theorem 14.15. *Let φ be the unique normalized solution to equation (14.4.1) and fix $a > 0$ such that $a\psi^- \in PSH(X, \omega_0/2)$. There exists $A > 0$ depending on C and $\int_X e^{-2\varphi/a}\omega^n$ such that*

$$\varphi \geq a\psi^- + \phi - A.$$

Proof. We observe first that it suffices to treat the case $\lambda = 0$. Indeed, assume that $\lambda > 0$ (the solution φ_λ is unique in this case) and let φ_0 denote the solution to the equation $(14.4.1)_0$ when $\lambda = 0$ that is normalized by $\sup_X \varphi_0 = 0$. The function φ_0 is a subsolution to $(14.4.1)_\lambda$, since

$$(\theta + dd^c\varphi_0)^n = f\omega^n \geq e^{\lambda\varphi_0} f\omega^n.$$

Since φ_λ is the envelope of subsolutions (see Chapter 13), we infer $\varphi_\lambda \geq \varphi_0$, hence a lower bound on φ_0 produces a lower bound on φ_λ.

From now on $\lambda = 0$ and φ denotes φ_0. Fix $a > 0$ small so that $a\psi^- \in PSH(X, \omega_0/2)$. Then

$$\psi := a\psi^- + \frac{\phi}{2} \in PSH(X, \theta/2)$$

with

$$\frac{\theta}{2} + dd^c\psi = \frac{\omega_0}{2} + dd^c(a\psi^-) + \frac{[E]}{2} \geq 0.$$

Recall that

$$\mathrm{Vol}_\omega \le C_1 \exp\left(\frac{-C_2}{\left[\mathrm{Cap}_{\theta/2}\right]^{1/n}}\right),$$

for some universal constants $C_1, C_2 > 0$. Arguing as in the proof of Lemma 12.15 we get

$$\mathrm{Cap}_{\theta/2} \le \mathrm{Cap}_\psi,$$

where Cap_ψ is the generalized capacity defined by the form θ and ψ:

$$\mathrm{Cap}_\psi(E) := \sup\left\{\int_E (\theta + dd^c u)^n \,\middle|\, u \in PSH(X,\theta),\ \psi - 1 \le u \le \psi\right\}.$$

We can apply the same arguments as in the proof of Theorem 12.16 to conclude. □

14.4.2 Laplacian estimate We now prove a C^2 a priori estimate in the semipositive and big case. Following [BEGZ10], we replace θ by the Kähler forms $\theta + t\omega$, $t > 0$, and try and establish a priori estimates that are locally uniform outside $D \cup E$.

Theorem 14.16. *Assume $\lambda \ge 0$ and $f = e^{\psi^+ - \psi^-}$, where ψ^+, ψ^- are smooth on X. Fix $t \in (0,1)$. Let $\varphi \in \mathcal{C}^\infty(X)$ be the unique normalized solution to*

$$(\theta + t\omega + dd^c\varphi)^n = e^{\lambda\varphi + \psi^+ - \psi^-}\omega^n.$$

Fix $C > 0$ such that $dd^c\psi^\pm \ge -C\omega$, $\sup_X \psi^+ \le C$, and the holomorphic bisectional curvature of ω is bounded from below by $-C$.

Then there exists $A = A\left(C, \int_X e^{-2(3C+1)\varphi}\omega^n\right) > 0$ such that

$$\Delta_\omega \varphi \le A e^{-\psi^- - (3C+1)\phi}.$$

Proof. We treat the case $\lambda = 0$ and let the reader deal with the case $\lambda > 0$ in Exercise 14.16.

We set $\omega_\varphi := \theta + t\omega + dd^c\varphi$ and consider

$$H := \log \mathrm{tr}_\omega(\omega_\varphi) + \psi^- - (3C+1)(\varphi - \phi).$$

Since ϕ goes to $-\infty$ on E, H attains its maximum on $X \setminus E$ at some point $x_0 \in X \setminus E$.

Arguing as earlier, we obtain

$$\Delta_{\omega_\varphi} \log \mathrm{tr}_\omega(\omega_\varphi) \ge -3C\ \mathrm{tr}_{\omega_\varphi}(\omega) - \Delta_{\omega_\varphi}\psi^-. \tag{14.4.3}$$

Observe that

$$dd^c(\varphi - \psi) = (\theta + t\omega + dd^c\varphi) - (\omega_0 + t\omega + [E]).$$

Hence $dd^c(\varphi - \psi) \le \omega_\varphi - \omega$ in $X \setminus E$ since $\omega_0 + t\omega \ge \omega$. Thus

$$\Delta_{\omega_\varphi}(\varphi - \phi) \le \mathrm{tr}_{\omega_\varphi}(\omega_\varphi - \omega) = n - \mathrm{tr}_{\omega_\varphi}(\omega). \tag{14.4.4}$$

It therefore follows from (14.4.3) and (14.4.4) that on $X \setminus E$,

$$\Delta_{\omega_\varphi} H \geq \mathrm{tr}_{\omega_\varphi}(\omega) - n(3C+1).$$

The maximum principle now yields

$$0 \geq \Delta_{\omega_\varphi} H(x_0) \geq \mathrm{tr}_{\omega_\varphi}(\omega)(x_0) - n(3C+1).$$

We infer that

$$\mathrm{tr}_\omega(\omega_\varphi)(x_0) \leq n e^{\psi^+ - \psi^-}(x_0) \left(\mathrm{tr}_{\omega_\varphi}(\omega)\right)^{n-1}(x_0) \leq A_1 e^{\psi^+ - \psi^-}(x_0),$$

hence

$$\log \mathrm{tr}_\omega(\omega_\varphi)(x_0) \leq \log A_1 + \psi^+(x_0) - \psi^-(x_0) \leq A_2 - \psi^-(x_0),$$

since $\sup_X \psi^+ \leq C$. Thus

$$H(x) \leq H(x_0) \leq A_2 + \psi^-(x_0) - (3C+1)(\varphi - \phi)(x_0).$$

It follows from Theorem 14.15 that

$$\varphi \geq \frac{1}{3C+1}\psi^- + \phi - A_3,$$

where A_3 depends on C and $\int_X e^{-2(3C+1)\varphi}\omega^n$. Thus

$$\log \mathrm{tr}_\omega(\omega_\varphi) \leq A_4 - \psi^- + (3C+1)(\varphi - \phi).$$

We finally get

$$\mathrm{tr}_\omega(\omega_\varphi) \leq A_5 e^{-\psi^- - (3C+1)\phi}. \qquad \square$$

Proof of Theorem 14.14. Let $\rho_\varepsilon(\psi^\pm)$ be the smoothing of $\psi^\pm$ obtained by Demailly's convolution technique. For each $\varepsilon > 0$ it follows from Yau's theorem that there exists a unique $\varphi_\varepsilon \in \mathcal{C}^\infty(X)$ such that $\sup_X \varphi_\varepsilon = 0$ and

$$(\theta + \varepsilon\omega + dd^c\varphi_\varepsilon)^n = c_\varepsilon e^{\lambda\varphi_\varepsilon + \rho_\varepsilon(\psi^+) - \rho_\varepsilon(\psi^-)}\omega^n,$$

where c_ε is a normalization constant (which converges to 1 as ε converges to zero). We have a uniform control on the right-hand side:

$$c_\varepsilon e^{\lambda\varphi_\varepsilon + \rho_\varepsilon(\psi^+) - \rho_\varepsilon(\psi^-)} \leq e^{C - \psi_\varepsilon^-}.$$

As we have obtained Laplacian estimates that are locally uniform in ε in $X \setminus E \cup D$, we can apply Evans–Krylov and Schauder estimates to conclude. $\square$

14.5 Exercises

Exercise 14.1. *Let (X, ω) be a compact Kähler manifold and set*

$$\Delta_\omega^+(h) := -\Delta_\omega(h) := -n\frac{dd^c h \wedge \omega^{n-1}}{\omega^n},$$

where $n = \dim_{\mathbb{C}} X$. Show that the eigenvalues of Δ_ω^+ are non-negative, that the eigenfunctions associated to the zero eigenvalue are the constants, and that the first non-zero eigenvalue is characterized by

$$\lambda_1 := \inf\left\{\frac{\|\nabla_\omega h\|_{L^2}^2}{\|h\|_{L^2}^2} \,\Big|\, h \in \mathcal{C}^\infty(X, \mathbb{R})\right\},$$

where

$$\|\nabla_\omega h\|_{L^2}^2 = n\int_X dh \wedge d^c h \wedge \omega^{n-1}.$$

Verify that the infimum is attained and show that given $f \in \mathcal{C}^\infty(X, \mathbb{R})$, there exists $h \in \mathcal{C}^\infty(X, \mathbb{R})$ such that $\Delta_\omega h = f$ iff $\int_X f\omega^n = 0$.

Exercise 14.2. *Let $\Omega \subset \mathbb{R}^k$ be an open set. Fix $1 < p \leq +\infty$ and let $W^{1,p}_{\mathrm{loc}}(\Omega)$ denote the Sobolev space of real functions that admit first derivatives in $L^p_{\mathrm{loc}}(\Omega)$.*

(i) *Show that $W^{1,p}_{\mathrm{loc}}(\Omega) \subset \mathcal{C}^{0,1-k/p}(\Omega)$ if $p > k$.*

(ii) *Check that $W^{1,\infty}_{\mathrm{loc}}(\Omega)$ is the set of Lipschitz functions.*

(iii) *Show that functions in $W^{1,p}_{\mathrm{loc}}(\Omega)$ are differentiable a.e. if $p > k$.*

Exercise 14.3. *Fix $1 < p < +\infty$ and $u \in W^{1,p}(X, \mathbb{R})$, the Sobolev space of functions that admit first derivatives in $L^p(X)$.*

(i) *Show that $|u| \in W^{1,p}(X)$.*

(ii) *Show that $u^+ := \max(u, 0) \in W^{1,p}(X)$, with*

$$\nabla u^+ = \mathbf{1}_{\{u>0\}}\nabla u$$

and $\nabla u = 0$ almost everywhere on $\{u = 0\}$.

Exercise 14.4. *We refer the reader to* [GT83] *for the proof of*

$$\Delta u \in L^p_{\mathrm{loc}} \Longrightarrow u \in W^{2,p}_{\mathrm{loc}} \text{ for all } 1 \leq p < +\infty,$$

where

$$W^{2,p}_{\mathrm{loc}} = \{f \mid D^\alpha f \in L^2_{\mathrm{loc}} \text{ for all } |\alpha| \leq 2\}.$$

By using these implications and the Sobolev embedding theorem, show that

$$\Delta u \in L^\infty_{\mathrm{loc}} \Longrightarrow u \in \mathcal{C}^{1,\gamma} \text{ for all } 0 < \gamma < 1.$$

Explain why this fails for $p = +\infty$ (resp. $\gamma = 1$).

Exercise 14.5. *Check that the function*

$$u(x,y) := |xy| \log[|x| + |y|]$$

has locally bounded Laplacian in $\mathbb{R}^2$*, but does not belong to* $\mathcal{C}^{1,1}$.

Exercise 14.6. *Let* (X,ω) *be a compact Kähler manifold of complex dimension* $n \geq 2$*. Let* α, β *be smooth* $(1,1)$*-forms. Show that*

$$n(n-1)\alpha \wedge \beta \wedge \omega^{n-2} = [\mathrm{tr}_\omega(\alpha)\,\mathrm{tr}_\omega(\beta) - \langle \alpha, \beta \rangle_\omega]\,\omega^n,$$

where

$$\langle \alpha, \beta \rangle_\omega := \sum g^{jk} g^{pq} \alpha_{jq} \beta_{pk},$$

with obvious notations.

Exercise 14.7. *Let* $H \in \mathcal{H}$ *be a small Hermitian matrix. Show that*

$$\log\det(\mathrm{Id} + H) = \mathrm{tr}(H) - \frac{1}{2}\,\mathrm{tr}(H\,{}^t\overline{H}) + o(H^2)$$

and conclude that $A \in \mathcal{H}_+ \mapsto \log\det A \in \mathbb{R}$ *is concave.*

Exercise 14.8. *For* $F(D^2u) = \det(u_{i\bar{j}})$*, show that*

$$\lambda_{\min}(\partial F/\partial u_{pq}) = \frac{\det(u_{i\bar{j}})}{4\lambda_{\max}(u_{i\bar{j}})}, \quad \lambda_{\max}(\partial F/\partial u_{pq}) = \frac{\det(u_{i\bar{j}})}{4\lambda_{\min}(u_{i\bar{j}})},$$

where $\lambda_{\min}(A)$ *and* $\lambda_{\max}(A)$ *denote respectively the minimal and the maximal eigenvalues of* A (*see* [Blo99] *for more details*).

Exercise 14.9. *By using the solution to* (CY) *and a stability property, show that the a priori estimate in Theorem* 14.3 *continues to hold when* φ *is an* ω*-psh function which is merely continuous.*

Exercise 14.10. *Set* $\psi(z) := (\max\{0, |z|^2 - 1/2\})^2$*. Check that*

$$\phi(z,w) = (|z|^2 - 1/2)^2 = (|w|^2 - 1/2)^2$$

is real-analytic on the unit sphere $\partial\mathbb{B} \subset \mathbb{C}^2$*. Show that*

$$u(z,w) := \max\{\psi(z), \psi(w)\}$$

is plurisubharmonic in $\mathbb{B}$*,* $\mathcal{C}^{1,1}$*-smooth, but not* $\mathcal{C}^2$*-smooth, and verifies*

$$(d\,d^c u)^2 \equiv 0 \text{ in } \mathbb{B} \quad \text{with} \quad u|_{\partial\mathbb{B}} \equiv \phi$$

(*see* [GS80] *for more details*).

Exercise 14.11. *Let $u \in PSH(\mathbb{B})$ be the unique continuous psh function in the unit ball $\mathbb{B} \subset \mathbb{C}^n$ such that $u_{|\partial\mathbb{B}} \equiv 0$ and*

$$(dd^c u)^n = A\chi\left(\frac{|z_n|}{|z'|^\alpha}\right)|z'|^\beta,$$

where $z = (z', z_n) \in \mathbb{B}$, $\alpha > 2$, and $\beta = 2(n-1)(\alpha-1) - 1 > 2$, $A > 0$ and

$$\chi(t) = \begin{cases} \exp(-[1-t^2]^{-1}), & \text{if } |t| < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Show that one can adjust the value of $\alpha > 2$ such that $u \notin \mathcal{C}^{1,1}$ (see [Plis05] for more details).

Exercise 14.12. *Let (X, ω) be a compact Kähler manifold. Let φ be a bounded ω-plurisubharmonic function such that*

$$(\omega + dd^c\varphi)^n = e^g \omega^n.$$

Show that if $\operatorname{Osc}_X(\varphi) \leq C$ and $\Delta_\omega g \geq -C$, then

$$\Delta_\omega \varphi \leq A,$$

where A depends only on C (and ω).

Exercise 14.13. *Let u be a smooth strongly plurisubharmonic function solution of $\det(u_{i\bar{j}}) = f$.*

(1) *Show that*

$$\left(\det\left(u_{i\bar{j}}\right)\right)_\zeta = \det\left(u_{i\bar{j}}\right)\operatorname{tr}\left(\left(\tilde{u}^{i\bar{j}}\right)\cdot\left(u_{i\bar{j}}\right)\right)$$

where $\left(\tilde{u}^{i\bar{j}}\right) = \left(u_{i\bar{j}}\right)^{-1}$ and deduce that

$$\left(\log\det\left(u_{i\bar{j}}\right)\right)_\zeta = u^{i\bar{j}}u_{\zeta i\bar{j}},$$

where $\left(u^{i\bar{j}}\right) = {}^t\left(u_{i\bar{j}}\right)^{-1}$

(2) *Observe that $\left(\log\det\left(u_{i\bar{j}}\right)\right)_{\zeta\bar{\zeta}} = u^{i\bar{j}}u_{\zeta\bar{\zeta}i\bar{j}} + \left(u^{i\bar{j}}\right)_{\bar{\zeta}}u_{\zeta i\bar{j}}$ and differentiate the identity*

$$\left(u^{i\bar{j}}\right){}^t\left(u_{i\bar{j}}\right) = \mathrm{Id}$$

to conclude that $\left(u^{i\bar{j}}\right)_{\bar{\zeta}}u_{\zeta i\bar{j}} = -u^{i\bar{k}}u^{h\bar{j}}u_{\zeta i\bar{j}}u_{\bar{\zeta}h\bar{k}}$, hence

$$u^{i\bar{j}}u_{\zeta\bar{\zeta}i\bar{j}} = (\log f)_{\zeta\bar{\zeta}} + u^{i\bar{l}}u^{k\bar{j}}u_{\zeta i\bar{j}}u_{\bar{\zeta}k\bar{l}}.$$

(3) *Use the concavity property of Exercise* 14.7 *to check that*

$$u^{i\bar{l}}u^{k\bar{j}}u_{\xi i\bar{j}}u_{\bar{\xi}k\bar{l}} \geq 0.$$

Exercise 14.14. *Let $\eta(r)$ denote the modulus of continuity of a continuous function u. Show that if there exists $\lambda > 1$, $\gamma > 0$, and $0 < \delta < 1$ such that for all $0 < r$ small enough,*

$$\eta(r) \leq \delta\eta(\lambda r) + r^{\gamma},$$

then u is Hölder continuous of exponent $\min(\alpha, \gamma^{-})$. We refer the reader to [GT83, Lemma 8.23] *for more details.*

Exercise 14.15. *Let (X, ω) be a compact n-dimensional Kähler manifold. Let $\theta = \omega + dd^{c}\chi$ be a smooth form, not necessarily positive. We set*

$$V_{\theta}(x) = \sup\{u(x)|\ u \in PSH(X, \omega) \text{ and } u \leq 0\}.$$

(1) *Show that V_{θ} has bounded Laplacian* (*see* [BD12] *for details*).

(2) *Show that for all $\beta > 0$, there exists a unique smooth θ-psh function u_{β} such that*

$$(\theta + dd^{c}u_{\beta})^{n} = e^{\beta u_{\beta}}\omega^{n}.$$

(3) *Show that $u_{\beta} \to V_{\theta}$ as $\beta \to +\infty$ in $\mathcal{C}^{1,\alpha}(X)$, for all $0 < \alpha < 1$* (*see* [Ber13] *for details*).

Exercise 14.16. *We use the notations of the last section. Assume $f = e^{\psi^{+}-\psi^{-}}$, where ψ^{+}, ψ^{-} are smooth on X. Fix $t \in (0, 1)$. Let $\varphi \in \mathcal{C}^{\infty}(X)$ be the unique solution to*

$$(\theta + t\omega + dd^{c}\varphi)^{n} = e^{\varphi+\psi^{+}-\psi^{-}}\omega^{n}.$$

Fix $C > 0$ such that $dd^{c}\psi^{\pm} \geq -C\omega$, $\sup_{X}\psi^{+} \leq C$, and the holomorphic bisectional curvature of ω is bounded from below by $-C$.

Show that there exists $A = A\left(C, \int_{X} e^{-2(3C+1)\varphi}\omega^{n}\right) > 0$ such that

$$\Delta_{\omega}\varphi \leq Ae^{-\psi^{-}-(3C+1)\phi}$$

(*see* [DiNL14] *for more details*).

Part IV

Singular Kähler–Einstein metrics

15 Canonical metrics

A basic problem in differential geometry is to find canonical metrics on smooth manifolds. The most famous example is the uniformization theorem which guarantees that every compact manifold of real dimension two admits a metric with constant curvature, the latter being (essentially) unique in its conformal class.

There are many ways one can generalize the uniformization theorem to higher-dimensional manifolds. The purpose of this chapter is to study the canonical metrics of Kähler geometry, following the introduction of extremal metrics by Calabi in [Cal82].

Among them the notion of Kähler–Einstein metric is of highest importance. Constructing them boils down to solving certain complex Monge–Ampère equations that we have been studying in the previous chapters.

In the sequel we let X be a compact n-dimensional Kähler manifold and fix a Kähler class $\alpha \in H^{1,1}(X, \mathbb{R})$.

15.1 Canonical metrics

15.1.1 Extremal metrics

15.1.1.1 The Calabi functional Recall that if a Kähler form $\omega \in \alpha$ is locally written as

$$\omega = \sum_{\alpha,\beta} \omega_{\alpha\beta} i dz_\alpha \wedge d\overline{z}_\beta,$$

its Ricci curvature is given by

$$\mathrm{Ric}(\omega) = -c \sum_{\alpha,\beta} \frac{\partial^2 \log \det(\omega_{pq})}{\partial z_\alpha \partial \overline{z}_\beta} i dz_\alpha \wedge d\overline{z}_\beta,$$

where $c > 0$ is a normalizing constant such that $\mathrm{Ric}(\omega)$ represents the first Chern class $c_1(X)$ of X. The scalar curvature of ω is

$$\mathrm{Scal}(\omega) = n \frac{\mathrm{Ric}(\omega) \wedge \omega^{n-1}}{\omega^n}.$$

We let $\mathcal{H}_\alpha$ denote the set of all Kähler forms representing α and set

$$\mathrm{Ca} : \omega \in \mathcal{H}_\alpha \longmapsto \int_X \mathrm{Scal}(\omega)^2 dV_\omega,$$

where $dV_\omega := V_\alpha^{-1} \omega^n$, $V_\alpha = \alpha^n = \int_X \omega^n$. This is the Calabi functional.

Definition 15.1. *A metric $\omega \in \mathcal{H}_\alpha$ is extremal if it is a critical point of the Calabi functional* Ca.

A metric $\omega \in \mathcal{H}_\alpha$ is a constant scalar curvature Kähler metric (cscK) *if its scalar curvature* $\mathrm{Scal}(\omega)$ *is constant.*

Observe that if $\mathrm{Scal}(\omega) \equiv \lambda \in \mathbb{R}$ is constant, then the constant λ can be computed in cohomology, namely

$$\lambda = \lambda_\alpha := n \frac{c_1(X) \cdot \alpha^{n-1}}{\alpha^n}.$$

These notions have been introduced by Calabi in [Cal82] as possible generalization of metrics of constant Gaussian curvature on Riemann surfaces. Calabi showed the following:

Lemma 15.2. *Let $\omega \in \mathcal{H}_\alpha$ be a Kähler metric with constant scalar curvature. Then ω is a (global) minimizer of* Ca*, hence it is extremal.*

Conversely, if X does not admit any non-zero holomorphic vector field, then any extremal metric has constant scalar curvature.

Proof. Set $\lambda = nc_1(X) \cdot \alpha^{n-1}/\alpha^n$. It follows from the Cauchy–Schwarz inequality that

$$\lambda^2 = \left[\int_X \mathrm{Scal}(\omega) dV_\omega \right]^2 \leq \int_X \mathrm{Scal}(\omega)^2 dV_\omega.$$

Therefore $\mathrm{Ca}(\omega) \geq \lambda^2$, with equality if ω has constant scalar curvature.

To show the converse we need to compute the Euler–Lagrange equation for the Calabi functional Ca. We fix a Kähler representative $\omega = \omega_0$ of α, another Kähler form ω_1 in α, and consider a path ω_t of Kähler forms in $\mathcal{H}_\alpha$, joining ω_0 to ω_1.

It follows from the $\partial\overline{\partial}$-lemma that $\omega_1 = \omega + dd^c\varphi$, hence we can consider $\omega_t := \omega + dd^c(t\varphi)$. We let the reader check that

$$s(\omega_t) = s(\omega) + t \left(- \sum_{\alpha,\beta} g^{\alpha\beta} \frac{\partial^2 \Delta_\omega \varphi}{\partial z_\alpha \partial \overline{z}_\beta} - \sum_{\alpha,\beta,\gamma,\delta} g^{\alpha\delta} \frac{\partial^2 \varphi}{\partial z_\gamma \partial \overline{z}_\delta} g^{\gamma\beta} R_{\alpha\beta} \right) + O(t^2),$$

where $s(\omega)$ denotes the scalar curvature of ω.

Plugging this into the definition of Ca, integrating by parts, and using Bianchi's identity, we end up[1] with the fact that ω is extremal if and only if

$$\frac{\partial^2 \mathrm{Scal}(\omega)}{\partial z_\alpha \partial \overline{z}_\beta} = 0.$$

Set $V := \sum_\alpha g^{\alpha\beta} \frac{\partial \mathrm{Scal}(\omega)}{\partial z_\alpha}$. Then $\overline{\partial} V = 0$, i.e., V is a holomorphic vector field. When there is no non-zero holomorphic vector field, i.e., $H^0(X, TX) = \{0\}$, we conclude that $V = 0$, hence $\mathrm{Scal}(\omega)$ is constant, as claimed. □

[1] These manipulations are far from being trivial! See [T] for more details.

15.1.1.2 Mabuchi functional The Mabuchi functional was introduced by Mabuchi in [Mab86]. It is the functional

$$\mathcal{M}_\alpha : \mathcal{H}_\alpha \ni \omega \longmapsto \mathcal{M}_\alpha(\omega) \in \mathbb{R}$$

defined on the set of Kähler metrics ω with fixed cohomology class α, by setting $\mathcal{M}_\alpha(\omega_0) = 0$ for a reference metric $\omega_0 \in \alpha$ and

$$\frac{d\mathcal{M}_\alpha(\omega_t)}{dt} = -\frac{1}{V_\alpha}\int_X [\mathrm{Scal}(\omega_t) - \lambda_\alpha]\,\dot\varphi_t\,\omega_t^n,$$

for any path $\omega_t = \omega + dd^c\varphi_t$ of Kähler forms in α. Recall that

$$\lambda_\alpha = n\frac{c_1(X)\cdot\alpha^{n-1}}{\alpha^n}.$$

Note that a critical point of the Mabuchi functional is a metric with constant scalar curvature, hence extremizing $\mathcal{M}_\alpha$ is a possible way of constructing such canonical metrics.

One can integrate $\mathcal{M}_\alpha$ along a path to obtain the following formula established by Chen in [Che00b] (see also [T]):

Proposition 15.3. *Fix a Kähler form $\omega \in \alpha$ and let $\omega_\varphi = \omega + dd^c\varphi$ denote any other Kähler form in α. Then*

$$\mathcal{M}_\alpha(\omega_\varphi) = \frac{1}{V_\alpha}\int_X \log\left(\frac{\omega_\varphi^n}{\omega^n}\right)\omega_\varphi^n - \frac{1}{V_\alpha}\sum_{j=0}^{n-1}\int_X \varphi\,\mathrm{Ric}(\omega)\wedge\omega_\varphi^j\wedge\omega^{n-1-j} + \lambda_\alpha E(\varphi).$$

Recall that

$$E(\varphi) = \frac{1}{(n+1)V_\alpha}\sum_{j=0}^{n}\int_X \varphi\,\omega_\varphi^j\wedge\omega^{n-j}$$

is a primitive of the Monge–Ampère operator (see Chapter 11).

Proof. We let $\omega_t = \omega + dd^c\varphi_t$ denote a path of Kähler metrics in α such that $\frac{d\omega_t}{dt} = dd^c\dot\varphi_t$. By definition,

$$\frac{dE(\varphi_t)}{dt} = \int_X \dot\varphi_t\,MA(\varphi_t) = \int_X \dot\varphi_t\,\frac{\omega_t^n}{V_\alpha}.$$

A straightforward computation yields

$$\frac{d}{dt}\left\{\frac{1}{V_\alpha}\int_X \log\left(\frac{\omega_t^n}{\omega^n}\right)\omega_t^n\right\} = -\frac{n}{V_\alpha}\int_X \dot\varphi_t\,dd^c\log\left(\frac{\omega_t^n}{\omega^n}\right)\wedge\omega_t^{n-1},$$

while

$$\frac{d}{dt}\left\{\frac{1}{V_\alpha}\sum_{j=0}^{n-1}\int_X \varphi_t\,\mathrm{Ric}(\omega)\wedge\omega_t^j\wedge\omega^{n-1-j}\right\} = \frac{n}{V_\alpha}\int_X \dot\varphi_t\,\mathrm{Ric}(\omega)\wedge\omega_t^{n-1},$$

as the reader will check in Exercise 15.2.

Recall now that $\mathrm{Ric}(\omega_t) = \mathrm{Ric}(\omega) - dd^c \log(\omega_t^n/\omega^n)$ and use the definition of the Mabuchi functional

$$\frac{d\mathcal{M}_\alpha(\omega_t)}{dt} = \frac{1}{V_\alpha}\int_X \left[n\frac{\mathrm{Ric}(\omega_t)\wedge\omega_t^{n-1}}{\omega_t^n} - \lambda_\alpha\right]\dot{\varphi}_t\,\omega_t^n$$

to conclude. □

When $\alpha = c_1(X)$ is the first Chern class of X (which is henceforth assumed to be Fano) the formula simplifies and we recover the formula for $\mathcal{M}_\mu$ used in Chapter 11 with $\mu = \omega^n/V_\alpha$. We let the reader check this fact in Exercise 15.3.

Remark 15.4. *The constant scalar curvature equation is a non-linear* PDE *of order* 4. *Finding a metric of constant scalar curvature in a given Kähler class is thus a delicate problem that has attracted much attention in recent years* (*see* [Don02, ACGTF08, Don09, AP09, RS09, Sz12]).

The uniqueness problem is also quite interesting, it is related to the convexity properties of the Mabuchi functional, as we briefly explain in Section 15.2.

15.1.2 The Calabi–Yau theorem

15.1.2.1 Prescribing the Ricci curvature We have seen earlier that the Ricci curvature $\mathrm{Ric}(\omega)$ of a Kähler form ω is a closed $(1,1)$-form that represents the first Chern class $c_1(X)$.

Conversely, given a closed $(1,1)$-form η representing $c_1(X)$ and $\alpha \in H^{1,1}(X,\mathbb{R})$ a Kähler class, Calabi asked in the 50's [Cal57] whether there exists a Kähler form ω in α such that $\mathrm{Ric}(\omega) = \eta$. He showed that if the answer is positive, then the solution is unique and proposed a (continuity) method to prove the existence.

The problem remained open for two decades and was known as the *Calabi conjecture*; it was finally solved by Yau in [Yau78] and is now known as the Calabi–Yau theorem. The theory developed in the previous chapters has been strongly influenced by the Calabi–Yau theorem.

The Calabi conjecture reduces to solving a complex Monge–Ampère equation, as we know explain.

15.1.2.2 Reduction to a Monge–Ampère equation Fix $\eta \in c_1(X)$ and ω a Kähler form in α. Since $\mathrm{Ric}(\omega)$ also represents $c_1(X)$, it follows from the $\partial\overline{\partial}$-lemma that there exists $h \in \mathcal{C}^\infty(X,\mathbb{R})$ such that

$$\mathrm{Ric}(\omega) = \eta + dd^c h.$$

We now seek for $\omega_\varphi := \omega + dd^c\varphi$ a new Kähler form such that $\mathrm{Ric}(\omega_\varphi) = \eta$. Since

$$\mathrm{Ric}(\omega_\varphi) = \mathrm{Ric}(\omega) - dd^c \log\left(\frac{(\omega + dd^c\varphi)^n}{\omega^n}\right),$$

the equation $\mathrm{Ric}(\omega_\varphi) = \eta$ is thus equivalent to

$$dd^c \left\{ h - \log\left(\frac{(\omega + dd^c\varphi)^n}{\omega^n} \right) \right\} = 0.$$

The function inside the brackets is pluriharmonic, hence constant since X is compact: shifting initially h by a constant, our problem is thus equivalent to solving the complex Monge–Ampère equation

$$(\omega + dd^c\varphi)^n = e^h \omega^n. \tag{CY}$$

Note that h necessarily satisfies the normalizing condition

$$\int_X e^h \omega^n = \int_X \omega^n = V_\alpha.$$

This is precisely the equation we have solved in Theorem 14.1 when $F \equiv 0$. Thus Theorem 14.1 contains the Calabi–Yau theorem as a particular case:

Theorem 15.5 (Yau78). *Let X be a compact Kähler manifold and fix a Kähler class $\alpha \in H^{1,1}(X, \mathbb{R})$. Given a smooth closed $(1,1)$-form $\eta \in c_1(X)$ representing $c_1(X)$, there exists a unique Kähler form $\omega \in \alpha$ such that $\mathrm{Ric}(\omega) = \eta$.*

15.1.3 Kähler–Einstein metrics

15.1.3.1 Kähler–Einstein versus constant scalar curvature The following metrics are the main objects of interest for us in the sequel:

Definition 15.6. *A Kähler metric ω is Kähler–Einstein if there exists $\lambda \in \mathbb{R}$ such that*

$$\mathrm{Ric}(\omega) = \lambda\omega.$$

The existence of such a metric requires $c_1(X)$ to have a definite sign, since $c_1(X) \in \mathbb{R}\{\omega\}$.

This is always satisfied in dimension $n = 1$: either $c_1(X) > 0$ ($X = \mathbb{P}^1$ is the Riemann sphere), or $c_1(X) = 0$ (elliptic curves), or else $c_1(X) < 0$ (X is hyperbolic).

This is not necessarily the case in dimension 2: in the simple case where $X = S_1 \times S_2$ is the product of two compact Riemann surfaces, $c_1(X)$ is proportional to a Kähler class if and only if S_1 and S_2 are of the same type.

Remark 15.7. *Note that $\mathrm{Ric}(\varepsilon\omega) = \mathrm{Ric}(\omega)$ for any $\varepsilon > 0$, hence by rescaling there are essentially three cases to be considered, depending on the sign of $c_1(X)$: $\lambda \in \{-1, 0, +1\}$.*

Kähler–Einstein metrics obviously have constant scalar curvature. The converse turns out to be true under the appropriate cohomological assumption:

Lemma 15.8. *Let ω be a Kähler metric such that $c_1(X) \in \mathbb{R}\{\omega\}$. Then ω has constant scalar curvature if and only if it is Kähler–Einstein.*

Proof. Fix $\lambda \in \mathbb{R}$ such that $\lambda\{\omega\} = c_1(X)$ and $h \in \mathcal{C}^\infty(X, \mathbb{R})$ such that

$$\mathrm{Ric}(\omega) = \lambda\omega + dd^c h.$$

If ω is Kähler–Einstein, $\mathrm{Ric}(\omega) = \lambda\omega$, then $\mathrm{Scal}(\omega) \equiv n\lambda$, hence ω has constant scalar curvature.

Conversely, assume that ω has constant scalar curvature $\mathrm{Scal}(\omega) \equiv c$. Necessarily $c = n\lambda$, as can be seen by integrating over X. Now

$$c\omega^n + n\, dd^c h \wedge \omega^{n-1} = n\mathrm{Ric}(\omega) \wedge \omega^{n-1} = c\omega^n,$$

therefore

$$\Delta_\omega h := n\frac{dd^c h \wedge \omega^{n-1}}{\omega^n} = 0,$$

i.e., h is harmonic. Since X is compact, we infer that h is constant, hence $\mathrm{Ric}(\omega) = \lambda\omega$. □

15.1.3.2 Reduction to a Monge–Ampère equation Fix $\lambda \in \mathbb{R}$ such that $\lambda\{\omega\} = c_1(X)$ and $h \in \mathcal{C}^\infty(X, \mathbb{R})$ such that

$$\mathrm{Ric}(\omega) = \lambda\omega + dd^c h.$$

We seek for a Kähler form $\omega_\varphi := \omega + dd^c\varphi$ such that $\mathrm{Ric}(\omega_\varphi) = \lambda\omega_\varphi$. Since

$$\mathrm{Ric}(\omega_\varphi) = \mathrm{Ric}(\omega) - dd^c \log\left(\frac{(\omega + dd^c\varphi)^n}{\omega^n}\right),$$

the equation $\mathrm{Ric}(\omega_\varphi) = \lambda\omega_\varphi$ is thus equivalent to

$$dd^c\left\{h - \lambda\varphi - \log\left(\frac{(\omega + dd^c\varphi)^n}{\omega^n}\right)\right\} = 0.$$

The function inside the brackets is pluriharmonic hence constant since X is compact: shifting initially h by a constant, our problem is thus equivalent to solving the complex Monge–Ampère equation

$$(\mathrm{MA})_\lambda \qquad (\omega + dd^c\varphi)^n = e^{-\lambda\varphi + h}\omega^n.$$

15.1.3.3 Known results It follows from Chapter 14 that one can always solve $(\mathrm{MA})_\lambda$ when $\lambda < 0$:

Theorem 15.9. [Aub78, Yau78] *Let X be a compact projective algebraic manifold with ample canonical line bundle $K_X > 0$. Then there exists a unique Kähler–Einstein metric $\omega \in c_1(K_X)$ with negative Ricci curvature* $\mathrm{Ric}(\omega) = -\omega$.

It follows again from Chapter 14 that one can always solve $(\mathrm{MA})_0$:

Theorem 15.10. [Yau78] *Let X be a compact Kähler manifold with $c_1(X) = 0$ and fix a Kähler class $\alpha \in H^{1,1}(X, \mathbb{R})$. There exists a unique Kähler form $\omega \in \alpha$ which is Ricci flat, i.e., such that* $\mathrm{Ric}(\omega) = 0$.

The situation is much more complicated when $\lambda > 0$ (Fano manifolds): there are obstructions to the existence of Kähler–Einstein metrics in this case and the solutions — when they exist — are moreover not necessarily unique. A sufficient condition was proposed in Chapter 11. Together with the regularity theory developed in Chapter 14, this yields the following result of Tian [Tian87]:

Theorem 15.11. *Let X be a Fano manifold of dimension n. If $\alpha(X) > n/(n+1)$, then X admits a Kähler–Einstein metric.*

This criterion has been used by Tian and Yau [TY87] to give a simple proof that certain Fano surfaces admit a Kähler–Einstein metric (see also [Siu88, Nad90]). The proof goes by showing that the numerical condition $\alpha(X) > n/(n+1)$ implies that the functional $\mathcal{F}_1$ is proper, where

$$\mathcal{F}_1(\varphi) = E(\varphi) + \log\left[\int_X e^{-\varphi} d\mu\right],$$

with E a primitive of the Monge–Ampère operator and μ is an appropriate smooth volume form.

It follows from the work of Ding and Tian [DT92, Tian97] that the properness of the functional $\mathcal{F}_1$ is equivalent to the existence of a Kähler–Einstein metric, when the Fano manifold X has no non-zero holomorphic vector field. The properness is then equivalent to a stronger coercivity property, as was established by Phong, Song, Sturm, and Weinkove:

Theorem 15.12. *[PSSW08] Let X be a Kähler–Einstein Fano manifold with no non-zero holomorphic vector field. Then there exists $A, B > 0$ such that*

$$\mathcal{F}_1(\varphi) \leq AE(\varphi) + B.$$

The proof of this Moser–Trudinger type inequality is quite involved, it uses a smoothing property of the Kähler–Ricci flow.

We take a closer look at the Kähler–Einstein problem on Fano manifolds in the next sections.

15.2 The Riemannian space of Kähler metrics

Fix $\omega \in \alpha$ a Kähler form and let $\mathcal{H}$ be the space of *Kähler potentials*

$$\mathcal{H} = \left\{\varphi \in C^\infty(X) |\ \omega_\varphi = \omega + dd^c\varphi > 0\right\}.$$

This is a convex open subset of the Fréchet vector space $C^\infty(X)$, thus itself a Fréchet manifold, which is moreover parallelizable : $T\mathcal{H} = \mathcal{H} \times C^\infty(X)$. Each tangent space is identified with $C^\infty(X)$.

As two Kähler potentials define the same metric when (and only when) they differ by an additive constant, we have

$$\mathcal{H}_\alpha = \mathcal{H}/\mathbb{R},$$

where $\mathbb{R}$ acts on $\mathcal{H}$ by addition. The set $\mathcal{H}_\alpha$ is the space of Kähler metrics on X in the cohomology class $\alpha := \{\omega\} \in H^{1,1}(X, \mathbb{R})$.

We review in this section known facts about the Riemannian structure of this space, as introduced by Mabuchi in [Mab87] and further investigated by Semmes [Sem92], Donaldson [Don99], Chen [Che00, CC02].

We also indicate the recent results of Berman and Berndtsson [BerBer14] who established the convexity of the Mabuchi functional along weak Mabuchi geodesics and proved the uniqueness of cscK metrics modulo holomorphic vector fields.

15.2.1 The Riemannian structure

Definition 15.13. *[Mab87] The* Mabuchi metric *is the L^2 Riemannian metric on $\mathcal{H}$. It is defined by*

$$\langle \psi_1, \psi_2 \rangle_\varphi = \int_X \psi_1 \psi_2 \frac{(\omega + dd^c\varphi)^n}{V_\alpha},$$

where $\varphi \in \mathcal{H}$, $\psi_1, \psi_2 \in C^\infty(X)$ and $(\omega + dd^c\varphi)^n/V_\alpha$ is the volume element, normalized so that it is a probability measure.

Here $V_\alpha := \alpha^n = \int_X \omega^n$ denotes the volume of the class α and we shall also use the notations $\omega_\varphi := \omega + dd^c\varphi$ and

$$MA(\varphi) := \frac{\omega_\varphi^n}{V_\alpha}.$$

15.2.1.1 The geodesic equation Geodesics between two points φ_0, φ_1 in $\mathcal{H}$ are defined as the extremals of the energy functional

$$\varphi \longmapsto H(\varphi) = \frac{1}{2} \int_0^1 \int_X (\dot\varphi_t)^2 \, MA(\varphi_t) \, dt,$$

where $\varphi = \varphi_t$ is a path in $\mathcal{H}$ joining φ_0 and φ_1. The geodesic equation is obtained by computing the Euler–Lagrange equation for this energy functional (with fixed end points):

Lemma 15.14. *The geodesic equation is*

$$\ddot\varphi = \|\nabla\dot\varphi\|_\varphi^2, \tag{15.2.1}$$

where the gradient is relative to the metric ω_φ. This identity also writes

$$\ddot\varphi \, MA(\varphi) = \frac{n}{V_\alpha} d\dot\varphi \wedge d^c\dot\varphi \wedge \omega_\varphi^{n-1}.$$

Proof. We need to compute the Euler–Lagrange equation of the energy functional H. Let $(\phi_{s,t})$ be a variation of (φ_t) with fixed end points:

$$\phi_{0,t} \equiv \varphi_t \text{ and } \phi_{s,0} \equiv \varphi_0,\ \phi_{s,1} \equiv \varphi_1.$$

Set $\psi_t := \frac{\partial \phi}{\partial s}|_{s=0}$ and observe that $\psi_0 \equiv \psi_1 \equiv 0$. Thus

$$\phi_{s,t} = \varphi_t + s\psi_t + o(s) \text{ and } \frac{\partial \phi_{s,t}}{\partial t} = \dot{\varphi}_t + s\dot{\psi}_t + o(s).$$

A direct computation yields

$$\begin{aligned} H(\phi_{s,t}) &= \frac{1}{2}\int_0^1 \int_X (\dot{\phi_{s,t}})^2\, MA(\phi_{s,t})\, dt \\ &= H(\varphi_t) + s\int_0^1 \int_X \dot{\varphi}_t \dot{\psi}_t MA(\varphi_t) dt + \frac{ns}{2V}\int_0^1 \int_X (\dot{\varphi}_t)^2 dd^c\psi_t \wedge \omega_{\varphi_t}^{n-1} dt, \end{aligned}$$

noting that

$$MA(\phi_{s,t}) = MA(\varphi_t) + \frac{ns}{V} dd^c\psi_t \wedge \omega_{\varphi_t}^{n-1}.$$

Integrating by parts yields, since $\psi_0 \equiv \psi_1 \equiv 0$,

$$\int_0^1 \int_X \dot{\varphi}_t \dot{\psi}_t MA(\varphi_t) dt = -\int_0^1 \int_X \psi_t \left\{ \ddot{\varphi}_t MA(\varphi_t) + \frac{n}{V}\dot{\varphi}_t dd^c\dot{\varphi}_t \wedge \omega_{\varphi_t}^{n-1} \right\} dt,$$

while

$$\int_0^1 \int_X (\dot{\varphi}_t)^2 dd^c\psi_t \wedge \omega_{\varphi_t}^{n-1} dt = 2\int_0^1 \int_X \psi_t \left\{ d\dot{\varphi}_t \wedge d^c\dot{\varphi}_t + \dot{\varphi}_t \wedge dd^c\dot{\varphi}_t \right\} \wedge \omega_{\varphi_t}^{n-1} dt,$$

hence

$$H(\phi_{s,t}) = H(\varphi_t) + s\int_0^1 \int_X \psi_t \left\{ -\ddot{\varphi}_t MA(\varphi_t) + \frac{n\, d\dot{\varphi}_t \wedge d^c\dot{\varphi}_t \wedge \omega_{\varphi_t}^{n-1}}{V} \right\} dt + o(s).$$

Therefore, (φ_t) is a critical point of H if and only if

$$\ddot{\varphi}\, MA(\varphi) = \frac{n}{V_\alpha} d\dot{\varphi} \wedge d^c\dot{\varphi} \wedge \omega_\varphi^{n-1}. \qquad \square$$

15.2.1.2 $\mathcal{H}$ as a symmetric space As for Riemannian manifolds of finite dimension, one can find the local expression of the Levi-Civita connection by polarizing the geodesic equation. We define the covariant derivative of the vector field ψ_t along the path φ_t in $\mathcal{H}$ by the formula

$$\frac{D\psi}{Dt} = \partial_t \psi - \langle \nabla\psi, \nabla\dot{\varphi} \rangle_\varphi .$$

This covariant derivative is symmetric by its very definition, i.e.,

$$\frac{D}{Ds}\partial_t\varphi = \frac{D}{Dt}\partial_s\varphi,$$

for every family $\varphi = \varphi_{s,t}$, and it can be checked directly that it preserves the metric,

$$\frac{d}{dt}\langle\psi_1,\psi_2\rangle_\varphi = \left\langle\frac{D\psi_1}{Dt},\psi_2)\right\rangle_\varphi + \left\langle\psi_1,\frac{D\psi_2}{Dt}\right\rangle_\varphi.$$

Consider a 2-parameters family $\varphi(s,t)\in\mathcal{H}$ and a vector field $\psi(s,t)\in C^\infty(X)$ defined along φ. The curvature tensor of the Mabuchi metric on $\mathcal{H}$ is defined by

$$R_\varphi(\varphi_s,\varphi_t)\psi = (D_sD_t - D_tD_s)\,\psi,$$

where φ_s and φ_t denote the s and t derivatives of φ and

$$D_t\psi = \psi_t + \Gamma_\varphi(\varphi_t,\psi) = \psi_t - \langle\nabla\psi,\nabla\varphi_t\rangle_\varphi$$

is the covariant derivative of ψ.

Proposition 15.15. *The curvature tensor on $\mathcal{H}$ can be expressed as*

$$R_\varphi(\varphi_s,\varphi_t)\psi = -\{\{\varphi_s,\varphi_t\},\psi\},$$

where $\{\varphi_1,\varphi_2\}$ is the Poisson bracket *associated with the symplectic structure ω_φ. Furthermore, the covariant derivative $D_rR(\varphi_s,\varphi_t)$ of the curvature tensor R vanishes.*

We refer the reader to [Kol12, Proposition 6.27] for a proof of this result which essentially says that $\mathcal{H}$ is a *locally symmetric space*:

$$\nabla R = 0.$$

In finite dimension, such a manifold is characterized by the fact that in the neighborhood of each point x, the (local) *geodesic symmetry* of center x, $\exp_x(X)\mapsto\exp_x(-X)$, is an isometry (see [Hel01]). Note however that the exponential map is not well defined in our infinite-dimensional setting.

15.2.2 Dirichlet problem for the Monge–Ampère equation

15.2.2.1 Semmes trick We are interested in the boundary value problem for the geodesic equation: given two distinct points φ_0,φ_1 in $\mathcal{H}$, can one find a path $(\varphi(t))_{0\le t\le1}$ in $\mathcal{H}$ which is a solution of (15.2.1) with end points $\varphi(0)=\varphi_0$ and $\varphi(1)=\varphi_1$?

It has been observed by Semmes [Sem92] that this can be re-formulated as a homogeneous complex Monge–Ampère equation.

For each path $(\varphi_t)_{t\in[0,1]}$ in $\mathcal{H}$, we set

$$\varphi(x,z)=\varphi_t(x),\qquad x\in X,\quad e^{t+is}\in A=[1,e]\times\mathbb{S}^1;$$

i.e., we associate to each path (φ_t) a function φ on the complex manifold $X\times A$ which is radial in the annulus coordinate: we consider the annulus A as a Riemann surface with boundary and use the complex coordinate $z=e^{t+is}$ to parametrize it. Set $\omega(x,z):=\omega(x)$.

Proposition 15.16. [Sem92] *The path φ_t is a geodesic in $\mathcal{H}$ if and only if the associated radial function φ on $X\times A$ is a solution of the homogenous complex Monge–Ampère equation in $X\times\dot{A}$,*

$$(\omega+dd^c_{x,z}\varphi)^{n+1}=0.$$

We have stressed the fact that we take derivatives in all variables x,z.

Proof. Write $d_{x,z}\varphi=d_x\varphi+d_z\varphi$ and observe that $d_z\varphi=\dot\varphi_t(dz+d\overline{z})$ (by invariance), $(\omega+d_xd^c_x\varphi)^{n+1}=0$, and

$$\omega+dd^c_{x,z}\varphi=\omega_{\varphi_t}+R,$$

where $R=d_xd^c_z\varphi+d_zd^c_x\varphi+d_zd^c_z\varphi$ is such that $R^3=0$. This yields

$$(\omega+dd^c_{x,z}\varphi)^{n+1}=(n+1)\omega^n_{\varphi_t}\wedge R+\frac{n(n+1)}{2}\omega^{n-1}_{\varphi_t}\wedge R^2.$$

Now $d_zd^c_z\varphi=\frac{i}{\pi}\ddot\varphi_t dz\wedge d\overline{z}$ and

$$(d_xd^c_z\varphi+d_zd^c_x\varphi)^2=-2d_x\dot\varphi_t\wedge d^c_x\dot\varphi_t\wedge\frac{i}{\pi}dz\wedge d\overline{z},$$

thus

$$(\omega+dd^c_{x,z}\varphi)^{n+1}=(n+1)\omega^n_{\varphi_t}\wedge\frac{i}{\pi}dz\wedge d\overline{z}\left\{\ddot\varphi_t-\frac{n\,d\dot\varphi\wedge d^c\dot\varphi\wedge\omega^{n-1}_\varphi}{\omega^n_\varphi}\right\},$$

and the result follows. □

15.2.2.2 "Smooth" geodesics We have noted in Chapter 5 that homogeneous complex Monge–Ampère equations in bounded domains of $\mathbb{C}^n$ admit a unique solution which is at most $\mathcal{C}^{1,1}$-smooth.

A fundamental result of Chen [Che00] shows the existence (and uniqueness) of almost $\mathcal{C}^{1,1}$-solutions to the geodesic equation[2].

[2] Chen shows that the function $(x,z)\mapsto\varphi(x,z)$ has bounded Laplacian, hence it is in particular $\mathcal{C}^{1,\alpha}$ for all $0<\alpha<1$.

It was expected that Chen's regularity result is essentially optimal[3]. This has been confirmed by Lempert, Vivas and Darvas in [LV13, DL12]. We summarize this discussion in the following:

Theorem 15.17 (Chen, Lempert–Vivas). *Given $\varphi_0, \varphi_1 \in \mathcal{H}$, there exist $C > 0$ and a continuous geodesic path $(\varphi_t)_{0\leq t\leq 1}$ in $\overline{\mathcal{H}}$ joining φ_0 to φ_1 such that*

$$|\Delta_{x,z}\varphi| \leq C.$$

Moreover, smooth geodesics do not exist in general, and this estimate is optimal: there exists geodesics such that $\Delta_{x,z}\varphi$ is bounded, but not continuous.

The proof of Chen's result goes by approximating the degenerate homogeneous complex Monge–Ampère equation by smoother approximate geodesics

$$(\omega + d\,d^c\varphi_\varepsilon)^{n+1} = \varepsilon dV,$$

and establishing a priori estimates as $\varepsilon > 0$ decreases to zero. We refer the reader to [Blo12, Bou12] for a neat presentation of this result.

It follows from the Frobenius theorem that if smooth geodesics existed, one could find a holomorphic foliation of $X \times A$ by complex curves (the Monge–Ampère foliation) along which φ is harmonic. When $X = \mathbb{C}/\mathbb{Z}[\tau]$ is an elliptic curve, the holomorphic involution $x \mapsto f(x) = -x$ has four fixed points p_i and one can then check that $\{p_i\} \times A$ are necessarily leaves of the Monge–Ampère foliation (assuming φ_0, φ_1 are f-invariant). Analyzing the regularity of φ near such leaves allowed Lempert and Vivas to derive a contradiction in [LV13]. We refer the reader to [DL12, Ber14] for further improvements.

15.2.2.3 Bounded geodesics On the other hand, generalized geodesics with lower regularity are easy to construct by using Perron's method, and turn out to be quite useful:

Proposition 15.18. *Assume φ_0, φ_1 are bounded ω-psh functions. Then*

$$\Phi(x,z) := \sup\{\psi(x,z) \,|\, \psi \in PSH(X \times A, \omega) \text{ with } \lim_{t\to 0,1} \psi \leq \varphi_{0,1}\}.$$

is the unique bounded ω-psh function on $X \times A$ solution of the Dirichlet problem $\Phi|_{X\times\partial A} = \varphi_{0,1}$ with

$$(\omega + d\,d^c_{x,z}\Phi)^{n+1} = 0 \text{ in } X \times \dot{A}.$$

Moreover, $\Phi(x,z) = \Phi(x,t)$ only depends on $|z|$ and

$$|\dot{\Phi}| \leq \|\varphi_1 - \varphi_0\|_{L^\infty(X)}.$$

[3] Chen and Tian have proposed in [CT08] some improvements, but some technical details have been overlooked, see [RWN15].

Proof. Recall that $A = [1, e] \times S^1$ is an annulus in $\mathbb{C}$ with complex coordinate $z = e^{t+is}$. The proof follows from the classical balayage technique, together with a barrier argument: observe that for $A = \|\varphi_1 - \varphi_0\|_{L^\infty(X)}$, the function

$$\chi_t(x) = \max\{\varphi_0(x) - A \log|z|, \varphi_1(x) + A(\log|z| - 1)\}$$

belongs to the family and has the right boundary values. Using that $t \mapsto \Phi_t$ is convex (by subharmonicity in z), one easily deduces the bound on $|\dot{\Phi}|$. We refer the reader to [PSS12, Bern13] for more details. □

It is also interesting to consider (weak) *subgeodesics*: these are paths $(\varphi_t(x))$ of functions in $\mathcal{H}$ (or in larger classes of ω-psh functions) such that the associated invariant functions are ω-psh functions on $X \times A$. It follows from the computation made in the proof of Proposition 15.16 that this is equivalent to

$$\ddot{\varphi}_t MA(\varphi_t) \geq \frac{n}{V} d\dot{\varphi} \wedge d^c\dot{\varphi} \wedge \omega_\varphi^{n-1}.$$

15.2.3 The Mabuchi distance

15.2.3.1 Metric completion The length of an arbitrary path $(\varphi_t)_{t\in[0,1]}$ in $\mathcal{H}$ is defined in a standard way,

$$\ell(\varphi) := \int_0^1 |\dot{\varphi}_t| dt = \int_0^1 \sqrt{\int_X \dot{\varphi}_t^2 MA(\varphi_t)}\, dt.$$

The distance between two points in $\mathcal{H}$ is then

$$d(\varphi_0, \varphi_1) := \inf\{\ell(\varphi) \mid \varphi \text{ is a path joining } \varphi_0 \text{ to } \varphi_1\}.$$

It is easy to verify that d defines a semi-distance (i.e., non-negative, symmetric and satisfying the triangle inequality). It is however non-trivial, in this infinite dimensional context, to check that d is non-degenerate (see [MM05] for a striking example). This was proved by Chen in [Che00] (see also [CC02]), who established that

Theorem 15.19. [Che00, CC02]

- *$(\mathcal{H}, d)$ is a metric space of non-positive curvature;*
- *the geodesics (φ_t) are length minimizing;*
- *any sequence of asymptotically length-minimizing paths converges, in the Hausdorff topology, to the unique geodesic.*

Moreover, for all $t \in [0, 1]$,

$$d(\varphi_0, \varphi_1) = \sqrt{\int_X \dot{\varphi}_t^2 MA(\varphi_t)}.$$

Remark 15.20. *A delicate issue here is that it is not clear that the geodesics constructed by Chen lie in $\mathcal{H}$. This problem has been adressed by Berman and Demailly* [BD12] (*see also* [He15]), *who showed that the whole construction can be extended to*

$$\mathcal{H}^{1,1} := \{\varphi \in PSH(X,\omega) \mid \Delta\varphi \in L^\infty(X)\}.$$

Observe that d induces a distance on $\mathcal{H}_\alpha$ (that we abusively still denote by d) compatible with the Riemannian splitting $\mathcal{H} = \mathcal{H}_\alpha \times \mathbb{R}$, by setting

$$d(\omega_\varphi, \omega_\psi) := d(\varphi, \psi)$$

whenever the potentials φ, ψ of $\omega_\varphi, \omega_\psi$ are normalized by $E(\varphi) = E(\psi) = 0$ (see the comment after Lemma 15.23).

It is rather easy to check that $(\mathcal{H}_\alpha, d)$ is not a complete metric space (see Exercise 15.6). The metric completion $(\overline{\mathcal{H}}_\alpha, d)$ has been studied recently in [Dar14, G14]. We mention without proof the following characterization of $\overline{\mathcal{H}}_\alpha$ obtained in [Dar14]:

Theorem 15.21. *The metric completion $(\overline{\mathcal{H}}_\alpha, d)$ of $(\mathcal{H}_\alpha, d)$ is a Hadamard space which consists of finite energy currents, more precisely*

$$(\overline{\mathcal{H}}_\alpha, d) \text{ is isometric to } \mathcal{E}^2(\alpha).$$

Recall that a Hadamard space is a complete CAT(0) space, i.e., a complete geodesic space which has non-positive curvature in the sense of Alexandrov (see [BH99, Chapter II.1]).

The set $\mathcal{E}^2(\alpha)$ of finite energy currents has been defined in Chapter 10; it is isomorphic (modulo constants) to the space $\mathcal{E}^2(X,\omega)$ if we fix a base point ω in α.

15.2.3.2 Bergman geodesics When $\alpha = c_1(L)$ is the first Chern class of a holomorphic line bundle, one can approximate any function $\varphi \in \mathcal{H}$ by Bergman kernels

$$\varphi_j = \frac{1}{2}\log \sum_{\ell=1}^{s_j} \left|\sigma_\ell^{(j)}\right|^2 \in \mathcal{H},$$

where $\sigma_1^{(j)}, \ldots, \sigma_{s_j}^{(j)}$ is an orthonormal basis of the Bergman space

$$\mathcal{B}(j\varphi) := \left\{ s \in H^0(X, L^j) \mid \int_X |s|^2 e^{-2j\varphi} < +\infty \right\}.$$

We have shown in Theorem 8.9 that this convergence holds in the weak (L^1-)sense. The same proof actually yields uniform convergence as soon as the function φ is Hölder continuous. It turns out that the convergence holds in the smooth category if φ is smooth (see [Zel98]).

This approximation is canonical. We let $\mathcal{H}_j$ denote the finite-dimensional symmetric space of Hermitian norms on $H^0(X, L^j)$. Since φ_j induces a norm $h_j(\varphi)$ on $H^0(X, L^j)$, we have maps

$$\mathrm{Hilb}_j : \varphi \in \mathcal{H} \longmapsto h_j \in \mathcal{H}_j.$$

Conversely, any norm h_j on $H^0(X, L^j)$ allows one to define a smooth Kähler potential in $\mathcal{H}$ by considering an orthonormal basis $\sigma_1^{(h_j)}, \ldots, \sigma_{s_j}^{(h_j)}$ and its associated Bergman metric

$$\psi_j(h_j) = \frac{1}{2} \log \sum_{\ell=1}^{s_j} \left| \sigma_\ell^{(h_j)} \right|^2 \in \mathcal{H}.$$

We thus have natural maps

$$FS_j : h_j \in \mathcal{H}_j \longmapsto \psi_j(h_j) \in \mathcal{H}.$$

The convergence result of Zelditch [Zel98] can thus be reformulated as

$$\varphi_j := FS_j \circ \mathrm{Hilb}_j(\varphi) \xrightarrow{C^\infty} \varphi$$

It is natural to ask whether Mabuchi geodesics in $\mathcal{H}$ can also be approximated by geodesics coming from the finite-dimensional symmetric spaces $\mathcal{H}_j$. This question was first addressed by Phong and Sturm in [PS06b]; their positive answer was later on reinforced by works of Chen and Sun [CS12] and Berndtsson [Bern09b]. We summarize these results in the following:

Theorem 15.22. *The finite-dimensional geodesics uniformly converge towards the Mabuchi geodesics and the spaces $(\mathcal{H}_j, d_j)$ converge to $(\mathcal{H}, d)$ as metric spaces.*

We refer the reader to [Bern09b] for a proof of this result, as well as many additional information. The key point of his proof is the plurisubharmonic variation of Bergman kernels in families that Berndtsson establishes in [Bern09]. This property is also the key to the convexity property of the Mabuchi functional along geodesics.

15.2.4 Convexity of functionals along geodesics

15.2.4.1 The Aubin–Mabuchi functional Each tangent space $T_\varphi \mathcal{H}$ admits the orthogonal decomposition

$$T_\varphi \mathcal{H} = \left\{ \psi \in C^\infty(X) |\ \beta_\varphi(\psi) = 0 \right\} \oplus \mathbb{R},$$

where $\beta = MA$ is the 1-form defined on $\mathcal{H}$ by

$$\beta_\varphi(\psi) = \int_X \psi\, MA(\varphi).$$

It is a classical observation (see [Kol12, pp. 245–246]) that the 1-form β is closed. Therefore, there exists a unique function E defined on the convex open set $\mathcal{H}$, such that $\beta = dE$ and $E(0) = 0$: this is the energy functional that we have considered extensively in previous chapters (see notably Chapter 10 and Chapter 11).

It is often called the *Aubin–Mabuchi functional* and can be expressed (after integration along affine paths) by

$$E(\varphi) = \frac{1}{(n+1)V_\alpha} \sum_{j=0}^{n} \int_X \varphi \, (\omega + dd^c\varphi)^j \wedge \omega^{n-j}. \tag{15.2.2}$$

Lemma 15.23. *The Aubin–Mabuchi functional E is concave, non-decreasing and satisfies the cocycle condition*

$$E(\varphi) - E(\psi) = \frac{1}{(n+1)V_\alpha} \sum_{j=0}^{n} \int_X (\varphi - \psi) \, (\omega + dd^c\varphi)^j \wedge (\omega + dd^c\psi)^{n-j}.$$

It is affine along geodesics in $\mathcal{H}$.

Proof. The monotonicity property follows from the definition, since the first derivative of E is $dE = \beta = MA \geq 0$, a probability measure.

The behavior of E along geodesics is easily understood:

$$\begin{aligned} \frac{d^2}{dt^2} E(\varphi_t) &= \int_X \ddot{\varphi} \, MA(\varphi) + \frac{n}{V_\alpha} \int_X \dot{\varphi} \, dd^c\dot{\varphi} \wedge \omega_\varphi^{n-1} \\ &= \int_X \left\{ \ddot{\varphi} \, MA(\varphi) - \frac{n}{V_\alpha} d\dot{\varphi} \wedge d^c\dot{\varphi} \wedge \omega_\varphi^{n-1} \right\} = 0 \end{aligned}$$

when φ_t is a Mabuchi geodesic. The concavity follows from the same computation, since for $\varphi_t = t\varphi + (1-t)\psi$ we get $\ddot{\varphi}_t = 0$, hence

$$\frac{d^2}{dt^2} E(\varphi_t) = -\int_X \frac{n}{V_\alpha} d\dot{\varphi} \wedge d^c\dot{\varphi} \wedge \omega_\varphi^{n-1} \leq 0.$$

The cocycle condition is obtained similarly by differentiating the function $t \mapsto E(t\varphi + (1-t)\psi)$. □

Since each path $\varphi_t = \varphi + t$ in $\mathcal{H}$ is a geodesic, we obtain in particular

$$E(\varphi + t) = E(\varphi) + t. \tag{15.2.3}$$

Given $\varphi \in \mathcal{H}$ there exists a unique $c \in \mathbb{R}$ such that $E(\varphi + c) = 0$. The restriction of the Mabuchi metric to the fiber $E^{-1}(0)$ induces a Riemannian structure on the quotient space $\mathcal{H}_\alpha = \mathcal{H}/\mathbb{R}$ and allows to decompose

$$\mathcal{H} = \mathcal{H}_\alpha \times \mathbb{R}$$

as a product of Riemannian manifolds.

15.2.4.2 The Mabuchi and $\mathcal{F}_\lambda$ functionals Recall from Chapter 11 that the variational approach is an efficient method to solve the Kähler–Einstein equation. It amounts to try and maximize the functionals

$$\mathcal{F}_\lambda(\varphi) = E(\varphi) + \frac{1}{\lambda}\log\left[\int_X e^{-\lambda\varphi}\, d\mu\right],$$

where $\varphi \in \mathcal{H}$, μ is a smooth probability volume form, and λ is a real parameter whose sign coincides with that of $c_1(X)$.

The properness of $\mathcal{F}_\lambda$ is related to the existence of critical points: we have already observed that $\mathcal{F}_\lambda$ is always proper when $\lambda \leq 0$, but not necessarily when $\lambda > 0$.

The uniqueness of solutions is related to the convexity properties of $\mathcal{F}_\lambda$. When $\lambda \leq 0$, the functional $\mathcal{F}_\lambda$ is a sum of two functionals E and $\frac{1}{\lambda}\log\left[\int_X e^{-\lambda\varphi}\, d\mu\right]$ which are both concave along affine segments in $\mathcal{H}$: the computation made in Lemma 15.23 indeed shows that

$$\frac{d^2}{dt^2}E(\varphi_t) = \int_X -\frac{n}{V_\alpha} d\dot{\varphi} \wedge d^c\dot{\varphi} \wedge \omega_\varphi^{n-1} \leq 0$$

when $\ddot{\varphi}_t = 0$, while it follows from Hölder inequality that

$$t \longmapsto \log\left[\int_X e^{-\lambda\varphi_t}\, d\mu\right]$$

is convex when $\ddot{\varphi}_t = 0$ and $\lambda \leq 0$.

On the other hand, $\mathcal{F}_\lambda$ is rather a difference of concave functionals when $\lambda > 0$. Looking at Mabuchi geodesics rather than Euclidean ones, the functional E becomes affine (by Lemma 15.23), while the remaining term becomes concave, as was established by Berndtsson in [Bern09b]:

Theorem 15.24. *Assume X is Fano and $\alpha = c_1(X)$. The functional $\mathcal{F}_1$ is concave along Mabuchi geodesics.*

The proof relies on the plurisubharmonic variation of Bergman kernels in families [Bern09].

It is natural to wonder whether similar results hold for the Mabuchi and/or for more general Kähler classes. It has been shown by Mabuchi in [Mab87] that $\mathcal{M}_\alpha$ is convex along *smooth* geodesics, and this was one of his motivations to introduce this Riemannian structure on the space $\mathcal{H}$. As already explained, smooth geodesics unfortunately do not exist in general, but the convexity property has been extended to weak geodesics by Berman and Berndtsson [BerBer14]:

Theorem 15.25. *The Mabuchi functional $\mathcal{M}_\alpha$ is convex along weak (continuous) geodesics.*

The proof uses (once again!) the plurisubharmonic variation of Bergman kernels in families, this time in a local context [Bern06], together with an approximation process. The authors use this to prove the uniqueness of constant scalar curvature Kähler metrics, a result that has been obtained by Donaldson in [Don01] when $\alpha = c_1(L)$ and $\mathrm{Aut}_0(X)$ (the connected component of the identity in the group of biholomorphisms of X) is trivial:

Theorem 15.26. *Assume $\omega_0, \omega_1 \in \alpha$ are constant scalar curvature metrics. Then there exists $f \in \mathrm{Aut}_0(X)$ such that $\omega_1 = f^*\omega_0$.*

15.3 Fano manifolds

A compact complex manifold is called *Fano* if the anticanonical bundle K_X^{-1} is ample. This equivalently means that the first Chern class of X is positive,

$$c_1(X) = -c_1(K_X) > 0.$$

Note that Fano manifolds are necessarily projective algebraic.

15.3.1 Classification The only Fano Riemann surface is the Riemann sphere $\mathbb{P}^1$. If X is a 2-dimensional Fano manifold, then it is isomorphic to

- either $\mathbb{P}^1 \times \mathbb{P}^1$
- or $\mathbb{P}^2$ blown-up at r points in general position, $0 \leq r \leq 8$.

The points are in general position if there are no three points on the same line, no six points on the same conic, and no eight points on a singular cubic with one of these points at the singularity. Fano manifolds of dimension two are called *Del Pezzo surfaces* (see [Dolgachev]).

Fano manifolds have also been classified in dimension 3, there are 105 families! There are finitely many families in any dimension, but this number is becoming very large with n (see [Deb97]).

Example 15.27. *Basic examples of Fano manifolds are provided by smooth hypersurfaces $X_d \subset \mathbb{CP}^{n+1}$ of degree d with $1 \leq d \leq n+1$. It follows indeed from the adjunction formula that*

$$K_{X_d}^* = \mathcal{O}(n+2-d).$$

15.3.2 Bochner formula Given $u \in \mathcal{C}^\infty(X, \mathbb{R})$, we let V_u denote the $\mathcal{C}^\infty$-smooth vector field such that

$$\delta_{V_u}\omega = \overline{\partial} u,$$

where $\delta_V \omega$ denotes the contraction of ω with respect to V.

Proposition 15.28. *The following Bochner-type formula holds:*

$$\|\Delta_\omega u\|^2_{L^2(\omega)} = \|\overline{\partial} V_u\|^2_{L^2(\omega)} + \int_X \mathrm{Ric}(\omega)(V_u, V_u)\omega^n.$$

We encourage the reader to prove this formula in Exercise 15.10.

15.3.2.1 Eigenvalues of the Laplacian The Bochner formula yields information on the eigenvalues of Δ_ω:

Lemma 15.29. *If* $\mathrm{Ric}(\omega) > t\omega$, $t > 0$, *then the operator* $\Delta_\omega + t\,\mathrm{Id}$ *is invertible.*
If ω *is Kähler–Einstein,* $\mathrm{Ric}(\omega) = \omega$, *then*

$$\mathrm{Ker}(\Delta_\omega + \mathrm{Id}) \simeq H^0(X, TX).$$

A motivation here comes from the following continuity method to solve $(\mathrm{MA})_1$ on a Fano manifold: following Aubin [Aub84] one considers, for $0 \leq t \leq 1$, the path

$$\mathrm{Ric}(\omega_t) = t\omega_t + (1-t)\omega,$$

so that $\mathrm{Ric}(\omega_t) > t\omega_t$ when $t < 1$. Solving this equation for $t = 1$ is then equivalent to solving $(\mathrm{MA})_1$.

Proof. The first eigenvalue λ_1 of $-\Delta_\omega$ can be characterized variationally as

$$\lambda_1 = \inf\left\{ \frac{\int_X |\nabla u|^2 \omega^n}{\int_X |u|^2} \;\Big|\; \int_X u\,\omega^n = 0 \text{ and } u \not\equiv 0 \right\}.$$

It is also classical that there exists an eigenfunction u such that $\Delta_\omega u = -\lambda_1 u$. When $\mathrm{Ric}(\omega) \geq t\omega$, we obtain

$$\begin{aligned}\int_X \mathrm{Ric}(\omega)(V_u, V_u)\omega^n &\geq t\|V_u\|^2 = t\|\overline{\partial} u\|^2 \\ &= -t\langle \Delta_\omega u, u\rangle = t\lambda_1^{-1}\|\Delta_\omega u\|^2.\end{aligned}$$

It follows therefore from the Bochner formula that

$$(1 - t\lambda_1^{-1})\|\Delta_\omega u\|^2_{L^2(\omega)} \geq \|\overline{\partial} V_u\|^2_{L^2(\omega)},$$

with strict inequality if $\mathrm{Ric}(\omega) > t\omega$. This shows in this case that $\lambda_1 > t$, hence the operator $\Delta_\omega + t\,\mathrm{Id}$ is invertible.

When $\mathrm{Ric}(\omega) = \omega$, we obtain an equality

$$(1 - \lambda_1^{-1})\|\Delta_\omega u\|^2_{L^2(\omega)} = \|\overline{\partial} V_u\|^2_{L^2(\omega)}$$

which shows that $\lambda_1 \geq 1$, with equality if and only if V_u is a holomorphic vector field, yielding an injective morphism

$$\mathrm{Ker}(\Delta_\omega + \mathrm{Id}) \xrightarrow{\Phi} H^0(X, TX).$$

It turns out that Φ is also surjective: given $V \in H^0(X, TX)$ a holomorphic vector field, we let α denote the holomorphic $(0,1)$-form such that $\delta_V \omega = \alpha$. Since X is Fano, it follows from the Kodaira vanishing theorem that

$$H^1(X, \mathcal{O}_X) = H^1(X, K_X + (-K_X)) = \{0\},$$

hence there exists $u \in \mathcal{C}^\infty(X, \mathbb{R})$ such that $\overline{\partial} u = \alpha$. The Bochner formula now yields (since $\mathrm{Ric}(\omega) = \omega$) $\Delta_\omega u = -u$. □

15.3.3 Holomorphic vector fields Holomorphic vector fields play an important role in the Kähler–Einstein problem on Fano manifolds as we now indicate.

15.3.3.1 Futaki invariants Fix a Kähler class $\alpha \in H^{1,1}(X, \mathbb{R})$ (the case of most interest for us here is $\alpha = c_1(X)$, X being Fano). For $\omega \in \mathcal{K}_\alpha$ a Kähler class in α, we fix $h_\omega \in \mathcal{C}^\infty(X, \mathbb{R})$ such that

$$\mathrm{Scal}(\omega) - \gamma = \Delta_\omega h_\omega,$$

where

$$\gamma = n \frac{c_1(X) \cdot \alpha^{n-1}}{\alpha^n} = \int_X \mathrm{Scal}(\omega) \omega^n / V_\alpha.$$

Definition 15.30. *For $V \in H^0(X, TX)$ a holomorphic vector field, we set*

$$\mathrm{Fut}_\omega(V) := \int_X V(h_\omega)\, \omega^n,$$

where $V(h)$ is the Lie derivative with respect to V of the function h.

This is the Futaki character.

Let Φ_s denote the holomorphic flow of the holomorphic vector field V: this is a one-parameter subgroup of $\mathrm{Aut}(X)$, the group of biholomorphic automorphisms of X. Setting $\omega_s := \Phi_s^* \omega$, we have:

Lemma 15.31. *The Futaki character is related to the Mabuchi functional by*

$$\left. \frac{d\mathcal{M}(\omega_s)}{ds} \right|_{s=0} = \mathrm{Fut}_\omega(V).$$

Here (and in the sequel), we let $\mathcal{M} = \mathcal{M}_\alpha$ denote the Mabuchi functional associated to the Kähler cohomology class α, which is fixed.

Proof. Since Φ_s is homotopic to the identity, $\Phi_s^*\omega$ is cohomologous to ω, hence $\omega_s := \Phi_s^*\omega = \omega + dd^c\varphi_s$, by the $\partial\overline{\partial}$-lemma. Thus

$$\partial\omega_s/\partial s = dd^c\dot{\varphi}_s, \text{ where } \dot{\varphi}_s = \partial\varphi_s/\partial s.$$

Now

$$dd^c\dot{\varphi}_s = \partial\omega_s/\partial s = L_V\omega_s = \partial\theta_s, \text{ with } \theta_s = \delta_V\omega_s,$$

since $\omega_s = \Phi_s^*\omega$ is the pull-back of ω by the flow of V. Let h_s be such that

$$\mathrm{Scal}(\omega_s) - \gamma = \Delta_\omega h_s.$$

It follows from the definition of the Mabuchi functional that

$$\frac{d\mathcal{M}(\omega_s)}{ds} = \int_X \dot{\varphi}_s \Delta_\omega h_s \, \omega_s^n = n\int_X h_s dd^c\dot{\varphi}_s \wedge \omega_s^{n-1} = \int_X V(h_s)\omega_s^n,$$

from which the lemma follows. □

We now prove the following result, originally due to Futaki [Fut83]:

Theorem 15.32. *The Futaki character* Fut_ω *depends only on the cohomology class* $\alpha = \{\omega\}$ *and will thus be denoted by* Fut_α. *In particular, if there is one metric with constant scalar curvature in* α, *then* $\mathrm{Fut}_\alpha \equiv 0$.

Proof. Let $t \mapsto \omega_t$ be a curve of Kähler forms in α. The result will follow if we can prove that for all $V \in H^0(X, TX)$, $t \mapsto \mathrm{Fut}_{\omega_t}(V)$ is constant. Set

$$g(s,t) := \mathcal{M}(\Phi_s^*\omega_t),$$

where Φ_s denotes as above the flow of the holomorphic vector field V. We will be done if we show that $\partial^2 g/\partial s\partial t = 0$ at $(s,t) = (0,0)$.

We claim that $(s,t) \mapsto \mathcal{M}(\Phi_s^*\omega_t)$ is convex at $(0,0)$ and $s \mapsto g(s,t)$ is linear. This shows that $g_{ss} = 0$ and the convexity property now yields

$$D^2 g = \begin{pmatrix} 0 & g_{st} \\ g_{ts} & g_{tt} \end{pmatrix} \geq 0,$$

hence $g_{st} = 0$ at the origin as desired. We let the reader complete the proof of the claim as an Exercise 15.11. □

The importance of the convexity properties of the Mabuchi functional in this context has been emphasized by Donaldson (see [Don99, Che00]).

We let Fut_X denote the Futaki invariant when $\alpha = c_1(X)$. We stress that there cannot exist a Kähler–Einstein metric when $\mathrm{Fut}_X \not\equiv 0$:

Corollary 15.33. *Let* X *be a Fano manifold. If* $\mathrm{Fut}_X \not\equiv 0$, *then* X *does not admit a Kähler–Einstein metric.*

15.3.3.2 The Bando–Mabuchi uniqueness theorem The following important result is due to Bando–Mabuchi [BM87] :

Theorem 15.34. *Let X be a Fano manifold and assume ω_0 and ω_1 are Kähler–Einstein metrics. Then there exists $V \in H^0(X, TX)$, a holomorphic vector field whose flow Φ_t connects ω_0 to $\omega_1 = \Phi_1^* \omega_0$.*

The proof of Bando and Mabuchi uses an inverse continuity method. We sketch an alternative and powerful proof due to Berndtsson [Bern15].

Proof. Fix a reference Kähler form $\omega \in c_1(X)$ and write $\omega_0 = \omega + dd^c\varphi_0$, $\omega_1 = \omega + dd^c\varphi_1$. It follows from from Chapter 11 that the potentials φ_0 and φ_1 are Kähler–Einstein potentials if and only if they maximize the functional

$$\mathcal{F}(\varphi) = E(\varphi) + \log\left(\int_X e^{-\varphi}\, d\mu\right),$$

where

$$E(\varphi) = \frac{1}{(n+1)V} \sum_{j=0}^{n} \int_X \varphi\, \omega_\varphi^j \wedge \omega^{n-j}$$

and $V = c_1(X)^n$, $\mu = e^h\, \omega^n / V$, where $\mathrm{Ric}(\omega) = \omega + dd^c h$ with h normalized so that μ is a probability measure.

We join φ_0 to φ_1 by a weak geodesic, i.e., a continuous path $(\varphi_t)_{0 \le t \le 1}$ of ω-psh potentials which satisfy

$$\frac{n}{V} d\dot{\varphi}_t \wedge d^c\dot{\varphi}_t \wedge (\omega + dd^c\varphi_t)^{n-1} = \frac{\ddot{\varphi}_t}{V} (\omega + dd^c\varphi_t)^n$$

in the weak sense. Recall that $t \mapsto E(\varphi_t)$ is affine along geodesics (by Lemma 15.23).

It follows from the work of Berndtsson [Bern09] that

$$t \longmapsto \log \int_X e^{-\varphi_t}\, d\mu$$

is concave along geodesics and moreover strictly concave unless φ_0 and φ_1 are joined by the flow of a holomorphic vector field.

We infer that $t \mapsto \mathcal{F}(\varphi_t)$ is concave. It follows from the variational characterization of Kähler–Einstein potentials (Chapter 11) that $t \mapsto \mathcal{F}(\varphi_t)$ is actually constant since the end points φ_0, φ_1 are both Kähler–Einstein, and hence maximize $\mathcal{F}$. They are therefore joined by the flow of a holomorphic vector field. □

15.4 The Yau–Tian–Donaldson conjecture

15.4.1 Del Pezzo surfaces It is a classical fact that holomorphic vector fields on the complex projective plane $\mathbb{P}^2$ can be identified with the set of trace-free complex matrices,

$$H^0(\mathbb{P}^2, T\mathbb{P}^2) \simeq \{A \in \mathcal{M}(3,\mathbb{C}) \mid \operatorname{tr} A = 0\}.$$

We let the reader describe similarly holomorphic vector fields on $\mathbb{P}^1 \times \mathbb{P}^1$ and focus on the remaining Del Pezzo surfaces:

Lemma 15.35. *Let X_r denote the blow-up of $\mathbb{P}^2$ at r points $p_1, \dots, p_r$ in general position. Then*

- $H^0(X_1, TX_1) \simeq \{X \in H^0(\mathbb{P}^2, T\mathbb{P}^2) \mid X(p_1) = 0\}$;
- $H^0(X_2, TX_2) \simeq \{X \in H^0(\mathbb{P}^2, T\mathbb{P}^2) \mid X(p_1) = X(p_2) = 0\}$;
- $H^0(X_3, TX_3) \simeq \mathbb{C}^3$;
- $H^0(X_r, TX_r) = \{0\}$ *if* $r \geq 4$.

The Futaki character vanishes identically if and only if $r \neq 1, 2$.

We let the reader check these assertions as Exercise 15.9.

An important work of Tian [Tian90] settles the Kähler–Einstein Fano problem in dimension 2:

Theorem 15.36. *A smooth Del Pezzo surface admits a Kähler–Einstein metric unless it is biholomorphic to $\mathbb{P}^2$ blown-up at one or two points. Thus a Fano surface is Kähler–Einstein if and only if its Futaki invariant vanishes identically.*

We will not prove this theorem in this book (see [Tos10] for an updated treatment), however the interested reader is now well prepared for it.

It was conjectured for some time that the non-vanishing of the Futaki invariant was the only obstruction to the existence of a Kähler–Einstein metric. A special case is well understood and confirms this guess, as was shown by Wang and Zhu in [WZ04]:

Theorem 15.37. *A toric Fano manifold admits a (toric) Kähler–Einstein metric if and only if the Futaki invariant vanishes .*

It turns out however that the Futaki character is not the only obstruction to the existence of Kähler–Einstein metrics of positive curvature (see [T, Section 7.1] for a detailed example). The right obstruction is encoded in the notion of K-stability that we now introduce.

15.4.2 K-stability It has been conjectured by Yau (see [Yau93, Yau96]) that a Fano manifold admits a Kähler–Einstein metric if and only if it is "stable" in some algebro-geometric sense. The conjecture was later on refined by Tian and Donaldson, who extended it to the context of constant scalar curvature metrics.

The leading idea is to characterize the analytic non-linear problem of solving $(\mathrm{MA})_1$ by an algebro-geometric criterion, in the spirit of the solution to the Hermite–Einstein equation (see [Siu87]).

There are many possible notions of stability and many reasons to expect that such a result should hold. We refer the reader to the survey of Phong and Sturm [PS10] for an introduction to these ideas and motivations. In this section we very briefly introduce the stability notion that works well for the Kähler–Einstein problem.

After many attempts, the right notion of stability (in the Kähler–Einstein context) turned out to be the K-stability introduced by Tian in [Tian97]. This notion was generalized by Donaldson [Don02] to pairs (X, L) where $L \to X$ is an ample line bundle (not necessarily equal to the anticanonical bundle). Donaldson formulated a more algebraic version of K-stability that we introduce below.

Definition 15.38. *A test configuration for (X, L) of exponent r is a $\mathbb{C}^*$-equivariant, flat, polarized family $(\mathcal{X}, \mathcal{L})$ over $\mathbb{C}$ whose generic fiber is isomorphic to (X, L^r).*

Following [LX14] we also assume that the total space $\mathcal{X}$ is normal and say the test configuration is trivial if it is equivariantly isomorphic to a product test configuration $X \times \mathbb{C}$.

The central fiber (X_0, L_0) of a test configuration $(\mathcal{X}, \mathcal{L})$ has an induced $\mathbb{C}^*$-action. We let A_k denote the infinitesimal generator of this action on the N_k-dimensional vector space $H^0(X_0, L_0^k)$, so that the eigenvalues of A_k are the weights of the action. There are expansions

$$\dim H^0(X_0, L_0^k) = a_0 k^n + a_1 k^{n-1} + O(k^{n-2})$$

and

$$\mathrm{tr}(A_k) = b_0 k^{n+1} + b_1 k^n + O(k^{n-1}).$$

Definition 15.39. *The Donaldson–Futaki invariant of a test configuration $(\mathcal{X}, \mathcal{L})$ is*

$$\mathrm{DF}(\mathcal{X}, \mathcal{L}) := \frac{a_1 b_0 - a_0 b_1}{a_0^2}.$$

Example 15.40. *Assume $X = \mathbb{P}^2$ is the complex projective plane and $L = \mathcal{O}(1)$ is the hyperplane bundle. The family of conics*

$$\{[x : y : z] \in \mathbb{P}^2 \,\big|\, xy = tz^2\} \quad \text{for } t \in \mathbb{C}$$

defines a test configuration $(\mathcal{X}, \mathcal{L})$ for (X, L) of exponent 2 which degenerates the smooth conic into a union of two lines $\{x = 0\} \cup \{y = 0\}$. We let the reader check in Exercise 15.13 that $\mathrm{DF}(\mathcal{X}, \mathcal{L}) = 1/8 > 0$.

Definition 15.41. *The pair (X, L) is K-stable if $\mathrm{DF}(\mathcal{X}, \mathcal{L}) > 0$ for all non-trivial test configurations $(\mathcal{X}, \mathcal{L})$.*

The original definition of K-stability by Tian [Tian97] is slightly weaker as it only requires $\mathrm{DF}(\mathcal{X}, \mathcal{L}) > 0$ for special test configurations with mild singularities (in the sense of the Minimal Model Program, see the next chapter). It follows from the work of Li and Xu [LX14] that these two definitions are equivalent in the Fano case when $L = -K_X$ is the anticanonical bundle. A very influential conjecture in the field is:

Conjecture 15.42 (Yau, Tian, and Donaldson). *Let (X, L) be a polarized projective algebraic manifold with no non-zero holomorphic vector field. There exists a Kähler metric with constant scalar curvature in $c_1(L)$ if and only if (X, L) is K-stable.*

When X admits holomorphic vector fields which can be lifted to L, then (X, L) is never K-stable, as one can find in that case test configurations with total space $\mathcal{X} = X \times \mathbb{C}$ and non-positive Futaki invariant. One has to use a refined notion of stability (K-polystability), we refer the reader to [PS10, Sz14] for a discussion of the corresponding subtleties, as well as much more information on stability notions.

15.4.3 The Kähler–Einstein case

Tian has shown in [Tian97] that a Kähler–Einstein metric is necessarily K-stable (this is somehow the easy part of the conjecture, see [Ber16] for a recent account).

A major recent breakthrough is the converse result proved by Chen, Donaldson, and Sun in [CDS1, CDS2, CDS3]:

Theorem 15.43. *Let X be a Fano manifold. Then X admits a Kähler–Einstein metric if and only if it is K-stable.*

The proof is a real tour de force. It would require a whole book to explain it in details, so we just say a few words about the method.

The starting point is to fix a smooth divisor $D \in -mK_X$ (if there exists one such in $-K_X$, just take $m = 1$). The authors consider a continuity method for metrics with cone singularities along D,

$$\mathrm{Ric}(\omega_t) = t\omega_t + \frac{1-t}{m}[D],$$

where $[D]$ denotes the current of integration along the divisor D.

Donaldson has shown in [Don12] that the set of parameters t for which one can solve this equation is open. One can show the existence of Kähler–Einstein metrics with small cone angle, so these equations are solvable on some maximal interval $[\varepsilon, T_{\max})$ and the goal is to show that $T_{\max} = 1$ and $\omega_t \to \omega_{KE}$ as $t \to T_{\max} = 1$.

A key step in the proof of Chen, Donaldson, and Sun is to show that along a subsequence t_i, the manifolds (X, ω_{t_i}) converge in the Gromov–Hausdorff sense to a $\mathbb{Q}$-Fano variety V and that the divisor $D \subset X$ converges to a (Weil) divisor $\Delta \subset V$ in such a way that V admits a singular Kähler–Einstein metric with cone singularities along Δ where (V, Δ) is log smooth. Moreover, there are embeddings

$F_i : X \to \mathbb{P}^N$ and $F : V \to \mathbb{P}^N$, N large enough, such that $(F_i(X), F_i(D))$ converge to $(F(V), F(\Delta))$ in an algebro-geometric sense. A previous important result along these lines had previously been achieved by Donaldson and Sun in [DS14].

One then runs into the following alternative: either $(F(V), F(\Delta))$ belongs to the $\mathrm{SL}(N+1,\mathbb{C})$-orbit of $(F_i(X), F_i(D))$ and we can solve the equation for $t = T_{\max}$, or there exists a test configuration $(\mathcal{X}, \mathcal{L}_D)$ with central fiber (V, Δ) with $\mathrm{DF}(\mathcal{X}, \mathcal{L}_D) \leq 0$, showing that $(X, -K_X)$ is not K-stable.

We are going to discuss the definition and existence of $\mathbb{Q}$-Fano varieties and singular Kähler–Einstein metrics in the next chapter. Needless to say, there are many other difficulties one has to overcome which are beyond the scope of this book.

Remark 15.44. *Kähler metrics with conic singularities have attracted much attention, especially after Donaldson's work* [Don12]. *We refer the reader to* [Bre13, JMR16, CGP13, DaSo15, Rub14] *and the references therein.*

Apart from the case of toric surfaces [Don09] and the Kähler–Einstein cases, the Yau–Tian–Donaldson conjecture is still fully open in the more general context of constant scalar curvature metrics. We refer the reader to [PS10, Sz14] for recent surveys on this important conjecture.

15.5 Exercises

Exercise 15.1. *Let $\omega_t = \omega + dd^c(t\varphi)$ be a smooth perturbation of a Kähler form ω. Show that the scalar curvature $s(\omega_t)$ of ω_t admits the asymptotic expansion*

$$s(\omega_t) = s(\omega) + t\left(-\sum_{\alpha,\beta} g^{\alpha\beta}\frac{\partial^2 \Delta_\omega \varphi}{\partial z_\alpha \partial \overline{z}_\beta} - \sum_{\alpha,\beta,\gamma,\delta} g^{\alpha\delta}\frac{\partial^2 \varphi}{\partial z_\gamma \partial \overline{z}_\delta} g^{\gamma\beta} R_{\alpha\beta}\right) + O(t^2).$$

Exercise 15.2. *Let (X,ω) be a compact Kähler manifold of dimension n and let φ_t be a path of Kähler potentials in $\mathcal{H}$. Check that*

$$\frac{d}{dt}\left\{\sum_{j=0}^{n-1}\int_X \varphi_t \,\mathrm{Ric}(\omega)\wedge\omega_t^j\wedge\omega^{n-1-j}\right\} = n\int_X \dot{\varphi}_t \,\mathrm{Ric}(\omega)\wedge\omega_t^{n-1}.$$

Exercise 15.3. *Assume X is Fano and set $\alpha = c_1(X)$. Pick a Kähler form ω in α and set $\mu = \omega^n/V_\alpha$. Show that the formula defining $\mathcal{M}_\mu$ in Chapter 11 coincides with the one obtained in Proposition 15.3.*

Exercise 15.4. *Let (X,ω) be a compact Kähler manifold and fix two continuous ω-psh functions φ_0, φ_1. Let $(\varphi_t)_{0\leq t\leq 1}$ denote the continuous Mabuchi geodesic joining φ_0 to φ_1.*

(1) *Show that* $\sup_{t\in[0,1]}\varphi_t = \max(\varphi_0,\varphi_1)$.

(2) *Use Kiselman's minimum principle* (*Theorem* 2.44 *in Chapter* 2) *to show that*

$$\inf_{t\in[0,1]}\varphi_t = P(\min(\varphi_0,\varphi_1))$$

coincides with the ω*-psh envelope of the minimum of* φ_0 *and* φ_1 (*see* [DR16] *for more details*).

Exercise 15.5. *Let* X *be a compact Kähler manifold with no holomorphic vector field. Assume there exists an extremal Kähler metric* ω. *Show that there is an extremal Kähler metric in any Kähler class sufficiently close to* $\{\omega\}$ (*see* [LS94] *and* [Szek] *for some help*).

Exercise 15.6. *Assume* $X = \mathbb{CP}^1$ *is the Riemann sphere and* ω *is the Fubini–Study Kähler form. Consider*

$$\varphi_0[z] = \max(\log|z_1|,\log|z_0|) - \log\|z\| \in PSH(\mathbb{P}^1,\omega_{\mathrm{FS}})$$

and

$$\varphi_j[z] = (1-\varepsilon_j)\varphi_0[z] + \varepsilon\max(\log|z_1|,\log|z_0| - C_j) \in PSH(\mathbb{P}^1,\omega_{\mathrm{FS}}),$$

where ε_j *decreases to* 0, *while* C_j *increases to* $+\infty$. *Show that the Mabuchi distance between* φ_j *and* φ_0 *is*

$$d(\varphi_j,\varphi_0) = \frac{C_j\varepsilon_j^{3/2}}{\sqrt{3}}$$

and deduce that, as $j\to+\infty$,

- $\varphi_j\to\varphi_0$ *in* L^1 *iff* $\varepsilon_j\to 0$;
- $\varphi_j\to\varphi_0$ *in* L^∞ *iff* $\varepsilon_j C_j\to 0$;
- $\varphi_j\to\varphi_0$ *in* $(\mathcal{E}^1(X,\omega)^1, I_1)$ *iff* $\varepsilon_j^2 C_j\to 0$;
- $\varphi_j\to\varphi_0$ *in* $(\mathcal{E}^2(X,\omega),d)$ *iff* $\varepsilon_j^3 C_j^2\to 0$.

The convergence in $(\mathcal{E}^1(X,\omega)^1, I_1)$ *is here* (*i.e., in dimension* $n=1$) *the convergence in the Sobolev norm* $W^{1,2}$ (*see* [G14] *for more details*).

Exercise 15.7. *Let* X_r *be the blow-up of* $\mathbb{P}^2$ *at finitely many points* $p_1,\dots,p_r$. *Show that* X *is not Fano if* $r\geq 9$, *or if the points are not in general position. Justify conversely that a Fano surface is isomorphic to* X_r *or to* $\mathbb{P}^1\times\mathbb{P}^1$ (*see* [Dolgachev] *for some help*).

Exercise 15.8. *Let* X_1 *be the blow-up of* $\mathbb{P}^2$ *at the point* $[1:0:0]$ *and let* V *be the holomorphic vector field on* X_1 *whose flow corresponds to*

$$\Phi_s[1:x:y] = [1:e^s x:e^s y].$$

Check that $\mathrm{Fut}_{X_1}(V) = -\frac{4\pi^2}{3}$ (*see* [T, Chapter 3] *for some help*).

Exercise 15.9. *Let X_r denote the blow-up of $\mathbb{P}^2$ at r points $p_1, \dots, p_r$ in general position. Show that*

- $H^0(X_1, TX_1) \simeq \{X \in H^0(\mathbb{P}^2, T\mathbb{P}^2) \mid X(p_1) = 0\}$;
- $H^0(X_2, TX_2) \simeq \{X \in H^0(\mathbb{P}^2, T\mathbb{P}^2) \mid X(p_1) = X(p_2) = 0\}$;
- $H^0(X_3, TX_3) \simeq \mathbb{C}^3$;
- $H^0(X_r, TX_r) = \{0\}$ *if* $r \geq 4$,

and check that the Futaki character vanishes identically iff $r \neq 1, 2$ (see [T, Chapter 3] *for some help*).

Exercise 15.10. *For $u \in \mathcal{C}^\infty(X, \mathbb{R})$ let V_u denote the $\mathcal{C}^\infty$-smooth vector field such that*

$$\delta_{V_u}\omega = \overline{\partial} u,$$

where $\delta_V \omega$ denotes the contraction (interior product) of ω with respect to V.

Show that the following Bochner-type formula holds:

$$\|\Delta_\omega u\|^2_{L^2(\omega)} = \|\overline{\partial} V_u\|^2_{L^2(\omega)} + \int_X \mathrm{Ric}(\omega)(V_u, V_u)\omega^n.$$

Exercise 15.11. *Let X be a compact Kähler manifold and pick a Kähler class $\alpha \in H^{1,1}(X, \mathbb{R})$. Let $t \mapsto \omega_t$ be a curve of Kähler forms in α. For $V \in H^0(X, TX)$ we consider*

$$g(s, t) := \mathcal{M}(\Phi_s^* \omega_t),$$

where Φ_s denotes the flow of the holomorphic vector field V.

Show that $(s, t) \mapsto \mathcal{M}(\Phi_s^* \omega_t)$ is convex at $(0, 0)$ and $s \mapsto g(s, t)$ is linear (this completes the proof of Theorem 15.32).

Exercise 15.12. *Let X be a Fano manifold with no holomorphic vector field. Show that the set of Kähler–Einstein metrics is discrete.*

Exercise 15.13. *Assume $X = \mathbb{P}^2$ is the complex projective plane and $L = \mathcal{O}(1)$ is the hyperplane bundle. The family of conics $\{[x : y : z] \in \mathbb{P}^2 \mid xy = tz^2\}$ for $t \in \mathbb{C}$ defines a test configuration $(\mathcal{X}, \mathcal{L})$ for (X, L) of exponent 2 which degenerates the smooth conic into a union of two lines $\{x = 0\} \cup \{y = 0\}$. Show that*

$$\mathrm{DF}(\mathcal{X}, \mathcal{L}) = 1/8$$

(*see* [Szek] *for some help*).

Exercise 15.14. *Let X be the double cover of $\mathbb{P}^n$ ramified along a smooth hypersurface of degree $2n$. Show that X is a Kähler–Einstein Fano manifold* (*see* [AGP06] *for help and further similar constructions*).

Exercise 15.15. *Let X be a Fano manifold and let*

$$\alpha(X) = \alpha(c_1(-K_X))$$

denote the uniform integrability index associated to the anticanonical bundle $-K_X$.

(i) *Show that $\alpha(X) > 2/3$ when $X = X_r$ is the blow-up of $\mathbb{P}^2$ at r points in general position, with $7 \leq r \leq 8$.*
(ii) *Construct a smooth cubic in $\mathbb{P}^3$ such that $\alpha(X) = 2/3$ (see* [Chel08] *for more computations of these invariants).*
(iii) *Show that a smooth cubic surface in $\mathbb{P}^3$ is isomorphic to the blow-up of $\mathbb{P}^2$ at six points in general position.*

Exercise 15.16. *Show that the blow-up X of $\mathbb{P}^3$ at two points is not a Fano manifold. What can you say about the positivity properties of the anticanonical bundle K_X^*?*

16 Singularities and the Minimal Model Program

The classification of higher-dimensional complex algebraic manifolds (the so called Mori program or MMP=Minimal Model Program) requires one to work on varieties that are "mildly singular". In this context one can still make sense of the canonical bundle, the Ricci curvature, the Kähler–Einstein equation.

The latter again boils down to solving a complex Monge–Ampère equation which however becomes degenerate, due to the presence of singularities. Working in a desingularisation, one is thus led to study complex Monge–Ampère equations of the type

$$(\omega + dd^c\varphi)^n = e^{\lambda\varphi}\mu,$$

where

- ω is a smooth closed semi-positive differential $(1,1)$-form representing a cohomology class α that is *big* ($\int_X \omega^n > 0$), rather than Kähler,
- and $\mu = fdV$ is absolutely continuous with respect to Lebesgue measure with density $0 \leq f \in L^p$ for some $p > 1$ that looks locally like

$$f = \frac{\prod_i |f_i|^{2r_i}}{\prod_j |g_j|^{2s_j}},$$

  where the f_i's and g_j's are holomorphic functions.

We can thus apply the techniques that we have developed in previous chapters to show the existence of weak solutions and study their regularity properties.

16.1 Positivity properties of cohomology classes

In this section we review the various notions of positivity for cohomology classes of type $(1,1)$. Much more information (and proofs) can be found in [Dem, Dem13, Laz].

16.1.1 Algebraic notions Let $L \to X$ be a holomorphic line bundle on a compact projective algebraic manifold X. Recall that $H^0(X, L)$ denotes the finite-dimensional vector space of holomorphic sections of L. We fix $s_0, \dots s_N$ a basis of $H^0(X, L)$ and consider

$$\Phi_L : X \ni x \longmapsto [s_0(x) : \cdots : s_N(x)] \in \mathbb{CP}^N = \mathbb{P}(H^0(X, L)).$$

Definition 16.1. *The line bundle L is very ample if Φ_L is an embedding. The line bundle L is ample if L^k is very ample for $k \gg 1$ sufficiently large.*

The line bundle L is big if Φ_{L^k} is merely a birational embedding for $k \gg 1$ sufficiently large.

The important Nakai–Moishezon criterion asserts that L is ample if and only if for every $1 \leq j \leq n = \dim_{\mathbb{C}} X$ and every $Y \subset X$ subvariety of X of dimension j, one has

$$c_1(L)^j \cdot Y > 0.$$

The proof proceeds by induction on the dimension and uses the Riemann–Roch theorem (see [Laz, Theorem 1.2.23]).

Remark 16.2. *When $X = \mathbb{CP}^n$ is the complex projective space, every line bundle L is isomorphic to a multiple $\mathcal{O}(k)$ of the hyperplane bundle $\mathcal{O}(1)$. In this case $L = \mathcal{O}(k)$ is ample if and only if it is very ample if and only if $k \geq 1$.*

Note however that there are ample line bundles such that $H^0(X, L) = \{0\}$, hence which are far from being very ample. We let the reader construct an example on a compact Riemann surface in Exercise 16.1.

Definition 16.3. *A line bundle $L \to X$ is semi-ample if there exist $k \in \mathbb{N}^*$, a surjective holomorphic morphism $f : X \to Y$, with Y a projective variety, and an ample line bundle $A \to Y$ on Y such that $L^k = f^*A$.*

We also say that L^k is *globally generated*, i.e., the section ring

$$R(X, L) = \bigoplus_{m \geq 0} H^0(X, L^{mk})$$

is finitely generated. It follows from [Laz, Theorem 2.1.27] that these two notions are equivalent.

Definition 16.4. *A line bundle $L \to X$ is* nef *if for all irreducible curves $\mathcal{C} \subset X$, $L \cdot \mathcal{C} \geq 0$.*

Kleiman criterion (see [Laz, Corollary 1.4.10]) asserts that L is nef iff $c_1(L)$ is a limit of ample classes, iff $L^k \otimes A$ is ample for all $k \geq 1$ and every ample line bundle A.

It follows from this criterion that a line bundle L associated to an irreducible curve $\mathcal{C}$ on a projective surface X is nef if an only if $\mathcal{C}^2 \geq 0$. Similarly one can deduce the following useful invariance property: if $f : X \to Y$ is a proper holomorphic map and $L \to Y$ is given, then

$$L \text{ is nef} \iff f^*L \text{ is nef}.$$

Definition 16.5. *A line bundle $L \to X$ is effective if $H^0(X, L) \neq 0$, i.e., there exists a non-zero holomorphic section $s \in H^0(X, L)$. In this case one often identifies L and the divisor $D = (s = 0)$.*

16.1.2 Metric notions We assume here that X is a compact Kähler manifold equipped with a Kähler form ω_X, and fix $\alpha \in H^{1,1}(X, \mathbb{R})$.

Definition 16.6. *The cohomology class α is Kähler if it can be represented by a Kähler form.*

The class α is big if it can be represented by a Kähler current, i.e., a positive closed current which dominates a Kähler form.

We let $H^{1,1}_{\mathrm{Kahl}}(X, \mathbb{R})$ and $H^{1,1}_{\mathrm{big}}(X, \mathbb{R})$ denote the sets of Kähler and big classes. These are open convex cones.

When $\alpha = c_1(L)$ is the first Chern class of a holomorphic line bundle[1], Kodaira's embedding theorem ensures that

$$L \text{ is ample if and only if } c_1(L) \text{ is a Kähler class.}$$

It was similarly shown by Demailly that

$$L \text{ is big if and only if } c_1(L) \text{ is big.}$$

The proof goes roughly as follows: fix an ample divisor A and assume L is big. The bigness assumption ensures that $\dim_{\mathbb{C}} H^0(X, L^k)$ grows like k^n as $k \to +\infty$, while $\dim_{\mathbb{C}} H^0(A, L^k_{|A})$ grows like k^{n-1}. It therefore follows from the exact sequence

$$0 \longrightarrow H^0(X, L^k \otimes A^{-1}) \longrightarrow H^0(X, L^k) \longrightarrow H^0(A, L^k_{|A})$$

that $H^0(X, L^k \otimes A^{-1}) \neq \{0\}$ if k is large enough (this argument is usually called Kodaira's lemma). Pick $s \in H^0(X, L^k \otimes A^{-1}) \setminus \{0\}$ and fix a Kähler form ω_A representing $c_1(A)$. The Kähler current

$$T = \frac{1}{k}[s = 0] + \frac{1}{k}\omega_A$$

represents $c_1(L)$, hence the latter is a big cohomology class.

The converse implication can be seen to be a consequence of Hörmander's L^2-estimates (see [Dem13, Theorem 3.4]).

Definition 16.7. *The cohomology class α is* nef *if it is the limit of Kähler classes, i.e., if for all $\varepsilon > 0$, there exists a smooth closed differential form θ_ε representing α such that $\theta_\varepsilon \geq -\varepsilon\omega_X$.*

If we can set $\varepsilon = 0$, we then say that α is semi-positive: in this case α can be represented by a smooth semi-positive form $\theta \geq 0$.

It follows from the Kleiman criterion that when $\alpha = c_1(L)$ is the first Chern class of a holomorphic line bundle, α is nef if and only if L is nef.

[1] The first Chern class is an element of $H^2(X, \mathbb{Z})$, one actually means here the image of the first Chern class in $H^2(X, \mathbb{R})$ induced by the inclusion map $\mathbb{Z} \to \mathbb{R}$.

Definition 16.8. *The cohomology class α is* psef (*pseudoeffective*) *if it can be represented by a positive closed current of bidegree* $(1,1)$.

We let $H^{1,1}_{\text{nef}}(X,\mathbb{R})$ and $H^{1,1}_{\text{psef}}(X,\mathbb{R})$ denote the sets of nef and psef classes. These are closed convex cones, since

$$H^{1,1}_{\text{nef}}(X,\mathbb{R}) = \overline{H^{1,1}_{\text{Kahl}}(X,\mathbb{R})} \text{ and } H^{1,1}_{\text{psef}}(X,\mathbb{R}) = \overline{H^{1,1}_{\text{big}}(X,\mathbb{R})}.$$

Note also that

$$H^{1,1}_{\text{Kahl}}(X,\mathbb{R}) \subset H^{1,1}_{\text{nef}}(X,\mathbb{R}) \subset H^{1,1}_{\text{psef}}(X,\mathbb{R})$$

and

$$H^{1,1}_{\text{Kahl}}(X,\mathbb{R}) \subset H^{1,1}_{\text{big}}(X,\mathbb{R}) \subset H^{1,1}_{\text{psef}}(X,\mathbb{R}).$$

Remark 16.9. *Notice that when L is a holomorphic line bundle, L semi-ample implies $c_1(L)$ semi-positive, and L effective implies $c_1(L)$* psef*, but there is no equivalence in general.*

Demailly and Păun have established in [DP04] a Nakai–Moishezon criterion for transcendental classes:

Theorem 16.10. *Let (X,ω) be a compact Kähler manifold. Then*

$$H^{1,1}_{\text{Kahl}}(X,\mathbb{R}) = \{\alpha \in H^{1,1}(X,\mathbb{R}) |\, \alpha^j \cdot \{\omega\}^{k-j} \cdot Y > 0,\ \forall 1 \le j \le k \le n, \\ \forall Y \text{ analytic subvariety of dim } k.\}$$

The delicate proof uses in an essential way Yau's theorem!

16.1.3 Positivity of the canonical bundle As we have already noticed, the positivity properties of the canonical bundle play a fundamental role in the classification of projective algebraic varieties (both from a birational and a differential point of view).

We briefly mention in this paragraph some cornerstone notions and results that illustrate this. Recall first from Chapter 7 that the Kodaira dimension $\text{kod}(X)$ of a manifold X is the integer $\nu_X \in \{-\infty, 0, \dots, n\}$, where $n = \dim_{\mathbb{C}} X$, such that there exists $C > 0$ such that

$$\frac{1}{C} k^{\nu_X} \le \dim_{\mathbb{C}} H^0(X, K_X^k) \le C k^{\nu_X}$$

for all $k \in \mathbb{N}$ sufficiently divisible. The convention here is that $\text{kod}(X) = -\infty$ if $H^0(X, K_X^k) = 0$ for all k.

The greater ν_X, the more positive K_X. Manifolds with $\text{kod}(X) = n = \dim_{\mathbb{C}} X$ are those manifolds such that K_X is big, they are called manifolds of general type.

It has been shown by Birkar, Cascini, Hacon, and McKernan [BCHM10] that the canonical ring

$$R_X = \bigoplus_{m\geq 0} H^0(X, K_X^m)$$

is finitely generated. This allows, in particular, to produce a canonical model for manifolds of general type (see the corresponding discussion in Section 16.2.

Let us mention the following beautiful result of Boucksom, Demailly, Păun, and Peternell [BDPP13], whose proof can also be found in [Laz, Corollary 11.4.20]:

Theorem 16.11. *Let X be a projective manifold. Either K_X is pseudoeffective, or X is uniruled, i.e., covered by rational curves.*

The Minimal Model Program (MMP) aims at producing a birational model V of a given manifold X such that K_V is nef. This is stronger than asking for K_V to being psef. The *abundance conjecture* (one of the main open problems in this area) asks furthermore whether K_V nef implies K_V semi-ample.

16.1.4 Examples When X is a *complex homogeneous manifold*, i.e., when the automorphism group $\mathrm{Aut}(X)$ acts transitively on X, one can regularize positive currents inside their cohomology class using mollifiers. In this case Kähler $\iff$ big and

$$\text{psef} \iff \text{nef} \iff \text{semi-positive}.$$

These very special manifolds are completely classified (projective spaces, grassmanians, flag manifolds, complex tori and products of such). The positive cones are more delicate to describe in general, as the next examples suggest.

16.1.4.1 Blow-ups Let $\pi : X \to \mathbb{P}^2$ be the blow-up of $\mathbb{P}^2$ at one point p. This is a geometrically ruled surface over $\mathbb{P}^1$. We let F denote a generic fiber (the strict transform of a line passing through p) and E denote the exceptional divisor. The corresponding cohomology classes generate $H^{1,1}(X, \mathbb{R})$, we still denote them by F, E.

Proposition 16.12.

(i) $H^{1,1}_{\text{psef}}(X, \mathbb{R}) = \{aF + bE \,|\, 0 \leq a, b\}$;

(ii) $H^{1,1}_{\text{big}}(X, \mathbb{R}) = \{aF + bE \,|\, 0 < a, b\}$;

(iii) $H^{1,1}_{\text{nef}}(X, \mathbb{R}) = \{aF + bE \,|\, 0 \leq a \text{ and } 0 \leq b \leq a\}$;

(iv) $H^{1,1}_{\text{Kahl}}(X, \mathbb{R}) = \{aF + bE \,|\, 0 < a \text{ and } 0 < b < a\}$.

We let the reader establish this result as an Exercise 16.4.

One can also easily describe the positive cones of X_r, the blow-up of $\mathbb{P}^2$ at r generic points when r is small. The situation becomes however quite intricate as r increases, as the following celebrated problem of Nagata shows (see [Laz, Remark 5.1.14]):

Example 16.13 (Nagata conjecture). *Let $\pi : X_r \to \mathbb{P}^2$ be the blow-up of $\mathbb{P}^2$ at r generic points. Set $H = \pi^*\mathcal{O}(1)$, $E = \sum_{i=1}^r E_i$, and*

$$t_{\text{nef}} := \min\{t \mid tH - E \textit{ is } \text{nef}\}, \quad t_{\text{psef}} := \min\{t \mid tH - E \textit{ is } \text{psef}\}.$$

One can show that $t_{\text{nef}} t_{\text{psef}} = r$ and $t_{\text{psef}} \leq \sqrt{r} \leq t_{\text{nef}}$. All values of these numerical invariants are known for $r \leq 9$, but the following conjecture of Nagata is still largely open:

$$\textit{Is it true that } \ t_{\text{nef}} = t_{\text{psef}} = \sqrt{r} \ \textit{ for } r \geq 9?$$

The blow-up of $\mathbb{P}^2$ at one point is a particular example of a projective fiber bundle. Let $\mathcal{E} \to \mathcal{C}$ be a holomorphic vector bundle of rank $r \geq 2$ and let

$$X = \mathbb{P}(\mathcal{E}) \longrightarrow \mathcal{C}$$

be the corresponding projective bundle. Then $\dim_{\mathbb{C}} H^{1,1}(X, \mathbb{R}) = 2$ and one can give a reasonable description of the positive cones. The following result due to Miyaoka is particularly striking:

$$H^{1,1}_{\text{nef}}(X, \mathbb{R}) = H^{1,1}_{\text{psef}}(X, \mathbb{R}) \Longleftrightarrow \mathcal{E} \text{ is semi-stable.}$$

We refer to [Laz, Section 6.4.A] for the definition of semi-stability. In the next section we shall have a closer look at projective bundles over elliptic curves.

16.1.4.2 Demailly–Peternell–Schneider example

A nef class that is not semi-positive Consider the elliptic curve $C = \mathbb{C}/\mathbb{Z}[i]$ and the holomorphic vector bundle $E \to C$ of rank 2 defined as the quotient $E = \mathbb{C} \times \mathbb{C}^2/\mathbb{Z}[i]$, where the action of the lattice $\mathbb{Z}[i]$ is generated by the two automorphisms

$$g(x, z_1, z_2) = (x + 1, z_1, z_2) \quad \text{and} \quad h(x, z_1, z_2) = (x + i, z_1 + z_2, z_2).$$

and the projection is induced by $(x, z_1, z_2) \mapsto x$,

$$\pi : E \ni [x, z_1, z_2] \longmapsto [x] \in C.$$

We let $X = \mathbb{P}(E)$ denote the projective bundle associated to E. This is a geometrically ruled surface over C. Observe that

$$L_1 = \mathbb{C} \times \mathbb{C} \times \{0\}/\mathbb{Z}[i] \longrightarrow C$$

is a trivial sub-fiber-bundle of E (i.e., a holomorphic line bundle). Similarly,

$$E/L_1 = L_2 \simeq E \times \{0\} \times \mathbb{C} \longrightarrow C$$

is another trivial sub-bundle. Thus E is a non-trivial extension of trivial bundles,

$$0 \longrightarrow \mathcal{O} \longrightarrow E \longrightarrow \mathcal{O} \longrightarrow 0.$$

We finally consider $L = \mathcal{O}_{\mathbb{P}(E)}(1)$, the tautological quotient line bundle induced by

$$p^*E \longrightarrow \mathcal{O}_{\mathbb{P}(E)}(1) \longrightarrow 0,$$

where $p : X = \mathbb{P}(E) \to C$ is the canonical projection.

Note that L is nef (being the quotient of nef bundles), so one would expect to be able to represent its first Chern class by various positive closed $(1,1)$-currents with mild singularities. It has been however proved by Demailly, Peternell, and Schneider [DPS94] that this is not at all the case:

Proposition 16.14. *Let $D \subset X$ denote the closed curve, image of $(z_2 = 0)$. The current of integration along D is the only positive closed current representing $c_1(L)$.*

Proof. Observe that X is the quotient of $\mathbb{C} \times \mathbb{C}^2 \setminus \{0\}/G$, where the action of $G \simeq \mathbb{Z}[i] \times \mathbb{C}^*$ is generated by $g(x, z_1, z_2) = (x + 1, z_1, z_2)$,

$$h(x, z_1, z_2) = (x + i, z_1 + z_2, z_2), \quad \text{and} \quad \lambda(x, z_1, z_2) = (x, \lambda z_1, \lambda z_2).$$

We let $q : \mathbb{C} \times \mathbb{C}^2 \setminus \{0\} \to X$ denote the projection.

Observe that $D = q(z_2 = 0) \simeq C \times (1, 0)$ is isomorphic to the elliptic curve C. Let T be a positive closed current of bidegree $(1,1)$ cohomologous to the current of integration $[D]$. Its pull-back q^*T is well defined in $\mathbb{C} \times \mathbb{C}^2 \setminus \{0\}$ and trivially extends as a positive closed current of bidegree $(1,1)$ in $\mathbb{C}^3$, thus

$$q^*T = dd^c\psi, \text{ where } \psi \in PSH(\mathbb{C}^3)$$

is a plurisubharmonic function with various invariance properties.

It follows from the dd^c-lemma that $T = [D] + dd^c u$, hence

$$q^*T = [z_2 = 0] + u \circ q.$$

We can thus choose $\psi = \log|z_2| + u \circ q$, hence

$$\psi \circ g = \psi \circ h = \psi \quad \text{and} \quad \psi \circ \lambda = \log|\lambda| + \psi.$$

We are going to show that $\log|z_2| + cst$ are the only psh functions in $\mathbb{C}^3$ which satisfy these invariance properties. We let the reader check as Exercise 16.5 the following property:

Lemma 16.15. *Let v be a subharmonic function in $\mathbb{C}$ with logarithmic growth. If v is periodic, then it is constant.*

Fix $z = (z_1, z_2) \in \mathbb{C}^2$ and observe that $x \mapsto \psi(x, z_1, z_2)$ is 1-periodic by assumption. Now for all $a, b \in \mathbb{N}^*$,

$$\psi(x, z_1, z_2) = \log|b| + \psi(x - a - bi, z_1/b - z_2, z_2/b),$$

hence $x \mapsto \psi(x, z_1, z_2)$ has logarithmic growth. We infer that it is constant, and so $\psi(x, z_1, z_2) = \psi(z_1, z_2)$ is independent of x.

Fix $z_2 \neq 0$. The function $z_1 \mapsto \psi(z_1, z_2)$ is subharmonic with logarithmic growth and z_2-periodic, so it is constant as well, i.e., ψ only depends on z_2.

Thus $\psi = \psi(z_2)$ is such that $\psi(\lambda z_2) = \log|\lambda| + \psi(z_2)$, hence $\psi = \log|z_2| + cst$, and we are done. □

Nef and big class with singularities Following [BEGZ10, Example 5.4], we let now $Y := \mathbb{P}(V)$ be the projective bundle of hyperplanes in $V := E \oplus A$, where A is a given ample line bundle on C, and let $L' := \mathcal{O}_V(1)$ be the tautological bundle. The line bundle L' is nef since E and A are nef, and it is also big since A is big.

One can check that the non-ample locus of L' is exactly $X = \mathbb{P}(E) \subset \mathbb{P}(V) = Y$. But the restriction of L' to X is $\mathcal{O}_E(1)$, and positive currents can be restricted to any subvariety not entirely contained in their polar set. It follows that any positive current in the nef and big class $\alpha := c_1(L')$ has poles along D.

It follows from Exercise 16.3 that there are positive currents in $c_1(L')$ with zero Lelong number at all points; their potentials are nevertheless $-\infty$ along D.

16.2 Mildly singular varieties

16.2.1 The Minimal Model Program Let X be a manifold of general type, i.e., a compact Kähler manifold whose canonical bundle K_X is *big*: this means that there are asymptotically "many" pluricanonical sections (so many that one can birationally embed X into a complex projective space), or, equivalently, that one can find a positive singular current T which dominates a Kähler form and represents $c_1(K_X)$. This is the "right" higher-dimensional analogue of hyperbolic Riemann surfaces.

It has been shown recently by Birkar, Cascini, Hacon, and McKernan (see [BCHM10]) that the canonical ring

$$R_X = \bigoplus_{k \in \mathbb{N}} H^0(X, K_X^k)$$

is finitely generated, hence $X_{\text{can}} = \text{Proj}(R_X)$, the canonical model associated to X, has only canonical singularities (see definition below) and ample canonical bundle

$$K_{X_{\text{can}}} > 0.$$

It is natural to wonder whether there is a unique Kähler–Einstein metric on the regular part of X_{can}, generalizing the classical situation when K_X is ample (in which case $X = X_{\text{can}}$).

The Minimal Model Program (MMP for short, or Mori program) studies more generally the problem of classifying higher-dimensional algebraic manifolds up to birational equivalence. While the classification of algebraic surfaces (complex dimension two) has been understood for decades, the situation is much more complicated in (complex) dimension ≥ 3.

It has been realized long ago that one has to enlarge the picture and consider mildly singular varieties. The purpose of this section is to define the singularities we will have to deal with. We do not provide much detail, or proofs. For more information on singularities and the MMP, we refer the reader to [KM, Kollar, Deb01].

16.2.2 $\mathbb{Q}$-factorial varieties

16.2.2.1 Normal varieties

Definition 16.16. *A complex analytic space $(V, \mathcal{O})$ is reduced if one of the following equivalent conditions is satisfied:*

(1) *local holomorphic functions are continuous;*
(2) *for all $x \in V$, the local ring $\mathcal{O}_{V,x}$ has no nilpotent element.*

Definition 16.17. *A complex analytic space $(V, \mathcal{O})$ is normal if one of the following equivalent conditions is satisfied:*

(1) *for each open subset $U \subset V$, every meromorphic function in U which is holomorphic in U^{reg} extends continuously to U and defines a holomorphic function;*
(2) *each local ring $\mathcal{O}_{V,x}$ is integrally closed in his fractions field.*

The singular locus V^{sing} of a normal complex analytic space V has codimension at least 2; we set $V^{\mathrm{reg}} = V \setminus V^{\mathrm{sing}}$. A normal complex analytic space V is locally irreducible.

The sheaf of holomorphic functions $\mathcal{O}_V$ is the subsheaf of the sheaf of continuous functions on V consisting of the functions whose restriction to V^{reg} is holomorphic. It follows from Hartogs' extension theorem that any holomorphic function on V^{reg} extends to V, which means that $j_*\mathcal{O}_{V^{\mathrm{reg}}} = \mathcal{O}$, where $j : V^{\mathrm{reg}} \to V$ denotes the inclusion map.

16.2.2.2 Gorenstein varieties Every meromorphic n-form α on V^{reg} extends to V: if we let $\pi : X \to V$ be a resolution of singularities of V, then the meromorphic n-form $\pi^*\alpha$ defined on $\pi^{-1}V^{\mathrm{reg}}$ extends to a meromorphic n-form on X. Let $\omega_{V^{\mathrm{reg}}}$ be the canonical sheaf of the smooth variety V^{reg}. The sheaf $\omega_V = j_*\omega_{V^{\mathrm{reg}}}$ is a coherent analytic sheaf on V.

Similarly every meromorphic pluricanonical form on V^{reg} extends to V and $\omega_V^{[q]} = j_*\omega_{V^{\mathrm{reg}}}^q$, $q > 0$ is a coherent analytic sheaf on V.

Definition 16.18. *The variety V is 1-Gorenstein if one of the following equivalent conditions holds:*

(1) *Every $x \in V$ has an open neighborhood U such that U^{reg} carries an holomorphic n-form with an empty zero divisor.*
(2) *ω_V is a rank one locally free sheaf.*
(3) *Every $x \in V$ has an open neighborhood U such that $\omega_{U^{\mathrm{reg}}}$ is isomorphic to $\mathcal{O}_{V^{\mathrm{reg}}}|_U$.*

A local section of ω_V defining an holomorphic n-form without zeroes on V^{reg} will be called a local generator of ω_V. If furthermore V is Cohen–Macaulay, V is said to be Gorenstein.

One can check that any normal hypersurface is Gorenstein, as well as any quotient singularity of the type $\mathbb{C}^n/G$, where $G \subset \mathrm{SL}(n,\mathbb{C})$ is a finite subgroup. However, this class of singularities is not closed under taking quotient by cyclic groups, which motivates the introduction of $\mathbb{Q}$-Gorenstein varieties.

16.2.2.3 $\mathbb{Q}$-Gorenstein varieties

Definition 16.19. *The variety V is called $\mathbb{Q}$-Gorenstein if one of the following equivalent conditions is satisfied:*

(1) *Every $x \in V$ has an open neighborhood U such that U^{reg} carries a holomorphic pluricanonical form with an empty zero divisor.*
(2) *For every $x \in V$, there exist $N_x \in \mathbb{N}$ and an open neighborhood U of x such that $\omega_U^{[N_x]}$ is a rank one locally free sheaf.*
(3) *For every $x \in V$ there exist $N_x \in \mathbb{N}$ and an open neighborhood U of x such that $\omega_{U^{\mathrm{reg}}}^{N_x}$ is isomorphic to $\mathcal{O}_{V^{\mathrm{reg}}}|_U$.*

A local section of $\omega_V^{[N]}$ defining an holomorphic pluricanonical form without zeroes on V^{reg} will be called a local generator of $\omega_V^{[N]}$.

For every $x \in V$, the smallest N_x fulfilling condition (3) *near x is called the local index of V at x. The l.c.m. of all local indices, if finite, is called the index of V.*

Any quotient singularity of the type $\mathbb{C}^n/G$, where $G \subset GL(n,\mathbb{C})$ is a finite subgroup, is $\mathbb{Q}$-Gorenstein.

16.2.2.4 Weil and Cartier divisors A Weil divisor is a formal $\mathbb{Z}$-combination of subvarieties of codimension 1. A Cartier divisor is a Weil divisor which is locally defined by a single equation. Weil and Cartier divisors coincide on a smooth manifold, this is however not necessarily the case on a singular variety:

Example 16.20. *Consider*

$$V := \{(x,y,z) \in \mathbb{C}^3 |\ xy + z^2 = 0\}.$$

Then $D = \{(x,y,z) \in \mathbb{C}^3 |\ x = z^2 = 0\} \subset V$ is a Weil divisor which cannot be expressed by a single equation, hence it is not a Cartier divisor.

Note however that $2D$ is a Cartier divisor in the previous example: one then says that D is $\mathbb{Q}$-Cartier.

Definition 16.21. *A normal complex analytic space V is locally factorial* (*resp. $\mathbb{Q}$-factorial*) *if every Weil divisor is Cartier* (*resp. $\mathbb{Q}$-Cartier*).

Thus a normal variety which is locally factorial (resp. $\mathbb{Q}$-factorial) is Gorenstein (resp. $\mathbb{Q}$-Gorenstein). From a technical point of view, this notion is often used as follows: if $\pi : W \to V$ is a birational morphism with V an algebraic $\mathbb{Q}$-factorial, then the exceptional locus of π is a divisor (see [Deb01]).

16.2.2.5 Resolution of singularities

Definition 16.22. *Let V be a normal complex analytic space. A resolution (resp. log-resolution) of singularities of V is a surjective proper bimeromorphic morphism $\pi : X \to V$ with X smooth (resp. and $\mathrm{Exc}(\pi)$ a divisor with simple normal crossings).*

We let $\mathrm{Exc}(\pi)$ denote the exceptional locus of π, which is the set of points $x \in X$ at which π is not a local isomorphism.

A celebrated result of Hironaka asserts that an algebraic variety V admits (log-) resolutions (see [Kollar]). When $\dim_{\mathbb{C}} V \leq 2$ one can even find a unique resolution $\pi : X \to V$ that is *minimal*, i.e., such that any other resolution $\pi' : X' \to V$ factorizes through a morphism $f : X' \to X$ such that $\pi' = f \circ \pi$.

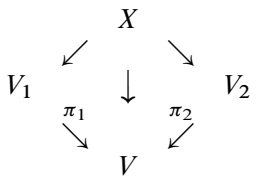
This is however no longer the case in dimension $\dim_{\mathbb{C}} V \geq 3$ and one rather seeks for a "minimal model" that is midly singular, as smooth minimal models need not exist (e.g., $\mathbb{C}^3/\pm 1$).

Example 16.23. *The quadratic cone*

$$V = \{(x, y, z, w) \in \mathbb{C}^4 |\ x^2 + y^2 + z^2 + w^2 = 0\}$$

is smooth, except at the origin. We blow-up the origin in $\mathbb{C}^4$ and consider the strict transform $X \xrightarrow{\pi} V$ of V.

One can check that the inverse image of $0 \in V$ under π is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$. It turns out that one can contract any of the two rulings of $\mathbb{P}^1 \times \mathbb{P}^1$ and obtain this way two models $V_1 \xrightarrow{\pi_1} V$, $V_2 \xrightarrow{\pi_2} V$ which are two different minimal resolutions of V:



The resolutions π_i are called *small resolutions*, as the exceptional loci $\pi_i^{-1}(0)$ are curves (here these are Riemann spheres $\mathbb{P}^1$), not divisors. Passing from V_1 to V_2 is called a *flop*.

The following result is often called "Zariski's main theorem", we will use it on several occasions.

Theorem 16.24. *Let $\pi : X \to V$ be a proper morphism, where X and V are normal complex analytic spaces. Then for all $p \in V$, $\pi^{-1}(p)$ is connected.*

16.2.3 Canonical and log terminal singularities The complex analytic varieties V we are considering in the sequel are *normal*, in particular their singular set V^{sing} is a complex subvariety of complex codimension ≥ 2. We let V^{reg} denote the set of smooth points.

16.2.3.1 Canonical singularities

Definition 16.25. *A variety V has only canonical singularities if and only if V is $\mathbb{Q}$-Gorenstein, of finite index N, and one of the following equivalent conditions is fulfilled:*

(1) *Let $\pi : X \to V$ be a resolution of singularities and let α be a local generator of $\omega_V^{[N]}$. Then the meromorphic pluricanonical form $\pi^*\alpha$ is holomorphic.*

(2) *Let $\pi : X \to V$ be a resolution. For every $m \in \mathbb{N}$,*

$$\pi_* \omega_X^{[Nm]} = \omega_V^{[Nm]}.$$

(3) (*Assuming V is an algebraic variety*) *Let $\pi : X \to V$ be a resolution. Then*

$$K_X \cong \pi^* K_V + \sum a_E E, \quad \text{with } a_E \geq 0,$$

where $\cong$ means numerical equivalence of $\mathbb{Q}$-Cartier divisors and the sum runs over the exceptional divisors of π.

Observe that it is enough to check the first two conditions for *some* resolution (to be done in Exercise 16.10). In the third condition Na_E is the order of vanishing of $\pi^*\alpha$ along the divisor E.

The importance of the class of canonical singularities comes from a theorem due to M. Reid [Reid79] (see also [Deb01, p. 174]):

Theorem 16.26. *Let X be a projective algebraic manifold of general type whose canonical ring $R = \bigoplus_{n\in\mathbb{N}} H^0(X, \omega_X^n)$ is of finite type. Then the canonical model of X, $X_{\text{can}} := \text{Proj}(R)$ has only canonical singularities. If $N = \text{Index}(X_{\text{can}})$, then $\omega_{X_{\text{can}}}^{[N]}$ is ample.*

The finiteness of the canonical ring has been established by Birkar, Cascini, Hacon, and McKernan [BCHM10]. X_{can} is a uniquely defined singular birational model of X.

16.2.3.2 Log-terminal singularities

Definition 16.27. *We say that V has only log-terminal singularities if V is $\mathbb{Q}$-Gorenstein, of finite index N, and the following holds: Let $\pi : X \to V$ be a log-resolution and let α be a local generator of $\omega_V^{[N]}$; then the pole along any component E of $\text{exc}(\pi)$ of the meromorphic N-canonical form $\pi^*\alpha$ on X is of order $< N - 1$.*

When V is algebraic, an equivalent formulation is, let $\pi : X \to V$ be a log-resolution. Then

$$K_X \cong \pi^* K_V + \sum_E a_E E, \quad \text{with } a_E > -1.$$

When $a_E > 0$, one says that the singularities are *terminal*. The Minimal Model Program predicts that terminal singularities are the right class of singularities for *minimal models* (a birational model V such that K_V is nef), while canonical singularities are those of canonical models.

Examples 16.28. *In higher dimension, quotient singularities are still log terminal. Fix $n > 0$ and let $H \subset \mathbb{CP}^{n+1}$ be a smooth degree d hypersurface. The affine cone over H has only canonical singularities if and only if $d \leq n + 1$.*

In particular, the ordinary double point $x^2 + y^2 + z^2 + t^2 = 0$ has only canonical singularities, but it is not a quotient singularity.

The hypersurface singularities of type A, D, or E are canonical.

16.2.4 Pairs It is important in practice to also consider *pairs*, i.e., a variety V together with a *boundary*, i.e., a sum of irreducible and reduced divisors with rational coefficients. All definitions seen so far extend to this context. The main definition of interest for us is that of Kawamata log terminal (klt) singularities:

Definition 16.29. *A pair (V, Δ) is* klt *if $K_V + \Delta$ is $\mathbb{Q}$-Cartier and for any log-resolution $\pi : X \to V$ of (V, Δ), we have the numerical equivalence of Cartier divisors:*

$$N(K_X + \Delta') \cong \pi^* N(K_V + \Delta) + \sum_{E \text{ exc.}} N a_E E$$

with $a_E > -1$, Δ' the proper transform of Δ in X (same multiplicities) and N an integer such that $N(K_V + \Delta)$ is Cartier.

Thus a variety V has only log terminal singularities if and only if $(V, \emptyset)$ is klt.

Example 16.30. *If V is smooth and $\Delta = \sum a_i \Delta_i$ is a simple normal crossing divisor, then (V, Δ) is* klt *if and only if $a_i < 1$.*

Example 16.31. *If $V = \mathbb{CP}^2$ and $\Delta = (xy^2 = w^3) \subset \mathbb{CP}^2$ is a cuspidal cubic, then $(V, a\Delta)$ is* klt *if and only if $a < 5/6$, as you will check in Exercise 16.11.*

It is also important, when compactifying the moduli space of canonically polarized varieties, to consider the following slightly more general type of singularities:

Definition 16.32. *A pair (V, Δ) is log canonical if $K_V + \Delta$ is $\mathbb{Q}$-Cartier and if for any log-resolution $\pi : X \to V$ of (V, Δ), we have the numerical equivalence of Cartier divisors:*

$$N(K_X + \Delta') \cong \pi^* N(K_V + \Delta) + \sum_{E \text{ exc.}} N a_E E$$

with $a_E \geq -1$, Δ' the proper transform of Δ in X (same multiplicities), and N an integer such that $N(K_V + \Delta)$ is Cartier.

16.2.5 Surfaces As already mentioned, a complex surface admits a unique minimal resolution $\pi : X \to V$. One can detect whether a model is minimal by using Castelnuovo's criterion: if $E \subset X$ is a smooth rational curve such that $E^2 = -1$, then one can contract E, i.e., there exists a morphism $\pi : X \to Y$ with Y smooth and $\pi : X \setminus E \to Y \setminus \{p\}$ an isomorphism, hence X is not minimal.

Singularities of surfaces have been studied long ago. Among them the Du Val singularities play a prominent role, these are the singularities isomorphic to one of the following A, D, or E types:

$$\begin{aligned}
A_n &\longleftrightarrow x y - z^{n+1} = 0 \\
D_n &\longleftrightarrow x^2 + y z^2 + y^{n-1} = 0 \quad (n \geq 4) \\
E_6 &\longleftrightarrow x^2 + y^3 + z^4 = 0 \\
E_7 &\longleftrightarrow x^2 + y^3 + y z^3 = 0 \\
E_8 &\longleftrightarrow x^2 + y^3 + z^5 = 0
\end{aligned}$$

We let the reader check (in Exercise 16.8) that these are precisely the canonical surface singularities:

Lemma 16.33. *Let S be a normal algebraic surface. The following are equivalent:*

(1) *S has only canonical singularities.*
(2) *S is locally analytically isomorphic to $X = \mathbb{C}^2/G$, where $G \subset \mathrm{SL}_2(\mathbb{C})$ is a finite subgroup.*
(3) *The exceptional divisors of the minimal resolution $\pi_{\min}$ of S, have simple normal crossings, their components are (-2) smooth rational curves, their incidence graphs are of type A, D, or E.*

Note that there are more finite subgroups of $\mathrm{GL}_2(\mathbb{C})$. We let the reader connect these to log terminal surface singularities:

Lemma 16.34. *The log terminal surface singularities are precisely the singularities of the form $X = \mathbb{C}^2/G$, $G \subset \mathrm{GL}_2(\mathbb{C})$ a finite subgroup.*

16.3 Kähler geometry on normal spaces

16.3.1 Kähler metrics on normal spaces We review in this section the basic notions of complex analysis on normal analytic spaces, following [EGZ09, Section 5].

16.3.1.1 Plurisubharmonic functions Let V be a normal analytic space of pure dimension n. A plurisubharmonic (psh) function φ on V is an upper semi-continuous function on V with values in $\mathbb{R} \cup \{-\infty\}$, which is not locally $-\infty$, and extends to a psh function in some local embedding $V \to \mathbb{C}^N$. The function φ is strongly psh (resp. $\mathcal{C}^0$, resp. $\mathcal{C}^\infty$) iff it extends to a strongly psh function (resp. $\mathcal{C}^0$, resp. $\mathcal{C}^\infty$) in some local embedding.

A classical result of Fornaess and Narasimhan ensures that a continuous function is psh iff its restriction to V^{reg} is so [FN80]. A bounded psh function on V^{reg} extends to V.

A pluriharmonic function on V is a real valued continuous function f on V such that one of the following equivalent conditions holds:

- f is locally the real part of a holomorphic function.
- Given a local embedding $V \to \mathbb{C}^N$, f extends locally to a pluriharmonic function on $\mathbb{C}^N$.
- $f|_{V^{\mathrm{reg}}}$ is pluriharmonic.

16.3.1.2 Kähler forms and positive currents

Definition 16.35. *A Kähler potential on V is a family $(U_i, \varphi_i)_{i\in I}$, where (U_i) is an open covering of V and φ_i a $\mathcal{C}^\infty$-smooth strictly psh function, such that $\varphi_i - \varphi_j$ is pluriharmonic on $U_i \cap U_j$.*

We define an equivalence relation on Kähler potentials by requiring that $(U_i, \varphi_i) \sim (V_j, \psi_j)$ iff $\varphi_i - \psi_j$ is pluriharmonic on $U_i \cap V_j$.

Definition 16.36. *A Kähler metric Ω on V is an equivalence class of Kähler potentials.*

Plurisubharmonic potentials are similarly defined as families $(U_i, \varphi_i)_{i\in I}$ where (U_i) is an open covering of V and φ_i a psh function, such that $\varphi_i - \varphi_j$ is pluriharmonic on $U_i \cap U_j$.

Definition 16.37. *A positive current on V is an equivalence classes of plurisubharmonic potentials.*

A positive current $\Omega = (U_i, \varphi_i)_{i\in I}/\sim$ is said to have L^∞_{loc} (resp. $\mathcal{C}^0$, resp. Hölder continuous, resp. $\mathcal{C}^\infty$) potentials if each φ_i is L^∞_{loc} (resp. $\mathcal{C}^0$, resp. Hölder continuous, resp. $\mathcal{C}^\infty$).

If a positive current Ω has locally bounded potentials, it is fully determined by the closed $(1,1)$ current Ω^{reg} on V^{reg} defined on U_i by $\Omega^{\mathrm{reg}} = dd^c\varphi_i$.

The notion of qpsh function is defined as in the smooth case:

Definition 16.38. *Let Ω be a Kähler metric on V with Kähler potential (U_i, φ_i). An upper semi-continuous function $\varphi : V \to \mathbb{R} \cup \{-\infty\}$ is said to be Ω-psh if for all i $\varphi_i + \varphi$ is psh on U_i for all i. The positive current whose potential is $(U_i, \varphi + \varphi_i)$ is denoted by $\Omega + dd^c\varphi$.*

Example 16.39. *Let $V = \mathbb{C}^2/\pm 1$. Let (x, y) be the usual affine coordinates on $\mathbb{C}^2$, (u, v, w) those on $\mathbb{C}^3$. One can realize V as the closed subscheme of $\mathbb{C}^3$ whose equation is*

$$V \simeq \{(u, v, w) \in \mathbb{C}^3 |\ u = x^2,\ v = y^2,\ w = xy\}.$$

We consider two natural Kähler metrics on V: the first one is smooth with potential

$$\varphi = |u|^2 + |v|^2 + 2|w|^2,$$

induced by a Euclidean type Kähler metric of $\mathbb{C}^3$, the second one is the Kähler current whose potential is

$$\psi = |u| + |v| = |x|^2 + |y|^2.$$

On V^{reg} it is the quotient of the Euclidean metric restricted to $\mathbb{C}^2 - \{0\}$.

Let $\pi : X \to V$ be the minimal resolution of $V = \mathbb{C}^2/\pm 1$ (blow-up $\mathbb{C}^3$ at the origin and take the strict tranform of V realized as the quadric cone). As it turns out, X can be identified with $\mathcal{O}_{\mathbb{P}^1}(-2)$ and π is the contraction of the zero section $\mathcal{C}$ (which has self-intersection -2). The holomorphic 2-form $dx \wedge dy$ is invariant under ± 1, hence descends to the quotient and yields a generator α of K_V such that $\pi^*\alpha$ does not vanish along $E = \pi^{-1}(0)$ and $dV = \pi^*\alpha \wedge \pi^*\alpha$ is the Lebesgue measure on X.

Observe that $(\pi^* dd^c\psi)^2 = c\, dV$, while $(\pi^* dd^c\varphi)^2 = c'|s|^2 dV$ vanishes along E (here $(s = 0) = \mathcal{C}$): this can be checked in local coordinates, observing that $\varphi = \psi^2$. The metric $dd^c\psi$ is thus the "right" one to consider on V, it is an example of an orbifold Kähler metric.

16.3.1.3 Chern–Weil forms and Hermitian metrics Let $\mathcal{PH}_V$ be the sheaf of real-valued pluriharmonic functions on V. By definition, a closed (1,1)-form on V is a section of the sheaf $\mathcal{C}^\infty_V/\mathcal{PH}_V$. We have the exact sequence:

$$\mathcal{C}^\infty(V) \longrightarrow \Gamma(V, \mathcal{C}^\infty_V/\mathcal{PH}_V) \xrightarrow{[\ \cdot\ ]} H^1(V, \mathcal{PH}_V) \longrightarrow 0.$$

A class in $H^1(X, \mathcal{PH}_X)$ will be called Kähler, if it is in the $[\ \cdot\ ]$ image of a smooth Kähler metric.

Let L be a holomorphic line bundle on V. The notion of smooth Hermitian metric on (V, L) is defined as in the smooth case. Let h be such a metric on (V, L).

Let $s \in H^0(U, L)$ be a nowhere zero local holomorphic section of L (a local generator of L) defined over the open subset $U \subset V$. Set $e^{-\varphi_s} := \|s\|_h^2$, where φ_s is a $\mathcal{C}^\infty$-smooth function on U. The current

$$dd^c\varphi_s$$

is a smooth closed $(1,1)$-form on V which does not depend on s; it is a positive current if φ_s is psh.

More generally, let $(U_i)_i$ be an open covering of V and let $s_i \in H^0(U_i, \mathcal{O}(L))$ be a local generator of L. Set $\varphi_i = \varphi_{s_i}$. The datum (U_i, φ_i) defines a smooth closed $(1,1)$-form on V.

Definition 16.40. *The Chern–Weil form of (V, L, h) (or of h) is the equivalence class of the datum (U_i, φ_i) constructed above. We will denote it by $c_1(L, h)$.*

It is immediate that $[c_1(L, h)]$ is independent of h. Hence there is a linear map

$$c_1 : \mathrm{Pic}(V) \longrightarrow H^1(V, \mathcal{PH}_V).$$

The connection with the smooth case is made by the observation that, if X is a compact Kähler manifold, $H^{1,1}(X, \mathbb{R}) = H^1(X, \mathcal{PH}_X)$.

Proposition 16.41. *Let V a compact normal complex analytic variety and let L a holomorphic line bundle on V.*

The space $H^1(V, \mathcal{PH}_V)$ is finite-dimensional. Every representative of $c_1(L)$ in $H^1(V, \mathcal{PH}_V)$ is the Chern–Weil form of a smooth Hermitian metric on L.

If there exists a smooth Hermitian metric h such that $c_1(L, h)$ is a Kähler form, then V is projective-algebraic and L is ample.

In proving this extension of Kodaira ampleness criterion, the most difficult task is to show that, in the last assertion, V is a Moishezon variety (i.e., bimeromorphic to a projective algebraic variety). This follows from Siu's solution of the Grauert–Riemenschneider conjecture [Siu84].

A singular metric on L is an expression $h = e^{-\varphi} h_{\mathrm{sm}}$, φ being qpsh (i.e., a locally smooth + psh function) and h_{sm} a smooth Hermitian metric. Its Chern–Weil form is the (quasi-)positive current

$$c_1(L, h_{\mathrm{sm}}) + dd^c\varphi.$$

16.3.2 Adapted volume forms

16.3.2.1 Monge–Ampère equations on normal Kähler spaces Let Ω be a Kähler metric on V. A classical result of P. Lelong states that if U is relatively compact in V, then U^{reg} is of finite volume with respect to the smooth volume form $\Omega^{\mathrm{reg}\, n}$.

As we have explained in the first part of this book, this has been generalized by E. Bedford and A. Taylor, who have studied Monge–Ampère measures for locally bounded psh functions. Since these measures do not charge proper analytic subsets, we obtain:

Proposition 16.42. *Let Ω be a positive current with L^∞_{loc} potentials on V. The Monge–Ampère measure $\Omega^{\mathrm{reg}\,n}$ is well defined on V^{reg} and satisfies $\int_{U^{\mathrm{reg}}} \Omega^{\mathrm{reg}\,n} < \infty$, for all relatively compact subset $U \subset V$.*

For any resolution $\pi : X \to V$, the Monge–Ampère measure $(\pi^*\Omega)^n$ is well defined on X and satisfies

$$\pi_*(\pi^*\Omega)^n = j_*\Omega^{\mathrm{reg}\,n}.$$

Moreover, if $\bar\pi : \bar X \to V$ is a resolution dominating π (i.e., $\bar\pi = \pi \circ \psi$ for some bimeromorphic proper holomorphic map $\psi : \bar X' \to X'$), then $\psi_*(\bar\pi^*\Omega)^n = (\pi^*\Omega)^n$.

The measure $\pi_*(\pi^*\Omega)^n$ is thus well defined on V and independent of the choice of resolution. We will call it the *Monge–Ampère measure of* Ω and denote it by Ω^n. The mass of this measure only depends on the cohomology class of Ω:

Lemma 16.43. *Assume V is compact. Let Ω_1, Ω_2 be two positive currents with L^∞_{loc} potential on V. If they are cohomologous, i.e., $\Omega_1 = \Omega_2 + dd^c\varphi$ for some $\varphi \in L^\infty(X)$, then $\int_V \Omega_1^n = \int_V \Omega_2^n$.*

More generally, all the basic facts of Bedford–Taylor's theory, the comparison principle included, are known to be valid for reduced complex analytic spaces [Bed81].

We can now reformulate some of the results we have obtained in previous chapters, for example:

Theorem 16.44. *Let V be a n-dimensional compact normal Kähler space and Ω be a smooth Kähler form on V. Then for every $f \in L^p(V, \Omega^n)$, $p > 1$, such that $\int_V f\,\Omega^n = \int_V \Omega^n$, there is a unique bounded Ω-psh function $\varphi : V \to \mathbb{R}$ such that*

$$(\Omega + dd^c\varphi)^n = f\,\Omega^n \text{ and } \sup_V \varphi = 0.$$

Proof. Let $\pi : X \to V$ be a resolution of V. We may define a semipositive big smooth form on X by $\omega = \pi^*\Omega$. It follows from Chapter 12 that we can solve uniquely

$$(\omega + dd^c\bar\varphi)^n = f \circ \pi\, \omega^n$$

where $\bar\varphi$ is a bounded function on X such that $\omega + dd^c\bar\varphi$ is a positive current.

Let F be a fiber of π and $i : F \to X$ the inclusion map. F is connected by Zariski's main theorem. Furthermore, $i^*\omega + dd^c i^*\bar\varphi$ is semipositive on F. Since $i^*\omega = 0$, it follows that $i^*\varphi$ is a bounded psh function on F. Hence, $i^*\bar\varphi$ is constant since F is compact. This implies that $\bar\varphi = \varphi \circ \pi$, where φ is a bounded function on V. We do have $(\Omega + dd^c\varphi)^n = f\,\Omega^n$. □

All results we have obtained for semi-positive and big forms on smooth manifolds can similarly be extended to this singular context (the nature of the singularity somehow matters, as it will soon become clear).

16.3.2.2 Adapted measures on log terminal Kähler spaces Let V be a n-dimensional Gorenstein Kähler space and Ω be a smooth Kähler form on V. Fix $x \in V$ and let α be a local generator of ω_V defined over an open subset $x \in U$; then

$$v = c_n \alpha \wedge \bar{\alpha}$$

is a positive definite volume form on U^{reg}, for an appropriate choice of the constant $c_n = i^n (-1)^{\frac{n(n+1)}{2}}$.

When V is merely $\mathbb{Q}$-Gorenstein of finite index N, we choose β to be a local generator of $\omega_V^{[N]}$ defined over an open subset $x \in U$ and we set

$$v = v_\beta = \left(i^{Nn} (-1)^{N \frac{n(n+1)}{2}} \beta \wedge \bar{\beta} \right)^{\frac{1}{N}} .$$

This is a positive definite volume form on U^{reg}.

An elementary but key observation is that log terminal singularities are the worst singularities we can allow in order for the volume form v to have finite mass:

Lemma 16.45. *For all $U_1 \Subset U$, $\int_{U_1^{\mathrm{reg}}} v < \infty$ iff V is log terminal.*

If V is log terminal, then the Radon measure $\mu = j_* v$ satisfies $\mu = f \Omega^n$ with $f \in L^{1+\varepsilon}(U_1, \Omega^n)$ for some $\varepsilon > 0$.

Proof. Let $\pi : X \to V$ be a log resolution. Write

$$K_X \cong \pi^* K_V + \sum a_E E.$$

Since $\mathrm{exc}(\pi)$ has simple normal crossings, at every $P \in E = \mathrm{exc}(\pi)$ there are local coordinates $(z^i)_{i=1,\dots,n}$ such that E is described by the equation $z^1 \cdots z^q = 0$. Let E_j be the divisor $z_j = 0$. We have:

$$\pi^* v = \prod_{j=1}^{q} |z^j|^{2a_{E_j}} \, d\lambda$$

where $d\lambda$ is a Lebesgue measure on X, hence the measure $\pi^* v$ has finite mass near P iff $\forall j, a_{E_j} > -1$. Thus $\int_{U_1^{\mathrm{reg}}} v < \infty$ iff $\forall E,\ a_E > -1$.

Let f_1 be the density of $\pi^* v$ with respect to $d\lambda$. Since f_1 is comparable to $\prod_{j=1}^{q} |z^j|^{2a_{E_j}}$ near P, it follows that f_1 belongs actually to $L^p(X, d\lambda)$ for some $p > 1$ when X is log terminal.

Let $D = 1/f$ be the density of Ω^n with respect to v. We will see below that D is bounded but it might have zeroes on E, hence f is unbounded in general. However, we will show that $f \circ \pi \in L^\alpha(X, d\lambda)$ for $\alpha > 0$ small enough, hence it follows from Hölder's inequality that

$$\int_{U_1^{\mathrm{reg}}} f^{1+\varepsilon} \Omega^n = \int_{\pi^{-1} U_1^{\mathrm{reg}}} f^\varepsilon f_1 d\lambda < +\infty$$

if $\varepsilon > 0$ is small enough.

Fix $x \in V$ and let $i : U_x \to \mathbb{C}^m$ be a local embedding of a neighborhood U_x of x. We consider the $\binom{m}{n}$ n-forms on U_x^{reg}

$$du^I = du^{i_1} \wedge \ldots \wedge du^{i_n},$$

where (u^i) is a set of affine coordinates on $\mathbb{C}^m$. Observe that Ω^n is comparable to $\sum_I v_{du^I}$ [2]. Since β is a local generator at x of $\omega_V^{[N]}$, we have $(du^I)^N = f_I \beta$, where $f_I \in \mathcal{O}_{V,x}$ is the germ of a holomorphic function at x. Therefore, Ω^n is comparable to $\sum_I |f_I|^{\frac{2}{N}} v$, hence D is comparable to $[\sum_I |f_I|^{\frac{2}{N}}]^{-1}$ near x.

The functions (f_I) generate an ideal $\mathcal{I}_x \subset \mathcal{O}_{V,x}$. Actually, the construction can be globalized to provide a coherent ideal sheaf $\mathcal{I} \subset \mathcal{O}_V$ cosupported on V^{sing}.

By [Hir64] we may assume that $\pi : X \to V$ is a log resolution of $(V, \mathcal{I})$, namely a log resolution of V with the additional property that the ideal sheaf $\pi^{-1}\mathcal{I}.\mathcal{O}_X$ which is the ideal sheaf of $\mathcal{O}_X$ generated by the family of holomorphic functions $(\pi^* f_I)_I$, satisfies

$$\pi^{-1}\mathcal{I}.\mathcal{O}_X = \mathcal{O}_X\left(-\sum N b_E E\right) \subset \mathcal{O}_X,$$

where $N b_E \in \mathbb{N}$ is a positive multiplicity attached to any exceptional divisor of π.

In local coordinates near $P \in X$, $\pi^* D$ is comparable to $\prod_j |z_j|^{2b_{E_j}}$, hence $\pi^*(f_1 D^{-\varepsilon})$ is comparable to $\prod_j |z_j|^{2(a_{E_j} - \varepsilon b_{E_j})}$. It follows that for every relatively compact subset $U_1 \Subset U$,

$$f \in L^{1+\varepsilon}(U_1, \Omega^n) \quad \text{iff } \forall E,\ \pi(E) \cap U \neq \emptyset \Longrightarrow a_E - \varepsilon b_E > -1. \qquad \square$$

Definition 16.46. *Assume V has only log terminal singularities. A positive definite adapted measure on V is a positive Radon measure locally of the form $e^f v$ where f is a bounded measurable function. A positive definite adapted measure has $\mathcal{C}^0$, $\mathcal{C}^\alpha$, $\mathcal{C}^\infty$ density if f does.*

Remark 16.47. *It follows from Lemma 16.45 that if a variety V is $\mathbb{Q}$-Gorenstein but has non-log terminal singularities, then v is a volume form on V^{reg} which does not extend to a measure on V.*

16.3.2.3 Adapted volume forms for klt pairs Let (V, Δ) be a kltpair. Let β be a local generator of $\mathcal{O}(N(K_V + \Delta))$. Then $\beta_{V^{\mathrm{reg}}}$ can be viewed as a meromorphic N-canonical form with a pole of order $N d_i$ on E_i, where $\Delta = \sum_i d_i E_i$ is the decomposition of Δ into prime divisors. Thus $v_{\beta_{V^{\mathrm{reg}}}}$ defines a volume form with poles on V^{reg}, namely $v_{\beta_{V^{\mathrm{reg}}}}$ is comparable to $\prod_i |s_i|^{-2d_i} d\lambda$, where s_i denotes the canonical section of $\mathcal{O}(E_i)$. If $v_{\beta_{V^{\mathrm{reg}}}}$ is a finite measure, then $d_i < 1$, but the converse is not true.

We have the following staightforward extension of Lemma 16.45:

[2] Note that the formula for v_β makes sense even if β is not a local generator.

Lemma 16.48. *Let $j' : V - \cup_i E_i \to V$ be the canonical inclusion. $j'_* v_\beta$ is a well-defined Radon measure on V iff (X, Δ) is* klt.

The definition of an adapted measure for a klt/lc pair is obtained similarly.

16.3.3 Estimating Lelong numbers in log-resolutions

It is important, especially when working on Fano varieties, to be able to compare the Lelong numbers of positive currents in various log-resolutions. This is the contents of this section which is taken from [BBEGZ11, Appendix A].

Let X be a normal complex space with a given point $x \in X$ and let φ be a psh function on X. Choose local generators (f_i) of the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_{X,x}$ and define the *slope* of φ at x by

$$s(\varphi, x) := \sup \left\{ s \geq 0 \mid \varphi \leq s \log \sum_i |f_i| + O(1) \right\} \in [0, +\infty[. \tag{16.3.1}$$

Since $\log \sum_i |f_i|$ only depends on the choice of generators up to a bounded term, it is clear that $s(\varphi, x)$ is independent of the choice of (f_i).

Example 16.49. *When X is smooth one can take $f_i = z_i$, a system of local coordinates, hence $s(\varphi, x)$ is the Lelong number of φ as defined in Chapter 2.*

For $f \in \mathcal{O}_{X,x}$ we have

$$s(\log |f|, x) = \overline{\mathrm{ord}}_x(f) := \lim_{m \to \infty} \tfrac{1}{m} \mathrm{ord}_x(f^m),$$

with

$$\mathrm{ord}_x(f) := \max \left\{ k \in \mathbb{N} \mid f \in \mathfrak{m}_x^k \right\}.$$

Remark 16.50. *The non-decreasing function*

$$\chi(t) := \sup_{\{\sum_i |f_i| < e^t\}} \varphi$$

is convex (generalized three-circle theorem), and we have

$$s(\varphi, x) = \lim_{t \to -\infty} \chi(t)/t.$$

This implies in particular that the supremum in (16.3.1) is attained.

Izumi's theorem [Izu81] states that for every resolution of singularities $\pi : \widetilde{X} \to X$ and every prime divisor $E \subset \widetilde{X}$ lying above $x \in X$, there exists a constant $C > 0$ such that

$$\mathrm{ord}_E(f \circ \pi) \leq C \, \overline{\mathrm{ord}}_x(f)$$

for all $f \in \mathcal{O}_{X,x}$. We extend here this result to psh functions:

Theorem 16.51. *Let $\pi : \widetilde{X} \to X$ be any resolution of singularities and let $E \subset \widetilde{X}$ be a prime divisor above $x \in X$. Then there exists $C > 0$ such that*

$$\nu(\varphi \circ \pi, E) \leq C s(\varphi, x)$$

for all psh functions φ on X.

Here

$$\nu(\varphi \circ \pi, E) = \min_{p \in E} \nu(\varphi \circ \pi, p)$$

is the generic Lelong number of $\varphi \circ \pi$ along E. Note that $\mathrm{ord}_E(f \circ \pi) = \nu(\varphi \circ \pi, E)$ with $\varphi = \log |f|$.

Corollary 16.52. *If φ is a psh function with $s(\varphi, x) = 0$ for some $x \in X$, then $\nu(\varphi \circ \pi, p) = 0$ for every resolution of singularities and every $p \in \pi^{-1}(x)$.*

Proof. Let $b : X' \to \widetilde{X}$ be the blow-up of $\widetilde{X}$ at the point $p \in \widetilde{X}$. Then $\pi' = \pi \circ b : X' \to X$ is yet another resolution of singularities. Set $E = \pi'^{-1}(p)$. This is a prime divisor to which we can apply Theorem 16.51 . The conclusion follows then by recalling the following classical interpretation of the Lelong number:

$$\nu(\varphi \circ \pi \circ b, E) = \nu(\varphi \circ \pi, p)$$

(see Exercise 16.13). □

Proof of Theorem 16.51. By Hironaka's theorem, we may assume that $\pi : \widetilde{X} \to X$ dominates the blow-up of X at x, so that the scheme-theoretic fiber $\pi^{-1}(x)$ is an effective divisor $\sum_i a_i E_i$. Note that $\sum_i E_i$ is connected by Zariski's "main theorem".

Set $b_i := \nu(\pi^* \varphi, E_i)$. Using the Siu decomposition of the positive current $T := dd^c \pi^* \varphi$ we may write $T = R + B$, where $B = \sum_i b_i E_i$ is an effective $\mathbb{R}$-divisor and R is a positive current such that $\nu(R, E_i) = 0$ for all i. We first claim that

$$s(\varphi, x) = \min_i \frac{b_i}{a_i}. \tag{16.3.2}$$

Indeed, if we write $\mathfrak{m}_x = (f_i)$ as above, then $\pi^* \log \sum_i |f_i|$ has analytic singularities described by the divisor $\pi^{-1}(x)$, i.e., locally on X we have

$$\pi^* \log \sum_i |f_i| = \sum_i a_i \log |z_i| + O(1),$$

where z_i is a local equation of E_i. We thus see that

$$\varphi \leq s \log \sum_i |f_i| + O(1) \iff \pi^* \varphi \leq \sum_i s a_i \log |z_i| + O(1)$$

locally on $\widetilde{X}$, and the positivity of $R = T - B$ shows that this holds iff $b_i = \nu(\pi^* \varphi, E_i) \geq s a_i$ for all i, hence the claim.

In view of (16.3.2), the desired statement amounts to an estimate $\max_i b_i \leq C \min_i b_i$ for some $C > 0$ independent of φ. Thanks to Lemma 16.53 below, this will hold if we can show that $-B|_{E_i}$ is pseudoeffective for all i. Since the restriction to each E_i of the cohomology class of $T = \pi^* dd^c\varphi$ is trivial, we are reduced to showing that $\{R\}|_{E_i}$ is pseudoeffective.

Since $\nu(R, E_i) = 0$, this follows from Demailly's regularization theorem. Let us recall the standard argument: by [Dem92], after perhaps shrinking X slightly about 0 we may write R as a weak limit of closed positive $(1,1)$-currents R_k with analytic singularities such that $\{R_k\} = \{R\}$, $R_k \geq -\varepsilon_k \omega$ for some $\varepsilon_k \to 0$, and R_k is less singular than R. In particular, we have $\nu(R_k, E_i) = 0$ for all i, which means that the local potentials of R_k are not entirely singular along E_i, so that $R_k|_{E_i}$ is a well-defined closed $(1,1)$-current. We thus see that $(\{R\} + \varepsilon_k\{\omega\})|_{E_i}$ is pseudoeffective for all k, and the claim follows. □

Lemma 16.53. *Let $E = \sum_i E_i$ be a reduced compact connected divisor on a Kähler manifold M. Let $B = \sum b_i E_i$ be an effective $\mathbb{R}$-divisor supported in E, and assume that $-B|_{E_i}$ is pseudoeffective for all i. Then there exists a constant $C > 0$ depending only on E such that $\max_i b_i \leq C \min_i b_i$.*

The proof to follow is directly inspired from [BFJ16, §6.1].

Proof. Let ω be a Kähler form on M. Thanks to the connectedness of E, we may index the E_i so that $B = \sum_{i=1}^N b_i E_i$ with $b_1 = \min_i b_i$, $b_r = \max_i b_i$ for $1 \leq r \leq N$, and $E_i \cap E_{i+1} \neq \emptyset$ for all $i = 1, \ldots, r-1$. For each i we have

$$(-D|_{E_i}) \cdot (\omega|_{E_i})^{n-2} = -\sum_j b_j c_{i,j} \geq 0,$$

with $c_{i,j} := \left(E_i \cdot E_j \cdot \omega^{n-2}\right)$, hence

$$\sum_{j \neq i} b_j c_{i,j} \leq b_i \, |c_{i,i}| \,. \tag{16.3.3}$$

Now $c_{i,j} \geq 0$ if $j \neq i$, and $c_{i,i+1} > 0$ for all i since E_i meets E_{i+1}. It follows that

$$b_{i+1} \leq \frac{|c_{i,i}|}{c_{i,i+1}} b_i$$

for all i, hence $\max_i b_i = b_r \leq C b_1 = \min_i b_i$ with $C := \prod_{i=1}^{r-1} \frac{|c_{i,i}|}{c_{i,i+1}}$ □

Remark 16.54. *Besides the slope $s(\varphi, x)$ considered above, Demailly introduced in [Dem85] a different generalization of Lelong numbers on normal complex spaces, defined as the intersection multiplicity*

$$\nu(\varphi, x) := (dd^c\varphi) \wedge \left(dd^c \log \sum_i |f_i| \right)^{n-1} (\{x\}) \,,$$

where (f_i) are generators of $\mathfrak{m}_x$, the definition being independent of that choice. When $\mathfrak{a} = (g_1, \dots, g_r)$ is an $\mathfrak{m}_x$-primary ideal and $\varphi = \log \sum_i |g_i|$, then $\nu(\varphi, x)$ computes the mixed (Hilbert–Samuel) multiplicity $\langle \mathfrak{a}, \mathfrak{m}_x, \dots, \mathfrak{m}_x \rangle$. In particular, for $\varphi = \psi$ we have $\nu(\psi, x) = m(X, x)$, the multiplicity of X at x. By Demailly's comparison theorem we have

$$\nu(\varphi, x) \geq s(\varphi, x) m(X, x),$$

and the inequality is strict in general. Using the notation of the proof of Theorem 16.51 and recalling that $-\pi^{-1}(x)$ is π-nef, we conjecture by analogy with the algebraic case that

$$\nu(\varphi, x) = (B \cdot (-\pi^{-1}(x))^{n-1}).$$

By Theorem 16.51, this would imply in particular that, conversely,

$$\nu(\varphi, x) \leq C s(\varphi, x)$$

for some $C > 0$ independent of φ.

16.4 Singular Kähler–Einstein metrics

We finally explain how the results obtained in Part 3 allow us to construct singular Kähler–Einstein metrics, following [EGZ09, BEGZ10, BBEGZ11, BG14].

16.4.1 Singular Ricci curvature Assume V is compact with only log terminal singularities and has index N, and let h^N be a smooth Hermitian metric on $\omega_V^{[N]}$. Let β be a local generator of $\omega_V^{[N]}$. Define $v_\beta(h)$ to be the volume form on V^{reg}:

$$v(h) = v_\beta(h) = \left(i^{Nn} (-1)^{N \frac{n(n+1)}{2}} \frac{\beta \wedge \bar{\beta}}{\|\beta\|^2_{h^N}} \right)^{\frac{1}{N}}$$

Since $v_\beta(h)$ is independent of β, this expression defines an adapted measure $v(h)$ with $\mathcal{C}^\infty$ density on V.

Let $h^{N\,\mathrm{sing}} = e^{-N\chi} h^N$ be a singular metric on $\omega_V^{[N]}$. The Chern–Weil form $c_1\left(\omega_X^N, h^{\mathrm{sing}\,N}\right)$ is then well defined as a (quasi-)positive current. Since $h^{\mathrm{sing}\,N}$ has locally $\mathcal{C}^\infty$+psh potentials χ, it is locally bounded above and the above formula defines a measure $v(h^{\mathrm{sing}}) = e^\chi v(h)$ on V such that $\frac{v(h^{\mathrm{sing}})}{v(h)} \in L^\infty_{\mathrm{loc}}$. In particular,

$$\frac{v(h^{\mathrm{sing}})}{\Omega^n} \in L^{1+\varepsilon}(V, \Omega^n) \text{ for } \varepsilon > 0 \text{ small enough.}$$

We have

$$c_1\left(K_X, h^{\mathrm{sing}}\right) = c_1\left(K_X, h\right) + dd^{\mathrm{c}}\chi,$$

where $c_1(K_X, h) := \frac{1}{N} c_1\left(\omega_X^N, h^N\right)$.

Definition 16.55. *Assume V has only log terminal singularities. An adapted measure on V is a positive Radon measure locally of the form $e^f v$, where f is locally given as the sum of a psh and a smooth function on V. An adapted measure is $\mathcal{C}^0$, $\mathcal{C}^\alpha$, $\mathcal{C}^\infty$ density if so is e^f.*

The definition has the virtue of generalizing the usual equivalence between smooth metrics on the canonical sheaf of a manifold and positive definite volume forms to singular metrics and log terminal spaces. This suggests the following:

Definition 16.56. *Let V be a $\mathbb{Q}$-Gorenstein Kähler normal n-dimensional complex space with only log terminal singularities. Let Ω be a semi-Kähler current with L^∞_{loc} potential and adapted Monge–Ampère measure. Let h be the singular metric on the canonical sheaf such that $\Omega^n = v(h)$. We define*

$$\mathrm{Ric}(\Omega) := -c_1(K_V, h),$$

where the equality is to be taken in the sense of currents.

Definition 16.57. *A Kähler metric Ω is a singular Kähler–Einstein metric if* $\mathrm{Ric}(\Omega) = c\Omega$ *for some $c \in \mathbb{R}$.*

16.4.2 Varieties of general type

Theorem 16.58. *Let V be a general type projective algebraic variety with only canonical singularities such that K_V is ample. Let h^N a smooth Hermitian metric on ω_V^N such that $\Omega = c_1(K_V, h)$ is a smooth Kähler form on V. There is a unique function $\varphi : V \to \mathbb{R}$ such that*

(1) *φ is Ω-psh.*

(2) *$\Omega + dd^c\varphi$ is a positive current with bounded potentials.*

(3) *$(\Omega + dd^c\varphi)^n = e^\varphi v(h)$.*

Thus $\Omega + dd^c\varphi$ is the unique singular Kähler–Einstein metric on V of negative curvature in the canonical class of V. The current $\Omega + dd^c\varphi$ is smooth on V^{reg}, where it defines a bona fide Kähler–Einstein metric.

Proof. Passing to a resolution $\pi : X \to V$, the problem boils down to solving a degenerate complex Monge–Ampère equation

$$(\omega + dd^c\psi)^n = e^\psi f dV,$$

where $\omega = \pi^*\Omega$, dV is a volume form on X, $f \in L^\infty(dV)$, and f can be locally written as $f = \prod_i |f_i|^{2r_i}$, where $(f_i = 0)$ are local equations for the exceptional divisors E_i.

We have seen in Chapter 12 that such an equation admits a unique bounded ω-psh solution ψ. Moreover, from Chapter it follows 14 that ψ is smooth in $X \setminus E$.

We infer as earlier that $\psi = \varphi \circ \pi$, where φ is bounded and Ω-psh on V, smooth in V^{reg} and is a singular Kähler–Einstein metric according to Definition 16.57. □

Remark 16.59. *Thanks to Theorem 16.26 and* [BCHM10]*, a projective algebraic manifold of general type X has a unique birational model V such that the above hypotheses hold. Thus we have a birational map $\pi : X \dashrightarrow V$ which is well defined outside an indeterminacy locus S of codimension ≤ 2. In particular, $\pi^*(\omega + dd^c\varphi)$ is a closed positive current on $X - S$ that extends to a closed positive current T on X itself. The current T defines a* KE *metric on $X - S$. It needs not be a singular* KE *metric on X though, since its potentials may have logarithmic poles on S, in fact algebraic singularities of the form $\alpha \log\left(\sum |f_i|^2\right) + O(1)$, with f_i holomorphic and $\alpha \in \mathbb{Q}_{>0}$. Moreover, T lies in the canonical class of X iff X is a smooth minimal model.*

Example 16.60. *A nodal sextic threefold in $\mathbb{P}^4$ is of general type, Gorenstein, terminal, is its own canonical model, has no smooth minimal model, and does not have quotient singularities.*

It is natural to study how such Kähler–Einstein metrics vary in holomorphic families. Schumacher has shown in [Sch12] that smooth Kähler–Einstein metrics vary plurisubharmonically in smooth families; it would be interesting to extend his result to the singular context.

16.4.3 Calabi–Yau varieties and pairs

16.4.3.1 $\mathbb{Q}$-Calabi–Yau varieties

Definition 16.61. *Let V be a Kähler space with only log terminal singularities. We say that V is a $\mathbb{Q}$-Calabi–Yau variety if there is some multiple N' of* index(X) *such that $H^0(V, \omega_V^{[N']}) = \mathbb{C}\alpha$, where α is a global generator of $\omega_V^{[N']}$.*

Theorem 16.62. *Assume V is a compact $\mathbb{Q}$-Calabi–Yau variety. Let Ω be a Kähler metric on V. Then there is a unique positive current with bounded potential $\Omega' = \Omega + dd^c\varphi$, such that*

$$(\Omega + dd^c\varphi)^n = C\, v_\alpha \text{ and } \sup_V \varphi = 0,$$

where $\int_V \Omega^n = C \int_V (-1)^n v_\alpha$.

Furthermore, if V is projective-algebraic and $[\Omega] \in NS_{\mathbb{R}}(V)$, then $\Omega + dd^c\varphi$ is smooth on V^{reg}, where it defines a bona fide Ricci flat metric.

Corollary 16.63. *In each Kähler class $\alpha \in H^{1,1}(V, \mathbb{R})$ there is a unique singular Ricci flat metric.*

Proof. This follows again from Chapter 12 and Chapter 14. □

Simple examples of such singular Kähler–Einstein metrics are to be found on midly singular hypersurfaces of the complex projective space:

Example 16.64. *A nodal quintic threefold in $\mathbb{P}^4$ is a $\mathbb{Q}$-Calabi–Yau variety which has canonical singularities* (*ordinary double points*).

16.4.3.2 Families of Calabi–Yau varieties The study of degenerations of families of Calabi–Yau manifolds is an interesting topic, with ramifications in several different branches of mathematics and physics (see [GHJ03], [KS01], and [YN10] for some motivations).

The following setting was considered by several authors (see notably [W97, RZ11, RZ13, Tos15]). Let

$$\pi : \mathcal{X} \to \Delta$$

be a flat projective family of n-dimensional varieties over the unit disk $\Delta \subset \mathbb{C}$, smooth over Δ^*, with $\mathcal{X}$ normal, $K_{\mathcal{X}}$ Cartier, and $K_{\mathcal{X}/\Delta} \sim \mathcal{O}_{\mathcal{X}}$.

It follows that $X_t = \pi^{-1}(t)$ is a Calabi–Yau manifold for $t \neq 0$. Let Ω be a nowhere vanishing holomorphic section of $K_{\mathcal{X}/\Delta}$ and let $\Omega_t = \Omega|_{X_t}$. Fix a relatively ample holomorphic line bundle $\mathcal{L} \to \mathcal{X}$ and set $L_t = \mathcal{L}|_{X_t}$.

It follows from the Calabi–Yau theorem that there exists a unique Kähler–Einstein metric in $c_1(L_t)$ for all $t \neq 0$, while Theorem 16.62 guarantees the existence of a unique singular Kähler–Einstein metric in $c_1(L_0)$, if the central fiber X_0 is not too singular.

It is natural to wonder what happens when $t \to 0$. The following is a combination of results from [W97, RZ11, RZ13, Tos15]:

Theorem 16.65. *Assume that the family $\pi : \mathcal{X} \to \Delta$ is relatively minimal. Then the following are equivalent:*

(1) *X_0 has at worst canonical singularities;*
(2) *there exists $C > 0$ such that for all $t \neq 0$, $\int_{X_t} i^{n^2/2} \Omega_t \wedge \overline{\Omega_t} \leq C$;*
(3) *there exists $C > 0$ such that for all $t \neq 0$, $i^{n^2/2} \Omega_t \wedge \overline{\Omega_t} \leq C \omega_t^n$.*

Any of these conditions implies that ω_t converges, as $t \to 0$, to the unique singular Ricci flat Kähler–Einstein $\omega_0 \in c_1(L_0)$.

The convergence here is smooth in X_0^{reg}. Rong and Zhang [RZ11] show that the metric spaces (X_t, ω_t) converge in the Gromov–Hausdorff sense to a compact metric space (Z, d_Z) which is isometric to the metric completion of $(X_0^{\mathrm{reg}}, \omega_0)$. It follows moreover from the work of Donaldson and Sun [DS14] that Z is homeomorphic to X_0.

In a related but slightly different direction, we mention the following problem: on a fixed compact Kähler manifold X, pick a nef class $\alpha_0 \in H^{1,1}_{\mathrm{nef}}(X, \mathbb{R})$ and approach it by a smooth path $(\alpha_t)_{0<t<1}$ of Kähler classes. Yau's theorem yields a unique Kähler–Einstein Ricci flat metric $\omega_t \in \alpha_t$ and the problem is to understand what happens as $t \to 0$.

There are many interesting results obtained by various authors, some of which can be proved using the tools we have developed in this book. We refer the interested reader to the articles [GW00, KS01, Tos09, Tos10b, GTZ13] for more information on this problem which is motivated by questions coming from mirror symmetry.

16.4.3.3 Singular Kähler–Einstein metrics on klt pairs

Definition 16.66. *Let* (V, Δ) *be a* klt *compact Kähler pair.*

The pair (V, Δ) *is said to be* $\mathbb{Q}$-CY*, if there is some multiple* N' *of* $\mathrm{index}(X, \Delta)$ *such that* $H^0(V, \mathcal{O}(N'(K_V + \Delta))) = \mathbb{C}\alpha$*, where* α *is a global generator of* $\mathcal{O}(N'(K_V + \Delta))$.

The pair (V, Δ) *is canonically polarized if* $K_V + \Delta$ *is ample.*

The existence of singular Kähler–Einstein metrics on klt pairs has been established in [EGZ09]:

Theorem 16.67. *Let* (V, Δ) *be a* klt *compact Kähler pair.*

If (V, Δ) *is* $\mathbb{Q}$-CY *it carries a singular Ricci flat metric with adapted volume form in any Kähler class of* V*, this current being smooth outside* $\Delta \cup V^{\mathrm{sing}}$ *if* V *is projective and the Kähler class is rational.*

If it is canonically polarized, it carries a unique singular KE *metric in the cohomology class of* $K_V + \Delta$*, regular outside* $\Delta \cup V^{\mathrm{sing}}$.

Proof. The existence and uniqueness of a weak solution follows, e.g., from the variational approach developed in Chapter 11. The regularity is a consequence of Theorem 14.14 (for a density $f \in L^p$, $p > 1$). □

We also recover the main result of Berman and Guenancia [BG14, Theorem A]:

Theorem 16.68. *Let* (V, Δ) *be a canonically polarized* (*semi*) *log canonical pair. It carries a unique singular Kähler–Einstein metric in the cohomology class of* $K_V + \Delta$*, regular outside* $\Delta \cup V^{\mathrm{sing}}$.

The proof is again a consequence of Theorem 14.14, but this time the density is no longer L^p (the KE potentials are singular along the log canonical locus).

One can similarly recover the construction of Kähler–Einstein metrics of Poincaré type (work of Auvray [Auv14]).

Berman and Guenancia conversely show that if a canonically polarized variety admits a singular Kähler–Einstein metric, then it has semi-log canonical singularities. On the other hand, Odaka [Oda12, Oda13] has shown that a canonically polarized variety has semi-log canonical singularities precisely when (X, K_X) is K-stable. Altogether this yields the following case of the Yau–Tian–Donaldson conjecture:

Theorem 16.69. [BG14, Theorem B] *Let* X *be a complex projective variety such that* K_X *is ample. Then* X *admits a Kähler–Einstein metric if and only if* (X, K_X) *is* K*-stable.*

16.4.4 Fano varieties We finally discuss briefly singular Kähler–Einstein metrics on Fano pairs, following [BBEGZ11].

Definition 16.70. *A* log Fano pair *is a* klt *pair* (X, D) *such that* X *is projective and* $-(K_X + D)$ *is ample.*

Let (X, D) be a log Fano pair. We fix a reference smooth strictly psh metric ϕ_0 on $-(K_X + D)$, with curvature ω_0 and adapted measure μ_0. We normalize ϕ_0 so that μ_0 is a probability measure. The volume of (X, D) is

$$V := c_1(X, D)^n = \int_X \omega_0^n.$$

For any current $\omega = \omega_0 + dd^c\varphi_\omega$ with finite energy, we set

$$\mu_\omega := \frac{e^{-\varphi_\omega}\mu_0}{\int_X e^{-\varphi_\omega}\mu_0}. \tag{16.4.1}$$

Definition 16.71. *A* Kähler–Einstein metric ω *for the log Fano pair* (X, D) *is a current with finite energy* ω *such that*

$$V^{-1}\omega^n = \mu_\omega. \tag{16.4.2}$$

Lemma 16.72. *A Kähler–Einstein metric* ω *is automatically smooth on* X_0*, with bounded potentials on* X*, and it satisfies*

$$\mathrm{Ric}(\omega) = \omega + [D]$$

on X_{reg}.

Here $[D]$ is the integration current on $D|_{X_{\mathrm{reg}}}$. Writing $\mathrm{Ric}(\omega)$ on X_{reg} implicitely means that the positive measure $\omega^n|_{X_{\mathrm{reg}}}$ corresponds to a singular metric on $-K_{X_{\mathrm{reg}}}$, whose curvature is then $\mathrm{Ric}(\omega)$ by definition.

Proof. Set $\varphi := \varphi_\omega$. The Kähler–Einstein equation (16.4.2) reads

$$(\omega_0 + dd^c\varphi)^n = e^{-\varphi+c}\mu_0$$

for some constant $c \in \mathbb{R}$. If we choose a log resolution $\pi : \widetilde{X} \to X$ of (X, D), the equation becomes

$$(\widetilde{\omega}_0 + dd^c\widetilde{\varphi})^n = e^{-\widetilde{\varphi}+c}\widetilde{\mu}_0,$$

where $\widetilde{\omega}_0 = \pi^*\omega_0$ is semipositive and big and $\widetilde{\varphi} = \varphi \circ \pi$ is such that $e^{-\widetilde{\varphi}} \in L^q$ for all finite q. It follows from Chapter 12 that $\widetilde{\varphi}$ is bounded on $\widetilde{X}$, and hence φ is bounded on X by the properness of π. The smoothness of φ on X_0 follows from Theorem 14.14.

Finally, on X_{reg} we have

$$\mathrm{Ric}(\omega) = -dd^c \log \omega^n = dd^c\varphi - dd^c \log \mu_0 = dd^c\varphi + \omega_0 + [D],$$

by the definition of the adapted measure μ_0 and the Lelong–Poincaré formula. □

The definition of a log Fano pair requires the singularities to be klt. This condition is in fact necessary to obtain Kähler–Einstein metrics on the regular part:

Proposition 16.73. *Let (X, D) be any pair with $-(K_X + D)$ ample. Let $\Omega \subset X_{\mathrm{reg}}$ be a Zariski open subset with complement of codimension at least 2, and assume the existence of a closed positive $(1,1)$-current ω on Ω with bounded potentials such that*

$$\mathrm{Ric}(\omega) = \omega + [D]$$

on Ω. Then (X, D) is necessarily klt. *We further have*

$$\int_\Omega \omega^n \leq c_1(X, D)^n,$$

with equality if and only if ω is the restriction to Ω of a Kähler–Einstein metric for (X, D).

Proof. Let ψ be the singular metric on $-K_\Omega$ corresponding to the measure ω^n, with curvature $dd^c\psi = \mathrm{Ric}(\omega)$. If we let ϕ_D be the canonical psh metric on the $\mathbb{Q}$-line bundle attached to $D|_\Omega$, such that $dd^c\phi_D = [D]$, we have by assumption $dd^c\psi = \omega + dd^c\phi_D$ on Ω, so that $\phi := \psi - \phi_D$ defines a psh metric on the $\mathbb{Q}$-line bundle $-(K_X + D)|_\Omega$, with curvature ω.

Now let σ be a local trivialisation of $r(K_X + D)$ for some positive $r \in \mathbb{N}$, defined on an open set $U \subset X$. If we denote by $|\sigma|_{r\phi}$ the length of σ with respect to the metric induced by $r\phi$, then $u := \log |\sigma|^2_{r\phi}$ is a psh function on $U \cap \Omega$, hence it automatically extends to a psh function on U by normality.

This means, on the one hand, that ψ extends to a globally defined psh metric on $-(K_X + D)$, so that its curvature ω satisfies

$$\int_\Omega \omega^n \leq c_1(X, D)^n.$$

On the other hand, unravelling the definitions yields

$$(\sigma \wedge \overline{\sigma})^{1/r} = e^{u/r}\omega^n \tag{16.4.3}$$

on $V \cap \Omega$. Since u is in particular bounded above, this shows that $(\sigma \wedge \overline{\sigma})^{1/r}$ has locally finite mass near singular points of U, so that (X, D) is klt.

The equality $\int_\Omega \omega^n = c_1(X, D)^n$ means that ω has finite energy. The last assertion follows from Lemma 16.72. □

There has been important recent works on degeneration of families of Kähler–Einstein metrics of positive curvature [CDS14, CDS1, CDS2, CDS3, Tian15], as well as on the study of compactified moduli spaces of Kähler–Einstein Fano manifolds [OSS16].

16.5 Exercises

Exercise 16.1. *Construct an example of a compact Riemann surface S and a holomorphic line bundle L on S which is ample and such that $H^0(S, L) = 0$.*

Exercise 16.2. *Let $X = \mathbb{C}^2/\Lambda$ be a compact complex torus of complex dimension 2. Show that one can choose Λ in such a way that there exists a class $\alpha \in H^{1,1}(X, \mathbb{R})$ which is semi-positive, but not semi-ample. Is this class big?*

Exercise 16.3. *Let X be a compact Kähler manifold and pick $\alpha \in H^{1,1}(X, \mathbb{R})$. Show that if α is* nef *and big, there exists a positive closed current $T \in \alpha$ with zero Lelong numbers at all points $x \in X$.*

Exercise 16.4. *Let $\pi : X \to \mathbb{P}^2$ be the blow-up of $\mathbb{P}^2$ at one point p. Let F denote the strict transform of a line passing through p and E denote the exceptional divisor. The corresponding cohomology classes generate $H^{1,1}(X, \mathbb{R})$, we still denote them by F, E. Show that*

(i) $H^{1,1}_{\text{psef}}(X, \mathbb{R}) = \{aF + bE,\ 0 \le a, b\}$;

(ii) $H^{1,1}_{\text{big}}(X, \mathbb{R}) = \{aF + bE,\ 0 < a, b\}$;

(iii) $H^{1,1}_{\text{nef}}(X, \mathbb{R}) = \{aF + bE,\ 0 \le a \text{ and } 0 \le b \le a\}$;

(iv) $H^{1,1}_{\text{Kahl}}(X, \mathbb{R}) = \{aF + bE,\ 0 < a \text{ and } 0 < b < a\}$.

Exercise 16.5. *Let v be a subharmonic function in $\mathbb{C}$ with logarithmic growth. Show that if v is periodic then it is constant.*

Exercise 16.6. *Let $\pi : X_r \to \mathbb{P}^2$ be the blow-up of $\mathbb{P}^2$ at r generic points. Set $H = \pi^*\mathcal{O}(1)$, $E = \sum_{i=1}^r E_i$ and*

$$t_{\text{nef}} := \min\{t \mid tH - E \text{ is nef}\},\quad t_{\text{psef}} := \min\{t \mid tH - E \text{ is psef}\}.$$

(1) *Show that $t_{\text{nef}} t_{\text{psef}} = r$ and $t_{\text{psef}} \le \sqrt{r} \le t_{\text{nef}}$.*
(2) *Compute the values of these numerical invariants for r "small".*

Exercise 16.7. *Let S be a complex surface with terminal singularities. Show that S is smooth (see* [Kollar] *for some help).*

Exercise 16.8. *Let S be a normal algebraic surface. Show that the following are equivalent:*

(1) *S has only canonical singularities.*
(2) *S is locally analytically isomorphic to $X = \mathbb{C}^2/G$, where $G \subset \mathrm{SL}_2(\mathbb{C})$ is a finite subgroup.*
(3) *The exceptional divisors of the minimal resolution $\pi_{\min}$ of S, have simple normal crossings, their components are (-2) smooth rational curves, their incidence graphs are of type A, D, or E.*

(*see* [Kollar] *for some help*).

Exercise 16.9. *Let $H \subset \mathbb{CP}^{n+1}$ be a smooth hypersurface of degree d. Let C denote the affine cone over H,*

$$C = \{z \in \mathbb{C}^{n+2} \setminus \{0\} \,\big|\, [z] \in H\} \cup \{0\}.$$

Show that C has only one singular point (the origin) and that it is a canonical singularity if and only if $d \leq n+1$.

Exercise 16.10. *Let V be an algebraic $\mathbb{Q}$-Gorenstein variety of finite index N. Show that the following conditions are equivalent:*

(1) *There exists a resolution $\pi : X \to V$ such that*

$$K_X \cong \pi^* K_V + \sum a_E E \quad \text{with } a_E \geq 0.$$

(2) *For all resolutions $\pi : X \to V$,*

$$K_X \cong \pi^* K_V + \sum a_E E \quad \text{with } a_E \geq 0.$$

Here $\cong$ means numerical equivalence of $\mathbb{Q}$-Cartier divisors and the sum runs over the exceptional divisors of π.

Exercise 16.11. *Assume $V = \mathbb{CP}^2$ and $\Delta = (xy^2 = w^3) \subset \mathbb{CP}^2$ is a cuspidal cubic. Show that $(V, a\Delta)$ is* klt *if and only if $a < 5/6$.*

Exercise 16.12. *Let X be a normal complex analytic space. A function is plurisubharmonic if it is locally the restriction of an ambient plurisubharmonic function in a local embedding of X in $\mathbb{C}^N$.*

A function φ is weakly plurisubharmonic if it is upper semi-continuous and for every holomorphic disk $f : \Delta \to X$, the function $\varphi \circ f$ is subharmonic (allowing here $\varphi \circ f \equiv -\infty$).

Show that a function is plurisubharmonic if and only if it is weakly plurisubharmonic (see [FN80, Section 5]).

Exercise 16.13. *Let Y be a smooth compact Kähler manifold. Let $X \to Y$ be the blow-up of Y at some point p. Let T be a positive closed current of bidegree $(1,1)$ on Y.*

(1) *Show that the Lelong numbers of the current $\pi^* T$ satisfy, for all $x \in X$,*

$$\nu(T, \pi(x)) \leq \nu(\pi^* T, x) \leq 2\nu(T, \pi(x)).$$

(2) *Show that for $x \in E = \pi^{-1}(p)$ generic,*

$$\nu(\pi^* T, x) = \inf_{y \in E} \nu(\pi^* T, y) = \nu(T, p).$$

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Index

(φ,ψ)-capacity, 335–337
  outer, 336
$(\mathrm{DMA}_v^\varepsilon)$, 343
$(\mathrm{DMA}_v^\varepsilon)_+$, 343
$(\mathrm{DMA}_{v|z}^\varepsilon)$, 343
$(\mathrm{DMA}_{v|z}^\varepsilon)_+$, 344
K-energy of Mabuchi, 300
K-stable, 412–414, 446
$PSH(X,\omega)$-polar, 240, 246–248, 317
Ω-psh, 434, 436, 443
$\mathbb{Q}$-CY, 446
$\mathbb{Q}$-Calabi–Yau variety, 444, 445
$\mathbb{Q}$-factorial, 428, 429
$\mathrm{Ric}(\Omega)$, 443
$\mathrm{Ric}(\omega)$, 210, 211, 367, 369, 389, 392–395, 407, 413, 447
$\mathcal{F}_{\sigma\delta}$-subset, 104
$\mathcal{F}_\sigma$-set, 104
$\mathcal{G}_\delta$-set, 104
$\mathcal{K}$-analytic subset, 105–107
$\mathcal{K}_{\sigma\delta}$ space, 104
$\mathcal{K}_\sigma$ space, 104
ω-psh, 215, 217, 218, 220, 224, 226–228, 236, 240–242, 252, 257, 258, 278, 280, 289, 298, 306, 310, 330, 345, 352, 354, 355, 378, 400, 401
$\mathbb{Q}$-Gorenstein variety, 428–430, 438
$\mathrm{DMA}(X,\omega)$, 280, 281
$\mathrm{DMA}_{\mathrm{loc}}(\mathbb{B})$, 279

abelian varieties, 193
abundance conjecture, 423
adapted measure, 438, 439, 442, 443, 447
adjunction formula, 202, 406
Alexander–Taylor capacity, 245, 248, 315
ample, 201, 202, 220, 231, 311, 332, 333, 356, 363, 394, 406, 412, 420, 426, 435, 445–447
ample locus, 231, 311, 332, 333, 356, 363, 426
Aubin–Mabuchi functional, 403, 404

base point free, 200
Bergman kernel, 402, 403, 405, 406
Bergman metric, 403
Bergman space, 221, 402
big, 215, 231, 240, 257, 289, 317, 333, 363, 419–422, 426
bimeromorphic, 195, 201, 429
blow-up, 193–195, 300, 411, 423
Bochner formula, 406, 407
Bombieri–Skoda–Hörmander theorem, 60
Borel measure, 14, 16–18, 26, 47, 71, 77, 84, 93, 101, 103, 108, 109, 140

Calabi conjecture, 379, 392
Calabi functional, 389, 390
Calabi's trick, 305
Calabi–Yau theorem, 392, 393, 445
canonical line bundle, 196, 202, 394
canonical singularities, 426, 430, 431, 443, 446
capacitable, 103, 108, 123
capacity, 101, 102, 107, 108, 110–114, 119, 123, 125, 126, 238–244, 248, 249, 278, 316, 322, 323, 381
Cartier divisor, 428, 430–432
Cauchy–Riemann equation, 3, 19, 61
Chebyshev constants, 250
Chern–Levine–Nirenberg inequality, 80, 108, 110, 114, 116, 236–238, 303, 331
Chern–Weil form, 434, 435, 442
Choquet capacity, 101–104, 119, 123, 243
Choquet's capacitability theorem, 104, 107

Choquet's lemma, 117, 237, 243, 244
classical solution, 161, 162, 165, 166, 346
compact singularity, 86, 95
comparison principle, 12, 92, 93, 120, 124, 131, 139, 149–155, 161, 166, 172, 173, 178, 179, 182, 183, 236, 257, 259, 281, 315, 343, 345, 346, 351, 358, 360, 436
complete pluripolar, 123
complex analytic space
  normal, 427–429
  reduced, 427, 436
complex manifold, 191, 192, 194, 195, 201, 205
complex Monge–Ampère operator, 75, 80, 86, 87, 89, 92, 96, 108, 110, 112, 114, 131, 139, 152, 174, 176, 179
convergence in energy, 273, 277, 278
current, 3, 14, 15, 39–43, 46–52, 55–57, 60, 64, 66, 68, 75–80, 82, 83, 85, 87, 89, 90, 111, 114, 116

degenerate ellipticity condition, 161, 163, 164, 166
Del Pezzo surface, 300, 406, 411
Demailly estimates, 228
determinent bundle, 196
Dirichlet problem, 7, 18, 108, 119, 131, 145, 147, 149–152, 154, 161, 162, 235, 246, 352, 398, 400
DirMA, 7, 18
divisor, 194, 197, 198, 206, 413, 420, 423, 427–431, 438, 441
domination principle, 95, 262, 297, 305, 309, 335, 337
Donaldson–Futaki invariant, 412

effective, 198, 206, 379, 420, 422, 441
elliptic curve, 193, 202, 393, 424
equilibrium measure, 123, 244
evaluation functional, 23

Fano manifold, 225, 299, 395, 406–411, 413
finite energy class, 173, 258, 335
first Chern class, 198, 206, 210, 219, 225, 250, 298, 299, 323, 367, 389, 392, 402, 406, 421
Fubini–Study Kähler form, 204, 218, 223
fundamental solution, 15, 86, 133
Futaki character, 408, 409, 411

geodesic equation, 396–399
geodesic symmetry, 398
globally generated, 420
gluing construction, 20
Gorenstein variety, 427, 428
Green function, 17, 68, 133
Green potential, 17, 108

Hadamard space, 402
Hamiltonian, 162, 164, 166, 177
harmonic function, 3–8, 17, 18, 34, 92, 132
Harnack's inequality, 5, 6, 30, 31
Hartogs lemma, 27, 28, 217
Hermitian metric, 207, 208, 380, 434, 435, 442, 443
Hodge index theorem, 206
holomorphic, 3, 50, 191, 192, 194, 196–201, 205, 207, 208, 210, 219, 221, 250, 380, 390, 400, 408, 410, 411, 420, 424, 427, 428, 430
holomorphic section, 199, 200, 219, 221, 225, 380, 419, 435, 445
holomorphic sectional curvature, 210
holomorphic tangent space, 191, 196
holomorphic vector bundle, 196, 209, 424
Hopf surface, 193, 206

Itaka dimension, 199, 201
Izumi's theorem, 439

Jensen–Lelong formula, 66–68
jet (2-jet), 167, 168
Josefson's theorem, 125, 240, 248, 317

Kähler metric, 203, 207, 211, 225, 228, 345, 390, 391, 393, 394, 396, 413, 414, 433–435, 443, 444
Kähler potential, 395, 396, 403, 433, 434
Kähler–Einstein metric, 333, 363, 389, 393–395, 409–411, 413, 414, 426, 442, 443, 446–448
Kähler form, 199, 203–205, 210, 218–220, 225, 228, 229, 267, 274, 299, 305, 323, 353, 367, 380, 381, 389, 391–395, 421, 426, 433, 437
Kiselman's minimum principle, 61
Kiselman–Legendre transform, 230, 231
klt, 431, 438, 439, 446–448
Kodaira embedding theorem, 200

Laplace equation, 3, 7, 131, 132, 134
Laplace operator, 3, 52, 86, 131–133, 136, 139
Lelong class, 95, 96
Lelong number, 52, 54–56, 58, 59, 63, 64, 69
line bundle, 198, 199, 201, 202, 219, 220, 225, 323, 412, 420, 425, 426
  holomorphic, 196–201, 380, 402, 419, 421, 422, 424, 434, 435, 445
linearly equivalent, 198
Liouville's Theorem, 54
locally factorial, 428, 429
locally symmetric space, 398
log canonical threshold, 225
log Fano pair, 447, 448
log-terminal singularities, 430
logarithmic potential, 16, 17
logarithmic potential theory, 13, 17
lower test, 165, 166, 168, 177, 182
Lundin's formula, 249

Mabuchi functional, 391, 392, 396, 403, 405, 408
maps, 191, 194–196, 403
maximal plurisubharmonic function, 92
maximum principle, 6, 7, 11, 18, 92, 94, 132, 133, 135, 137, 140, 161, 165, 166, 169, 180, 181
mean value, 3, 4, 10, 52, 132, 251, 262
meromorphic, 194, 195, 198, 427, 430, 438
metric
  extremal, 389, 390
  positive, 201, 219–221, 250
  singular, 219, 221
  smooth, 198, 199, 201, 219, 220, 225, 250
metric completion, 402, 445
Minimal Model Program, 202, 413, 419, 423, 426, 431
Moishezon variety, 435
Monge–Ampère capacity, 101, 108, 119, 124, 237, 238, 243, 315, 322, 324, 336
Monge–Ampère mass, 96, 119
Monge–Ampère measure, 75, 78, 84, 86, 88, 91, 92, 95, 96, 170
Montel property, 27, 29
Mori program, 419, 426

Nagata conjecture, 424
Nakai–Moishezon criterion, 420, 422
nef, 202, 420–426, 431, 445
Newton kernel, 133
normal coordinates, 208, 229

pairs, 412, 431, 438, 444, 446, 447
Perron–Bremermann envelope, 131, 135, 139–141, 143, 151
plurisubharmonic function, 2, 3, 13, 18–24, 27, 28, 30, 36, 49, 52, 54, 55, 60, 61, 64, 66, 68, 69, 78, 82–84, 86, 87, 89–92, 101, 108, 110, 112–116, 118, 119, 122–127, 129, 132, 140, 141, 150, 152, 164, 171, 173, 176
Poisson
  equation, 375
  formula, 5, 7, 17, 18
  kernel, 7, 18, 133

transform, 17, 131, 134
Poisson–Jensen formula, 17, 18, 30, 32, 57, 67, 68, 131
Poisson–Szegö formula, 68, 69
polar locus, 101, 110, 127
positive current, 50, 203–205, 207, 215, 235, 306, 333, 426, 433–436, 439, 442, 444
positive definite adapted measure, 438
precapacity, 102–104
projection theorem, 298
projective algebraic variety, 192, 435, 443
psef (pseudoeffective), 207, 422, 423, 441
pseudoconvex, 60

quasi-continuous function, 111, 114

rational, 195, 197, 423, 431, 432
relative (φ, ψ)-extremal function, 336
relative extremal function, 117–119, 242, 243
resolution of singularities, 427, 429, 430, 439, 440
Ricci curvature, 210, 211, 367, 389, 392, 394, 419, 442
Riemann surface, 193, 202, 269, 390, 393, 399, 426
Riemannian metric, 207, 229, 396
Riesz measure, 14, 15, 17, 30, 52, 56, 64, 71, 134
Riesz representation formula, 13, 16–18

scalar curvature, 211, 389–391, 393, 411, 414
semi-ample, 225, 420, 422, 423
set, 2, 8, 9, 21, 123
exceptional, 91, 115
level sets, 60
pluripolar, 21, 91, 101, 123–125, 127
Siciak's extremal function, 245, 249
Skoda's integrability theorem, 64, 69
slope, 53, 439, 441
strong topology, 276, 277
subharmonic function, 3, 8–13, 17–19, 27, 80, 86, 132–134
submean value inequality, 3, 8, 20, 21, 119, 135
supersolution, 161, 162, 165, 166, 177, 182, 345, 346, 351–353
symmetric space, 210, 397, 403

Tian's α-invariant, 224

uniformization theorem, 210, 389
uniruled, 423
upper test, 134, 135, 137–139, 165, 168

very ample, 201, 420
viscosity comparison principle, 179, 345, 346, 351, 359, 360
viscosity solution, 161, 165, 168, 170, 182, 183, 343, 345, 346, 351, 353, 356, 360
viscosity subsolution, 165, 166, 168, 171, 173, 344, 346, 352
viscosity supersolution, 165, 166, 168, 178

Weil divisor, 428

Tracts in Mathematics 26

Vincent Guedj
Ahmed Zeriahi

Degenerate Complex Monge–Ampère Equations

Complex Monge–Ampère equations have been one of the most powerful tools in Kähler geometry since Aubin and Yau's classical works, culminating in Yau's solution to the Calabi conjecture. A notable application is the construction of Kähler–Einstein metrics on some compact Kähler manifolds. In recent years degenerate complex Monge–Ampère equations have been intensively studied, requiring more advanced tools.

The main goal of this book is to give a self-contained presentation of the recent developments of pluripotential theory on compact Kähler manifolds and its application to Kähler–Einstein metrics on mildly singular varieties. After reviewing basic properties of plurisubharmonic functions, Bedford–Taylor's local theory of complex Monge–Ampère measures is developed. In order to solve degenerate complex Monge–Ampère equations on compact Kähler manifolds, fine properties of quasi-plurisubharmonic functions are explored, classes of finite energies defined and various maximum principles established. After proving Yau's celebrated theorem as well as its recent generalizations, the results are then used to solve the (singular) Calabi conjecture and to construct (singular) Kähler–Einstein metrics on some varieties with mild singularities.

The book is accessible to advanced students and researchers of complex analysis and differential geometry.

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